# MINIMIZERS OF THE NORMALIZED VOLUME ON SYMPLECTIC SINGULARITIES

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Abstract. We prove that the minimizer of the normalized volume on any symplectic singularity is, up to scaling, the valuation of a Kollár component whose log discrepancy equals the dimension of the singularity or half of it and whose log Fano pair is K-polystable. On the index-one cover of the Kollár component, the residue of the symplectic form is an orbifold contact form in the sense of reflexive differential forms, so that the Kollár component, with the orbifold structure given by the index-one cover, is a contact Fano orbifold. We thus solve Problem 25 of Xu–Zhuang's open problem list on K-stability of Fano varieties and confirm the conjecture of Namikawa and Odaka that the minimizer is a multiple of a divisorial valuation.

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## 1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

A *symplectic singularity* is a germ $(X \ni x)$ of a normal variety whose regular locus $X_{\mathrm{reg}}$ carries a closed non-degenerate holomorphic 2-form $\omega$ such that the pullback of $\omega$ to a resolution of $X$ extends to a holomorphic 2-form [Bea99, Definition 1.1] (see Definition 2.9). Symplectic singularities are rational and Gorenstein, and thus klt [Bea99, Proposition 1.3 and footnote 1]. An affine symplectic variety is *conical* if its coordinate ring $R$ carries a grading $R = \bigoplus_{j\geq 0} R_j$ with $R_0 = \mathbb{C}$ for which $\omega$ is homogeneous of some degree $r > 0$, called the *weight* of $\omega$ [XZ26a, Paragraph 19]. Among the klt singularities, symplectic singularities form a class on which the local theory of K-stability is expected to have a particularly rigid form. The normalized volume of valuations, introduced in [Li18], is the basic invariant of that theory.

Xu and Zhuang asked whether the minimizer of the normalized volume on a symplectic singularity is a Kollár component that carries the structure of a contact Fano orbifold [XZ26a,

*Date*: October 26, 2026.

2020 *Mathematics Subject Classification.* 14B05, 14J17, 14J45, 32Q20, 53C26, 53D10.

*Key words and phrases.* symplectic singularity, normalized volume, Kollár component, K-polystability, contact Fano orbifold, Ricci-flat Kähler cone metric.

Problem 25]. To formulate the question, we use the normalized volume $\widehat{\mathrm{vol}}$ and its minimizers (Definition 2.5), Kollár components, their extractions, and their index-one covers (Definition 2.2), orbifold contact forms and contact Fano orbifolds (Definition 2.12), and K-polystability of log Fano pairs [LWX18, Definitions 2.26–2.28] (see Section 2.4).

**Question 1.1** ([XZ26a, Problem 25])**.** Let $(X \ni x)$ be a symplectic singularity of dimension $2n$ with symplectic form $\omega$.

(1) (Rank one) Is every minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$ a positive multiple of $\mathrm{ord}_E$ for some Kollár component $E$ over $(X \ni x)$?
(2) (Weight) If so, is $A_X(E) = rn$ for some $r \in \{1, 2\}$?
(3) (Contact structure) For $E$ and $r$ as in (2), let $f : Y \to X$ be the extraction of $E$, let $q : \mathcal{Y} \to Y$ be the index-one cover of $E$, and let $\mathcal{E} := q^*E$ and $\pi := f \circ q$. Is $\pi^*\omega$ a section of $\Omega^2_{\mathcal{Y}}(\log \mathcal{E}) \otimes \mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$ whose residue along $\mathcal{E}$ is an orbifold contact form with contact line bundle $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}$?
(4) (K-polystability) For $E$ as in (1), is the log Fano pair $(E, \Delta_E)$ K-polystable?
(5) (Contact Fano orbifold) In particular, does $(X \ni x)$ admit a Kollár component $E$ such that $(\mathcal{E}, L, \theta)$ is a contact Fano orbifold for some line bundle $L$ on $\mathcal{E}$ and some orbifold contact form $\theta$ (Definition 2.12)?

Part (1) of Question 1.1 is a conjecture of Namikawa and Odaka [NO25, Conjecture 4.11], which predicts that the minimizer of $\widehat{\mathrm{vol}}$ on a symplectic singularity is a multiple of a divisorial valuation. In the discussion after [XZ26a, Problem 25], Xu and Zhuang explain the expectations behind parts (1) and (2). A minimizer as in part (1) corresponds to the case of a quasi-regular Reeb vector (Definition 2.6) in the literature on Sasaki–Einstein metrics, and part (1) is suggested by the expectation that the Sasaki–Einstein metric on the link of the K-polystable degeneration of the singularity (recalled below) is 3-Sasakian. On a compact 3-Sasakian manifold, the three Reeb vector fields generate a locally free action of $\mathrm{Sp}(1)$, and the space of leaves of each Reeb foliation is a compact orbifold [BG99, §1.2 and Theorem 2.2.2]. For a symplectic singularity, the link of that degeneration is compact but can be singular along a set of positive dimension (for $\mathbb{A}^2 \times (\mathbb{A}^2/\{\pm 1\})$, along the image of $\mathbb{A}^2 \times \{0\}$), so the regular part of the link is not compact and the argument of [BG99, §1.2] for compact 3-Sasakian manifolds does not carry over.

Part (2) predicts a canonical $\mathbb{G}_m$-action for which the weight of $\omega$ is 1 or 2 and which may differ from a given conical structure. Xu and Zhuang add that part (2) could be too optimistic, although they know no counterexample. For a surface singularity $\mathbb{C}^2/G$ with $G \subset \mathrm{SL}_2(\mathbb{C})$ finite, we have $r = 2/|I|$, where $I$ is the kernel of the homomorphism $G \to \mathrm{PGL}_2(\mathbb{C})$, see the discussion after [XZ26a, Problem 25].

Question 1.1 refines [XZ26a, Problem 20], which asks whether the stable degeneration of a symplectic singularity is again symplectic, with the degenerated form. A positive answer to Question 1.1 is also the starting point of [XZ26a, Problems 27–31] on Hamiltonian reduction, symmetry of weights, quaternionic structures, and short star-products, which assume the minimizer predicted by Question 1.1.

We now explain what the general theory of normalized volumes implies for a klt singularity $(X \ni x)$. The following results form the stable degeneration theorem [XZ26b, Theorem 2.1]:

(1) a minimizer of $\widehat{\mathrm{vol}}$ exists [Blu18, Main Theorem];
(2) for a prime divisor $E$ over $X$ with center $x$, the valuation $\mathrm{ord}_E$ is a minimizer if and only if $E$ is a Kollár component and $(E, \Delta_E)$ is K-semistable, and a divisorial minimizer is unique among divisorial valuations up to scaling [LX16, Theorem 1.2];
(3) any minimizer is quasi-monomial [Xu20, Theorem 1.2];
(4) the minimizer is unique up to scaling [XZ20, Theorem 1.1];
(5) for a minimizer $v$, the associated graded ring $\mathrm{gr}_v \mathcal{O}_{X,x}$ is finitely generated [XZ25, Theorems 1.1–1.2], so $X$ degenerates to the K-semistable Fano cone $W := \mathrm{Spec}\, \mathrm{gr}_v \mathcal{O}_{X,x}$ (Section 2.2).

A K-semistable Fano cone further degenerates to a unique K-polystable Fano cone [LWX18, Theorem 1.2]. We refer to [LLX20] for a survey of the theory, and to [Wan25, Corollary 1.5] for a comparison of K-polystability of log Fano cones with respect to special and to normal test configurations.

For Question 1.1, the general theory leaves two gaps. First, the minimizer is only known to be quasi-monomial, and the rational rank of the minimizer can be greater than one. Indeed, the cones of the irregular Sasaki–Einstein metrics found by Gauntlett, Martelli, Sparks, and Waldram, see [CS15, Section 1], are Ricci-flat Kähler cones with irrational Reeb vector fields. They are K-polystable by [LWX18, Corollary A.4], and the valuation induced by the Reeb vector field minimizes $\widehat{\mathrm{vol}}$ by [LX18, Theorem 3.5] (see also [LWX18, Theorem 2.5]). On a conical symplectic singularity, the canonical torus of [XZ26b, Theorem 1.2 and the paragraph after it], which is generated by the Reeb vector of the minimizer of $\widehat{\mathrm{vol}}$ and under which $\omega$ is homogeneous (see [XZ26b, Theorem 6.2]), can a priori have rank greater than one. Thus part (1) of Question 1.1 does not follow from local K-stability alone, and a proof has to use $\omega$. Second, the K-polystable degeneration of [LWX18, Theorem 1.2] need not preserve the symplectic form. We explain the second gap after recalling the known results on symplectic singularities.

Kaledin conjectured that any symplectic singularity is formally isomorphic to a conical symplectic singularity [Kal06, Remark 4.2, §4], [Kal09, Conjecture 1.8]. Namikawa and Odaka proved Kaledin's conjecture for germs on polarized projective symplectic varieties admitting a projective symplectic resolution, equivalently a polarized smoothing [NO25, Theorem 1.1], by combining metric tangent cones in the sense of Donaldson and Sun [DS15] with Poisson deformation theory. They also proved it for germs carrying a singular hyperkähler metric that satisfies [NO25, Conjecture 1.3], see [NO25, Theorem 1.4]. Their rigidity theorem [NO25, Theorems 3.3 and 3.5] is reproved algebraically in [XZ26b, Theorem 4.1] in the following form: if the degeneration of $X$ induced by a Kollár component carries a $\mathbb{G}_m$-equivariant relative 2-form that is symplectic on any fiber, then $(X \ni x)$ and the vertex of the central fiber are formally isomorphic as symplectic singularities.

The stable degeneration of symplectic singularities is studied in [XZ26b]. By [XZ26b, Theorem 5.11], the degeneration of $X$ to $W$ specializes any non-degenerate reflexive form to a non-degenerate reflexive form on $W$, which answers [XZ26a, Problem 20]. Moreover, Kaledin's conjecture holds for symplectic singularities whose symplectic form is algebraic: any such singularity is formally isomorphic to a conical symplectic singularity [XZ26b, Corollary 6.3]. Any conical symplectic singularity is a K-semistable Fano cone for the Reeb vector of the minimizer of $\widehat{\mathrm{vol}}$ [XZ26b, Corollary 6.3] (see [XZ26b, Theorem 6.2] for a stronger form), and two conical symplectic singularities that are formally isomorphic are algebraically isomorphic [XZ26b, Proposition 6.5]. It is further conjectured in [XZ26b, Conjecture 1.3] that any conical symplectic singularity is a hyperkähler cone.

The preservation of $\omega$ in [XZ26b, Theorem 5.11] depends on an equality case. [XZ26b, Theorem 5.10] bounds from below the order of a reflexive $p$-form, along a Kollár component $E$ with $(E, \Delta_E)$ K-semistable and at the minimizer of $\widehat{\mathrm{vol}}$, by $p/\dim X$ times the log discrepancy. For $\omega$, equality holds at the minimizer, and the equality persists for valuations near the minimizer in a cone of quasi-monomial valuations [XZ26b, (5.3)–(5.4)], [XZ26b, Proposition 3.12]. The equality controls only the first degeneration of $X$ to $W$. The second degeneration of $W$ to a K-polystable Fano cone [LWX18, Theorem 1.2] goes in another direction, and the limit of $\omega$ along it can be degenerate.

For example, consider the product degeneration of $\mathbb{A}^4$ induced by the $\mathbb{G}_m$-action with weights $(1,1,1,2)$ on the coordinates $x_1, \dots, x_4$, whose central fiber is again $\mathbb{A}^4$ (see Section 2.4 for degenerations of Fano cones). For a symplectic form $\omega = \sum_{i<j} a_{ij} dx_i \wedge dx_j$ with constant coefficients $a_{ij}$, the normalized limit of $\omega$ is the part of lowest weight, $\sum_{1\le i<j\le 3} a_{ij} dx_i \wedge dx_j$. The part of lowest weight is non-zero, since otherwise $\omega = \sum_{i\le 3} a_{i4} dx_i \wedge dx_4$ would have rank at most 2. As it involves only $dx_1$, $dx_2$, and $dx_3$, it has rank 2, and thus it is degenerate on $\mathbb{A}^4$.

Therefore tracking $\omega$ along the second degeneration and applying the rigidity theorem does not lead to part (4) of Question 1.1, and this is the main difficulty of part (4).

We now turn to contact geometry, whose relation with symplectic singularities is classical. LeBrun proved that a Fano contact manifold is a twistor space if and only if it admits a Kähler–Einstein metric, and he suggested that any Fano contact manifold is a twistor space [LeB95, Theorem A and the paragraph after it]. A Fano contact manifold for which the linear system of the contact line bundle defines a generically finite rational map and whose group of contact automorphisms is reductive is an adjoint variety [Bea98, Theorem 0.1].

For a contact variety $Z$ with contact line bundle $L$ in the sense of [Smi22, Definition 2.1], the complement of the zero section in the total space of $L^*$ is a symplectic variety on which the symplectic form has weight 1 for the natural $\mathbb{G}_m$-action, and conversely [Smi22, Theorem 2.5]. If moreover $\dim Z = 2n-1$, $Z$ is Fano with $-K_Z = nL$, and $L \sim rH$ for an ample Weil divisor $H$ on $Z$, then the affine cone over $Z$ with respect to $H$ is a conical symplectic singularity of weight $r$ [XZ26a, Paragraph 22]. Conversely, the quotient of a conical symplectic variety minus its vertex by $\mathbb{G}_m$ is a contact orbifold [Nam11, Theorem (4.4.1)], and over the open set where the quotient map is a $\mathbb{G}_m$-bundle its contact line bundle is the $r$-th power of the tautological line bundle, where $r$ is the weight of $\omega$ [Nam11, (4.3)].

A contact structure alone does not force K-stability. The weighted projective space $\mathbb{P}(a_1, \dots, a_{2n})$ with

$$a_1 + a_2 = a_3 + a_4 = \cdots = a_{2n-1} + a_{2n}$$

is a contact Fano orbifold, and it is not K-semistable unless $a_1 = \cdots = a_{2n}$ [XZ26a, Paragraph 22]. The smallest example is the case $n = 1$ and $(a_1, a_2) = (1, 2)$, that is, $\mathbb{P}(1, 2)$, which appears in the weighted blow-up of $\mathbb{A}^2$ with weights $(1, 2)$. The exceptional divisor $E$ of the weighted blow-up is a Kollár component over $(\mathbb{A}^2 \ni 0)$ which, with the orbifold structure given by its index-one cover, is the contact Fano orbifold $\mathbb{P}(1, 2)$, and $(E, \Delta_E)$ is $\mathbb{P}^1$ with one point of coefficient $\frac{1}{2}$, which is not K-semistable by the case $n = 1$ of [XZ26a, Paragraph 22]. Thus part (4) of Question 1.1 concerns the minimizer, while part (5) only asks for the existence of some Kollár component.

Before this paper, parts of Question 1.1 were known in the following cases.

(1) Quotient singularities $\mathbb{C}^{2n}/G$ with $G \subset \mathrm{Sp}_{2n}(\mathbb{C})$ finite. The minimizer of $\widehat{\mathrm{vol}}$ is induced by the quotient of the exceptional divisor of the blow-up of $0 \in \mathbb{C}^{2n}$ [LX16, Example 7.1(1)], see also [LX17, Theorem 2.7(1)].

(2) Normalizations of nilpotent orbit closures. The scaling action induces the minimizer, and the quotient $(Z, \Delta_Z)$ of the cone minus its vertex by the scaling action, with the boundary given by its orbifold structure (the quotient log Fano pair of Definition 2.6), is K-semistable [XZ26b, Theorem 7.5], by the criterion [XZ26b, Lemma 7.3] on the center of a reductive group acting on the cone. The weight of the Kirillov–Kostant form $\omega$ for the scaling action is 1 [XZ26b, Section 7.2, the paragraph before Lemma 7.4], and the scaling action on the normalization is faithful, since it is faithful on the orbit closure in the Lie algebra. The pair $(Z, \Delta_Z)$ is expected to be K-polystable, and for classical simple groups K-polystability of $(Z, \Delta_Z)$ follows from the hyperkähler metrics of [Kro90, Theorem 1 and Remark (ii)] and hyperkähler reduction, see [XZ26b, Remark 7.6].

(3) Hypertoric singularities. A hypertoric singularity is a Hamiltonian reduction of $\mathbb{C}^{2N}$ with its standard symplectic form by a subtorus of $\mathbb{G}_m^N$ [XZ26b, Definition 7.7]. The scaling action of $\mathbb{G}_m$ on $\mathbb{C}^{2N}$ descends to the hypertoric singularity and induces the minimizer [XZ26b, Theorem 7.10]. The form $\omega$ is induced by the standard symplectic form, so it has weight 2 for the descended action. The descended action need not be faithful: for $\mathbb{C}^2/\{\pm 1\}$ the element $-1$ acts trivially. For the faithful action the weight of $\omega$ is 1 or 2. For hypertoric cones of dimension at least 4 with an isolated singularity, K-polystability also follows from the conical hyperkähler metrics of [BD00, Theorem 4.1] together with [LWX18, Corollary A.4]. Here the rows of the matrix $B$ of [XZ26b, Definition 7.7] are

mutually nonparallel, as [BD00, Theorem 4.1] requires, since in dimension at least 4 two parallel rows would make the singular locus positive-dimensional.

(4) Germs satisfying one of the hypotheses (i)–(iv) of [NO25, Theorem 6.16]: an isolated singularity satisfying [NO25, Conjecture 1.3], a germ on a polarized smoothable projective irreducible symplectic variety $(V, L)$ such that $c_1(L)^\perp$ contains a non-zero isotropic class or $b_2(V) \geq 6$, or a hyperkähler quotient as in [NO25, Proposition 6.10]. For such germs, [NO25, Theorem 6.16] proves that the minimizer is quasi-regular and divisorial, that the minimizing Kollár component is extracted by a plt blow-up, that the weight of $\omega$ is 2 for a possibly non-faithful $\mathbb{G}_m$-action [NO25, Remark 6.17], and that the quotient is a log K-polystable contact orbifold with a conical Kähler–Einstein metric.

The instances of hypothesis (iv) of [NO25, Theorem 6.16] include nilpotent orbit closures and Slodowy slices of type $A$, many Nakajima quiver varieties, and many toric hyperkähler varieties [NO25, Example 6.11]. The conclusions of [NO25, Theorem 6.16] listed in item (4) also hold for any germ that carries a hyperkähler metric satisfying [NO25, Conjecture 1.3] and for which the rotated Reeb vector fields are complete [NO25, Theorem 6.19].

The conclusions of [NO25, Theorem 6.16] come from metric tangent cones in the sense of Donaldson and Sun and from hyperkähler metrics, and thus depend on global or metric hypotheses. A general symplectic singularity is not known to carry such a metric. In this paper we use instead the weak Ricci-flat Kähler cone metric of [Li21, Theorem 2.9] (see Theorem 2.27 and Remark 2.28) on the K-polystable degeneration of $W$. Such a metric exists on any K-polystable Fano cone, but a priori the potential of the metric is only known to be locally bounded and smooth on the regular locus. The correspondence between Ricci-flat Kähler cone metrics and K-stability of Fano cones goes back to [CS15, Theorem 1.1].

We do not pass to the metric completion of the link to use RCD or Cheeger–Colding theory: the RCD structure of singular Kähler metrics [GSS26, Theorem 1.1] is proved for compact normal Kähler spaces with klt singularities and Ricci curvature bounded below on the regular part, whereas the K-polystable degeneration of $W$ is not compact and the link is not a complex space. Moreover, the space of leaves of the Reeb foliation on the regular part of the link is Hausdorff if and only if the Reeb vector field is rational, which is part of what we prove. Thus the argument stays on the regular part of the link, see the Sketch of the proof below.

Algebraic bounds on the weight of $\omega$ are known under additional hypotheses. If the coordinate ring $R$ of a conical symplectic variety $X$ is generated in degree one, then the weight of $\omega$ is 1 or 2 [Nam18, Proposition 1], and $(X, \omega)$ is either $\mathbb{C}^{2n}$ with a constant symplectic form or a normal nilpotent orbit closure in a complex semisimple Lie algebra with the Kirillov–Kostant form [Nam18, the main theorem]. If $R$ is finite over the subalgebra generated by the elements of $R$ of degree equal to the weight of $\omega$, and the degrees of the grading have greatest common divisor 1, then the weight of $\omega$ is 1 or 2 [Nam17, Theorem in the introduction and Proposition 3]. For a K-semistable Fano cone with an isolated Gorenstein singularity, there is no non-constant function of normalized weight less than 1 [CS18, Theorem 7]. A general conical symplectic singularity need not satisfy the hypotheses of [Nam18, Proposition 1], [Nam17, Theorem in the introduction], or [CS18, Theorem 7].

Without such hypotheses, the algebraic data do not single out a rational Reeb vector, and we prove the rationality of the Reeb vector by a metric argument. Indeed, let $T$ be a torus with a good action on $X$ (Definition 2.6), for instance a torus acting faithfully on $X$ and containing the image of the $\mathbb{G}_m$-action given by the grading of the coordinate ring, and let $\omega$ be a $T$-eigenvector of character $\alpha$. Since $\omega^n$ generates $\Omega_X^{[2n]}$, [XZ26b, Lemma 4.2] (see Theorem 2.13(2)) and Lemma 2.14 imply that

$$A_\omega(\mathrm{wt}_\xi) = \langle \alpha, \xi \rangle \quad \text{and} \quad A_X(\mathrm{wt}_\xi) = n\langle \alpha, \xi \rangle$$

for any Reeb vector $\xi$ of $T$. Both equalities are linear in $\xi$ on the cone of Reeb vectors, and they do not distinguish rational Reeb vectors. On the metric side, the action of $\mathrm{Sp}(1)$ that makes the Reeb orbits of a compact 3-Sasakian manifold closed rotates the complex structure. Indeed,

on a hyperkähler cone with radial function $\rho$, Levi-Civita connection $\nabla$, and parallel complex structures $I$ and $J$ with $IJ = -JI$, we have $\nabla(J\rho\partial_\rho) = J$, and thus

$$L_{J\rho\partial_\rho} I = IJ - JI = 2IJ \neq 0.$$

Among the three Reeb vector fields $I\rho\partial_\rho$, $J\rho\partial_\rho$, and $IJ\rho\partial_\rho$, only $I\rho\partial_\rho$ is holomorphic for $I$. Therefore the action of $\mathrm{Sp}(1)$ is not an action by holomorphic automorphisms of the cone, and it has no algebraic counterpart.

Xu and Zhuang ask whether the quotient log Fano pairs of nilpotent orbit closures and of Slodowy slices by their natural conical $\mathbb{G}_m$-actions are K-polystable [XZ26a, Problem 23]. Problem 23 of [XZ26a] is treated in [Liu26a]. Among other recent results related to Question 1.1, [Liu26b, Theorem 1.1] compares the normalized volumes of klt singularities related by a finite crepant morphism, [Nam24, Theorem (5.8)] shows that a conical symplectic variety with a projective symplectic resolution and an effective Hamiltonian action of a torus of half dimension, compatible with the conical action, is equivariantly analytically isomorphic to a toric hyperkähler variety, and [Nam26b, Theorem in §1] characterizes hypertoric varieties among such conical symplectic varieties, assuming that the weight of $\omega$ is 2.

The goal of this paper is to answer Question 1.1 affirmatively for any symplectic singularity, with part (3) understood for reflexive differential forms, see Theorem 1.2 and Remark 1.4.

1.1. **Minimizers on symplectic singularities.** Let $(X \ni x)$ be a symplectic singularity of dimension $2n$ with symplectic form $\omega$ (see Definition 2.9). By [XZ26b, Theorem 2.1] (see Theorem 2.20), the normalized volume $\widehat{\mathrm{vol}}$ has a minimizer $v$ on the space $\mathrm{Val}_{X,x}$ of real valuations of $\mathbb{C}(X)$ centered at $x$, and any two minimizers differ by a positive multiple (see Definition 2.5). Besides the notions used in Question 1.1, Theorem 1.2 uses reflexive logarithmic forms (see Definition 2.8) and K-polystability of Fano cones for a torus (see Definitions 2.6 and 2.16).

**Theorem 1.2.** *Let $(X \ni x)$ be a symplectic singularity of dimension $2n$, $n \geq 1$, with symplectic form $\omega$, and let $v$ be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$. Then the following hold.*

1. *(Rank one) There exist a real number $c > 0$ and a Kollár component $E$ over $(X \ni x)$ such that $v = c\,\mathrm{ord}_E$.*
2. *(Weight) $A_X(E) = rn$ for some $r \in \{1, 2\}$.*
3. *(Contact structure) For the extraction $f : Y \to X$ of $E$, the index-one cover $q : \mathcal{Y} \to Y$ of $E$, $\mathcal{E} := q^*E$, and $\pi := f \circ q$, there is a neighborhood $U$ of $x$ such that*

$$\pi^*\omega \in H^0\big(\pi^{-1}(U), \Omega^{[2]}_{\mathcal{Y}}(\log \mathcal{E}) \otimes \mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})\big),$$

*understood analytically if $\omega$ is not algebraic, and the residue*

$$\theta \in H^0\big(\mathcal{E}, \Omega^{[1]}_{\mathcal{E}} \otimes \mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}\big)$$

*of $\pi^*\omega$ is an orbifold contact form with contact line bundle $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}$.*

4. *(K-polystability) The log Fano pair $(E, \Delta_E)$ is K-polystable. Equivalently, the Fano cone $(W, \zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$, where $W := \mathrm{Spec}\,\mathrm{gr}_v \mathcal{O}_{X,x}$ is graded by the values of $v/c$ and $\zeta$ is the $\mathbb{G}_m$-action defined by the grading.*
5. *(Contact Fano orbifold) In particular, $(\mathcal{E}, \mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}, \theta)$ is a contact Fano orbifold. Thus any symplectic singularity admits a Kollár component which, with the orbifold structure of the index-one cover, is a contact Fano orbifold.*

In [XZ26b, Theorem 6.2] the symplectic form is a reflexive differential form on the variety, and thus algebraic. Together with [XZ26b, Theorem 6.2], Theorem 1.2 implies the following description of the completion of $\mathcal{O}_{X,x}$ in terms of a conical symplectic variety (see Definition 2.9). We prove Corollary 1.3 in Section 11.

**Corollary 1.3.** *Let $(X \ni x)$ be a symplectic singularity of dimension $2n$, $n \geq 1$, whose symplectic form $\omega$ is algebraic, let $v$ be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, let $c > 0$ be the real number with $v(\mathbb{C}(X)^*) = c\mathbb{Z}$ (Theorem 1.2(1)), let $W := \mathrm{Spec}\,\mathrm{gr}_v \mathcal{O}_{X,x}$, graded by the values of $v/c$, and let*

*$\zeta$ be the $\mathbb{G}_m$-action defined by the grading. Then the completion of $\mathcal{O}_{X,x}$ is isomorphic to the completion of the local ring of $W$ at the vertex of $W$. Moreover, $W$ is a conical symplectic variety whose grading is induced by $\zeta$ and whose symplectic form has degree $r := A_X(v)/(cn) \in \{1,2\}$, and the Fano cone $(W,\zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$.*

**Remark 1.4.** In Theorem 1.2(3) the sheaves of reflexive differential forms cannot be replaced by sheaves of Kähler differentials. For the minimizing Kollár component $E$ of the symplectic quotient singularity $\mathbb{A}^4/\{1,g\}$ with $g(x_1,x_2,x_3,x_4) = (-x_1,-x_2,x_3,x_4)$, the residue $\theta$ is not locally in the image of $\Omega^1_{\mathcal{E}} \otimes L \to \Omega^{[1]}_{\mathcal{E}} \otimes L$, where $L = \mathcal{O}_{\mathcal{Y}}(-2\mathcal{E})|_{\mathcal{E}}$ and $\Omega^1_{\mathcal{E}}$ is the sheaf of Kähler differentials, see Example 6.5.

**Remark 1.5.** In Theorem 1.2(4), a product test configuration in the sense of Definition 2.16 may be induced by a $\mathbb{G}_m$-action on $W$ which commutes with $\zeta(\mathbb{G}_m)$ but does not factor through $\zeta(\mathbb{G}_m)$. If one requires the $\mathbb{G}_m$-action to factor through $\zeta(\mathbb{G}_m)$, as in [LWX18, Definition 2.14], then the conclusion of Theorem 1.2(4) for the cone fails already for $X = \mathbb{A}^2$ with $\omega = dx \wedge dy$, see Example 2.25.

**Sketch of the proof.** Theorem 1.2(1)(2) is Theorem 5.4, and (3) and (5) follow from Theorem 6.3, whose proof uses no metric. For $n \geq 2$, (4) follows from Corollary 3.13 and Theorem 10.5, which shows $W \cong X_0$ for the central fiber $X_0$ of a K-polystable degeneration of $W$ (Theorem 2.22). For $n \geq 2$, Theorem 5.4 is proved on $X_0$ by steps (i) and (ii) below, and Proposition 5.1 passes from $W$ to the germ. Theorem 10.5 is proved by steps (i)–(iii). For $n = 1$, Theorem 5.4 follows from Lemma 5.3, and (4) is treated in the last paragraph of the Sketch.

(i) The weak Ricci-flat cone metric of Theorem 2.27 has a continuous proper radius and $\Phi$-cutoffs on its link (Proposition 3.4, Theorem 3.9).

(ii) Reflexive 2-forms $\sigma$ with $L_{\rho\partial_\rho}\sigma = 2\sigma$, where $\rho$ is the radius function of the metric, are parallel (Proposition 4.3), $X_0$ carries a parallel symplectic form $\tau$ (Theorem 4.4), and the Reeb vector $\xi$ of the metric, normalized so that $A_{X_0}(\mathrm{wt}_\xi) = 2n$, satisfies

$$\xi = \frac{2}{r}\zeta_0, \qquad r \in \{1,2\},$$

for a primitive cocharacter $\zeta_0$ (Theorems 4.6 and 4.7).

(iii) A cover of $X_0$ by a product, étale outside codimension 4 (Theorem 8.8), extends through the degeneration to a product family (Theorem 10.4, Corollary 9.19), and the product structure descends (Lemma 9.4).

The key step, also for (4), is (i): cutting off only exceptional divisors over the quotient stack yields cutoffs whose energy, weighted by the logarithmic factor in $\Phi$, tends to 0. Such cutoffs suffice for the Bochner argument and for lifting paths to the link, which need not be complete (see (a) and (c) below). Apart from the existence of $W$, $X_0$, and the metric, which are due to [XZ26b, Theorems 2.1 and 5.11], [LWX18, Theorem 1.2], and [Li21, Theorem 2.9], the results in (i)–(iii) are new. The statements about $E$ and the isomorphism $W \cong X_0$ are algebraic, although for $n \geq 2$ the proofs of (1), (2), and (4) use the metric.

*(a) Logarithmic cutoffs, step (i).* The potential of the metric of Theorem 2.27 (see Remark 2.28) is only locally bounded and smooth over the regular locus [Li21, p. 34], so when $X_0$ is singular outside the vertex, integration by parts on the incomplete link needs cutoffs with vanishing energy. Vanishing Dirichlet energy does not suffice for the two-dimensional sup estimate behind the lifting of paths in (c), which, like the error terms of the Bochner argument, uses the logarithmic factor of $\Phi(t) = t^2 \log^2(e+t)$ (Definition 2.29, Remark 2.30).

The branch divisors of the quotient log Fano pair consist of smooth points of the quotient orbifold and cannot be cut off. Thus Theorem 3.9 resolves the quotient stack of the punctured cone, not its coarse space, and cuts off only the exceptional divisors over the singular locus of the stack. A polynomial lower bound for the metric along the exceptional divisors, proved for smooth

solutions of regularized weighted Monge–Ampère equations on a compact orbifold resolution (Theorem 3.9), turns their finite logarithmic energy (Lemma 3.5) into Φ-cutoffs.

*(b) A parallel symplectic form, step (ii).* As explained in the introduction before the example of $\mathbb{A}^4$, the limit on $X_0$ of the symplectic form of $W$, normalized by a power of the parameter $t$ of the degeneration, can be degenerate. Instead of the limit, Theorem 4.4 takes the part of the symplectic form on which $\xi$ acts with weight 2, the least weight allowed by Theorem 2.13, and filters the relative tangent vectors along $X_0$ by the power of $t$ dividing their pairings with it. The pairings induced on the steps of the filtration are reflexive 2-forms $\beta$ on $X_0$ with $L_{\rho\partial_\rho}\beta = 2\beta$, possibly degenerate, hence parallel, with algebraic kernel projectors (Lemma 4.2), and their orthogonal sum is $\tau$. Since $\tau$ is not known to extend to the degeneration, it yields (1) and (2) but not (4).

*(c) The period on an incomplete link, step (ii).* On a compact 3-Sasakian manifold, the three Reeb vector fields are complete and generate a locally free action of Sp(1), so the Reeb orbits are closed [BG99, §1.2 and Theorem 2.2.2]. The form $\tau$ yields three such vector fields on the link of $X_0$, but only the one generated by $\xi$ is holomorphic, and their flows may leave the regular locus in finite time. Instead, Lemma 4.5 lifts to the link, from almost every point, a null-homotopy in Sp(1) of the loop $\exp(si)$, $0 \leq s \leq 2\pi$, which acts through the Reeb flow. Volume preservation makes the Φ-energy of the cutoffs along the lifts tend to 0, and a two-dimensional sup estimate excludes escape. Thus $\exp(2\pi\xi)$ fixes almost every point, hence $\exp(2\pi\xi) = 1$, and the weight of $\tau$ forces $r \in \{1, 2\}$ (Theorem 4.6). The torus and $\xi$ of $X_0$ are those of $W$ (see Remark 2.23), so the conclusion holds on $W$.

*(d) Product covers and rigidity, step (iii).* The rigidity theorem [XZ26b, Theorem 4.1], after [NO25, Theorems 3.3 and 3.5], and its variant Theorem 9.2 need a relative 2-form whose restriction to $X_0$ is symplectic. The normalized limit of the symplectic form of $W$ is such a form when $\tau$ spans the reflexive 2-forms on $X_0$ of the degree of $\tau$ (Lemma 9.5), but not in general. Otherwise there exists a reflexive 2-form $\sigma$ on $X_0$ of the degree of $\tau$ which is not a multiple of $\tau$. The pencil spanned by $\sigma$ and $\tau$ contains a degenerate member, which is parallel by (ii), and its kernel splits $X_0$ into a product up to a finite quasi-étale morphism, which need not be an isomorphism (Theorems 8.2 and 8.7, Remark 8.3).

After at most $n-1$ splittings, Lemma 7.5 yields a cover of $X_0$ by $\mathbb{A}^{2b} \times X_1 \times \cdots \times X_k$, where $b, k \geq 0$ and $X_1, \ldots, X_k$ are singular conical symplectic varieties. The cover is étale outside codimension 4 (Theorem 8.8), since the cover constructed at each splitting is the quotient of a product by a finite group whose non-trivial elements act non-trivially on at least two factors (Lemma 7.5(1)). The closure of the branch locus under the $\mathbb{G}_m$-action of the degeneration has codimension at least 3, which suffices to extend the cover order by order in $t$ (Theorem 10.4).

On the extended family, Hamiltonian translations split off $\mathbb{A}^{2b}$ (Theorem 9.18), and the Euler actions of the $X_i$, whose symplectic forms can appear at different orders in the parameter, are lifted one at a time (Theorem 9.16). Although the cover need not be Galois, the product structure descends through the normalized trace (Lemma 9.4).

Finally, $X_0$ is K-polystable for a torus which may be larger than $\zeta(\mathbb{G}_m)$ (see Remark 2.23). Corollary 3.13 does not compare the test configurations for the larger torus with those for $\zeta(\mathbb{G}_m)$: the cone potential of the metric is homogeneous for $\zeta$, and it descends to a weak log Kähler–Einstein metric on the quotient pair, which is K-polystable by [Ber16, Theorem 4.8] (Proposition 3.12). For $n = 1$, (4) follows from Proposition 3.12, applied to the flat metric on $W$.

**Structure of the paper.** In Section 2 we recall preliminary material and fix notation. In particular, we introduce marked test configurations (Set-up 2.15), K-polystability of a Fano cone for a torus (Definition 2.16), Φ-cutoffs (Definition 2.29), and the hypotheses on cone metrics under which we work in Sections 4–5, 8, and 10 (Set-up 2.31), we fix the conventions on torus actions, gradings, and homogeneous forms (Definitions 2.6 and 2.8), and we recall the log discrepancy $A_\sigma$ of a reflexive form along a valuation (Definition 2.8, [XZ26b, Definition 3.6]).

In Section 3 we study the weak Ricci-flat Kähler cone metrics of [Li21, Theorem 2.9] on K-polystable Fano cones: we show that the radius function is continuous and proper, that the link has finite volume, and that logarithmic cutoffs exist on the quotient and on the link, so that the metric satisfies the conditions of Set-up 2.31 (Corollary 3.11), and we deduce K-polystability of the quotient log Fano pair from a cone potential (Proposition 3.12 and Corollary 3.13).

In Section 4 we use the cone metric to prove that reflexive 2-forms $\sigma$ with $L_{\rho\partial_\rho}\sigma = 2\sigma$, where $\rho$ is the radius function of the cone metric, are parallel (Proposition 4.3), reconstruct a parallel symplectic form on the central fiber of a K-polystable degeneration of a conical symplectic variety, and prove the period theorem, which implies that the Reeb vector of the cone metric, normalized so that the log discrepancy of the associated valuation is $2n$, is $2/r$ times a cocharacter for some $r \in \{1, 2\}$ (Theorem 4.7).

In Section 5 we use the cone metric of Section 3, through the results of Section 4, to prove Theorem 5.4, that is, parts (1) and (2) of Theorem 1.2, by passing from the associated graded cone of the minimizer back to the germ.

In Section 6 we prove that the pullback to the index-one cover of any reflexive two-form, whose log discrepancy along $E$ is at least $r$ by Theorem 2.13, is a reflexive logarithmic 2-form twisted by $\mathcal{O}_\mathcal{Y}(-r\mathcal{E})$, and that the residue of $\omega$ is an orbifold contact form, which implies parts (3) and (5) of Theorem 1.2, and we compute Example 6.5.

In Section 7, which is algebraic, we study homogeneous derivations of negative degree on conical symplectic varieties whose degree valuation minimizes the normalized volume, quasi-étale covers of products (Lemma 7.5), and graded quasi-étale covers of such varieties (Proposition 7.6).

In Section 8 we use the cone metric to split a K-polystable conical symplectic variety into a product up to a finite quasi-étale morphism, and we construct a finite cover of it by a product of an affine space and conical symplectic varieties which is étale outside a closed subset of codimension at least four.

In Section 9, which is algebraic, we prove that a marked test configuration of a cone whose regular locus carries a closed non-degenerate 2-form is itself a product if its central fiber is a product of an affine space and conical symplectic varieties whose symplectic forms have the degree of the 2-form, under conditions which hold for the products of Theorem 8.8 (Corollary 9.19).

In Section 10 we extend the product cover of Section 8 through a test configuration by an algebraic argument (Theorem 10.4), and, using the results of Sections 3 and 4 on the central fiber, we prove that the K-polystable degeneration of a K-semistable conical symplectic variety of dimension at least four is a product (Theorem 10.5).

In Section 11 we prove Theorem 1.2 and Corollary 1.3.

**Acknowledgements.** The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

## 2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

### 2.1. Pairs and Kollár components.

**Definition 2.1.** A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on $X$ such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of $z$. If $\pi$ is the identity morphism, then we may drop $Z$ and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that $(X, B)$ is a pair. If we do not require that $B \geq 0$, then we say that $(X, B)$ is a *sub-pair*. Let $h : X' \to X$ be any log resolution of $(X, B)$ and $E$ a prime divisor on

$X'$, and write $K_{X'}+B':=h^*(K_X+B)$. Then we define

$$a(E,X,B):=1-\operatorname{mult}_E B'$$

as the *log discrepancy* of $E$ with respect to $(X,B)$. For any divisorial valuation $v=c\operatorname{ord}_E$ for some real number $c>0$, we denote by $a(v,X,B):=ca(E,X,B)$.

For any (sub-)pair $(X,B)$, we say that $(X,B)$ is *(sub-)lc* if $a(E,X,B)\geq 0$ for any prime divisor $E$ over $X$ and *(sub-)klt* if $a(E,X,B)>0$ for any prime divisor $E$ over $X$. An *lc place* of $(X,B)$ is a prime divisor $E$ over $X$ such that $a(E,X,B)=0$. An *lc center* of $(X,B)$ is the center of an lc place of $(X,B)$ on $X$.

**Definition 2.2.** Let $(X\ni x)$ be a klt singularity. A *Kollár component* over $(X\ni x)$ is a prime divisor $E$ over $X$ with $\operatorname{center}_X E=x$ for which there exists a projective birational morphism $f:Y\to X$ from a normal variety $Y$ such that $f$ is an isomorphism over $X\backslash\{x\}$, $E$ is the only $f$-exceptional prime divisor, $-E$ is $\mathbb{Q}$-Cartier and $f$-ample, and $(Y,E)$ is plt [XZ26b, Definition 2.3]. We call $f$ the *extraction* of $E$, and we define $\Delta_E$ by adjunction

$$(K_Y+E)|_E=K_E+\Delta_E.$$

The *index-one cover* of $E$ is the Deligne–Mumford stack $q:\mathcal{Y}\to Y$ obtained by gluing the quotient stacks of the local index-one covers of the $\mathbb{Q}$-Cartier divisor $E$ [KM98, Definition 5.19], [XZ26b, Lemma 5.6], and we set $\mathcal{E}:=q^*E$. Here a Deligne–Mumford stack of finite type over $\mathbb{C}$ with finite inertia has a coarse space, it is tame since finite groups are linearly reductive over $\mathbb{C}$, and étale locally over its coarse space it is a quotient stack $[U/G]$ of a scheme $U$ by a linearly reductive group scheme $G$ [AOV08, Section 3, the paragraph before Definition 3.1, Definition 3.1 and Theorem 3.2].

**Lemma 2.3** ([XZ26b, Definition 2.3 and Lemma 5.6(1)] and [KM98, Definition 5.19])**.** *Let $E$ be a Kollár component over a klt singularity $(X\ni x)$, with extraction $f:Y\to X$ and index-one cover $q:\mathcal{Y}\to Y$, and let $\mathcal{E}:=q^*E$. Then $(E,\Delta_E)$ is a klt log Fano pair, $-E|_E$ is ample, $\mathcal{Y}$ is normal, and $\mathcal{E}$ is a normal Cartier divisor on $\mathcal{Y}$.*

**Definition 2.4.** A finite surjective morphism $h:X'\to X$ between normal varieties is *quasi-étale* if there exists a closed subset $\Sigma\subset X$ of codimension at least 2 such that $h$ is étale over $X\backslash\Sigma$.

## 2.2. Normalized volume and Fano cones.

**Definition 2.5.** Let $(X\ni x)$ be a klt singularity of dimension $d$. Let $\operatorname{Val}_{X,x}$ be the set of real valuations $v$ of $\mathbb{C}(X)$ with $\operatorname{center}_X v=x$, and for $v\in\operatorname{Val}_{X,x}$ let $A_X(v)\in(0,+\infty]$ be the log discrepancy of $v$ [Li18, Section 2.1], so that $A_X(c\operatorname{ord}_E)=ca(E,X,0)$. We write $A_X(E):=A_X(\operatorname{ord}_E)$. Let $h:Y\to X$ be a proper birational morphism from a smooth variety $Y$, let $E_1,\dots,E_k$ be prime divisors on $Y$ whose sum $E$ has simple normal crossing support, let $\eta$ be the generic point of an irreducible component of $E_1\cap\cdots\cap E_k$, and let $y_1,\dots,y_k$ be local equations of $E_1,\dots,E_k$ at $\eta$. For $\alpha\in\mathbb{R}^k_{\geq 0}\backslash\{0\}$ let $v_\alpha$ be the valuation with

$$v_\alpha(u)=\min\{\langle\alpha,\beta\rangle\mid c_\beta\neq 0\}\quad\text{for } u=\sum_{\beta\in\mathbb{N}^k}c_\beta y^\beta\in\widehat{\mathcal{O}}_{Y,\eta},$$

with coefficients $c_\beta$ in a coefficient field of $\widehat{\mathcal{O}}_{Y,\eta}$. We denote by $\operatorname{QM}(Y,E)$ the set of all such $v_\alpha$ and call a valuation *quasi-monomial* if it belongs to $\operatorname{QM}(Y,E)$ for some such $h$ and $E$. If moreover $\operatorname{Exc}(h)\subset\operatorname{Supp}E$ near $\eta$, then $A_X(v_\alpha)=\sum_i\alpha_iA_X(E_i)$ [XZ26b, Definition 2.11]. For $\lambda\geq 0$ let

$$\mathfrak{a}_\lambda(v):=\{f\in\mathcal{O}_{X,x}\mid v(f)\geq\lambda\}\quad\text{and}\quad\mathfrak{a}_{>\lambda}(v):=\{f\in\mathcal{O}_{X,x}\mid v(f)>\lambda\}.$$

The *volume* and the *associated graded ring* of $v$ are

$$\operatorname{vol}(v):=\lim_{m\to+\infty}\frac{d!\ell(\mathcal{O}_{X,x}/\mathfrak{a}_m(v))}{m^d},\qquad\operatorname{gr}_v\mathcal{O}_{X,x}:=\bigoplus_{\lambda\geq 0}\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v).$$

The *normalized volume* of $v$ is $\widehat{\mathrm{vol}}(v) := A_X(v)^d \mathrm{vol}(v)$ if $A_X(v) < +\infty$ and $\widehat{\mathrm{vol}}(v) := +\infty$ if $A_X(v) = +\infty$ [Li18, Section 4], and

$$\widehat{\mathrm{vol}}(X \ni x) := \inf_{v \in \mathrm{Val}_{X,x}} \widehat{\mathrm{vol}}(v).$$

A valuation $v \in \mathrm{Val}_{X,x}$ with $\widehat{\mathrm{vol}}(v) = \widehat{\mathrm{vol}}(X \ni x)$ is called a *minimizer* of $\widehat{\mathrm{vol}}$.

**Definition 2.6.** Let $X = \operatorname{Spec} R$ be a normal affine variety with a faithful action of a torus $T$. For a character $\alpha$ of $T$ let

$$R_\alpha := \{f \in R \mid f(t \cdot y) = \alpha(t) f(y) \text{ for } t \in T \text{ and } y \in X\},$$

so that $R = \bigoplus_\alpha R_\alpha$. The action is *good* if it has a unique fixed point $o$, called the *vertex*, which belongs to the closure of any $T$-orbit. A *Reeb vector* of $T$ is an element $\xi \in \mathrm{Hom}(\mathbb{G}_m, T) \otimes \mathbb{R}$ with $\langle \alpha, \xi \rangle > 0$ for any $\alpha \neq 0$ with $R_\alpha \neq 0$, and

$$\mathrm{wt}_\xi \Big( \sum_\alpha f_\alpha \Big) := \min\{\langle \alpha, \xi \rangle \mid f_\alpha \neq 0\}$$

is the valuation of $\xi$. For $\xi \in \mathrm{Hom}(\mathbb{G}_m, T) \otimes \mathbb{R}$ we denote by $\mathrm{eu}_\xi$ the derivation of $R$ with $\mathrm{eu}_\xi(f) = \langle \alpha, \xi \rangle f$ for $f \in R_\alpha$, and also the holomorphic vector field on $X_{\mathrm{reg}}$ defined by it. We identify $\mathrm{Hom}(\mathbb{G}_m, T) \otimes \mathbb{R}$ with the Lie algebra of the maximal compact subtorus $T_c \subset T$ so that

$$\mathrm{Hom}(\mathbb{G}_m, T) = \{y \mid \exp(2\pi y) = 1\}.$$

Then the real vector field on $X$ generated by $y$ sends $f \in R_\alpha$ to $\sqrt{-1}\langle \alpha, y \rangle f$. A *Fano cone* $(X, \xi)$ is such an $X$ with klt singularities and a good $T$-action, together with a Reeb vector $\xi$ of $T$. The Reeb vector $\xi$ is *quasi-regular* if $\xi = c\zeta$ for a real number $c > 0$ and a cocharacter $\zeta$ of $T$. If $\zeta$ is a primitive cocharacter which is a Reeb vector, let $Z := (X \backslash \{o\})/\zeta(\mathbb{G}_m)$ and

$$\Delta_Z := \sum_P \Big(1 - \frac{1}{m_P}\Big) P,$$

where $m_P$ is the order of the generic stabilizer of $\zeta(\mathbb{G}_m)$ on the preimage of a prime divisor $P \subset Z$ [Kol04, Definition 3, Theorem 7 and Section 4]. We call $(Z, \Delta_Z)$ the *quotient log Fano pair* of $(X, \zeta)$.

A grading $R = \bigoplus_{j \geq 0} R_j$ with $R_0 = \mathbb{C}$ is called a *good positive grading*, and the point defined by $\bigoplus_{j>0} R_j$ is called the *vertex* of the grading. If such a grading is induced by a $\mathbb{G}_m$-action $\zeta$, that is, $R_j = \{f \in R \mid f(\zeta(\lambda) \cdot y) = \lambda^j f(y)\}$, then the valuation $\mathrm{wt}_\zeta$, given by $\mathrm{wt}_\zeta(\sum_j f_j) = \min\{j \mid f_j \neq 0\}$, is called the *degree valuation* of the grading, and $\mathrm{eu} := \mathrm{eu}_\zeta$ is called the *Euler derivation* of the grading. For a grading $R = \bigoplus_j R_j$ and a subring $A \subset R_0$, a derivation $D \in \mathrm{Der}_A(R)$ is *homogeneous of degree* $d$ if $D(R_j) \subset R_{j+d}$ for any $j$, and $\mathrm{Der}_A(R)_d$ denotes the $A$-module of homogeneous derivations of degree $d$.

**Lemma 2.7** ([XZ26b, Paragraph 2.13], [Kol04, Definition 3, Theorem 7 and Section 4], and [LWX18, Definition 2.13])**.** *Let $X$ be a normal affine variety with klt singularities and a good faithful action of a torus $T$ with vertex $o$, let $\zeta$ be a primitive cocharacter of $T$ which is a Reeb vector of $T$, and let $(Z, \Delta_Z)$ be the quotient log Fano pair of $(X, \zeta)$. Then $Z$ is a normal projective variety, $(Z, \Delta_Z)$ is a klt log Fano pair, and there exists a unique Kollár component $F$ over $(X \ni o)$ with $\mathrm{ord}_F = \mathrm{wt}_\zeta$. Moreover, $(F, \Delta_F) \cong (Z, \Delta_Z)$.*

*Proof.* Write $X = \operatorname{Spec} R$ and $R = \bigoplus_{j \geq 0} R_j$ for the grading induced by $\zeta$. Since $\zeta$ is primitive, by [XZ26b, Paragraph 2.13 and (2.4)], the quotient morphism $\pi : X \backslash \{o\} \to Z$ is a Seifert $\mathbb{G}_m$-bundle, $Z$ is a normal projective variety, and there exists an ample $\mathbb{Q}$-divisor $L$ on $Z$ such that

$$R \cong \bigoplus_{m \in \mathbb{N}} H^0(Z, \mathcal{O}_Z(\lfloor mL \rfloor)).$$

Moreover, the valuation $\mathrm{wt}_\zeta$ is proportional to $\mathrm{ord}_F$, where $F$ is the zero section of $Y := \mathrm{Spec}_Z \bigoplus_{m\in\mathbb{N}} \mathcal{O}_Z(\lfloor mL\rfloor)$, the divisor $F$ is a Kollár component over $(X \ni o)$ with extraction $Y \to X$, $F$ is identified with $Z$, $L \sim_{\mathbb{Q}} -F|_F$, and

$$\pi^*(K_F + \Delta_F) = K_X|_{X\setminus\{o\}}.$$

Since $T$ acts faithfully and $\zeta$ is primitive, $\zeta(\mathbb{G}_m)$ acts faithfully on $X$. Thus the integers $j$ with $R_j \neq 0$ generate $\mathbb{Z}$, and the value group of $\mathrm{wt}_\zeta$ is $\mathbb{Z}$. The value group of $\mathrm{ord}_F$ is also $\mathbb{Z}$, so $\mathrm{wt}_\zeta = \mathrm{ord}_F$. A prime divisor over $X$ is determined by the associated divisorial valuation, hence $F$ is unique.

Let $P \subset Z$ be a prime divisor, which we also regard as a prime divisor on $F$, and let $M := m_P$, which is the multiplicity of the Seifert fiber of $\pi$ over a general point of $P$ [Kol04, Definition 3 and Theorem 7]. By [Kol04, Proposition 19], there is a neighborhood $V \subset Z$ of a general point of $P$ such that over $V$, $X\setminus\{o\}$ is the quotient of $Z' \times \mathbb{G}_m$ by a diagonal action of $\mu_M$, where $Z' \to Z$ is a cyclic cover with ramification index $M$ along $P$ and $\mu_M$ acts on $\mathbb{G}_m$ by multiplication. Since $X\setminus\{o\} = \mathrm{Spec}_Z \bigoplus_{m\in\mathbb{Z}} \mathcal{O}_Z(\lfloor mL\rfloor)$ [XZ26b, Paragraph 2.13] and $Y = \mathrm{Spec}_Z \bigoplus_{m\in\mathbb{N}} \mathcal{O}_Z(\lfloor mL\rfloor)$, computing the $\mu_M$-invariants of $\mathcal{O}_{Z'}[t]$ instead of $\mathcal{O}_{Z'}[t, t^{-1}]$, where $t$ is the coordinate of $\mathbb{G}_m$, shows that over $V$ we have $Y = (Z' \times \mathbb{A}^1)/\mu_M$ and $F$ is the image of $Z' \times \{0\}$. Let $g : Z' \times \mathbb{A}^1 \to Y$ be the quotient morphism, and let $P' \subset Z'$ be the reduced preimage of $P$. Non-trivial elements of $\mu_M$ have fixed points only on $P' \times \{0\}$, which has codimension two, so $g$ is quasi-étale and

$$g^*F = Z' \times \{0\} =: F'.$$

Since $X\setminus\{o\}$ is normal, its singular locus is a $\zeta(\mathbb{G}_m)$-invariant closed subset of codimension at least two, whose image in $Z$ has codimension at least two. Thus $X\setminus\{o\}$ is smooth along the fiber of $\pi$ over a general point of $P$, and $Z'$ is smooth over a general point of $P$ by [Kol04, Proposition 20].

Let $s$ be a rational section of $\omega_Y$ with a simple pole along $F$, and let $D := \mathrm{div}(s) + F$, so that $F \not\subset \mathrm{Supp}\, D$ and $D|_F = K_F + \Delta_F$ with $K_F := \mathrm{div}(\mathrm{Res}_F s)$. Since $g$ is quasi-étale and étale at the generic point of $F'$, we have $g^*D = \mathrm{div}(g^*s) + F'$ and $\mathrm{Res}_{F'} g^*s = (g|_{F'})^* \mathrm{Res}_F s$. As $Z' \times \mathbb{A}^1$ and $F'$ are smooth over a general point of $P$, computing both sides of adjunction with $\mathrm{Res}_F s$ and $\mathrm{Res}_{F'} g^*s$ gives, over a general point of $P$,

$$K_{F'} = (g^*D)|_{F'} = (g|_{F'})^*(D|_F) = (g|_{F'})^*(K_F + \Delta_F), \quad \text{where } K_{F'} := \mathrm{div}(\mathrm{Res}_{F'} g^*s).$$

On the other hand, $g|_{F'} : F' \cong Z' \to F \cong Z$ has ramification index $M$ along $P'$, so by the Hurwitz formula, over a general point of $P$,

$$K_{F'} = (g|_{F'})^*K_F + (M-1)P' \quad \text{and} \quad (g|_{F'})^*P = MP'.$$

Thus $(g|_{F'})^*\Delta_F = (M-1)P'$ over a general point of $P$, and the coefficient of $P$ in $\Delta_F$ is $1-1/M$, which is the coefficient of $P$ in $\Delta_Z$. Since $P$ is arbitrary, $\Delta_F = \Delta_Z$. Therefore $(F, \Delta_F) \cong (Z, \Delta_Z)$, and $(Z, \Delta_Z)$ is a klt log Fano pair by Lemma 2.3. □

We call $F$ the *Kollár component defined by* $\zeta$, or by the grading.

### 2.3. **Symplectic singularities and reflexive forms.**

**Definition 2.8.** Let $X$ be a normal variety and $\iota : X_{\mathrm{reg}} \to X$ the inclusion. For $p \geq 0$ let $\Omega^{[p]}_X := \iota_*\Omega^p_{X_{\mathrm{reg}}}$, whose sections are called *reflexive $p$-forms*. If a torus $T$ acts on $X$, a reflexive $p$-form $\sigma$ is a *$T$-eigenvector of character $\alpha$* if $t^*\sigma = \alpha(t)\sigma$ for $t \in T$. If a $\mathbb{G}_m$-action $\zeta$ induces a grading of $X$, a form $\sigma$ on a $\zeta(\mathbb{G}_m)$-invariant open subset $V \subset X$ is *homogeneous of ($\zeta$-)degree $j$* if $\zeta(\lambda)^*\sigma = \lambda^j\sigma$ for $\lambda \in \mathbb{C}^*$, so that $L_{\mathrm{eu}}\sigma = j\sigma$, and $H^0(V, \Omega^2)_j$ denotes the space of algebraic 2-forms on $V$, not necessarily closed, which are homogeneous of degree $j$. For example, if $x \in R_1$, then $dx$ is homogeneous of degree 1, and $dx \wedge dy$ on $\mathbb{A}^2$ with the standard grading is homogeneous of degree 2.

For a non-zero reflexive $p$-form $\sigma$ and a prime divisor $E$ over $X$, let $A_\sigma(E)$ be the order of vanishing along $E$ of $h^*\sigma$, regarded as a logarithmic $p$-form on a log resolution $h : X' \to X$ on

which $E$ is a divisor, and set $A_\sigma(c\operatorname{ord}_E) := cA_\sigma(E)$ for $c > 0$. If $X$ is klt, then $h^*\sigma$ extends to a regular $p$-form on $X'$ [GKKP11, Theorem 1.4]. For a quasi-monomial valuation $v \in \mathrm{QM}(Y, E)$ as in Definition 2.5, where $h : Y \to X$ is a log resolution of $X$, we set $A_\sigma(v) := v(h^*\sigma)$, where $h^*\sigma$ is regarded as a rational section of $\Omega^p_Y(\log E)$ and $v(h^*\sigma)$ is the minimum of the values of $v$ on the coefficients of $h^*\sigma$ with respect to a local basis of $\Omega^p_Y(\log E)$. The number $A_\sigma(v)$ does not depend on the choice of $h$ and $E$ [XZ26b, Definition 3.6 and Lemma 3.7]. We call $A_\sigma(v)$ the *log discrepancy of $\sigma$ along $v$*. In [XZ26b, Definition 3.6] the number $A_\sigma(v)$ has the name log discrepancy of $v$ with respect to $\sigma$. For example, $A_{dx\wedge dy}(\operatorname{ord}_0) = 2$ on $\mathbb{A}^2$, which equals the degree of $dx \wedge dy$.

For a normal Deligne–Mumford stack $\mathcal{Z}$ let $\Omega^{[p]}_{\mathcal{Z}}$ be the reflexive extension of the sheaf of $p$-forms on the smooth locus of $\mathcal{Z}$, and for a reduced Cartier divisor $\mathcal{E}$ on a normal Deligne–Mumford stack $\mathcal{Y}$ let $\Omega^{[p]}_{\mathcal{Y}}(\log \mathcal{E})$ be the reflexive extension of the sheaf of logarithmic $p$-forms on the open set where $\mathcal{Y}$ is smooth and $\mathcal{E}$ is a smooth divisor.

**Definition 2.9.** A *symplectic variety* is a normal variety $X$ together with a closed non-degenerate holomorphic 2-form $\omega$ on $X_{\mathrm{reg}}$ such that for any resolution $h : X' \to X$ the form $h^*\omega$ on $h^{-1}(X_{\mathrm{reg}})$ extends to a holomorphic 2-form on $X'$ [Bea99, Definition 1.1]. A *symplectic singularity* is a germ $(X \ni x)$ of a symplectic variety. An affine symplectic variety $X = \operatorname{Spec} R$ is *conical* if $R = \bigoplus_{j\geq 0} R_j$ is graded with $R_0 = \mathbb{C}$, and $\omega$ is algebraic and homogeneous of degree $r > 0$ with respect to the grading. The integer $r$ is called the *weight* of $\omega$. For regular functions $f, g$ on an open subset of $X$, let $H_f$ be the vector field on $X_{\mathrm{reg}}$ with $i_{H_f}\omega = -df$, and let $\{f, g\} := H_f(g)$, which is regular since $X$ is normal. We call $\{,\}$ the *Poisson bracket* of $\omega$ and $\{f, \cdot\}$ the *Hamiltonian derivation* of $f$.

**Lemma 2.10** ([Bea99, Proposition 1.3] and [KM98, Corollary 5.24])**.** *Let $X$ be a symplectic variety. Then $X$ has rational Gorenstein singularities, and in particular $X$ is canonical and klt. If moreover $X$ is conical with grading induced by a faithful $\mathbb{G}_m$-action $\zeta$, then $(X, \zeta)$ is a Fano cone.*

*Proof.* The first sentence follows from [Bea99, Proposition 1.3] and [KM98, Corollary 5.24]. The second sentence follows from the first one and Definition 2.6: for $y \in X$ and $f \in R_j$ with $j > 0$, we have $f(\zeta(\lambda)\cdot y) = \lambda^j f(y) \to 0$ as $\lambda \to 0$, so the vertex of the grading is the unique fixed point and belongs to the closure of any orbit, and $\zeta$ is a Reeb vector since $j > 0$ for any $j \neq 0$ with $R_j \neq 0$. □

Finite quasi-étale covers of symplectic varieties with algebraic symplectic forms are again symplectic varieties.

**Lemma 2.11.** *Let $(X, \omega)$ be a symplectic variety with $\omega$ algebraic, and let $h : X' \to X$ be a finite surjective quasi-étale morphism from a normal variety $X'$. Then $h^*\omega$ extends to an algebraic 2-form $\omega'$ on $X'_{\mathrm{reg}}$, and $(X', \omega')$ is a symplectic variety.*

*Proof.* By Definition 2.4, and since $X$ is normal, there exists a closed subset $\Sigma \subset X$ of codimension at least 2 such that $\operatorname{Sing} X \subset \Sigma$ and $h$ is étale over $X\backslash\Sigma$. Let $V := h^{-1}(X\backslash\Sigma)$. Then $V \subset X'_{\mathrm{reg}}$, and $X'\backslash V$ has codimension at least 2 in $X'$ since $h$ is finite.

By Lemma 2.10, $X$ is klt. Since $h$ is étale over $X\backslash\Sigma$, the Hurwitz formula implies $K_{X'} = h^*K_X$, so $X'$ is klt by [KM98, Proposition 5.20].

The form $h^*\omega$ is closed and non-degenerate on $V$. By [Har77, Proposition II.6.3A], $h^*\omega|_V$ extends to an algebraic 2-form $\omega'$ on $X'_{\mathrm{reg}}$, and $d\omega' = 0$ since $d\omega'|_V = 0$. Let $2n := \dim X$. Then $\omega'^{\wedge n}$ is a section of the canonical bundle of $X'_{\mathrm{reg}}$ which vanishes nowhere on $V$. By Krull's principal ideal theorem [Har77, Theorem I.1.11A], the zero locus of $\omega'^{\wedge n}$ is empty or of pure codimension 1, so it is empty. Thus $\omega'$ is non-degenerate on $X'_{\mathrm{reg}}$.

Since the extension of $f^*\omega'$ to a resolution $f : Z \to X'$ can be checked over an affine open cover of $X'$, we may assume that $X'$ is affine, in particular quasi-projective. Since $X'$ is klt, for

any log resolution $\pi : \widetilde{X} \to X'$, $\pi^*\omega'$ extends to a 2-form on $\widetilde{X}$ by [GKKP11, Theorem 1.4]. Let $f : Z \to X'$ be a resolution, and let $\pi = f \circ g : \widetilde{X} \to X'$ be a log resolution which factors through a morphism $g : \widetilde{X} \to Z$. Since $Z$ is smooth and $g$ is proper and birational, $g$ is an isomorphism over an open subset of $Z$ whose complement has codimension at least 2. Thus $f^*\omega'$ extends to a 2-form on $Z$ by [Har77, Proposition II.6.3A], and $(X', \omega')$ is a symplectic variety. □

**Definition 2.12.** Let $\mathcal{E}$ be a normal Deligne–Mumford stack of dimension $2n-1$ and $L$ a line bundle on $\mathcal{E}$. An $L$-valued reflexive 1-form $\theta \in H^0(\mathcal{E}, \Omega^{[1]}_{\mathcal{E}} \otimes L)$ is an *orbifold contact form* if $\theta \wedge (d\theta)^{n-1}$, computed in local trivializations of $L$, is a nowhere vanishing section of $\Omega^{2n-1}_{\mathcal{E}_{\mathrm{reg}}} \otimes L^n$, and we call $L$ the *contact line bundle* of $\theta$ (cf. [Nam11, (4.4)] and [Smi22, Definition 2.1]). We say that $(\mathcal{E}, L, \theta)$ is a *contact Fano orbifold* if moreover $\theta$ induces a surjection $T_{\mathcal{E}} \to L$ whose kernel $\mathcal{K}$ is a reflexive sheaf with $\mathcal{K} \cong \mathcal{K}^* \otimes L$ induced by $d\theta$, a positive power of $L$ descends to an ample line bundle on the coarse space of $\mathcal{E}$, and $\mathcal{E}$ has an étale atlas by varieties with klt singularities.

The conditions in the second sentence of Definition 2.12, that $\theta$ induces a surjection $T_{\mathcal{E}} \to L$ with reflexive kernel $\mathcal{K}$ and that $d\theta$ induces $\mathcal{K} \cong \mathcal{K}^* \otimes L$, hold for any orbifold contact form on a normal Deligne–Mumford stack $\mathcal{E}$. Indeed, on $\mathcal{E}_{\mathrm{reg}}$ they say that $\ker \theta$ is a contact distribution, which is equivalent to $\theta \wedge (d\theta)^{n-1}$ being nowhere vanishing [Smi22, Definition 2.1]. Across the singular locus, in a local trivialization of $L$, the vector field $v$ on $\mathcal{E}_{\mathrm{reg}}$ with $\theta(v) = 1$ and $i_v d\theta = 0$ extends to a section of the reflexive sheaf $T_{\mathcal{E}}$ and splits $T_{\mathcal{E}} \to L$, so $\mathcal{K}$ is a direct summand of $T_{\mathcal{E}}$ and hence reflexive. Then $\mathcal{K} \to \mathcal{K}^* \otimes L$ is a morphism of reflexive sheaves which is an isomorphism on $\mathcal{E}_{\mathrm{reg}}$, hence an isomorphism. Thus what a contact Fano orbifold requires beyond an orbifold contact form is that a positive power of $L$ descends to an ample line bundle on the coarse space of $\mathcal{E}$, and that $\mathcal{E}$ has an étale atlas by varieties with klt singularities.

We require the charts of the atlas to be klt, and not smooth, because the index-one charts can be singular: for the minimizing Kollár component $E$ of $X = \mathbb{A}^4/\{1, g\}$ in Remark 1.4, $E$ is Cartier on an open subset of $\mathrm{Bl}_0 \mathbb{A}^4/\{1, g\}$ meeting $E$, the index-one cover is an isomorphism there, and over it $\mathcal{E}$ is $V \times \mathbb{A}^1$, where $V$ is the $A_1$ surface singularity (Example 6.5). On such a chart the residues are reflexive forms and not Kähler differentials (Remark 1.4), which is why Definition 2.12 uses reflexive forms.

When $\mathcal{E}$ is the divisor of Definition 2.2, the local index-one charts of $\mathcal{E}$ form an étale atlas of $\mathcal{E}$. In the situation of Set-up 6.1, the local index-one charts of $\mathcal{E}$ are klt by Theorem 6.3(3). Definition 2.12 requires klt singularities on an atlas, and not on the coarse space with a boundary, because the coarse space of a general stack carries no given boundary.

By [XZ26b, Theorem 2.1 and Definition 2.11] (see Theorem 2.20), any minimizer $v$ of $\widehat{\mathrm{vol}}$ belongs to $\mathrm{QM}(Y, E)$ for a log resolution $h : Y \to X$ and some $E$ as in Definition 2.5, so $A_\sigma(v)$ below is defined.

**Theorem 2.13** ([XZ26b, Theorem 5.10(2) and Lemma 4.2])**.** *The following hold.*

*(1) Let $(X \ni x)$ be a klt singularity of dimension $d$, let $v$ be a minimizer of $\widehat{\mathrm{vol}}$, and let $\sigma$ be a non-zero reflexive $p$-form on a neighborhood of $x$ with $p \geq 1$. Then*

$$\frac{A_X(v)}{d} \leq \frac{A_\sigma(v)}{p}.$$

*(2) Let $X = \operatorname{Spec} R$ be a normal affine variety with a good action of a torus $T$, and let $\sigma \in H^0(X, \Omega^{[p]}_X)$ be a non-zero $T$-eigenvector of character $\alpha$. Then $A_\sigma(\mathrm{wt}_\xi) = \langle \alpha, \xi \rangle$ for any Reeb vector $\xi$ of $T$.*

In [XZ26b, Theorem 5.10] the form $\sigma$ is any non-zero reflexive $p$-form, not necessarily closed or non-degenerate. Definition-Lemma 6.2 uses Theorem 2.13(1) for any reflexive 2-form on $X$, and Theorem 4.4 for any $T$-eigencomponent of $\omega$.

**Lemma 2.14.** *Let $X = \operatorname{Spec} R$ be a normal affine variety of dimension $d$ with klt singularities and a good action of a torus $T$, and assume that $\Omega_X^{[d]}$ is generated by a $T$-eigenvector of character $\kappa$. Then $A_X(\mathrm{wt}_\xi) = \langle \kappa, \xi \rangle$ for any Reeb vector $\xi$ of $T$.*

*Proof.* Let $\sigma$ be a $T$-eigenvector of character $\kappa$ which generates $\Omega_X^{[d]}$, and let $\xi$ be a Reeb vector of $T$. By [LX18, Lemma 2.16], $\mathrm{wt}_\xi$ is a quasi-monomial valuation centered at $o$. By [LX18, Definition 2.8], there exists a log resolution $h : Y \to X$ such that $\mathrm{wt}_\xi$ is monomial in local equations of prime divisors $E_1, \dots, E_r$ over $o$ at the generic point of a component of $E_1 \cap \cdots \cap E_r$. Let $D$ be the reduced divisor with support $\mathrm{Exc}(h) \cup E_1 \cup \cdots \cup E_r$. If $d \geq 2$, then $\operatorname{Supp} D = \mathrm{Exc}(h)$, since each $E_i$ maps to $o$ and so is $h$-exceptional. If $d = 1$, then $h$ is an isomorphism and $D = E_1$. In both cases $D$ is a simple normal crossing divisor, and $\mathrm{wt}_\xi \in \mathrm{QM}(Y, D)$ in the sense of [JM12, Definition 3.2 and the paragraph after Remark 3.3], applied on the regular scheme $Y$. Let $E_1, \dots, E_k$ be the components of $D$ containing the center of $\mathrm{wt}_\xi$, and let $E := \sum_i E_i$, so that $\mathrm{wt}_\xi = v_\alpha \in \mathrm{QM}(Y, E)$ for some $\alpha \in \mathbb{R}^k_{>0}$. Let $\eta$ and $y_1, \dots, y_k$ be the generic point and the local equations of Definition 2.5. Then no component of $D$ other than $E_1, \dots, E_k$ contains $\eta$.

Since $\sigma$ generates $\Omega_X^{[d]}$, $K_X$ is Cartier and, by the definition of log discrepancies, near $\eta$ we have

$$h^*\sigma = w \cdot \prod_{i=1}^{k} y_i^{A_X(E_i)} \cdot \theta,$$

where $w$ is a unit at $\eta$ and $\theta$ is a local generator of $\Omega_Y^d(\log E)$ at $\eta$. Thus, by [XZ26b, Definition 3.6] and [XZ26b, Definition 2.11],

$$A_\sigma(\mathrm{wt}_\xi) = v_\alpha\Big(\prod_{i=1}^{k} y_i^{A_X(E_i)}\Big) = \sum_{i=1}^{k} \alpha_i A_X(E_i) = A_X(\mathrm{wt}_\xi).$$

By Theorem 2.13(2) applied to $\sigma$ with $p = d$, $A_\sigma(\mathrm{wt}_\xi) = \langle \kappa, \xi \rangle$. Hence $A_X(\mathrm{wt}_\xi) = \langle \kappa, \xi \rangle$. □

### 2.4. **Test configurations and K-polystability.**

**Set-up 2.15.** Let $X = \operatorname{Spec} R$, $T$, $\zeta$, $\Lambda$, and $\mathcal{X} = \operatorname{Spec} \mathcal{R}$ be as follows.

(1) $X = \operatorname{Spec} R$ is a normal affine variety with an action of a torus $T$, and $\zeta$ is a cocharacter of $T$ inducing a grading $R = \bigoplus_{j \geq 0} R_j$ with $R_0 = \mathbb{C}$.
(2) $\Lambda := \mathbb{G}_m$, and $\mathcal{R}$ is a finitely generated normal domain which is flat over $\mathbb{C}[t]$, with $\mathcal{X} := \operatorname{Spec} \mathcal{R} \to \mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[t]$.
(3) $T$ and $\Lambda$ act algebraically on $\mathcal{R}$ with commuting actions, $T$ fixes $t$, $\Lambda$ acts on $t$ with weight $-1$, and the $\zeta$-grading $\mathcal{R} = \bigoplus_{j \geq 0} \mathcal{R}_j$ has degree-zero part $\mathbb{C}[t]$.
(4) A $T \times \Lambda$-equivariant isomorphism $\mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$ is fixed, where $\Lambda$ acts trivially on $R$.

We call $(\mathcal{X}, T, \Lambda, \zeta)$ a *marked test configuration* of $X$ with *central fiber* $X_0 := \operatorname{Spec}(\mathcal{R}/t\mathcal{R})$. We let $T$ and $\Lambda$ act on $(\mathcal{R}/t\mathcal{R})[t]$ through their actions on $X_0$, with $T$ fixing $t$ and $\Lambda$ acting on $t$ with weight $-1$, and equivariance of an isomorphism $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ refers to the actions on $(\mathcal{R}/t\mathcal{R})[t]$ defined in this way.

The marking turns forms on $X$ into relative forms on the smooth locus of $\mathcal{X}$ over $\mathbb{A}^1$, which can be normalized by powers of $t$ (Definition-Lemma 4.1). Let $\pi : \mathcal{X} \to \mathbb{A}^1$ be the projection and let $\eta$ be the vector field generating $\Lambda$. Since $\Lambda$ acts on $t$ with weight $-1$, we have $\eta(t) = -t$, that is, $\pi_*\eta = -t\partial_t$, which is the sign convention for the vector field of a test configuration in [LWX18, Definition 2.14].

**Definition 2.16.** Let $(X, \xi)$ be a Fano cone and let $T_0$ be a torus acting faithfully on $X$ such that $\xi$ is a Reeb vector of $T_0$. Let $(\mathcal{X}, \eta)$ be a $T_0$-equivariant test configuration of $(X, \xi)$ in the sense of [LWX18, Definition 2.14], where $\eta$ is the vector field generating the $\mathbb{G}_m$-action on $\mathcal{X}$. We say that $(\mathcal{X}, \eta)$ is a *product* if there exist a $T_0$-equivariant isomorphism $\kappa : \mathcal{X} \to X \times \mathbb{A}^1$ over

$\mathbb{A}^1$ and an action $\lambda$ of $\mathbb{G}_m$ on $X$ commuting with $T_0$ such that $\kappa$ maps $\eta$ to $\eta_\lambda - t\partial_t$, where $\eta_\lambda$ is the vector field generating $\lambda$ and $t$ is the coordinate of $\mathbb{A}^1$.

We say that $(X, \xi)$ is *K-semistable for* $T_0$ if any $T_0$-equivariant special test configuration of $(X, \xi)$ has non-negative Futaki invariant [LWX18, Definitions 2.16, 2.22, and 2.23], that $(X, \xi)$ is *K-polystable for the torus* $T_0$ if moreover any $T_0$-equivariant special test configuration of $(X, \xi)$ with Futaki invariant 0 is a product, and that $(X, \xi)$ is *K-semistable* if it is K-semistable for some torus $T_0$ acting faithfully on $X$ such that $\xi$ is a Reeb vector of $T_0$.

**Remark 2.17.** In Definition 2.16 the action $\lambda$ need not factor through $T_0$. A product in the sense of [LWX18, Definition 2.14] is the case where $\eta_\lambda$ is a coweight vector of $T_0$. Thus K-polystability for $T_0$ in the sense of [LWX18, Definition 2.23] implies K-polystability for $T_0$ in the sense of Definition 2.16. We use Definition 2.16 for two reasons. First, with the convention of Definition 2.16, a Fano cone $(X, c\zeta)$, where $c > 0$ and $\zeta$ is a primitive cocharacter, is K-polystable for $\zeta(\mathbb{G}_m)$ if and only if the quotient log Fano pair of $(X, \zeta)$ is K-polystable (Lemma 2.24), and Theorem 1.2(4) concerns the log Fano pair $(E, \Delta_E)$. Second, with the convention of [LWX18, Definition 2.14], the cone $\mathbb{A}^2$ with the scalar action $\zeta$ is not K-polystable for $\zeta(\mathbb{G}_m)$ (Example 2.25).

The rigidity results of the later sections are equivariant splittings of marked test configurations, and the following lemma shows that such a splitting is a product.

**Lemma 2.18.** *Notation and conditions as in Set-up 2.15, and let $\eta$ be the vector field generating $\Lambda$. Assume that $(X, \xi)$ is a Fano cone, that $T$ acts faithfully on $X$ and $\xi$ is a Reeb vector of $T$, that $(\mathcal{X}, \eta)$ is a $T$-equivariant test configuration of $(X, \xi)$ in the sense of* [LWX18, Definition 2.14]*, and that there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ over $\mathbb{C}[t]$. Then $(\mathcal{X}, \eta)$ is a product in the sense of Definition 2.16.*

*Proof.* Let $\lambda$ be the action of $\Lambda$ on $X_0$, and let $\eta_\lambda$ be the vector field generating $\lambda$. By the definition of the actions on $(\mathcal{R}/t\mathcal{R})[t]$ in Set-up 2.15, $\Lambda$ acts on $X_0 \times \mathbb{A}^1$ through $\lambda$ and with weight $-1$ on $t$, so the vector field generating $\Lambda$ on $X_0 \times \mathbb{A}^1$ is $\eta_\lambda - t\partial_t$. Thus the given isomorphism defines a $T$-equivariant isomorphism $\kappa_0 : \mathcal{X} \to X_0 \times \mathbb{A}^1$ over $\mathbb{A}^1$ which maps $\eta$ to $\eta_\lambda - t\partial_t$. Since $T$ fixes $t$, setting $t = 1$ in the given isomorphism and in the marking $\mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$ yields $T$-equivariant isomorphisms $\mathcal{R}/t\mathcal{R} \cong \mathcal{R}/(t-1)\mathcal{R} \cong R$, that is, a $T$-equivariant isomorphism $\iota : X \to X_0$. Since the actions of $T$ and $\Lambda$ on $\mathcal{R}$ commute and preserve $t\mathcal{R}$, $\lambda$ commutes with the action of $T$ on $X_0$. Let $\lambda'$ be the action of $\mathbb{G}_m$ on $X$ such that $\iota$ is $\mathbb{G}_m$-equivariant for $\lambda'$ and $\lambda$. Then $\lambda'$ commutes with $T$. Let

$$\kappa := (\iota^{-1} \times \mathrm{id}_{\mathbb{A}^1}) \circ \kappa_0 : \mathcal{X} \to X \times \mathbb{A}^1.$$

Then $\kappa$ is a $T$-equivariant isomorphism over $\mathbb{A}^1$ which maps $\eta$ to $\eta_{\lambda'} - t\partial_t$. □

**Lemma 2.19.** *Let $(X, \xi)$ be a Fano cone, let $T_0$ be a torus acting faithfully on $X$ such that $\xi$ is a Reeb vector of $T_0$, and let $c > 0$ be a real number. Then $(X, \xi)$ is K-semistable for $T_0$ if and only if $(X, c\xi)$ is K-semistable for $T_0$, and $(X, \xi)$ is K-polystable for $T_0$ if and only if $(X, c\xi)$ is K-polystable for $T_0$.*

*Proof.* Let $d := \dim X$. Since $c\xi$ is a Reeb vector of a torus if and only if $\xi$ is, the conditions defining $T_0$-equivariant test configurations and special test configurations [LWX18, Definitions 2.14 and 2.16] do not change when $\xi$ is replaced with $c\xi$, and the definition of a product in Definition 2.16 does not involve $\xi$.

Let $(\mathcal{X}, \eta)$ be a $T_0$-equivariant special test configuration of $(X, \xi)$ with central fiber $X_0$. Let $A$ be the linear function on the Lie algebra of the torus $T_0 \times \mathbb{G}_m$, which acts on $X_0$ through $T_0$ and through the $\mathbb{G}_m$-action of $\mathcal{X}$ generated by $\eta$, with $A(\xi') = A_{X_0}(\mathrm{wt}_{\xi'})$ for any Reeb vector $\xi'$, and let $\mathrm{vol}_{X_0}(\xi') := \mathrm{vol}(\mathrm{wt}_{\xi'})$ [LWX18, Definition 2.10, the sentence after (7), the paragraph before (8), and the paragraph after Definition 2.23]. Let

$$P_\xi(\eta) := \frac{A(\xi)\eta - A(\eta)\xi}{d}$$

[LWX18, (8)], where it is denoted $T_\xi(\eta)$, and for a vector $u$ let $D_u \operatorname{vol}_{X_0}(\xi) := \frac{d}{d\epsilon}\big|_{\epsilon=0} \operatorname{vol}_{X_0}(\xi + \epsilon u)$ [LWX18, Definition 2.22]. Since $\operatorname{wt}_{c\xi'} = c \operatorname{wt}_{\xi'}$ and $\mathfrak{a}_m(c \operatorname{wt}_{\xi'}) = \mathfrak{a}_{m/c}(\operatorname{wt}_{\xi'})$, we have $A(c\xi') = cA(\xi')$ and $\operatorname{vol}_{X_0}(c\xi') = c^{-d} \operatorname{vol}_{X_0}(\xi')$. Thus the derivative of $\operatorname{vol}_{X_0}$ in any direction is homogeneous of degree $-d-1$, and

$$P_{c\xi}(\eta) = \frac{A(c\xi)\eta - A(\eta)c\xi}{d} = cP_\xi(\eta).$$

By [LWX18, Definition 2.22], we obtain

$$\operatorname{Fut}(\mathcal{X}, c\xi; \eta) = \frac{D_{cP_\xi(\eta)} \operatorname{vol}_{X_0}(c\xi)}{\operatorname{vol}_{X_0}(c\xi)} = \frac{c \cdot c^{-d-1} D_{P_\xi(\eta)} \operatorname{vol}_{X_0}(\xi)}{c^{-d} \operatorname{vol}_{X_0}(\xi)} = \operatorname{Fut}(\mathcal{X}, \xi; \eta).$$

Thus the conditions in Definition 2.16 for $(X, \xi)$ and for $(X, c\xi)$ coincide. □

**Theorem 2.20** ([XZ26b, Theorem 2.1 and Paragraph 2.14])**.** *Let $(X \ni x)$ be a klt singularity. Then there exists a minimizer $v$ of $\widehat{\operatorname{vol}}$, the valuation $v$ is quasi-monomial, and any minimizer is a positive multiple of $v$. The ring $\operatorname{gr}_v \mathcal{O}_{X,x}$ is finitely generated, and $W := \operatorname{Spec} \operatorname{gr}_v \mathcal{O}_{X,x}$ is a normal affine variety with klt singularities and a good action of the torus $T$ whose character lattice is the value group $v(\mathbb{C}(X)^*)$. The grading by the values of $v$ defines a Reeb vector $\xi_v$ of $T$, and $(W, \xi_v)$ is a K-semistable Fano cone.*

Theorem 2.20 was conjectured in [Li18, Conjecture 7.1] and [LX18, Conjecture 1.2]. The existence of a minimizer is proved in [Blu18, Main Theorem], the quasi-monomiality in [Xu20, Theorem 1.2], the uniqueness in [XZ20], the finite generation in [XZ25], and the K-semistability of the Fano cone in [LX18]. The description of $T$ through the value group is [XZ26b, Paragraph 2.14]. Here the K-semistability of $(W, \xi_v)$ is in the sense of Definition 2.16, that is, in the sense of [LWX18, Definition 2.23] for some torus. [XZ26b, Paragraph 2.14] formulates it as the condition that $\operatorname{wt}_{\xi_v}$ minimizes $\widehat{\operatorname{vol}}$, which is equivalent by Theorem 2.21 below.

**Theorem 2.21** ([LWX18, Theorem 2.5])**.** *Let $(X, \xi)$ be a Fano cone with vertex $o$, and let $T_0$ be a torus acting faithfully on $X$ such that $\xi$ is a Reeb vector of $T_0$. Then $(X, \xi)$ is K-semistable for $T_0$ if and only if $\operatorname{wt}_\xi$ is a minimizer of $\widehat{\operatorname{vol}}$ on $\operatorname{Val}_{X,o}$. In particular, $(X, \xi)$ is K-semistable if and only if it is K-semistable for $T_0$.*

The condition that $\operatorname{wt}_\xi$ minimizes $\widehat{\operatorname{vol}}$ does not involve $T_0$, which implies the equivalence in the last sentence of Theorem 2.21.

**Theorem 2.22** ([LWX18, Theorem 1.2])**.** *Let $(X, \xi)$ be a K-semistable Fano cone with a good faithful action of a torus $T$ such that $\xi$ is a Reeb vector of $T$. Then there exists a $T$-equivariant special test configuration of $(X, \xi)$ with Futaki invariant 0 whose central fiber $(X_0, \xi)$ is a Fano cone which is K-polystable for some torus which acts faithfully on $X_0$ and contains $T$.*

**Remark 2.23.** The $T$-equivariance and the torus in Theorem 2.22 do not appear in the statement of [LWX18, Theorem 1.2]. They are obtained in [LWX18, Section 4.3], where [LWX18, Theorem 1.2] is proved: the central fiber is taken among the K-semistable special degenerations of $(X, \xi)$ as one admitting a torus of maximal dimension, the torus contains $T$, and all test configurations there are $T$-equivariant in the sense of [LWX18, Definition 2.14]. The Futaki invariant is 0 since $(X, \xi)$ and the central fiber are K-semistable and the Futaki invariant is linear in the vector field of the test configuration [LWX18, Lemma 2.25]: the product test configuration of the central fiber defined by the inverse of the $\mathbb{G}_m$-action of the test configuration has Futaki invariant of the opposite sign. By Remark 2.17, the central fiber is then K-polystable for the torus of maximal dimension in the sense of Definition 2.16.

The torus of $X_0$ in Theorem 2.22 may be larger than $T$, which is the reason for Corollary 3.13.

For log Fano pairs we use test configurations, products, Futaki invariants, K-semistability, and K-polystability in the sense of [LWX18, Definitions 2.26–2.28]. See also [Ber16, Section 4.3].

**Lemma 2.24.** *Let $(X,\xi)$ be a Fano cone with $\xi = c\zeta$, where $c>0$ and $\zeta$ is a primitive cocharacter of a torus acting faithfully on $X$, and let $(Z,\Delta_Z)$ be the quotient log Fano pair of $(X,\zeta)$. Then $(X,\xi)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$ if and only if $(Z,\Delta_Z)$ is K-polystable.*

*Proof.* By Lemma 2.19, K-polystability for $\zeta(\mathbb{G}_m)$ does not change when $\xi$ is replaced with $c^{-1}\xi = \zeta$. Thus we may assume that $\xi = \zeta$. Let $T_0 := \zeta(\mathbb{G}_m)$, write $X = \operatorname{Spec} R$ with the $\zeta$-grading $R = \bigoplus_{j\geq 0} R_j$, let $o$ be the vertex, and let $q : X\backslash\{o\} \to Z$ be the quotient map, which is a Seifert $\mathbb{G}_m$-bundle [Kol04, Definition 1 and the paragraph after it, and Paragraph 14]. Then $\mathbb{C}(Z)$ consists of the fractions $f/g$ with $f,g\in R_j$, $g\neq 0$, for some $j$, and eu is the Euler derivation of the $\zeta$-grading. Since $\zeta$ is a non-trivial cocharacter of a torus acting faithfully on $X$, we have $R_j\neq 0$ for some $j>0$.

***Step 1***. By [Li21, Section 2.5], taking the quotient by $T_0$ and the cone construction define a one-to-one correspondence between $T_0$-equivariant special test configurations of $(X,\zeta)$ and special test configurations of $(Z,\Delta_Z)$. For $\xi=\zeta$ the weight function $g$ of [Li21, Section 2.5] is constant, so by [Li21, Section 2.5] the non-Archimedean Ding invariants of corresponding test configurations differ by a positive factor. On special test configurations the non-Archimedean Ding invariant is a positive multiple of the Futaki invariant, by [LWX18, Remark A.2] for $(X,\zeta)$ and by [BHJ17, Definitions 7.13 and 7.31 and Proposition 7.32] for $(Z,\Delta_Z)$, since a special test configuration of $(Z,\Delta_Z)$ defines an anti-canonical non-Archimedean metric and has an integral central fiber. Thus the Futaki invariants of corresponding special test configurations also differ by a positive factor. If $\zeta(\mathbb{G}_m)$ acts freely on $X\backslash\{o\}$ and $X$ is the cone over $Z$ with respect to a positive rational multiple of $-(K_Z+\Delta_Z)$, the comparison of the Futaki invariants is also [LWX18, Definition 2.13, the paragraph before Lemma 2.30, and Lemma 2.30]. The vector field $\eta$ of a $T_0$-equivariant test configuration of $(X,\zeta)$ commutes with eu, so $\eta$ descends to the quotient. Since $\pi_*\eta = -t\partial_t$ in [LWX18, Definition 2.14] while the vector field of a product in [LWX18, Definition 2.26] has the form $\eta_0 + t\partial_t$, the vector field of the corresponding test configuration of $(Z,\Delta_Z)$ is the negative of the descended vector field. In particular, $(X,\zeta)$ is K-semistable for $T_0$ if and only if $(Z,\Delta_Z)$ is K-semistable. Thus it suffices to show that a $T_0$-equivariant special test configuration $(\mathcal{X},\eta)$ of $(X,\zeta)$ is a product in the sense of Definition 2.16 if and only if the corresponding test configuration of $(Z,\Delta_Z)$ is a product in the sense of [LWX18, Definition 2.26].

***Step 2***. We show that if $\delta$ is a derivation of $R$ with $\delta(R_j)\subset R_j$ for any $j$ and $\delta(\mathbb{C}(Z))=0$, then $\delta = c'\operatorname{eu}$ for a constant $c'\in\mathbb{C}$. The field $\mathbb{C}(X)$ has transcendence degree 1 over $\mathbb{C}(Z)$, and eu is a non-zero derivation of $\mathbb{C}(X)$ over $\mathbb{C}(Z)$. Thus the derivations of $\mathbb{C}(X)$ over $\mathbb{C}(Z)$ form a $\mathbb{C}(X)$-vector space of dimension 1, and $\delta = f\operatorname{eu}$ for some $f\in\mathbb{C}(X)$. The vertex $o$ is the only fixed point of $T_0$, so the stabilizers of $T_0$ on $X\backslash\{o\}$ are finite. If eu vanished at a point $y\in X_{\mathrm{reg}}\backslash\{o\}$, then the orbit map $\mathbb{G}_m\to X$, $s\to\zeta(s)\cdot y$, would have zero differential at 1, hence everywhere by equivariance, and $y$ would be fixed by $T_0$. Thus eu is nowhere zero on $X_{\mathrm{reg}}\backslash\{o\}$. Near any $y\in X_{\mathrm{reg}}\backslash\{o\}$ there exists a 1-form $\phi$ with $\phi(\operatorname{eu})=1$, so $f=\phi(\delta)$ is regular near $y$. Since $X$ is normal and $\operatorname{codim}_X X_{\mathrm{sing}}\geq 2$, $f$ extends to a regular function on $X\backslash\{o\}$. Since $\delta$ preserves the grading, $[\operatorname{eu},\delta]=0$, hence

$$0 = [\operatorname{eu}, f\operatorname{eu}] = \operatorname{eu}(f)\operatorname{eu},$$

and $\operatorname{eu}(f)=0$. Thus $f$ is $T_0$-invariant and descends to a regular function on $Z$. Since $Z$ is a normal projective variety (Lemma 2.7), $f$ is constant. The argument does not use $\dim X\geq 2$: if $\dim X = 1$, then $Z$ is a point.

***Step 3***. Assume that $(\mathcal{X},\eta)$ is a product in the sense of Definition 2.16, with $\kappa$ and $\lambda$ as there. Since $\lambda$ commutes with $T_0$, $\lambda$ descends to a $\mathbb{G}_m$-action $\mu$ on $Z$. Since $\lambda$ preserves the orders of the generic stabilizers of $T_0$ on the preimages of prime divisors of $Z$, $\mu$ preserves $\Delta_Z$. The quotient of $(X\backslash\{o\})\times\mathbb{A}^1$ by $T_0$ is $Z\times\mathbb{A}^1$, and $\eta_\lambda - t\partial_t$ descends to $\eta_\mu - t\partial_t$, where $\eta_\mu$ is the vector field generating $\mu$. By **Step 1**, the test configuration of $(Z,\Delta_Z)$ corresponding to $(\mathcal{X},\eta)$ is $(Z,\Delta_Z)\times\mathbb{A}^1$ with the vector field $-\eta_\mu + t\partial_t$. Since $-\eta_\mu$ is a coweight vector of the torus $\mu(\mathbb{G}_m)$ acting on $(Z,\Delta_Z)$, it is a product in the sense of [LWX18, Definition 2.26].

***Step 4***. Assume that the test configuration of $(Z, \Delta_Z)$ corresponding to $(\mathcal{X}, \eta)$ is a product, that is, $(Z, \Delta_Z) \times \mathbb{A}^1$ with the vector field $\eta_0 + t\partial_t$, where $\eta_0$ is a coweight vector of a torus acting on $(Z, \Delta_Z)$. The cone over $(Z, \Delta_Z) \times \mathbb{A}^1$ is $X \times \mathbb{A}^1 = \operatorname{Spec} R[t]$, with $T_0$ acting through the grading of $R$ and fixing $t$, so we may assume that $\mathcal{X} = \operatorname{Spec} R[t]$. By **Step 1**, $\eta$ descends to $-\eta_0 - t\partial_t$. The vector field $\eta$ is a derivation of $R[t]$ with $\eta(t) = -t$, and $\eta(R_j[t]) \subset R_j[t]$ for any $j$ since $\eta$ commutes with eu. Let $\delta := \eta + t\partial_t$. Then $\delta$ is a $\mathbb{C}[t]$-linear derivation of $R[t]$ which descends to $-\eta_0$. Since $R$ is finitely generated, there exist derivations $\delta_0, \dots, \delta_m$ of $R$ with $\delta_k(R_j) \subset R_j$ for any $j$ and $k$, such that

$$\delta(f) = \sum_{k=0}^{m} t^k \delta_k(f) \quad \text{for any } f \in R.$$

For $g \in \mathbb{C}(Z)$ we have $\sum_k t^k \delta_k(g) = -\eta_0(g)$, which does not depend on $t$. Thus $\delta_k(\mathbb{C}(Z)) = 0$ for $k \geq 1$, and by **Step 2** there exist constants $c_k \in \mathbb{C}$ with $\delta_k = c_k \operatorname{eu}$ for $1 \leq k \leq m$.

We show that $c_k = 0$ for $1 \leq k \leq m$. Since $\eta$ generates an algebraic $\mathbb{G}_m$-action on $R[t]$, $R[t]$ is the direct sum of the eigenspaces $V_n$ of $\eta$ with eigenvalues $n \in \mathbb{Z}$. If $W \subset R[t]$ is a subspace with $\eta(W) \subset W$ and $w = \sum_n w_n \in W$ with $w_n \in V_n$, then $\eta^l(w) = \sum_n n^l w_n \in W$ for any $l \geq 0$, so $w_n \in W$ by the Vandermonde determinant. Thus any such $W$ is the direct sum of the subspaces $W \cap V_n$, and $\mathbb{G}_m$ acts on $W$. Suppose that $c_k \neq 0$ for some $k \geq 1$, and let $K$ be the largest such $k$. Let $j > 0$ with $R_j \neq 0$. Since $\eta(R_j[t]) \subset R_j[t]$, there exists an eigenvector $p = \sum_{i=0}^{M} t^i p_i$ of $\eta$ with $p_i \in R_j$ and $p_M \neq 0$. For $f \in R_j$ we have

$$\eta(t^i f) = t^i \delta_0(f) - i t^i f + j \sum_{k=1}^{K} c_k t^{i+k} f,$$

so the coefficient of $t^{M+K}$ in $\eta(p)$ is $j c_K p_M \neq 0$. Since $p$ has degree $M < M + K$ in $t$, we obtain $\eta(p) \notin \mathbb{C}p$, a contradiction. Thus $c_k = 0$ for $k \geq 1$, and $\eta = \delta_0 - t\partial_t$, where $\delta_0$ is extended $\mathbb{C}[t]$-linearly.

In particular $\eta(R) \subset R$, so $\mathbb{G}_m$ acts on $R \subset R[t]$, which defines a $\mathbb{G}_m$-action $\lambda$ on $X$ with $\eta_\lambda = \delta_0$. Since $\delta_0(R_j) \subset R_j$ for any $j$, $\lambda$ commutes with $T_0$. Thus $(\mathcal{X}, \eta) = (X \times \mathbb{A}^1, \eta_\lambda - t\partial_t)$ is a product in the sense of Definition 2.16, with $\kappa$ the identity. Together with **Step 1** and **Step 3**, we obtain the lemma. □

**Example 2.25.** Let $X := \mathbb{A}^2$, let $\zeta$ be the scalar $\mathbb{G}_m$-action on $X$, and let $\lambda$ be the $\mathbb{G}_m$-action on $X$ of weights $(1, -1)$. Then $X \times \mathbb{A}^1$ with the vector field $\eta_\lambda - t\partial_t$, where $\eta_\lambda$ is the vector field generating $\lambda$, is a $\zeta(\mathbb{G}_m)$-equivariant special test configuration of the Fano cone $(X, \zeta)$ with Futaki invariant 0. It is a product in the sense of Definition 2.16, but not in the sense of [LWX18, Definition 2.14] for the torus $\zeta(\mathbb{G}_m)$. The Fano cone $(X, \zeta)$ is K-polystable for $\zeta(\mathbb{G}_m)$ in the sense of Definition 2.16, and it is not K-polystable for $\zeta(\mathbb{G}_m)$ in the sense of [LWX18, Definition 2.23].

*Proof.* Let $x_1, x_2$ be the coordinates of $X$, and let $T := \mathbb{G}_m^2$ act on $X$ by $(s_1, s_2) \cdot (x_1, x_2) = (s_1 x_1, s_2 x_2)$. We identify the cocharacters of $T$ with $\mathbb{Z}^2$, so that $\zeta = (1, 1)$ and $\lambda = (1, -1)$. The Reeb vectors of $T$ are the vectors $\xi = (\xi_1, \xi_2)$ with $\xi_1, \xi_2 > 0$, and $\operatorname{wt}_\xi$ is the monomial valuation with $\operatorname{wt}_\xi(x_i) = \xi_i$. Thus

$$A(\xi) := A_X(\operatorname{wt}_\xi) = \xi_1 + \xi_2, \qquad \operatorname{vol}_X(\xi) := \operatorname{vol}(\operatorname{wt}_\xi) = \frac{1}{\xi_1 \xi_2}.$$

The family $X \times \mathbb{A}^1 \to \mathbb{A}^1$ is flat, $\eta_\lambda - t\partial_t$ generates the $\mathbb{G}_m$-action $s \cdot (y, t) = (\lambda(s) \cdot y, s^{-1}t)$, and $\zeta(\mathbb{G}_m)$ acts on the first factor and commutes with $\lambda$. Thus $(X \times \mathbb{A}^1, \eta_\lambda - t\partial_t)$ is a $\zeta(\mathbb{G}_m)$-equivariant test configuration of $(X, \zeta)$ [LWX18, Definition 2.14]. Since $X \times \mathbb{A}^1$ is smooth and the central fiber $X$ is smooth, it is special [LWX18, Definition 2.16]. The restriction of $\eta_\lambda - t\partial_t$ to the central fiber is $\eta_\lambda$. The linear extension of $A$ satisfies $A(\eta_\lambda) = 1 - 1 = 0$ and $A(\zeta) = 2$, so with $P_\zeta$ as defined in the proof of Lemma 2.19

$$P_\zeta(\eta_\lambda) = \frac{A(\zeta)\eta_\lambda - A(\eta_\lambda)\zeta}{2} = \eta_\lambda.$$

By [LWX18, Definition 2.22], the Futaki invariant is

$$\frac{D_{\eta_\lambda}\operatorname{vol}_X(\zeta)}{\operatorname{vol}_X(\zeta)} = \frac{d}{d\epsilon}\Big|_{\epsilon=0}\frac{1}{(1+\epsilon)(1-\epsilon)} = 0.$$

The test configuration is a product in the sense of Definition 2.16, with $\kappa$ the identity and the action $\lambda$, which commutes with $\zeta(\mathbb{G}_m)$. Suppose that it is a product in the sense of [LWX18, Definition 2.14] for $\zeta(\mathbb{G}_m)$. Then there exists a $\zeta(\mathbb{G}_m)$-equivariant isomorphism $\psi : X \times \mathbb{A}^1 \to X \times \mathbb{A}^1$ over $\mathbb{A}^1$ mapping $\eta_\lambda - t\partial_t$ to $m\zeta - t\partial_t$ for some $m$. The restriction $\psi_0$ of $\psi$ to the central fiber is a $\zeta(\mathbb{G}_m)$-equivariant automorphism of $X$ mapping $\eta_\lambda$ to $m\zeta$, so $\psi_0$ fixes the vertex 0, the only fixed point of $\zeta(\mathbb{G}_m)$. The linear parts of $\eta_\lambda$ and $m\zeta$ at 0 have eigenvalues $\{1,-1\}$ and $\{m,m\}$, and $d\psi_0(0)$ conjugates the first linear part to the second, a contradiction.

The group $\zeta(\mathbb{G}_m)$ acts freely on $X\backslash\{0\}$, so the quotient log Fano pair of $(X,\zeta)$ is $(\mathbb{P}^1, 0)$. The Fubini–Study metric is a Kähler–Einstein metric, so $\mathbb{P}^1$ is K-polystable by [Ber16, Theorem 4.8] with $D = 0$. The K-polystability of [Ber16] is tested on all test configurations, among them the special ones, so $\mathbb{P}^1$ is K-polystable in the sense of [LWX18, Definition 2.28]. By Lemma 2.24, $(X,\zeta)$ is K-polystable for $\zeta(\mathbb{G}_m)$ in the sense of Definition 2.16. The test configuration above is special with Futaki invariant 0 and is not a product in the sense of [LWX18, Definition 2.14], so $(X,\zeta)$ is not K-polystable for $\zeta(\mathbb{G}_m)$ in the sense of [LWX18, Definition 2.23]. □

### 2.5. **Ricci-flat Kähler cone metrics.**

**Definition 2.26.** We use Kähler cone metrics, radius functions, Reeb vector fields, and contact forms in the sense of [Li21, Definition 2.1 and (61)–(62)]. We write $dd^c := \sqrt{-1}\partial\bar\partial$. The Kähler form of a Kähler cone metric $g = d\rho^2 + \rho^2 h$ with radius function $\rho$ is $\frac{1}{2}dd^c\rho^2$ [CS15, Section 3, p. 8].

**Theorem 2.27** ([Li21, Theorem 2.9])**.** *Let $(X,\xi)$ be a Fano cone with $X = \operatorname{Spec} R$ of dimension $d \geq 2$, with vertex $o$ and a good faithful action of a torus $T$ such that $\xi$ is a Reeb vector of $T$ with $A_X(\operatorname{wt}_\xi) = d$, and assume that $(X,\xi)$ is K-polystable for $T$. Let $T_c \subset T$ be the maximal compact subtorus, let $M := X_{\mathrm{reg}}\backslash\{o\}$, let $\chi$ be a primitive cocharacter of $T$ which is a Reeb vector, and let $(Z,\Delta_Z)$ be the quotient log Fano pair of $(X,\chi)$. Then there exist $T_c$-invariant smooth functions $\rho > 0$ and $\rho_\chi > 0$ on $M$ with the following properties.*

1. *The function $\rho^2$ extends to a locally bounded plurisubharmonic function on $X$, the form $\frac{1}{2}dd^c\rho^2$ defines a Ricci-flat Kähler cone metric $g = d\rho^2 + \rho^2 h$ on $M$, and $\rho\partial_\rho s = \langle\alpha,\xi\rangle s$ for any $s \in R_\alpha$. If $\Omega_X^{[d]}$ is generated by a $T$-eigenvector $\Omega$, then $(dd^c\rho^2)^d = C(\sqrt{-1})^{d^2}\Omega\wedge\bar\Omega$ on $M$ for a constant $C > 0$.*
2. *The function $\rho_\chi$ is the radius function of a Kähler cone metric on $M$ whose Reeb vector field is the vector field generated by $\chi$, and $\{\rho_\chi = 1\} = \{\rho = 1\}$. The form $dd^c\log\rho_\chi^2$ descends to a Kähler form on the quotient orbifold $[M/\chi(\mathbb{G}_m)]$ which satisfies the weighted soliton equation of* [Li21, Section 2.4] *and is equal to $\theta + dd^c\varphi$ on $[M/\chi(\mathbb{G}_m)]$, where $\theta$ is a positive multiple of the restriction of a Fubini–Study form under a projective embedding of $Z$ and $\varphi$ is a bounded $\theta$-plurisubharmonic function on $Z$.*
3. *Let $\eta_\chi$ be the contact form of $\rho_\chi$, that is, the 1-form on $M$ with $\eta_\chi(\chi) = 1$ which vanishes on $\rho_\chi\partial_{\rho_\chi}$ and on the orthogonal complement, with respect to the Kähler cone metric of $\rho_\chi$, of the span of $\rho_\chi\partial_{\rho_\chi}$ and $\chi$. Then there exist constants $0 < a \leq b$ such that $a \leq \eta_\chi(\xi) \leq b$ on $M$.*

**Remark 2.28.** Item (1) of Theorem 2.27 is [Li21, Theorem 2.9] together with the definition of a weak Ricci-flat Kähler cone potential before it [Li21, p. 34]: such a potential is a locally bounded function on $X$ which is a smooth radius function over $X_{\mathrm{reg}}$ and satisfies the Ricci-flat Kähler cone equation [Li21, Definition 2.2 and (73)] in the pluripotential sense. The volume form in the equation is defined by a $T$-equivariant generator of $\omega_X^{[m]}$ for some $m > 0$ [Li21, Definition 2.2 and (72)–(75)], and two $T$-eigenvectors generating $\omega_X^{[m]}$ differ by a non-zero constant factor, which

implies the formula for $(dd^c\rho^2)^d$. The identity $\rho\partial_\rho s = \langle\alpha,\xi\rangle s$ follows from $J(\rho\partial_\rho) = \xi$ [Li21, Definition 2.1] and the holomorphy of $s$. By [Li21, p. 34], the weighted soliton potential on the quotient is smooth on the orbifold locus. By the slice theorem, the image of $M$ in the quotient is contained in the orbifold locus, so the metrics in (1) and (2) are smooth on $M$.

The hypothesis of [Li21, Theorem 2.9] is that $(X,\xi)$ is Ding-polystable, and by the last sentence of [Li21, Theorem 2.9] it suffices to test Ding-polystability on special test configurations. A product in the sense of Definition 2.16 has central fiber isomorphic to $X$, so K-polystability for $T$ implies $T$-equivariant K-polystability in the sense of [CS15, Definition 2.2], which is a condition on special test configurations. In [CS15] the singularity is assumed to be isolated, and [Li21, Section 2.5] uses [CS15, Definition 2.2], with special test configurations in the sense of Collins–Székelyhidi, for arbitrary Fano cones. On special test configurations the non-Archimedean Ding invariant of [Li21, Section 2.5, p. 33] equals the Futaki invariant [LWX18, Remark A.2], so by [Li21, Section 2.5] this is Ding-polystability of $(X,\xi)$ tested on special test configurations.

Items (2) and (3) do not appear in the statement of [Li21, Theorem 2.9]. They are proved on [Li21, pp. 34–35], in the proof of [Li21, Theorem 2.9]: $\rho_\chi$ is the radius function defined by the bounded weighted soliton potential on the quasi-regular quotient [HL20, Corollary 3.6], which [Li21, p. 34] cites under a different numbering, $\rho$ is the deformation of $\rho_\chi$ with respect to $\xi$ [Li21, Section 2.2], the equality $\{\rho_\chi = 1\} = \{\rho = 1\}$ is [Li21, p. 35, footnote 3], and the bounds in (3) come from the moment polytope [Li21, p. 35], where $\eta_\chi(\xi)$ is the positive function in the decomposition of $\xi$ in [Li21, Section 2.2, (79)–(83)]. The potential of [HL20, Corollary 3.6] is bounded with respect to a smooth Kähler form $\omega_0$ in $c_1$ of the polarization, for instance a positive multiple of the restriction of a Fubini–Study form, and $\theta - \omega_0$ is $dd^c$ of a continuous function on the compact space $Z$, so $\varphi$ is bounded.

**Definition 2.29.** Let $\Phi(t) := t^2\log^2(e+t)$ for $t \geq 0$. Let $Q$ be a Riemannian manifold or orbifold with metric $h$. A sequence $(\beta_j)_{j\geq 1}$ of smooth compactly supported functions on $Q$ is a sequence of $\Phi$-*cutoffs* if $0 \leq \beta_j \leq 1$, if for any compact subset $W \subset Q$ we have $\beta_j|_W = 1$ for $j \gg 0$, and if

$$\lim_{j\to+\infty} \int_Q \Phi(|d\beta_j|_h)\, d\operatorname{Vol}_h = 0.$$

**Remark 2.30.** Since $\Phi(t) \geq t^2$, the integrals $\int_Q |d\beta_j|_h^2 d\operatorname{Vol}_h$ of a sequence of $\Phi$-cutoffs also tend to 0. For $c \geq 1$ and $t \geq 0$ we have $\log(e+ct) \leq \log c + \log(e+t) \leq (1+\log c)\log(e+t)$, since $\log(e+t) \geq 1$. For $0 < c \leq 1$ we have $\log(e+ct) \leq \log(e+t)$. Thus, for any real numbers $a > 0$ and $t \geq 0$,

$$\Phi(at) \leq a^2\big(1+\log\max\{1,a\}\big)^2\Phi(t). \tag{2.1}$$

The logarithmic factor in $\Phi$ is used for two estimates. First, on a bounded domain in $\mathbb{R}^2$ the Orlicz–Sobolev space defined by $\Phi$ embeds into $C^0$ [Cia96], because

$$\int^{+\infty} t\Phi(t)^{-1}dt = \int^{+\infty} t^{-1}\log^{-2}(e+t)dt < +\infty.$$

For the function $t^2$ in place of $\Phi$ the corresponding integral diverges, and $\log\log(1/|y|)$ belongs to $W^{1,2}$ near $0 \in \mathbb{R}^2$ but is unbounded. Second, the weighted error terms in the Bochner argument of Section 4 are controlled by the factor $\log^2(e+t)$ in $\Phi$. Theorem 3.9 constructs $\Phi$-cutoffs on the quotient orbifold and on the link for the metric of Theorem 2.27.

**Set-up 2.31.** Let $X = \operatorname{Spec} R$, $T$, $o$, $T_c$, $\xi$, and $M$ be as follows: $X$ is an irreducible normal affine variety of dimension $d \geq 2$ with klt singularities and a good faithful action of the torus $T$ with vertex $o$, $T_c \subset T$ is the maximal compact subtorus, $\xi$ is a Reeb vector of $T$, and $M := X_{\mathrm{reg}}\backslash\{o\}$. We consider the following conditions.

(1) The manifold $M$ carries a $T_c$-invariant smooth Ricci-flat Kähler cone metric $g = d\rho^2 + \rho^2 h$, where $M = (0,+\infty)\times N$ with $N = \{\rho = 1\}$ connected, and $\rho\partial_\rho s = \langle\alpha,\xi\rangle s$ for any $s \in R_\alpha$.

(2) The function $\rho$ extends to a continuous proper function $X \to [0, +\infty)$ with $\rho^{-1}(0) = \{o\}$, and $\rho^2$ extends to a locally bounded plurisubharmonic function on $X$.
(3) $\mathrm{Vol}_h(N) < +\infty$.
(4) The link $N$ admits a sequence of $T_c$-invariant $\Phi$-cutoffs.

Conditions (1)–(4) are the hypotheses of Proposition 4.3, Theorem 4.7, and Theorem 8.2. Theorem 4.6 and Lemma 4.5 have their own hypotheses, and where they are applied, conditions (1), (3), and (4) imply the hypotheses on the metric, on the volume of $N$, and on the cutoffs. Condition (1) gives the Bochner formula in Proposition 4.3, conditions (2) and (3) give the finiteness of the integrals over compact annuli and over $N$, and condition (4) enters Proposition 4.3, Lemma 4.5, and Theorems 4.6, 4.7, and 8.2 only through the existence of a sequence of $\Phi$-cutoffs on $N$. The $T_c$-invariance in condition (4) is used to construct the cutoffs: in Theorem 3.9 they are built on the quotient and pulled back to $N$ by Lemma 3.3. Corollary 3.11 verifies conditions (1)–(4) for the metric of Theorem 2.27.

## 3. Weak Ricci-flat cone metrics

In this section, we prove Corollary 3.11 and Corollary 3.13. For the weak metric of Theorem 2.27 on a K-polystable Fano cone, we check the analytic conditions of Set-up 2.31, and at the end of the section, we deduce K-polystability of the quotient log Fano pair from a continuous cone potential.

3.1. **The radius and the link.** In this subsection, we fix notation for the weak metric of Theorem 2.27, and we show that the radius function of the metric is continuous and proper on the cone and that the link has finite volume.

**Set-up 3.1.** Let $(X, \xi)$, $T$, $T_c$, $M$, $\chi$, $\rho$, $\rho_\chi$, $g = d\rho^2 + \rho^2 h$, $N$, $\eta_\chi$, $U$, $g_U$, $(Z, \Delta_Z)$, $f$, $a$, and $b$ be as follows.

(1) $(X, \xi)$ is a Fano cone of dimension $d \geq 2$ with vertex $o$ and a good faithful action of a torus $T$ such that $\xi$ is a Reeb vector of $T$ with $A_X(\mathrm{wt}_\xi) = d$, and $(X, \xi)$ is K-polystable for $T$.
(2) $T_c \subset T$ is the maximal compact subtorus, and $M := X_{\mathrm{reg}} \backslash \{o\}$.
(3) $\chi$ is a primitive cocharacter of $T$ which is a Reeb vector.
(4) $\rho > 0$ and $\rho_\chi > 0$ are $T_c$-invariant smooth functions on $M$ satisfying (1)–(3) of Theorem 2.27 for $\chi$ (see Remark 2.28), $g = d\rho^2 + \rho^2 h$ is the Kähler cone metric on $M$ defined by $\frac{1}{2} dd^c \rho^2$, and $N := \{\rho = 1\}$, so that $M = (0, +\infty) \times N$.
(5) $\eta_\chi$ is the contact form of $\rho_\chi$, as in Theorem 2.27(3).
(6) $U := [M/\chi(\mathbb{G}_m)]$ is the quotient orbifold, and $g_U$ is the Kähler metric on $U$ whose pullback under the quotient map to the contact distribution $\ker \eta_\chi \cap TN$ of $N$ is $\frac{1}{2} d\eta_\chi(\cdot, J\cdot)$.
(7) $(Z, \Delta_Z)$ is the quotient log Fano pair of $(X, \chi)$.
(8) $f$ is the function on $U$ induced by $\eta_\chi(\xi)$, and $a, b$ are constants with $0 < a \leq f \leq b$.

**Lemma 3.2.** *Let $(X, \xi)$, $T$, $T_c$, $M$, $\chi$, and $(Z, \Delta_Z)$ be as in Set-up 3.1(1)–(3) and (7). Then there exist functions $\rho$ and $\rho_\chi$ as in Set-up 3.1(4). Moreover, for any such $\rho$ and $\rho_\chi$ and for $\eta_\chi$ as in Set-up 3.1(5), denoting also by $\xi$ and $\chi$ the vector fields on $M$ generated by $\xi$ and $\chi$, the following hold.*

*(1) We have $\xi = J(\rho \partial_\rho)$ and $\chi = J(\rho_\chi \partial_{\rho_\chi})$. The flow of $\rho_\chi \partial_{\rho_\chi}$ at time $s$ is the action of $\chi(e^s)$, and $\rho_\chi(\chi(\lambda) \cdot y) = |\lambda| \rho_\chi(y)$ for $\lambda \in \mathbb{C}^*$ and $y \in M$.*
*(2) We have $\eta_\chi(W) = -d \log \rho_\chi(JW)$ for any tangent vector $W$ of $M$, and $d\eta_\chi = dd^c \log \rho_\chi^2$ on $M$.*
*(3) The function $\eta_\chi(\xi)$ on $M$ is invariant under $\chi(\mathbb{G}_m)$, and there exist constants $0 < a \leq b$ such that $a \leq \eta_\chi(\xi) \leq b$ on $M$.*
*(4) There exists a unique Kähler metric $g_U$ on $U$ as in Set-up 3.1(6), and the Kähler form of $g_U$ is $\frac{1}{2}$ times the Kähler form on $U$ to which $dd^c \log \rho_\chi^2$ descends (Theorem 2.27(2)).*

*In particular, objects as in Set-up 3.1 exist.*

*Proof.* By Theorem 2.27, functions $\rho$ and $\rho_\chi$ as in Set-up 3.1(4) exist.

We prove (1). By Theorem 2.27(1) and Definition 2.6, both $\rho\partial_\rho$ and $-J\xi$ act on $R_\alpha$ by multiplication by $\langle\alpha,\xi\rangle$. Since $X$ embeds into an affine space by finitely many $T$-eigenfunctions, a real vector field on $M$ is determined by the derivatives of the functions in $R$ along it. Thus $\xi = J(\rho\partial_\rho)$. By Theorem 2.27(2), the Reeb vector field of $\rho_\chi$ is the field generated by $\chi$, that is, $\chi = J(\rho_\chi\partial_{\rho_\chi})$ [Li21, Section 2.1, Definition 2.1 and (61)–(62)]. Hence $\rho_\chi\partial_{\rho_\chi} = -J\chi$ acts on $R_\alpha$ by multiplication by $\langle\alpha,\chi\rangle$, so by the convention $\exp(2\pi y) = 1$ of Definition 2.6, the flow of $\rho_\chi\partial_{\rho_\chi}$ at time $s$ is the action of $\chi(e^s)$. Since $(\rho_\chi\partial_{\rho_\chi})\rho_\chi = \rho_\chi$ and $\rho_\chi$ is $T_c$-invariant, we have $\rho_\chi(\chi(\lambda)\cdot y) = |\lambda|\rho_\chi(y)$ for $\lambda\in\mathbb{C}^*$ and $y\in M$.

Statement (2) is [Li21, Section 2.4, (97) and the line after it].

We prove (3). By Theorem 2.27(3), there exist constants $0 < a \le b$ such that $a \le \eta_\chi(\xi) \le b$ on $M$. By (1), $d\log\rho_\chi$ is invariant under $\chi(\mathbb{G}_m)$. Since $\chi(\lambda)$ is holomorphic for $\lambda\in\mathbb{C}^*$, it commutes with $J$, so $\eta_\chi$ is invariant under $\chi(\mathbb{G}_m)$ by (2). Since $T$ is commutative, the vector field $\xi$ is invariant under $\chi(\mathbb{G}_m)$. Thus the function $\eta_\chi(\xi)$ is invariant under $\chi(\mathbb{G}_m)$, and it induces a function $f$ on the quotient orbifold $[M/\chi(\mathbb{G}_m)]$ with $a\le f\le b$.

We prove (4). Let $N := \{\rho = 1\}$ as in Set-up 3.1(4). Since $\eta_\chi(\chi) = 1$, the distribution $\ker\eta_\chi\cap TN$ does not contain $\chi$, so the quotient map $N\to U$ maps it isomorphically onto the tangent spaces of $U$ in uniformizing charts. By (2), $d\eta_\chi = dd^c\log\rho_\chi^2$ is a $J$-invariant real $(1,1)$-form which descends to the Kähler form of Theorem 2.27(2). Thus, with the convention of Definition 2.26, the form $\frac12 d\eta_\chi(\cdot, J\cdot)$ on $\ker\eta_\chi\cap TN$ is the pullback of the Kähler metric on $U$ whose Kähler form is $\frac12$ times the descended form. Since a metric on $U$ is determined by its pullback to $\ker\eta_\chi\cap TN$, the metric $g_U$ exists and is unique. □

**Lemma 3.3.** *Notation and conditions as in Set-up 3.1, and let $p: N\to U$ be the map induced by the quotient map $M\to U$. Then there exists a constant $\ell > 0$ such that $p_*(d\operatorname{Vol}_h) = \ell f^{-d} d\operatorname{Vol}_{g_U}$, and any $T_c$-invariant smooth function $u$ on $U$ satisfies $|d(u\circ p)|_h^2 = (f|du|_{g_U}^2)\circ p$. In particular, $\operatorname{Vol}_h(N) < +\infty$ if $\operatorname{Vol}_{g_U}(U) < +\infty$, and for any sequence $(\beta_j)$ of $T_c$-invariant $\Phi$-cutoffs on $U$, the sequence $(\beta_j\circ p)$ is a sequence of $T_c$-invariant $\Phi$-cutoffs on $N$.*

*Proof.* We also write $\xi$ and $\chi$ for the vector fields on $M$ generated by $\xi$ and $\chi$. By Lemma 3.2(1), $\xi = J(\rho\partial_\rho)$, $\chi = J(\rho_\chi\partial_{\rho_\chi})$, and $\rho_\chi(\chi(\lambda)\cdot y) = |\lambda|\rho_\chi(y)$ for $\lambda\in\mathbb{C}^*$ and $y\in M$. Let $S^1\subset\mathbb{C}^*$ be the unit circle. Since $N = \{\rho=1\} = \{\rho_\chi = 1\}$ by Theorem 2.27(2), any $\chi(\mathbb{G}_m)$-orbit in $M$ intersects $N$ in exactly one $\chi(S^1)$-orbit. Thus $p$ is the quotient map of $N$ by the compact group $\chi(S^1)$. In particular, $p$ is proper and $T_c$-equivariant, and the kernel of $dp$ is spanned by $\chi$.

Let $\eta$ be the contact form of $\rho$, that is, the 1-form on $M$ with $\eta(\xi) = 1$ which vanishes on $\rho\partial_\rho$ and on the orthogonal complement, with respect to $g$, of the span of $\rho\partial_\rho$ and $\xi$. By [Li21, Section 2.1, (61)–(62)] and Lemma 3.2(2),

$$\eta(W) = -d\log\rho(JW) \quad\text{and}\quad \eta_\chi(W) = -d\log\rho_\chi(JW)$$

for any tangent vector $W$ of $M$. Along $N$ we have $TN = \ker d\rho = \ker d\rho_\chi$, so a vector $W\in TN$ belongs to $\ker\eta$ if and only if $JW\in TN$, and $W$ belongs to $\ker\eta_\chi$ if and only if $JW\in TN$. Thus $\eta|_N$ and $\eta_\chi|_N$ both have kernel

$$D := TN\cap J(TN).$$

Since $d\log\rho(\xi) = -\eta(\rho\partial_\rho) = 0$, the vector field $\xi$ is tangent to $N$, and since $\eta(\xi) = 1$, we have $TN = \mathbb{R}\xi\oplus D$. On $N$ we have $\eta_\chi(\xi) = f\circ p$, hence

$$\eta|_N = (f\circ p)^{-1}\eta_\chi|_N.$$

By $g = d\rho^2 + \rho^2 h$ and the Kähler form $\frac12 dd^c\rho^2$ of $g$, the vector field $\xi$ is $h$-orthogonal to $D$, $h(\xi,\xi) = 1$, and $h|_D = \frac12 d\eta(\cdot, J\cdot)|_D$ [Li21, Section 2.2, equations (79)–(83)]. Since $\eta_\chi$ vanishes on $D$, differentiating $\eta|_N = (f\circ p)^{-1}\eta_\chi|_N$ implies

$$d\eta(W_1,W_2) = f(p(y))^{-1} d\eta_\chi(W_1,W_2) \quad\text{for } y\in N \text{ and } W_1,W_2\in D_y.$$

Since $\eta_\chi(\chi) = 1$, the vector $\chi$ does not belong to $D$, so $dp$ maps $D_y$ isomorphically onto the tangent space of $U$ at $p(y)$, in uniformizing charts. Thus, by Set-up 3.1(6),

$$h(\xi,\xi) = 1, \qquad h(\xi, D) = 0, \qquad h|_D = (f\circ p)^{-1}(p^* g_U)|_D. \tag{3.1}$$

We prove the volume formula. Let $y \in N$, and let $e_1, \dots, e_{2d-2}$ be an $h$-orthonormal basis of $D_y$. By (3.1), $(\xi, e_1, \dots, e_{2d-2})$ is an $h$-orthonormal basis of $T_y N$, and the vectors $dp(e_k)$ are $g_U$-orthogonal with $|dp(e_k)|^2_{g_U} = f(p(y))$. Since $\eta_\chi(e_k) = 0$ and $\eta_\chi(\xi) = f(p(y))$,

$$\big|(\eta_\chi \wedge p^* d\operatorname{Vol}_{g_U})(\xi, e_1, \dots, e_{2d-2})\big| = f(p(y)) \cdot f(p(y))^{d-1} = f(p(y))^d.$$

Thus

$$d\operatorname{Vol}_h = (f\circ p)^{-d} |\eta_\chi \wedge p^* d\operatorname{Vol}_{g_U}|.$$

By Definition 2.6, the flow of the vector field generated by $\chi$ at time $t$ is $\chi(e^{\sqrt{-1}t})$, and since $\chi$ is primitive, $\chi(e^{\sqrt{-1}t}) = 1$ if and only if $t \in 2\pi\mathbb{Z}$. Since $\eta_\chi(\chi) = 1$, the integral of $\eta_\chi$ over any $\chi(S^1)$-orbit in $N$ with trivial stabilizer is $\ell := 2\pi$. In a uniformizing chart $(\tilde V \times S^1)/\Gamma$ of $p$, where $\Gamma$ is a finite group, the integral of $\eta_\chi$ along each circle $\{v\} \times S^1$ is $\ell$, and the integrals on both $N$ and $U$ carry the factor $1/|\Gamma|$. Using a partition of unity on $U$, we obtain

$$\int_N (G\circ p) d\operatorname{Vol}_h = \ell \int_U G f^{-d} d\operatorname{Vol}_{g_U} \tag{3.2}$$

for any non-negative measurable function $G$ on $U$, that is, $p_*(d\operatorname{Vol}_h) = \ell f^{-d} d\operatorname{Vol}_{g_U}$.

We prove the gradient formula. By (3.1), any covector $\vartheta$ at $y \in N$ satisfies

$$|\vartheta|_h^2 = \vartheta(\xi)^2 + f(p(y))\big|\vartheta|_{D_y}\big|^2_{p^* g_U}.$$

Let $u$ be a $T_c$-invariant smooth function on $U$. Since $p$ is $T_c$-equivariant and $\xi$ belongs to the Lie algebra of $T_c$, the function $u\circ p$ is $T_c$-invariant and $\xi(u\circ p) = 0$. Since the map $dp$ from $(D_y, p^* g_U)$ to $(T_{p(y)}U, g_U)$ is an isometry, $|d(u\circ p)|_{D_y}|_{p^* g_U} = |du|_{g_U}(p(y))$. Thus $|d(u\circ p)|_h^2 = (f|du|^2_{g_U})\circ p$.

By (3.2) with $G = 1$ and $f \geq a$,

$$\operatorname{Vol}_h(N) = \ell \int_U f^{-d} d\operatorname{Vol}_{g_U} \leq \ell a^{-d} \operatorname{Vol}_{g_U}(U),$$

so $\operatorname{Vol}_h(N) < +\infty$ if $\operatorname{Vol}_{g_U}(U) < +\infty$.

Finally, let $(\beta_j)$ be a sequence of $T_c$-invariant $\Phi$-cutoffs on $U$, and we pass to the functions $\beta_j \circ p$ on $N$. They are smooth, $T_c$-invariant, and $0 \leq \beta_j \circ p \leq 1$. Since $p$ is proper, $\beta_j \circ p$ has compact support. For a compact subset $W \subset N$, the set $p(W)$ is compact, so $\beta_j = 1$ on $p(W)$ and $\beta_j \circ p = 1$ on $W$ for $j \gg 0$. By (2.1), $\Phi(ct) \leq C_c \Phi(t)$ for $c \geq 1$ and $t \geq 0$, where $C_c := c^2(1 + \log c)^2$. Let $c_0 := \max(1, \sqrt{b})$. Since $\sqrt{f} \leq c_0$ and $\Phi$ is increasing,

$$\Phi\big(\sqrt{f}|d\beta_j|_{g_U}\big) \leq \Phi\big(c_0|d\beta_j|_{g_U}\big) \leq C_{c_0}\Phi\big(|d\beta_j|_{g_U}\big).$$

By the gradient formula,

$$\Phi(|d(\beta_j\circ p)|_h) = \Phi(\sqrt{f}|d\beta_j|_{g_U})\circ p.$$

Thus, by (3.2) and $f \geq a$,

$$\int_N \Phi\big(|d(\beta_j\circ p)|_h\big) d\operatorname{Vol}_h \leq \ell a^{-d} C_{c_0} \int_U \Phi\big(|d\beta_j|_{g_U}\big) d\operatorname{Vol}_{g_U},$$

which tends to 0 as $j \to +\infty$. Hence $(\beta_j \circ p)$ is a sequence of $T_c$-invariant $\Phi$-cutoffs on $N$. □

Theorem 2.27 defines $\rho$ only on $M$, and it only implies that the $\theta$-plurisubharmonic function $\varphi$ of Theorem 2.27(2) is bounded. The continuity of $\rho$ follows from the continuity of $\varphi$ [CC24, Theorem 1.3], which only uses that the density of an adapted measure of the klt pair $(Z, \Delta_Z)$ belongs to $L^{1+\epsilon}$ for some $\epsilon > 0$. In particular, we do not assume that $\Delta_Z = 0$.

**Proposition 3.4.** *Notation and conditions as in Set-up 3.1. Then $\rho$ extends to a continuous proper function $X \to [0, +\infty)$ with $\rho^{-1}(0) = \{o\}$, the link $N$ is relatively compact in $X\backslash\{o\}$, and $\mathrm{Vol}_h(N) < +\infty$.*

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step, we show that the function $\varphi$ of Theorem 2.27(2) is continuous.

By Theorem 2.27(2), there exist a positive multiple $\theta$ of the restriction of a Fubini–Study form under a projective embedding of $Z$ and a bounded $\theta$-plurisubharmonic function $\varphi$ on $Z$ such that $dd^c \log \rho_\chi^2$ descends to $\theta + dd^c\varphi$ on $U$. Since $Z = (X\backslash\{o\})/\chi(\mathbb{G}_m)$, we have $\dim Z = d - 1$. Let $Z^\circ \subset Z$ be the image of $M$. Then $Z^\circ$ is the complement of the image of $X_{\mathrm{sing}}\backslash\{o\}$, so $Z^\circ$ is a dense Zariski open subset of $Z$, and $Z^\circ$ is the coarse space of $U$. By Theorem 2.27(2), [Li21, Proposition 2.7 and equation (105)], and the expression of $f$ by the Hamiltonian in [Li21, (101)–(103)], the weighted soliton equation on $U$ reads

$$(\theta + dd^c\varphi)^{d-1} = c_0 f^{d+1} e^{-\gamma\varphi} G\nu \quad \text{on } Z^\circ,$$

where $c_0 > 0$ and $\gamma$ are real constants, $\nu$ is an adapted measure of $(Z, \Delta_Z)$ [Ber+11, Definition 3.1], and $G$ is a function on $Z^\circ$ with $C^{-1} \le G \le C$ for a constant $C > 0$.

The factor $G$ appears when the reference form and the volume form of [Li21, p. 30] are replaced by $\theta$ and $\nu$. Fix a closed embedding $X \to \mathbb{C}^I$ by finitely many generators of the maximal ideal of $o$ which are $T$-eigenvectors, and let $\rho_{\mathcal{P}}$ be the radius function on $\mathbb{C}^I\backslash\{0\}$ such that $\{\rho_{\mathcal{P}} = 1\}$ is the unit sphere and the Reeb vector field of $\rho_{\mathcal{P}}$ generates the action of $S^1$ on $\mathbb{C}^I$ whose weights are the $\chi$-weights of the generators [Li21, Section 2.2], that is, the function of (3.15). The action of $S^1$ restricts on $X$ to the action of $\chi(S^1)$, so the restriction of $\rho_{\mathcal{P}}$ to $M$ is a smooth radius function for $\chi$, and $\rho_{\mathcal{P}}$ is continuous and positive on $X\backslash\{o\}$. We take Li's reference Hermitian metric $h_0$ [Li21, Section 2.4, p. 29] to be the one induced by $\rho_{\mathcal{P}}|_M$. Since the radius functions of [Li21, p. 29] are positive constant powers of the Hermitian metrics they induce, the potential of [Li21, p. 30] with respect to $h_0$ is a positive constant multiple of

$$\log \rho_\chi^2 - \log \rho_{\mathcal{P}}^2 = \varphi - \log \frac{\rho_{\mathcal{P}}^2}{\rho_\theta^2},$$

where $\rho_\theta := \rho_\chi e^{-\varphi/2}$ is the function of **Step 2**. By the argument of **Step 2**, which does not use the continuity of $\varphi$, the function $\rho_\theta$ extends to a continuous positive function on $X\backslash\{o\}$. Thus the $\chi(\mathbb{G}_m)$-invariant function $\log(\rho_{\mathcal{P}}^2/\rho_\theta^2)$ extends to a continuous function on $X\backslash\{o\}$, which descends to the compact space $Z$, so it is bounded, and passing from one potential to the other changes the exponential factor only by a factor bounded above and below by positive constants. The volume form of [Li21, p. 30] and $\nu$ are, up to positive constant factors, the measures defined as in [Ber+11, Definition 3.1] by two Hermitian metrics on $-(K_Z + \Delta_Z)$, namely the metric induced by $h_0$ and a smooth metric. For an étale chart $V$ of the quotient stack of $X\backslash\{o\}$ by $\chi(\mathbb{G}_m)$, the morphism $V \to Z$ is ramified with index $m_P$ along any prime divisor of $V$ over a prime divisor $P \subset Z$ (Definition 2.6), so $K_Z + \Delta_Z$ pulls back to $K_V$ by the Hurwitz formula, and on $V$ both measures are positive multiples of $(\sqrt{-1}^{m(d-1)^2} \varpi \wedge \bar{\varpi})^{1/m}$, where $m$ is a positive integer such that $m(K_Z + \Delta_Z)$ is Cartier and $\varpi$ is a local generator of the reflexive sheaf $\mathcal{O}_V(mK_V)$. The metric induced by $h_0$ is continuous over all of $Z$, since $\rho_{\mathcal{P}}$ is continuous and positive on $X\backslash\{o\}$. Thus the ratio of the volume form of [Li21, p. 30] to $\nu$ is a positive constant multiple of the ratio of two continuous metrics on $-(K_Z + \Delta_Z)$, so it is a continuous positive function on $Z$. Let $\lambda$ be the function on $Z$ which is equal to $c_0 f^{d+1} e^{-\gamma\varphi} G$ on $Z^\circ$ and to 1 on $Z\backslash Z^\circ$. Since $a \le f \le b$ and $\varphi$ is bounded, there exists a constant $C > 0$ such that $C^{-1} \le \lambda \le C$. The Monge–Ampère measure of the bounded $\theta$-plurisubharmonic function $\varphi$ does not charge the proper analytic subset $Z\backslash Z^\circ$ [Ber+11, Section 1.2], and neither does $\nu$. Thus

$$(\theta + dd^c\varphi)^{d-1} = \lambda\nu \quad \text{on } Z.$$

We show that $\nu$ has a density in $L^{1+\epsilon}$ with respect to $\theta^{d-1}$. Let $\mu : \tilde{Z} \to Z$ be a log resolution of $(Z, \Delta_Z)$ which also principalizes the ideal generated by the maximal minors of the Jacobian matrix of the projective embedding defining $\theta$. Locally on $\tilde{Z}$, there exist coordinates $z_1, \dots, z_{d-1}$ such that, up to factors bounded above and below by positive constants,

$$\mu^*\nu = \prod_i |z_i|^{2c_i} dV, \qquad \mu^*\theta^{d-1} = \prod_i |z_i|^{2e_i} dV,$$

where $dV$ is the Lebesgue measure and $e_i \geq 0$. The second expression holds because $\mu^*\theta^{d-1}$ is, up to factors bounded above and below by positive constants, the sum of the squared absolute values of the pullbacks of the maximal Jacobian minors times $dV$. Since $(Z, \Delta_Z)$ is klt (Lemma 2.7, see also [LWX18, Definition 2.13]) and $\nu$ is an adapted measure of $(Z, \Delta_Z)$, we have $c_i > -1$ for any $i$ [Ber+11, (3.2) and the paragraph containing it]. Since $\tilde{Z}$ is compact, there exists $\epsilon > 0$ such that $(1+\epsilon)c_i - \epsilon e_i > -1$ in any of finitely many such coordinate charts. Write $\nu = F\theta^{d-1}$ on $Z_{\mathrm{reg}}$. Then

$$\int_Z F^{1+\epsilon}\theta^{d-1} = \int_{\tilde{Z}} \Big(\frac{\mu^*\nu}{\mu^*\theta^{d-1}}\Big)^{1+\epsilon} \mu^*\theta^{d-1},$$

and the integrand on the right is locally bounded by a constant multiple of $\prod_i |z_i|^{2((1+\epsilon)c_i - \epsilon e_i)} dV$, which is integrable. Thus $(\theta + dd^c\varphi)^{d-1} = \lambda F \theta^{d-1}$ with $\lambda F \in L^{1+\epsilon}(Z, \theta^{d-1})$.

Since $Z$ is normal, $Z$ is a reduced and locally irreducible compact Kähler space with Kähler form $\theta$. By [CC24, Theorem 1.3], $\varphi$ is continuous.

**Step 2.** In this step, we show that $\rho_\chi$ extends to a continuous positive function on $X\backslash\{o\}$.

Let $\rho_\theta := \rho_\chi e^{-\varphi/2}$, where $\varphi$ is pulled back to $M$. Since $\varphi$ is invariant under $\chi(\mathbb{G}_m)$, we have $\rho_\theta(\chi(\lambda)\cdot y) = |\lambda|\rho_\theta(y)$ for $\lambda \in \mathbb{C}^*$ and $y \in M$, and $dd^c \log \rho_\theta^2$ is the pullback of $\theta$. Let $\pi : X\backslash\{o\} \to Z$ be the quotient map and $x \in X\backslash\{o\}$. Since $\theta$ is a positive multiple of the restriction of a Fubini–Study form, there exist an open neighborhood $W$ of $\pi(x)$ and a continuous function $v_0$ on $W$, a positive multiple of the restriction of $\log(1 + \sum_j |z_j|^2)$ in an affine chart, such that $\theta = dd^c v_0$ on $W \cap Z_{\mathrm{reg}}$. Let $V_W := \pi^{-1}(W)$ and $w := \log \rho_\theta^2 - v_0 \circ \pi$ on $V_W \cap M$. Then $dd^c w = 0$ on $V_W \cap M$. Since $X$ is normal, $V_W\backslash M$ is an analytic subset of $V_W$ of codimension at least 2, so $w$ and $-w$ extend to plurisubharmonic functions $w_1$ and $w_2$ on $V_W$ [GR56]. The function $w_1 + w_2$ is plurisubharmonic on $V_W$ and zero on $V_W \cap M$.

Since $w_1 + w_2$ is upper semicontinuous and vanishes on the dense subset $V_W \cap M$, we have $w_1 + w_2 \geq 0$ on $V_W$. Conversely, for $x' \in V_W\backslash M$, there exists a holomorphic map $\sigma$ from the unit disc $\Delta$ to $V_W$ with $\sigma(0) = x'$ and $\sigma(\Delta) \not\subset V_W\backslash M$, namely the normalization of an irreducible curve germ through $x'$ not contained in $V_W\backslash M$. Then $(w_1 + w_2) \circ \sigma$ is subharmonic on $\Delta$ and vanishes outside the discrete set $\sigma^{-1}(V_W\backslash M)$, so by the sub-mean value inequality on small circles avoiding $\sigma^{-1}(V_W\backslash M)$, $(w_1 + w_2)(x') \leq 0$. Thus $w_1 + w_2 = 0$ on $V_W$, and $w_1 = -w_2$ is both upper and lower semicontinuous, so $w$ extends to a continuous function on $V_W$, and $\rho_\theta = \exp((w + v_0 \circ \pi)/2)$ extends to a continuous positive function on $V_W$. By **Step 1**, $\rho_\chi = \rho_\theta e^{\varphi/2}$ extends to a continuous positive function on $X\backslash\{o\}$.

**Step 3.** In this step, we show that $\rho$ extends to a continuous positive function on $X\backslash\{o\}$ with $\rho(\Psi_s(y)) = e^s\rho(y)$, where $\Psi_s$ is defined below.

For $s \in \mathbb{R}$, let $\Psi_s : X \to X$ be the action of the element of $T$ determined by $u \circ \Psi_s = e^{s\langle\alpha,\xi\rangle}u$ for any character $\alpha$ and $u \in R_\alpha$. Then $(y, s) \to \Psi_s(y)$ is continuous on $X \times \mathbb{R}$, $\Psi_s \circ \Psi_t = \Psi_{s+t}$, $\Psi_s(o) = o$, and $\Psi_s(M) = M$. By Theorem 2.27(1), $\rho\partial_\rho$ acts on $R_\alpha$ by multiplication by $\langle\alpha, \xi\rangle$, so $(\Psi_s)$ is the flow of $\rho\partial_\rho$ on $M$, and

$$\rho(\Psi_s(y)) = e^s\rho(y) \quad \text{for } y \in M.$$

By Lemma 3.2(2), $d\log\rho_\chi(W) = \eta_\chi(JW)$ for any tangent vector $W$ of $M$. Since $J(\rho\partial_\rho)$ is the vector field generated by $\xi$ (Lemma 3.2(1)), for $y \in M$,

$$\frac{d}{ds}\log\rho_\chi(\Psi_s(y)) = \eta_\chi(\xi)(\Psi_s(y)) \in [a, b].$$

Thus, for $y \in M$ and $s_1 \le s_2$,

$$e^{a(s_2-s_1)} \le \frac{\rho_\chi(\Psi_{s_2}(y))}{\rho_\chi(\Psi_{s_1}(y))} \le e^{b(s_2-s_1)}. \tag{3.3}$$

Since $X$ is normal, $M$ is dense in $X\backslash\{o\}$. By **Step 2**, both sides of (3.3) are continuous in $y \in X\backslash\{o\}$, so (3.3) holds for any $y \in X\backslash\{o\}$. In particular, for $y \in X\backslash\{o\}$, the function $s \to \rho_\chi(\Psi_s(y))$ is continuous and strictly increasing, and it tends to 0 as $s \to -\infty$ and to $+\infty$ as $s \to +\infty$. Thus there exists a unique $s_*(y) \in \mathbb{R}$ such that $\rho_\chi(\Psi_{s_*(y)}(y)) = 1$.

We show that $s_*$ is continuous on $X\backslash\{o\}$. Let $y \in X\backslash\{o\}$ and $\delta > 0$. By (3.3),

$$\rho_\chi(\Psi_{s_*(y)-\delta}(y)) \le e^{-a\delta} < 1 \quad \text{and} \quad \rho_\chi(\Psi_{s_*(y)+\delta}(y)) \ge e^{a\delta} > 1.$$

Since $(y', s) \to \rho_\chi(\Psi_s(y'))$ is continuous, any $y'$ in a neighborhood of $y$ satisfies

$$\rho_\chi(\Psi_{s_*(y)-\delta}(y')) < 1 < \rho_\chi(\Psi_{s_*(y)+\delta}(y')),$$

and thus $|s_*(y') - s_*(y)| < \delta$ for such $y'$.

Let $\tilde\rho := e^{-s_*}$ on $X\backslash\{o\}$. Then $\tilde\rho$ is continuous and positive. By uniqueness of $s_*$, we have $s_*(\Psi_t(y)) = s_*(y) - t$, that is, $\tilde\rho(\Psi_t(y)) = e^t\tilde\rho(y)$. For $y \in M$, the point $\Psi_s(y)$ belongs to $N = \{\rho = 1\} = \{\rho_\chi = 1\}$ if and only if $e^s\rho(y) = 1$, so $s_*(y) = -\log\rho(y)$ and $\tilde\rho(y) = \rho(y)$. Thus $\tilde\rho$ extends $\rho$, and we denote $\tilde\rho$ again by $\rho$.

**Step 4.** In this step, we show that $\rho$ is proper, that $N$ is relatively compact in $X\backslash\{o\}$, and that $\mathrm{Vol}_h(N) < +\infty$.

Let $u_1, \dots, u_K$ be generators of the $\mathbb{C}$-algebra $R$ with $u_k \in R_{\alpha_k}$ and $\alpha_k \ne 0$, and let $w_k := \langle \alpha_k, \xi\rangle$. Since $\xi$ is a Reeb vector, $w_k > 0$ for any $k$. Let

$$\rho_{\mathrm{ref}} := \max_k |u_k|^{1/w_k}.$$

Then $\rho_{\mathrm{ref}}$ is continuous and proper on $X$, $\rho_{\mathrm{ref}}^{-1}(0) = \{o\}$, and $\rho_{\mathrm{ref}}(\Psi_s(y)) = e^s\rho_{\mathrm{ref}}(y)$. The set $\{\rho_{\mathrm{ref}} = 1\}$ is a compact subset of $X\backslash\{o\}$, so by **Step 3**, there exists a constant $C > 1$ such that $C^{-1} \le \rho \le C$ on $\{\rho_{\mathrm{ref}} = 1\}$. For $y \in X\backslash\{o\}$, the point $\Psi_{-\log\rho_{\mathrm{ref}}(y)}(y)$ belongs to $\{\rho_{\mathrm{ref}} = 1\}$, so the homogeneity of $\rho$ and $\rho_{\mathrm{ref}}$ implies

$$C^{-1}\rho_{\mathrm{ref}} \le \rho \le C\rho_{\mathrm{ref}} \quad \text{on } X\backslash\{o\}.$$

Thus $\rho$ extends continuously to $X$ by $\rho(o) = 0$, $\rho^{-1}(0) = \{o\}$, and $\{\rho \le c\} \subset \{\rho_{\mathrm{ref}} \le Cc\}$ is compact for any $c \ge 0$, that is, $\rho$ is proper. Moreover, $N = \{\rho = 1\}$ is contained in the compact subset $\{C^{-1} \le \rho_{\mathrm{ref}} \le C\}$ of $X\backslash\{o\}$, so $N$ is relatively compact in $X\backslash\{o\}$.

By Lemma 3.2(4) and Theorem 2.27(2), the Kähler form of $g_U$ is $\frac{1}{2}(\theta + dd^c\varphi)$ on $U$. Since integration on the orbifold $U$ is integration on the coarse space $Z^\circ$,

$$\mathrm{Vol}_{g_U}(U) = \frac{2^{-(d-1)}}{(d-1)!}\int_{Z^\circ}(\theta + dd^c\varphi)^{d-1} \le \frac{2^{-(d-1)}}{(d-1)!}\int_Z \theta^{d-1} < +\infty,$$

where we use that the Bedford–Taylor measure $(\theta + dd^c\varphi)^{d-1}$ of the bounded $\theta$-plurisubharmonic function $\varphi$ has total mass $\int_Z \theta^{d-1}$ [Ber+11, Section 1.2]. By Lemma 3.3, $\mathrm{Vol}_h(N) < +\infty$. □

3.2. **Logarithmic energy on compact orbifolds.** In this subsection, we prove Lemma 3.5 below. Let $\mathcal{Y}$, $\alpha$, $\alpha_u = \alpha + dd^c u$, $E_i$, and $x_i$ be as in Lemma 3.5, where $m = \dim\mathcal{Y}$. We call the integral in Lemma 3.5(1) the *logarithmic energy* of $E_i$ with respect to $\alpha_u$. Lemma 3.5 shows that the logarithmic energy of $E_i$ is finite if $\int_{E_i}\alpha^{m-1} = 0$, and constructs $\Phi$-cutoffs from it.

If $\alpha = \pi^*\alpha'$ for a morphism $\pi$ from $\mathcal{Y}$ to an orbifold or a Deligne–Mumford stack with $\dim\pi(E_i) \le m-2$ and a smooth closed real $(1,1)$-form $\alpha'$, then $\alpha^{m-1}|_{E_i} = 0$, so $\int_{E_i}\alpha^{m-1} = 0$. This is the reason why exceptional divisors over subsets of codimension at least two can be cut off. A branch divisor of a quotient orbifold does not satisfy $\int_{E_i}\alpha^{m-1} = 0$, and in Theorem 3.9 such divisors stay in the orbifold and are not cut off.

**Lemma 3.5.** *Let $\mathcal{Y}$ be a compact Kähler orbifold of dimension $m \geq 2$ with Kähler form $\theta$, let $\alpha$ be a smooth closed semipositive real $(1,1)$-form on $\mathcal{Y}$, and let $\alpha_u := \alpha + dd^c u$ with $u$ a bounded $\alpha$-plurisubharmonic function. Let $E_1, \dots, E_r$ be prime divisors on $\mathcal{Y}$ whose union $E$ is a simple normal crossing divisor, and let $x_i = |s_i|^2 \leq e^{-2}$ for smooth Hermitian norms of defining sections $s_i$ of $E_i$. Assume that $\int_{E_i} \alpha^{m-1} = 0$ for any $i$. Then the following hold.*

*(1) For any $i$,*

$$\int_{\mathcal{Y}\backslash E} \sqrt{-1}\partial \log x_i \wedge \bar{\partial} \log x_i \wedge \alpha_u^{m-1} < +\infty.$$

*(2) If moreover $\alpha_u$ is a smooth Kähler form on $\mathcal{Y}\backslash E$ and $\alpha_u \geq C^{-1} \prod_i x_i^q \theta$ on $\mathcal{Y}\backslash E$ for constants $C, q > 0$, then $\mathcal{Y}\backslash E$ admits a sequence of $\Phi$-cutoffs with respect to $\alpha_u$, and the cutoffs can be taken invariant under any compact group acting on $\mathcal{Y}$ and preserving $\theta$, $\alpha_u$, and $E$.*

*Proof.* Integration on $\mathcal{Y}$ is integration in a finite atlas of uniformizing charts, with the factor $1/|\Gamma|$ for the chart group $\Gamma$ and a partition of unity. A compact orbifold admits such a finite atlas. Wedge products of $\alpha_u$ with itself and with smooth forms are the Bedford–Taylor products of bounded potentials in uniformizing charts [BT82]. They are invariant under the chart groups and coincide on overlaps, since Bedford–Taylor products are local. Thus they define closed positive currents on $\mathcal{Y}$, and the Stokes formula for such a current and a smooth form holds on $\mathcal{Y}$, since the boundary terms from the partition of unity cancel. The integral over $\mathcal{Y}$ of a wedge product of $\alpha_u$, $\alpha$, and smooth closed forms depends only on cohomology classes [GZ17].

The proof is long, so we divide it into several steps. In **Steps 1–3**, we fix $i$. Let $\Theta_i$ be the smooth closed real $(1,1)$-form with

$$dd^c \log x_i = 2\pi[E_i] - \Theta_i$$

(Poincaré–Lelong formula, [GH78, Chapter 3, Section 2]), and choose a constant $C_0 > 0$ with $-C_0\theta \leq \Theta_i \leq C_0\theta$. Let $S := \alpha_u^{m-1}$ and

$$A := \sum_{k=0}^{m-2} \alpha_u^k \wedge \alpha^{m-2-k},$$

a closed positive $(m-2, m-2)$-current, so that $S - \alpha^{m-1} = dd^c u \wedge A$. For $0 < \epsilon < e^{-2}$, let

$$\psi_\epsilon := \log(x_i + \epsilon), \qquad a_\epsilon := \frac{x_i}{x_i + \epsilon}, \qquad P_\epsilon := dd^c \psi_\epsilon + a_\epsilon \Theta_i.$$

The functions $\psi_\epsilon$, $a_\epsilon$, and the form $P_\epsilon$ are smooth on $\mathcal{Y}$. A direct calculation shows that

$$P_\epsilon = \frac{\epsilon x_i}{(x_i + \epsilon)^2} \sqrt{-1}\partial \log x_i \wedge \bar{\partial} \log x_i \geq 0 \quad \text{on } \mathcal{Y}\backslash E_i,$$

so $P_\epsilon \geq 0$ on $\mathcal{Y}$. Moreover, $0 \leq -\psi_\epsilon \leq -\log x_i$, $-\psi_\epsilon \leq |\log \epsilon|$, and $-\psi_\epsilon$ increases to $-\log x_i$ as $\epsilon$ decreases to 0.

**Step 1.** In this step, we show that

$$\int_{\mathcal{Y}} (-\log x_i)\theta \wedge S < +\infty. \tag{3.4}$$

Testing $S - \alpha^{m-1} = dd^c(uA)$ against the smooth form $(-\psi_\epsilon)\theta$ and using $dd^c(-\psi_\epsilon) = a_\epsilon \Theta_i - P_\epsilon$, we obtain

$$\int_{\mathcal{Y}} (-\psi_\epsilon)\theta \wedge S = \int_{\mathcal{Y}} (-\psi_\epsilon)\theta \wedge \alpha^{m-1} + \int_{\mathcal{Y}} u(a_\epsilon \Theta_i - P_\epsilon) \wedge \theta \wedge A.$$

By the Stokes formula,

$$0 \leq \int_{\mathcal{Y}} P_\epsilon \wedge \theta \wedge A = \int_{\mathcal{Y}} a_\epsilon \Theta_i \wedge \theta \wedge A \leq C_0 \int_{\mathcal{Y}} \theta^2 \wedge A.$$

Since $0 \le a_\epsilon \le 1$ and $\pm\Theta_i \wedge \theta \wedge A \le C_0\theta^2 \wedge A$, the absolute value of the last integral in the previous identity is at most $2C_0\|u\|_\infty \int_{\mathcal{Y}} \theta^2 \wedge A$, which is finite and independent of $\epsilon$. Since $\log x_i$ is integrable with respect to the smooth volume $\theta \wedge \alpha^{m-1}$, the first integral on the right is bounded by $\int_{\mathcal{Y}}(-\log x_i)\theta \wedge \alpha^{m-1} < +\infty$. By the monotone convergence theorem, (3.4) holds. In particular, the measure $\theta \wedge S$ does not charge $E_i$.

**Step 2.** In this step, we show that $|\log\epsilon| \int_{\mathcal{Y}} P_\epsilon \wedge S \to 0$ as $\epsilon \to 0$.

By the Stokes formula, $\int_{\mathcal{Y}} \Theta_i \wedge S = \int_{\mathcal{Y}} \Theta_i \wedge \alpha^{m-1}$, and by the Poincaré–Lelong formula and the assumption,

$$\int_{\mathcal{Y}} \Theta_i \wedge \alpha^{m-1} = 2\pi \int_{E_i} \alpha^{m-1} - \int_{\mathcal{Y}} \log x_i dd^c \alpha^{m-1} = 0.$$

Since $\int_{\mathcal{Y}} dd^c\psi_\epsilon \wedge S = 0$ and $1 - a_\epsilon = \epsilon/(x_i + \epsilon)$, we obtain

$$0 \le \int_{\mathcal{Y}} P_\epsilon \wedge S = \int_{\mathcal{Y}} a_\epsilon \Theta_i \wedge S = -\int_{\mathcal{Y}} \frac{\epsilon}{x_i + \epsilon} \Theta_i \wedge S.$$

We show that

$$|\log\epsilon| \frac{\epsilon}{x_i + \epsilon} \le 1 - \log x_i \quad \text{on } \mathcal{Y}\backslash E_i.$$

If $x_i \le \epsilon$, then $\epsilon/(x_i + \epsilon) \le 1$ and $|\log\epsilon| \le -\log x_i$. If $x_i > \epsilon$, let $t := x_i/\epsilon > 1$. Then $|\log\epsilon| = -\log x_i + \log t$ and $\epsilon/(x_i+\epsilon) = 1/(1+t)$, so the left side is at most $-\log x_i + (\log t)/(1+t) \le 1 - \log x_i$. Moreover, the left side tends to 0 at any point of $\mathcal{Y}\backslash E_i$. Since $|\Theta_i \wedge S| \le C_0\theta \wedge S$, the measure $\theta \wedge S$ does not charge $E_i$, and $\int_{\mathcal{Y}}(1 - \log x_i)\theta \wedge S < +\infty$ by (3.4), the dominated convergence theorem implies the claim of this step.

**Step 3.** In this step, we prove (1).

Since $\psi_\epsilon$ is smooth and $S$ is closed, the Stokes formula implies

$$\int_{\mathcal{Y}} \sqrt{-1}\partial\psi_\epsilon \wedge \bar\partial\psi_\epsilon \wedge S = -\int_{\mathcal{Y}} \psi_\epsilon P_\epsilon \wedge S + \int_{\mathcal{Y}} \psi_\epsilon a_\epsilon \Theta_i \wedge S.$$

Since $0 \le -\psi_\epsilon \le |\log\epsilon|$, the first term on the right tends to 0 by **Step 2**. Since $|\psi_\epsilon a_\epsilon| \le -\log x_i$ and $\psi_\epsilon a_\epsilon \to \log x_i$ on $\mathcal{Y}\backslash E_i$, the second term converges to $\int_{\mathcal{Y}} \log x_i \Theta_i \wedge S$ by (3.4) and the dominated convergence theorem. Thus the left side has a finite limit as $\epsilon \to 0$. The measures $\sqrt{-1}\partial\psi_\epsilon \wedge \bar\partial\psi_\epsilon \wedge S$ are bounded by constant multiples of $\theta \wedge S$, which does not charge $E$ by **Step 1** applied to each component of $E$. On $\mathcal{Y}\backslash E$, we have $\partial\psi_\epsilon = a_\epsilon \partial \log x_i$, and $a_\epsilon^2$ increases to 1 as $\epsilon$ decreases to 0. By the monotone convergence theorem,

$$\int_{\mathcal{Y}\backslash E} \sqrt{-1}\partial \log x_i \wedge \bar\partial \log x_i \wedge \alpha_u^{m-1} = \lim_{\epsilon\to 0} \int_{\mathcal{Y}} \sqrt{-1}\partial\psi_\epsilon \wedge \bar\partial\psi_\epsilon \wedge S < +\infty.$$

**Step 4.** In this step, we prove (2).

If $E = \emptyset$, then $\mathcal{Y}$ is compact and the constant functions $\beta_j = 1$ form a sequence of invariant $\Phi$-cutoffs. Assume that $E \ne \emptyset$. All norms below are with respect to $\alpha_u$, and $d\,\mathrm{Vol} := \alpha_u^m/m!$ on $\mathcal{Y}\backslash E$. Let

$$\tau := 1 - \sum_i \log x_i \ge 1 \quad \text{on } \mathcal{Y}\backslash E.$$

There exists a constant $c_m > 0$ depending only on $m$ such that

$$|dv|^2 \alpha_u^m = c_m \sqrt{-1}\partial v \wedge \bar\partial v \wedge \alpha_u^{m-1}$$

for any smooth real function $v$ on $\mathcal{Y}\backslash E$. By the Cauchy–Schwarz inequality,

$$\sqrt{-1}\partial\tau \wedge \bar\partial\tau \le r \sum_i \sqrt{-1}\partial \log x_i \wedge \bar\partial \log x_i.$$

Thus, by (1),

$$\int_{\mathcal{Y}\backslash E} |d\tau|^2 d\,\mathrm{Vol} < +\infty. \tag{3.5}$$

We bound $|d\tau|$. Cover $\mathcal{Y}$ by finitely many uniformizing charts with coordinates in which $E$ is a union of coordinate hyperplanes. In such a chart, $x_i$ is either a smooth positive function, or the product of $|z_k|^2$ and a smooth positive function for a coordinate $z_k$, so, for a constant $C_1 > 0$,

$$|d\log x_i|_\theta \le C_1 x_i^{-1/2} \le C_1 x_i^{-1}.$$

Since $\alpha_u \ge C^{-1}\prod_i x_i^q \theta$, any covector $\vartheta$ on $\mathcal{Y}\backslash E$ satisfies $|\vartheta|^2 \le C\prod_i x_i^{-q}|\vartheta|_\theta^2$. Thus $|d\tau| \le C_2\prod_i x_i^{-q-1}$ for a constant $C_2 > 0$, and since $x_i \le e^{-2} < 1$,

$$\log(e + |d\tau|) \le \log(e + C_2) + (q+1)\sum_i (-\log x_i) \le C_3\tau$$

for a constant $C_3 > 0$.

Let $\kappa : \mathbb{R} \to [0,1]$ be a smooth function with $\kappa = 1$ on $(-\infty, 0]$ and $\kappa = 0$ on $[1, +\infty)$, and let $C_4 := \max(1, \sup|\kappa'|)$. For integers $j \ge 1$, let

$$\beta_j := \kappa(\log\tau - j).$$

Since $\tau \to +\infty$ near $E$, the support of $\beta_j$ is contained in the compact subset $\{\tau \le e^{j+1}\}$ of $\mathcal{Y}\backslash E$. Since $\tau$ is bounded on any compact subset $W$ of $\mathcal{Y}\backslash E$, we have $\beta_j = 1$ on $W$ for $j \gg 0$. The form $d\beta_j$ is supported in $\{e^j \le \tau \le e^{j+1}\}$, and $|d\beta_j| \le C_4|d\tau|/\tau$. Since $\Phi$ is increasing and $\tau \ge 1$, (2.1) with $a = C_4$ implies

$$\Phi(|d\beta_j|) \le C_4^2(1 + \log C_4)^2 \frac{|d\tau|^2}{\tau^2}\log^2(e + |d\tau|).$$

Thus, by the bound $\log(e + |d\tau|) \le C_3\tau$,

$$\Phi(|d\beta_j|) \le C_5|d\tau|^2$$

on $\{e^j \le \tau \le e^{j+1}\}$, where $C_5 := C_3^2 C_4^2(1 + \log C_4)^2$. Thus

$$\int_{\mathcal{Y}\backslash E} \Phi(|d\beta_j|)d\,\mathrm{Vol} \le C_5 \int_{\{\tau \ge e^j\}} |d\tau|^2 d\,\mathrm{Vol},$$

which tends to 0 as $j \to +\infty$ by (3.5). Hence $(\beta_j)$ is a sequence of $\Phi$-cutoffs on $\mathcal{Y}\backslash E$ with respect to $\alpha_u$.

Let $G$ be a compact group acting on $\mathcal{Y}$ and preserving $\theta$, $\alpha_u$, and $E$, and let $dt$ be the Haar measure on $G$ with total mass 1. We replace $\beta_j$ with the function

$$\bar\beta_j : y \to \int_G \beta_j(t\cdot y)dt.$$

Then $\bar\beta_j$ is smooth and $G$-invariant, and $0 \le \bar\beta_j \le 1$. The support of $\bar\beta_j$ is contained in the compact subset $G\cdot\mathrm{Supp}\,\beta_j$ of $\mathcal{Y}\backslash E$. For a compact subset $W \subset \mathcal{Y}\backslash E$, the set $G\cdot W$ is a compact subset of $\mathcal{Y}\backslash E$, so $\bar\beta_j = 1$ on $W$ for $j \gg 0$. Since $G$ preserves $\alpha_u$, we have

$$|d\bar\beta_j|(y) \le \int_G |d\beta_j|(t\cdot y)dt.$$

The function $\Phi$ is increasing and convex on $[0, +\infty)$, because $\Phi'(t) = 2t\log^2(e+t) + 2t^2\log(e+t)/(e+t)$ is nondecreasing. By Jensen's inequality,

$$\Phi(|d\bar\beta_j|)(y) \le \int_G \Phi(|d\beta_j|)(t\cdot y)dt.$$

Since $G$ preserves $d\,\mathrm{Vol}$, the Fubini theorem implies

$$\int_{\mathcal{Y}\backslash E} \Phi(|d\bar\beta_j|)d\,\mathrm{Vol} \le \int_{\mathcal{Y}\backslash E} \Phi(|d\beta_j|)d\,\mathrm{Vol} \to 0.$$

Hence $(\bar{\beta}_j)$ is a sequence of $G$-invariant $\Phi$-cutoffs on $\mathcal{Y}\backslash E$ with respect to $\alpha_u$. □

3.3. **Logarithmic cutoffs.** In this subsection, we prove Theorem 3.9. We first prove three analytic lemmas. They are used only to bound the Hamiltonian function of the Reeb vector, and the gradient of the Hamiltonian function, uniformly along the smooth approximating solutions constructed for Theorem 3.9, where the Hamiltonian function appears in the drift term of the weighted equation.

The first lemma is a pointwise inequality on a Kähler manifold, and no compactness is used.

**Lemma 3.6.** *Let $(Y, g, J)$ be a Kähler manifold, let $f > 0$ be a smooth function on $Y$, and let $k \geq 0$ be a real number. Let $F := -k \log f$, let $\Delta_F := \Delta + \langle \nabla F, \nabla \cdot \rangle$ be the weighted Laplace operator acting on functions, where $\Delta = \operatorname{div} \nabla$ is the Laplace–Beltrami operator, and let $V := \nabla f$, $B := \Delta_F f$, $w := |\nabla f|^2$, and $H := \log(1 + w) + F$. Assume that $J\nabla f$ is a Killing vector field. Then*

$$\Delta_F H \geq \frac{V(B)}{1+w} - \frac{k}{f} B.$$

*Proof.* We regard $\operatorname{Hess} f$ as the symmetric endomorphism of $T_Y$ given by $\operatorname{Hess} f(X) = \nabla_X V$ for any tangent vector $X$, and we regard $\operatorname{Ric}_g$ also as a symmetric endomorphism of $T_Y$ through $g$. For a vector field $W$ and a point $y \in Y$, we write $\operatorname{tr} \nabla^2 W := \sum_i \nabla^2_{e_i,e_i} W$, where $e_1, \dots, e_{2n}$ is an orthonormal basis of $T_{Y,y}$ and $2n = \dim_{\mathbb{R}} Y$. Then $\operatorname{div} \operatorname{Hess} f = \operatorname{tr} \nabla^2 V$.

**Step 1.** In this step, we show that

$$d\Delta f = -2 \operatorname{Ric}_g(V, \cdot). \tag{3.6}$$

Since $J$ is parallel, $\nabla(JV) = J \circ \operatorname{Hess} f$. Since $JV$ is a Killing vector field, $J \circ \operatorname{Hess} f$ is skew-symmetric. For any point $y \in Y$ and any $X, Z \in T_{Y,y}$, we have

$$g(J \operatorname{Hess} f(X), Z) = -g(\operatorname{Hess} f(X), JZ) = -g(X, \operatorname{Hess} f(JZ)),$$

and skew-symmetry says that the left-hand side equals $-g(X, J \operatorname{Hess} f(Z))$. Thus

$$\operatorname{Hess} f \circ J = J \circ \operatorname{Hess} f. \tag{3.7}$$

Recall that any Killing vector field $W$ satisfies $\operatorname{tr} \nabla^2 W = -\operatorname{Ric}_g(W)$. We take $W = JV$. Since $J$ is parallel, $\operatorname{tr} \nabla^2 (JV) = J \operatorname{div} \operatorname{Hess} f$. Since $g$ is Kähler, $\operatorname{Ric}_g$ commutes with $J$. Thus

$$J \operatorname{div} \operatorname{Hess} f = -\operatorname{Ric}_g(JV) = -J \operatorname{Ric}_g(V),$$

hence $\operatorname{div} \operatorname{Hess} f = -\operatorname{Ric}_g(V)$. On the other hand, the Ricci identity reads

$$\operatorname{div} \operatorname{Hess} f = \nabla \Delta f + \operatorname{Ric}_g(V).$$

Subtracting $\operatorname{div} \operatorname{Hess} f = -\operatorname{Ric}_g(V)$ from the Ricci identity, we obtain $\nabla \Delta f = -2 \operatorname{Ric}_g(V)$, which is (3.6).

**Step 2.** In this step, we compute $\Delta_F w$ and bound $|\nabla w|^2$.

With the sign convention $\Delta = \operatorname{div} \nabla$, the Bochner formula reads

$$\frac{1}{2} \Delta w = |\operatorname{Hess} f|^2 + V(\Delta f) + \operatorname{Ric}_g(V, V).$$

By (3.6), $V(\Delta f) = -2 \operatorname{Ric}_g(V, V)$. Thus

$$\Delta w = 2 |\operatorname{Hess} f|^2 + V(\Delta f).$$

Since $\nabla F = -\frac{k}{f} V$, we have

$$\Delta_F u = \Delta u - \frac{k}{f} V(u) \tag{3.8}$$

for any smooth function $u$ on $Y$. In particular, $\Delta f = B + \frac{k}{f} w$. Since $V(f) = w$, we have

$$V(\Delta f) = V(B) + \frac{k}{f} V(w) - \frac{k}{f^2} w^2.$$

Therefore, by (3.8),

$$\Delta_F w = \Delta w - \frac{k}{f} V(w) = 2|\mathrm{Hess}\, f|^2 + V(B) - \frac{k}{f^2} w^2. \tag{3.9}$$

Next we show that

$$|\nabla w|^2 \le 2w|\mathrm{Hess}\, f|^2. \tag{3.10}$$

For any tangent vector $X$, we have $X(w) = 2g(\nabla_X V, V) = 2g(X, \mathrm{Hess}\, f(V))$, hence $\nabla w = 2\,\mathrm{Hess}\, f(V)$. At a point where $w = 0$, we have $V = 0$, so $\nabla w = 0$ and (3.10) holds. At a point where $w > 0$, let $e_1 := V/\sqrt{w}$ and $e_2 := Je_1$, and extend $e_1, e_2$ to an orthonormal basis $e_1, \dots, e_{2n}$ of the tangent space. By (3.7), $\mathrm{Hess}\, f(e_2) = J\,\mathrm{Hess}\, f(e_1)$, so $|\mathrm{Hess}\, f(e_2)| = |\mathrm{Hess}\, f(e_1)|$. Thus

$$|\mathrm{Hess}\, f|^2 \ge |\mathrm{Hess}\, f(e_1)|^2 + |\mathrm{Hess}\, f(e_2)|^2 = 2|\mathrm{Hess}\, f(e_1)|^2,$$

and

$$|\nabla w|^2 = 4|\mathrm{Hess}\, f(V)|^2 = 4w|\mathrm{Hess}\, f(e_1)|^2 \le 2w|\mathrm{Hess}\, f|^2.$$

**Step 3.** In this step, we conclude the proof.

By (3.8), $\Delta_F$ differs from $\Delta$ by a first-order operator. Hence, for any smooth function $u$ on $Y$ and any smooth function $\varphi$ on an open interval containing $u(Y)$,

$$\Delta_F(\varphi \circ u) = \varphi'(u)\Delta_F u + \varphi''(u)|\nabla u|^2.$$

With $\varphi = \log$, we obtain

$$\Delta_F \log(1+w) = \frac{\Delta_F w}{1+w} - \frac{|\nabla w|^2}{(1+w)^2}, \qquad \Delta_F F = -k\Big(\frac{B}{f} - \frac{w}{f^2}\Big) = -\frac{k}{f}B + \frac{k}{f^2}w.$$

Combining the equalities for $\Delta_F \log(1+w)$ and $\Delta_F F$ with (3.9) and the equality

$$\frac{k}{f^2}\Big(w - \frac{w^2}{1+w}\Big) = \frac{kw}{f^2(1+w)},$$

we obtain

$$\Delta_F H = \frac{V(B)}{1+w} - \frac{k}{f}B + \Big(\frac{2|\mathrm{Hess}\, f|^2}{1+w} - \frac{|\nabla w|^2}{(1+w)^2}\Big) + \frac{kw}{f^2(1+w)}. \tag{3.11}$$

By (3.10),

$$\frac{2|\mathrm{Hess}\, f|^2}{1+w} - \frac{|\nabla w|^2}{(1+w)^2} \ge \frac{2|\mathrm{Hess}\, f|^2}{1+w} - \frac{2w|\mathrm{Hess}\, f|^2}{(1+w)^2} = \frac{2|\mathrm{Hess}\, f|^2}{(1+w)^2} \ge 0.$$

Since $k \ge 0$, we have $\frac{kw}{f^2(1+w)} \ge 0$. Thus Lemma 3.6 follows from (3.11). □

The second lemma is a form of the Moser iteration [Mos60]. We give the proof, since we use the dependence of the bound on the constants.

**Lemma 3.7.** *Let $\mathcal{Y}$ be a compact Riemannian orbifold with a smooth positive measure $\nu$, let $q > 1$, $S > 0$, and $C_1 > 0$ be constants, and let $H \ge 1$ be a smooth function on $\mathcal{Y}$. Assume that*

$$\Big(\int_{\mathcal{Y}} |u|^{2q} d\nu\Big)^{1/q} \le S\Big(\int_{\mathcal{Y}} |du|^2 d\nu + \int_{\mathcal{Y}} |u|^2 d\nu\Big)$$

*for any smooth function $u$ on $\mathcal{Y}$, and that*

$$\int_{\mathcal{Y}} |d(e^{aH/2})|^2 d\nu \le C_1 a^2 \int_{\mathcal{Y}} e^{(a-1)H} d\nu$$

*for any real number $a \ge 1$. Then $\sup_{\mathcal{Y}} H \le C$, where $C$ depends only on $q$, $S$, $C_1$, and $\int_{\mathcal{Y}} e^H d\nu$.*

*Proof.* For any real number $p \geq 1$, let $\|e^H\|_p := \big(\int_{\mathcal{Y}} e^{pH} d\nu\big)^{1/p}$.

Let $a \geq 1$ be a real number. Since $e^{aH/2}$ is a smooth function on $\mathcal{Y}$, the assumptions of the lemma imply

$$\Big(\int_{\mathcal{Y}} e^{aqH} d\nu\Big)^{1/q} \leq S\Big(C_1 a^2 \int_{\mathcal{Y}} e^{(a-1)H} d\nu + \int_{\mathcal{Y}} e^{aH} d\nu\Big).$$

Since $H \geq 1 > 0$, we have $e^{(a-1)H} \leq e^{aH}$. Thus

$$\Big(\int_{\mathcal{Y}} e^{aqH} d\nu\Big)^{1/q} \leq S(1 + C_1 a^2) \int_{\mathcal{Y}} e^{aH} d\nu.$$

Taking the $a$-th root of both sides, we obtain

$$\|e^H\|_{aq} \leq \big(S(1 + C_1 a^2)\big)^{1/a} \|e^H\|_a. \tag{3.12}$$

For any integer $l \geq 0$, we have $q^l \geq 1$. Applying (3.12) with $a = q^l$, we obtain

$$\|e^H\|_{q^{l+1}} \leq \big(S(1 + C_1 q^{2l})\big)^{q^{-l}} \|e^H\|_{q^l}.$$

By induction on $l$, for any integer $l \geq 0$,

$$\|e^H\|_{q^l} \leq \prod_{j=0}^{l-1} \big(S(1 + C_1 q^{2j})\big)^{q^{-j}} \int_{\mathcal{Y}} e^H d\nu. \tag{3.13}$$

We bound the product in (3.13) independently of $l$. Let $C_2 := \max\{\log(S(1 + C_1)), 0\}$. Since $q^{2j} \geq 1$, we have $S(1 + C_1 q^{2j}) \leq S(1 + C_1) q^{2j}$, hence

$$q^{-j} \log\big(S(1 + C_1 q^{2j})\big) \leq q^{-j}(C_2 + 2j \log q) \leq (C_2 + 2 \log q)(1 + j) q^{-j}.$$

Since $q > 1$, we have $\sum_{j \geq 0} (1 + j) q^{-j} < +\infty$. Let

$$C_3 := (C_2 + 2 \log q) \sum_{j \geq 0} (1 + j) q^{-j}.$$

Then $C_3$ depends only on $q$, $S$, and $C_1$, and (3.13) implies

$$\|e^H\|_{q^l} \leq e^{C_3} \int_{\mathcal{Y}} e^H d\nu \tag{3.14}$$

for any integer $l \geq 0$.

Since $H$ is continuous and $\nu$ is a smooth positive measure on the compact orbifold $\mathcal{Y}$, we have $\|e^H\|_{q^l} \to \sup_{\mathcal{Y}} e^H$ as $l \to +\infty$, so (3.14) implies

$$\sup_{\mathcal{Y}} e^H \leq e^{C_3} \int_{\mathcal{Y}} e^H d\nu.$$

Therefore

$$\sup_{\mathcal{Y}} H \leq C_3 + \log \int_{\mathcal{Y}} e^H d\nu,$$

and the lemma holds with $C := C_3 + \log \int_{\mathcal{Y}} e^H d\nu$. □

The third lemma bounds the range of a Hamiltonian function of $\xi$ after the Kähler form is changed by $dd^c$ of an invariant function.

**Lemma 3.8.** *Let $\mathcal{Y}$ be a compact complex orbifold with a holomorphic action of a compact torus $T_c$, let $\xi \in \mathrm{Lie}(T_c)$, and denote also by $\xi$ the vector field on $\mathcal{Y}$ induced by $\xi$. Let $\omega_1$ be a $T_c$-invariant Kähler form on $\mathcal{Y}$, let $\psi$ be a smooth $T_c$-invariant function on $\mathcal{Y}$ such that $\omega_2 := \omega_1 + dd^c\psi$ is a Kähler form, let $c \in \mathbb{R}$, and let $\vartheta_1, \vartheta_2$ be smooth functions on $\mathcal{Y}$. Assume that $d\vartheta_1 = i_\xi \omega_1$, $d\vartheta_2 = i_\xi \omega_2$, and $\vartheta_2 = \vartheta_1 + c(J\xi)(\psi)$. Then*

$$\min_{\mathcal{Y}} \vartheta_1 \leq \vartheta_2 \leq \max_{\mathcal{Y}} \vartheta_1.$$

*Proof.* Let $y \in \mathcal{Y}$ be a point where $\vartheta_2$ attains the maximum. Let $\tilde{y}$ be a preimage of $y$ in an orbifold chart around $y$. The lift of $d\vartheta_2 = i_\xi \omega_2$ vanishes at $\tilde{y}$, and $\omega_2$ is non-degenerate at $\tilde{y}$, so the lift of $\xi$ vanishes at $\tilde{y}$. Thus $(J\xi)(\psi)(y) = 0$, and

$$\max_{\mathcal{Y}} \vartheta_2 = \vartheta_2(y) = \vartheta_1(y) \leq \max_{\mathcal{Y}} \vartheta_1.$$

Replacing the maximum with the minimum, we obtain $\min_{\mathcal{Y}} \vartheta_2 \geq \min_{\mathcal{Y}} \vartheta_1$. □

The Bochner argument and the lifting of paths from almost every point (Lemma 4.5) in the later sections use $\Phi$-cutoffs on the link (Definition 2.29). The branch divisors of $(Z, \Delta_Z)$ consist of smooth points of the orbifold $U$, so the cutoffs are not allowed to cut them off. For this reason we resolve the quotient stack $[(X\backslash\{o\})/\chi(\mathbb{G}_m)]$ itself rather than $Z$, and we cut off only the exceptional divisors over the singular locus of $[(X\backslash\{o\})/\chi(\mathbb{G}_m)]$. In particular, we need no assumption on $\Delta_Z$.

The cutoffs come from the finite logarithmic energy of the exceptional divisors (Lemma 3.5) together with a polynomial lower bound of the metric along the exceptional divisors. We prove the lower bound for smooth solutions of regularized weighted Monge–Ampère equations on a compact orbifold resolution of $U$, pass to the limit, and identify the limit by [HL20, Proposition 2.14]. On $U$ itself, the integration by parts behind the gradient bound of the Hamiltonian function of $\xi$ would need the cutoffs we are constructing, and the maximum principle in the trace estimate would need a prior control of the metric near the exceptional divisors.

**Theorem 3.9.** *Notation and conditions as in Set-up 3.1. Then $U$ admits a sequence of $T_c$-invariant $\Phi$-cutoffs with respect to $g_U$, smooth in orbifold charts, and $N$ admits a sequence of $T_c$-invariant $\Phi$-cutoffs with respect to $h$.*

*Proof.* The proof is long, so we divide it into several steps. Let $m := d - 1$ and $k := d + 1$.

**Step 1.** In this step we treat the case $d = 2$, and for $d \geq 3$ we construct an orbifold resolution $\pi : \mathcal{Y} \to \mathcal{B}$ of the quotient stack, a reference form $\alpha$, a Kähler form $\theta_{\mathcal{Y}}$, and a barrier $\varphi_E$ on $\mathcal{Y}$.

Assume that $d = 2$. Then $X$ is a normal surface, so $X_{\mathrm{sing}}$ is a finite $T$-invariant set. Since $\chi$ is a Reeb vector, any orbit of $\chi(\mathbb{G}_m)$ in $X\backslash\{o\}$ is one-dimensional, so $X_{\mathrm{sing}} \subset \{o\}$ and $M = X\backslash\{o\}$. Thus $U = [(X\backslash\{o\})/\chi(\mathbb{G}_m)]$ is a compact orbifold, and $N$ is compact by Proposition 3.4. The constant functions $\beta_j := 1$ on $U$ and on $N$ are $T_c$-invariant $\Phi$-cutoffs, since $\Phi(0) = 0$. From now on we assume that $d \geq 3$, so $m \geq 2$.

Let $z_n$, $n \in I$, be finitely many homogeneous generators of the maximal ideal of $o$ in $R$ which are $T$-eigenvectors, and regard $z_n$ as the restriction to $X$ of the $n$-th coordinate of $\mathbb{C}^I$ under the closed embedding $X \to \mathbb{C}^I$ defined by $(z_n)_{n\in I}$. Let $m_n$ be the character of $T$ with $z_n \in R_{m_n}$, and let $w_n := \langle m_n, \chi\rangle$ be the $\chi$-weight of $z_n$. Since the action of $T$ is good, $R_0 = \mathbb{C}$, so the maximal ideal of $o$ is $\bigoplus_{\alpha\neq 0} R_\alpha$ and $m_n \neq 0$. Since $\chi$ is a Reeb vector, $w_n > 0$ for any $n$ (Definition 2.6). Let

$$\mathcal{B} := [(X\backslash\{o\})/\chi(\mathbb{G}_m)], \qquad Q_n := \{z_n = 1\} \subset X.$$

The morphism $Q_n \times \mathbb{G}_m \to \{z_n \neq 0\}$, $(y, s) \to \chi(s) \cdot y$, is finite étale of degree $w_n$, and the open substacks $[Q_n/\mu_{w_n}]$, $n \in I$, cover $\mathcal{B}$. Since $X$ is normal and klt (Definition 2.6), $Q_n \times \mathbb{G}_m$ is normal, being étale over the normal variety $\{z_n \neq 0\}$, and klt by [KM98, Proposition 5.20(4)], hence so is $Q_n$. The morphism $(z_n)_{n\in I}$ embeds $\mathcal{B}$ as a closed substack of the weighted projective stack $\mathcal{P} := \mathcal{P}(w_n; n \in I)$, so $\mathcal{B}$ is a proper Deligne–Mumford stack. It is effective, since $T$ acts faithfully and $\chi(\mathbb{G}_m)$ acts on $X\backslash\{o\}$ with finite stabilizers. The coarse space of $\mathcal{B}$ is $Z$, $U \subset \mathcal{B}$ is open, and the singular locus $\mathcal{B}_{\mathrm{sing}} = [(X_{\mathrm{sing}}\backslash\{o\})/\chi(\mathbb{G}_m)]$ satisfies $\mathrm{codim}_{\mathcal{B}} \mathcal{B}_{\mathrm{sing}} \geq 2$, since $X$ is normal.

By [BM07, Theorem 1.1, its condition (2′), Definitions 2.1(2), and Theorem 1.3], there exists a projective resolution $\sigma : \tilde{X} \to X$, obtained as a sequence of blow-ups with smooth centers and functorial with respect to smooth morphisms, such that $\mathrm{Exc}(\sigma)$ is a simple normal crossing divisor and $\mathrm{Exc}(\sigma) = \sigma^{-1}(X_{\mathrm{sing}})$. Here $\mathrm{Exc}(\sigma)$ has simple normal crossings by [BM07, Definitions 2.1(2) and 2.2(2)], $\mathrm{Exc}(\sigma) \subset \sigma^{-1}(X_{\mathrm{sing}})$ by condition (2′) of [BM07, Theorem 1.1],

and $\sigma^{-1}(X_{\mathrm{sing}}) \subset \mathrm{Exc}(\sigma)$ since $\sigma$ is an open immersion off $\mathrm{Exc}(\sigma)$ and $\tilde{X}$ is smooth. In particular, $\sigma$ is an isomorphism over $X_{\mathrm{reg}}$. Applying the functoriality to the action and projection morphisms $T \times X \to X$, the action of $T$ lifts to $\tilde{X}$. Let

$$\mathcal{Y} := [(\tilde{X} \backslash \sigma^{-1}(o))/\chi(\mathbb{G}_m)], \qquad \pi : \mathcal{Y} \to \mathcal{B}$$

the induced morphism. Then $\pi$ is proper, representable, and an isomorphism over $U$. The stabilizer of a point of $\tilde{X} \backslash \sigma^{-1}(o)$ is contained in the stabilizer of the image point, so it is finite. An element of $\chi(\mathbb{G}_m)$ which acts trivially near a point of $\tilde{X} \backslash \sigma^{-1}(o)$ acts trivially on the dense open subset $M$, hence it is trivial. Thus $\mathcal{Y}$ is a compact smooth effective orbifold. Let $E := \mathcal{Y} \backslash U = \pi^{-1}(\mathcal{B}_{\mathrm{sing}})_{\mathrm{red}}$ and let $E_1, \dots, E_r$ be the irreducible components of $E$. Then $E$ is a simple normal crossing divisor and $\dim \pi(E_i) \le m-2$ for any $i$.

Let $\rho_{\mathcal{P}} : \mathbb{C}^I \backslash \{0\} \to (0, +\infty)$ be the function determined by

$$\sum_{n \in I} |z_n|^2 \rho_{\mathcal{P}}^{-2w_n} = 1, \tag{3.15}$$

where $(z_n)_{n \in I}$ are the coordinates of $\mathbb{C}^I$. For fixed $z \neq 0$, the left side is strictly decreasing in $\rho_{\mathcal{P}} \in (0, +\infty)$ with limits $+\infty$ and 0. Then $\{\rho_{\mathcal{P}} = 1\}$ is the unit sphere of $\mathbb{C}^I$, $\rho_{\mathcal{P}}((\lambda^{w_n} z_n)_n) = \lambda \rho_{\mathcal{P}}(z)$ for $\lambda > 0$, and $\rho_{\mathcal{P}}$ is the radius function of the Kähler cone metric of [Li21, Section 2.2] whose Reeb vector field generates the action of $S^1$ on $\mathbb{C}^I$ with weights $(w_n)_{n \in I}$. Since the $z_n$ are $T$-eigenvectors, $\rho_{\mathcal{P}}$ restricted to $X \backslash \{o\}$ is $T_c$-invariant, and $\rho_{\mathcal{P}}(\chi(s) \cdot y) = |s| \rho_{\mathcal{P}}(y)$. The form $dd^c \log \rho_{\mathcal{P}}^2$ descends to a $T_c$-invariant orbifold Kähler form $\alpha_{\mathcal{P}}$ on $\mathcal{P}$ [Li21, Section 2.4, (97) and the line after it]. Let $\alpha_{\mathcal{B}} := \alpha_{\mathcal{P}}|_{\mathcal{B}}$. Let

$$\alpha := \pi^* \alpha_{\mathcal{B}}, \qquad u := \log \rho_\chi^2 - \log \rho_{\mathcal{P}}^2 \text{ on } M.$$

Then $\alpha$ is a smooth closed semipositive real $(1,1)$-form on $\mathcal{Y}$. By Lemma 3.2(1), $\rho_\chi(\chi(s) \cdot y) = |s| \rho_\chi(y)$ for $s \in \mathbb{C}^*$ and $y \in M$. Thus the function $u$ is $T_c$-invariant and $\chi(\mathbb{G}_m)$-invariant, so it descends to $U$. Any $\chi(\mathbb{G}_m)$-orbit in $M$ contains a point of $\{\rho_\chi = 1\} = N$ (Theorem 2.27(2)), and on $N$ we have $u = -\log \rho_{\mathcal{P}}^2$, which is bounded since $N$ is relatively compact in $X \backslash \{o\}$ (Proposition 3.4). Thus $u$ is bounded. On $U$, the form

$$\alpha_u := \alpha + dd^c u$$

is the descent of $dd^c \log \rho_\chi^2$ from Theorem 2.27(2), so $u$ is a bounded $\alpha$-plurisubharmonic function on $U$. Since $\mathrm{codim}_{\mathcal{B}} \mathcal{B}_{\mathrm{sing}} \ge 2$ and $E$ is a proper analytic subset of $\mathcal{Y}$, $u$ extends to a bounded $\alpha_{\mathcal{B}}$-plurisubharmonic function on $\mathcal{B}$ and to a bounded $\alpha$-plurisubharmonic function on $\mathcal{Y}$ [GR56], and we denote both extensions by $u$.

By Lemma 3.2(4), the Kähler form of $g_U$ is $\frac{1}{2}\alpha_u$ on $U$. Hence $|d\beta|^2_{g_U} = 2|d\beta|^2_{\alpha_u}$ for any smooth function $\beta$ on $U$, where $|\cdot|_{\alpha_u}$ denotes the norm of the metric of $\alpha_u$, and $d\,\mathrm{Vol}_{g_U} = 2^{-m} d\,\mathrm{Vol}_{\alpha_u}$. Since $\Phi(\sqrt{2}s) \le 2(1 + \frac{1}{2}\log 2)^2 \Phi(s) \le 4\Phi(s)$ for $s \ge 0$ by (2.1), any sequence of $\Phi$-cutoffs on $U$ with respect to the metric of $\alpha_u$ is a sequence of $\Phi$-cutoffs with respect to $g_U$.

We show that there exist rational numbers $c_1, \dots, c_r > 0$ such that $-\sum_i c_i E_i$ is $\pi$-ample. By [BM07, Theorem 1.1], $\sigma = \sigma_1 \circ \cdots \circ \sigma_{l_0}$, where $\sigma_l : X_l \to X_{l-1}$ is the blow-up of a smooth center, $X_0 = X$, and $X_{l_0} = \tilde{X}$. The inverse image ideal sheaf of the center of $\sigma_l$ is the invertible sheaf $\mathcal{O}_{X_l}(1)$ [Har77, Proposition II.7.13 and its proof]. We construct ample line bundles $L_l$ on $X_l$ by induction on $l$. Let $L_0 := \mathcal{O}_X$, which is ample since $X$ is affine. If $L_{l-1}$ is ample, then $L_l := \mathcal{O}_{X_l}(1) \otimes \sigma_l^* L_{l-1}^{\otimes \ell}$ is very ample over $X$ for $\ell \gg 0$ [Har77, Exercise II.7.14(b)], hence ample [Har77, Theorem II.7.6]. By condition $(2')$ of [BM07, Theorem 1.1] and induction on $l$, the center of $\sigma_l$ is contained in the singular locus of $X_{l-1}$ or in the exceptional locus of $X_{l-1} \to X$, so the image in $X$ of any center is contained in $X_{\mathrm{sing}}$. Thus $L_{l_0} \cong \mathcal{O}_{\tilde{X}}(-\sum_i c_i' F_i)$, where the $F_i$ are the exceptional prime divisors of $\sigma$ and the $c_i'$ are positive integers. Each $F_i$ is $T$-invariant, and the restriction of $-\sum_i c_i' F_i$ to $\tilde{X} \backslash \sigma^{-1}(o) = \mathcal{Y} \times_{\mathcal{B}} (X \backslash \{o\})$ is ample over $X \backslash \{o\}$. Thus $-\sum_i c_i E_i$ is $\pi$-ample, where $c_i := c_i'$ if $E_i$ is the image of $F_i$ in $\mathcal{Y}$. We now show that, after multiplying the $c_i$ by a positive rational number, there exist Hermitian metrics $h_i^0$ on $\mathcal{O}_{\mathcal{Y}}(E_i)$ with curvature forms

$\Theta_i^0$ such that $\alpha - \sum_i c_i\Theta_i^0$ is a Kähler form. Since the $F_i$ are $T$-invariant, $L_{l_0}$ is $T$-linearized. By [Har77, Theorem II.7.6], $L_{l_0}^{r'}$ is very ample over $X$ for a positive integer $r'$. Since $X$ is affine and $T$ acts rationally on $H^0(\tilde{X}, L_{l_0}^{r'})$, finitely many $T$-eigensections $\varsigma_1, \dots, \varsigma_{l_1}$ generate $L_{l_0}^{r'}$ and define a $T$-equivariant closed embedding of $\tilde{X}$ into $X \times \mathbb{P}^{l_1-1}$ over $X$. Let $w_l'$ be the $\chi$-weight of $\varsigma_l$. On $\tilde{X}\backslash\sigma^{-1}(o)$, let $h_L$ be the Hermitian metric on $L_{l_0}^{r'}$ with

$$|v|_{h_L}^{-2} = \sum_{l=1}^{l_1} |\varsigma_l/v|^2(\rho_{\mathcal{P}} \circ \sigma)^{-2w_l'}$$

for a nowhere vanishing local section $v$, where the sign of $w_l'$ is fixed by the convention of the linearization so that $h_L$ is invariant under $\chi(\mathbb{G}_m)$. Such a sign exists since $\rho_{\mathcal{P}}(\chi(s)\cdot y) = |s|\rho_{\mathcal{P}}(y)$. The metric $h_L$ is smooth, $T_c$-invariant, and $\chi(\mathbb{G}_m)$-invariant, so it descends to a $T_c$-invariant metric $\bar{h}_L$ on the orbifold line bundle $\mathcal{O}_{\mathcal{Y}}(-r'\sum_i c_iE_i)$. Let $\Theta_L$ be the curvature form of $\bar{h}_L$. Since $\alpha = \pi^*\alpha_{\mathcal{B}}$ and $\alpha_{\mathcal{B}}$ is the restriction of an orbifold Kähler form on $\mathcal{P}$, a tangent vector $v'$ of $\mathcal{Y}$ with $\alpha(v', Jv') = 0$ satisfies $d\pi(v') = 0$, so $v'$ is the image of a tangent vector $\tilde{v}$ of $\tilde{X}\backslash\sigma^{-1}(o)$ with $d\sigma(\tilde{v}) = 0$. Since $d\sigma(\tilde{v}) = 0$, $\Theta_L(\tilde{v}, J\tilde{v})$ is the value of the pullback under $(\varsigma_l)_l$ of the Fubini–Study form of the Hermitian inner product $\sum_l(\rho_{\mathcal{P}} \circ \sigma)^{-2w_l'}|\cdot|^2$ on $\mathbb{C}^{l_1}$, with $\rho_{\mathcal{P}} \circ \sigma$ evaluated at the base point of $\tilde{v}$. Since $\tilde{X} \to X \times \mathbb{P}^{l_1-1}$ is a closed embedding, $\Theta_L(\tilde{v}, J\tilde{v}) > 0$ if $v' \neq 0$. By compactness of $\mathcal{Y}$, there exists a rational number $C_L > 0$ such that $\Theta_L + C_L\alpha$ is a Kähler form. Taking smooth Hermitian metrics on the $\mathcal{O}_{\mathcal{Y}}(E_i)$ and multiplying one of them by a smooth positive function, we obtain metrics $h_i^0$ on $\mathcal{O}_{\mathcal{Y}}(E_i)$ with curvature forms $\Theta_i^0$ such that $-r'\sum_i c_i\Theta_i^0 = \Theta_L$. Replacing $c_i$ with $r'c_i/C_L$, we obtain that $\alpha - \sum_i c_i\Theta_i^0 = C_L^{-1}(\Theta_L + C_L\alpha)$ is a Kähler form. Let $dy$ be the Haar measure on $T_c$ of total mass 1, and let $h_i$ be the Hermitian metric on $\mathcal{O}_{\mathcal{Y}}(E_i)$ with $\log h_i = \int_{T_c} y^* \log h_i^0 dy$. The curvature form of $h_i$ is $\Theta_i := \int_{T_c} y^*\Theta_i^0 dy$, and $h_i$ is $T_c$-invariant. Since $\alpha$ is $T_c$-invariant,

$$\theta_{\mathcal{Y}} := \alpha - \sum_i c_i\Theta_i = \int_{T_c} y^*\Big(\alpha - \sum_i c_i\Theta_i^0\Big)dy$$

is a $T_c$-invariant Kähler form on $\mathcal{Y}$. Let $s_i$ be a defining section of $E_i$, and, after multiplying $h_i$ by a positive constant, let $x_i := |s_i|_{h_i}^2 \leq e^{-2}$. By the Poincaré–Lelong formula, $dd^c \log x_i = -\Theta_i$ on $\mathcal{Y}\backslash E_i$. Thus the $T_c$-invariant function

$$\varphi_E := \sum_i c_i \log x_i \leq 0$$

satisfies

$$\alpha + dd^c\varphi_E = \theta_{\mathcal{Y}} \quad \text{on } \mathcal{Y}\backslash E. \tag{3.16}$$

This is the barrier of [GSS26, Section 4.1, (4.3)]. The divisors $E_i$ are $T$-invariant, so the holomorphic vector field induced by $\xi$ is tangent to $E_i$. Thus, denoting also by $\xi$ the real vector field on $\mathcal{Y}$ induced by $\xi$, the function $(J\xi)(\log x_i)$ extends to a smooth function on $\mathcal{Y}$ for any $i$.

**Step 2.** In this step we write the equation of $\alpha_u$ on $\mathcal{Y}$ and regularize the right-hand side.

By [Li21, Proposition 2.7 and (105)], applied in dimension $\dim U = d-1$ and with Li's reference Hermitian metric $h_0$ taken to be the one induced by $\rho_{\mathcal{P}}$, there exist constants $C_U, \gamma_U > 0$ such that

$$f^{-k}\alpha_u^m = C_U e^{-\gamma_U u}\Omega_0 \quad \text{on } U. \tag{3.17}$$

We may take $h_0$ in this way, since in [Li21, Section 2.4, p. 29], $h_0$ is induced by a smooth radius function of the Reeb vector field, and the Reeb vector field of $\rho_{\mathcal{P}}$ on $\mathbb{C}^I\backslash\{0\}$ generates the action of $S^1$ with weights $(w_n)$, which restricts on $X$ to the action of $\chi(S^1)$. The Reeb vector field in [Li21, Section 2.4] is $c_0\chi$, where $c_0 > 0$ is the constant with $A_X(\mathrm{wt}_{c_0\chi}) = d$, since Li normalizes Reeb vector fields $\hat{\xi}$ by $A_X(\mathrm{wt}_{\hat{\xi}}) = d$ [Li21, (75) and the sentence after it]. The radius function of [Li21, p. 29] is a positive constant power of Li's Hermitian metric. Thus Li's potential $\varphi_{\mathrm{Li}}$ of

the Hermitian metric $h_0 e^{\varphi_{\mathrm{Li}}}$ is a positive constant multiple of $u$, and Li's function $n+1+\theta_{v_\xi}$, where Li's $n+1$ is $d$, is a positive constant multiple of $f$. The exponent in [Li21, Proposition 2.7 and (105)] is $-\gamma\varphi_{\mathrm{Li}}$, where $\gamma>0$ is the constant for which the anti-canonical $\mathbb{Q}$-line bundle of Li's log Fano pair is $\gamma$ times Li's polarization [Li21, Section 2.4, p. 29]. Hence the exponent in (3.17) is $-\gamma_U u$ for a constant $\gamma_U>0$. Here $\Omega_0$ is the smooth positive orbifold volume form on $\mathcal{B}$ induced by $h_0$ [Li21, p. 30]. Write

$$K_{\mathcal{Y}} = \pi^* K_{\mathcal{B}} + \sum_i a_i E_i.$$

We compute $\pi^*\Omega_0$ in the étale charts $Q_n \to \mathcal{B}$. The base change of $\pi$ to $Q_n$ is a log resolution of $Q_n$, by the functoriality of [BM07, Theorem 1.1(3)] for étale and smooth morphisms (**Step 1**). Let $r_n$ be a positive integer such that $r_n K_{Q_n}$ is Cartier, and let $\varpi$ be a local generator of $\mathcal{O}(r_n K_{Q_n})$. Since $\Omega_0$ is induced by $h_0$ [Li21, p. 30], $\Omega_0$ is a smooth positive multiple of $(\sqrt{-1}^{r_n m^2}\varpi\wedge\bar{\varpi})^{1/r_n}$. By the definition of discrepancies [KM98, Definition 2.25 and Notation 2.26],

$$\pi^*\varpi = \prod_i s_i^{r_n a_i}\varpi',$$

where $\varpi'$ is a local generator of $\mathcal{O}(r_n K_{\mathcal{Y}})$ and the defining section $s_i$ of $E_i$ is written as a function in a local trivialization of $\mathcal{O}_{\mathcal{Y}}(E_i)$. Thus $\pi^*\Omega_0 = \prod_i x_i^{a_i} dV$, where $dV$ is a smooth positive $T_c$-invariant orbifold volume form on $\mathcal{Y}$. Since $Q_n$ is klt, $a_i > -1$ for any $i$ [KM98, Definition 2.34]. Since $\mathcal{Y}\times_{\mathcal{B}} Q_n$ is smooth and étale over $\mathcal{Y}$, and the open substacks $[(\mathcal{Y}\times_{\mathcal{B}} Q_n)/\mu_{w_n}]$, $n\in I$, cover $\mathcal{Y}$, the computation above determines $\pi^*\Omega_0$ in a finite atlas of orbifold charts of $\mathcal{Y}$. Thus

$$f^{-k}\alpha_u^m = C_U e^{-\gamma_U u}\prod_i x_i^{a_i} dV \quad \text{on } U=\mathcal{Y}\backslash E. \tag{3.18}$$

The current $\alpha+dd^c u$ on $\mathcal{Y}$ has bounded potentials, so the Lelong numbers of $\alpha+dd^c u$ vanish. By [Dem92, Main Theorem 1.1], there exist smooth functions $\tilde{u}_j$ on $\mathcal{Y}$ decreasing to $u$ and numbers $\epsilon_j\in(0,\frac{1}{2}]$ decreasing to 0 such that $\alpha+dd^c\tilde{u}_j \geq -\epsilon_j\theta_{\mathcal{Y}}$. The cited theorem concerns compact complex manifolds. We omit the adaptation to the orbifold $\mathcal{Y}$, since the regularization is carried out in finite orbifold charts, compatibly with the finite groups, and replacing $\tilde{u}_j$ with $\int_{T_c} y^*\tilde{u}_j dy$ we may assume that $\tilde{u}_j$ is $T_c$-invariant. Then $\sup_{\mathcal{Y}}|\tilde{u}_j|\leq C$ and $dd^c\tilde{u}_j\geq -C\theta_{\mathcal{Y}}$ for a constant $C$ independent of $j$. Let

$$\psi_j^+ := \sum_{a_i>0} a_i\log(x_i+\epsilon_j), \qquad \psi_j^- := \gamma_U\tilde{u}_j + \sum_{a_i<0}(-a_i)\log(x_i+\epsilon_j).$$

By the calculation of $P_\epsilon$ in the proof of Lemma 3.5, which holds for any $\epsilon>0$, applied with $\epsilon=\epsilon_j$ and the curvature form $\Theta_i$ of $h_i$,

$$dd^c\log(x_i+\epsilon_j) + \frac{x_i}{x_i+\epsilon_j}\Theta_i \geq 0 \quad \text{on } \mathcal{Y}.$$

Thus $dd^c\psi_j^\pm \geq -C\theta_{\mathcal{Y}}$. Since $x_i+\epsilon_j\leq 1$, we have $\psi_j^+\leq 0$, and $(x_i+\epsilon_j)^{a_i}\leq x_i^{a_i}$ for $a_i<0$. Thus

$$e^{\psi_j^+-\psi_j^-} \leq e^{\gamma_U C}\prod_{a_i<0} x_i^{a_i}. \tag{3.19}$$

**Step 3.** In this step we introduce the Hamiltonians of $\xi$ and the continuity path, and we prove uniform bounds for the Hamiltonians, for the normalizing constants, and for the derivatives of $\psi_j^\pm$ along the gradient field.

Denote also by $\xi$ the real vector field on $U$ and on $\mathcal{Y}$ induced by $\xi$, and let $V:=-J\xi$. Since $\rho_\chi$ and $J$ are $T_c$-invariant, $L_\xi\eta_\chi=0$, so the Cartan formula implies $d(\eta_\chi(\xi)) = -i_\xi d\eta_\chi$. The form $d\eta_\chi = dd^c\log\rho_\chi^2$ descends to $\alpha_u$ on $U$ [Li21, Section 2.4, (97) and the line after it]. Thus

$$df = i_{-\xi}\alpha_u \quad \text{on } U.$$

Since $g_U$ is $T_c$-invariant and $\xi \in \mathrm{Lie}(T_c)$, $\xi$ is a Killing vector field for $g_U$ and for any $T_c$-invariant Kähler metric on $\mathcal{Y}$. If $\omega_1$ is a $T_c$-invariant Kähler form and $\vartheta_1$ is a smooth function with $d\vartheta_1 = i_{-\xi}\omega_1$, then $\nabla_{\omega_1}\vartheta_1 = V$, where $\nabla_{\omega_1}$ is the gradient of the metric of $\omega_1$, and $J\nabla_{\omega_1}\vartheta_1 = \xi$ is a Killing vector field. Since $dd^c = \sqrt{-1}\partial\bar{\partial}$, we have $d^c w(X) = -\frac{1}{2}dw(JX)$ for a smooth function $w$ and a real vector field $X$. If $w$ is $T_c$-invariant, then $L_\xi d^c w = 0$, since $\xi$ is a holomorphic Killing vector field, and the Cartan formula implies

$$i_{-\xi}dd^c w = d\big(d^c w(\xi)\big) = d\big(\tfrac{1}{2}V(w)\big). \tag{3.20}$$

Let $\eta_{\mathcal{P}}$ be the contact form of $\rho_{\mathcal{P}}$, that is, $\eta_{\mathcal{P}}(W) = -d\log\rho_{\mathcal{P}}(JW)$ for any tangent vector $W$ of $\mathbb{C}^I\backslash\{0\}$. This is the convention of $\eta_\chi$ in Set-up 3.1(5) [Li21, Section 2.4, (97)]. The torus $T$ acts on $\mathbb{C}^I$ by the characters $m_n$, and the function $\eta_{\mathcal{P}}(\xi)$ on $\mathbb{C}^I\backslash\{0\}$ is smooth, $\chi(\mathbb{G}_m)$-invariant, and $T_c$-invariant. Let $f_\alpha$ be the pullback to $\mathcal{Y}$ of the function on $\mathcal{B}$ induced by $\eta_{\mathcal{P}}(\xi)$. Since $\rho_{\mathcal{P}}$ is $T_c$-invariant, the Cartan formula implies $d(\eta_{\mathcal{P}}(\xi)) = -i_\xi d\eta_{\mathcal{P}}$, so

$$df_\alpha = i_{-\xi}\alpha.$$

Since $(\eta_\chi - \eta_{\mathcal{P}})(W) = -\frac{1}{2}du(JW)$ for any tangent vector $W$ of $M$ and $V = -J\xi$, we have

$$f = f_\alpha + \tfrac{1}{2}V(u) \quad \text{on } U.$$

We show that $\min_{\mathcal{Y}} f_\alpha > 0$. The flow of $V = -J\xi$ multiplies $z_n$ by $e^{t\langle m_n,\xi\rangle}$ (Definition 2.6). Since $\eta_{\mathcal{P}}(\xi) = V(\log\rho_{\mathcal{P}})$, differentiating (3.15) along the flow of $V$ implies

$$f_\alpha = \frac{\sum_n |z_n|^2\rho_{\mathcal{P}}^{-2w_n}\langle m_n,\xi\rangle}{\sum_n |z_n|^2\rho_{\mathcal{P}}^{-2w_n}w_n}.$$

Thus

$$\min_n \frac{\langle m_n,\xi\rangle}{w_n} \le f_\alpha \le \max_n \frac{\langle m_n,\xi\rangle}{w_n}.$$

Since $m_n \neq 0$ (**Step 1**) and $\xi$ is a Reeb vector, $\langle m_n,\xi\rangle > 0$. Thus $\min_{\mathcal{Y}} f_\alpha > 0$. By (3.16) and (3.20), the function

$$f_\theta := f_\alpha + \tfrac{1}{2}V(\varphi_E)$$

extends to a smooth $T_c$-invariant function on $\mathcal{Y}$ with $df_\theta = i_{-\xi}\theta_{\mathcal{Y}}$ (**Step 1**). Let $\alpha_j := \alpha + \epsilon_j\theta_{\mathcal{Y}}$, which is a $T_c$-invariant Kähler form. For a smooth $T_c$-invariant function $u_{j,t}$ on $\mathcal{Y}$ with $\alpha_j + dd^c u_{j,t} > 0$ let

$$f_{j,t} := f_\alpha + \epsilon_j f_\theta + \tfrac{1}{2}V(u_{j,t}),$$

so that $df_{j,t} = i_{-\xi}(\alpha_j + dd^c u_{j,t})$ by (3.20). The continuity path, of the form [HL20, (3.14)], is

$$f_{j,t}^{-tk}(\alpha_j + dd^c u_{j,t})^m = \kappa_{j,t}e^{\psi_j^+ - \psi_j^-}\,dV, \qquad t \in [0,1], \tag{3.21}$$

where $u_{j,t}$ is a smooth $T_c$-invariant function with $\alpha_{j,t} := \alpha_j + dd^c u_{j,t} > 0$ and $\kappa_{j,t} > 0$ is a constant. In the a priori estimates below, we fix $j$ and $t$ and let $(u_{j,t},\kappa_{j,t})$ be a solution of (3.21). Constants denoted by $C$ and $c$ are independent of $j$ and $t$ and may change from line to line.

In each application of Lemma 3.8 below, we take $-\xi \in \mathrm{Lie}(T_c)$ in place of $\xi$ and $c = \frac{1}{2}$. By Lemma 3.8 with $\omega_1 = \alpha_j$, $\vartheta_1 = f_\alpha + \epsilon_j f_\theta$, and $\psi = u_{j,t}$,

$$\min_{\mathcal{Y}} f_\alpha - \epsilon_j \max_{\mathcal{Y}} |f_\theta| \le f_{j,t} \le \max_{\mathcal{Y}} f_\alpha + \epsilon_j \max_{\mathcal{Y}} |f_\theta|.$$

Possibly omitting finitely many $j$, we may assume that $\epsilon_j \max_{\mathcal{Y}} |f_\theta| \le \frac{1}{2}\min_{\mathcal{Y}} f_\alpha$. Thus there exist constants $0 < a_0 \le b_0$ with

$$a_0 \le f_{j,t} \le b_0 \quad \text{on } \mathcal{Y}. \tag{3.22}$$

Moreover $|V(u_{j,t})| = 2|f_{j,t} - f_\alpha - \epsilon_j f_\theta| \le C$. Let $C'$ be a constant with $dd^c\tilde{u}_j \ge -(C'-1)\theta_{\mathcal{Y}}$ for any $j$. Lemma 3.8 with $\omega_1 = C'\theta_{\mathcal{Y}}$, $\vartheta_1 = C'f_\theta$, and $\psi = \tilde{u}_j$ implies that $|V(\tilde{u}_j)| \le C$. Since $V(\log(x_i+\epsilon_j)) = \frac{x_i}{x_i+\epsilon_j}V(\log x_i)$ and $V(\log x_i)$ is a smooth function on $\mathcal{Y}$ (**Step 1**), we obtain

$$|V(\psi_j^+)| + |V(\psi_j^-)| + |V(u_{j,t})| + |V(\varphi_E)| \le C \quad \text{on } \mathcal{Y}\backslash E. \tag{3.23}$$

We show that the constant $\kappa_{j,t}$ is bounded above and below. By (3.21),

$$\kappa_{j,t}\int_{\mathcal{Y}} e^{\psi_j^+-\psi_j^-}\,dV=\int_{\mathcal{Y}} f_{j,t}^{-tk}\alpha_{j,t}^m.$$

By (3.22), the right-hand side is at least $\min(1,b_0^{-k})\int_{\mathcal{Y}}\alpha_{j,t}^m$ and at most $\max(1,a_0^{-k})\int_{\mathcal{Y}}\alpha_{j,t}^m$, and $\int_{\mathcal{Y}}\alpha_{j,t}^m=\int_{\mathcal{Y}}\alpha_j^m$. Thus the right-hand side is bounded above and below by positive constants, since

$$\int_{\mathcal{Y}}\alpha^m\le\int_{\mathcal{Y}}\alpha_j^m\le\int_{\mathcal{Y}}(\alpha+\theta_{\mathcal{Y}})^m,$$

where $\int_{\mathcal{Y}}\alpha^m>0$ since $\alpha_{\mathcal{B}}$ is Kähler. The integral on the left-hand side is bounded above by (3.19) and $a_i>-1$, and bounded below since $\psi_j^\pm$ are uniformly bounded on a fixed compact subset of $\mathcal{Y}\backslash E$.

Let $F_{j,t}:=-tk\log f_{j,t}$, and let

$$\nu_{j,t}:=e^{F_{j,t}}d\operatorname{Vol}_{\alpha_{j,t}}=\frac{\kappa_{j,t}}{m!}e^{\psi_j^+-\psi_j^-}\,dV,$$

where $d\operatorname{Vol}_{\alpha_{j,t}}=\alpha_{j,t}^m/m!$. Let $\Delta_{F_{j,t}}$ be the weighted Laplace operator of Lemma 3.6 for the metric of $\alpha_{j,t}$. Since $\nabla_{\alpha_{j,t}}f_{j,t}=V$, we have $\Delta_{F_{j,t}}f_{j,t}=\operatorname{div}_{\nu_{j,t}}V$. Taking the Lie derivative of $\nu_{j,t}$ along $V$, we obtain

$$B_{j,t}:=\Delta_{F_{j,t}}f_{j,t}=\operatorname{div}_{dV}V+V(\psi_j^+)-V(\psi_j^-),\qquad |B_{j,t}|\le C,\tag{3.24}$$

by (3.23), since $V$ and $dV$ are fixed smooth data on $\mathcal{Y}$.

**Step 4.** In this step we prove a uniform Sobolev inequality for $\nu_{j,t}$ and a uniform bound for $u_{j,t}$, using an auxiliary resolution of the coarse space of $\mathcal{Y}$.

Let $Y$ be the coarse space of $\mathcal{Y}$, and let $\mu:Y'\to Y$ be a projective log resolution of the pair consisting of $Y$, the image of $E$, and the branch divisor of $\mathcal{Y}\to Y$. Then $Y'$ is a compact connected Kähler manifold with $\dim Y'=m$. Fix a Kähler form $\theta_{Y'}$ and a smooth positive volume form $\Omega_{Y'}$ on $Y'$.

We first show that the measure $\prod_{a_i<0}x_i^{a_i}dV$ on $\mathcal{Y}$, viewed as a measure on $Y'$, has density in $L^{p_0}(\Omega_{Y'})$ for some $p_0>1$. Let $W\to W/\Gamma$ be a finite orbifold chart of $\mathcal{Y}$, where $W$ is smooth and $\Gamma$ is finite, and let $D_W:=\sum_{a_i<0}(-a_i)E_i|_W$. Then $(W,D_W)$ is klt, since $E|_W$ has simple normal crossings and the coefficients of $D_W$ are smaller than 1. By the Hurwitz formula, $K_W+D_W$ is the pullback of $K_{W/\Gamma}+D_{W/\Gamma}$ under the quotient map $W\to W/\Gamma$, where a prime divisor of $W/\Gamma$ over which the quotient map has ramification index $e$ and below a component of $D_W$ of coefficient $c$ (with $c=0$ if no such component exists) has coefficient $1-(1-c)/e<1$ in $D_{W/\Gamma}$. By [KM98, Proposition 5.20], $(W/\Gamma,D_{W/\Gamma})$ is klt. Up to a smooth positive factor, the measure $\prod_{a_i<0}x_i^{a_i}dV$ on $W/\Gamma$ is the measure defined by a local generator of $\mathcal{O}(r(K_{W/\Gamma}+D_{W/\Gamma}))$ for $r$ divisible enough. Thus, on $\mu^{-1}(W/\Gamma)$, the pullback of $\prod_{a_i<0}x_i^{a_i}dV$ is a bounded function times $\prod_l|z_l|^{2b_l}\Omega_{Y'}$ in local coordinates $z_l$ adapted to a simple normal crossing divisor, with $b_l>-1$ since $(W/\Gamma,D_{W/\Gamma})$ is klt [Ber+11, Section 3.1, (3.2) and Lemma 3.2(i)]. Since $Y$ is covered by finitely many such charts, there exists $p_0>1$ as claimed.

Let $r_0$ be a positive integer such that the $r_0$-th power of the ample $\mathbb{Q}$-line bundle on $Z$ induced by $\alpha_{\mathcal{B}}$ is a very ample line bundle, and let $\beta'_Z$ be $1/r_0$ times the pullback to $Y'$ of the Fubini–Study form of the corresponding projective embedding of $Z$. Let $r_1$ be a positive integer such that the $r_1$-th power of the ample $\mathbb{Q}$-line bundle on $Y$ induced by the class of $\theta_{\mathcal{Y}}$ is a very ample line bundle, and let $\beta'_Y$ be $1/r_1$ times the pullback to $Y'$ of the Fubini–Study form of the corresponding projective embedding of $Y$. Let $\beta'_j:=\beta'_Z+\epsilon_j\beta'_Y$. The $r_0$-th, respectively $r_1$-th, powers of the corresponding orbifold line bundles on $\mathcal{B}$ and $\mathcal{Y}$ descend to the very ample line bundles above on $Z$ and $Y$, and on each of them the ratio of the Fubini–Study metric to the metric with curvature $r_0\alpha_{\mathcal{B}}$, respectively $r_1\theta_{\mathcal{Y}}$, is a positive continuous function on a compact space. Let $Y'_0\subset Y'$ be the complement of the exceptional locus of $\mu$, of the preimage of the

image of $E$, and of the preimage of the branch divisor of $\mathcal{Y} \to Y$. Then $Y_0'$ is a dense Zariski open subset, and, on $Y_0'$,

$$\mu^*\alpha_{j,t} = \beta_j' + dd^c v_{j,t}, \qquad v_{j,t} := \mu^*(u_{j,t} + v_j),$$

where the function $v_j$ on $Y$ is $\frac{1}{r_0}$ times the logarithm of the first ratio plus $\frac{\epsilon_j}{r_1}$ times the logarithm of the second ratio, so $v_j$ is continuous and $|v_j| \leq C$. The function $v_{j,t}$ is bounded and $\beta_j'$-plurisubharmonic, so it extends across $Y' \backslash Y_0'$, and we denote also by $\mu^*\alpha_{j,t}$ the closed positive current $\beta_j' + dd^c v_{j,t}$ on $Y'$. By (3.21), (3.22), (3.19), and the bounds for $\kappa_{j,t}$ in **Step 3**,

$$(\mu^*\alpha_{j,t})^m = F_{j,t}^{Y'} \Omega_{Y'}, \qquad \|F_{j,t}^{Y'}\|_{L^{p_0}(\Omega_{Y'})} \leq C, \tag{3.25}$$

where we use that $(\mu^*\alpha_{j,t})^m$ puts no mass on proper analytic subsets, since $v_{j,t}$ is bounded [BT82].

We check that $\mu^*\alpha_{j,t} \in W_{\mathrm{big}}(Y', \theta_{Y'}, m, p_0, A, B)$ for constants $A, B$ independent of $j, t$. By definition [NV24, pp. 2–3], $W_{\mathrm{big}}(Y', \theta_{Y'}, m, p_0, A, B)$ is the set of closed positive $(1,1)$-currents $\mathfrak{T}$ on $Y'$ such that (i) $\mathfrak{T}$ is a smooth Kähler form on a dense Zariski open subset, $\{\mathfrak{T}\}$ is big, and $\mathfrak{T}$ has minimal singularities; (ii) $\{\mathfrak{T}\} \cdot \{\theta_{Y'}\}^{m-1} \leq A$; (iii) $(\{\mathfrak{T}\}^m)^{-1}\mathfrak{T}^m = f_{\mathfrak{T}}\theta_{Y'}^m$ with $\|f_{\mathfrak{T}}\|_{L^{p_0}(\theta_{Y'}^m)} \leq B$, where $\{\mathfrak{T}\}^m$ is the volume of $\{\mathfrak{T}\}$. Condition (i): $\mu^*\alpha_{j,t}$ is a smooth Kähler form on $Y_0'$, the class $\{\mu^*\alpha_{j,t}\} = \{\beta_j'\}$ is big since $\int_{Y'}(\beta_j')^m \geq \int_{Y'}(\beta_Z')^m > 0$, and $\mu^*\alpha_{j,t}$ has minimal singularities since $v_{j,t}$ is bounded and $\beta_j'$ is smooth. Condition (ii):

$$\{\mu^*\alpha_{j,t}\} \cdot \{\theta_{Y'}\}^{m-1} = \{\beta_Z' + \epsilon_j \beta_Y'\} \cdot \{\theta_{Y'}\}^{m-1} \leq A.$$

Condition (iii): the density of $(\int_{Y'}(\beta_j')^m)^{-1}(\mu^*\alpha_{j,t})^m$ with respect to $\theta_{Y'}^m$ has norm at most $B$ in $L^{p_0}(\theta_{Y'}^m)$, by (3.25), $\int_{Y'}(\beta_j')^m \geq \int_{Y'}(\beta_Z')^m$, and since $\theta_{Y'}^m/\Omega_{Y'}$ is bounded above and below by positive constants. Fix $q$ with $1 < q < (d-1)/(d-2)$. By [NV24, Theorem 1.2(ii)], applied in dimension $m = d-1$, there exists $S > 0$ independent of $j, t$ such that

$$\Big(\int_{Y'} |w|^{2q}(\mu^*\alpha_{j,t})^m\Big)^{1/q} \leq S\Big(\int_{Y'} |dw|^2_{\mu^*\alpha_{j,t}}(\mu^*\alpha_{j,t})^m + \int_{Y'} |w|^2(\mu^*\alpha_{j,t})^m\Big)$$

for any Borel function $w$ on $Y'$ such that $w$ and $\nabla w$ are locally square integrable on $Y_0'$ and

$$\int_{Y_0'} |w|^2(\mu^*\alpha_{j,t})^m < +\infty, \qquad \int_{Y_0'} dw \wedge d^c w \wedge (\mu^*\alpha_{j,t})^{m-1} < +\infty.$$

The inequality of [NV24, Theorem 1.2(ii)] is normalized by $\int_{Y'}(\beta_j')^m$, and the energy term there is $\int_{Y_0'} dw \wedge d^c w \wedge (\mu^*\alpha_{j,t})^{m-1}$. Removing the normalization and rewriting the energy term changes only the constant $S$, since $\int_{Y'}(\beta_Z')^m \leq \int_{Y'}(\beta_j')^m \leq \int_{Y'}(\beta_Z' + \beta_Y')^m$ and $dw \wedge d^c w \wedge (\mu^*\alpha_{j,t})^{m-1}$ is a constant multiple of $|dw|^2_{\mu^*\alpha_{j,t}}(\mu^*\alpha_{j,t})^m$. Let $w$ be a smooth function on $\mathcal{Y}$. Then $w$ is a function on the coarse space $Y$, its pullback to $Y'$, which we also denote by $w$, is bounded and smooth on $Y_0'$, and $\int_{Y_0'} |dw|^2_{\mu^*\alpha_{j,t}}(\mu^*\alpha_{j,t})^m$ is the integral over $\mathcal{Y}$ of the smooth form $|dw|^2_{\alpha_{j,t}}\alpha_{j,t}^m$, which is finite. Thus $w$ satisfies the conditions above. All the integrals above are integrals over the common dense open subset $Y_0'$, which has full measure for $\nu_{j,t}$. Since $\nu_{j,t} = e^{F_{j,t}}\alpha_{j,t}^m/m!$ and $|F_{j,t}| \leq C$ by (3.22), we obtain, after enlarging $S$,

$$\Big(\int_{\mathcal{Y}} |w|^{2q} d\nu_{j,t}\Big)^{1/q} \leq S\Big(\int_{\mathcal{Y}} |dw|^2_{\alpha_{j,t}} d\nu_{j,t} + \int_{\mathcal{Y}} |w|^2 d\nu_{j,t}\Big) \tag{3.26}$$

for any smooth function $w$ on $\mathcal{Y}$.

Next we bound $u_{j,t}$. The function $v_{j,t}$ solves $(\beta_j' + dd^c v_{j,t})^m = F_{j,t}^{Y'}\Omega_{Y'}$, where $\beta_j'$ is a smooth semipositive closed form in a big class with locally bounded potentials, uniformly in $j$. We check the hypotheses of [DP10, Theorem 2.2(A) and (C)] for the family $(\beta_j')_{j\geq 1}$, indexed by $j$ in place of the index $t$ of the family in [DP10, Theorem 2.2(C)]. For a form $\gamma$ of the family, let

$$I_{\gamma,\epsilon}(F) := \{\gamma\}^{-m} \int_{Y'} F \log^{m+\epsilon}(e + \{\gamma\}^{-m} F)\Omega_{Y'}$$

[DP10, Theorem 2.2(A)]. The integrals $I_{\beta'_j,\epsilon}(F^{Y'}_{j,t})$ are uniformly bounded by (3.25), since an $L^{p_0}$ bound with $p_0 > 1$ bounds $F\log^{m+\epsilon}(e+cF)$ in $L^1$ and $\{\beta'_j\}^{-m} \le (\int_{Y'}(\beta'_Z)^m)^{-1}$. Condition (C1) asks that the local potentials of the forms of the family on a fixed finite covering by coordinate balls, normalized to have supremum 0, are bounded uniformly in the index. Condition (C1) holds since the local potentials of $\beta'_j = \beta'_Z + \epsilon_j\beta'_Y$ are bounded uniformly in $j$. Condition (C2a) asks that each form $\gamma$ of the family be the sum of a smooth form $\theta_\gamma$ and $dd^c\psi_\gamma$ for a function $\psi_\gamma$ with $\min_{Y'}\psi_\gamma = 0$, such that $\max_{Y'}\psi_\gamma$ is bounded uniformly in the index and $\theta_\gamma \le (\{\gamma\}^m)^{1/m}\omega$ for one Kähler form $\omega$ independent of the index. For $\gamma = \beta'_j$ we take $\theta_\gamma := \beta'_j$ and $\psi_\gamma := 0$. Then

$$\beta'_j \le C\theta_{Y'} \le \Big(\int_{Y'}(\beta'_j)^m\Big)^{1/m} C\Big(\int_{Y'}(\beta'_Z)^m\Big)^{-1/m}\theta_{Y'}.$$

Thus, by [DP10, Theorem 2.2(A) and (C)], the oscillation of $v_{j,t}$ is bounded independently of $j,t$, and so is the oscillation of $u_{j,t} = v_{j,t} - v_j$ on $Y'_0$, hence on $\mathcal{Y}$. Since (3.21) does not change when a constant is added to $u_{j,t}$, we normalize $\sup_{\mathcal{Y}} u_{j,t} = 0$ and obtain

$$\sup_{\mathcal{Y}} |u_{j,t}| \le C. \tag{3.27}$$

**Step 5.** In this step we bound the gradient of $f_{j,t}$, prove a trace estimate with a barrier, and solve (3.21) for $t = 1$.

Let $w_{j,t} := |\nabla_{\alpha_{j,t}} f_{j,t}|^2 = |V|^2_{\alpha_{j,t}}$, let $C_0$ be a constant with $F_{j,t} + C_0 \ge 1$, and let

$$H := \log(1 + w_{j,t}) + F_{j,t} + C_0 \ge 1.$$

Since $J\nabla_{\alpha_{j,t}} f_{j,t} = \xi$ is a Killing vector field, Lemma 3.6, applied in orbifold charts with $tk$ in place of $k$, implies

$$\Delta_{F_{j,t}} H \ge \frac{V(B_{j,t})}{1 + w_{j,t}} + F'_{j,t}B_{j,t}, \qquad F'_{j,t} := -\frac{tk}{f_{j,t}}.$$

Let $\lambda \ge 1$. Since $\Delta_{F_{j,t}}$ is symmetric with respect to $\nu_{j,t}$,

$$\int_{\mathcal{Y}} e^{\lambda H}\Delta_{F_{j,t}} H d\nu_{j,t} = -\lambda\int_{\mathcal{Y}} e^{\lambda H}|dH|^2 d\nu_{j,t}.$$

Since $\operatorname{div}_{\nu_{j,t}} V = B_{j,t}$ and $V(\log(1 + w_{j,t})) = V(H) - F'_{j,t}w_{j,t}$, integrating $V$ by parts over the compact orbifold $\mathcal{Y}$ implies

$$\int_{\mathcal{Y}} e^{\lambda H}\frac{V(B_{j,t})}{1 + w_{j,t}} d\nu_{j,t} = -\int_{\mathcal{Y}} e^{\lambda H}\frac{B^2_{j,t} + (\lambda - 1)B_{j,t}V(H) + F'_{j,t}w_{j,t}B_{j,t}}{1 + w_{j,t}} d\nu_{j,t}.$$

Multiplying the inequality for $\Delta_{F_{j,t}} H$ by $e^{\lambda H}$, integrating against $\nu_{j,t}$, and using the formulas for $\int_{\mathcal{Y}} e^{\lambda H}\Delta_{F_{j,t}} H d\nu_{j,t}$ and $\int_{\mathcal{Y}} e^{\lambda H}V(B_{j,t})(1 + w_{j,t})^{-1} d\nu_{j,t}$ together with

$$F'_{j,t}B_{j,t} - \frac{F'_{j,t}w_{j,t}B_{j,t}}{1 + w_{j,t}} = \frac{F'_{j,t}B_{j,t}}{1 + w_{j,t}},$$

we obtain

$$\lambda\int_{\mathcal{Y}} e^{\lambda H}|dH|^2 d\nu_{j,t} \le \int_{\mathcal{Y}} e^{\lambda H}\frac{B^2_{j,t} - F'_{j,t}B_{j,t}}{1 + w_{j,t}} d\nu_{j,t} + (\lambda - 1)\int_{\mathcal{Y}} e^{\lambda H}\frac{B_{j,t}V(H)}{1 + w_{j,t}} d\nu_{j,t}.$$

Since $|V(H)| \le w^{1/2}_{j,t}|dH|$, Young's inequality, (3.22), and (3.24) bound the last integral by

$$\frac{\lambda}{2}\int e^{\lambda H}|dH|^2 d\nu_{j,t} + C\lambda\int e^{\lambda H}(1 + w_{j,t})^{-1} d\nu_{j,t}.$$

Since $|B_{j,t}| \le C$ and $|F'_{j,t}| \le k/a_0$ by (3.24) and (3.22), the first integral on the right-hand side of the previous inequality is at most $C\int e^{\lambda H}(1 + w_{j,t})^{-1} d\nu_{j,t}$. Thus, for $\lambda \ge 1$,

$$\frac{\lambda}{2}\int_{\mathcal{Y}} e^{\lambda H}|dH|^2 d\nu_{j,t} \le C\lambda\int_{\mathcal{Y}} e^{\lambda H}(1 + w_{j,t})^{-1} d\nu_{j,t}.$$

Since $|d(e^{\lambda H/2})|^2 = \frac{\lambda^2}{4} e^{\lambda H}|dH|^2$ and

$$e^{\lambda H}(1+w_{j,t})^{-1} = e^{(\lambda-1)H} e^H (1+w_{j,t})^{-1} = e^{(\lambda-1)H} e^{F_{j,t}+C_0} \le C e^{(\lambda-1)H},$$

we obtain

$$\int_{\mathcal{Y}} |d(e^{\lambda H/2})|^2 d\nu_{j,t} \le C\lambda^2 \int_{\mathcal{Y}} e^{(\lambda-1)H} d\nu_{j,t}$$

for any real number $\lambda \ge 1$. Moreover,

$$\int_{\mathcal{Y}} w_{j,t} d\nu_{j,t} = \int_{\mathcal{Y}} V(f_{j,t}) d\nu_{j,t} = -\int_{\mathcal{Y}} f_{j,t} B_{j,t} d\nu_{j,t} \le C,$$

since $\nu_{j,t}(\mathcal{Y}) \le C\int_{\mathcal{Y}} \alpha_j^m \le C$. Thus $\int_{\mathcal{Y}} e^H d\nu_{j,t} \le C$. By Lemma 3.7 with the Sobolev inequality (3.26), $\sup_{\mathcal{Y}} H \le C$. Thus

$$|\nabla_{\alpha_{j,t}} f_{j,t}|_{\alpha_{j,t}} \le C. \tag{3.28}$$

**Claim 3.10.** *Let $\mathfrak{b} \le 0$ be a $T_c$-invariant function on $\mathcal{Y}$, smooth on $\mathcal{Y}\backslash E$, and let $c_{\mathfrak{b}} > 0$ be a constant. Assume that $\alpha_j + dd^c\mathfrak{b} \ge c_{\mathfrak{b}}\theta_{\mathcal{Y}}$ on $\mathcal{Y}\backslash E$, that $|V(\mathfrak{b})| \le C_{\mathfrak{b}}$ on $\mathcal{Y}\backslash E$ for a constant $C_{\mathfrak{b}}$, and that either $\mathfrak{b}$ is smooth on $\mathcal{Y}$ or $\mathfrak{b}(y) \to -\infty$ as $y \to E$. Then*

$$\mathrm{tr}_{\theta_{\mathcal{Y}}} \alpha_{j,t} \le C_1 \exp\big(-\psi_j^- - D_0\mathfrak{b}\big) \quad \text{on } \mathcal{Y}\backslash E,$$

*where $D_0$ depends only on $c_{\mathfrak{b}}$ and the constants $C$ above, and $C_1$ depends only on $c_{\mathfrak{b}}$, $C_{\mathfrak{b}}$, and the constants $C$ above.*

*Proof.* Let $A_1 := \mathrm{tr}_{\theta_{\mathcal{Y}}} \alpha_{j,t}$ and $A_2 := \mathrm{tr}_{\alpha_{j,t}} \theta_{\mathcal{Y}}$, let $\Delta^c_{\alpha_{j,t}} := \mathrm{tr}_{\alpha_{j,t}} dd^c$, and let $\mathcal{L} := \Delta^c_{\alpha_{j,t}} + \frac{F'_{j,t}}{2} V$, where $F'_{j,t} = -tk/f_{j,t}$. Since $df_{j,t} = i_{-\xi}\alpha_{j,t}$ and $\alpha_{j,t}$ is $J$-invariant, $i_V\alpha_{j,t} = 2d^c f_{j,t}$, so

$$dd^c f_{j,t} = \tfrac{1}{2} d i_V \alpha_{j,t} = \tfrac{1}{2} L_V \alpha_{j,t}.$$

Since $V$ is a fixed smooth vector field on $\mathcal{Y}$, $\mathrm{tr}_{\theta_{\mathcal{Y}}} L_V\alpha_{j,t} \ge V(A_1) - CA_1$. Together with (3.28), we obtain

$$\Delta^c_{\theta_{\mathcal{Y}}} f_{j,t} \ge \frac{1}{2} V(A_1) - CA_1, \qquad |\partial f_{j,t}|^2_{\theta_{\mathcal{Y}}} \le A_1 |\partial f_{j,t}|^2_{\alpha_{j,t}} \le CA_1.$$

By the Aubin–Yau inequality [Aub78, Yau78], with a constant depending only on the curvature of $\theta_{\mathcal{Y}}$,

$$\Delta^c_{\alpha_{j,t}} \log A_1 \ge \frac{\Delta^c_{\theta_{\mathcal{Y}}} \log(\alpha^m_{j,t}/\theta^m_{\mathcal{Y}})}{A_1} - CA_2.$$

By (3.21),

$$\log(\alpha^m_{j,t}/\theta^m_{\mathcal{Y}}) = tk \log f_{j,t} + \log \kappa_{j,t} + \psi_j^+ - \psi_j^- + \log(dV/\theta^m_{\mathcal{Y}}).$$

Using $A_1A_2 \ge m^2$, $dd^c\psi_j^+ \ge -C\theta_{\mathcal{Y}}$, and the inequalities $\Delta^c_{\theta_{\mathcal{Y}}} f_{j,t} \ge \frac{1}{2}V(A_1) - CA_1$ and $|\partial f_{j,t}|^2_{\theta_{\mathcal{Y}}} \le CA_1$, we obtain

$$\mathcal{L}\log A_1 \ge -\frac{\Delta^c_{\theta_{\mathcal{Y}}}\psi_j^-}{A_1} - CA_2 - C.$$

Since $dd^c\psi_j^- + C\theta_{\mathcal{Y}} \ge 0$, comparing the eigenvalues of $\alpha_{j,t}$ with respect to $\theta_{\mathcal{Y}}$ implies

$$\Delta^c_{\alpha_{j,t}}\psi_j^- - \Delta^c_{\theta_{\mathcal{Y}}}\psi_j^-/A_1 \ge -CA_2.$$

By assumption,

$$\Delta^c_{\alpha_{j,t}}(u_{j,t} - \mathfrak{b}) = m - \mathrm{tr}_{\alpha_{j,t}}(\alpha_j + dd^c\mathfrak{b}) \le m - c_{\mathfrak{b}}A_2.$$

By the last two inequalities, $|F'_{j,t}| \le k/a_0$, (3.23), and $|V(\mathfrak{b})| \le C_{\mathfrak{b}}$, for any constant $D_0 > 0$,

$$\mathcal{L}\psi_j^- \ge \frac{\Delta^c_{\theta_{\mathcal{Y}}}\psi_j^-}{A_1} - CA_2 - C, \qquad \mathcal{L}\big(-D_0(u_{j,t} - \mathfrak{b})\big) \ge D_0 c_{\mathfrak{b}} A_2 - D_0 m - CD_0(1 + C_{\mathfrak{b}}).$$

Let

$$\mathcal{G} := \log A_1 + \psi_j^- - D_0(u_{j,t} - \mathfrak{b}).$$

Adding the last two inequalities to the lower bound for $\mathcal{L}\log A_1$ cancels $\Delta^c_{\theta_\mathcal{Y}}\psi_j^-/A_1$, and

$$\mathcal{L}\mathcal{G} \geq (D_0 c_\mathfrak{b} - 2C)A_2 - 2C - D_0 m - CD_0(1+C_\mathfrak{b}),$$

where $C$ is the constant of the lower bounds for $\mathcal{L}\log A_1$ and $\mathcal{L}\psi_j^-$. We choose $D_0$ with $D_0 c_\mathfrak{b} \geq 2C+1$. Thus

$$\mathcal{L}\mathcal{G} \geq A_2 - 2C - D_0 m - CD_0(1+C_\mathfrak{b}).$$

For fixed $j$ the function $\mathcal{G}$ is bounded above on $\mathcal{Y}\backslash E$, and $\mathcal{G}(y)\to-\infty$ as $y\to E$ unless $\mathfrak{b}$ is smooth on $\mathcal{Y}$. Thus $\mathcal{G}$ attains a maximum at a point $y_0$ of $\mathcal{Y}\backslash E$, or of $\mathcal{Y}$ if $\mathfrak{b}$ is smooth. In an orbifold chart at $y_0$, $d\mathcal{G}=0$ and $\Delta^c_{\alpha_{j,t}}\mathcal{G}\leq 0$, so $\mathcal{L}\mathcal{G}\leq 0$ and $A_2(y_0)\leq C_2$, where $C_2$ depends only on $c_\mathfrak{b}$, $C_\mathfrak{b}$, and $C$. Then the eigenvalues of $\alpha_{j,t}$ with respect to $\theta_\mathcal{Y}$ at $y_0$ are bounded below by $C_2^{-1}$, so $A_1 \leq C\alpha_{j,t}^m/\theta_\mathcal{Y}^m \leq Ce^{-\psi_j^-}$ at $y_0$ by (3.21), (3.22), and $\psi_j^+\leq 0$. Since $\mathfrak{b}\leq 0$ and $|u_{j,t}|\leq C$ by (3.27), $\mathcal{G}(y_0)\leq C$. Thus $\mathcal{G}\leq C$ on $\mathcal{Y}\backslash E$, which is the claim. □

We now solve (3.21). Fix $j$. For $t=0$, (3.21) has a smooth $T_c$-invariant solution by [Fau17, Theorem 1.1], with $\kappa_{j,0}$ determined by the total masses. Let $\mathcal{T}\subset[0,1]$ be the set of $t$ for which (3.21) has a smooth $T_c$-invariant solution. We use the Hölder spaces and the elliptic theory on compact orbifolds of [Fau17, Sections 2–5]. At a solution, the linearization of the logarithm of (3.21) in $u_{j,t}$ is

$$\delta \to \mathrm{tr}_{\alpha_{j,t}}\, dd^c\delta - \frac{tk}{2f_{j,t}}V(\delta) = \frac{1}{2}\Delta_{F_{j,t}}\delta,$$

since $\mathrm{tr}_{\alpha_{j,t}}\, dd^c = \frac{1}{2}\Delta$ and $\nabla F_{j,t} = -\frac{tk}{f_{j,t}}V$ for the metric of $\alpha_{j,t}$. The operator $\frac{1}{2}\Delta_{F_{j,t}}$ is symmetric with respect to $\nu_{j,t}$, Fredholm of index 0, and has the constants as kernel by the maximum principle. Thus the operator $(\delta,s)\to\frac{1}{2}\Delta_{F_{j,t}}\delta - s$, acting on $T_c$-invariant functions $\delta$ with $\int_\mathcal{Y}\delta d\nu_{j,t}=0$ and on $s\in\mathbb{R}$, is invertible, and $\mathcal{T}$ is open by the implicit function theorem. For closedness, let $t\in\mathcal{T}$. The function $\mathfrak{b}=0$ satisfies the assumptions of Claim 3.10 with $c_\mathfrak{b}=\epsilon_j$, and $\psi_j^-$ is smooth on $\mathcal{Y}$ for fixed $j$. Thus $\mathrm{tr}_{\theta_\mathcal{Y}}\alpha_{j,t}\leq C_j$, and (3.21) implies $\alpha_{j,t}\geq C_j^{-1}\theta_\mathcal{Y}$, where $C_j$ depends on $j$ but not on $t$. Together with (3.27) and (3.28), the interior Evans–Krylov estimate and the Schauder estimates [GT01, Section 17.4 and Theorem 6.17] in a finite atlas of orbifold charts bound $u_{j,t}$ in any $C^l$, independently of $t$. Thus $\mathcal{T}$ is closed, and $1\in\mathcal{T}$. Let $u_j := u_{j,1}$, $f_j := f_{j,1}$, and $\kappa_j := \kappa_{j,1}$.

**Step 6.** In this step we prove a polynomial lower bound for $\alpha_u$ whose factors are only the functions $x_i$, by passing to the limit $j\to+\infty$.

By (3.16), $\alpha_j + dd^c\varphi_E = (1+\epsilon_j)\theta_\mathcal{Y} \geq \theta_\mathcal{Y}$ on $\mathcal{Y}\backslash E$, $\varphi_E\leq 0$, $\varphi_E(y)\to-\infty$ as $y\to E$, and $|V(\varphi_E)|\leq C$ by (3.23). Thus Claim 3.10 with $\mathfrak{b}=\varphi_E$ and $c_\mathfrak{b}=1$ implies

$$\mathrm{tr}_{\theta_\mathcal{Y}}\,\alpha_{j,1} \leq C\exp(-\psi_j^- - D_0\varphi_E),$$

with $C$ and $D_0$ independent of $j$. Since $|\tilde{u}_j|\leq C$ and $(x_i+\epsilon_j)^{a_i}\leq x_i^{a_i}$ for $a_i<0$, we have $e^{-\psi_j^-}\leq C\prod_{a_i<0}x_i^{a_i}$. Thus there exists $N_1>0$ such that

$$\mathrm{tr}_{\theta_\mathcal{Y}}\,\alpha_{j,1} \leq C\prod_i x_i^{-N_1} \quad \text{on } \mathcal{Y}\backslash E.$$

By (3.21) and (3.22),

$$\alpha_{j,1}^m/\theta_\mathcal{Y}^m \geq ce^{\psi_j^+-\psi_j^-} \geq c\prod_{a_i>0}x_i^{a_i},$$

since $\psi_j^+\geq\sum_{a_i>0}a_i\log x_i$ and $-\psi_j^-\geq-\gamma_U C$. Comparing eigenvalues,

$$\mathrm{tr}_{\alpha_{j,1}}\theta_\mathcal{Y} \leq m(\mathrm{tr}_{\theta_\mathcal{Y}}\alpha_{j,1})^{m-1}\theta_\mathcal{Y}^m/\alpha_{j,1}^m.$$

Thus there exists $N_0>0$ such that

$$\alpha_{j,1} \geq C^{-1}\prod_i x_i^{N_0}\theta_\mathcal{Y} \quad \text{on } \mathcal{Y}\backslash E. \tag{3.29}$$

On any compact subset $U' \subset U = \mathcal{Y}\backslash E$, the bound $\mathrm{tr}_{\theta_\mathcal{Y}} \alpha_{j,1} \leq C \prod_i x_i^{-N_1}$ and (3.29) imply $C_{U'}^{-1}\theta_\mathcal{Y} \leq \alpha_{j,1} \leq C_{U'}\theta_\mathcal{Y}$, so $dd^c u_j$ and $u_j$ are bounded on $U'$ independently of $j$, and $u_j$ is bounded in $C^{1,\beta}(U')$ for any $\beta \in (0,1)$ in orbifold charts. After passing to a subsequence, $u_j$ converges locally uniformly on $U$, together with first derivatives, to a $T_c$-invariant function $u_\infty$ with $|u_\infty| \leq C$ and $\alpha + dd^c u_\infty \geq 0$ on $U$. Since $\mathrm{codim}_\mathcal{B} \mathcal{B}_{\mathrm{sing}} \geq 2$, $u_\infty$ extends to a bounded $\alpha_\mathcal{B}$-plurisubharmonic function on $\mathcal{B}$ [GR56]. The Hamiltonians $f_j = f_\alpha + \epsilon_j f_\theta + \frac{1}{2}V(u_j)$ converge locally uniformly on $U$ to $f_\infty := f_\alpha + \frac{1}{2}V(u_\infty)$, and $a_0 \leq f_\infty \leq b_0$. By the continuity of Monge–Ampère measures of bounded plurisubharmonic functions along locally uniformly convergent sequences [BT82], by $\tilde{u}_j \to u$, and since $\kappa_j$ converges after passing to a subsequence to some $\kappa_\infty > 0$, we obtain

$$f_\infty^{-k}(\alpha + dd^c u_\infty)^m = \kappa_\infty e^{-\gamma_U u} \prod_i x_i^{a_i} dV \quad \text{on } U$$

in the sense of measures. Comparing with (3.18), the measures $f_\infty^{-k}(\alpha + dd^c u_\infty)^m$ and $f^{-k}\alpha_u^m$ on $U$ are proportional.

We show that $u_\infty - u$ is constant. By Theorem 2.27(2), $\alpha_u = \theta + dd^c\varphi$ on $U$, where $\theta$ is a positive multiple of the restriction of a Fubini–Study form under a projective embedding of $Z$ and $\varphi$ is a bounded $\theta$-plurisubharmonic function on $Z$. Let $T_Z$ be the image of $T$ in the automorphism group of $Z$. Then $T_Z$ is a torus acting effectively on $Z$, and $T_Z$ preserves $\Delta_Z$, since the action of $T$ commutes with $\chi(\mathbb{G}_m)$ and so preserves the orders of the stabilizers of the $\chi(\mathbb{G}_m)$-orbits in $X\backslash\{o\}$. Let $\lambda_\theta > 0$ be the constant with $[\theta] = \lambda_\theta c_1(-(K_Z + \Delta_Z))$ [Li21, Section 2.4], and let $r_Z$ be a positive integer such that $-r_Z(K_Z + \Delta_Z)$ is a very ample Cartier divisor. The torus $T$ acts on $H^0(Z, \mathcal{O}_Z(-r_Z(K_Z + \Delta_Z)))$ by pullback of forms, so the corresponding projective embedding $\iota_Z : Z \to \mathbb{P}(H^0(Z, \mathcal{O}_Z(-r_Z(K_Z + \Delta_Z)))^\vee)$ is $T$-equivariant. Let $\theta'$ be $\lambda_\theta/r_Z$ times the pullback under $\iota_Z$ of the Fubini–Study form of a $T_c$-invariant Hermitian inner product. Since $\lambda_\theta^{-1}\theta$ and $\lambda_\theta^{-1}\theta'$ are curvature forms of continuous Hermitian metrics on the $\mathbb{Q}$-line bundle $-(K_Z + \Delta_Z)$, $\theta' = \theta + dd^c w$ for a continuous function $w$ on $Z$. Possibly replacing $\theta$ with $\theta'$ and $\varphi$ with $\varphi - w$, we may assume that $\theta$ is $T_c$-invariant. For $y \in T_c$, the function $y^*\varphi - \varphi$ is bounded and pluriharmonic on the image of $U$ in $Z$, since $\alpha_u$ and $\theta$ are $T_c$-invariant, so it extends to a pluriharmonic function on $Z$ [GR56], which is a constant $c_y$. The map $y \to c_y$ is a continuous homomorphism from the compact group $T_c$ to $\mathbb{R}$, so $c_y = 0$ for any $y$, and $\varphi$ is $T_c$-invariant. Let $\varphi_\infty := \varphi + u_\infty - u$. Then $\varphi_\infty$ is bounded, $T_c$-invariant, and $\theta + dd^c\varphi_\infty = \alpha + dd^c u_\infty \geq 0$ on $U$. In the charts $Q_n$, the functions $\varphi$ and $\varphi_\infty$ are $\mu_{w_n}$-invariant, so they descend to bounded $\theta$-plurisubharmonic functions on the complement of a proper Zariski closed subset of $Z$, hence on $Z$ [GR56].

In [HL20, Section 1, pp. 2–3], the data are a projective variety with a $\mathbb{Q}$-divisor $\Delta \geq 0$ whose log canonical divisor is $\mathbb{Q}$-Cartier, a $\mathbb{Q}$-line bundle with a possibly singular Hermitian metric whose curvature current is denoted $\Theta$, such that the anti-canonical $\mathbb{Q}$-line bundle of the pair minus this $\mathbb{Q}$-line bundle is ample, and a faithful holomorphic action of a complex torus preserving the divisor. We apply it to the pair $(Z, \Delta_Z)$, which is a klt log Fano pair [LWX18, Definition 2.13], with the trivial $\mathbb{Q}$-line bundle and the trivial metric, so that $\Theta = 0$ and the ample $\mathbb{Q}$-line bundle there is $-(K_Z + \Delta_Z)$, to the torus $T_Z$, and to the image of $\xi$ in the Lie algebra of the maximal compact subtorus $T_{Z,c} \subset T_Z$, which is the image of $T_c$. Since $[\theta] = \lambda_\theta c_1(-(K_Z + \Delta_Z))$, $\lambda_\theta^{-1}\theta$ is a Kähler form in $c_1(-(K_Z + \Delta_Z))$, and the moment polytope of $\lambda_\theta^{-1}\theta$ is $\lambda_\theta^{-1}P$, where $P$ is the moment polytope of $\theta$. Since $f$ and $f_\infty$ are Hamiltonians of $-\xi$ for $\theta + dd^c\varphi$ and $\theta + dd^c\varphi_\infty$ with $f_\infty - f = \frac{1}{2}V(\varphi_\infty - \varphi)$, there exists one affine function whose compositions with the moment maps of $\theta + dd^c\varphi$ and $\theta + dd^c\varphi_\infty$ are $f$ and $f_\infty$, and the weight $g_\xi := f^{-k}$, as a function on the moment polytope, is smooth and positive there, since $\xi$ is a Reeb vector [Li21, (103)].

The weighted Monge–Ampère measures of $\varphi$ and $\varphi_\infty$ are $g_\xi$-Monge–Ampère measures of bounded potentials. They vanish on the complement of the image of $U$ in $Z$ [BT82], they are proportional since $f_\infty^{-k}(\alpha + dd^c u_\infty)^m$ and $f^{-k}\alpha_u^m$ are proportional on $U$, and they have total mass 1 after the normalization [HL20, (2.10)–(2.11)], which does not depend on the potential. Thus

they are equal. They are $T_c$-invariant, and $E^*_{g_\xi} < +\infty$ for them by [HL20, Lemma 2.15], since $\varphi$ and $\varphi_\infty$ are bounded. Thus the common measure is in $\mathcal{M}^1_{T_{Z,c}}(Z)$ [HL20, Definition 2.13]. By [HL20, Proposition 2.14], $\varphi_\infty - \varphi$ is constant. Thus $u_\infty - u$ is constant on $U$, and $\alpha + dd^c u_\infty = \alpha_u$.

The currents $\alpha_{j,1} = \alpha_j + dd^c u_j$ converge weakly to $\alpha_u$ on $U$, and (3.29) is preserved under weak limits. Since $\alpha_u$ is smooth on $U$,

$$\alpha_u \geq C^{-1} \prod_i x_i^{N_0} \theta_{\mathcal{Y}} \quad \text{on } U = \mathcal{Y} \backslash E. \tag{3.30}$$

**Step 7.** In this step we construct the cutoffs on $U$ and on $N$.

We check the hypotheses of Lemma 3.5 in dimension $m = d - 1 \geq 2$, with the Kähler form $\theta_{\mathcal{Y}}$, the form $\alpha$, the function $u$, the divisors $E_1, \dots, E_r$, and the functions $x_i \leq e^{-2}$ of **Step 1**. Since $\alpha = \pi^* \alpha_{\mathcal{B}}$ and $\dim \pi(E_i) \leq m - 2$, we have $\alpha^{m-1}|_{E_i} = 0$ in orbifold charts, so $\int_{E_i} \alpha^{m-1} = 0$ for any $i$. Moreover $\alpha_u$ is a smooth Kähler form on $\mathcal{Y} \backslash E = U$, and (3.30) is the lower bound of Lemma 3.5(2) with exponent $N_0$. The group $T_c$ preserves $\theta_{\mathcal{Y}}$, $\alpha_u$, and $E$. By Lemma 3.5(2), $U$ admits a sequence $(\beta_j)$ of $T_c$-invariant $\Phi$-cutoffs, smooth in orbifold charts, with respect to the metric of $\alpha_u$. By **Step 1**, $(\beta_j)$ is a sequence of $\Phi$-cutoffs with respect to $g_U$. Finally, the sequence $(\beta_j \circ p)$ passes to $N$ by Lemma 3.3: it is a sequence of $T_c$-invariant $\Phi$-cutoffs on $N$ with respect to $h$. □

**Corollary 3.11.** *Notation and conditions as in Set-up 3.1. Then $X$, $\xi$, and $g$ satisfy the conditions of Set-up 2.31.*

*Proof.* By Definition 2.6, $X = \operatorname{Spec} R$ is a normal affine variety with klt singularities and a good action of $T$ with vertex $o$. Since the action is good, the closure of any $T$-orbit contains $o$, so $X$ is connected. Since $X$ is Noetherian, normal, and connected, it is irreducible [Sta, Tag 033M]. Thus $X_{\mathrm{reg}}$ is a connected complex manifold, $M = X_{\mathrm{reg}} \backslash \{o\}$ is connected since $d \geq 2$, and $N$ is connected since $M \cong (0, +\infty) \times N$. Conditions (1) and (2) of Set-up 2.31 hold by Theorem 2.27(1) and Proposition 3.4, condition (3) holds by Proposition 3.4, and condition (4) holds by Theorem 3.9. □

3.4. **From cone potentials to K-polystability.** In this subsection, we show that a continuous homogeneous potential which solves the Ricci-flat cone equation on a dense Zariski open subset implies K-polystability of the quotient log Fano pair (Proposition 3.12), and we deduce Corollary 3.13.

For a log Fano pair, a weak log Kähler–Einstein metric is a Kähler–Einstein metric in the sense of [Ber+11, Definition 3.5], which is defined through the adapted measure of [Ber+11, Definition 3.1]. If the potential in Proposition 3.12 is smooth on $X_{\mathrm{reg}} \backslash \{o\}$ and solves the Ricci-flat cone equation there, then the conclusion also follows from [Li21, Theorem 2.9] and Lemma 2.24. Later, Proposition 3.12 is used for a potential pulled back under a finite quasi-étale cover, which is smooth only over the étale locus of the cover. Thus we only require the equation on a dense Zariski open subset, and the continuity of the potential carries the conclusion across the complement.

**Proposition 3.12.** *Let $X$ be a normal affine variety of dimension $d \geq 2$ with klt singularities and a good positive grading induced by a faithful $\mathbb{G}_m$-action $\zeta$, with vertex $o$ and quotient log Fano pair $(Z, \Delta_Z)$ of $(X, \zeta)$, and let $\Omega$ be a nowhere vanishing reflexive $d$-form on $X$ which is homogeneous of degree $b > 0$. Let $p$ be a continuous positive function on $X \backslash \{o\}$ with $p(\lambda \cdot y) = |\lambda|^{2b/d} p(y)$ for any $\lambda \in \mathbb{C}^*$ and $y \in X \backslash \{o\}$. Assume that there exists a non-empty $\mathbb{G}_m$-invariant Zariski open subset $U \subset X_{\mathrm{reg}} \backslash \{o\}$ on which $p$ is smooth, $dd^c p$ is a Kähler form, $dd^c \log p \geq 0$, and*

$$(dd^c p)^d = C(\sqrt{-1})^{d^2} \Omega \wedge \bar{\Omega}$$

*for a constant $C > 0$. Then $(Z, \Delta_Z)$ admits a weak log Kähler–Einstein metric with continuous potentials and is K-polystable.*

*Proof.* The proof is long, so we divide it into several steps. Write $X = \operatorname{Spec} R$ with $R = \bigoplus_{j\geq 0} R_j$, let $\pi : X\backslash\{o\} \to Z$ be the quotient map, and let eu be the Euler derivation of the grading, viewed as a holomorphic vector field on $X_{\mathrm{reg}}$.

**Step 1.** In this step, we construct a reference metric on $-(K_Z + \Delta_Z)$ and a bounded continuous function $\varphi$ on $Z$.

By Lemma 2.7, $Z$ is a normal projective variety of dimension $d-1$ and $(Z, \Delta_Z)$ is a klt log Fano pair. By [XZ26b, Paragraph 2.13], $\pi$ is a Seifert $\mathbb{G}_m$-bundle whose zero section is the Kollár component $F$ with $\operatorname{ord}_F = \operatorname{wt}_\zeta$, and there exists an ample $\mathbb{Q}$-divisor $H \sim_{\mathbb{Q}} -F|_F$ on $Z \cong F$ such that $R \cong \bigoplus_{j\geq 0} H^0(Z, \lfloor jH \rfloor)$ as graded rings. Since $\zeta$ is faithful, there exists a homogeneous rational function $t$ of degree 1 on $X$. Under the isomorphism

$$X\backslash\{o\} \cong \operatorname{Spec}_Z \bigoplus_{j\in\mathbb{Z}} \mathcal{O}_Z(\lfloor jH \rfloor)$$

of [XZ26b, Paragraph 2.13], the rational section 1 of $\mathcal{O}_Z(\lfloor H \rfloor)$ is a homogeneous rational function $t_1$ of degree 1 with $\operatorname{div}(t_1) = \pi^* H$ on $X\backslash\{o\}$, where for a prime divisor $P$ of $Z$ over which the fibers of $\pi$ have multiplicity $a$, $\pi^* P$ is $a$ times the reduced preimage of $P$ (cf. [Kol04, Example 12 and Proposition 16]). Since $t/t_1$ is $\mathbb{G}_m$-invariant, $t = t_1 \cdot \pi^* q$ for some $q \in \mathbb{C}(Z)^*$. Replacing $H$ with $H + \operatorname{div}(q)$, we may assume that $\operatorname{div}(t) = \pi^* H$ on $X\backslash\{o\}$ and that $f \to f/t^j$ is the isomorphism $R_j \cong H^0(Z, \lfloor jH \rfloor)$ for any $j \geq 0$.

Since $\Omega$ is homogeneous of degree $b$, we have $L_{\mathrm{eu}}\Omega = b\Omega$. Hence the rational $(d-1)$-form $i_{\mathrm{eu}}\Omega/t^b$ satisfies $L_{\mathrm{eu}}(i_{\mathrm{eu}}\Omega/t^b) = 0$ and $i_{\mathrm{eu}}(i_{\mathrm{eu}}\Omega/t^b) = 0$, so $i_{\mathrm{eu}}\Omega/t^b = \pi^*\nu$ for a rational $(d-1)$-form $\nu$ on $Z$. Since $i_{\mathrm{eu}}(dt/t) = 1$, the top form $\Omega - t^b \frac{dt}{t} \wedge \pi^*\nu$ is annihilated by $i_{\mathrm{eu}}$, and thus

$$\Omega = t^b \frac{dt}{t} \wedge \pi^*\nu.$$

By the Hurwitz formula for the Seifert $\mathbb{G}_m$-bundle $\pi$ ([XZ26b, (2.4)] and [Kol04, Proposition 40 and Corollary 41]), $\operatorname{div}(\frac{dt}{t} \wedge \pi^*\nu) = \pi^*(\operatorname{div}(\nu) + \Delta_Z)$ on $X\backslash\{o\}$. Since $\Omega$ is nowhere vanishing, we obtain $\pi^*(bH + \operatorname{div}(\nu) + \Delta_Z) = 0$, and since $\pi$ is surjective, $\pi^*$ is injective on $\mathbb{Q}$-divisors. Thus, with the canonical divisor $K_Z := \operatorname{div}(\nu)$ determined by $\Omega$,

$$-(K_Z + \Delta_Z) = bH. \tag{3.31}$$

Let $e$ be a positive integer such that $eH$ is a very ample Cartier divisor, and let $s_0, \ldots, s_k$ be a basis of $R_e$. Then $\operatorname{div}(s_i) = \pi^*(\operatorname{div}(s_i/t^e) + eH)$ on $X\backslash\{o\}$, so $s_0, \ldots, s_k$ have no common zero on $X\backslash\{o\}$. Let $\iota : Z \to \mathbb{P}^k$ be the morphism defined by $s_0/t^e, \ldots, s_k/t^e$, so $\iota$ is a closed embedding. Let $\theta_0 := \frac{b}{e}\iota^*\omega_{\mathrm{FS}}$, where $\omega_{\mathrm{FS}} = dd^c \log \sum_i |x_i|^2$ is the Fubini–Study form, and let $h_0$ be the metric on the $\mathbb{Q}$-line bundle $-(K_Z + \Delta_Z)$ induced by the Fubini–Study metric on $\mathcal{O}_Z(eH)$ and (3.31). Let

$$p_0 := \Big(\sum_{i=0}^{k} |s_i|^2\Big)^{b/(de)}.$$

Then $p_0$ is a smooth positive function on $X\backslash\{o\}$ with $p_0(\lambda\cdot y) = |\lambda|^{2b/d} p_0(y)$. Hence $\log\big((p/p_0)^d\big)$ is a continuous $\mathbb{G}_m$-invariant function on $X\backslash\{o\}$. Since $\pi$ is open and surjective, $\log\big((p/p_0)^d\big) = \varphi\circ\pi$ for a continuous function $\varphi$ on $Z$, and since $Z$ is compact, $\varphi$ is bounded.

**Step 2.** In this step, we show that $\varphi$ is $\theta_0$-plurisubharmonic.

Let $Z_0 \subset Z$ be a non-empty Zariski open subset such that $Z_0$ is smooth, $Z_0 \cap \operatorname{Supp}\Delta_Z = \emptyset$, $Z_0 \cap \operatorname{Supp} H = \emptyset$, $\mathbb{G}_m$ acts freely on $\pi^{-1}(Z_0)$, and $\pi^{-1}(Z_0) \subset U$. Such a $Z_0$ exists: $\operatorname{Supp} H$ is a proper closed subset of $Z$, $\pi$ maps the $\mathbb{G}_m$-invariant closed subset $(X\backslash\{o\})\backslash U$ onto a proper closed subset of $Z$, and since $\zeta$ is faithful, the points of $X\backslash\{o\}$ with non-trivial stabilizer form a proper closed $\mathbb{G}_m$-invariant subset.

Let $W \subset Z_0$ be a small open ball in the Euclidean topology. Then $\pi^{-1}(W) \cong \mathbb{C}^* \times W$, and there exists a nowhere vanishing holomorphic function $s$ on $\pi^{-1}(W)$ with $s(\lambda \cdot y) = \lambda s(y)$. Let $u$

be the holomorphic function on $W$ with $\pi^*u = s/t$, and let $\nu_W := u^{-b}\nu$. Since $\frac{dt}{t} = \frac{ds}{s} - \pi^*\frac{du}{u}$ and $\pi^*(\frac{du}{u} \wedge \nu) = 0$, we have

$$\Omega = s^{b-1} ds \wedge \pi^*\nu_W \quad \text{on } \pi^{-1}(W), \tag{3.32}$$

and $\nu_W$ is a nowhere vanishing holomorphic $(d-1)$-form on $W$. Let $\psi_0$ and $\psi$ be the functions on $W$ with

$$\pi^*\psi_0 = \frac{b}{e}\log\sum_{i=0}^{k}\left|\frac{s_i}{s^e}\right|^2 \quad \text{and} \quad p = |s|^{2b/d}\exp(\pi^*\psi/d).$$

Then $\psi$ is smooth, $p_0 = |s|^{2b/d}\exp(\pi^*\psi_0/d)$, $\theta_0 = dd^c\psi_0$, and $\varphi = \psi - \psi_0$ on $W$. Since $dd^c\log|s|^2 = 0$, we have $dd^c\log p = \frac{1}{d}\pi^*dd^c\psi$, so $dd^c\psi \geq 0$. Thus $\theta_0 + dd^c\varphi = dd^c\psi \geq 0$ on $W$. Since $W \subset Z_0$ is arbitrary, $\theta_0 + dd^c\varphi \geq 0$ on $Z_0$.

Let $g : Z' \to Z$ be a resolution, let $A := g^{-1}(Z\backslash Z_0)$, let $V \subset Z'$ be a coordinate ball on which $g^*\theta_0 = dd^c v_0$ for a smooth function $v_0$, and let $v := v_0 + \varphi\circ g$. Then $v$ is continuous on $V$ and plurisubharmonic on $V\backslash A$. Let $D \subset V$ be a closed disc contained in a complex affine line $\ell$. If $D \not\subset A$, then $V_\ell := \ell \cap V$ is an open disc, and $V_\ell \cap A$ is a proper analytic subset of $V_\ell$, hence a closed discrete subset of $V_\ell$. Since a point $z_0$ is the $-\infty$ pole of $\log|z - z_0|$, $V_\ell \cap A$ is a polar set. Since $v$ is bounded near $V_\ell \cap A$ and subharmonic on $V_\ell\backslash A$, by [Dem12, Chapter I, Theorem 5.23], $v|_{V_\ell\backslash A}$ extends uniquely to a subharmonic function on $V_\ell$. Since $v$ is continuous, the extension is equal to $v|_{V_\ell}$, so $v|_D$ is subharmonic. If $D \subset A$, then $D$ is the limit of translates $D_k \not\subset A$, and since $v$ is continuous, the sub-mean value inequality for $v$ on $D$ follows from the sub-mean value inequalities on the $D_k$. Thus $v$ is plurisubharmonic on $V$ (cf. [GR56, Satz 3]). Hence $\varphi\circ g$ is $g^*\theta_0$-plurisubharmonic on $Z'$, and by [Ber+11, Section 1.1], $\varphi$ is $\theta_0$-plurisubharmonic on $Z$. Let

$$\theta := \theta_0 + dd^c\varphi.$$

Then $\theta$ is a closed positive $(1,1)$-current on $Z$ with continuous potentials.

**Step 3.** In this step, we show that $\theta$ solves the log Kähler–Einstein equation on $Z_0$.

Let $W$, $s$, $\nu_W$, $\psi_0$, and $\psi$ be as in **Step 2**. Since $\log p = \frac{b}{d}\log|s|^2 + \frac{1}{d}\pi^*\psi$, a direct calculation shows

$$dd^c p = p\Big(\frac{1}{d}\pi^*dd^c\psi + \sqrt{-1}\alpha\wedge\bar\alpha\Big), \qquad \alpha := \frac{b}{d}\frac{ds}{s} + \frac{1}{d}\pi^*\partial\psi.$$

Since $(\pi^*dd^c\psi)^d = 0$, $(\alpha\wedge\bar\alpha)^2 = 0$, and $\pi^*\partial\psi \wedge (\pi^*dd^c\psi)^{d-1} = 0$, we obtain

$$\begin{aligned}(dd^c p)^d &= b^2 d^{-d} p^d |s|^{-2}\sqrt{-1}ds\wedge d\bar s\wedge\pi^*(dd^c\psi)^{d-1}\\ &= b^2 d^{-d}|s|^{2b-2}\exp(\pi^*\psi)\sqrt{-1}ds\wedge d\bar s\wedge\pi^*(dd^c\psi)^{d-1}.\end{aligned}$$

On the other hand, by (3.32),

$$(\sqrt{-1})^{d^2}\Omega\wedge\bar\Omega = |s|^{2b-2}\sqrt{-1}ds\wedge d\bar s\wedge\pi^*\big((\sqrt{-1})^{(d-1)^2}\nu_W\wedge\bar\nu_W\big).$$

Since $\pi^{-1}(W) \subset U$, the equation $(dd^c p)^d = C(\sqrt{-1})^{d^2}\Omega\wedge\bar\Omega$ implies

$$(dd^c\psi)^{d-1} = Cb^{-2}d^d\exp(-\psi)(\sqrt{-1})^{(d-1)^2}\nu_W\wedge\bar\nu_W \quad \text{on } W.$$

We compare the right-hand side with the adapted measure $\mu_0$ of $h_0$ [Ber+11, Definition 3.1]. On $W$, $u^e$ is a local generator of $\mathcal{O}_Z(eH)$ with Fubini–Study norm $|u^e|^2 = (\sum_i|s_i/s^e|^2)^{-1}$, and under (3.31) the local generator $\nu_W = u^{-b}\nu$ of $\mathcal{O}_Z(K_Z)|_W$ corresponds to $u^{-b}$. Thus $\mu_0 = \exp(-\psi_0)(\sqrt{-1})^{(d-1)^2}\nu_W\wedge\bar\nu_W$ on $W$. Since $\psi = \psi_0 + \varphi$ and $\theta = dd^c\psi$ on $W$, we obtain

$$\theta^{d-1} = C'\exp(-\varphi)\mu_0 \quad \text{on } Z_0, \qquad C' := Cb^{-2}d^d. \tag{3.33}$$

**Step 4.** In this step, we extend (3.33) to $Z$ and conclude the proof.

Since $(Z,\Delta_Z)$ is klt, $\mu_0$ has finite total mass and charges no proper analytic subset of $Z$, by the log resolution formula for $\mu_0$ in [Ber+11, Definition 3.1 and Lemma 3.2]. Since $\varphi$ is bounded, the measure $\exp(-\varphi)\mu_0$ has finite total mass and charges no proper analytic subset of $Z$. Since $\theta$ has bounded potentials, the Bedford–Taylor product $\theta^{d-1}$ has full Monge–Ampère mass and

charges no pluripolar set, in particular no proper analytic subset [Ber+11, Section 1.2]. Since $Z\backslash Z_0$ is a proper analytic subset, (3.33) holds on $Z$ as an equality of Radon measures, and the constant $C'$ is the ratio of the total masses of $\theta^{d-1}$ and $\exp(-\varphi)\mu_0$. Another normalization of $dd^c$ differs from ours by a positive constant, which changes $\theta$ by a positive multiple and only changes $C'$.

Thus $\theta$ is a Kähler–Einstein metric of $(Z, \Delta_Z)$ in the sense of [Ber+11, Definition 3.5]. By [Ber+11, Lemma 3.6], $\theta$ is smooth on $Z_{\mathrm{reg}}\backslash \operatorname{Supp}\Delta_Z$, where it is a Kähler form since $\theta^{d-1}$ has a smooth positive density there, and $\operatorname{Ric}(\theta) = \theta + [\Delta_Z]$ on $Z_{\mathrm{reg}}$. By **Step 2**, $\theta$ has continuous potentials. Hence $\theta$ is a log Kähler–Einstein metric of $(Z, \Delta_Z)$ in the sense of [Ber16, Section 4.3]. Since $\Delta_Z \geq 0$, $(Z, \Delta_Z)$ is K-polystable by [Ber16, Theorem 4.8]. □

Corollary 3.13 is applied in Theorem 10.5 to a cone which is K-polystable for a torus $T$ that may be larger than $\zeta(\mathbb{G}_m)$ (Definition 2.16).

**Corollary 3.13.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety of dimension $2n$, $n \geq 1$, whose symplectic form has degree $r$ and whose grading is induced by a faithful $\mathbb{G}_m$-action $\zeta$, and assume that the Fano cone $(X, \zeta)$ is K-polystable for a torus $T$ which acts faithfully on $X$ and contains $\zeta(\mathbb{G}_m)$. Then the quotient log Fano pair of $(X, \zeta)$ is K-polystable, and $(X, \zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$.*

*Proof.* Let $\omega$ be the symplectic form, let $o$ be the vertex, let $(Z, \Delta_Z)$ be the quotient log Fano pair of $(X, \zeta)$, and let $\xi := \frac{2}{r}\zeta$. By Lemma 2.10, $X$ has klt singularities and $(X, \zeta)$ is a Fano cone.

We show that the $T$-action is good and that $\zeta$ is a Reeb vector of $T$. Let $\alpha$ be a character of $T$ with $R_\alpha \neq 0$, and let $j := \langle \alpha, \zeta \rangle$. Since $\zeta(\mathbb{G}_m) \subset T$, any $f \in R_\alpha$ satisfies $f(\zeta(\lambda) \cdot y) = \lambda^j f(y)$, so $R_\alpha \subset R_j$, so $j \geq 0$, and if $j = 0$, then $R_\alpha \subset R_0 = \mathbb{C}$, hence $\alpha = 0$. Thus $\zeta$, and hence $\xi$, is a Reeb vector of $T$. If $y \in X$ is fixed by $\zeta(\mathbb{G}_m)$, then $f(y) = \lambda^j f(y)$ for any $f \in R_j$ and $\lambda \in \mathbb{C}^*$, so $f(y) = 0$ for $j > 0$, hence $y = o$. The vertex $o$ is fixed by $T$ since the ideal $\bigoplus_{j>0} R_j = \bigoplus_{\alpha \neq 0} R_\alpha$ is $T$-stable. Thus $o$ is the unique fixed point of $T$. Since $f(\zeta(\lambda) \cdot y) = \lambda^j f(y) \to 0$ as $\lambda \to 0$ for any $f \in R_j$ with $j > 0$, we have $\lim_{\lambda \to 0} \zeta(\lambda) \cdot y = o$ for any $y \in X$, so the vertex $o$ belongs to the closure of any $T$-orbit. Thus $(X, \xi)$ is a Fano cone with torus $T$. Since $\zeta$ is faithful, $\zeta$ is a primitive cocharacter of $T$.

Let $\Omega := \omega^n$. Then $\Omega$ generates $\Omega_X^{[2n]}$. For $t \in T$, $t^*\Omega$ is also a generator of $\Omega_X^{[2n]}$, so $t^*\Omega = u_t \Omega$ for a unit $u_t$ of $R$. Since $R$ is positively graded with $R_0 = \mathbb{C}$, we have $u_t \in \mathbb{C}^*$, and $t \to u_t$ is a character $\kappa$ of $T$. Thus $\Omega$ is a $T$-eigenvector. Since $\Omega$ is homogeneous of degree $nr$, we have $\langle \kappa, \zeta \rangle = nr$, hence $\langle \kappa, \xi \rangle = 2n$, and by Lemma 2.14,

$$A_X(\mathrm{wt}_\xi) = 2n.$$

By Lemma 2.19, $(X, \xi)$ is K-polystable for $T$.

Let $M := X_{\mathrm{reg}}\backslash\{o\}$. By Theorem 2.27, applied to $(X, \xi)$, $T$, and $\chi := \zeta$ (see Remark 2.28), there exists a $T_c$-invariant smooth function $\rho > 0$ on $M$ such that $\frac{\sqrt{-1}}{2}\partial\bar\partial\rho^2$ defines a Ricci-flat Kähler cone metric $g = d\rho^2 + \rho^2 h$ on $M$, $\rho\partial_\rho s = \langle \alpha, \xi \rangle s$ for any $s \in R_\alpha$, and

$$(dd^c\rho^2)^{2n} = C(\sqrt{-1})^{4n^2}\Omega \wedge \bar\Omega \quad \text{on } M$$

for a constant $C > 0$. By Proposition 3.4, applied with $\chi = \zeta$, $\rho$ extends to a continuous function on $X$ with $\rho^{-1}(0) = \{o\}$. Let $p := \rho^2$.

We check the hypotheses of Proposition 3.12 for $p$. The function $p$ is continuous and positive on $X\backslash\{o\}$. Since $\rho\partial_\rho$ acts on holomorphic functions as $\mathrm{eu}_\xi = \frac{2}{r}\mathrm{eu}_\zeta$, the flow of $\rho\partial_\rho$ at time $\tau$ is the action of $\zeta(e^{2\tau/r})$. Since $\rho\partial_\rho(\rho^2) = 2\rho^2$ and $\rho$ is invariant under $\zeta(S^1) \subset T_c$, we obtain $p(\zeta(\lambda) \cdot y) = |\lambda|^r p(y)$ for any $\lambda \in \mathbb{C}^*$ and $y \in M$, hence for any $y \in X\backslash\{o\}$ by continuity. Since $\Omega$ is homogeneous of degree $nr$, $\dim X = 2n$, and $r = 2nr/(2n)$, the function $p$ has the homogeneity of Proposition 3.12. The form $dd^c p$ is a Kähler form on $M$. A direct calculation shows

$$dd^c \log \rho^2 = \rho^{-2}\big(dd^c\rho^2 - \rho^{-2}\sqrt{-1}\partial\rho^2 \wedge \bar\partial\rho^2\big).$$

Since $|d\rho|_g = 1$, the norm of $\partial\rho^2$ with respect to the hermitian form $(v,w) \to \partial\bar\partial\rho^2(v,\bar w)$ on $T^{1,0}M$ is $\rho$, and by the Cauchy–Schwarz inequality,

$$|\partial\rho^2(v)|^2 \leq \rho^2\partial\bar\partial\rho^2(v,\bar v)$$

for any $v \in T^{1,0}M$. Thus $dd^c \log p \geq 0$ on $M$, with equality in the directions of $\rho\partial_\rho$ and $J(\rho\partial_\rho)$. Moreover, $M$ is a non-empty $\mathbb{G}_m$-invariant Zariski open subset of $X_{\mathrm{reg}}\backslash\{o\}$, and $\Omega$ is a nowhere vanishing reflexive $2n$-form, homogeneous of degree $nr > 0$.

By Proposition 3.12, applied with $U := M$, $(Z, \Delta_Z)$ is K-polystable. By Lemma 2.24, applied to $(X,\zeta)$ with $c = 1$, $(X,\zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$. □

## 4. Parallel forms and the period theorem

The goal of this section is to prove Theorem 4.7. This section is the central step in the proof of parts (1) and (2) of Theorem 1.2: on the K-polystable central fiber $X_0$ of a degeneration, we construct a parallel symplectic form and compute the period of the Reeb vector. The analysis enters only through Set-up 2.31.

### 4.1. Closed two-forms on test configurations.

In this subsection, we study closed relative 2-forms on a marked test configuration.

**Definition-Lemma 4.1.** Notation and conditions as in Set-up 2.15. Assume that $\dim X = 2n$, that $X_0$ is integral and normal, and that $X_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form $\sigma$ of $\zeta$-degree $r > 0$. Let $\mathcal{U} \subset \mathcal{X}$ be the smooth locus of $\mathcal{X} \to \mathbb{A}^1$, regard forms on $X_{\mathrm{reg}}$ as relative forms on $\mathcal{U}\backslash X_0$ through the marking, and let $\mathcal{C}_r$ be the space of closed reflexive 2-forms on $X$ of $\zeta$-degree $r$. Then the following hold.

(1) We have $\dim \mathcal{C}_r < +\infty$, and for every $0 \neq \alpha \in \mathcal{C}_r$ there exists a unique integer $b(\alpha)$ such that $t^{-b(\alpha)}\alpha$ extends to a regular relative 2-form on $\mathcal{U}$ whose restriction to $(X_0)_{\mathrm{reg}}$ is non-zero.

(2) The closed relative 2-forms on $\mathcal{U}$ of $\zeta$-degree $r$ form a free $\mathbb{C}[t]$-module with basis $t^{-b(\alpha_1)}\alpha_1, \dots, t^{-b(\alpha_N)}\alpha_N$ for any basis $\alpha_1, \dots, \alpha_N$ of $\mathcal{C}_r$ such that, for every integer $b$, the $\alpha_i$ with $b(\alpha_i) \geq b$ span $\{\alpha \in \mathcal{C}_r \mid \alpha = 0 \text{ or } b(\alpha) \geq b\}$.

(3) Let $\nu := \sigma^n/n!$ and let $a$ be the integer such that $t^{-a}\nu$ extends to a regular relative top form on $\mathcal{U}$ with non-zero restriction to $(X_0)_{\mathrm{reg}}$. Then $t^{-a}\nu$ vanishes nowhere on $\mathcal{U}$.

(4) Every closed relative 2-form $\gamma$ on $\mathcal{U}$ of $\zeta$-degree $r$ satisfies $\gamma = \frac{1}{r} d_{\mathrm{rel}}(i_{\mathrm{eu}}\gamma)$, where eu is the Euler vector field of $\zeta$.

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we describe $\mathcal{U}$, and we determine, in terms of the order along $X_0$, when a relative form multiplied by a power of $t$ extends over $X_0$.

Since $\mathcal{X} \to \mathbb{A}^1$ is flat, a point $y \in \mathcal{X}$ belongs to $\mathcal{U}$ if and only if the fiber through $y$ is smooth at $y$ [Gro67, Théorème 17.5.1]. Thus $\mathcal{U} \cap X_0 = (X_0)_{\mathrm{reg}}$, and under the marking $\mathcal{U}\backslash X_0 = X_{\mathrm{reg}} \times \mathbb{G}_m$. Since $X_0$ is integral, $(X_0)_{\mathrm{reg}}$ is a non-empty open subset of $X_0$, hence a prime divisor on $\mathcal{U}$, and it is the only prime divisor on $\mathcal{U}$ which is disjoint from $\mathcal{U}\backslash X_0$. Let $\eta_0$ be the generic point of $X_0$. Since $\mathcal{U}$ is smooth, $\mathcal{O}_{\mathcal{U},\eta_0}$ is a discrete valuation ring, and since $X_0 = \mathrm{Spec}(\mathcal{R}/t\mathcal{R})$ is reduced, $t$ is a uniformizer of $\mathcal{O}_{\mathcal{U},\eta_0}$. The stalk $\Omega^p_{\mathcal{U}/\mathbb{A}^1,\eta_0}$ is a free $\mathcal{O}_{\mathcal{U},\eta_0}$-module. For a non-zero relative $p$-form $\tau$ on $\mathcal{U}\backslash X_0$, we denote by $\mathrm{ord}_{X_0}(\tau)$ the largest integer $c$ such that $t^{-c}\tau \in \Omega^p_{\mathcal{U}/\mathbb{A}^1,\eta_0}$.

Let $c$ be an integer. We show the following.

(a) The form $t^{-c}\tau$ extends to a regular relative $p$-form on $\mathcal{U}$ if and only if $c \leq \mathrm{ord}_{X_0}(\tau)$.

(b) If $c \leq \mathrm{ord}_{X_0}(\tau)$, then the restriction of $t^{-c}\tau$ to $(X_0)_{\mathrm{reg}}$ is non-zero if and only if $c = \mathrm{ord}_{X_0}(\tau)$.

Since $t$ is invertible on $\mathcal{U}\backslash X_0$, $t^{-c}\tau$ is regular on $\mathcal{U}\backslash X_0$. Since $\mathcal{U}$ is normal and $\Omega^p_{\mathcal{U}/\mathbb{A}^1}$ is locally free, a rational section of $\Omega^p_{\mathcal{U}/\mathbb{A}^1}$ is regular if and only if it is regular at the generic point of any

prime divisor on $\mathcal{U}$ [Har77, Proposition II.6.3A]. The only prime divisor at which $t^{-c}\tau$ may fail to be regular is $(X_0)_{\mathrm{reg}}$, which implies (a). Since $\Omega^p_{\mathcal{U}/\mathbb{A}^1}|_{(X_0)_{\mathrm{reg}}} = \Omega^p_{(X_0)_{\mathrm{reg}}}$ is locally free on the integral scheme $(X_0)_{\mathrm{reg}}$, the restriction of $t^{-c}\tau$ is zero if and only if it vanishes at $\eta_0$, that is, if and only if $t^{-c-1}\tau \in \Omega^p_{\mathcal{U}/\mathbb{A}^1,\eta_0}$. This holds if and only if $c+1 \leq \mathrm{ord}_{X_0}(\tau)$, which implies (b).

**Step 2.** In this step we prove (1).

Since $X$ is normal, $\Omega^{[2]}_X$ is the reflexive hull of $\Omega^2_X$, hence coherent, and $H^0(X, \Omega^{[2]}_X)$ is a finitely generated $R$-module. The space $H^0(X, \Omega^{[2]}_X)$ is a rational representation of $\zeta(\mathbb{G}_m)$ [Bor91, Chapter III, §8.4, Proposition], so it is a graded $R$-module whose degree $j$ part is $H^0(X_{\mathrm{reg}}, \Omega^2)_j$. As $R$ is generated by finitely many elements of positive degree and $R_0 = \mathbb{C}$, any graded piece of a finitely generated graded $R$-module has finite dimension. Since $\mathcal{C}_r \subset H^0(X_{\mathrm{reg}}, \Omega^2)_r$, we have $\dim \mathcal{C}_r < +\infty$.

Let $0 \neq \alpha \in \mathcal{C}_r$. Then $\alpha$ is a non-zero relative 2-form on $\mathcal{U}\backslash X_0 = X_{\mathrm{reg}} \times \mathbb{G}_m$. By **Step 1**, $b(\alpha) := \mathrm{ord}_{X_0}(\alpha)$ is the unique integer such that $t^{-b(\alpha)}\alpha$ extends to a regular relative 2-form on $\mathcal{U}$ whose restriction to $(X_0)_{\mathrm{reg}}$ is non-zero. This proves (1).

**Step 3.** In this step we prove (2).

For any integer $b$ let

$$F^b := \{\alpha \in \mathcal{C}_r \mid \alpha = 0 \text{ or } b(\alpha) \geq b\}.$$

Since $b(\alpha) = \mathrm{ord}_{X_0}(\alpha)$, we have $b(c\alpha) = b(\alpha)$ for $c \in \mathbb{C}^*$, and $b(\alpha + \alpha') \geq \min\{b(\alpha), b(\alpha')\}$ if $\alpha + \alpha' \neq 0$. Thus any $F^b$ is a linear subspace of $\mathcal{C}_r$, and $F^{b+1} \subset F^b$. Since $\dim \mathcal{C}_r < +\infty$, we have $F^b = \mathcal{C}_r$ for $b \ll 0$ and $F^b = 0$ for $b \gg 0$. In particular, bases $\alpha_1, \ldots, \alpha_N$ as in (2) exist: take a basis of the smallest non-zero $F^b$ and extend it successively to bases of the larger $F^b$. Fix such a basis and let $b_i := b(\alpha_i)$, so that $F^b$ is spanned by the $\alpha_i$ with $b_i \geq b$.

Let $\mathcal{M}$ be the $\mathbb{C}[t]$-module of closed relative 2-forms on $\mathcal{U}$ of $\zeta$-degree $r$. Since $\mathcal{U}$ is integral and $\Omega^2_{\mathcal{U}/\mathbb{A}^1}$ is locally free, an element of $\mathcal{M}$ is determined by the restriction to $\mathcal{U}\backslash X_0$. Let $0 \neq \gamma \in \mathcal{M}$. Under the marking, $\Omega^2_{(\mathcal{U}\backslash X_0)/\mathbb{G}_m}$ is the pullback of $\Omega^2_{X_{\mathrm{reg}}}$ under the projection $X_{\mathrm{reg}} \times \mathbb{G}_m \to X_{\mathrm{reg}}$. Since $\mathbb{G}_m = \mathrm{Spec}\, \mathbb{C}[t, t^{-1}]$ is affine, we may write

$$\gamma|_{\mathcal{U}\backslash X_0} = \sum_{j \in \mathbb{Z}} t^j \theta_j,$$

where $\theta_j \in H^0(X_{\mathrm{reg}}, \Omega^2_{X_{\mathrm{reg}}}) = H^0(X, \Omega^{[2]}_X)$ and only finitely many $\theta_j$ are non-zero. Since $T$ fixes $t$, any $\theta_j$ has $\zeta$-degree $r$, and since $d_{\mathrm{rel}}(t^j\theta_j) = t^j d\theta_j$, any $\theta_j$ is closed. Thus $\theta_j \in \mathcal{C}_r$ for any $j$.

We show that $t^j\theta_j \in \mathcal{M}$ for any $j$. Since $\Lambda$ acts on $\mathcal{X}$ over the action of $\Lambda$ on $\mathbb{A}^1$, $\Lambda$ preserves $\mathcal{U}$. Since the actions of $\Lambda$ and $T$ commute and $\lambda^*$ commutes with $d_{\mathrm{rel}}$, we have $\lambda^*\mathcal{M} = \mathcal{M}$ for $\lambda \in \Lambda$. Under the marking, $\Lambda$ acts trivially on $X$ and with weight $-1$ on $t$, so $t^j\theta_j$ is a $\Lambda$-eigenvector of weight $-j$. Let $V \subset \mathcal{M}$ be the subspace spanned by the $\lambda^*\gamma$ with $\lambda \in \Lambda$. Then $V$ is $\Lambda$-stable and $V \subset \bigoplus_j \mathbb{C}t^j\theta_j$, so $\dim V < +\infty$. By [Bor91, Chapter III, §8.4, Proposition], $V$ is the direct sum of the subspaces $V \cap \mathbb{C}t^j\theta_j$. Since $\gamma \in V$, we have $t^j\theta_j \in V \subset \mathcal{M}$ for any $j$.

For $0 \neq \theta \in \mathcal{C}_r$ and $j \in \mathbb{Z}$, the form $t^j\theta$ is closed of $\zeta$-degree $r$, and $\mathrm{ord}_{X_0}(t^j\theta) = j + b(\theta)$. By **Step 1**(a), $t^j\theta \in \mathcal{M}$ if and only if $j + b(\theta) \geq 0$, that is, if and only if $\theta \in F^{-j}$. Thus $\theta_j \in F^{-j}$ for any $j$, and we may write

$$t^j\theta_j = \sum_{i : b_i \geq -j} c_{ij} t^{j+b_i} \cdot t^{-b_i}\alpha_i$$

with $c_{ij} \in \mathbb{C}$ and $j + b_i \geq 0$. Summing over $j$, we obtain $\gamma \in \sum_i \mathbb{C}[t]t^{-b_i}\alpha_i$. Conversely, $t^{-b_i}\alpha_i \in \mathcal{M}$ by (1). Since $\alpha_1, \ldots, \alpha_N$ are linearly independent over $\mathbb{C}$, the forms $t^{-b_1}\alpha_1, \ldots, t^{-b_N}\alpha_N$ are linearly independent over $\mathbb{C}[t]$ in $\mathcal{C}_r \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}]$. Thus

$$\mathcal{M} = \bigoplus_{i=1}^{N} \mathbb{C}[t]t^{-b_i}\alpha_i.$$

This proves (2).

**Step 4.** In this step we prove (3).

Since $\sigma$ is non-degenerate on $X_{\mathrm{reg}}$, $\nu$ vanishes nowhere on $X_{\mathrm{reg}}$, so $\nu$ is a nowhere vanishing relative top form on $\mathcal{U}\backslash X_0 = X_{\mathrm{reg}} \times \mathbb{G}_m$. By **Step 1**, $a = \mathrm{ord}_{X_0}(\nu)$. Since $\mathcal{U}$ is smooth over $\mathbb{A}^1$ of relative dimension $2n$, $\Omega^{2n}_{\mathcal{U}/\mathbb{A}^1}$ is a line bundle, and $t^{-a}\nu$ is a rational section of $\Omega^{2n}_{\mathcal{U}/\mathbb{A}^1}$. The divisor of $t^{-a}\nu$ is supported on $(X_0)_{\mathrm{reg}}$, and the coefficient of $(X_0)_{\mathrm{reg}}$ in it is $\mathrm{ord}_{X_0}(t^{-a}\nu) = 0$. Thus the divisor of $t^{-a}\nu$ is zero. By [Har77, Proposition II.6.3A], $t^{-a}\nu$ and the rational section $(t^{-a}\nu)^{-1}$ of the dual line bundle are regular on $\mathcal{U}$. Thus $t^{-a}\nu$ vanishes nowhere on $\mathcal{U}$. This proves (3).

**Step 5.** In this step we prove (4).

Since $T$ fixes $t$, $\mathrm{eu}(t) = 0$, so eu is a vertical vector field on $\mathcal{U}$, and the contraction $i_{\mathrm{eu}}$ acts on $\Omega^\bullet_{\mathcal{U}/\mathbb{A}^1}$. Let $L_{\mathrm{eu}}$ be the derivative at $\lambda = 1$ of $\zeta(\lambda)^*$ acting on $\Omega^\bullet_{\mathcal{U}/\mathbb{A}^1}$. Since $\zeta(\lambda)^*$ commutes with $d_{\mathrm{rel}}$, so does $L_{\mathrm{eu}}$. Since $d^2_{\mathrm{rel}} = 0$, the operator $d_{\mathrm{rel}} i_{\mathrm{eu}} + i_{\mathrm{eu}} d_{\mathrm{rel}}$ also commutes with $d_{\mathrm{rel}}$. Both $L_{\mathrm{eu}}$ and $d_{\mathrm{rel}} i_{\mathrm{eu}} + i_{\mathrm{eu}} d_{\mathrm{rel}}$ are derivations of degree 0 of $\Omega^\bullet_{\mathcal{U}/\mathbb{A}^1}$, and both send a function $f$ to $\mathrm{eu}(f)$ and $d_{\mathrm{rel}} f$ to $d_{\mathrm{rel}}(\mathrm{eu}(f))$. Since functions and their relative differentials generate $\Omega^\bullet_{\mathcal{U}/\mathbb{A}^1}$ locally, we have

$$L_{\mathrm{eu}} = d_{\mathrm{rel}} i_{\mathrm{eu}} + i_{\mathrm{eu}} d_{\mathrm{rel}}$$

on $\Omega^\bullet_{\mathcal{U}/\mathbb{A}^1}$ (cf. [KN63, Chapter I, Proposition 3.10] for the absolute version). Let $\gamma$ be a closed relative 2-form on $\mathcal{U}$ of $\zeta$-degree $r$. Then $\zeta(\lambda)^*\gamma = \lambda^r \gamma$ for $\lambda \in \mathbb{C}^*$, so $L_{\mathrm{eu}}\gamma = r\gamma$. Since $d_{\mathrm{rel}}\gamma = 0$, we obtain

$$r\gamma = L_{\mathrm{eu}}\gamma = d_{\mathrm{rel}}(i_{\mathrm{eu}}\gamma) + i_{\mathrm{eu}}(d_{\mathrm{rel}}\gamma) = d_{\mathrm{rel}}(i_{\mathrm{eu}}\gamma).$$

Since $r > 0$, (4) follows. □

4.2. **Parallel two-forms.** In this subsection, we prove that grading-invariant holomorphic endomorphisms of the tangent bundle of the punctured regular locus of a cone are algebraic (Lemma 4.2), which is used for the reconstruction in Subsection 4.3, and that under the conditions of Set-up 2.31, any reflexive 2-form $\sigma$ on $X$ with $L_{\rho\partial_\rho}\sigma = 2\sigma$ on $M$ is parallel (Proposition 4.3).

The orthogonal projector onto a parallel subbundle is a priori only a holomorphic section over the punctured regular locus, whereas the induction and the splittings in Subsection 4.3 take place in algebraic reflexive sheaves. The grading leaves only finitely many terms in the weight-zero part of a formal expansion at the vertex, so the weight-zero part is algebraic.

**Lemma 4.2.** *Let $X = \operatorname{Spec} R$ be a normal affine variety of dimension at least 2 with a good positive grading induced by a $\mathbb{G}_m$-action, with vertex $o$, and let $M := X_{\mathrm{reg}}\backslash\{o\}$. Let $\varpi$ be a holomorphic section of $\mathcal{H}om(T_M, T_M)$ on $M$ which is invariant under the $\mathbb{G}_m$-action. Then $\varpi$ is the restriction to $M$ of a unique section of the coherent sheaf $\mathcal{H}om(T_X, T_X)$ on $X$, where $T_X$ is the reflexive tangent sheaf.*

*Proof.* The proof is long, so we divide it into several steps. Let $\mathcal{H} := \mathcal{H}om(T_X, T_X)$, and let $\mathfrak{m} := \bigoplus_{j>0} R_j$ be the maximal ideal of $o$.

**Step 1.** In this step we write $\mathcal{H}$ as the kernel of a morphism of free sheaves and extend $\varpi$ to a section of $\mathcal{H}^{\mathrm{an}}$ on $X^{\mathrm{an}}$.

Since $X$ is affine, there exist exact sequences

$$\mathcal{O}_X^b \xrightarrow{\delta_1} \mathcal{O}_X^a \to \Omega_X \to 0 \quad \text{and} \quad \mathcal{O}_X^d \xrightarrow{\delta_2} \mathcal{O}_X^c \to T_X \to 0.$$

Since $T_X = \mathcal{H}om(\Omega_X, \mathcal{O}_X)$, dualizing the first sequence implies that

$$0 \to T_X \to \mathcal{O}_X^a \xrightarrow{{}^t\delta_1} \mathcal{O}_X^b$$

is exact, and applying $\mathcal{H}om(-,T_X)$ to the second sequence implies that $0\to\mathcal{H}\to T_X^{\oplus c}\to T_X^{\oplus d}$ is exact. Thus

$$0\to\mathcal{H}\to\mathcal{O}_X^a\otimes\mathcal{O}_X^c\xrightarrow{\psi}\left(\mathcal{O}_X^b\otimes\mathcal{O}_X^c\right)\oplus\left(\mathcal{O}_X^a\otimes\mathcal{O}_X^d\right),\qquad \psi:=\left({}^t\delta_1\otimes\mathrm{id}_{\mathcal{O}_X^c},\ \mathrm{id}_{\mathcal{O}_X^a}\otimes{}^t\delta_2\right), \tag{4.1}$$

is exact. In particular, locally on $X$ the reflexive sheaf $\mathcal{H}$ is the kernel of a morphism $\mathcal{O}^{ac}\to\mathcal{O}^{bc+ad}$ of free sheaves.

By [Ser56, §2, no. 6, Proposition 3 and Corollary 1], $\mathcal{O}_{X,x}^{\mathrm{an}}$ is flat over $\mathcal{O}_{X,x}$ for any $x\in X$. Thus the analytification

$$0\to\mathcal{H}^{\mathrm{an}}\to\left(\mathcal{O}_{X^{\mathrm{an}}}\right)^{ac}\xrightarrow{\psi}\left(\mathcal{O}_{X^{\mathrm{an}}}\right)^{bc+ad}$$

of (4.1) is exact. Since $\mathcal{H}^{\mathrm{an}}|_M=\mathcal{H}om(T_M,T_M)$, the section $\varpi$ is a tuple $(\varpi_1,\dots,\varpi_{ac})$ of holomorphic functions on $M$ with $\psi(\varpi_1,\dots,\varpi_{ac})=0$.

Since $X$ is normal and $\dim X\geq 2$, the complex space $X^{\mathrm{an}}$ is normal [Gro03, Exposé XII, Proposition 2.1] and $X^{\mathrm{an}}\backslash M=\operatorname{Sing}X\cup\{o\}$ is an analytic subset of codimension at least 2. By the second Riemann removable singularity theorem [GR84, Chapter 7], each $\varpi_i$ extends to a holomorphic function $\tilde{\varpi}_i$ on $X^{\mathrm{an}}$. Since $M$ is dense in $X^{\mathrm{an}}$, we have $\psi(\tilde{\varpi}_1,\dots,\tilde{\varpi}_{ac})=0$. Thus $\tilde{\varpi}:=(\tilde{\varpi}_1,\dots,\tilde{\varpi}_{ac})$ is a section of $\mathcal{H}^{\mathrm{an}}$ on $X^{\mathrm{an}}$ with $\tilde{\varpi}|_M=\varpi$, and any section of $\mathcal{H}^{\mathrm{an}}$ on $X^{\mathrm{an}}$ whose restriction to $M$ is $\varpi$ coincides with $\tilde{\varpi}$. For any $\lambda\in\mathbb{C}^*$, the section $\lambda\cdot\tilde{\varpi}$ of $\mathcal{H}^{\mathrm{an}}$ restricts to $\lambda\cdot\varpi=\varpi$ on $M$, hence $\lambda\cdot\tilde{\varpi}=\tilde{\varpi}$.

**Step 2.** In this step we show that the image of the germ of $\tilde{\varpi}$ at $o$ in the formal completion is an algebraic section of $\mathcal{H}$ of weight zero.

Let $Q:=H^0(X,\mathcal{H})$. The $\mathbb{G}_m$-action on $\mathcal{H}$ induces a grading $Q=\bigoplus_{j\in\mathbb{Z}}Q_j$, where $\lambda\in\mathbb{C}^*$ acts on $Q_j$ by multiplication by $\lambda^j$ and $R_iQ_j\subset Q_{i+j}$, and $Q$ is a finitely generated graded $R$-module. Choose homogeneous generators of $Q$ of degrees $e_1,\dots,e_s$, and homogeneous generators of $\mathfrak{m}$ of degrees in $\{1,\dots,\ell\}$. Let $e':=\min_i e_i$ and $e'':=\max_i e_i$. Any element of $R_j$ with $j\geq k\ell$ is a sum of monomials in at least $k$ generators of $\mathfrak{m}$, so

$$\bigoplus_{j\geq k\ell+e''}Q_j\subset\mathfrak{m}^kQ\subset\bigoplus_{j\geq k+e'}Q_j$$

for any integer $k\geq 1$. Thus the $\mathfrak{m}$-adic completion of $Q$ is

$$\widehat{Q}:=\varprojlim_k Q/\mathfrak{m}^kQ=\prod_{j\in\mathbb{Z}}Q_j,$$

and $Q_j=0$ for $j<e'$.

By [Ser56, §2, no. 6, Proposition 3], the natural morphism $\widehat{\mathcal{O}_{X,o}}\to\widehat{\mathcal{O}_{X,o}^{\mathrm{an}}}$ of completions is an isomorphism. Since $\mathcal{H}_o^{\mathrm{an}}=\mathcal{H}_o\otimes_{\mathcal{O}_{X,o}}\mathcal{O}_{X,o}^{\mathrm{an}}$, and since the completion of a finite module $L$ over a Noetherian local ring $A$ is $L\otimes_A\widehat{A}$ [Mat89, Theorem 8.7], we have

$$\widehat{\mathcal{H}_o^{\mathrm{an}}}=\mathcal{H}_o\otimes_{\mathcal{O}_{X,o}}\widehat{\mathcal{O}_{X,o}}=\widehat{Q}.$$

Let $\hat{\varpi}=(\hat{\varpi}_j)_{j\in\mathbb{Z}}\in\prod_j Q_j$ be the image of the germ of $\tilde{\varpi}$ at $o$. Since $o$ is fixed by $\mathbb{G}_m$ and $\mathcal{H}$ is $\mathbb{G}_m$-equivariant, the morphism $\mathcal{H}_o^{\mathrm{an}}\to\widehat{Q}$ is $\mathbb{C}^*$-equivariant, where $\lambda\in\mathbb{C}^*$ acts on $\widehat{Q}$ by $\lambda^j$ on $Q_j$. By **Step 1**, $\lambda\cdot\tilde{\varpi}=\tilde{\varpi}$, so $\lambda^j\hat{\varpi}_j=\hat{\varpi}_j$ for any $\lambda\in\mathbb{C}^*$ and any $j$. Thus $\hat{\varpi}_j=0$ for $j\neq 0$, and

$$\hat{\varpi}=\hat{\varpi}_0\in Q_0.$$

Let $\varpi':=\hat{\varpi}_0$. Then $\varpi'$ is a $\mathbb{G}_m$-invariant section of $\mathcal{H}$ on $X$.

**Step 3.** In this step we show that $\tilde{\varpi}=\varpi'$ on $X^{\mathrm{an}}$ and conclude the proof.

Since $\mathcal{H}_o^{\mathrm{an}}$ is a finite module over the Noetherian local ring $\mathcal{O}_{X,o}^{\mathrm{an}}$, the morphism $\mathcal{H}_o^{\mathrm{an}}\to\widehat{\mathcal{H}_o^{\mathrm{an}}}=\widehat{Q}$ is injective by the Krull intersection theorem [Mat89, Theorem 8.10]. By **Step 2**, the germs of $\tilde{\varpi}$ and $\varpi'$ at $o$ both map to $\varpi'\in Q_0\subset\widehat{Q}$. Thus the germs of $\tilde{\varpi}$ and $\varpi'$ at $o$ coincide.

Let $U\subset X^{\mathrm{an}}$ be the set of points at which the germs of $\tilde{\varpi}$ and $\varpi'$ coincide. Then $U$ is open and contains $o$. Since $\tilde{\varpi}$ and $\varpi'$ are $\mathbb{G}_m$-invariant, $U$ is $\mathbb{C}^*$-stable. Since the grading is positive,

for any $x \in X$ the closure of $\mathbb{C}^* \cdot x$ contains $o$, so $\mathbb{C}^* \cdot x$ intersects $U$, and $x \in U$. Thus $U = X^{\mathrm{an}}$, and $\varpi = \tilde{\varpi}|_M = \varpi'|_M$.

Finally, by (4.1), $\mathcal{H}$ is a subsheaf of $\mathcal{O}_X^{ac}$, and $M$ is dense in $X$. Thus a section of $\mathcal{H}$ on $X$ is determined by the restriction to $M$, and $\varpi'$ is the unique section of $\mathcal{H}$ with $\varpi'|_M = \varpi$. □

We prove parallelism for any reflexive 2-form $\sigma$ with $L_{\rho\partial_\rho}\sigma = 2\sigma$ on $M$, degenerate forms included, since each pairing in the reconstruction of Subsection 4.3 is such a form, possibly degenerate. The link $N$ is not assumed to be complete, and for such a form $\sigma$ the function $\log(1 + |\sigma|_g^2)$ is only known to be exponentially integrable on $N$, so we use the factor $\log^2(e + t)$ of $\Phi$ in the $\Phi$-cutoffs (Remark 2.30).

**Proposition 4.3.** *Notation and conditions as in Set-up 2.31. Then any reflexive 2-form $\sigma$ on $X$ with $L_{\rho\partial_\rho}\sigma = 2\sigma$ on $M$ is parallel with respect to $g$ on $M$.*

*Proof.* The proof is long, so we divide it into several steps. Let $u := |\sigma|_g^2$, a smooth function on $M$.

**Step 1.** In this step we show that $u$ is the pullback of a function on $N$, still denoted by $u$, and that

$$\int_N u d\,\mathrm{Vol}_h < +\infty.$$

Since $g$ is a Kähler cone metric with radius function $\rho$, the Kähler form of $g$ is $\frac{\sqrt{-1}}{2}\partial\bar{\partial}\rho^2 = dd^c\varphi$ on $M$ [CS15, Section 3, p. 8], where $\varphi := \frac{1}{2}\rho^2$. When $g$ is the metric of Theorem 2.27, the equality is also part of Theorem 2.27(1). By Set-up 2.31(2), $\varphi$ extends to a locally bounded plurisubharmonic function on $X$, still denoted by $\varphi$.

Let $\pi : Z \to X$ be a log resolution which is an isomorphism over $X_{\mathrm{reg}}$, and let $E := \pi^{-1}(X \backslash M)$. Since $X$ is klt, $\pi_*\Omega_Z^2$ is reflexive [GKKP11, Theorem 1.4], hence $\pi_*\Omega_Z^2 = \Omega_X^{[2]}$. Thus $\pi^*\sigma$ extends to a holomorphic 2-form $\sigma_Z$ on $Z$. Since $\rho$ and $\pi$ are proper, $\pi^{-1}(\rho^{-1}([\frac{1}{2}, 2]))$ is compact. Cover it by finitely many coordinate balls $\mathbb{B}_1, \dots, \mathbb{B}_m$, where each $\mathbb{B}_k$ is relatively compact in a coordinate ball $\mathbb{B}'_k$ with coordinates $z = (z_1, \dots, z_d)$, and $\mathbb{B}'_k$ is relatively compact in $Z$. Let $v := \pi^*\varphi$. Then $v$ is smooth and plurisubharmonic on $\mathbb{B}'_k \backslash E$ and bounded on $\mathbb{B}'_k$. Since $E$ is an analytic subset, it is a closed pluripolar set, so $v|_{\mathbb{B}'_k \backslash E}$ extends uniquely to a plurisubharmonic function $\tilde{v}$ on $\mathbb{B}'_k$ [Dem12, Chapter I, Theorem 5.24]. The function $\tilde{v}$ is bounded and differs from $v$ at most on $E$, a set of measure zero, and we still denote $\tilde{v}$ by $v$. The form $\sigma_Z$ has bounded coefficients on each $\mathbb{B}'_k$.

We show that for each $k$,

$$\int_{\mathbb{B}_k \backslash E} (dd^c|z|^2)^2 \wedge (dd^c v)^{d-2} < +\infty. \tag{4.2}$$

If $d = 2$, then $(dd^c v)^{d-2} = 1$ and (4.2) is immediate. Suppose that $d \geq 3$. Choose a coordinate ball $\mathbb{B}''_k$ with $\mathbb{B}_k \Subset \mathbb{B}''_k \Subset \mathbb{B}'_k$, and for $0 < \epsilon \ll 1$ let $v_\epsilon$ be the convolution of $v$ with a standard smoothing kernel of radius $\epsilon$ on $\mathbb{B}''_k$. Then $v_\epsilon$ is smooth and plurisubharmonic on $\mathbb{B}''_k$, $\sup_{\mathbb{B}''_k}|v_\epsilon| \leq \sup_{\mathbb{B}'_k}|v|$, and $v_\epsilon \to v$ in the $C^\infty$ topology on compact subsets of $\mathbb{B}''_k \backslash E$. By the Chern–Levine–Nirenberg inequality [CLN69], there exists $C_k > 0$, depending only on $\mathbb{B}_k \Subset \mathbb{B}''_k$, such that

$$\int_{\mathbb{B}_k} (dd^c|z|^2)^2 \wedge (dd^c v_\epsilon)^{d-2} \leq C_k \Big( \sup_{\mathbb{B}'_k} |v| \Big)^{d-2}$$

for $0 < \epsilon \ll 1$. The bound does not depend on $\epsilon$, and the integrands converge pointwise on $\mathbb{B}_k \backslash E$. Thus Fatou's lemma implies (4.2).

Since $\sigma_Z$ has bounded coefficients on $\mathbb{B}'_k$, there exists $C'_k > 0$ such that $C'_k(dd^c|z|^2)^2 - \sigma_Z \wedge \bar{\sigma}_Z$ is a positive $(2,2)$-form on $\mathbb{B}_k$. Since $(dd^c v)^{d-2}$ is a strongly positive form on $\mathbb{B}_k \backslash E$ as a product of positive $(1,1)$-forms [Dem12, Chapter III, Corollary 1.9 and Proposition 1.11], and since the

wedge product of a positive form with a strongly positive form is positive [Dem12, Chapter III, Proposition 1.11], the forms

$$\sigma_Z \wedge \bar{\sigma}_Z \wedge (dd^c v)^{d-2} \quad \text{and} \quad \left(C_k'(dd^c|z|^2)^2 - \sigma_Z \wedge \bar{\sigma}_Z\right) \wedge (dd^c v)^{d-2}$$

are positive on $\mathbb{B}_k \backslash E$. Thus (4.2) implies

$$0 \leq \int_{\mathbb{B}_k \backslash E} \sigma_Z \wedge \bar{\sigma}_Z \wedge (dd^c v)^{d-2} \leq C_k' \int_{\mathbb{B}_k \backslash E} (dd^c|z|^2)^2 \wedge (dd^c v)^{d-2} < +\infty.$$

A direct calculation in unitary coordinates at a point shows that

$$u(dd^c\varphi)^d = c_d \sigma \wedge \bar{\sigma} \wedge (dd^c\varphi)^{d-2} \quad \text{on } M$$

for a constant $c_d > 0$ depending only on $d$. Since $\pi$ is an isomorphism over $M$, $\pi^{-1}(M) = Z \backslash E$, and the balls $\mathbb{B}_k$ cover $\pi^{-1}(\rho^{-1}([\frac{1}{2}, 2]))$, we obtain

$$\int_{\rho^{-1}([\frac{1}{2},2]) \cap M} u(dd^c\varphi)^d \leq c_d \sum_{k=1}^{m} \int_{\mathbb{B}_k \backslash E} \sigma_Z \wedge \bar{\sigma}_Z \wedge (dd^c v)^{d-2} < +\infty. \tag{4.3}$$

Let $\phi_s$ be the flow of $\rho\partial_\rho$ on $M = (0, +\infty) \times N$, that is, $\phi_s(\rho, y) = (e^s\rho, y)$. Since $g = d\rho^2 + \rho^2 h$ and $L_{\rho\partial_\rho}\sigma = 2\sigma$, we have $\phi_s^* g = e^{2s} g$ and $\phi_s^*\sigma = e^{2s}\sigma$, hence

$$u \circ \phi_s = |\phi_s^*\sigma|^2_{\phi_s^* g} = e^{4s} \cdot e^{-4s}|\sigma|_g^2 = u.$$

Thus $u$ is the pullback of a function on $N$. Since $(dd^c\varphi)^d$ is a positive constant multiple of

$$d\operatorname{Vol}_g = \rho^{2d-1} d\rho \wedge d\operatorname{Vol}_h,$$

(4.3) implies

$$\int_{1/2}^{2} \rho^{2d-1} d\rho \cdot \int_N u \, d\operatorname{Vol}_h < +\infty,$$

so $\int_N u \, d\operatorname{Vol}_h < +\infty$.

**Step 2.** In this step we show that $H := \log(1 + u)$, a smooth non-negative function on $N$, satisfies $\Delta_h H \geq 0$ and $\int_N e^H d\operatorname{Vol}_h < +\infty$.

Let $D$ be the Chern connection of the holomorphic vector bundle $\Lambda^2 T_M^*$ with the hermitian metric induced by $g$. Since $g$ is Kähler, $D$ is induced by the Levi-Civita connection $\nabla$ of $g$, and since $\sigma$ is holomorphic, $\nabla\sigma = D\sigma$ is of type $(1, 0)$. In local holomorphic coordinates, let

$$\Delta^c f := \sum_{i,j} g^{i\bar{j}} \partial_i \partial_{\bar{j}} f \quad \text{and} \quad |D\sigma|_g^2 := \sum_{i,j} g^{i\bar{j}} \langle D_i\sigma, D_j\sigma\rangle_g$$

for a smooth function $f$ on $M$, where $D_i$ is the covariant derivative in the direction of $\partial/\partial z_i$. Since $g$ is Kähler, $\Delta_g$ is a positive constant multiple of $\Delta^c$. By the Bochner identity [KW70, pp. 1–2],

$$\Delta^c u = |D\sigma|_g^2 - (\Theta\sigma, \sigma)_g,$$

where $\Theta$ is the mean curvature of $\Lambda^2 T_M^*$. Since $g$ is Ricci-flat, $\Theta = 0$, so $\Delta^c u = |D\sigma|_g^2$. Since $D$ is compatible with the metric and $\sigma$ is holomorphic, $\partial_i u = \langle D_i\sigma, \sigma\rangle_g$, and the Cauchy–Schwarz inequality implies

$$|\partial u|_g^2 := \sum_{i,j} g^{i\bar{j}} \partial_i u \partial_{\bar{j}} u \leq u|D\sigma|_g^2.$$

Thus

$$\Delta^c \log(1 + u) = \frac{\Delta^c u}{1 + u} - \frac{|\partial u|_g^2}{(1 + u)^2} \geq \frac{|D\sigma|_g^2}{1 + u} - \frac{u|D\sigma|_g^2}{(1 + u)^2} = \frac{|D\sigma|_g^2}{(1 + u)^2} \geq 0,$$

so $\Delta_g \log(1 + u) \geq 0$ on $M$. On $M = (0, +\infty) \times N$ with $g = d\rho^2 + \rho^2 h$ and $\dim_{\mathbb{R}} N = 2d - 1$,

$$\Delta_g f = \partial_\rho^2 f + \frac{2d - 1}{\rho} \partial_\rho f + \rho^{-2} \Delta_h f$$

for any smooth function $f$. By **Step 1**, $\log(1+u)$ is the pullback of $H$, so $\Delta_g \log(1+u) = \rho^{-2}\Delta_h H$. Thus $\Delta_h H \geq 0$ on $N$. Moreover, $e^H = 1+u$, so **Step 1** and Set-up 2.31(3) imply

$$\int_N e^H d\operatorname{Vol}_h = \operatorname{Vol}_h(N) + \int_N u d\operatorname{Vol}_h < +\infty.$$

**Step 3.** In this step we show that

$$\lim_{j\to+\infty} \int_N H^2 |d\beta_j|_h^2 d\operatorname{Vol}_h = 0, \tag{4.4}$$

where $(\beta_j)_{j\geq 1}$ is a sequence of $\Phi$-cutoffs on $N$ as in Set-up 2.31(4).

Let $a_j := |d\beta_j|_h$ and $N_j := \{y \in N \mid H(y) \leq 4\log(e + a_j(y))\}$. On $N_j$,

$$H^2 a_j^2 \leq 16 a_j^2 \log^2(e + a_j) = 16\Phi(a_j).$$

Thus Definition 2.29 implies

$$\int_{N_j} H^2 a_j^2 d\operatorname{Vol}_h \leq 16 \int_N \Phi(a_j) d\operatorname{Vol}_h \to 0.$$

On $N\backslash N_j$, we have $a_j < e + a_j < e^{H/4}$, so $a_j^2 < e^{H/2}$ and

$$H^2 a_j^2 \leq H^2 e^{H/2} \leq 16 e^{-2} e^H,$$

since $\sup_{t\geq 0} t^2 e^{-t/2} = 16e^{-2}$. By **Step 2**, $16e^{-2}e^H$ is integrable on $N$, and it does not depend on $j$. For any $y \in N$, $\beta_j = 1$ on a compact neighborhood of $y$ for $j \gg 0$, so $a_j(y) = 0$ for $j \gg 0$. By the dominated convergence theorem,

$$\int_{N\backslash N_j} H^2 a_j^2 d\operatorname{Vol}_h \to 0.$$

Thus (4.4) holds.

**Step 4.** In this step we show that $H$ is constant and conclude the proof.

Since $\beta_j^2 H$ is smooth with compact support in $N$, integration by parts is valid on $N$, although $N$ need not be complete. By **Step 2**,

$$0 \leq \int_N \beta_j^2 H \Delta_h H d\operatorname{Vol}_h = -\int_N \beta_j^2 |dH|_h^2 d\operatorname{Vol}_h - 2\int_N \beta_j H \langle d\beta_j, dH\rangle_h d\operatorname{Vol}_h.$$

By the Cauchy–Schwarz inequality,

$$\int_N \beta_j^2 |dH|_h^2 d\operatorname{Vol}_h \leq 2\int_N \beta_j H |d\beta_j|_h |dH|_h d\operatorname{Vol}_h \leq \frac{1}{2}\int_N \beta_j^2 |dH|_h^2 d\operatorname{Vol}_h + 2\int_N H^2 |d\beta_j|_h^2 d\operatorname{Vol}_h.$$

Thus

$$\int_N \beta_j^2 |dH|_h^2 d\operatorname{Vol}_h \leq 4\int_N H^2 |d\beta_j|_h^2 d\operatorname{Vol}_h,$$

which tends to 0 by (4.4). On any compact subset of $N$, $\beta_j = 1$ for $j \gg 0$, so $dH = 0$ there. Thus $dH = 0$ on $N$. Since $N$ is connected, $H$ is constant, hence $u$ is constant on $M = (0, +\infty) \times N$. By **Step 2**,

$$|D\sigma|_g^2 = \Delta^c u = 0.$$

Thus $\nabla\sigma = D\sigma = 0$ on $M$, that is, $\sigma$ is parallel with respect to $g$ on $M$. □

4.3. **Reconstruction of a parallel symplectic form.** In this subsection, we prove Theorem 4.4. Under the hypotheses of Theorem 4.4, the restriction of $t^{-b(\omega)}\omega$ to $(X_0)_{\mathrm{reg}}$ in Definition-Lemma 4.1(1) can be degenerate. Theorem 4.4 assembles a parallel symplectic form $\tau$ on $X_0$ from the $T$-eigencomponents $\omega_\alpha$ of $\omega$ with $\langle\alpha,\xi\rangle = 2$, where $\xi$ is the Reeb vector of the theorem, by filtering relative tangent vectors along $X_0$ according to the power of $t$ that divides their pairings with $\sum_{\langle\alpha,\xi\rangle=2}\omega_\alpha$. The form $\tau$ need not be a limit of $\omega$.

**Theorem 4.4.** *Notation and conditions as in Set-up 2.15. Assume that* $\dim X = 2n$ *with* $n \geq 2$*, that* $X$ *is a symplectic variety whose symplectic form* $\omega$ *is algebraic of* $\zeta$*-degree* $r > 0$*, that* $X_0$ *is integral and normal with klt singularities, that the* $\zeta$*-grading of* $X_0$ *is good with vertex* $o$*, and that* $T$ *acts faithfully on* $X_0$*. Since the* $\zeta$*-degree* $nr$ *part of* $H^0(X, \Omega_X^{[2n]})$ *is* $\mathbb{C}\omega^n$*, the torus* $T$ *acts on* $\omega^n$ *by a character* $\kappa$*. Let* $\xi$ *be a Reeb vector of* $T$ *on* $X_0$ *such that* $(X_0,\xi)$ *is a K-semistable Fano cone with* $A_{X_0}(\mathrm{wt}_\xi) = 2n$*, and let* $S \subset T$ *be the smallest subtorus with* $\xi \in \mathrm{Hom}(\mathbb{G}_m, S)\otimes\mathbb{R}$*. Assume that* $M_0 := (X_0)_{\mathrm{reg}}\backslash\{o\}$ *carries a smooth Ricci-flat Kähler cone metric* $g = d\rho^2 + \rho^2 h$ *with connected link such that* $\rho\partial_\rho f = \langle\alpha,\xi\rangle f$ *for every regular function* $f$ *on* $X_0$ *of* $T$*-character* $\alpha$*, and such that every reflexive 2-form* $\beta$ *on* $X_0$ *with* $L_{\rho\partial_\rho}\beta = 2\beta$ *is parallel on* $M_0$*. Then* $X_0$ *carries a closed reflexive 2-form* $\tau$ *which is non-degenerate on* $(X_0)_{\mathrm{reg}}$*, parallel on* $M_0$*, and extends regularly to a resolution of* $X_0$*, and which is an* $S$*-eigenvector of a character* $\chi$ *with* $n\chi = \kappa|_S$ *and* $\langle\chi,\xi\rangle = 2$*.*

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step, we construct the sum $\omega_*$ of the $T$-eigencomponents $\omega_\alpha$ of $\omega$ with $\langle\alpha,\xi\rangle = 2$, we show that $\omega_*$ is an $S$-eigenvector of a character $\chi$ with $n\chi = \kappa|_S$ and $\langle\chi,\xi\rangle = 2$, and we choose a cocharacter $\zeta'$ of $S$ inducing a good positive grading of $X_0$.

Let $\mathcal{U}\subset\mathcal{X}$ be the smooth locus of $\mathcal{X}\to\mathbb{A}^1$. Since $\mathcal{X}\to\mathbb{A}^1$ is flat, a point of $X_0$ belongs to $\mathcal{U}$ if and only if $X_0$ is smooth at the point [Gro67, Théorème 17.5.1]. Thus $\mathcal{U}\cap X_0 = (X_0)_{\mathrm{reg}}$. Let $\nu := \omega^n/n!$, and let $a$ be the integer of Definition-Lemma 4.1(3) for $\sigma = \omega$, so that $t^{-a}\nu$ extends to a regular relative top form on $\mathcal{U}$ with non-zero restriction to $(X_0)_{\mathrm{reg}}$. By Definition-Lemma 4.1(3), $t^{-a}\nu$ vanishes nowhere on $\mathcal{U}$. Thus $\nu_0 := (t^{-a}\nu)|_{(X_0)_{\mathrm{reg}}}$ is a nowhere vanishing $2n$-form on $(X_0)_{\mathrm{reg}}$, and $\nu_0$ generates $\Omega_{X_0}^{[2n]}$. Since $T$ fixes $t$ and the marking is $T$-equivariant, $\nu_0$ is a $T$-eigenvector of character $\kappa$. By Lemma 2.14,

$$\langle\kappa,\xi\rangle = A_{X_0}(\mathrm{wt}_\xi) = 2n.$$

Let $\mathcal{C}_r$ be the space of closed reflexive 2-forms on $X$ of $\zeta$-degree $r$. Since $\zeta$ is a cocharacter of the torus $T$, the action of $T$ preserves $\mathcal{C}_r$, and $\dim\mathcal{C}_r < +\infty$ by Definition-Lemma 4.1(1). Thus

$$\omega = \sum_\alpha \omega_\alpha,$$

where $\alpha$ runs through finitely many characters of $T$ and $\omega_\alpha\in\mathcal{C}_r$ is a $T$-eigenvector of character $\alpha$. Let $\alpha$ be a character with $\omega_\alpha\neq 0$. By Definition-Lemma 4.1(1), $t^{-b(\omega_\alpha)}\omega_\alpha$ extends to a regular relative 2-form on $\mathcal{U}$ with non-zero restriction to $(X_0)_{\mathrm{reg}}$. Since $X_0$ is normal, the restriction extends to a non-zero reflexive 2-form $\beta_\alpha$ on $X_0$, and $\beta_\alpha$ is a $T$-eigenvector of character $\alpha$ because $T$ fixes $t$. Since $(X_0,\xi)$ is K-semistable, $\mathrm{wt}_\xi$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_0,o}$ (Theorem 2.21). By Theorem 2.13(1) and (2), applied to $\beta_\alpha$ with $d = 2n$ and $p = 2$,

$$1 = \frac{A_{X_0}(\mathrm{wt}_\xi)}{2n} \leq \frac{A_{\beta_\alpha}(\mathrm{wt}_\xi)}{2} = \frac{\langle\alpha,\xi\rangle}{2}.$$

Thus $\langle\alpha,\xi\rangle\geq 2$ for any $\alpha$ with $\omega_\alpha\neq 0$.

Let

$$\omega_* := \sum_{\langle\alpha,\xi\rangle=2}\omega_\alpha\in\mathcal{C}_r.$$

We show that $\omega_*^n = \omega^n$. Expand $\omega^n$ as the sum of the products $\omega_{\alpha_1} \wedge \cdots \wedge \omega_{\alpha_n}$. Such a product is a $T$-eigenvector of character $\alpha_1 + \cdots + \alpha_n$, and

$$\langle \alpha_1 + \cdots + \alpha_n, \xi \rangle \geq 2n,$$

with equality if and only if $\langle \alpha_k, \xi \rangle = 2$ for $1 \leq k \leq n$. For a character $\gamma$ of $T$, let $(\omega^n)_\gamma$ be the component of $\omega^n$ of character $\gamma$. Grouping the products by characters, we obtain

$$\omega_*^n = \sum_{\langle \gamma, \xi \rangle = 2n} (\omega^n)_\gamma.$$

Since $\omega^n$ is a $T$-eigenvector of character $\kappa$ and $\langle \kappa, \xi \rangle = 2n$, the right-hand side is $\omega^n$. Thus $\omega_*^n = \omega^n$. In particular, $\omega_*$ is a closed reflexive 2-form of $\zeta$-degree $r$ which is non-degenerate on $X_{\mathrm{reg}}$.

If $\chi'$ is a character of $S$ with $\langle \chi', \xi \rangle = 0$, then the identity component of $\mathrm{Ker}(\chi')$ is a subtorus of $S$ whose real cocharacter space contains $\xi$. By minimality of $S$, the identity component of $\mathrm{Ker}(\chi')$ is $S$, so $\chi' = 0$. Since $\xi \in \mathrm{Hom}(\mathbb{G}_m, S) \otimes \mathbb{R}$, we have $\langle \alpha|_S, \xi \rangle = \langle \alpha, \xi \rangle$ for any character $\alpha$ of $T$. Thus all characters $\alpha$ with $\langle \alpha, \xi \rangle = 2$ restrict to a single character $\chi$ of $S$, the form $\omega_*$ is an $S$-eigenvector of character $\chi$, and $\langle \chi, \xi \rangle = 2$. Since $\omega_*^n = \omega^n \neq 0$ is an $S$-eigenvector of character $n\chi$ and of character $\kappa|_S$, we have $n\chi = \kappa|_S$.

Finally we choose $\zeta'$. Let $f_1, \ldots, f_m$ be non-constant generators of the $\mathbb{C}$-algebra $\mathcal{O}(X_0)$ which are $T$-eigenvectors, of characters $\alpha_1, \ldots, \alpha_m$. Since the $\zeta$-grading of $X_0$ is good, the degree-zero part of $\mathcal{O}(X_0)$ is $\mathbb{C}$, so $\alpha_k \neq 0$. Since $\xi$ is a Reeb vector, $\langle \alpha_k, \xi \rangle > 0$ for $1 \leq k \leq m$. The set

$$\{ y \in \mathrm{Hom}(\mathbb{G}_m, S) \otimes \mathbb{R} \mid \langle \alpha_k, y \rangle > 0 \text{ for } 1 \leq k \leq m \}$$

is an open cone defined by finitely many linear inequalities with rational coefficients, and it contains $\xi$. Thus it contains a cocharacter $\zeta'$ of $S$. Any non-constant monomial in $f_1, \ldots, f_m$ has positive $\zeta'$-degree, so $\zeta'$ induces a good positive grading of $\mathcal{O}(X_0)$ with vertex $o$.

**Step 2.** In this step, we show the following. Let $\beta$ be a reflexive 2-form on $X_0$ which is an $S$-eigenvector of character $\chi$. Then $\beta$ is parallel on $M_0$, $\mathrm{Ker}(\beta)$ is a parallel subbundle of $T_{M_0}$, and the orthogonal projector $\varpi_\beta$ onto $\mathrm{Ker}(\beta)$ is the restriction to $M_0$ of an $S$-invariant idempotent endomorphism of the reflexive tangent sheaf $T_{X_0}$, again denoted by $\varpi_\beta$. In particular, the image and the kernel of $\varpi_\beta$ are reflexive direct summands of $T_{X_0}$.

For $s \in \mathbb{R}$, let $\phi_s, \psi_s \in S$ be the elements with $\alpha(\phi_s) = e^{s\langle \alpha, \xi \rangle}$ and $\alpha(\psi_s) = e^{\sqrt{-1}s\langle \alpha, \xi \rangle}$ for any character $\alpha$ of $S$. Then $\psi_s$ belongs to the maximal compact subtorus $S_c$ of $S$, and $\psi_s = \exp(s\xi)$ under the identification of Definition 2.6. A real vector field on $M_0$ is determined by the values on regular functions, since regular functions on $X_0$ define local holomorphic coordinates on $M_0$. For $f \in \mathcal{O}(X_0)$ of character $\alpha$ and $y \in M_0$, we have

$$\frac{d}{ds}\Big|_{s=0} f(\phi_s \cdot y) = \langle \alpha, \xi \rangle f(y) = (\rho \partial_\rho f)(y).$$

Thus the flow of $\rho\partial_\rho$ at time $s$ is the action of $\phi_s$. Since $\beta$ is an $S$-eigenvector of character $\chi$, $\phi_s^* \beta = e^{s\langle \chi, \xi \rangle} \beta = e^{2s}\beta$, that is, $L_{\rho\partial_\rho}\beta = 2\beta$. By the hypothesis of Theorem 4.4, $\beta$ is parallel on $M_0$.

Since $M_0 \cong (0, +\infty) \times \{\rho = 1\}$ and the link $\{\rho = 1\}$ is connected, $M_0$ is connected. Thus the rank of $\beta$ is constant on $M_0$, and $\mathrm{Ker}(\beta) \subset T_{M_0}$ is a parallel complex subbundle. Hence $\varpi_\beta$ is a parallel complex-linear endomorphism of $T_{M_0}$. On a Kähler manifold, the Levi-Civita connection on the holomorphic tangent bundle is the Chern connection [Huy05, Proposition 4.A.9]. Thus $\bar{\partial}\varpi_\beta$ is the $(0,1)$-part of $\nabla \varpi_\beta = 0$, and $\varpi_\beta$ is holomorphic.

Next we show that $\varpi_\beta$ is $S$-invariant. A direct calculation from $g = d\rho^2 + \rho^2 h$ shows that $\nabla(\rho\partial_\rho) = \mathrm{id}_{T_{M_0}}$. Since the complex structure $I$ is parallel, $\nabla(I\rho\partial_\rho) = I$, which is skew-symmetric, so $I\rho\partial_\rho$ is a Killing vector field. For $f \in \mathcal{O}(X_0)$ of character $\alpha$, $f$ is holomorphic, so

$$(I\rho\partial_\rho)f = \sqrt{-1}(\rho\partial_\rho f) = \sqrt{-1}\langle \alpha, \xi \rangle f = \frac{d}{ds}\Big|_{s=0} f(\psi_s \cdot y).$$

Thus the flow of $I\rho\partial_\rho$ at time $s$ is the action of $\psi_s$, and $\psi_s$ acts on $M_0$ by holomorphic isometries of $g$. Since $\psi_s^*\beta = e^{2\sqrt{-1}s}\beta$, the action of $\psi_s$ preserves $\mathrm{Ker}(\beta)$, hence preserves $\varpi_\beta$. The closure of $\{\psi_s \mid s \in \mathbb{R}\}$ in $S_c$ is a connected compact subgroup, hence a subtorus. The complexification of the closure is an algebraic subtorus of $S$ whose real cocharacter space contains $\xi$, so it is $S$ by minimality of $S$. Thus the closure is $S_c$, and $\varpi_\beta$ is $S_c$-invariant by continuity. Since the action of $S$ and $\varpi_\beta$ are holomorphic, invariance under the Lie algebra of $S_c$ implies invariance under the complexification of the Lie algebra, which is the Lie algebra of $S$. Since $S$ is connected, $\varpi_\beta$ is $S$-invariant.

By **Step 1**, $\zeta'$ induces a good positive grading of $\mathcal{O}(X_0)$ with vertex $o$, and $\varpi_\beta$ is invariant under $\zeta'(\mathbb{G}_m) \subset S$. Since $\dim X_0 = 2n \geq 2$, Lemma 4.2 implies that $\varpi_\beta$ is the restriction to $M_0$ of a unique endomorphism of $T_{X_0}$. By uniqueness, the extension is $S$-invariant. The endomorphism $\varpi_\beta^2 - \varpi_\beta$ of the torsion-free sheaf $T_{X_0}$ vanishes on $M_0$, so $\varpi_\beta^2 = \varpi_\beta$. Thus $T_{X_0}$ is the direct sum of the image and the kernel of $\varpi_\beta$, and both are reflexive, being direct summands of a reflexive sheaf.

**Step 3.** In this step, we construct subsheaves $\mathcal{F}_0 \supset \mathcal{F}_1 \supset \cdots$ of $T_{M_0}$, the pairings $c_i$ on $\mathcal{F}_i$, and the reflexive 2-forms $\beta_i$ on $X_0$. The sheaf $\mathcal{F}_i$ consists of the reductions modulo $t$ of the relative tangent vectors whose pairings with $t^{-b(\omega_*)}\omega_*$ are divisible by $t^i$.

Recall from **Step 1** that $\mathcal{U} \cap X_0 = (X_0)_{\mathrm{reg}}$, so $M_0$ is a closed subset of $\mathcal{U}\backslash\{o\}$ defined by $t = 0$. Let $\widehat{\mathcal{U}}$ be the formal completion of $\mathcal{U}\backslash\{o\}$ along $M_0$, and let $\mathcal{V}$ be the restriction of $T_{\mathcal{U}/\mathbb{A}^1}$ to $\widehat{\mathcal{U}}$. Then $\mathcal{V}$ is locally free of rank $2n$ and $\mathcal{V}/t\mathcal{V} = T_{M_0}$. Let $b := b(\omega_*)$ be the integer of Definition-Lemma 4.1(1), and let $q$ be the regular relative 2-form on $\mathcal{U}$ extending $t^{-b}\omega_*$. Since $T$ fixes $t$, $q$ is an $S$-eigenvector of character $\chi$. We regard $q$ as an alternating pairing $\mathcal{V} \times \mathcal{V} \to \mathcal{O}_{\widehat{\mathcal{U}}}$. For $i \geq 0$, let

$$\mathcal{V}^i := \{v \in \mathcal{V} \mid q(v, \mathcal{V}) \subset t^i\mathcal{O}_{\widehat{\mathcal{U}}}\},$$

and let $\mathcal{F}_i \subset T_{M_0}$ be the image of $\mathcal{V}^i$ in $\mathcal{V}/t\mathcal{V}$. For local sections $v, w$ of $\mathcal{F}_i$ with lifts $\tilde{v}, \tilde{w} \in \mathcal{V}^i$, let

$$c_i(v, w) := t^{-i}q(\tilde{v}, \tilde{w}) \bmod t.$$

The pairing $c_i$ does not depend on the lifts: if $\tilde{v}' \in \mathcal{V}^i$ is another lift of $v$, then $\tilde{v}' - \tilde{v} = tu$ for a local section $u$ of $\mathcal{V}$, and $q(u, \tilde{w}) = -q(\tilde{w}, u) \in t^i\mathcal{O}_{\widehat{\mathcal{U}}}$, so

$$q(\tilde{v}', \tilde{w}) - q(\tilde{v}, \tilde{w}) = tq(u, \tilde{w}) \in t^{i+1}\mathcal{O}_{\widehat{\mathcal{U}}}.$$

Thus $c_i$ is an alternating bilinear pairing on $\mathcal{F}_i$. Since $S$ fixes $t$ and $q$ is an $S$-eigenvector of character $\chi$, $S$ preserves $\mathcal{V}^i$ and $\mathcal{F}_i$, and $c_i$ is an $S$-eigenvector of character $\chi$. Since $t^i\mathcal{V} \subset \mathcal{V}^i$, the sheaf $\mathcal{F}_i$ and the pairing $c_i$ are determined by $q$ modulo $t^{i+1}$, and they are obtained as kernels, images, and reductions of maps between algebraic coherent sheaves on $(\mathcal{U}\backslash\{o\}) \times_{\mathbb{A}^1} \mathrm{Spec}\,\mathbb{C}[t]/(t^{i+1})$. Thus $\mathcal{F}_i$ is a coherent subsheaf of $T_{M_0}$, and $c_i$ is a regular alternating pairing on $\mathcal{F}_i$ over the algebraic open subset $M_0$ of $X_0$.

We prove by induction on $i \geq 0$ the following two properties.

- (a)$_i$ There exists an $S$-invariant idempotent endomorphism $\varpi_i$ of $T_{X_0}$ whose restriction to $M_0$ is the orthogonal projector onto $\mathcal{F}_i$. In particular, $\mathcal{F}_i$ is a parallel subbundle of $T_{M_0}$.
- (b)$_i$ Any point of $M_0$ has an affine open neighborhood $U' \subset M_0$ such that $\mathcal{V}|_{U'}$ has a basis in which the matrix of $q$ is block diagonal of the form

$$q = A_0 \oplus tA_1 \oplus \cdots \oplus t^{i-1}A_{i-1} \oplus t^iC_i, \tag{4.5}$$

  where $A_0, \ldots, A_{i-1}$ are invertible alternating matrices and $C_i$ is an alternating matrix.

For $i = 0$, we have $\mathcal{V}^0 = \mathcal{V}$ and $\mathcal{F}_0 = T_{M_0}$, so (a)$_0$ holds with $\varpi_0 = \mathrm{id}_{T_{X_0}}$, and (b)$_0$ holds with $C_0 = q$ in any basis of $\mathcal{V}|_{U'}$.

First we compute $\mathcal{F}_i$ and $c_i$ from (4.5). Write a local section $v$ of $\mathcal{V}|_{U'}$ as $v = v_0 + \cdots + v_{i-1} + v'$ according to the blocks of (4.5). Since $A_j$ is invertible, $q(v, \mathcal{V}) \subset t^i\mathcal{O}_{\widehat{\mathcal{U}}}$ if and only if $v_j \in t^{i-j}\mathcal{V}$ for $0 \leq j < i$. For $v, w \in \mathcal{V}^i$ and $j < i$, the term $t^jA_j(v_j, w_j)$ belongs to $t^{2i-j}\mathcal{O}_{\widehat{\mathcal{U}}} \subset t^{i+1}\mathcal{O}_{\widehat{\mathcal{U}}}$. Thus

$\mathcal{F}_i|_{U'}$ has a basis given by the reductions modulo $t$ of the basis vectors of the last block, and in the basis of $\mathcal{F}_i|_{U'}$ so obtained

$$c_i = C_i \bmod t. \tag{4.6}$$

Assume that $(\mathrm{a})_i$ and $(\mathrm{b})_i$ hold. Let $\beta_i$ be the reflexive 2-form on $X_0$ extending the regular 2-form $(v,w) \to c_i(\varpi_i v, \varpi_i w)$ on $M_0$. The extension exists because $X_0$ is normal and $\operatorname{codim}_{X_0}(X_0 \backslash M_0) \geq 2$. Since $\varpi_i$ is $S$-invariant and $c_i$ is an $S$-eigenvector of character $\chi$, $\beta_i$ is an $S$-eigenvector of character $\chi$. By **Step 2**, $\beta_i$ is parallel on $M_0$. Let $W_i := \mathcal{F}_i \cap \operatorname{Ker}(\beta_i)$. Then $W_i$ is the pointwise radical of $c_i$ on $\mathcal{F}_i$, and

$$\operatorname{Ker}(\beta_i) = \mathcal{F}_i^{\perp} \oplus W_i$$

is an orthogonal decomposition on $M_0$. By **Step 2**, $\varpi_{\beta_i}$ is an $S$-invariant idempotent endomorphism of $T_{X_0}$, and on $M_0$ it is the sum of $\mathrm{id}_{T_{X_0}} - \varpi_i$ and the orthogonal projector onto $W_i$. Let

$$\varpi_{i+1} := \varpi_{\beta_i} - \mathrm{id}_{T_{X_0}} + \varpi_i.$$

Then $\varpi_{i+1}$ is an $S$-invariant endomorphism of $T_{X_0}$ whose restriction to $M_0$ is the orthogonal projector onto $W_i$. Thus $\varpi_{i+1}^2 - \varpi_{i+1}$ vanishes on $M_0$, so $\varpi_{i+1}$ is idempotent, and $W_i$ is a parallel subbundle of $T_{M_0}$. Let $\mathcal{Q}_i$ be the image of $\varpi_i - \varpi_{i+1}$. On $M_0$, $\mathcal{Q}_i$ is the orthogonal complement of $W_i$ in $\mathcal{F}_i$, so $\mathcal{F}_i = \mathcal{Q}_i \oplus W_i$ on $M_0$, and $c_i$ is non-degenerate on $\mathcal{Q}_i$ at any point of $M_0$.

Next we prove $(\mathrm{b})_{i+1}$ and $\mathcal{F}_{i+1} = W_i$. Fix a point of $M_0$, and let $U'$ be as in $(\mathrm{b})_i$. Shrinking $U'$, we may assume that $\mathcal{Q}_i|_{U'}$ and $W_i|_{U'}$ are trivial. The reductions modulo $t$ of the vectors of the last block of (4.5) form a basis of $\mathcal{F}_i|_{U'}$. Choose a basis $\bar{u}_1, \dots, \bar{u}_p$ of $\mathcal{Q}_i|_{U'}$ and a basis $\bar{w}_1, \dots, \bar{w}_s$ of $W_i|_{U'}$. Together they form a basis of $\mathcal{F}_i|_{U'}$, so there is an invertible matrix $G$ with entries in $\mathcal{O}_{U'}$ which carries the reductions modulo $t$ of the vectors of the last block to $\bar{u}_1, \dots, \bar{u}_p, \bar{w}_1, \dots, \bar{w}_s$. Let $\widetilde{G}$ be a matrix with entries in $\mathcal{O}_{\widehat{\mathcal{U}}}|_{U'}$ lifting $G$, and let $u_1, \dots, u_p, w_1, \dots, w_s$ be the images of the vectors of the last block under $\widetilde{G}$. Then $u_k \equiv \bar{u}_k$ and $w_k \equiv \bar{w}_k \bmod t$. Replace the vectors of the last block by $u_1, \dots, u_p, w_1, \dots, w_s$. The matrix $\widetilde{G}$ is invertible, since $\widehat{\mathcal{U}}$ is $t$-adically complete and the determinant is a unit modulo $t$. In the new basis,

$$C_i = \begin{pmatrix} A_i & B_i \\ -B_i^{\top} & D_i \end{pmatrix},$$

where $A_i$ is the block on $u_1, \dots, u_p$. By (4.6), $A_i \bmod t$ is the matrix of $c_i$ on $\mathcal{Q}_i$, which is invertible, and $B_i \equiv 0$, $D_i \equiv 0 \bmod t$, since $W_i$ is the radical of $c_i$. Since $\widehat{\mathcal{U}}$ is $t$-adically complete, $A_i$ is invertible. Replace $(w_1, \dots, w_s)$ by $(w_1, \dots, w_s) - (u_1, \dots, u_p)A_i^{-1}B_i$. The new vectors are $q$-orthogonal to $u_1, \dots, u_p$, and they are congruent to $w_1, \dots, w_s$ modulo $t$ because $B_i \equiv 0 \bmod t$. The matrix of $C_i$ on the new vectors is

$$D_i + B_i^{\top} A_i^{-1} B_i \equiv 0 \bmod t,$$

so it equals $tC_{i+1}$ for an alternating matrix $C_{i+1}$, since $t$ is a nonzerodivisor on $\mathcal{O}_{\widehat{\mathcal{U}}}$. Thus $t^i C_i = t^i A_i \oplus t^{i+1} C_{i+1}$, which proves $(\mathrm{b})_{i+1}$. If $\mathcal{Q}_i = 0$, the block $A_i$ is empty. By the computation leading to (4.6) for $i+1$, $\mathcal{F}_{i+1}|_{U'}$ is spanned by the reductions modulo $t$ of the new vectors, so $\mathcal{F}_{i+1} = W_i$. Hence $\varpi_{i+1}$ is the orthogonal projector onto $\mathcal{F}_{i+1}$ on $M_0$, which proves $(\mathrm{a})_{i+1}$. This finishes the induction.

**Step 4.** In this step, we show that $\mathcal{F}_i = 0$ for $i \gg 0$, and we construct $\tau$.

Let $\widehat{\mathcal{O}}_{\mathcal{U},X_0}$ be the completion of the local ring of $\mathcal{U}$ at the generic point of $X_0$. Since $\mathcal{X}$ is normal and $X_0$ is integral, $\widehat{\mathcal{O}}_{\mathcal{U},X_0}$ is a complete discrete valuation ring with uniformizer $t$. Over $\mathcal{U} \backslash X_0$, $q = t^{-b}\omega_*$ is non-degenerate because $\omega_*$ is non-degenerate on $X_{\mathrm{reg}}$ (**Step 1**). Thus the matrix of $q$ over $\widehat{\mathcal{O}}_{\mathcal{U},X_0}$ is invertible over $\widehat{\mathcal{O}}_{\mathcal{U},X_0}[t^{-1}]$, and there exists an integer $\ell \geq 0$ such that $t^{\ell} q^{-1}$ has entries in $\widehat{\mathcal{O}}_{\mathcal{U},X_0}$. By the definition of $\mathcal{V}^i$ in **Step 3**, $\mathcal{F}_i$ is the image modulo $t$ of the kernel of the map $v \to tq(v,\cdot)$ from $\mathcal{V}/t^{i+1}\mathcal{V}$ to the dual of $\mathcal{V}/t^{i+1}\mathcal{V}$ over $\mathcal{O}_{\widehat{\mathcal{U}}}/(t^{i+1})$. The map $v \to tq(v,\cdot)$ is a map of locally free sheaves on $(\mathcal{U}\backslash\{o\}) \times_{\mathbb{A}^1} \operatorname{Spec} \mathbb{C}[t]/(t^{i+1})$ defined by $q$

modulo $t^{i+1}$. Kernels and images commute with localization at the generic point of $X_0$, and $\mathcal{O}_{\mathcal{U},X_0}/(t^{i+1}) = \widehat{\mathcal{O}}_{\mathcal{U},X_0}/(t^{i+1})$. Thus the rank of $\mathcal{F}_i$ at the generic point of $X_0$ is the rank over $\widehat{\mathcal{O}}_{\mathcal{U},X_0}/(t)$ of the image modulo $t$ of $\{v \in \mathcal{V}\otimes\widehat{\mathcal{O}}_{\mathcal{U},X_0} \mid q(v,\cdot) \in t^i\widehat{\mathcal{O}}_{\mathcal{U},X_0}\}$. If $v \in \mathcal{V}\otimes\widehat{\mathcal{O}}_{\mathcal{U},X_0}$ satisfies $q(v,\cdot) = t^i\lambda$ with $\lambda$ integral, then

$$v = t^{i-\ell}\cdot(t^\ell q^{-1})(\lambda) \in t^{i-\ell}(\mathcal{V}\otimes\widehat{\mathcal{O}}_{\mathcal{U},X_0}).$$

Thus $\mathcal{F}_i$ has rank zero at the generic point of $X_0$ for $i > \ell$. Since $\mathcal{F}_i$ is a subbundle of $T_{M_0}$ and $M_0$ is connected, $\mathcal{F}_i = 0$ for $i > \ell$. Thus $\varpi_{\ell+1}$ vanishes on $M_0$, so $\varpi_{\ell+1} = 0$.

Since $\mathcal{F}_{i+1} = W_i \subset \mathcal{F}_i$, the projectors $\varpi_0, \dots, \varpi_{\ell+1}$, where $\varpi_0 = \mathrm{id}_{T_{X_0}}$ and $\varpi_{\ell+1} = 0$, satisfy

$$\varpi_i\varpi_{i+1} = \varpi_{i+1}\varpi_i = \varpi_{i+1} \quad \text{for } 0 \le i \le \ell$$

on $M_0$, hence on $X_0$. Thus

$$T_{X_0} = \bigoplus_{i=0}^{\ell} \mathcal{Q}_i,$$

where $\mathcal{Q}_i$ is the image of $\varpi_i - \varpi_{i+1}$ (**Step 3**), and the summands are mutually orthogonal and parallel on $M_0$. Let

$$\tau := \sum_{i=0}^{\ell} \beta_i,$$

where $\beta_i$ is the reflexive 2-form of **Step 3** extending $(v,w) \to c_i(\varpi_i v, \varpi_i w)$. Let $v$ be a local section of $\mathcal{Q}_j|_{M_0}$. If $j < i$, then $\varpi_i v = 0$. If $j > i$, then $v \in \mathcal{F}_{i+1} = W_i$ and $\varpi_i v = v$. In both cases $\beta_i(v,\cdot) = 0$, since $W_i$ is the radical of $c_i$. Thus, on $M_0$, $\tau$ is the orthogonal sum of the restrictions $c_i|_{\mathcal{Q}_i}$, each of which is non-degenerate at any point of $M_0$. Hence $\tau$ is non-degenerate on $M_0$. Each $\beta_i$ is parallel on $M_0$ and is an $S$-eigenvector of character $\chi$, so $\tau$ is parallel on $M_0$ and is an $S$-eigenvector of character $\chi$. Since the Levi-Civita connection is torsion-free, $d\tau$ is the alternation of $\nabla\tau = 0$ on $M_0$. Thus $d\tau = 0$ on $M_0$, and $d\tau = 0$ as a reflexive 3-form on $X_0$.

Recall from **Step 1** that $\nu_0$ generates $\Omega_{X_0}^{[2n]}$ and is an $S$-eigenvector of character $\kappa|_S = n\chi$. Thus $\tau^n/n! = u\nu_0$ for a regular function $u$ on $X_0$, and $u$ is $S$-invariant. Since $\zeta'(\mathbb{G}_m) \subset S$ and $\zeta'$ induces a good positive grading of $\mathcal{O}(X_0)$ (**Step 1**), $u$ is a constant. Since $\tau$ is non-degenerate on $M_0$, $u \neq 0$. Thus $\tau^n$ vanishes nowhere on $(X_0)_{\mathrm{reg}}$, and $\tau$ is non-degenerate on $(X_0)_{\mathrm{reg}}$.

Finally, let $f : Y \to X_0$ be a log resolution. Since $X_0$ is an affine variety with klt singularities, $f_*\Omega_Y^2$ is reflexive by [GKKP11, Theorem 1.4], so $f_*\Omega_Y^2 = \Omega_{X_0}^{[2]}$. Thus $f^*\tau$ extends to a regular 2-form on $Y$. By **Step 1**, $n\chi = \kappa|_S$ and $\langle\chi,\xi\rangle = 2$. This finishes the proof. □

4.4. **The period theorem.** In this subsection, we prove Theorem 4.6: a parallel non-degenerate holomorphic 2-form on a Kähler cone whose link admits a sequence of $\Phi$-cutoffs forces the Reeb vector to be integral, and the weight of the form is 1 or 2.

Unlike the compact 3-Sasakian case [BG99, §1.2], where the three Reeb vector fields are complete, the link here need not be complete, and a flow line may leave the regular locus in finite time. Lemma 4.5 replaces completeness by $\Phi$-cutoffs: it lifts one fixed two-parameter family of paths in the group, starting at almost every point.

**Lemma 4.5.** *Let $(P, h_P)$ be a smooth Riemannian manifold without boundary of finite volume which admits a sequence of $\Phi$-cutoffs. Let $\mathfrak{g}$ be a real Lie algebra with $\dim\mathfrak{g} < +\infty$, let $\iota : \mathfrak{g} \to \Gamma(TP)$ be a Lie algebra homomorphism whose values on a basis of $\mathfrak{g}$ are bounded divergence-free smooth vector fields, and let $G$ be a Lie group with Lie algebra $\mathfrak{g}$. Let $T_0 > 0$ and let $\gamma : [0,1]\times[0,T_0] \to G$ be a smooth map, that is, the restriction of a smooth map on an open neighborhood of $[0,1]\times[0,T_0]$, with $\gamma(s,0) = e$ for any $s$, and let $\gamma^{-1}\partial_t\gamma, \gamma^{-1}\partial_s\gamma : [0,1]\times[0,T_0] \to \mathfrak{g}$ be the maps obtained from $\partial_t\gamma$ and $\partial_s\gamma$ by the left translation by $\gamma^{-1}$. Then for almost every $y \in P$ there exists a classical solution $c_y : [0,1]\times[0,T_0] \to P$ of*

$$\partial_t c_y(s,t) = \iota(\gamma^{-1}\partial_t\gamma)(c_y(s,t)), \qquad c_y(s,0) = y,$$

*and it satisfies* $\partial_s c_y = \iota(\gamma^{-1}\partial_s\gamma)(c_y)$. *If moreover* $\gamma(s,T_0)$ *does not depend on* $s$, *then* $c_y(s,T_0)$ *does not depend on* $s$ *for almost every* $y \in P$.

*Proof.* We fix an extension of $\gamma$ to an open neighborhood of $[0,1]\times[0,T_0]$. The proof is long, so we divide it into several steps.

**Step 1.** In this step we show the following estimate in two parameters. Let $q > 0$ and $D_q := (0,1)\times(0,q)$. For a smooth function $v$ on $D_q$ let

$$\mathcal{E}_q(v) := \int_{D_q} \big(|v|^2 + \Phi(|\nabla v|)\big) ds dt,$$

where $\nabla$ is the Euclidean gradient. Then any sequence $(v_j)_{j\ge1}$ of smooth functions on $D_q$ with $\lim_{j\to+\infty}\mathcal{E}_q(v_j) = 0$ satisfies $\lim_{j\to+\infty}\sup_{D_q}|v_j| = 0$. The statement is a borderline case of the embedding of Orlicz–Sobolev spaces into continuous functions (cf. [Cia96]).

Let $v$ be a smooth function on an open neighborhood of $[0,1]^2$, and let

$$\tilde v(x_1,x_2) := v\big(1-|1-|x_1||, 1-|1-|x_2||\big) \quad \text{for } (x_1,x_2)\in[-1,2]^2,$$

that is, $\tilde v$ is the extension of $v$ by the reflections $x_i \to -x_i$ and $x_i \to 2-x_i$ across the sides of $[0,1]^2$. Then $\tilde v$ is continuous, $\tilde v$ is smooth on each of the nine closed unit squares of $[-1,2]^2$, and

$$\int_{[-1,2]^2} \big(|\tilde v|^2 + \Phi(|\nabla\tilde v|)\big) = 9\int_{[0,1]^2}\big(|v|^2+\Phi(|\nabla v|)\big). \tag{4.7}$$

Fix $z\in[0,1]^2$. The disk $B(z,1)$ of radius 1 centered at $z$ is contained in $[-1,2]^2$. For any unit vector $\theta$ outside a finite set and any $0<a<1$,

$$\tilde v(z) - \tilde v(z+a\theta) = -\int_0^a \nabla\tilde v(z+b\theta)\cdot\theta\, db.$$

We multiply by $a$, integrate over $0<a<1$ and over $\theta$, and interchange the integrals in $a$ and $b$. Since $\int_b^1 a\,da \le \frac12$, we obtain

$$\Big|\pi\tilde v(z) - \int_{B(z,1)}\tilde v(y)dy\Big| \le \frac12\int_{B(z,1)}\frac{|\nabla\tilde v(y)|}{|y-z|}dy.$$

By the Cauchy–Schwarz inequality,

$$\Big|\int_{B(z,1)}\tilde v\Big| \le \pi^{1/2}\|\tilde v\|_{L^2([-1,2]^2)}.$$

Thus

$$|v(z)| \le \pi^{-1/2}\|\tilde v\|_{L^2([-1,2]^2)} + \frac{1}{2\pi}\int_{B(z,1)}\frac{|\nabla\tilde v(y)|}{|y-z|}dy. \tag{4.8}$$

Let $u(y) := |\nabla\tilde v(y)|$ and $E := \int_{[-1,2]^2}\Phi(u)$, and fix $0<\epsilon<1$. We split the integral in (4.8) into three parts. On the set of $y$ with $|y-z|<\epsilon$ and $u(y)\le|y-z|^{-1/2}$, the integrand is at most $|y-z|^{-3/2}$, and

$$\int_{B(z,\epsilon)}|y-z|^{-3/2}dy = 4\pi\epsilon^{1/2}.$$

On the set of $y$ with $|y-z|<\epsilon$ and $u(y)>|y-z|^{-1/2}$, we have $\log(e+u(y)) \ge \frac12\log(1/|y-z|) > 0$, hence

$$\frac{u(y)}{|y-z|} \le \frac{2u(y)\log(e+u(y))}{|y-z|\log(1/|y-z|)}.$$

By the Cauchy–Schwarz inequality, the integral over $\{y\in B(z,\epsilon)\mid u(y)>|y-z|^{-1/2}\}$ is at most

$$2E^{1/2}\Big(\int_{B(z,\epsilon)}\frac{dy}{|y-z|^2\log^2(1/|y-z|)}\Big)^{1/2} = 2E^{1/2}\Big(2\pi\int_0^\epsilon\frac{dr}{r\log^2(1/r)}\Big)^{1/2} = \frac{2(2\pi)^{1/2}E^{1/2}}{\log^{1/2}(1/\epsilon)}.$$

On the set of $y$ with $\epsilon \le |y-z| < 1$, we use $u^2 \le \Phi(u)$ and the Cauchy–Schwarz inequality, and the integral is at most $E^{1/2}(2\pi \log(1/\epsilon))^{1/2}$. By (4.7) and (4.8), there exists an absolute constant $C$ such that

$$\sup_{[0,1]^2} |v| \le C\Big(\epsilon^{1/2} + \mathcal{E}_1(v)^{1/2}\big(1 + \log^{1/2}(1/\epsilon) + \log^{-1/2}(1/\epsilon)\big)\Big),$$

where $\mathcal{E}_1(v)$ denotes the energy $\int_{[0,1]^2}(|v|^2 + \Phi(|\nabla v|))$.

Now let $v$ be a smooth function on $D_q$. For $0 < \delta < \frac{1}{4}\min\{1, q\}$, let

$$\lambda_\delta : [0,1]^2 \to [\delta, 1-\delta] \times [\delta, q-\delta], \qquad (x_1, x_2) \to (\delta + (1-2\delta)x_1, \delta + (q-2\delta)x_2),$$

be the affine map. Then $v \circ \lambda_\delta$ is smooth on an open neighborhood of $[0,1]^2$, $|\nabla(v\circ\lambda_\delta)| \le a_q |\nabla v| \circ \lambda_\delta$ with $a_q := \max\{1, q\}$, and the Jacobian determinant of $\lambda_\delta$ is at least $q/4$. By (2.1) of Remark 2.30,

$$\int_{[0,1]^2} \big(|v\circ\lambda_\delta|^2 + \Phi(|\nabla(v\circ\lambda_\delta)|)\big) \le \frac{4}{q} a_q^2 (1 + \log a_q)^2 \mathcal{E}_q(v).$$

Since $D_q$ is the union of the open rectangles $(\delta, 1-\delta) \times (\delta, q-\delta)$, there exists a constant $C_q$ depending only on $q$ such that for any $0 < \epsilon < 1$,

$$\sup_{D_q} |v| \le C_q\Big(\epsilon^{1/2} + \mathcal{E}_q(v)^{1/2}\big(1 + \log^{1/2}(1/\epsilon) + \log^{-1/2}(1/\epsilon)\big)\Big). \tag{4.9}$$

For a sequence $(v_j)$ with $\mathcal{E}_q(v_j) \to 0$, (4.9) implies $\limsup_{j\to+\infty} \sup_{D_q} |v_j| \le C_q \epsilon^{1/2}$ for any $0 < \epsilon < 1$. Thus $\sup_{D_q} |v_j| \to 0$.

**Step 2.** In this step we show that the local solutions satisfy the equation for $\partial_s$, and that the maps $y \to c_y(s,t)$ preserve volume.

Let $A := \gamma^{-1}\partial_t\gamma$ and $B := \gamma^{-1}\partial_s\gamma$, as in the statement, computed with the fixed extension. Choose $T_1 > T_0$ such that $\gamma$ is defined on an open neighborhood of $[0,1] \times [0, T_1]$. The coefficients of $A$ and $B$ in a basis of $\mathfrak{g}$ are bounded on $[0,1] \times [0, T_1]$, and the values of $\iota$ on the basis are bounded. Thus there exists $C_0 < +\infty$ with

$$|\iota(A(s,t))|_{h_P} \le C_0 \quad \text{and} \quad |\iota(B(s,t))|_{h_P} \le C_0 \quad \text{for } (s,t) \in [0,1] \times [0, T_1].$$

We have the Maurer–Cartan identity $\partial_s A - \partial_t B = [A, B]$, and $B(s,0) = 0$ since $\gamma(s,0) = e$.

For $s \in [0,1]$ and $y \in P$, let $t \to c_y(s,t)$ be the maximal solution on $[0, T_1)$ of

$$\partial_t c_y(s,t) = \iota(A(s,t))(c_y(s,t)), \qquad c_y(s,0) = y,$$

and let $\ell(s,y) \in (0, T_1]$ be the supremum of the times of existence. Since $\gamma$ is defined on an open neighborhood of $[0,1] \times [0, T_1]$, there exists $\epsilon_0 > 0$ such that $A$ is defined and smooth on $(-\epsilon_0, 1+\epsilon_0) \times (-\epsilon_0, T_1 + \epsilon_0)$. We apply the fundamental theorem on flows [Lee13, Theorem 9.12] to the vector field $\partial_t + \iota(A(s,t))$, with $\partial_s$-component 0, on the manifold without boundary $Q := (-\epsilon_0, 1+\epsilon_0) \times (-\epsilon_0, T_1 + \epsilon_0) \times P$, and restrict to $[0,1] \times [0, T_1) \times P$. The integral curve through $(s, 0, y)$ is $t \to (s, t, c_y(s,t))$. Thus the set $\mathcal{D}$ of $(s,t,y)$ with $0 \le t < \ell(s,y)$ is open in $[0,1] \times [0, T_1) \times P$, the map $(s,t,y) \to c_y(s,t)$ is smooth on $\mathcal{D}$, and $\ell$ is lower semicontinuous.

We claim that

$$\partial_s c_y(s,t) = \iota(B(s,t))(c_y(s,t)) \quad \text{for } (s,t,y) \in \mathcal{D}. \tag{4.10}$$

For a vector field $V$ let $V'$ denote the Jacobian matrix of $V$ in local coordinates, so that $[U, V] = V'U - U'V$. Let $Z := \partial_s c_y$ and $W := \iota(B)(c_y)$. In local coordinates along the curve $t \to c_y(s,t)$,

$$\partial_t Z = \iota(A)' Z + \iota(\partial_s A), \qquad \partial_t W = \iota(\partial_t B) + \iota(B)'\iota(A).$$

Since $\iota$ is a homomorphism, the Maurer–Cartan identity implies

$$\iota(\partial_s A) = \iota(\partial_t B) + \iota(B)'\iota(A) - \iota(A)'\iota(B),$$

so $\partial_t W = \iota(A)'W + \iota(\partial_s A)$, which is the linear equation for $Z$ with $W$ in place of $Z$. Moreover $Z(s,0) = 0$ and $W(s,0) = \iota(B(s,0))(y) = 0$. By the uniqueness of solutions of linear equations, $Z = W$, which proves (4.10).

For fixed $(s,t)$, let $P_{s,t} := \{y \in P \mid \ell(s,y) > t\}$, which is open. The map $\phi_{s,t} : P_{s,t} \to P$, $y \to c_y(s,t)$, is injective by the uniqueness of solutions of the backward equation, and it preserves the Riemannian volume by the Liouville formula, since $\operatorname{div} \iota(A(s,t)) = 0$. Hence for any non-negative measurable function $b$ on $P$ and any measurable subset $V \subset P_{s,t}$,

$$\int_V b(c_y(s,t)) d\operatorname{Vol}_{h_P}(y) \le \int_P b d\operatorname{Vol}_{h_P}. \tag{4.11}$$

**Step 3.** In this step we show that for almost every $y \in P$ no solution in the family leaves $P$ before $t = T_0$.

Let $\ell_*(y) := \min_{s\in[0,1]} \ell(s,y)$. The minimum is attained since $\ell$ is lower semicontinuous and $[0,1]$ is compact, and $\ell_*(y) > 0$. Moreover $\ell_*$ is lower semicontinuous: for any $a < \ell_*(y)$, each $(s,y)$ has a neighborhood on which $\ell > a$, and finitely many such neighborhoods cover $[0,1] \times \{y\}$, so $\ell_* > a$ on a neighborhood of $y$. Let $q(y) := \min\{\ell_*(y), T_0\}$. Then $c_y(s,t)$ is defined for $(s,t) \in [0,1] \times [0,q(y))$, and the set of $(s,t,y)$ with $t < q(y)$ is open.

Let $(\beta_j)_{j\ge1}$ be a sequence of $\Phi$-cutoffs on $P$. Since $0 \le \beta_j \le 1$, $\beta_j \to 1$ pointwise, and $\operatorname{Vol}_{h_P}(P) < +\infty$, the dominated convergence theorem implies $\int_P |1-\beta_j|^2 \to 0$. Since a subsequence of a sequence of $\Phi$-cutoffs is again a sequence of $\Phi$-cutoffs, possibly replacing $(\beta_j)$ with a subsequence, we may assume that

$$\sum_{j\ge1} \int_P \big(|1-\beta_j|^2 + \Phi(|d\beta_j|_{h_P})\big) d\operatorname{Vol}_{h_P} < +\infty. \tag{4.12}$$

For $y \in P$ let $v_{j,y}(s,t) := 1 - \beta_j(c_y(s,t))$ on $(0,1) \times (0,q(y))$. By the equation of $c_y$ and (4.10),

$$|\nabla_{(s,t)} v_{j,y}(s,t)| \le 2^{1/2} C_0 |d\beta_j|_{h_P}(c_y(s,t)).$$

Let $C_1 := 2C_0^2(1 + \log\max\{1, 2^{1/2}C_0\})^2$. By (2.1) of Remark 2.30, the Tonelli theorem, and (4.11) applied to $V = \{y \mid t < q(y)\} \subset P_{s,t}$,

$$\int_P \int_0^1 \int_0^{q(y)} \big(|v_{j,y}|^2 + \Phi(|\nabla_{(s,t)} v_{j,y}|)\big) dt ds d\operatorname{Vol}_{h_P}(y) \le T_0 \int_P \big(|1-\beta_j|^2 + C_1 \Phi(|d\beta_j|_{h_P})\big) d\operatorname{Vol}_{h_P}.$$

By (4.12), for almost every $y \in P$ the energies $\int_0^1 \int_0^{q(y)} (|v_{j,y}|^2 + \Phi(|\nabla_{(s,t)} v_{j,y}|)) dt ds$ form a summable sequence, and in particular they tend to zero. For such $y$, the number $q(y) > 0$ does not depend on $j$, and **Step 1** applied to the rectangle $(0,1) \times (0,q(y))$ implies

$$\lim_{j\to+\infty} \sup_{0\le s\le1, 0<t<q(y)} |1 - \beta_j(c_y(s,t))| = 0, \tag{4.13}$$

where the values $s = 0$ and $s = 1$ are included by continuity.

Fix $y$ satisfying (4.13), and suppose that $\ell_*(y) \le T_0$. Choose $s_0 \in [0,1]$ with $\ell(s_0,y) = \ell_*(y) = q(y)$. Since $q(y) < T_1$, the time $q(y)$ is the maximal time of existence of $t \to c_y(s_0,t)$. For any compact subset $W \subset P$, there exists $t_W < q(y)$ with $c_y(s_0,t) \notin W$ for $t_W < t < q(y)$. Otherwise there exist times $t_k \uparrow q(y)$ with $c_y(s_0,t_k) \in W$, and since the flow domain of the vector field on $Q$ above is open [Lee13, Theorem 9.12] and $\{s_0\} \times [0,T_0] \times W$ is compact, the solutions for $s_0$ starting at a point of $W$ at a time in $[0,T_0]$ exist on time intervals of a common length $\epsilon_W > 0$, so $t \to c_y(s_0,t)$ extends beyond $q(y)$. Taking $W = \operatorname{Supp}\beta_j$, we obtain $\beta_j(c_y(s_0,t)) = 0$ for $t$ close to $q(y)$, hence the supremum in (4.13) equals 1 for any $j$, a contradiction. Therefore $\ell_*(y) > T_0$ for almost every $y \in P$.

For such $y$, the map $c_y$ is a classical solution on $[0,1] \times [0,T_0]$, and (4.10) holds on $[0,1] \times [0,T_0]$. If $\gamma(s,T_0)$ does not depend on $s$, then $B(s,T_0) = 0$, and (4.10) implies $\partial_s c_y(s,T_0) = 0$. Thus $c_y(s,T_0)$ does not depend on $s$. □

**Theorem 4.6.** *Let $X$ be an irreducible normal affine variety of dimension $2n$, $n \geq 1$, with a faithful action of a torus $T$ fixing a point $o$, let $T_c \subset T$ be the maximal compact subtorus, and let $\xi \in \mathrm{Lie}(T_c)$ be a Reeb vector of $T$. Assume that $M := X_{\mathrm{reg}} \backslash \{o\}$ is connected and carries a smooth Kähler cone metric $g = d\rho^2 + \rho^2 h$ with complex structure $I$ and connected link $N$ such that $\rho\partial_\rho f = \langle \alpha, \xi \rangle f$ for any regular function $f$ on $X$ of $T$-character $\alpha$, that $\mathrm{Vol}_h(N) < +\infty$, that $N$ admits a sequence of $\Phi$-cutoffs, and that $M$ carries a parallel non-degenerate holomorphic 2-form $\sigma$. Then there exist a primitive element $\zeta$ of the lattice $\{y \in \mathrm{Lie}(T_c) \mid \exp(2\pi y) = 1\}$ and $r \in \{1, 2\}$ such that $\xi = \frac{2}{r}\zeta$ and, regarding $\zeta$ as a cocharacter of $T$ as in Definition 2.6, $\zeta(\lambda)^*\sigma = \lambda^r \sigma$ for $\lambda \in \mathbb{C}^*$. In particular, $\exp(2\pi\xi) = 1$ in $T_c$.*

*Proof.* Let $\nabla$ be the Levi-Civita connection of $g$. On the cone $M = (0, +\infty) \times N$ we have $\nabla_Y(\rho\partial_\rho) = Y$ for any tangent vector $Y$.

**Step 1.** In this step we construct three vector fields on $N$ which span a Lie algebra isomorphic to the imaginary quaternions, and we show that the first one is the vector field generated by $\xi$.

Let $\Psi$ be the real endomorphism of $TM$ defined by $g(\Psi v, w) = \mathrm{Re}\,\sigma(v, w)$. Then $\Psi$ is skew-adjoint and parallel. Since $\sigma$ is of type $(2, 0)$, $\sigma(Iv, w) = \sqrt{-1}\sigma(v, w)$, which implies

$$g(\Psi I v, w) = -\,\mathrm{Im}\,\sigma(v, w) = -g(I\Psi v, w),$$

that is, $\Psi I = -I\Psi$. If $\Psi v = 0$, then $\mathrm{Re}\,\sigma(v, w) = 0$ and $\mathrm{Im}\,\sigma(v, w) = -\,\mathrm{Re}\,\sigma(v, Iw) = 0$ for any $w$, hence $v = 0$ since $\sigma$ is non-degenerate. Thus $-\Psi^2 = \Psi^{\mathsf{T}}\Psi$ is positive definite and parallel, and it commutes with $I$. Let $C := (-\Psi^2)^{1/2}$ be the positive square root. Then $C$ is parallel and commutes with $\Psi$ and $I$. Let $J := C^{-1}\Psi$. Then $J^2 = C^{-2}\Psi^2 = -\,\mathrm{id}_{TM}$, $J^{\mathsf{T}}J = -\Psi C^{-2}\Psi = \mathrm{id}_{TM}$, and $IJ = -JI$. Thus $J$ and $IJ$ are parallel orthogonal complex structures with $IJ = -JI$.

Let $V_1 := I(\rho\partial_\rho)$, $V_2 := J(\rho\partial_\rho)$, and $V_3 := IJ(\rho\partial_\rho)$. For $Q \in \{I, J, IJ\}$, we have $\nabla_Y(Q\rho\partial_\rho) = QY$, and $Q$ is skew-adjoint, so $Q\rho\partial_\rho$ is a Killing vector field. Moreover $g(Q\rho\partial_\rho, \partial_\rho) = 0$ and $|Q\rho\partial_\rho|_g = \rho$. Thus $V_1$, $V_2$, and $V_3$ are tangent to the level sets of $\rho$, the local flows of $V_1$, $V_2$, and $V_3$ preserve $g$ and $\rho$, and the restrictions $V_1|_N$, $V_2|_N$, and $V_3|_N$ are smooth vector fields of length 1 whose local flows preserve $h$. In particular, the restrictions are bounded and divergence-free. Since $\nabla$ is torsion-free,

$$[V_1, V_2] = \nabla_{V_1}V_2 - \nabla_{V_2}V_1 = (JI - IJ)\rho\partial_\rho = -2V_3,$$

and $[V_2, V_3] = (IJJ - JIJ)\rho\partial_\rho = -2V_1$ and $[V_3, V_1] = (IIJ - IJI)\rho\partial_\rho = -2V_2$. Let $\mathrm{Im}\,\mathbb{H}$ be the Lie algebra of imaginary quaternions, with basis $i, j, k$ and $[i, j] = 2k$, $[j, k] = 2i$, $[k, i] = 2j$. Then

$$\iota : \mathrm{Im}\,\mathbb{H} \to \Gamma(TN), \qquad i \to -V_1|_N, \quad j \to -V_2|_N, \quad k \to -V_3|_N,$$

is a Lie algebra homomorphism. The group $\mathrm{Sp}(1)$ of unit quaternions has Lie algebra $\mathrm{Im}\,\mathbb{H}$.

For a regular function $f$ on $X$ of $T$-character $\alpha$, the function $f$ is holomorphic, so

$$V_1 f = I(\rho\partial_\rho)f = \sqrt{-1}\rho\partial_\rho f = \sqrt{-1}\langle \alpha, \xi \rangle f,$$

which is the value on $f$ of the vector field generated by $\xi$ (Definition 2.6). Since $X$ embeds into an affine space by finitely many $T$-eigenfunctions, the homogeneous regular functions on $X$ separate tangent vectors on $M$. Thus $V_1$ is the vector field generated by $\xi$.

**Step 2.** In this step we show that $\exp(2\pi\xi) = 1$.

For $(s, t) \in \mathbb{R}^2$ let $\vartheta := \pi s/2$ and

$$a(s) := \cos\vartheta + \sin\vartheta\, j, \qquad b(s, t) := \cos\vartheta(\cos t + \sin t\, i) + \sin\vartheta\, j, \qquad \gamma(s, t) := a(s)^{-1}b(s, t).$$

The real, $i$, and $j$ components of $b(s, t)$ have squares summing to 1, so $a$, $b$, and $\gamma$ take values in $\mathrm{Sp}(1)$, and $\gamma$ is smooth on $\mathbb{R}^2$. Moreover

$$\gamma(s, 0) = \gamma(s, 2\pi) = 1, \qquad \gamma(0, t) = \cos t + \sin t\, i = \exp(ti), \qquad \gamma(1, t) = 1.$$

By Lemma 4.5 applied to $(N, h)$, $\iota$, $\mathrm{Sp}(1)$, $T_0 = 2\pi$, and $\gamma$, and since $\gamma(s, 2\pi)$ does not depend on $s$, for almost every $y \in N$ we have $c_y(0, 2\pi) = c_y(1, 2\pi)$. For $s = 1$, $\gamma^{-1}\partial_t\gamma = 0$, so $c_y(1, t) = y$.

For $s = 0$, $\gamma^{-1}\partial_t\gamma = i$, so $t \to c_y(0,t)$ is an integral curve of $-V_1$. By **Step 1**, $V_1$ is the vector field generated by $\xi$. The flow of $V_1$ is $\theta \to \exp(\theta\xi)$, which is defined for any $\theta \in \mathbb{R}$ and preserves $N$ by **Step 1**. By the uniqueness of integral curves, $c_y(0,t) = \exp(-t\xi)\cdot y$. Thus $\exp(-2\pi\xi)\cdot y = y$ for almost every $y \in N$.

Any non-empty open subset of $N$ has positive volume, so the set of $y \in N$ with $\exp(-2\pi\xi)\cdot y = y$ is dense in $N$. Since $\exp(-2\pi\xi)$ acts continuously, $\exp(-2\pi\xi)$ fixes $N$ pointwise. We have

$$[V_1, \rho\partial_\rho] = \nabla_{V_1}(\rho\partial_\rho) - \nabla_{\rho\partial_\rho}V_1 = V_1 - I\rho\partial_\rho = 0,$$

so the flow of $V_1$ commutes with the flow of $\rho\partial_\rho$. Any point of $M$ is the image of a point of $N$ under the flow of $\rho\partial_\rho$, hence $\exp(-2\pi\xi)$ fixes $M$ pointwise. Since $X$ is irreducible, $M$ is dense in $X$, so $\exp(2\pi\xi)$ acts trivially on $X$. Since $T$ acts faithfully on $X$, we obtain $\exp(2\pi\xi) = 1$ in $T_c$.

**Step 3.** In this step we determine the weight of $\sigma$.

Since $|V_1|_g = \rho > 0$, $\xi \neq 0$. By **Step 2**, $\xi$ is a non-zero element of the lattice

$$\{y \in \mathrm{Lie}(T_c) \mid \exp(2\pi y) = 1\},$$

so $\xi = m\zeta$ for a positive integer $m$ and a primitive element $\zeta$ of the lattice. Since $\sigma$ is parallel and $\nabla(\rho\partial_\rho) = \mathrm{id}_{TM}$, we have $L_{\rho\partial_\rho}\sigma = 2\sigma$, and $d\sigma = 0$. Since $\sigma$ is of type $(2,0)$, $i_{I\rho\partial_\rho}\sigma = \sqrt{-1}i_{\rho\partial_\rho}\sigma$. By the Cartan formula,

$$L_{V_1}\sigma = di_{V_1}\sigma = \sqrt{-1}di_{\rho\partial_\rho}\sigma = \sqrt{-1}L_{\rho\partial_\rho}\sigma = 2\sqrt{-1}\sigma.$$

Thus $\exp(\theta\xi)^*\sigma = e^{2\sqrt{-1}\theta}\sigma$ for $\theta \in \mathbb{R}$. Since $\exp(\frac{2\pi}{m}\xi) = \exp(2\pi\zeta) = 1$, we obtain $e^{4\pi\sqrt{-1}/m} = 1$, that is, $2/m$ is a positive integer. Hence $m \in \{1,2\}$. Let $r := 2/m \in \{1,2\}$. Then $\xi = \frac{2}{r}\zeta$ and $\exp(\theta\zeta)^*\sigma = e^{\sqrt{-1}r\theta}\sigma$.

We regard $\zeta$ as a cocharacter of $T$ (Definition 2.6). For a regular function $f$ on $X$ of $T$-character $\alpha$, the vector field generated by $u \to \zeta(e^u)$, $u \in \mathbb{R}$, sends $f$ to $\langle\alpha,\zeta\rangle f = \frac{r}{2}\langle\alpha,\xi\rangle f = \frac{r}{2}\rho\partial_\rho f$ by assumption. Since the homogeneous regular functions on $X$ separate tangent vectors on $M$ (**Step 1**), the vector field generated by $u \to \zeta(e^u)$ is $\frac{r}{2}\rho\partial_\rho$, and $L_{\frac{r}{2}\rho\partial_\rho}\sigma = r\sigma$. Therefore

$$\zeta(\lambda)^*\sigma = \lambda^r\sigma \quad \text{for } \lambda \in \mathbb{C}^*,$$

where we use $\exp(\theta\zeta) = \zeta(e^{\sqrt{-1}\theta})$. □

4.5. **Rank and weight on the central cone.** In this subsection, we prove Theorem 4.7, which determines the Reeb vector of the central fiber of a marked test configuration of a symplectic variety, and the log discrepancy of the corresponding valuation on the symplectic variety.

We first construct the parallel symplectic form $\tau$ of Theorem 4.7(1) by Proposition 4.3 and Theorem 4.4, and then apply Theorem 4.6 to $\tau$, since Theorem 4.6 requires a parallel non-degenerate form.

**Theorem 4.7.** *Notation and conditions as in Set-up 2.15. Assume that* $\dim X = 2n$ *with* $n \geq 2$, *that $X$ is a symplectic variety whose symplectic form $\omega$ is algebraic of $\zeta$-degree $w > 0$, that $T$ acts faithfully on $X$ and on $X_0$, that $X_0$ is integral and normal with klt singularities, and that the $\zeta$-grading of $X_0$ is good. Let $\xi$ be a Reeb vector of $T$ on $X_0$ such that $(X_0,\xi)$ is a K-semistable Fano cone with $A_{X_0}(\mathrm{wt}_\xi) = 2n$, and assume that $X_0$, $\xi$, and a metric $g$ satisfy the conditions of Set-up 2.31. Then the following hold.*

1. *The variety $X_0$ carries a closed reflexive 2-form $\tau$ which is non-degenerate on $(X_0)_{\mathrm{reg}}$, parallel on $(X_0)_{\mathrm{reg}}\backslash\{o\}$, and extends regularly to a resolution of $X_0$.*
2. *There exist a primitive cocharacter $\zeta_0$ of $T$ and $r \in \{1,2\}$ with $\xi = \frac{2}{r}\zeta_0$. The cocharacter $\zeta_0$ is a Reeb vector on $X$ and on $X_0$, and $\zeta_0(\lambda)^*\tau = \lambda^r\tau$ for $\lambda \in \mathbb{C}^*$.*
3. *We have $A_X(\mathrm{wt}_{\zeta_0}) = rn$, and if $(X,\xi)$ is K-semistable, then $\mathrm{wt}_{\zeta_0}$ is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex of $X$.*

*Proof.* Let $M_0 := (X_0)_{\mathrm{reg}} \backslash \{o\}$, where $o$ is the vertex of $X_0$, and let $\rho$ and $h$ be as in Set-up 2.31 for $X_0$, $\xi$, and $g$. Let $\kappa$ be the character by which $T$ acts on $\omega^n$, and let $S \subset T$ be the smallest subtorus with $\xi \in \mathrm{Hom}(\mathbb{G}_m, S) \otimes \mathbb{R}$. The proof is long, so we divide it into several steps.

**Step 1.** In this step we prove (1).

By Proposition 4.3 applied to $X_0$, any reflexive 2-form $\beta$ on $X_0$ with $L_{\rho\partial_\rho}\beta = 2\beta$ on $M_0$ is parallel on $M_0$. By Set-up 2.31(1), $g$ is a smooth Ricci-flat Kähler cone metric on $M_0$ with connected link, and $\rho\partial_\rho f = \langle \alpha, \xi \rangle f$ for any regular function $f$ on $X_0$ of $T$-character $\alpha$. Thus Theorem 4.4, applied with $w$ in place of $r$, implies that $X_0$ carries a closed reflexive 2-form $\tau$ which is non-degenerate on $(X_0)_{\mathrm{reg}}$, parallel on $M_0$, and extends regularly to a resolution of $X_0$, and which is an $S$-eigenvector of a character $\chi$ with

$$n\chi = \kappa|_S \qquad \text{and} \qquad \langle \chi, \xi \rangle = 2. \tag{4.14}$$

This proves (1).

**Step 2.** In this step we prove (2) on $X_0$.

We verify the conditions of Theorem 4.6 for $X_0$, $\xi$, and $\tau$. Since $\mathcal{X} \to \mathbb{A}^1$ is flat, $\dim X_0 = 2n$, and $X_0$ is an irreducible normal affine variety with a faithful action of $T$ fixing $o$. The manifold $M_0 = (0, +\infty) \times N$ is connected since $N$ is connected, $\mathrm{Vol}_h(N) < +\infty$, and $N$ admits a sequence of $\Phi$-cutoffs by Set-up 2.31(3) and (4). The condition on $\rho\partial_\rho$ in Theorem 4.6 is condition (1) of Set-up 2.31. By **Step 1**, $\tau|_{M_0}$ is a parallel non-degenerate holomorphic 2-form.

By Theorem 4.6 applied to $X_0$, $\xi$, and $\tau$, there exist a primitive element $\zeta_0$ of the lattice $\{y \in \mathrm{Lie}(T_c) \mid \exp(2\pi y) = 1\}$ and $r \in \{1, 2\}$ such that $\xi = \frac{2}{r}\zeta_0$ and $\zeta_0(\lambda)^*\tau = \lambda^r \tau$ for $\lambda \in \mathbb{C}^*$. Since $T$ acts faithfully on $X_0$, the lattice $\{y \in \mathrm{Lie}(T_c) \mid \exp(2\pi y) = 1\}$ is $\mathrm{Hom}(\mathbb{G}_m, T)$ by Definition 2.6, so $\zeta_0$ is a primitive cocharacter of $T$. For any character $\alpha \neq 0$ of $T$ with $(\mathcal{R}/t\mathcal{R})_\alpha \neq 0$, we have $\langle \alpha, \zeta_0 \rangle = \frac{r}{2}\langle \alpha, \xi \rangle > 0$. Thus $\zeta_0$ is a Reeb vector on $X_0$.

**Step 3.** In this step we show that for any character $\alpha$ of $T$, $R_\alpha \neq 0$ if and only if $(\mathcal{R}/t\mathcal{R})_\alpha \neq 0$. In particular, $\zeta_0$ and $\xi$ are Reeb vectors on $X$.

For a character $\alpha$ of $T$, let $\mathcal{R}_\alpha \subset \mathcal{R}$ be the $T$-eigenspace of character $\alpha$, as in Definition 2.6. Since $T$ fixes $t$, each $\mathcal{R}_\alpha$ is a $\mathbb{C}[t]$-submodule, and for $j \geq 0$ the $\zeta$-degree $j$ part $\mathcal{R}_j$ is the direct sum of the $\mathcal{R}_\alpha$ with $\langle \alpha, \zeta \rangle = j$. The $\mathbb{C}$-algebra $\mathcal{R}$ is generated by finitely many $T$-eigenvectors. Those of $\zeta$-degree 0 belong to the $\zeta$-degree 0 part $\mathbb{C}[t]$ of $\mathcal{R}$, and the others have positive $\zeta$-degree. Hence $\mathcal{R}_j$ is generated as a $\mathbb{C}[t]$-module by finitely many monomials in the generators. Since $\mathcal{R}$ is a domain, $\mathcal{R}_j$ is a finitely generated torsion-free module over the principal ideal domain $\mathbb{C}[t]$, hence free. Thus the direct summand $\mathcal{R}_\alpha$ of $\mathcal{R}_j$ is a free $\mathbb{C}[t]$-module of finite rank. The $T \times \Lambda$-equivariant isomorphism $\mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$ implies

$$\mathcal{R}_\alpha[t^{-1}] \cong R_\alpha \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}].$$

Thus

$$\dim_{\mathbb{C}} R_\alpha = \mathrm{rank}_{\mathbb{C}[t]} \mathcal{R}_\alpha = \dim_{\mathbb{C}} \mathcal{R}_\alpha / t\mathcal{R}_\alpha = \dim_{\mathbb{C}} (\mathcal{R}/t\mathcal{R})_\alpha.$$

Hence $R_\alpha \neq 0$ if and only if $(\mathcal{R}/t\mathcal{R})_\alpha \neq 0$. Since $\zeta_0$ and $\xi$ are Reeb vectors on $X_0$, they are Reeb vectors on $X$. This finishes the proof of (2).

**Step 4.** In this step we prove (3).

Since $X$ is a symplectic variety, $X$ has rational Gorenstein singularities [Bea99, Proposition 1.3], hence klt singularities. Let $o_X$ be the vertex of the $\zeta$-grading of $R$. Since the $\zeta$-degree 0 part of $R$ is $\mathbb{C}$, $\zeta(\lambda) \cdot y$ tends to $o_X$ as $\lambda \to 0$ for any $y \in X$. Thus the $T$-action on $X$ is good with vertex $o_X$. The form $\omega^n$ is nowhere zero on $X_{\mathrm{reg}}$, and the reflexive sheaf $\Omega_X^{[2n]}$ is determined by the restriction $\Omega_X^{[2n]}|_{X_{\mathrm{reg}}}$. Thus $\Omega_X^{[2n]}$ is generated by $\omega^n$, which is a $T$-eigenvector of character $\kappa$. By **Step 3**, $\zeta_0$ is a Reeb vector on $X$, so Lemma 2.14 implies $A_X(\mathrm{wt}_{\zeta_0}) = \langle \kappa, \zeta_0 \rangle$. Since $\zeta_0 = \frac{r}{2}\xi \in \mathrm{Hom}(\mathbb{G}_m, S) \otimes \mathbb{R}$, (4.14) implies

$$A_X(\mathrm{wt}_{\zeta_0}) = \langle \kappa, \zeta_0 \rangle = \frac{r}{2}\langle \kappa|_S, \xi \rangle = \frac{r}{2} n \langle \chi, \xi \rangle = rn.$$

Assume that $(X,\xi)$ is K-semistable. By **Step 3**, $\xi$ is a Reeb vector of $T$ on $X$, so $(X,\xi)$ is a Fano cone, since $T$ acts faithfully on $X$ by assumption. By Theorem 2.21, $\mathrm{wt}_\xi$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,o_X}$. By Definition 2.6, $\mathrm{wt}_{c\xi} = c\,\mathrm{wt}_\xi$ for $c>0$. For a valuation $v$ and $c>0$, we have $A_X(cv) = cA_X(v)$ and $\mathfrak{a}_m(cv) = \mathfrak{a}_{m/c}(v)$, where $\mathfrak{a}_m(v) := \{f \in R \mid v(f) \geq m\}$, hence $\mathrm{vol}(cv) = c^{-2n}\,\mathrm{vol}(v)$ and $\widehat{\mathrm{vol}}(cv) = \widehat{\mathrm{vol}}(v)$. Since $\zeta_0 = \frac{r}{2}\xi$, $\mathrm{wt}_{\zeta_0}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,o_X}$. This proves (3). □

## 5. Rank one and weight of the minimizer

In this section, we prove that a minimizer of $\widehat{\mathrm{vol}}$ on a symplectic singularity of dimension $2n$ is a positive multiple of $\mathrm{ord}_E$ for a Kollár component $E$ with $A_X(E) \in \{n, 2n\}$ (Theorem 5.4). Proposition 5.1 brings the divisoriality of $\mathrm{wt}_{\xi_v}$ on $W = \operatorname{Spec} \mathrm{gr}_v \mathcal{O}_{X,x}$ back to the germ $(X \ni x)$: the minimizer is $c\,\mathrm{ord}_E$ for a Kollár component $E$ over $(X \ni x)$, and $(E,\Delta_E)$ is isomorphic to the pair $(F,\Delta_F)$ on $W$ defined by the grading.

### 5.1. The associated graded cone and the germ.

In this subsection, we prove Proposition 5.1, which passes from the cone $W$ to the germ $(X \ni x)$, and Lemma 5.2, which shows that Kollár components are preserved by isomorphisms of completions.

**Proposition 5.1.** *Let $(X \ni x)$ be a klt singularity of dimension $d \geq 2$, let $v$ be a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, let $W := \operatorname{Spec} \mathrm{gr}_v \mathcal{O}_{X,x}$ with vertex $o$, and let $\xi_v$ be the Reeb vector on $W$ defined by the grading by the values of $v$ (Theorem 2.20). Assume that the valuation $\mathrm{wt}_{\xi_v}$ on $W$ is divisorial. Then there exist $c>0$ and a Kollár component $E$ over $(X \ni x)$ such that the following hold, where $\zeta$ is the $\mathbb{G}_m$-action defined by the grading of $W$ by the values of $v/c$, $F$ is the Kollár component over $(W \ni o)$ with $\mathrm{ord}_F = \mathrm{wt}_\zeta$, the maps $q : \mathcal{Y} \to Y$ and $q' : \mathcal{Y}' \to Y'$ are the index-one covers of $E$ and $F$, $\mathcal{E} := q^*E$, $\mathcal{E}' := q'^*F$, $L_1 := \mathcal{O}_{\mathcal{Y}}(-\mathcal{E})|_{\mathcal{E}}$, and $L_1' := \mathcal{O}_{\mathcal{Y}'}(-\mathcal{E}')|_{\mathcal{E}'}$.*

1. *We have $v = c\,\mathrm{ord}_E$, the grading of $W$ by the values of $v/c$ is a positive $\mathbb{Z}$-grading induced by the faithful $\mathbb{G}_m$-action $\zeta$, and $\mathrm{wt}_{\xi_v} = c\,\mathrm{ord}_F$.*
2. *There exists an isomorphism of graded rings $\mathrm{gr}_v \mathcal{O}_{X,x} \cong \bigoplus_{m\geq 0} H^0(\mathcal{E}, L_1^m)$, where the left side is graded by the values of $v/c$, which sends the class of $h \in \mathcal{O}_{X,x}$ with $v(h) \geq cm$ to the restriction to $\mathcal{E}$ of the pullback of $h$ to $\mathcal{Y}$, regarded as a section of $\mathcal{O}_{\mathcal{Y}}(-m\mathcal{E})$.*
3. *There exist isomorphisms $\mathcal{E} \cong [(W\backslash\{o\})/\zeta(\mathbb{G}_m)] \cong \mathcal{E}'$, the first one induced by the isomorphism in (2), under which $L_1$ corresponds to $L_1'$ and the homogeneous functions of degree $m$ on $W\backslash\{o\}$ correspond to the sections of $L_1^m$.*
4. *There exists an isomorphism $(E,\Delta_E) \cong (F,\Delta_F)$ under which $-E|_E$ corresponds to $-F|_F$, and $A_X(E) = A_W(F)$.*

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step, we prove (1).

Let $\Gamma := v(\mathbb{C}(X)^*)$ be the value group of $v$. By Theorem 2.20, $(W,\xi_v)$ is a Fano cone with vertex $o$, the torus $T$ acting on $W$ has character lattice $\Gamma$, and for $\lambda \in \Gamma$ the eigenspace of $\mathrm{gr}_v \mathcal{O}_{X,x}$ of character $\lambda$ is $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v)$, with $\langle \lambda, \xi_v\rangle = \lambda$. For any non-zero $h \in \mathcal{O}_{X,x}$, the class of $h$ in $\mathfrak{a}_{v(h)}(v)/\mathfrak{a}_{>v(h)}(v)$ is non-zero. Conversely, if $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) \neq 0$, then $\lambda = v(h)$ for some $h \in \mathcal{O}_{X,x}$. Since $\mathrm{wt}_{\xi_v}(b)$ is the least $\lambda$ such that the homogeneous component of $b$ of degree $\lambda$ is non-zero, we obtain

$$\mathrm{wt}_{\xi_v}\left(\mathrm{gr}_v \mathcal{O}_{X,x}\backslash\{0\}\right) = v\big(\mathcal{O}_{X,x}\backslash\{0\}\big).$$

Any element of $\mathbb{C}(X)^*$, respectively of $\mathbb{C}(W)^*$, is a quotient of two non-zero elements of $\mathcal{O}_{X,x}$, respectively of $\mathrm{gr}_v \mathcal{O}_{X,x}$. Thus the value group of $\mathrm{wt}_{\xi_v}$ is $\Gamma$.

By assumption, $\mathrm{wt}_{\xi_v} = c\,\mathrm{ord}_S$ for a real number $c>0$ and a prime divisor $S$ over $W$. The value group of $\mathrm{ord}_S$ is $\mathbb{Z}$, so $\Gamma = c\mathbb{Z}$. Thus $T$ is a one-dimensional torus, and the grading of $W$ by the values of $v/c$ is a $\mathbb{Z}$-grading induced by a faithful $\mathbb{G}_m$-action $\zeta$ with $\zeta(\mathbb{G}_m) = T$. The grading is positive: $v(h)/c \in \mathbb{Z}_{\geq 0}$ for any non-zero $h \in \mathcal{O}_{X,x}$, and since $\mathrm{center}_X\, v = x$, we have $\mathfrak{a}_{>0}(v) = \mathfrak{m}_x$ and the piece of degree 0 is $\mathcal{O}_{X,x}/\mathfrak{m}_x = \mathbb{C}$. Since $\langle \lambda, \xi_v\rangle = \lambda = c\cdot(\lambda/c)$ for $\lambda \in \Gamma$,

we have $\xi_v = c\zeta$, hence $\mathrm{wt}_{\xi_v} = c\,\mathrm{wt}_\zeta$. The cocharacter $\zeta$ of $T$ is primitive and is a Reeb vector, so by Lemma 2.7 there exists a unique Kollár component $F$ over $(W \ni o)$ with $\mathrm{ord}_F = \mathrm{wt}_\zeta$. In particular, $\mathrm{wt}_{\xi_v} = c\,\mathrm{ord}_F$.

By Theorem 2.20, $v$ is quasi-monomial. Thus there exist a proper birational morphism $\mu : X' \to X$ from a smooth variety $X'$, prime divisors $D_1, \dots, D_k$ on $X'$ whose sum has simple normal crossing support, and $\alpha \in \mathbb{R}^k_{\geq 0}\backslash\{0\}$ such that $v = v_\alpha$ as in Definition 2.5. For a local equation $y_i$ of $D_i$ we have $v(y_i) = \alpha_i$, so $\alpha_i \in \Gamma = c\mathbb{Z}$ for any $i$. Thus $v/c = v_{\alpha/c}$ is a quasi-monomial valuation with integral weights, and its value group $\mathbb{Z}$ has rational rank one, hence $v/c = t\,\mathrm{ord}_E$ for a real number $t > 0$ and a prime divisor $E$ over $X$ [JM12, Remark 3.9]. The value groups of $v/c$ and of $\mathrm{ord}_E$ are both $\mathbb{Z}$, so $t = 1$ and $v = c\,\mathrm{ord}_E$. In particular, $\mathrm{center}_X E = \mathrm{center}_X v = x$. For any $w \in \mathrm{Val}_{X,x}$ and $t' > 0$, we have $A_X(t'w) = t'A_X(w)$ and $\mathrm{vol}(t'w) = t'^{-d}\,\mathrm{vol}(w)$, so $\widehat{\mathrm{vol}}(t'w) = \widehat{\mathrm{vol}}(w)$. Thus $\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, and $E$ is a Kollár component over $(X \ni x)$ by [LX16, Theorem 1.2]. This proves (1).

**Step 2.** In this step, we prove (2).

Let $f : Y \to X$ be the extraction of $E$, and let $i : \mathcal{E} \to \mathcal{Y}$ be the inclusion. For $m \geq 0$ write $\mathfrak{a}_m := \mathfrak{a}_m(\mathrm{ord}_E)$. Since $Y$ is normal, $f$ is proper birational, and $E$ is the only $f$-exceptional prime divisor, a function $h \in \mathcal{O}_{X,x}$ is a section of $f_*\mathcal{O}_Y(-mE)$ if and only if $\mathrm{ord}_E(h) \geq m$, that is,

$$f_*\mathcal{O}_Y(-mE) = \mathfrak{a}_m.$$

Since $v = c\,\mathrm{ord}_E$ and $\mathrm{ord}_E$ takes integer values, $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) = \mathfrak{a}_m/\mathfrak{a}_{m+1}$ if $\lambda = cm$ with $m \in \mathbb{Z}_{\geq 0}$, and $\mathfrak{a}_\lambda(v)/\mathfrak{a}_{>\lambda}(v) = 0$ if $\lambda \notin c\mathbb{Z}$. Thus the piece of degree $m$ of $\mathrm{gr}_v \mathcal{O}_{X,x}$, graded by the values of $v/c$, is $\mathfrak{a}_m/\mathfrak{a}_{m+1}$.

Since $(Y, E)$ is plt and $E$ is $\mathbb{Q}$-Cartier, $Y$ is klt. Since $K_Y = f^*K_X + (A_X(E) - 1)E$, for any $m \geq 0$ we have

$$-(m+1)E - K_Y \equiv_f -(m + A_X(E))E,$$

which is $f$-ample because $A_X(E) > 0$ and $-E$ is $f$-ample. By the relative Kawamata–Viehweg vanishing theorem [KMM87, Theorem 1-2-5], $R^1 f_*\mathcal{O}_Y(-(m+1)E) = 0$. Applying $f_*$ to the exact sequence

$$0 \to \mathcal{O}_Y(-(m+1)E) \to \mathcal{O}_Y(-mE) \to \mathcal{O}_Y(-mE)/\mathcal{O}_Y(-(m+1)E) \to 0,$$

we obtain

$$\mathfrak{a}_m/\mathfrak{a}_{m+1} = H^0\big(Y, \mathcal{O}_Y(-mE)/\mathcal{O}_Y(-(m+1)E)\big) \tag{5.1}$$

for any $m \geq 0$ (cf. [LZ22, Proposition 2.10]).

Since $\mathcal{E}$ is a Cartier divisor on $\mathcal{Y}$ (Lemma 2.3), for any $m \in \mathbb{Z}$ there exists an exact sequence

$$0 \to \mathcal{O}_\mathcal{Y}(-(m+1)\mathcal{E}) \to \mathcal{O}_\mathcal{Y}(-m\mathcal{E}) \to i_*L_1^m \to 0.$$

The Deligne–Mumford stack $\mathcal{Y}$ has finite stabilizers over $\mathbb{C}$, so it is tame and $q_*$ is exact [AOV08, Definition 3.1 and Theorem 3.2]. By the definition of the index-one cover [KM98, Definitions 2.52 and 5.19], on a chart $[U'/\mu_l]$ of $\mathcal{Y}$ the sheaf $\mathcal{O}_Y(-mE)$ is the $\mu_l$-invariant part of $\mathcal{O}_{U'}(-mq^*E)$, that is, $q_*\mathcal{O}_\mathcal{Y}(-m\mathcal{E}) = \mathcal{O}_Y(-mE)$. Thus

$$q_*i_*L_1^m = \mathcal{O}_Y(-mE)/\mathcal{O}_Y(-(m+1)E) \quad \text{for any } m \in \mathbb{Z}. \tag{5.2}$$

For $m = 0$, (5.2) reads $q_*i_*\mathcal{O}_\mathcal{E} = \mathcal{O}_E$. Hence $q$ induces a morphism $p : \mathcal{E} \to E$ with $p_*\mathcal{O}_\mathcal{E} = \mathcal{O}_E$, which is the coarse space of $\mathcal{E}$, and $p_*L_1^m = \mathcal{O}_Y(-mE)/\mathcal{O}_Y(-(m+1)E)$ as sheaves on $E$. By (5.1) and (5.2), for any $m \geq 0$ we have

$$\mathfrak{a}_m/\mathfrak{a}_{m+1} = H^0\big(Y, q_*i_*L_1^m\big) = H^0(\mathcal{E}, L_1^m). \tag{5.3}$$

The equality sends the class of $h \in \mathfrak{a}_m$, that is, of $h \in \mathcal{O}_{X,x}$ with $v(h) \geq cm$, to the class of $f^*h$ in $\mathcal{O}_Y(-mE)/\mathcal{O}_Y(-(m+1)E)$, which is the restriction to $\mathcal{E}$ of $q^*f^*h \in H^0(\mathcal{Y}, \mathcal{O}_\mathcal{Y}(-m\mathcal{E}))$. The multiplications on both sides are induced by products of functions, so the direct sum over $m \geq 0$ of the equalities (5.3) is an isomorphism of graded rings. This proves (2).

**Step 3.** In this step, we prove (3).

For $m \in \mathbb{Z}$, let $(\mathrm{gr}_v \mathcal{O}_{X,x})_m$ denote the piece of degree $m$ for the grading by the values of $v/c$. Since $\mathrm{gr}_v \mathcal{O}_{X,x}$ is finitely generated (Theorem 2.20) and $-E|_E$ is ample (Lemma 2.3), there exists an integer $k > 0$ such that $kE$ is Cartier, $H := \mathcal{O}_Y(-kE)|_E$ is very ample, and $\bigoplus_{j\geq 0}(\mathrm{gr}_v \mathcal{O}_{X,x})_{jk}$ is generated by $(\mathrm{gr}_v \mathcal{O}_{X,x})_k$ as a $\mathbb{C}$-algebra [Bou89, Ch. III, §1, no. 3, Prop. 3]. Since $kE$ is Cartier, $\mathcal{O}_\mathcal{Y}(-k\mathcal{E}) = q^*\mathcal{O}_Y(-kE)$, so $L_1^k = p^*H$. By the projection formula,

$$p_*L_1^{m+jk} = p_*L_1^m \otimes H^j \quad \text{for any } m, j \in \mathbb{Z},$$

and $(\mathrm{gr}_v \mathcal{O}_{X,x})_k = H^0(\mathcal{E}, L_1^k) = H^0(E, H)$ by (2).

We describe $\mathcal{E}$ and $L_1$ on charts. By Definition 2.2, any point of $E$ has an affine open neighborhood $U \subset Y$ such that $\mathcal{Y} \times_Y U = [U'/\mu_l]$, where $l$ is the index of $E$ at the point and $U' \to U$ is the index-one cover of $E|_U$. After shrinking $U$, the ideal of $q^*E$ in $\mathcal{O}_{U'}$ is generated by a function on which $\mu_l$ acts through a character of order $l$ [KM98, Definitions 2.52 and 5.19]. Let $E' \subset U'$ be the preimage of $E$. Then $\mathcal{E} \times_E (E \cap U) = [E'/\mu_l]$, the line bundle $L_1$ is trivial on $E'$, and $\mu_l$ acts on a generator of $L_1|_{E'}$ through a character of order $l$. Hence

$$\mathrm{Spec}_\mathcal{E}\Big(\bigoplus_{m\in\mathbb{Z}} L_1^m\Big) \times_E (E\cap U) = [(E' \times \mathbb{G}_m)/\mu_l], \qquad \mathrm{Spec}_\mathcal{E}\Big(\bigoplus_{m\geq 0} L_1^m\Big) \times_E (E\cap U) = [(E' \times \mathbb{A}^1)/\mu_l],$$

where $\mu_l$ acts on the coordinate of $\mathbb{G}_m$ and of $\mathbb{A}^1$ through a character of order $l$. Since $\mu_l$ acts freely on $E' \times \mathbb{G}_m$, the first stack is the affine scheme $(E' \times \mathbb{G}_m)/\mu_l$ over $E \cap U$, and the coarse space of the second stack is $(E' \times \mathbb{A}^1)/\mu_l$. Taking $\mu_l$-invariants and gluing, we see that $\mathrm{Spec}_\mathcal{E}(\bigoplus_{m\in\mathbb{Z}} L_1^m)$ is the scheme $\mathrm{Spec}_E(\bigoplus_{m\in\mathbb{Z}} p_*L_1^m)$, and that the coarse space of

$$\mathcal{V} := \mathrm{Spec}_\mathcal{E}\Big(\bigoplus_{m\geq 0} L_1^m\Big)$$

is $\mathrm{Spec}_E(\bigoplus_{m\geq 0} p_*L_1^m)$. The stack $\mathcal{V}$ is the total space of $L_1^{-1}$, and we denote by $\mathcal{E}_0 \subset \mathcal{V}$ the zero section of $\mathcal{V} \to \mathcal{E}$, whose ideal is $\bigoplus_{m\geq 1} L_1^m$.

The isomorphism in (2) defines a homomorphism $\mathrm{gr}_v \mathcal{O}_{X,x} \to \bigoplus_{m\in\mathbb{Z}} H^0(E, p_*L_1^m)$, hence a $\mathbb{G}_m$-equivariant morphism

$$\gamma : \mathrm{Spec}_E\Big(\bigoplus_{m\in\mathbb{Z}} p_*L_1^m\Big) \to W,$$

where $\mathbb{G}_m$ acts on $p_*L_1^m$ with weight $m$ and on $W$ through $\zeta$. We show that $\gamma$ is an isomorphism onto $W\backslash\{o\}$. Let $s \in (\mathrm{gr}_v \mathcal{O}_{X,x})_k = H^0(E, H)$ be non-zero, let $E_s \subset E$ be the open subset where $s$ does not vanish, and let $D(s) \subset W$ be the open subset where $s$ does not vanish. Since $H$ is very ample, $E_s$ is affine, and $\gamma^{-1}(D(s))$ is the preimage of $E_s$, whose ring of functions is $\bigoplus_{m\in\mathbb{Z}} H^0(E_s, p_*L_1^m)$. On the other hand, the piece of degree $m$ of $\mathrm{gr}_v \mathcal{O}_{X,x}[s^{-1}]$ is

$$\varinjlim_j (\mathrm{gr}_v \mathcal{O}_{X,x})_{m+jk} = \varinjlim_j H^0(E, p_*L_1^m \otimes H^j) = H^0(E_s, p_*L_1^m),$$

where the transition maps are multiplication by $s$ and the last equality holds by [Har77, Lemma II.5.14] applied to the coherent sheaf $p_*L_1^m$. Thus $\gamma$ induces an isomorphism from the preimage of $E_s$ onto $D(s)$. Since $H$ is very ample, the sets $E_s$ cover $E$. For a homogeneous element $b$ of degree $m > 0$, the power $b^k$ has degree $mk$, so $b^k$ belongs to the ideal generated by $(\mathrm{gr}_v \mathcal{O}_{X,x})_k$ by the choice of $k$. Thus the radical of the ideal generated by $(\mathrm{gr}_v \mathcal{O}_{X,x})_k$ is $\bigoplus_{m>0}(\mathrm{gr}_v \mathcal{O}_{X,x})_m$, which is the ideal of $o$, and the sets $D(s)$ cover $W\backslash\{o\}$. Hence $\gamma$ is an isomorphism onto $W\backslash\{o\}$.

The morphism $\mathrm{Spec}_\mathcal{E}(\bigoplus_{m\in\mathbb{Z}} L_1^m) \to \mathcal{E}$ is a $\mathbb{G}_m$-torsor, and $\gamma$ is $\mathbb{G}_m$-equivariant. Thus $\gamma$ induces an isomorphism

$$\mathcal{E} \cong \Big[\mathrm{Spec}_\mathcal{E}\Big(\bigoplus_{m\in\mathbb{Z}} L_1^m\Big)\Big/\mathbb{G}_m\Big] \cong [(W\backslash\{o\})/\zeta(\mathbb{G}_m)],$$

under which the sections of $L_1^m$ correspond to the homogeneous functions of degree $m$ on $W\backslash\{o\}$. Since $\gamma$ is defined by the isomorphism in (2), the isomorphism $\mathcal{E} \cong [(W\backslash\{o\})/\zeta(\mathbb{G}_m)]$ is induced by the isomorphism in (2).

Via $\gamma$, the quotient $(W\backslash\{o\})/\zeta(\mathbb{G}_m)$ is $E$, and the pushforward of $\mathcal{O}_{W\backslash\{o\}}$ to $E$ is $\bigoplus_{m\in\mathbb{Z}} p_*L_1^m$. By [XZ26b, Paragraph 2.13] (see also the proof of Lemma 2.7), the extraction of the Kollár component over $(W \ni o)$ whose valuation is $\mathrm{wt}_\zeta$ is

$$\mathrm{Spec}_E\Big(\bigoplus_{m\geq 0} p_*L_1^m\Big) \to W,$$

and the Kollár component is its zero section. Since $F$ is the unique Kollár component with $\mathrm{ord}_F = \mathrm{wt}_\zeta$ (Lemma 2.7), the extraction $g : Y' \to W$ of $F$ satisfies $Y' = \mathrm{Spec}_E(\bigoplus_{m\geq 0} p_*L_1^m)$, and $F$ is the zero section of $Y' \to E$. In particular, $Y'$ is the coarse space of $\mathcal{V}$, and $F$ is the coarse space of $\mathcal{E}_0$.

We show that $\mathcal{V} \to Y'$ is the index-one cover of $F$. On the chart above, $\mathcal{V} \times_E (E \cap U) = [(E' \times \mathbb{A}^1)/\mu_l]$. Since $Y$ is normal, $E$ is Cartier at the generic point of $E$, so $q$ is an isomorphism over the generic point of $E$ and $\mu_l$ acts freely on a dense open subset of $E'$. Together with the free action on $E' \times \mathbb{G}_m$, this shows that $E' \times \mathbb{A}^1 \to (E' \times \mathbb{A}^1)/\mu_l$ is étale in codimension one. The preimage of $F$ is the Cartier divisor $E' \times \{0\}$, defined by the coordinate of $\mathbb{A}^1$, on which $\mu_l$ acts through a character of order $l$. Hence $E' \times \mathbb{A}^1 \to (E' \times \mathbb{A}^1)/\mu_l$ is the index-one cover of $F$ over $(E' \times \mathbb{A}^1)/\mu_l$ [KM98, Definitions 2.52 and 5.19], and $l$ is the index of $F$ at the corresponding points. Gluing, we obtain an isomorphism $(\mathcal{Y}', \mathcal{E}') \cong (\mathcal{V}, \mathcal{E}_0)$ over $Y'$. Since the ideal of $\mathcal{E}_0$ is $\bigoplus_{m\geq 1} L_1^m$, we have

$$L_1' = \mathcal{O}_{\mathcal{V}}(-\mathcal{E}_0)|_{\mathcal{E}_0} = L_1$$

under the isomorphism $\mathcal{E}_0 \cong \mathcal{E}$ given by the zero section. Thus $\mathcal{E} \cong \mathcal{E}_0 \cong \mathcal{E}'$, and $L_1$ corresponds to $L_1'$. Composing with the isomorphism $\mathcal{E} \cong [(W\backslash\{o\})/\zeta(\mathbb{G}_m)]$, we obtain (3).

**Step 4.** In this step, we prove (4).

Let $p' : \mathcal{E}' \to F$ be the coarse space. By **Step 3**, the isomorphism $\mathcal{E} \cong \mathcal{E}'$ induces the isomorphism $E \cong F$ given by the zero section of $Y' \to E$, and it is compatible with $p$ and $p'$. Let $P$ be a prime divisor on $E$, and let $P'$ be the corresponding prime divisor on $F$. The coefficients of $\Delta_E$ along $P$ and of $\Delta_F$ along $P'$ are $1 - 1/b_P$ and $1 - 1/b_{P'}$, where $b_P$ and $b_{P'}$ are the indices of $E$ and of $F$ at the generic points of $P$ and of $P'$ [Kol13, Section 4.1], which by the description of the index-one covers $q$ and $q'$ in **Step 3** are the orders of the stabilizers of $\mathcal{E}$ and of $\mathcal{E}'$ at the points over the generic points of $P$ and of $P'$, respectively. Since $\mathcal{E} \cong \mathcal{E}'$ over $E \cong F$, we have $b_P = b_{P'}$. Thus $(E, \Delta_E) \cong (F, \Delta_F)$.

By **Step 3**, the index of $F$ at any point is the index of $E$ at the corresponding point, so $kF$ is Cartier. Let $H' := \mathcal{O}_{Y'}(-kF)|_F$. Then $H = p_*L_1^k$ and $H' = p'_*L_1'^k$, so $H$ corresponds to $H'$ under $E \cong F$, because $L_1$ corresponds to $L_1'$. Since $-E|_E \sim_{\mathbb{Q}} \frac{1}{k}H$ and $-F|_F \sim_{\mathbb{Q}} \frac{1}{k}H'$, the class $-E|_E$ corresponds to $-F|_F$.

It remains to prove $A_X(E) = A_W(F)$. Since $K_X$ is $\mathbb{Q}$-Cartier, a Cartier multiple of $K_X$ is trivial on a neighborhood of $x$, so $(f^*K_X)|_E \sim_{\mathbb{Q}} 0$. Hence, by adjunction,

$$K_E + \Delta_E = (K_Y + E)|_E = \big(f^*K_X + A_X(E)E\big)|_E \sim_{\mathbb{Q}} -\frac{A_X(E)}{k}H.$$

Since a Cartier multiple of $K_W$ is trivial on a neighborhood of $o$, by adjunction on $Y'$ we have

$$K_F + \Delta_F = \big(g^*K_W + A_W(F)F\big)|_F \sim_{\mathbb{Q}} -\frac{A_W(F)}{k}H'.$$

Since $\dim E = d-1$ and $H$ is ample, $(H^{d-1}) > 0$. Using the isomorphism $(E, \Delta_E) \cong (F, \Delta_F)$, under which $H$ corresponds to $H'$, we obtain

$$\begin{aligned}\frac{A_X(E)}{k}(H^{d-1}) &= -\big((K_E + \Delta_E)\cdot H^{d-2}\big) = -\big((K_F + \Delta_F)\cdot H'^{d-2}\big)\\ &= \frac{A_W(F)}{k}(H'^{d-1}) = \frac{A_W(F)}{k}(H^{d-1}).\end{aligned}$$

Thus $A_X(E) = A_W(F)$, which proves (4). □

Lemma 5.2 is used in Theorem 5.4 and Theorem 1.2 for $n = 1$ and in Corollary 1.3, where it carries Kollár components along isomorphisms of completions.

**Lemma 5.2.** *Let $(X_1 \ni x_1)$ and $(X_2 \ni x_2)$ be klt singularities of dimension $d \geq 2$, and let $\phi : \widehat{\mathcal{O}}_{X_1,x_1} \to \widehat{\mathcal{O}}_{X_2,x_2}$ be an isomorphism of $\mathbb{C}$-algebras. Let $E_1$ be a Kollár component over $(X_1 \ni x_1)$ with extraction $f_1 : Y_1 \to X_1$. Then there exists a Kollár component $E_2$ over $(X_2 \ni x_2)$ with extraction $f_2 : Y_2 \to X_2$ such that the following hold.*

*(1) There exists an integer $k > 0$ such that $I := \mathfrak{a}_k(\mathrm{ord}_{E_1})$ satisfies $\mathfrak{a}_{jk}(\mathrm{ord}_{E_1}) = I^j$ for any $j \geq 0$ and $Y_1 = \mathrm{Bl}_I X_1$, and the ideal $J := \phi(I\widehat{\mathcal{O}}_{X_1,x_1}) \cap \mathcal{O}_{X_2,x_2}$ satisfies $J\widehat{\mathcal{O}}_{X_2,x_2} = \phi(I\widehat{\mathcal{O}}_{X_1,x_1})$, $Y_2 = \mathrm{Bl}_J X_2$, and $\mathfrak{a}_{jk}(\mathrm{ord}_{E_2}) = J^j$ for any $j \geq 0$.*

*(2) The isomorphism $\phi$ induces an isomorphism of the completions of $(Y_1, E_1)$ and $(Y_2, E_2)$ along the fibers over $x_1$ and $x_2$. Moreover, there exists an isomorphism $(E_1, \Delta_{E_1}) \cong (E_2, \Delta_{E_2})$ under which $-E_1|_{E_1}$ corresponds to $-E_2|_{E_2}$, we have $A_{X_1}(E_1) = A_{X_2}(E_2)$ and $\mathrm{vol}(\mathrm{ord}_{E_1}) = \mathrm{vol}(\mathrm{ord}_{E_2})$, and $\phi$ induces an isomorphism of graded $\mathbb{C}$-algebras $\mathrm{gr}_{\mathrm{ord}_{E_1}} \mathcal{O}_{X_1,x_1} \cong \mathrm{gr}_{\mathrm{ord}_{E_2}} \mathcal{O}_{X_2,x_2}$.*

*(3) If $\mathrm{ord}_{E_1}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_1,x_1}$, then $\mathrm{ord}_{E_2}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_2,x_2}$.*

*Proof.* The proof is long, so we divide it into several steps. For $i = 1, 2$, write $\mathcal{O}_i := \mathcal{O}_{X_i,x_i}$ with maximal ideal $\mathfrak{m}_i$, and $\widehat{\mathcal{O}}_i := \widehat{\mathcal{O}}_{X_i,x_i}$ with maximal ideal $\widehat{\mathfrak{m}}_i = \mathfrak{m}_i\widehat{\mathcal{O}}_i$. We identify an $\mathfrak{m}_i$-primary ideal $\mathfrak{b} \subset \mathcal{O}_i$ with the ideal sheaf $\mathcal{I}$ on $X_i$ such that $\mathcal{I}_{x_i} = \mathfrak{b}$ and $\mathcal{I}|_{X_i\setminus\{x_i\}} = \mathcal{O}_{X_i\setminus\{x_i\}}$. For such $\mathfrak{b}$ we have $\mathcal{O}_i/\mathfrak{b} = \widehat{\mathcal{O}}_i/\mathfrak{b}\widehat{\mathcal{O}}_i$ and $\mathfrak{b}\widehat{\mathcal{O}}_i \cap \mathcal{O}_i = \mathfrak{b}$, and $\widehat{\mathcal{O}}_i$ is faithfully flat over $\mathcal{O}_i$ [Mat89, Theorems 8.7, 8.8 and 8.14]. Since $\phi$ is an isomorphism of local rings, $\phi(\widehat{\mathfrak{m}}_1) = \widehat{\mathfrak{m}}_2$.

**Step 1.** In this step, we construct $k$ and $I$.

For $m \geq 0$, since $Y_1$ is normal, $f_1$ is proper birational, and $E_1$ is the only $f_1$-exceptional prime divisor, we have $f_{1*}\mathcal{O}_{Y_1}(-mE_1) = \mathfrak{a}_m(\mathrm{ord}_{E_1})$. For $m > 0$, the ideal $\mathfrak{a}_m(\mathrm{ord}_{E_1})$ is $\mathfrak{m}_1$-primary because $\mathrm{center}_{X_1} E_1 = x_1$. Let $k_0 > 0$ be an integer such that $k_0E_1$ is Cartier. Then $\mathcal{O}_{Y_1}(-k_0E_1)$ is $f_1$-ample, and $\mathcal{O}_{Y_1}(-(jk_0 + r)E_1) = \mathcal{O}_{Y_1}(-rE_1) \otimes \mathcal{O}_{Y_1}(-k_0E_1)^{\otimes j}$ for $j \geq 0$ and $0 \leq r < k_0$. Choose $c > 0$ such that $\mathcal{O}_{Y_1}(-ck_0E_1)$ is relatively very ample. Shrinking $X_1$ around $x_1$, we may assume that $X_1 = \mathrm{Spec}\, R$ is affine, and we embed $Y_1$ into a projective space over $X_1$ by $\mathcal{O}_{Y_1}(-ck_0E_1)$. By [Har77, Theorem III.5.2(a) and Exercise II.5.9(b)], for any coherent sheaf $\mathcal{F}$ on $Y_1$ the module $\bigoplus_{j\geq 0} H^0(Y_1, \mathcal{F} \otimes \mathcal{O}_{Y_1}(-k_0E_1)^{\otimes j})$ is finitely generated over the finitely generated $R$-algebra $\bigoplus_{j\geq 0} H^0(Y_1, \mathcal{O}_{Y_1}(-jk_0E_1))$ (apply the theorem to the finitely many sheaves $\mathcal{F} \otimes \mathcal{O}_{Y_1}(-tk_0E_1)$, $0 \leq t < c$, and to $\mathcal{F} = \mathcal{O}_{Y_1}$). Taking $\mathcal{F} = \mathcal{O}_{Y_1}(-rE_1)$, $0 \leq r < k_0$, and localizing at $x_1$, we see that each $\bigoplus_{j\geq 0} \mathfrak{a}_{jk_0+r}(\mathrm{ord}_{E_1})$ is a finitely generated module over the finitely generated $\mathcal{O}_1$-algebra $A := \bigoplus_{j\geq 0} \mathfrak{a}_{jk_0}(\mathrm{ord}_{E_1})$, so $\bigoplus_{m\geq 0} \mathfrak{a}_m(\mathrm{ord}_{E_1})$ is a finitely generated $\mathcal{O}_1$-algebra. By [Bou89, Ch. III, §1, no. 3, Prop. 3], there exists an integer $k > 0$ divisible by $k_0$ such that $\bigoplus_{j\geq 0} \mathfrak{a}_{jk}(\mathrm{ord}_{E_1})$ is generated by $I := \mathfrak{a}_k(\mathrm{ord}_{E_1})$, that is, $\mathfrak{a}_{jk}(\mathrm{ord}_{E_1}) = I^j$ for any $j \geq 0$. Since $\mathcal{O}_{Y_1}(-kE_1)$ is $f_1$-ample and $f_{1*}\mathcal{O}_{Y_1}(-jkE_1) = I^j$ for any $j \geq 0$, we have

$$Y_1 = \mathrm{Proj}_{X_1} \bigoplus_{j\geq 0} I^j = \mathrm{Bl}_I X_1,$$

and $I\mathcal{O}_{Y_1} = \mathcal{O}_{Y_1}(-kE_1)$. Since $I$ is $\mathfrak{m}_1$-primary, we may fix an integer $N \geq 1$ such that $\mathfrak{m}_1^N \subset I$.

**Step 2.** In this step, we construct $J$ and $Y_2$, and we prove that $\phi$ induces an isomorphism of the completions of $Y_1$ and $Y_2$ along the fibers over $x_1$ and $x_2$.

Since $\widehat{\mathfrak{m}}_1^N \subset I\widehat{\mathcal{O}}_1$, we have $\widehat{\mathfrak{m}}_2^N \subset \phi(I\widehat{\mathcal{O}}_1)$. Since $\mathcal{O}_2/\mathfrak{m}_2^N = \widehat{\mathcal{O}}_2/\widehat{\mathfrak{m}}_2^N$, the ideals of $\widehat{\mathcal{O}}_2$ containing $\widehat{\mathfrak{m}}_2^N$ correspond to the ideals of $\mathcal{O}_2$ containing $\mathfrak{m}_2^N$, via $\mathfrak{c} \to \mathfrak{c} \cap \mathcal{O}_2$ and $\mathfrak{b} \to \mathfrak{b}\widehat{\mathcal{O}}_2$. Thus $J = \phi(I\widehat{\mathcal{O}}_1) \cap \mathcal{O}_2$ satisfies $\mathfrak{m}_2^N \subset J$ and $J\widehat{\mathcal{O}}_2 = \phi(I\widehat{\mathcal{O}}_1)$. Let $f_2 : Y_2 := \mathrm{Bl}_J X_2 \to X_2$. Then $f_2$ is projective birational and is an isomorphism over $X_2 \backslash \{x_2\}$. Since completion is flat, $I^j \otimes_{\mathcal{O}_1} \widehat{\mathcal{O}}_1 = (I\widehat{\mathcal{O}}_1)^j$ and $J^j \otimes_{\mathcal{O}_2} \widehat{\mathcal{O}}_2 = (J\widehat{\mathcal{O}}_2)^j = \phi((I\widehat{\mathcal{O}}_1)^j)$ for any $j \geq 0$. Thus $\phi$ induces an isomorphism of the Rees algebras $\bigoplus_{j\geq 0}(I\widehat{\mathcal{O}}_1)^j$ and $\bigoplus_{j\geq 0}(J\widehat{\mathcal{O}}_2)^j$, hence an isomorphism

$$Y_1 \times_{X_1} \operatorname{Spec} \widehat{\mathcal{O}}_1 \cong Y_2 \times_{X_2} \operatorname{Spec} \widehat{\mathcal{O}}_2 \tag{5.4}$$

over $\operatorname{Spec} \phi$, under which $I\mathcal{O}$ corresponds to $J\mathcal{O}$. Since $\mathcal{O}_i/\mathfrak{m}_i^n = \widehat{\mathcal{O}}_i/\widehat{\mathfrak{m}}_i^n$ for any $n \geq 1$, the subschemes $f_i^{-1}(\operatorname{Spec} \mathcal{O}_i/\mathfrak{m}_i^n)$ of $Y_i$ are the corresponding subschemes of $Y_i \times_{X_i} \operatorname{Spec} \widehat{\mathcal{O}}_i$. Thus (5.4) induces an isomorphism of the completions of $Y_1$ and $Y_2$ along $f_1^{-1}(x_1)$ and $f_2^{-1}(x_2)$, which maps the reduced fiber $E_1$ of $f_1$ over $x_1$ to the reduced fiber of $f_2$ over $x_2$. We let $E_2$ be the reduced fiber of $f_2$ over $x_2$.

**Step 3.** In this step, we construct a common étale neighborhood of $x_1$ and $x_2$ over which $Y_1$ and $Y_2$ become isomorphic.

Let $M := N + 1$, so that $M \geq 2$ and $M \geq N$. The composition $\mathcal{O}_2 \to \widehat{\mathcal{O}}_2 \xrightarrow{\phi^{-1}} \widehat{\mathcal{O}}_1$ defines a morphism $\operatorname{Spec} \widehat{\mathcal{O}}_1 \to X_2$. By [Art69, Corollary (2.5)], there exist an étale neighborhood $(U, u)$ of $x_1$ in $X_1$ with residue field $k(u) = \mathbb{C}$ and a morphism $h : U \to X_2$ such that the induced homomorphism $\hat{h} : \widehat{\mathcal{O}}_2 \to \widehat{\mathcal{O}}_{U,u} = \widehat{\mathcal{O}}_1$ satisfies $\hat{h} \equiv \phi^{-1} \pmod{\widehat{\mathfrak{m}}_1^M}$. Here we identify $\widehat{\mathcal{O}}_{U,u}$ with $\widehat{\mathcal{O}}_1$ via the étale morphism $U \to X_1$, which is possible since $k(u) = \mathbb{C}$. In particular, $h(u) = x_2$. The endomorphism $\phi \circ \hat{h}$ of $\widehat{\mathcal{O}}_2$ is the identity modulo $\widehat{\mathfrak{m}}_2^M$, so it induces the identity on $\widehat{\mathfrak{m}}_2/\widehat{\mathfrak{m}}_2^2$. Since $\widehat{\mathcal{O}}_2$ is complete, $\phi \circ \hat{h}$ is surjective, and a surjective endomorphism of a Noetherian ring is injective. Thus $\hat{h}$ is an isomorphism, and $h$ is étale at $u$ [Gro67, Proposition 17.6.3]. Shrinking $U$, we may assume that $h$ is étale and that $u$ is the only point of $U$ over $x_1$ and the only point of $U$ over $x_2$.

Let $\psi := \hat{h}^{-1} : \widehat{\mathcal{O}}_1 \to \widehat{\mathcal{O}}_2$. Then $\psi \equiv \phi \pmod{\widehat{\mathfrak{m}}_2^M}$. Both $\psi(I\widehat{\mathcal{O}}_1)$ and $\phi(I\widehat{\mathcal{O}}_1)$ contain $\widehat{\mathfrak{m}}_2^N$, and their images in $\widehat{\mathcal{O}}_2/\widehat{\mathfrak{m}}_2^N$ coincide because $M \geq N$. Thus $\psi(I\widehat{\mathcal{O}}_1) = \phi(I\widehat{\mathcal{O}}_1) = J\widehat{\mathcal{O}}_2$. Applying $\hat{h}$, we see that $I\widehat{\mathcal{O}}_{U,u} = J\widehat{\mathcal{O}}_{U,u}$, both equal to $I\widehat{\mathcal{O}}_1$ under the identification $\widehat{\mathcal{O}}_{U,u} = \widehat{\mathcal{O}}_1$. Since $\widehat{\mathcal{O}}_{U,u}$ is faithfully flat over $\mathcal{O}_{U,u}$, we have $I\mathcal{O}_{U,u} = J\mathcal{O}_{U,u}$. Both $I\mathcal{O}_U$ and $J\mathcal{O}_U$ are trivial outside $u$, so $I\mathcal{O}_U = J\mathcal{O}_U$. Since $U$ is flat over $X_1$ and over $X_2$, we have $I^j\mathcal{O}_U = I^j \otimes_{\mathcal{O}_{X_1}} \mathcal{O}_U$ and $J^j\mathcal{O}_U = J^j \otimes_{\mathcal{O}_{X_2}} \mathcal{O}_U$ for any $j \geq 0$. Thus

$$Y_U := Y_1 \times_{X_1} U = \mathrm{Bl}_{I\mathcal{O}_U} U = Y_2 \times_{X_2} U.$$

Let $p_i : Y_U \to Y_i$ be the projections and $f_U : Y_U \to U$ the blow-up. Then $p_1$ and $p_2$ are étale. Since $k(u) = \mathbb{C}$, the fiber $f_U^{-1}(u)$ maps isomorphically onto $f_i^{-1}(x_i)$ for $i = 1, 2$, so the image of $p_2$ is an open subset of $Y_2$ containing $f_2^{-1}(x_2)$.

**Step 4.** In this step, we prove that $E_2$ is a Kollár component with extraction $f_2$, and we prove (1).

Let $E_U$ be the reduced fiber of $f_U$ over $u$. By **Step 3**, $p_i$ induces an isomorphism $E_U \cong E_i$ for $i = 1, 2$. Since $u$ is the only point of $U$ over $x_i$ and $p_i$ is étale, $E_U = p_i^{-1}(E_i)$ as reduced subschemes. In particular, $E_2 \cong E_1$ is a normal projective variety with $\dim E_2 = d - 1$. Since $Y_U$ is étale over the normal variety $Y_1$, $Y_U$ is normal, and hence the image of $p_2$ is normal [Mat89, Corollary to Theorem 23.9]. Since $Y_2$ is isomorphic to $X_2$ outside $f_2^{-1}(x_2)$, $Y_2$ is normal, and $E_2 = f_2^{-1}(x_2)_{\mathrm{red}}$ is the only $f_2$-exceptional prime divisor.

The ideal $J\mathcal{O}_{Y_2}$ is invertible and defines a Cartier divisor $D_2$ with $\operatorname{Supp} D_2 = E_2$. We have

$$p_2^*\mathcal{O}_{Y_2}(-D_2) = J\mathcal{O}_{Y_U} = I\mathcal{O}_{Y_U} = p_1^*\mathcal{O}_{Y_1}(-kE_1) = \mathcal{O}_{Y_U}(-kE_U).$$

Since $p_2$ is étale, the ramification index of $p_2$ along $E_U$ is 1, so $D_2 = kE_2$. Thus $kE_2$ is Cartier, and $\mathcal{O}_{Y_2}(-kE_2) = J\mathcal{O}_{Y_2}$ is $f_2$-ample. Since $K_{X_2}$ is $\mathbb{Q}$-Cartier and $K_{Y_2} - f_2^*K_{X_2}$ is supported on $E_2$, the divisor $K_{Y_2} + E_2$ is $\mathbb{Q}$-Cartier.

We show that $(Y_2, E_2)$ is plt. The base change of a log resolution along an étale morphism is a log resolution, and the relative log canonical divisor of the base change is the pullback of the relative log canonical divisor of the log resolution, since étale morphisms have ramification index 1 along any prime divisor. Since $(Y_1, E_1)$ is plt and $p_1$ is étale, $(Y_U, E_U)$ is plt. Since $(Y_U, E_U)$ is plt and the base change along $p_2$ of a log resolution of $(Y_2, E_2)$ is a log resolution of $(Y_U, E_U)$ whose relative log canonical divisor is the pullback of that of the log resolution of $(Y_2, E_2)$, $(Y_2, E_2)$ is plt over the image of $p_2$. Since $Y_2 \backslash f_2^{-1}(x_2) \cong X_2 \backslash \{x_2\}$ is klt and does not meet $E_2$, $(Y_2, E_2)$ is plt. Thus $E_2$ is a Kollár component over $(X_2 \ni x_2)$ with extraction $f_2$.

For $m \geq 0$, since $Y_2$ is normal and $E_2$ is the only $f_2$-exceptional prime divisor, we have $f_{2*}\mathcal{O}_{Y_2}(-mE_2) = \mathfrak{a}_m(\mathrm{ord}_{E_2})$. Since $p_2^*\mathcal{O}_{Y_2}(-jkE_2) = \mathcal{O}_{Y_U}(-jkE_U) = p_1^*\mathcal{O}_{Y_1}(-jkE_1)$, flat base change [Har77, Proposition III.9.3] implies

$$\mathfrak{a}_{jk}(\mathrm{ord}_{E_2})\mathcal{O}_U = f_{U*}\mathcal{O}_{Y_U}(-jkE_U) = \mathfrak{a}_{jk}(\mathrm{ord}_{E_1})\mathcal{O}_U = I^j\mathcal{O}_U = J^j\mathcal{O}_U$$

for any $j \geq 0$. Since $\mathcal{O}_{U,u}$ is faithfully flat over $\mathcal{O}_2$, we have $\mathfrak{a}_{jk}(\mathrm{ord}_{E_2}) = J^j$ for any $j \geq 0$. Together with **Steps 1 and 2**, this proves (1).

**Step 5.** In this step, we prove (2) and (3).

By **Step 2**, $\phi$ induces an isomorphism of the completions of $(Y_1, E_1)$ and $(Y_2, E_2)$ along the fibers over $x_1$ and $x_2$.

Let $P_U$ be a prime divisor on $E_U$, and let $P_i := p_i(P_U)$, a prime divisor on $E_i$. The coefficient of $\Delta_{E_i}$ along $P_i$ is $1 - 1/b_i$, where $b_i$ is the index of $E_i$ at the generic point of $P_i$ [Kol13, Section 4.1]. Since $p_i$ is étale and $E_U = p_i^*E_i$, the sheaf $\mathcal{O}_{Y_U}(lE_U)$ is the pullback of $\mathcal{O}_{Y_i}(lE_i)$ for any $l \in \mathbb{Z}$, and invertibility of a coherent sheaf descends along faithfully flat morphisms. Thus $b_i$ is the index of $E_U$ at the generic point of $P_U$ for $i = 1, 2$, so $b_1 = b_2$. Hence the isomorphism $E_1 \cong E_U \cong E_2$ of **Step 4** induces $(E_1, \Delta_{E_1}) \cong (E_2, \Delta_{E_2})$, under which $\mathcal{O}_{Y_1}(-kE_1)|_{E_1}$ and $\mathcal{O}_{Y_2}(-kE_2)|_{E_2}$ correspond, since both pull back to $\mathcal{O}_{Y_U}(-kE_U)|_{E_U}$. Thus $-E_1|_{E_1}$ corresponds to $-E_2|_{E_2}$.

Since $p_i$ and $U \to X_i$ are étale and $E_U = p_i^*E_i$, pulling back $K_{Y_i} = f_i^*K_{X_i} + (A_{X_i}(E_i) - 1)E_i$ along $p_i$ implies

$$K_{Y_U} = f_U^*K_U + (A_{X_i}(E_i) - 1)E_U \quad \text{for } i = 1, 2.$$

Thus $A_{X_1}(E_1) = A_{X_2}(E_2)$.

Write $I_1 := I$ and $I_2 := J$. For $m \geq 0$, $b \in \mathcal{O}_i$ satisfies $\mathrm{ord}_{E_i}(b) \geq m$ if and only if $\mathrm{ord}_{E_i}(b^k) \geq mk$, so by (1),

$$\mathfrak{a}_m(\mathrm{ord}_{E_i}) = \{b \in \mathcal{O}_i \mid b^k \in I_i^m\}.$$

We claim that

$$\mathfrak{a}_m(\mathrm{ord}_{E_i})\widehat{\mathcal{O}}_i = \{g \in \widehat{\mathcal{O}}_i \mid g^k \in I_i^m\widehat{\mathcal{O}}_i\}. \tag{5.5}$$

For $m = 0$ both sides are $\widehat{\mathcal{O}}_i$, so we may assume that $m > 0$. Since $I_i^m$ is $\mathfrak{m}_i$-primary, there exists $n \geq 1$ such that $\widehat{\mathfrak{m}}_i^n \subset I_i^m\widehat{\mathcal{O}}_i$. Let $g \in \widehat{\mathcal{O}}_i$. Since $\mathcal{O}_i/\mathfrak{m}_i^n = \widehat{\mathcal{O}}_i/\widehat{\mathfrak{m}}_i^n$, we may write $g = b + e$ with $b \in \mathcal{O}_i$ and $e \in \widehat{\mathfrak{m}}_i^n$. Then $g^k - b^k \in \widehat{\mathfrak{m}}_i^n \subset I_i^m\widehat{\mathcal{O}}_i$, and $e \in I_i^m\widehat{\mathcal{O}}_i \subset \mathfrak{a}_m(\mathrm{ord}_{E_i})\widehat{\mathcal{O}}_i$ because $I_i^m = \mathfrak{a}_{mk}(\mathrm{ord}_{E_i}) \subset \mathfrak{a}_m(\mathrm{ord}_{E_i})$. Since $I_i^m\widehat{\mathcal{O}}_i \cap \mathcal{O}_i = I_i^m$ and $\mathfrak{a}_m(\mathrm{ord}_{E_i})\widehat{\mathcal{O}}_i \cap \mathcal{O}_i = \mathfrak{a}_m(\mathrm{ord}_{E_i})$, we obtain

$$g^k \in I_i^m\widehat{\mathcal{O}}_i \iff b^k \in I_i^m \iff b \in \mathfrak{a}_m(\mathrm{ord}_{E_i}) \iff g \in \mathfrak{a}_m(\mathrm{ord}_{E_i})\widehat{\mathcal{O}}_i,$$

which proves (5.5). Since $\phi(I^m\widehat{\mathcal{O}}_1) = J^m\widehat{\mathcal{O}}_2$ by (1), (5.5) implies

$$\phi\big(\mathfrak{a}_m(\mathrm{ord}_{E_1})\widehat{\mathcal{O}}_1\big) = \mathfrak{a}_m(\mathrm{ord}_{E_2})\widehat{\mathcal{O}}_2 \quad \text{for any } m \geq 0.$$

For $m > 0$, $\mathfrak{a}_m(\mathrm{ord}_{E_i})$ is $\mathfrak{m}_i$-primary, so $\mathcal{O}_i/\mathfrak{a}_m(\mathrm{ord}_{E_i}) = \widehat{\mathcal{O}}_i/\mathfrak{a}_m(\mathrm{ord}_{E_i})\widehat{\mathcal{O}}_i$. Thus $\phi$ induces isomorphisms of $\mathbb{C}$-algebras

$$\mathcal{O}_1/\mathfrak{a}_m(\mathrm{ord}_{E_1}) \cong \mathcal{O}_2/\mathfrak{a}_m(\mathrm{ord}_{E_2}) \quad \text{for any } m \geq 1. \tag{5.6}$$

Taking lengths in (5.6), we obtain $\ell(\mathcal{O}_1/\mathfrak{a}_m(\mathrm{ord}_{E_1})) = \ell(\mathcal{O}_2/\mathfrak{a}_m(\mathrm{ord}_{E_2}))$ for any $m \geq 1$, hence $\mathrm{vol}(\mathrm{ord}_{E_1}) = \mathrm{vol}(\mathrm{ord}_{E_2})$. Since $\mathrm{ord}_{E_i}$ takes integer values,

$$\mathrm{gr}_{\mathrm{ord}_{E_i}} \mathcal{O}_i = \bigoplus_{m\geq 0} \mathfrak{a}_m(\mathrm{ord}_{E_i})/\mathfrak{a}_{m+1}(\mathrm{ord}_{E_i}),$$

and $\mathfrak{a}_m(\mathrm{ord}_{E_i})/\mathfrak{a}_{m+1}(\mathrm{ord}_{E_i})$ is the kernel of $\mathcal{O}_i/\mathfrak{a}_{m+1}(\mathrm{ord}_{E_i}) \to \mathcal{O}_i/\mathfrak{a}_m(\mathrm{ord}_{E_i})$. Thus the isomorphisms (5.6) induce $\mathbb{C}$-linear isomorphisms

$$\mathfrak{a}_m(\mathrm{ord}_{E_1})/\mathfrak{a}_{m+1}(\mathrm{ord}_{E_1}) \cong \mathfrak{a}_m(\mathrm{ord}_{E_2})/\mathfrak{a}_{m+1}(\mathrm{ord}_{E_2})$$

compatible with multiplication, that is, an isomorphism of graded $\mathbb{C}$-algebras $\mathrm{gr}_{\mathrm{ord}_{E_1}} \mathcal{O}_{X_1,x_1} \cong \mathrm{gr}_{\mathrm{ord}_{E_2}} \mathcal{O}_{X_2,x_2}$. This proves (2).

Assume that $\mathrm{ord}_{E_1}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_1,x_1}$. By [LX16, Theorem 1.2], $(E_1, \Delta_{E_1})$ is K-semistable. By (2), $(E_2, \Delta_{E_2}) \cong (E_1, \Delta_{E_1})$ is K-semistable. Since $E_2$ is a Kollár component, [LX16, Theorem 1.2] implies that $\mathrm{ord}_{E_2}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_2,x_2}$. This proves (3). □

5.2. **Proof of the rank one theorem.** In this subsection, we prove Theorem 5.4. We first prove a lemma on symplectic singularities of dimension 2, which is also used in Section 11.

**Lemma 5.3.** *Let $(X \ni x)$ be a symplectic singularity of dimension 2. Then there exist a finite subgroup $G \subset \mathrm{SU}(2)$ and a Kollár component $E$ over $(X \ni x)$ with the following properties, where $X_1 := \mathbb{A}^2/G$, $x_1$ is the image of 0, $\ell := |G \cap \mathbb{G}_m \cdot \mathrm{id}_{\mathbb{A}^2}|$, and $\mathbb{C}[X_1]_j$ is the space of $G$-invariant homogeneous polynomials of degree $\ell j$ for $j \geq 0$.*

*(1) We have $\ell \in \{1, 2\}$, the quotient morphism $\mathbb{A}^2 \to X_1$ is étale over $X_1\backslash\{x_1\}$, and the $\mathbb{G}_m$-action $\zeta_1$ on $X_1$ defined by the grading $\mathbb{C}[X_1] = \bigoplus_{j\geq 0} \mathbb{C}[X_1]_j$ is faithful, with $\zeta_1(\mu^\ell)$ induced by the scalar multiplication by $\mu$ on $\mathbb{A}^2$ for $\mu \in \mathbb{C}^*$.*

*(2) The valuation $\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, and $A_X(E) = A_{X_1}(\mathrm{wt}_{\zeta_1}) = 2/\ell$.*

*(3) There exists an isomorphism of graded $\mathbb{C}$-algebras $\mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x} \cong \mathbb{C}[X_1]$, where $\mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x}$ is graded by the values of $\mathrm{ord}_E$.*

*Proof.* By [Bea99, (2.1)], $(X \ni x)$ is either smooth or a rational double point. By [KM98, Remark 4.21(1)], $(X \ni x)$ is analytically isomorphic to $(X_1 \ni x_1)$, where $X_1 := \mathbb{A}^2/G$ for a finite subgroup $G \subset \mathrm{SL}(2, \mathbb{C})$ and $x_1$ is the image of 0. We take $G = \{1\}$ when $x$ is a smooth point. Since $G$ is finite, the sum over $g \in G$ of the pullbacks by $g$ of the standard Hermitian inner product on $\mathbb{C}^2$ is a $G$-invariant Hermitian inner product. Thus, after conjugation in $\mathrm{SL}(2, \mathbb{C})$, we may assume that $G \subset \mathrm{SU}(2)$. There exists an isomorphism $\phi : \widehat{\mathcal{O}}_{X_1,x_1} \to \widehat{\mathcal{O}}_{X,x}$ of $\mathbb{C}$-algebras [Ser56, §2, no. 6, Proposition 3].

We prove (1). Let $I := G \cap \mathbb{G}_m \cdot \mathrm{id}_{\mathbb{A}^2}$, so that $\ell = |I|$. Since $\det(\mu \cdot \mathrm{id}_{\mathbb{A}^2}) = \mu^2$, we have $I \subset \{\pm \mathrm{id}_{\mathbb{A}^2}\}$, so $\ell \in \{1, 2\}$ and $I = \{\mu \cdot \mathrm{id}_{\mathbb{A}^2} \mid \mu^\ell = 1\}$. Since $G$ acts linearly on $\mathbb{A}^2$, the homogeneous components of a $G$-invariant polynomial are $G$-invariant. If $\ell = 2$, then $-\mathrm{id}_{\mathbb{A}^2} \in G$ acts on homogeneous polynomials of degree $k$ by $(-1)^k$. Thus any non-zero homogeneous $G$-invariant polynomial has degree divisible by $\ell$, and

$$\mathbb{C}[X_1] = \bigoplus_{j\geq 0} \mathbb{C}[X_1]_j, \qquad \mathbb{C}[X_1]_0 = \mathbb{C}.$$

For $\mu \in \mathbb{C}^*$, the element $\zeta_1(\mu^\ell)$ acts on $\mathbb{C}[X_1]_j$ by $\mu^{\ell j}$, which is the action induced by the scalar multiplication by $\mu$ on $\mathbb{A}^2$. We show that $\zeta_1$ is faithful. Let $\lambda \in \mathbb{C}^*$ be such that $\zeta_1(\lambda)$ acts trivially on $X_1$, and let $\mu \in \mathbb{C}^*$ with $\mu^\ell = \lambda$. Then $\mathbb{A}^2 = \bigcup_{g\in G} \mathrm{Ker}(g - \mu \cdot \mathrm{id}_{\mathbb{A}^2})$, so $\mathrm{Ker}(g - \mu \cdot \mathrm{id}_{\mathbb{A}^2}) = \mathbb{A}^2$ for some $g \in G$, and $\mu \cdot \mathrm{id}_{\mathbb{A}^2} = g \in I$. Thus $\mu^\ell = 1$, that is, $\lambda = 1$. Since $\mathbb{C}[X_1]_0 = \mathbb{C}$, the $\zeta_1$-grading of $\mathbb{C}[X_1]$ is good. Any $g \in G\backslash\{1\}$ has finite order and $\det g = 1$, so $g$ is diagonalizable

with eigenvalues $a, a^{-1}$ for some $a \neq 1$. Thus $G$ acts freely on $\mathbb{A}^2\backslash\{0\}$, and the quotient morphism $\mathbb{A}^2 \to X_1$ is étale over $X_1\backslash\{x_1\}$. This proves (1).

We prove (2) and (3). By [KM98, Proposition 5.20(4)], $X_1$ is klt. Since $G \subset \mathrm{SL}(2,\mathbb{C})$ and $\mathbb{A}^2 \to X_1$ is étale over $X_1\backslash\{x_1\}$, the $G$-invariant form $dx \wedge dy$ descends to a generator of $\Omega^{[2]}_{X_1}$, which has $\zeta_1$-weight $2/\ell$ by (1). By Lemma 2.14, applied to the torus $\zeta_1(\mathbb{G}_m)$,

$$A_{X_1}(\mathrm{wt}_{\zeta_1}) = \frac{2}{\ell}.$$

By [LX16, Example 7.1(1)], $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_1,x_1}$ is minimized at $\mathrm{ord}_S$, where $S$ is the quotient by $G$ of the exceptional divisor of the blow-up of $0 \in \mathbb{A}^2$. The order of vanishing at 0, restricted to $\mathbb{C}(X_1)$, is $\ell\,\mathrm{wt}_{\zeta_1}$, so $\mathrm{ord}_S$ is a positive multiple of $\mathrm{wt}_{\zeta_1}$. Since $\zeta_1$ is faithful, the value group of $\mathrm{wt}_{\zeta_1}$ is $\mathbb{Z}$, as is the value group of $\mathrm{ord}_S$. Thus $\mathrm{ord}_S = \mathrm{wt}_{\zeta_1}$ and $A_{X_1}(S) = 2/\ell$. By [LX16, Theorem 1.2], $S$ is a Kollár component over $(X_1 \ni x_1)$. Since $\mathrm{ord}_S = \mathrm{wt}_{\zeta_1}$, the class of a function in $\mathrm{gr}_{\mathrm{ord}_S}\mathcal{O}_{X_1,x_1}$ is determined by the component of lowest $\zeta_1$-weight, so

$$\mathrm{gr}_{\mathrm{ord}_S}\mathcal{O}_{X_1,x_1} \cong \mathbb{C}[X_1]$$

as graded $\mathbb{C}$-algebras, where $\mathbb{C}[X_1]$ carries the $\zeta_1$-grading.

By Lemma 2.10, $(X \ni x)$ is klt. By Lemma 5.2, applied to $\phi$, with $(X \ni x)$ in place of $(X_2 \ni x_2)$ and $S$ in place of $E_1$, there exists a Kollár component $E$ over $(X \ni x)$ such that $A_X(E) = A_{X_1}(S) = 2/\ell$, $\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$, and $\phi$ induces an isomorphism of graded $\mathbb{C}$-algebras

$$\mathrm{gr}_{\mathrm{ord}_S}\mathcal{O}_{X_1,x_1} \cong \mathrm{gr}_{\mathrm{ord}_E}\mathcal{O}_{X,x}.$$

Since $A_{X_1}(S) = A_{X_1}(\mathrm{wt}_{\zeta_1})$, this proves (2). Composing the inverse of the isomorphism induced by $\phi$ with the isomorphism $\mathrm{gr}_{\mathrm{ord}_S}\mathcal{O}_{X_1,x_1} \cong \mathbb{C}[X_1]$ proves (3). □

**Theorem 5.4.** *Let $(X \ni x)$ be a symplectic singularity of dimension $2n$, $n \geq 1$, whose symplectic form need not be algebraic, and let $v$ be a minimizer of $\widehat{\mathrm{vol}}$. Then there exist $c > 0$ and a Kollár component $E$ over $(X \ni x)$ with $v = c\,\mathrm{ord}_E$ and*

$$A_X(E) \in \{n, 2n\}.$$

*Moreover, let $W := \mathrm{Spec}\,\mathrm{gr}_v\mathcal{O}_{X,x}$ be graded by the values of $v/c$, and let $F$ be the Kollár component over the vertex of $W$ defined by the grading. Then the grading is induced by a faithful $\mathbb{G}_m$-action, $A_X(E) = A_W(F)$, and $(E, \Delta_E)$ is K-semistable.*

For $n \geq 2$, the metric argument takes place on the central fiber of the K-polystable degeneration of $\mathrm{Spec}\,\mathrm{gr}_v\mathcal{O}_{X,x}$ given by Theorem 2.22. Since the degeneration is torus-equivariant and does not change which characters occur (**Step 3**), the conclusions on weight and rank obtained there return to $\mathrm{Spec}\,\mathrm{gr}_v\mathcal{O}_{X,x}$, and then to the germ by Proposition 5.1.

*Proof of Theorem 5.4.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we fix the associated graded cone $W$ and the torus $T$ acting on it.

By Lemma 2.10, $(X \ni x)$ is canonical, and in particular klt. By Theorem 2.20, $v$ is quasi-monomial, the ring $R := \mathrm{gr}_v\mathcal{O}_{X,x}$ is finitely generated, and $W = \mathrm{Spec}\,R$ is a normal affine variety with klt singularities and a good action of the torus $T$ whose character lattice is the value group $\Gamma := v(\mathbb{C}(X)^*)$. Let $o$ be the vertex of $W$, and let $\xi_v$ be the Reeb vector of $T$ defined by the grading by the values of $v$. By Theorem 2.20, $(W, \xi_v)$ is a K-semistable Fano cone. The characters of $T$ occurring in $R$ are the values $v(h)$ for $h \in \mathcal{O}_{X,x}\backslash\{0\}$, and they generate $\Gamma$ since $\mathbb{C}(X)$ is the fraction field of $\mathcal{O}_{X,x}$. Thus $T$ acts faithfully on $W$. We denote by $X_*(T)$ the cocharacter lattice of $T$, and by $\langle\cdot,\cdot\rangle$ the pairing between the characters of $T$ and $X_*(T)_{\mathbb{R}}$.

If $n = 1$, we go to **Step 6**. In **Steps 2–5** we assume that $n \geq 2$.

**Step 2.** In this step we construct a primitive cocharacter $\zeta$ of $T$ which is a Reeb vector of $T$, and a closed non-degenerate reflexive 2-form $\sigma$ on $W$ which is homogeneous of positive $\zeta$-degree $w$.

Choose a log smooth model $(Y, D)$ over $X$, $D = D_1 + \cdots + D_k$, such that $v = v_\alpha \in \mathrm{QM}(Y, D)$ for some $\alpha = (\alpha_1, \ldots, \alpha_k) \in \mathbb{R}^k_{>0}$ with $\alpha_1, \ldots, \alpha_k$ linearly independent over $\mathbb{Q}$ [XZ26b, Section 2.2, before Theorem 2.17]. Let $z_1, \ldots, z_k$ be local equations of $D_1, \ldots, D_k$ at the generic point of $D_1 \cap \cdots \cap D_k$. Then $\Gamma = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_k$, since $v_\alpha$ takes the value $\sum_i \beta_i \alpha_i$ on the monomial $z_1^{\beta_1} \cdots z_k^{\beta_k}$ for $\beta \in \mathbb{Z}^k$, and the value of any function is the minimum of the values of the monomials in the expansion of the function. Via the isomorphism $\mathbb{Z}^k \to \Gamma$, $\beta \to \sum_i \beta_i \alpha_i$, we regard the characters of $T$ as elements of $\mathbb{Z}^k$ and $X_*(T)_\mathbb{R}$ as $\mathbb{R}^k$, so that $\langle \beta, \alpha' \rangle = \sum_i \beta_i \alpha'_i$ and $\xi_v = \alpha$.

Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{X,x}$. Choose $f_1, \ldots, f_p \in \mathcal{O}_{X,x}$ whose classes $\bar{f}_1, \ldots, \bar{f}_p$ are $T$-homogeneous generators of $R$, of weights $\beta^{(1)}, \ldots, \beta^{(p)}$. Since $\alpha_1, \ldots, \alpha_k$ are linearly independent over $\mathbb{Q}$, the value $v_\alpha(f_i) = \langle \beta^{(i)}, \alpha \rangle$ is attained at the single exponent $\beta^{(i)}$ in the monomial expansion of $f_i$ at the generic point of $D_1 \cap \cdots \cap D_k$.

We next fix an algebraic reflexive 2-form $\sigma'$ near $x$ which is non-degenerate on $X_{\mathrm{reg}}$. If $\omega$ is algebraic, we let $\sigma' := \omega$.

Suppose that $\omega$ is only holomorphic. Since $X$ is Gorenstein (Lemma 2.10), possibly shrinking $X$ around $x$, we may assume that $\Omega_X^{[2n]}$ is generated by an algebraic form $\Omega$. Let $\mu_1, \ldots, \mu_l$ be generators of the stalk of $\Omega_X^{[2]}$ at $x$. The ring $\mathcal{O}^{\mathrm{an}}_{X,x}$ is flat over $\mathcal{O}_{X,x}$, and the completion of $\mathcal{O}_{X,x}$ maps isomorphically onto the completion of $\mathcal{O}^{\mathrm{an}}_{X,x}$ [Ser56, §2, no. 6, Proposition 3 and Corollary 1]. Since Hom of finitely presented modules commutes with flat base change [Mat89, Theorem 7.11], the stalk at $x$ of the sheaf of holomorphic reflexive 2-forms is generated by $\mu_1, \ldots, \mu_l$ over $\mathcal{O}^{\mathrm{an}}_{X,x}$, and the stalk of holomorphic reflexive $2n$-forms is generated by $\Omega$. Thus

$$\omega = \sum_{i=1}^{l} h_i \mu_i, \qquad \omega^n = u\Omega$$

for some $h_1, \ldots, h_l, u \in \mathcal{O}^{\mathrm{an}}_{X,x}$, and $u$ is a unit since $X$ is normal and $\omega^n$ has no zero on $X_{\mathrm{reg}}$ near $x$. Let $N > 0$ be an integer, which we fix below. Since the completions of $\mathcal{O}_{X,x}$ and $\mathcal{O}^{\mathrm{an}}_{X,x}$ coincide, there exist $h'_1, \ldots, h'_l \in \mathcal{O}_{X,x}$ with $h_i - h'_i \in \mathfrak{m}^N \mathcal{O}^{\mathrm{an}}_{X,x}$ for any $i$. Let $\sigma' := \sum_i h'_i \mu_i$, and let $a_1, \ldots, a_q$ be generators of $\mathfrak{m}^N$. Then

$$\omega - \sigma' = \sum_{i,j} g_{ij} a_j \mu_i, \qquad (\sigma')^n = u'\Omega$$

for some $g_{ij} \in \mathcal{O}^{\mathrm{an}}_{X,x}$ and $u' \in \mathcal{O}_{X,x}$. Since $\omega^n - (\sigma')^n$ is a sum of wedge products each of which has a factor $\omega - \sigma'$, we have $u - u' \in \mathfrak{m}^N \mathcal{O}^{\mathrm{an}}_{X,x}$, so $u'(x) = u(x) \neq 0$. Possibly shrinking $X$ around $x$, we may assume that $u'$ is a unit. Thus $\sigma'$ is an algebraic reflexive 2-form, possibly not closed, which is non-degenerate on $X_{\mathrm{reg}}$, and $(\sigma')^n$ generates $\Omega_X^{[2n]}$.

For an algebraic reflexive form $\theta$ and $v'' \in \mathrm{QM}(Y, D)$, $A_\theta(v'')$ is as in Definition 2.8. For a holomorphic reflexive form $\theta$ on a neighborhood of $x$, such as $\omega$ and $\omega - \sigma'$, we define $A_\theta(P)$ only for a prime divisor $P$ over $X$ with center $x$, as the logarithmic vanishing order along $P$ of the pullback of $\theta$ to the preimage of that neighborhood on a higher model on which $P$ is a divisor. Let $P$ be a prime divisor over $X$ with center $x$. By [XZ26b, Theorem 3.10], $A_{\mu_i}(P) \geq 0$ for any $i$. Since $\mathrm{ord}_P(g_{ij}) \geq 0$ and $\mathrm{ord}_P(a_j) \geq N \,\mathrm{mult}_P \mathfrak{m}$ for any $i, j$, we obtain

$$A_{\omega - \sigma'}(P) \geq N \,\mathrm{mult}_P \mathfrak{m}.$$

Since $(\sigma')^n$ generates $\Omega_X^{[2n]}$, we have $A_{(\sigma')^n} = A_X$, and since the logarithmic vanishing order of a wedge product is at least the sum of the orders of the factors, $nA_{\sigma'} \leq A_X$ on $\mathrm{QM}(Y, D)$. We fix $N$ with

$$Nv(\mathfrak{m}) > \frac{A_X(v)}{n}.$$

By [XZ26b, Lemma 3.9], applied to $\sigma'$, to $\Omega$, and to the generators of $\mathfrak{m}$, the functions $v'' \to A_X(v'')$ and $v'' \to v''(\mathfrak{m})$ are continuous on a neighborhood of $v$ in $\mathrm{QM}(Y, D)$.

In both cases, by [LX18, Lemma 2.10 and its proof], there exists $\epsilon > 0$ such that for any $\alpha' \in \mathbb{Q}^k$ with $|\alpha - \alpha'| < \epsilon$, there exists an isomorphism $\psi_{\alpha'} : R \to \mathrm{gr}_{v_{\alpha'}} \mathcal{O}_{X,x}$ of $\mathbb{C}$-algebras which sends $\bar{f}_i$ to the class of $f_i$ in $\mathrm{gr}_{v_{\alpha'}} \mathcal{O}_{X,x}$ for any $i$: the proof starts from arbitrary homogeneous generators $\bar{f}_i$ and arbitrary lifts $f_i$, and the isomorphism it constructs sends $\bar{f}_i$ to the class of $f_i$ in $\mathrm{gr}_{v_{\alpha'}} \mathcal{O}_{X,x}$. Possibly shrinking $\epsilon$, for any such $\alpha'$ the exponent $\beta^{(i)}$ also attains $v_{\alpha'}(f_i)$, so $v_{\alpha'}(f_i) = \langle \beta^{(i)}, \alpha' \rangle$ for any $i$. If $\omega$ is only holomorphic, then, by the continuity above and the choice of $N$, possibly shrinking $\epsilon$, for any $\alpha' \in \mathbb{R}^k$ with $|\alpha - \alpha'| < \epsilon$,

$$N v_{\alpha'}(\mathfrak{m}) > \frac{A_X(v_{\alpha'})}{n} \geq A_{\sigma'}(v_{\alpha'}).$$

For any $\alpha' \in \mathbb{Q}^k$ with $|\alpha - \alpha'| < \epsilon$, the valuation $v_{\alpha'}$ takes values in $\mathbb{Z}\alpha'_1 + \cdots + \mathbb{Z}\alpha'_k \subset \mathbb{Q}$, so the value group of $v_{\alpha'}$ has rational rank one, and $v_{\alpha'}$ is divisorial [JM12, Remark 3.9]. Possibly shrinking $\epsilon$, by [XZ26b, Theorem 2.17] and by [XZ26b, Theorem 5.11] applied to $\sigma'$, for any such $\alpha'$ the prime divisor $E'$ over $X$ with $v_{\alpha'} = c' \operatorname{ord}_{E'}$ for some $c' > 0$ is a Kollár component over $(X \ni x)$, and the test configuration of $E'$ specializes $\sigma'$ to a non-degenerate reflexive 2-form on $\operatorname{Spec} \mathrm{gr}_{\operatorname{ord}_{E'}} \mathcal{O}_{X,x}$, which is $W$ via $\psi_{\alpha'}$.

We fix $\alpha' \in \mathbb{Q}^k$ with $|\alpha - \alpha'| < \epsilon$, and we let $v' := v_{\alpha'}$ and $\xi_{v'} := \alpha' \in X_*(T)_{\mathbb{R}}$. Since $\bar{f}_1, \ldots, \bar{f}_p$ generate $R$, $\psi_{\alpha'}$ maps $R_\beta$ into the homogeneous piece of $\mathrm{gr}_{v_{\alpha'}} \mathcal{O}_{X,x}$ of degree $\langle \beta, \alpha' \rangle$ for any character $\beta$ of $T$. Thus the grading which $\psi_{\alpha'}$ induces on $R$ is the grading by $\langle \cdot, \alpha' \rangle$. Since $v'(\mathfrak{m}) > 0$, we have $\langle \beta, \xi_{v'} \rangle > 0$ for any non-zero character $\beta$ of $T$ occurring in $R$. Thus $\xi_{v'}$ is a Reeb vector of $T$. Since $\alpha' \in \mathbb{Q}^k$, we have $v' = c' \operatorname{ord}_{E'}$, where $c' > 0$ and the Kollár component $E'$ over $(X \ni x)$ are as in the previous paragraph. Let $\zeta$ be the primitive element of $X_*(T)$ on the ray $\mathbb{R}_{>0}\xi_{v'}$. Then $\zeta$ is a Reeb vector of $T$, and the $\zeta$-grading $R = \bigoplus_{j \geq 0} R_j$ satisfies $R_0 = \mathbb{C}$. Since the grading by $\langle \cdot, \zeta \rangle$ is proportional to the grading by the values of $\operatorname{ord}_{E'} = v'/c'$, an element $f \in R$ is $\zeta$-homogeneous if and only if $\psi_{\alpha'}(f)$ is homogeneous for the grading of $\mathrm{gr}_{v'} \mathcal{O}_{X,x} = \mathrm{gr}_{\operatorname{ord}_{E'}} \mathcal{O}_{X,x}$ by the values of $\operatorname{ord}_{E'}$. We let $\sigma$ be the specialization of $\sigma'$ [XZ26b, Definition 3.3]. By [XZ26b, Definition 3.3], $\sigma$ is the restriction to the central fiber of $t^m$ times the pullback of $\sigma'$ to the test configuration of $E'$, for some integer $m$. The pullback of $\sigma'$ is invariant under the $\mathbb{G}_m$-action on the test configuration, and $t^m$ is semi-invariant. Thus $\sigma$ is homogeneous for the $\mathbb{G}_m$-action on $W$, that is, for the grading by the values of $\operatorname{ord}_{E'}$, hence $\zeta$-homogeneous. Let $w \in \mathbb{Z}$ be the $\zeta$-degree of $\sigma$.

If $\omega$ is algebraic, then $\sigma$ is closed by [XZ26b, Proposition 3.5(1)]. Suppose that $\omega$ is only holomorphic. Since $v' = c' \operatorname{ord}_{E'}$,

$$A_{\omega - \sigma'}(E') \geq N \operatorname{mult}_{E'} \mathfrak{m} > A_{\sigma'}(E').$$

Let $y_0$ be a general point of $E'$ on a higher model of $X$, and let $z, y_1, \ldots, y_{2n-1}$ be local coordinates at $y_0$ such that $z$ is a local equation of $E'$. Near $y_0$, we expand the pullbacks of $\omega$ and $\sigma'$ as convergent sums of terms which are homogeneous in $z$, with respect to $dz/z$ and $dy_1, \ldots, dy_{2n-1}$. By [XZ26b, Lemma 2.9], near the generic point of $E'$ the test configuration of $E'$ is the deformation to the normal cone of $E'$, given near $y_0$ by $z = ts$, where $s$ is the fiber coordinate of the normal bundle of $E'$. We write the pullback of $\sigma'$ as $\sum_{j \geq A_{\sigma'}(E')} z^j \theta_j$, where $\theta_j$ is a 2-form in $dz/z$ and $dy_1, \ldots, dy_{2n-1}$ whose coefficients do not depend on $z$, and $\theta_{A_{\sigma'}(E')} \neq 0$. Substituting $z = ts$, the term $z^j \theta_j$ becomes $t^j s^j \theta_j$. Thus the smallest integer $m$ in [XZ26b, Definition 3.3] is $-A_{\sigma'}(E')$, and, on the open subset of $W$ lying over a neighborhood of $y_0$ in $E'$,

$$\sigma = s^{A_{\sigma'}(E')} \theta_{A_{\sigma'}(E')}.$$

Since $A_{\omega - \sigma'}(E') > A_{\sigma'}(E')$, the term of lowest degree of the pullback of $\omega$ is the term of lowest degree of the pullback of $\sigma'$. The exterior derivative maps a term which is homogeneous of degree $j$ in $z$ to a term which is homogeneous of degree $j$. Thus the term of lowest degree of $d\omega = 0$ is the exterior derivative of the term of lowest degree of $\omega$, and $\sigma$ is closed on a non-empty open subset of $W_{\mathrm{reg}}$. Hence $\sigma$ is closed on $W_{\mathrm{reg}}$.

Since $\sigma$ is non-degenerate on $W_{\mathrm{reg}}$, $\sigma^n$ generates $\Omega_W^{[2n]}$ and has $\zeta$-degree $nw$. Since the $\zeta$-grading of $R$ is good, Lemma 2.14, applied to the torus $\zeta(\mathbb{G}_m)$, implies that

$$A_W(\mathrm{wt}_\zeta) = nw.$$

Since $W$ is klt, $A_W(\mathrm{wt}_\zeta) > 0$. Thus $w > 0$. Since $W$ is klt, the pullback of $\sigma$ to a log resolution of $W$ extends to a regular 2-form by [GKKP11, Theorem 1.4]. Thus $W$ together with $\sigma$ is a symplectic variety (Definition 2.9 and [Bea99, (1.2)]).

**Step 3.** In this step we degenerate $W$ to a K-polystable Fano cone $X_0$, and we show that the degeneration is a marked test configuration of $W$.

By Theorem 2.22, applied to $(W, \xi_v)$ and $T$, there exists a $T$-equivariant special test configuration of $(W, \xi_v)$ with Futaki invariant 0 whose central fiber $(X_0, \xi_v)$ is a Fano cone which is K-polystable for some torus $T'$ which acts faithfully on $X_0$ and contains $T$. By [LWX18, Definition 2.14, items (1)–(3) and the last sentence, and Definition 2.16], the test configuration is $\mathcal{X} = \operatorname{Spec} \mathcal{R} \to \mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[t]$, where $\mathcal{R}$ is a finitely generated normal domain which is flat over $\mathbb{C}[t]$, the torus $\Lambda := \mathbb{G}_m$ acts on $\mathcal{R}$ with weight $-1$ on $t$, the torus $T$ acts on $\mathcal{R}$ fixing $t$ and commuting with $\Lambda$, and there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$, where $\Lambda$ acts trivially on $R$. The central fiber $X_0 = \operatorname{Spec}(\mathcal{R}/t\mathcal{R})$ is integral and normal with klt singularities.

Let $\mathcal{R} = \bigoplus_{j \in \mathbb{Z}} \mathcal{R}_j$ be the $\zeta$-grading. Since $\mathcal{R}$ is flat over $\mathbb{C}[t]$, $\mathcal{R} \subset \mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$. Since $T$ fixes $t$ and $R_0 = \mathbb{C}$, we have $\mathcal{R}_j = 0$ for $j < 0$ and $\mathcal{R}_0 \subset \mathbb{C}[t, t^{-1}]$. Since $\Lambda$ commutes with $T$, $\mathcal{R}_0$ is $\Lambda$-stable, and hence $\mathcal{R}_0$ is spanned by monomials in $t$. If $\mathcal{R}$ contains a negative power of $t$, then it contains $t^{-1}$, since $t \in \mathcal{R}$, and $\mathcal{R}/t\mathcal{R} = 0$, which is impossible. Thus

$$\mathcal{R}_0 = \mathbb{C}[t].$$

The algebra $\mathcal{R}$ is generated by finitely many $T$-homogeneous elements of non-negative $\zeta$-degrees, and the generators of $\zeta$-degree 0 belong to $\mathbb{C}[t]$. Thus $\mathcal{R}_j$ is a finitely generated $\mathbb{C}[t]$-module for any $j \geq 0$. Since $\mathcal{R}_j$ is torsion free, $\mathcal{R}_j$ is free. For a character $\beta$ of $T$, the $\beta$-component $\mathcal{R}_\beta$ is a direct summand of $\mathcal{R}_{\langle \beta, \zeta \rangle}$, so $\mathcal{R}_\beta$ is a free $\mathbb{C}[t]$-module of finite rank. Comparing $\mathcal{R}_\beta[t^{-1}] \cong R_\beta[t, t^{-1}]$ with $\mathcal{R}_\beta/t\mathcal{R}_\beta$, we obtain

$$\dim_{\mathbb{C}}(\mathcal{R}/t\mathcal{R})_\beta = \operatorname{rank}_{\mathbb{C}[t]} \mathcal{R}_\beta = \dim_{\mathbb{C}} R_\beta.$$

Thus, for any character $\beta$ of $T$, the multiplicity of $\beta$ on $X_0$ is $\dim_{\mathbb{C}} R_\beta$, which is the multiplicity of $\beta$ on $W$. Since $T$ acts faithfully on $W$ (**Step 1**), the characters occurring on $W$ generate the character lattice of $T$, and hence $T$ acts faithfully on $X_0$. The $\zeta$-grading of $\mathcal{R}/t\mathcal{R}$ has non-negative degrees and degree-zero part $\mathbb{C}[t]/(t) = \mathbb{C}$, so it is good. Moreover, $\xi_v$, $\xi_{v'}$, and $\zeta$ are Reeb vectors of $T$ on $X_0$. Thus $(\mathcal{X}, T, \Lambda, \zeta)$ is a marked test configuration of $W$ with central fiber $X_0$ (Set-up 2.15, applied with $W$ in place of $X$ and $R$ the coordinate ring of $W$).

**Step 4.** In this step we normalize the Reeb vector and construct the metric on $X_0$.

Since $X_0$ is klt, $A_{X_0}(\mathrm{wt}_{\xi_v}) > 0$. Since $\mathrm{wt}_{\lambda\xi_v} = \lambda \, \mathrm{wt}_{\xi_v}$ for $\lambda > 0$, there exists a positive multiple $\xi$ of $\xi_v$ with

$$A_{X_0}(\mathrm{wt}_\xi) = 2n.$$

By Lemma 2.19, $(X_0, \xi)$ is K-polystable for $T'$, in particular K-semistable for $T'$, and hence $(X_0, \xi)$ is a K-semistable Fano cone by Theorem 2.21. By Theorem 2.20 and Theorem 2.21, $\mathrm{wt}_{\xi_v}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{W,o}$. Since $\widehat{\mathrm{vol}}$ is invariant under multiplying a valuation by a positive number [LWX18, Definition-Proposition 2.3 and the sentence after it], $\mathrm{wt}_\xi$ is also a minimizer of $\widehat{\mathrm{vol}}$. Thus $(W, \xi)$ is K-semistable by Theorem 2.21.

We show that $\xi$ and $\zeta$ are Reeb vectors of $T'$ on $X_0$. Let $\gamma$ be a non-zero character of $T'$ occurring on $X_0$. The $T$-eigenspaces of $\mathcal{R}/t\mathcal{R}$ are sums of $T'$-eigenspaces, so the restriction $\gamma|_T$ is a character of $T$ occurring on $X_0$, hence on $W$ (**Step 3**). If $\gamma|_T = 0$, then a non-zero function

of $T'$-weight $\gamma$ has $\zeta$-degree 0, so it is a constant by **Step 3**, and $\gamma = 0$, a contradiction. Thus $\gamma|_T \neq 0$. Since $\xi, \zeta \in X_*(T)_{\mathbb{R}}$ are Reeb vectors of $T$ on $W$,

$$\langle \gamma, \xi \rangle = \langle \gamma|_T, \xi \rangle > 0, \qquad \langle \gamma, \zeta \rangle = \langle \gamma|_T, \zeta \rangle > 0.$$

We also show that $\zeta$ is primitive in $X_*(T')$. Let $\chi \in X_*(T')$ and $e > 0$ be an integer with $e\chi \in X_*(T)$. Since $\lambda \to \lambda^e$ is surjective on $\mathbb{G}_m$, $\chi(\mathbb{G}_m) = (e\chi)(\mathbb{G}_m) \subset T$, so $\chi \in X_*(T)$. Thus $X_*(T)$ is saturated in $X_*(T')$, and the primitive element $\zeta$ of $X_*(T)$ is primitive in $X_*(T')$.

We show that the action of $T'$ on $X_0$ is good. Let $o_0$ be the vertex of the good action of $T$ on $X_0$ (**Step 3**). Since $T \subset T'$, the closure of any $T'$-orbit contains the closure of a $T$-orbit, and hence contains $o_0$. A fixed point of $T'$ is fixed by $T$, and hence equals $o_0$. Since $T'$ is commutative and contains $\zeta(\mathbb{G}_m)$, $T'$ preserves $(\mathcal{R}/t\mathcal{R})_j$ for any $j$, and hence preserves the ideal $\bigoplus_{j>0}(\mathcal{R}/t\mathcal{R})_j$ of $o_0$. Thus $o_0$ is fixed by $T'$, and the action of $T'$ on $X_0$ is good.

The hypotheses of Theorem 2.27 hold for $(X_0, \xi)$ with the torus $T'$ and $\zeta$ in place of $\chi$: $(X_0, \xi)$ is a Fano cone of dimension $2n \geq 2$; $T'$ acts faithfully on $X_0$ (Theorem 2.22) and the action of $T'$ is good (previous paragraph); $\xi$ and $\zeta$ are Reeb vectors of $T'$ and $\zeta$ is primitive in $X_*(T')$ (previous paragraphs); $A_{X_0}(\mathrm{wt}_\xi) = 2n$; and $(X_0, \xi)$ is K-polystable for $T'$. By Theorem 2.27 (see Remark 2.28), there exist functions $\rho$ and $\rho_\zeta$ and the Kähler cone metric $g = d\rho^2 + \rho^2 h$ defined by $\frac{\sqrt{-1}}{2}\partial\bar{\partial}\rho^2$, where $h$ is the induced metric on the link $\{\rho = 1\}$, such that the conditions of Set-up 3.1 hold with $X_0$, $T'$, $\zeta$ in place of $X$, $T$, $\chi$. Here the quotient log Fano pair $(Z, \Delta_Z)$ of $(X_0, \zeta)$ is a klt log Fano pair by Lemma 2.7. By Corollary 3.11, $X_0$, $\xi$, and $g$ satisfy the conditions of Set-up 2.31 for the torus $T'$. Let $T_c \subset T$ and $T'_c \subset T'$ be the maximal compact subtori, so that $T_c \subset T'_c$. The conditions of Set-up 2.31 also hold for the torus $T$, since $T$ acts faithfully on $X_0$ and the action is good (**Step 3**), $\xi$ is a Reeb vector of $T$ on $X_0$, and the metric $g$ and the $\Phi$-cutoffs are $T'_c$-invariant, hence $T_c$-invariant. Moreover, the last part of condition (1) of Set-up 2.31 holds for $T$: a function $\varphi$ of $T$-weight $\delta$ is a sum of functions of $T'$-weights $\gamma$ with $\gamma|_T = \delta$, for which $\langle \gamma, \xi \rangle = \langle \delta, \xi \rangle$, so $\rho\partial_\rho\varphi = \langle \delta, \xi \rangle\varphi$.

**Step 5.** In this step we prove Theorem 5.4 when $n \geq 2$.

Recall that $\zeta$ is the primitive cocharacter of $T$ of **Step 2**, and that $\sigma$ is the closed non-degenerate reflexive 2-form on $W$ of $\zeta$-degree $w > 0$. We check the hypotheses of Theorem 4.7 with $W$ in place of $X$ and $\sigma$ in place of $\omega$. By **Step 3**, $(\mathcal{X}, T, \Lambda, \zeta)$ is a marked test configuration of $W$, $T$ acts faithfully on $W$ and on $X_0$, $X_0$ is integral and normal with klt singularities, and the $\zeta$-grading of $X_0$ is good. By **Step 2**, $\dim W = 2n$ with $n \geq 2$, and $W$ together with $\sigma$ is a symplectic variety, where $\sigma$ is algebraic of $\zeta$-degree $w > 0$. By **Step 4**, $\xi$ is a Reeb vector of $T$ on $X_0$, $(X_0, \xi)$ is a K-semistable Fano cone with $A_{X_0}(\mathrm{wt}_\xi) = 2n$, and $X_0$, $\xi$, and $g$ satisfy the conditions of Set-up 2.31. Thus, by Theorem 4.7(2), there exist a primitive cocharacter $\zeta_0$ of $T$ and $r \in \{1, 2\}$ with

$$\xi = \frac{2}{r}\zeta_0,$$

and $\zeta_0$ is a Reeb vector on $W$ and on $X_0$. Since $(W, \xi)$ is K-semistable (**Step 4**), Theorem 4.7(3) implies that $\mathrm{wt}_{\zeta_0}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{W,o}$ and

$$A_W(\mathrm{wt}_{\zeta_0}) = rn.$$

Since $(W, \xi_v)$ is a Fano cone and $\zeta_0$ is a primitive cocharacter of $T$ which is a Reeb vector, Lemma 2.7 implies that $\mathrm{wt}_{\zeta_0} = \mathrm{ord}_{F_0}$ for a Kollár component $F_0$ over $(W \ni o)$. Since $\xi_v$ is a positive multiple of $\xi$, hence of $\zeta_0$, the valuation $\mathrm{wt}_{\xi_v}$ is a positive multiple of $\mathrm{ord}_{F_0}$, so $\mathrm{wt}_{\xi_v}$ is divisorial. By Proposition 5.1, there exist $c > 0$ and a Kollár component $E$ over $(X \ni x)$ with $v = c\,\mathrm{ord}_E$ such that the grading of $W$ by the values of $v/c$ is a positive $\mathbb{Z}$-grading induced by a faithful $\mathbb{G}_m$-action, and

$$A_X(E) = A_W(F),$$

where $F$ is the Kollár component over $(W \ni o)$ whose valuation $\mathrm{ord}_F$ is the weight valuation of the grading by the values of $v/c$.

The grading of $W$ by the values of $v/c$ is the grading by $\langle\cdot,\xi_v/c\rangle$. The values $\langle\beta,\xi_v/c\rangle$ for the characters $\beta$ occurring in $R$ are integers, and they generate $\mathbb{Z}$ since the $\mathbb{G}_m$-action is faithful. Since the characters occurring in $R$ generate the character lattice of $T$ (**Step 1**), $\xi_v/c$ is a primitive element of $X_*(T)$. Since $\xi_v/c$ and $\zeta_0$ are primitive elements of $X_*(T)$ on the ray $\mathbb{R}_{>0}\xi_v$, $\xi_v/c=\zeta_0$. Thus $\mathrm{ord}_F=\mathrm{wt}_{\zeta_0}=\mathrm{ord}_{F_0}$, so $F=F_0$, and

$$A_X(E)=A_W(F)=A_W(\mathrm{wt}_{\zeta_0})=rn\in\{n,2n\}.$$

Since $\widehat{\mathrm{vol}}$ is invariant under multiplying a valuation by a positive number, $\mathrm{ord}_E=v/c$ is a minimizer of $\widehat{\mathrm{vol}}$. Thus $(E,\Delta_E)$ is K-semistable by [LX16, Theorem 1.2]. This proves Theorem 5.4 when $n\geq 2$.

**Step 6.** In this step we prove Theorem 5.4 when $n=1$.

By Lemma 5.3, there exist a finite subgroup $G\subset\mathrm{SU}(2)$ and a Kollár component $E$ over $(X\ni x)$ such that, with $X_1:=\mathbb{A}^2/G$, $\ell:=|G\cap\mathbb{G}_m\cdot\mathrm{id}_{\mathbb{A}^2}|$, and the faithful $\mathbb{G}_m$-action $\zeta_1$ on $X_1$ of Lemma 5.3(1), the valuation $\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$,

$$A_X(E)=A_{X_1}(\mathrm{wt}_{\zeta_1})=\frac{2}{\ell}\in\{1,2\},$$

and there exists an isomorphism of graded $\mathbb{C}$-algebras $\mathrm{gr}_{\mathrm{ord}_E}\mathcal{O}_{X,x}\cong\mathbb{C}[X_1]$, where $\mathbb{C}[X_1]$ carries the $\zeta_1$-grading. By [XZ20, Theorem 1.1], $v=c\,\mathrm{ord}_E$ for some $c>0$. Thus $(E,\Delta_E)$ is K-semistable by [LX16, Theorem 1.2]. Since $\mathrm{gr}_v\mathcal{O}_{X,x}=\mathrm{gr}_{\mathrm{ord}_E}\mathcal{O}_{X,x}$, where the grading by the values of $v/c$ is the grading by the values of $\mathrm{ord}_E$, the graded isomorphism of Lemma 5.3(3) induces an isomorphism $W\to X_1$ which carries the grading of $W$ by the values of $v/c$ to the $\zeta_1$-grading. In particular, the grading of $W$ is induced by a faithful $\mathbb{G}_m$-action, and under the isomorphism $W\to X_1$ the valuation $\mathrm{ord}_F$ corresponds to $\mathrm{wt}_{\zeta_1}$. Hence

$$A_W(F)=A_{X_1}(\mathrm{wt}_{\zeta_1})=\frac{2}{\ell}=A_X(E)\in\{1,2\}.$$

This proves Theorem 5.4 when $n=1$. □

## 6. The contact structure

In this section, we prove Theorem 6.3, which answers Question 1.1(3)(5) for the Kollár component $E$ of Theorem 5.4: the residue of $\omega$ along $E$ is an orbifold contact form on $E$, with the orbifold structure given by the index-one cover of $E$.

### 6.1. **Logarithmic extension.**

In this subsection, we extend the pullbacks of reflexive 2-forms on $X$ to the index-one cover of the minimizing Kollár component as logarithmic 2-forms twisted by $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$, and we define the residue $\theta_\omega$ of the symplectic form $\omega$.

**Set-up 6.1.** Let $(X\ni x)$, $n$, $\omega$, $v$, $c$, $E$, $r$, $f:Y\to X$, $q:\mathcal{Y}\to Y$, $\mathcal{E}$, $\pi$, $L$, and $\mathcal{F}_r$ be as follows.

(1) $(X\ni x)$ is a symplectic singularity of dimension $2n$ with symplectic form $\omega$.
(2) $v$ is a minimizer of $\widehat{\mathrm{vol}}$.
(3) $E$ is the Kollár component over $(X\ni x)$ with $v=c\,\mathrm{ord}_E$ for some $c>0$ given by Theorem 5.4.
(4) $r:=A_X(E)/n$.
(5) $f:Y\to X$ is the extraction of $E$.
(6) $q:\mathcal{Y}\to Y$ is the index-one cover of $E$, and $\mathcal{E}:=q^*E$.
(7) $\pi:=f\circ q$.
(8) $L:=\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}$.
(9) $\mathcal{F}_r:=\Omega^{[2]}_{\mathcal{Y}}(\log\mathcal{E})\otimes\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$.
(10) $X$ is replaced by a sufficiently small neighborhood of $x$ whenever necessary, and when $\omega$ is not algebraic, sections on $\mathcal{Y}$ are analytic sections over the preimage of a sufficiently small analytic neighborhood of $x$.

The germ $(X \ni x)$ is not a cone, so there is no Euler field to contract $\omega$ with. Instead, we use the bound of Theorem 2.13 for all non-zero reflexive 2-forms, not only for $\omega$, to extend their pullbacks to $\mathcal{Y}$ as sections of $\mathcal{F}_r$ in Definition-Lemma 6.2.

**Definition-Lemma 6.2.** Notation and conditions as in Set-up 6.1. Let $\mathfrak{m} \subset \mathcal{O}_{X,x}$ be the maximal ideal and $M_x := (\Omega^{[2]}_X)_x$. Then the following hold.

(1) There exists a unique $\mathcal{O}_{\mathcal{Y}}$-linear morphism $\mathfrak{p} : \pi^*\Omega^{[2]}_X \to \mathcal{F}_r$ which coincides with the pullback of forms over the open subset of $X_{\mathrm{reg}}$ over which $\pi$ is an isomorphism.
(2) The form $\pi^*\omega$ extends uniquely to a section $s_\omega$ of the analytic sheaf $\mathcal{F}_r^{\mathrm{an}}$ on $\mathcal{Y}$.
(3) The residue of logarithmic 2-forms along $\mathcal{E}$, tensored with $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$, on the open substack where $\mathcal{Y}$ is smooth and $\mathcal{E}$ is a smooth divisor, extends uniquely to a morphism $\mathrm{res} : \mathcal{F}_r|_{\mathcal{E}} \to \Omega^{[1]}_{\mathcal{E}} \otimes L$, and $\mathrm{res} \circ \mathfrak{p}|_{\mathcal{E}}$ factors through $(M_x/\mathfrak{m}M_x) \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{E}}$.
(4) Near $x$ we can write $\omega = \sum_i h_i\mu_i$ with $\mu_i \in M_x$ and holomorphic functions $h_i$, and the section $\theta_\omega := \mathrm{res}(s_\omega|_{\mathcal{E}})$, which we call the *residue* of $\omega$, is the image of the class of $\sum_i h_i(x)\mu_i$ under the morphism $\mathrm{res} \circ \mathfrak{p}|_{\mathcal{E}}$ of (3). In particular, $\theta_\omega$ is an algebraic section of $\Omega^{[1]}_{\mathcal{E}} \otimes L$.

*Proof.* **Proof of (1).** By Theorem 5.4, $A_X(E) \in \{n, 2n\}$, so $r \in \{1, 2\}$. By Lemma 2.3, $\mathcal{Y}$ is normal and $\mathcal{E}$ is a Cartier divisor on $\mathcal{Y}$. Hence $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$ is invertible, and $\mathcal{F}_r$ is reflexive, in particular torsion-free. Over the locus where $E$ is Cartier, $E$ has index one, so the local index-one cover of $E$ in Definition 2.2 is the identity, and $q$ is an isomorphism over that locus. Since $f$ is an isomorphism over $X\backslash\{x\}$ and $E$ is the only $f$-exceptional prime divisor, $\pi$ restricts to an isomorphism

$$\mathcal{Y}\backslash\mathcal{E} \to X\backslash\{x\}.$$

Over $\mathcal{Y}\backslash\mathcal{E}$, the sheaf $\mathcal{F}_r$ is $\Omega^{[2]}_{\mathcal{Y}}$. Thus, for any local section $\mu$ of $\Omega^{[2]}_X$, $\pi^*\mu$ is a section of $\mathcal{F}_r$ over $\mathcal{Y}\backslash\mathcal{E}$, and over $\pi^{-1}(X_{\mathrm{reg}}\backslash\{x\})$ it is the pullback of forms.

Let $\mu$ be a non-zero local section of $\Omega^{[2]}_X$ near $x$. Since $X$ has symplectic singularities, $X$ is canonical, hence klt (Lemma 2.10). By Theorem 2.13(1) with $d = 2n$, $p = 2$, and $v = c\,\mathrm{ord}_E$, we have

$$\frac{cA_X(E)}{2n} \le \frac{cA_\mu(E)}{2}, \quad \text{that is,} \quad A_\mu(E) \ge \frac{A_X(E)}{n} = r.$$

Let $\eta$ be the generic point of $E$. Since $Y$ is normal, $Y$ is regular at $\eta$ and $E$ is Cartier at $\eta$. Hence $q$ is an isomorphism over a neighborhood of $\eta$, and near the generic point of $\mathcal{E}$ the morphism $\pi$ is $f$. The local rings at the generic points of $E$ of $Y$ and of a log resolution of $X$ on which $E$ is a divisor are both the valuation ring of $\mathrm{ord}_E$. Therefore $A_\mu(E)$ is the order along $E$ of $f^*\mu$, regarded as a rational section of $\Omega^2_Y(\log E)$ near $\eta$. If $z$ is a local equation of $E$ near $\eta$, then $A_\mu(E) \ge r$ means that

$$f^*\mu \in z^r \cdot \Omega^2_Y(\log E) \quad \text{near } \eta,$$

that is, $\pi^*\mu$ is a section of $\mathcal{F}_r$ near the generic point of $\mathcal{E}$. Thus $\pi^*\mu$ is a section of $\mathcal{F}_r$ over an open substack of $\mathcal{Y}$ whose complement has codimension at least 2. Since $\mathcal{F}_r$ is reflexive on the normal stack $\mathcal{Y}$, $\pi^*\mu$ extends uniquely to a section of $\mathcal{F}_r$ over $\mathcal{Y}$ by [Har80, Proposition 1.6], applied on an étale atlas of $\mathcal{Y}$.

Possibly shrinking $X$ around $x$, let $\mu_1, \dots, \mu_k$ be generators of $\Omega^{[2]}_X$. Since $\pi^*$ is right exact, $\pi^*\Omega^{[2]}_X$ is generated by the $\pi^*\mu_i$, with relations generated by the pullbacks of relations $\sum_i a_i\mu_i = 0$, $a_i \in \mathcal{O}_X$. For such a relation, $\sum_i (a_i \circ \pi) \cdot \pi^*\mu_i$, with $\pi^*\mu_i$ extended to $\mathcal{F}_r$, is a section of $\mathcal{F}_r$ which vanishes on $\mathcal{Y}\backslash\mathcal{E}$, hence is zero because $\mathcal{F}_r$ is torsion-free. Therefore sending $\pi^*\mu_i$ to the extension of $\pi^*\mu_i$ defines an $\mathcal{O}_{\mathcal{Y}}$-linear morphism $\mathfrak{p} : \pi^*\Omega^{[2]}_X \to \mathcal{F}_r$, which coincides with the pullback of forms over $\pi^{-1}(X_{\mathrm{reg}}\backslash\{x\})$. If $\mathfrak{p}'$ is another such morphism, then $\mathfrak{p}(\pi^*\mu_i) - \mathfrak{p}'(\pi^*\mu_i)$ vanishes on the dense open subset $\pi^{-1}(X_{\mathrm{reg}}\backslash\{x\})$ of $\mathcal{Y}$ for any $i$, hence is zero since $\mathcal{F}_r$ is torsion-free, so $\mathfrak{p} = \mathfrak{p}'$. Uniqueness also implies that $\mathfrak{p}$ does not depend on the generators $\mu_i$ and is compatible with shrinking $X$. This proves (1).

**Proof of (2) and of the first statement of (4).** Analytic sheaves on $\mathcal{Y}$ and their sections are taken on an étale atlas of $\mathcal{Y}$ and are compatible with the transition maps. Locally $\mathcal{Y} = [\tilde{Y}/\mu_m]$, and over such a chart a section of $\mathcal{F}_r^{\mathrm{an}}$ is a $\mu_m$-invariant section of the analytification of the corresponding sheaf on $\tilde{Y}$.

We first show two facts.

(a) Analytification commutes with duals of coherent sheaves, because for a scheme $Z$ of finite type over $\mathbb{C}$ and $z \in Z$, $\mathcal{O}_{Z,z}^{\mathrm{an}}$ is flat over $\mathcal{O}_{Z,z}$ [Ser56, §2, no. 6, Proposition 3 and Corollary 1], and Hom of a finitely presented module commutes with flat base change [Mat89, Theorem 7.11]. In particular, the analytification of a reflexive coherent sheaf is reflexive.

(b) On a normal complex space, a section of a reflexive coherent analytic sheaf defined outside a closed analytic subset of codimension at least 2 extends uniquely, by the second Riemann removable singularity theorem [GR84, Chapter 7]: locally a reflexive coherent sheaf is the kernel of a morphism $\mathcal{O}^a \to \mathcal{O}^b$ (dualize a presentation of the dual sheaf), the components of a section extend as holomorphic functions across a closed analytic subset of codimension at least 2 of a normal space, and the extension is a section of the kernel by the identity theorem.

By (a), $(\Omega_X^{[2]})^{\mathrm{an}}$ is a reflexive coherent analytic sheaf on the normal complex space $X^{\mathrm{an}}$ ([Gro03, Exposé XII, Proposition 2.1]), and it is the sheaf of holomorphic 2-forms over $X_{\mathrm{reg}}^{\mathrm{an}}$. Let $U$ be an analytic neighborhood of $x$ on which $\omega$ is defined. Since $\mathrm{codim}(X \backslash X_{\mathrm{reg}}) \geq 2$, by (b), $\omega$ is a section of $(\Omega_X^{[2]})^{\mathrm{an}}$ over $U$. The stalk of $(\Omega_X^{[2]})^{\mathrm{an}}$ at $x$ is

$$(\Omega_X^{[2]})_x^{\mathrm{an}} = M_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X,x}^{\mathrm{an}}.$$

Hence we can write $\omega = \sum_i h_i \mu_i$ near $x$, with $\mu_i \in M_x$ and $h_i \in \mathcal{O}_{X,x}^{\mathrm{an}}$. This is the first statement of (4). Possibly shrinking $U$, we may assume that the $h_i$ are holomorphic functions on $U$ and that the $\mu_i$ are sections of $\Omega_X^{[2]}$ over a Zariski open neighborhood of $x$ containing $U$.

Let

$$s_\omega := \sum_i (h_i \circ \pi) \cdot \mathfrak{p}(\pi^* \mu_i),$$

a section of $\mathcal{F}_r^{\mathrm{an}}$ over $\pi^{-1}(U)$. By (1), over $\pi^{-1}(U \cap X_{\mathrm{reg}} \backslash \{x\})$ we have

$$s_\omega = \sum_i (h_i \circ \pi) \cdot \pi^* \mu_i = \pi^* \omega.$$

Thus $s_\omega$ extends $\pi^*\omega$. By (a), applied on an étale atlas of $\mathcal{Y}$, $\mathcal{F}_r^{\mathrm{an}}$ is reflexive, hence torsion-free. Since $\pi^{-1}(U \cap X_{\mathrm{reg}} \backslash \{x\})$ is dense in $\pi^{-1}(U)$, $s_\omega$ is the unique section of $\mathcal{F}_r^{\mathrm{an}}$ over $\pi^{-1}(U)$ extending $\pi^*\omega$. In particular, $s_\omega$ does not depend on the expression $\omega = \sum_i h_i \mu_i$. This proves (2).

**Proof of (3).** Let $\mathcal{Y}^\circ \subset \mathcal{Y}$ be the open substack where $\mathcal{Y}$ is smooth and $\mathcal{E}$ is a smooth divisor. Near a point of $\mathcal{E} \cap \mathcal{Y}^\circ$, let $z$ be a local equation of $\mathcal{E}$. Then $z^r$ is a local generator of $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$, any local section of $\mathcal{F}_r$ is of the form $z^r\left(\frac{dz}{z} \wedge \alpha + \beta\right)$, where $\alpha$ is a 1-form and $\beta$ is a 2-form, both regular, and the residue tensored with $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$ is

$$z^r\left(\frac{dz}{z} \wedge \alpha + \beta\right) \to \alpha|_{\mathcal{E}} \otimes z^r|_{\mathcal{E}}.$$

Therefore the residue tensored with $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$ is a morphism

$$\mathcal{F}_r|_{\mathcal{E} \cap \mathcal{Y}^\circ} \to \Omega^1_{\mathcal{E} \cap \mathcal{Y}^\circ} \otimes L.$$

We show that $\mathcal{E}_{\mathrm{reg}} \subset \mathcal{Y}^\circ$. Let $y$ be a point of $\mathcal{E}_{\mathrm{reg}}$, let $A$ be the local ring of an étale atlas of $\mathcal{Y}$ at a point over $y$, $d := \dim A$, and let $z \in A$ be a local equation of $\mathcal{E}$. Then $A/(z)$ is regular of dimension $d-1$. Lifts of $d-1$ regular parameters of $A/(z)$ together with $z$ generate the maximal

ideal of $A$, so the embedding dimension of $A$ is at most $d$, and $A$ is regular. Thus $\mathcal{Y}$ is smooth at $y$ and $\mathcal{E}$ is a smooth divisor near $y$, that is, $y \in \mathcal{Y}^\circ$.

By Lemma 2.3, $\mathcal{E}$ is normal, so the complement of $\mathcal{E}_{\mathrm{reg}}$ in $\mathcal{E}$ has codimension at least 2. Let $j : \mathcal{E}_{\mathrm{reg}} \to \mathcal{E}$ be the inclusion. Since $L$ is invertible, $\Omega^{[1]}_{\mathcal{E}} \otimes L$ is reflexive, so

$$\Omega^{[1]}_{\mathcal{E}} \otimes L = j_* j^* (\Omega^{[1]}_{\mathcal{E}} \otimes L)$$

by [Har80, Proposition 1.6], applied on an étale atlas of $\mathcal{E}$. Hence the residue over $\mathcal{E}_{\mathrm{reg}}$ extends uniquely to a morphism

$$\mathrm{res} : \mathcal{F}_r|_{\mathcal{E}} \to j_* j^* (\mathcal{F}_r|_{\mathcal{E}}) \to j_* j^* (\Omega^{[1]}_{\mathcal{E}} \otimes L) = \Omega^{[1]}_{\mathcal{E}} \otimes L,$$

since a morphism to $j_* j^* (\Omega^{[1]}_{\mathcal{E}} \otimes L)$ is determined by the restriction to $\mathcal{E}_{\mathrm{reg}}$.

Let $i : \mathcal{E} \to \mathcal{Y}$ be the inclusion. Since $\mathrm{center}_X E = x$, we have $\pi(\mathcal{E}) = \{x\}$. Since $\mathcal{E}$ is normal, hence reduced, $\pi \circ i$ factors through the closed point $x \in X$. Thus

$$(\pi^* \Omega^{[2]}_X)|_{\mathcal{E}} = (\pi \circ i)^* \Omega^{[2]}_X \cong (M_x / \mathfrak{m} M_x) \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{E}},$$

and $\mathrm{res} \circ \mathfrak{p}|_{\mathcal{E}}$ is a morphism

$$(M_x / \mathfrak{m} M_x) \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{E}} \to \Omega^{[1]}_{\mathcal{E}} \otimes L.$$

This proves (3).

**Proof of the second statement of (4).** Recall that $\omega = \sum_i h_i \mu_i$ on $U$ and $s_\omega = \sum_i (h_i \circ \pi) \cdot \mathfrak{p}(\pi^* \mu_i)$. Since $\pi(\mathcal{E}) = \{x\}$, the restriction of $h_i \circ \pi$ to $\mathcal{E}$ is the constant $h_i(x)$. Hence

$$s_\omega|_{\mathcal{E}} = \sum_i h_i(x) \cdot \mathfrak{p}(\pi^* \mu_i)|_{\mathcal{E}}, \qquad \theta_\omega = \mathrm{res}(s_\omega|_{\mathcal{E}}) = \sum_i h_i(x) \cdot \mathrm{res}\big(\mathfrak{p}(\pi^* \mu_i)|_{\mathcal{E}}\big).$$

By (3), the right-hand side is the image of the class of $\sum_i h_i(x) \mu_i$ in $M_x / \mathfrak{m} M_x$ under the morphism $\mathrm{res} \circ \mathfrak{p}|_{\mathcal{E}}$ of (3). Since the $h_i(x)$ are complex numbers and $\mathrm{res} \circ \mathfrak{p}|_{\mathcal{E}}$ is algebraic, $\theta_\omega$ is an algebraic section of $\Omega^{[1]}_{\mathcal{E}} \otimes L$.

Moreover, $\mathcal{O}^{\mathrm{an}}_{X,x} / \mathfrak{m} \mathcal{O}^{\mathrm{an}}_{X,x} = \mathbb{C}$, so

$$(\Omega^{[2]}_X)^{\mathrm{an}}_x / \mathfrak{m} (\Omega^{[2]}_X)^{\mathrm{an}}_x = M_x / \mathfrak{m} M_x,$$

and the class of $\omega$ on the left-hand side is the class of $\sum_i h_i(x) \mu_i$. Thus $\theta_\omega$ only depends on the class of $\omega$ modulo $\mathfrak{m} (\Omega^{[2]}_X)^{\mathrm{an}}_x$. □

6.2. **The residue is a contact form.** In this subsection, we prove that the residue $\theta_\omega$ of Definition-Lemma 6.2 is an orbifold contact form, that $(\mathcal{E}, L, \theta_\omega)$ is a contact Fano orbifold, and that $\theta_\omega$ is induced by the leading form of $\omega$ on the associated graded cone.

The proof of Theorem 6.3(1) uses two facts. First, the top form $\omega^n$ has log discrepancy exactly $nr$ along $E$, which forces $A_\omega(E) = r$ and excludes further zeros of the top-degree term of $\pi^* \omega$. Second, $\omega$ is closed, which implies a linear relation between the restrictions to $\mathcal{E}$ of the coefficients of $\pi^* \omega$ in local coordinates (see (6.3) below).

**Theorem 6.3.** *Notation and conditions as in Set-up 6.1. Let $\theta_\omega$ be the residue of $\omega$ in Definition-Lemma 6.2, let $W := \mathrm{Spec}\, \mathrm{gr}_{\mathrm{ord}_E} \mathcal{O}_{X,x}$ with the grading defined by $\mathrm{ord}_E$, Euler derivation $\mathrm{eu}$, and vertex $o$, let $\nu$ be an algebraic reflexive 2-form on a neighborhood of $x$ with $\omega - \nu \in \mathfrak{m} \cdot (\Omega^{[2]}_X)^{\mathrm{an}}_x$, where $\mathfrak{m} \subset \mathcal{O}_{X,x}$ is the maximal ideal, and let $\omega_0$ be the specialization of $\nu$ along the degeneration of $X$ to $W$ defined by $E$ [XZ26b, Definition 3.3], regarded as a reflexive 2-form on $W$. Then the following hold.*

*(1) We have $A_\omega(E) = r$, and $\theta_\omega \wedge (d\theta_\omega)^{n-1}$, computed in local trivializations of $L$, vanishes nowhere on $\mathcal{E}_{\mathrm{reg}}$.*

*(2) The form $\theta_\omega$ induces an isomorphism $\Omega^{[2n-1]}_{\mathcal{E}} \otimes L^n \cong \mathcal{O}_{\mathcal{E}}$ and a surjection $T_{\mathcal{E}} \to L$ whose kernel $\mathcal{K}$ is a reflexive sheaf of rank $2n-2$, and $d\theta_\omega$ induces an isomorphism $\mathcal{K} \cong \mathcal{K}^* \otimes L$.*

(3) *Let* $H := -E|_E$. *Then* $-(K_E + \Delta_E) \sim_{\mathbb{Q}} nrH$, *the line bundle* $L$ *corresponds to the* $\mathbb{Q}$*-divisor class* $rH$, *a positive power of* $L$ *descends to an ample line bundle on* $E$, *and the local index-one charts of* $\mathcal{E}$ *have klt singularities. In particular,* $(\mathcal{E}, L, \theta_\omega)$ *is a contact Fano orbifold.*

(4) *The form* $\omega_0$ *does not depend on* $\nu$, *it is a closed non-degenerate reflexive* 2*-form on* $W$ *of degree* $r$ *which extends regularly to a resolution of* $W$, *and under the isomorphism* $\mathcal{E} \cong [(W\backslash\{o\})/\mathbb{G}_m]$ *of Proposition 5.1,* $\theta_\omega$ *is the* $L$*-valued form induced by* $i_{\mathrm{eu}}\omega_0$.

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we show that $K_X$ is Cartier and compute the divisor of $\pi^*(\omega^n)$.

Since $\omega^n$ vanishes nowhere on $X_{\mathrm{reg}}$ and $X$ is normal, $\omega^n$ generates the stalk at $x$ of the analytic sheaf $(\omega_X)^{\mathrm{an}}$. Since $\mathcal{O}_{X,x} \to \mathcal{O}^{\mathrm{an}}_{X,x}$ is faithfully flat [Ser56, §2, no. 6, Proposition 3 and Corollary 1], the $\mathcal{O}_{X,x}$-module $\omega_{X,x}$ is free of rank one. Possibly shrinking $X$ around $x$, we may assume that $\omega_X$ has a generator $\Omega$ and that $\omega^n = u\Omega$ for a holomorphic function $u$ without zeros. In particular, $K_X$ is Cartier and $K_X \sim 0$. By Lemma 2.10, $X$ is klt.

Since $E$ is the only $f$-exceptional prime divisor and $A_X(E) = nr$, the divisor of the rational form $f^*\Omega$ on $Y$ is $(nr-1)E$. The morphism $q$ is an isomorphism over the open subset $Y^\circ \subset Y$ where $E$ is Cartier, and $Y^\circ$ contains all codimension one points of $Y$. Thus the divisor of $\pi^*\Omega$ is $(nr-1)\mathcal{E}$. Since $\pi^*u$ has no zeros, we obtain

$$\mathrm{div}\left(\pi^*(\omega^n)\right) = 0, \quad \text{where } \pi^*(\omega^n) \text{ is regarded as a section of } \Omega^{[2n]}_{\mathcal{Y}}(\log \mathcal{E}) \otimes \mathcal{O}_{\mathcal{Y}}(-nr\mathcal{E}). \tag{6.1}$$

**Step 2.** In this step we show that $\mathcal{Y}$ is regular along $\mathcal{E}_{\mathrm{reg}}$.

Let $Y'$ be a variety in an étale atlas of $\mathcal{Y}$, let $E'$ be the preimage of $\mathcal{E}$ in $Y'$, and let $y'$ be a regular point of $E'$. By Lemma 2.3, $E'$ is a Cartier divisor on $Y'$. Let $z$ be a local equation of $E'$ at $y'$, and let $d := \dim \mathcal{O}_{Y',y'}$. Since $\mathcal{O}_{E',y'} = \mathcal{O}_{Y',y'}/(z)$ is regular of dimension $d-1$, lifts of a regular system of parameters of $\mathcal{O}_{E',y'}$ together with $z$ generate the maximal ideal of $\mathcal{O}_{Y',y'}$ by Nakayama's lemma. Thus $\mathcal{O}_{Y',y'}$ is regular, and $E'$ is a smooth divisor on a smooth neighborhood of $y'$ in $Y'$.

**Step 3.** In this step we compute $\pi^*\omega$ near $\mathcal{E}_{\mathrm{reg}}$ and prove (1).

Notation as in **Step 2**. By **Step 2**, there exist regular functions $z, y_1, \dots, y_{2n-1}$ on a neighborhood of $y'$ in $Y'$ which form étale coordinates, with $E' = (z = 0)$. Near $y'$, the sheaf $\mathcal{F}_r$ is free with basis $z^r \frac{dz}{z} \wedge dy_i$ and $z^r dy_i \wedge dy_j$. By Definition-Lemma 6.2(2), we can uniquely write

$$\pi^*\omega = z^r\Big(\frac{dz}{z} \wedge \alpha + \beta\Big), \tag{6.2}$$

where $\alpha$ is a 1-form in $dy_1, \dots, dy_{2n-1}$ and $\beta$ is a 2-form in $dy_1, \dots, dy_{2n-1}$, both with holomorphic coefficients in $z, y_1, \dots, y_{2n-1}$. Let $\alpha_0 := \alpha|_{E'}$ and $\beta_0 := \beta|_{E'}$. By the definition of the residue of logarithmic forms, near $y'$ we have

$$\theta_\omega = \alpha_0 \otimes z^r,$$

where $z^r$ also denotes the local generator of $L$ given by the image of $z^r$.

Let $d_y$ be the exterior derivative in the variables $y_1, \dots, y_{2n-1}$. Since $\omega$ is closed, we have

$$0 = d(\pi^*\omega) = z^r\Big(\frac{dz}{z} \wedge \big(r\beta - d_y\alpha + z\partial_z\beta\big) + d_y\beta\Big).$$

Thus $r\beta - d_y\alpha + z\partial_z\beta = 0$ and $d_y\beta = 0$. Restricting to $E'$, we obtain

$$d\alpha_0 = r\beta_0 \quad \text{and} \quad d\beta_0 = 0. \tag{6.3}$$

Since $\beta$ is a 2-form in $2n-1$ variables, $\beta^n = 0$. Thus

$$(\pi^*\omega)^n = nz^{nr}\frac{dz}{z} \wedge \alpha \wedge \beta^{n-1}.$$

Near $y'$, the sheaf $\Omega^{[2n]}_{\mathcal{Y}}(\log \mathcal{E}) \otimes \mathcal{O}_{\mathcal{Y}}(-nr\mathcal{E})$ is free with generator $z^{nr}\frac{dz}{z} \wedge dy_1 \wedge \cdots \wedge dy_{2n-1}$, and a section of a line bundle on a normal variety whose divisor is zero generates the line bundle.

By (6.1), the coefficient of $\alpha \wedge \beta^{n-1}$ with respect to $dy_1 \wedge \cdots \wedge dy_{2n-1}$ is a unit near $y'$. Thus $\alpha_0 \wedge \beta_0^{n-1}$ vanishes nowhere on a neighborhood of $y'$ in $E'$. In particular, $\alpha_0 \neq 0$.

Since $q$ is an isomorphism at the generic point of $\mathcal{E}$, $A_\omega(E)$ is the order along $\mathcal{E}$ of $\pi^*\omega$, regarded as a logarithmic 2-form on the preimage of a small analytic neighborhood of $x$. By Definition-Lemma 6.2(2), $A_\omega(E) \geq r$. Taking $y'$ over a general point of $E$, (6.2) and $\alpha_0 \neq 0$ imply that $A_\omega(E) = r$.

By (6.3),

$$\alpha_0 \wedge (d\alpha_0)^{n-1} = r^{n-1}\alpha_0 \wedge \beta_0^{n-1},$$

which vanishes nowhere near $y'$ since $r > 0$. For $n = 1$, the form $\alpha_0$ itself vanishes nowhere. If $e = uz^r$ is another local generator of $L$, where $u$ is a unit, then $\theta_\omega = u^{-1}\alpha_0 \otimes e$. Since $d(u^{-1}\alpha_0) = u^{-1}d\alpha_0 + d(u^{-1}) \wedge \alpha_0$ and $\alpha_0 \wedge \alpha_0 = 0$, we have

$$u^{-1}\alpha_0 \wedge \big(d(u^{-1}\alpha_0)\big)^{n-1} = u^{-n}\alpha_0 \wedge (d\alpha_0)^{n-1}.$$

Thus the local forms $\alpha_0 \wedge (d\alpha_0)^{n-1} \otimes z^{nr}$ glue to a nowhere vanishing section $\theta_\omega \wedge (d\theta_\omega)^{n-1}$ of $\Omega^{2n-1}_{\mathcal{E}_{\mathrm{reg}}} \otimes L^n$. This proves (1).

**Step 4.** In this step we prove (2).

By Lemma 2.3, $\mathcal{E}$ is normal, so $\mathcal{E}\backslash\mathcal{E}_{\mathrm{reg}}$ has codimension at least two in $\mathcal{E}$. By **Step 3**, the morphism $\mathcal{O}_\mathcal{E} \to \Omega^{[2n-1]}_\mathcal{E} \otimes L^n$ defined by $\theta_\omega \wedge (d\theta_\omega)^{n-1}$ is an isomorphism over $\mathcal{E}_{\mathrm{reg}}$. Since both sheaves are reflexive, it is an isomorphism [Har80, Proposition 1.6].

Since $T_\mathcal{E}$ is the dual of $\Omega^{[1]}_\mathcal{E}$, the form $\theta_\omega$ induces a morphism $T_\mathcal{E} \to L$. Let $V \subset \mathcal{E}$ be an open substack on which $L$ has a generator $e$, and write $\theta_\omega|_V = \alpha_V \otimes e$ with $\alpha_V \in H^0(V, \Omega^{[1]}_\mathcal{E})$. At any point of $V \cap \mathcal{E}_{\mathrm{reg}}$, (1) implies that $(d\alpha_V)^{n-1} \neq 0$, so the alternating form $d\alpha_V$ on the tangent space, which is of dimension $2n-1$, has rank $2n-2$ and a kernel of dimension one, and $\alpha_V$ is non-zero on the kernel of $d\alpha_V$. Thus there exists a unique vector field $v$ on $V \cap \mathcal{E}_{\mathrm{reg}}$ with

$$\alpha_V(v) = 1 \quad \text{and} \quad i_v d\alpha_V = 0.$$

At every point of $V \cap \mathcal{E}_{\mathrm{reg}}$, the linear map $v \to i_v d\alpha_V + \alpha_V(v)\alpha_V$ from the tangent space to the cotangent space is invertible: if $i_v d\alpha_V + \alpha_V(v)\alpha_V = 0$, then evaluating at $v$, we obtain $\alpha_V(v)^2 = 0$, so $i_v d\alpha_V = 0$ and $v = 0$. Since $v$ satisfies $i_v d\alpha_V + \alpha_V(v)\alpha_V = \alpha_V$, the vector field $v$ is the unique solution of the square linear system $i_v d\alpha_V + \alpha_V(v)\alpha_V = \alpha_V$ in $v$, so $v$ is regular by Cramer's rule. Since $T_\mathcal{E}$ is reflexive, $v$ extends to a section of $T_\mathcal{E}$ over $V$ [Har80, Proposition 1.6], and $\theta_\omega(v) = e$ on $V$ since the equality holds on a dense open subset and $L$ is torsion free. Thus $T_\mathcal{E} \to L$ is surjective and locally split, and the kernel $\mathcal{K}$ is locally a direct summand of $T_\mathcal{E}$. Hence $\mathcal{K}$ is reflexive of rank $2n-2$.

For local sections $w_1, w_2$ of $\mathcal{K}$ over $V$, let $b_e(w_1, w_2) := d\alpha_V(w_1, w_2) \otimes e$, defined over $V \cap \mathcal{E}_{\mathrm{reg}}$ and extended to $V$ [Har80, Proposition 1.6]. If $e' = ue$ for a unit $u$, then $\theta_\omega|_V = u^{-1}\alpha_V \otimes e'$, and

$$d(u^{-1}\alpha_V)(w_1, w_2) = u^{-1}d\alpha_V(w_1, w_2)$$

since $\alpha_V(w_1) = \alpha_V(w_2) = 0$. Thus $b_e$ does not depend on $e$, and the morphisms $\mathcal{K}|_V \to \mathcal{K}^* \otimes L|_V$ defined by $b_e$ glue to a morphism $\mathcal{K} \to \mathcal{K}^* \otimes L$ induced by $d\theta_\omega$. Over $\mathcal{E}_{\mathrm{reg}}$, the restriction of $d\alpha_V$ to the kernel of $\alpha_V$ is non-degenerate by (1), so $\mathcal{K} \to \mathcal{K}^* \otimes L$ is an isomorphism over $\mathcal{E}_{\mathrm{reg}}$. Since $\mathcal{K}$ and $\mathcal{K}^* \otimes L$ are reflexive, the inverse over $\mathcal{E}_{\mathrm{reg}}$ extends to $\mathcal{E}$ [Har80, Proposition 1.6], and $\mathcal{K} \cong \mathcal{K}^* \otimes L$. This proves (2).

**Step 5.** In this step we prove (3).

Let $g : \mathcal{E} \to E$ be the morphism induced by $q$. Since $E$ is the only $f$-exceptional prime divisor and $A_X(E) = nr$, we have $K_Y + E = f^*K_X + nrE$. By **Step 1**, $K_X \sim 0$. By Definition 2.2, $(K_Y + E)|_E = K_E + \Delta_E$, so

$$K_E + \Delta_E \sim_\mathbb{Q} nrE|_E = -nrH.$$

Let $m$ be a positive integer such that $mE$ is Cartier near $E$. Then $\mathcal{O}_{\mathcal{Y}}(-mr\mathcal{E}) = q^*\mathcal{O}_Y(-mrE)$, so

$$L^m = g^*\big(\mathcal{O}_Y(-mrE)|_E\big),$$

and $L$ corresponds to the $\mathbb{Q}$-divisor class $rH$. By Lemma 2.3, $H$ is ample, so $\mathcal{O}_Y(-mrE)|_E$ is an ample line bundle on $E$.

We show that $E$ is the coarse space of $\mathcal{E}$. Let $\overline{\mathcal{E}}$ be the coarse space of $\mathcal{E}$. Since $\mathcal{E}$ is normal (Lemma 2.3), $\overline{\mathcal{E}}$ is normal. Locally on $Y$, $\mathcal{Y}$ is the quotient stack of the local index-one cover $Y' \to Y$ of $E$ by a finite cyclic group $\Gamma$ with $Y'/\Gamma = Y$, and $\overline{\mathcal{E}}$ is locally $E'/\Gamma$, where $E'$ is the preimage of $E$ in $Y'$. Thus $\overline{\mathcal{E}} \to E$ is finite and bijective. Since $q$ is an isomorphism over the codimension one points of $Y$, $\overline{\mathcal{E}} \to E$ is an isomorphism over the generic point of $E$. Since $(E, \Delta_E)$ is a klt pair (Lemma 2.3), $E$ is normal, and $\overline{\mathcal{E}} \to E$ is an isomorphism by Zariski's main theorem. Thus a positive power of $L$ descends to an ample line bundle on the coarse space $E$.

Since $q$ is an isomorphism over the codimension one points of $Y$, we have $K_{\mathcal{Y}} + \mathcal{E} = q^*(K_Y + E)$. Since $\mathcal{E}$ is a normal Cartier divisor on $\mathcal{Y}$ (Lemma 2.3) and $\mathcal{Y}$ is regular at the codimension one points of $\mathcal{E}$ (**Step 2**), $K_{\mathcal{E}} = (K_{\mathcal{Y}} + \mathcal{E})|_{\mathcal{E}}$. Since $(K_Y + E)|_E = K_E + \Delta_E$ (see the first paragraph of this step), we obtain

$$K_{\mathcal{E}} = g^*(K_E + \Delta_E) \sim_{\mathbb{Q}} -nrg^*H.$$

Let $h : Y' \to Y$ be a local index-one cover of $E$, and let $E' := h^*E$. The morphism $E' \to \mathcal{E}$ is étale, so $K_{E'} = (h|_{E'})^*(K_E + \Delta_E)$ since $K_{\mathcal{E}} = g^*(K_E + \Delta_E)$, and $h|_{E'} : E' \to E$ is finite. Since $E'$ is étale over the normal stack $\mathcal{E}$, $E'$ is normal, and $(E, \Delta_E)$ is klt (Lemma 2.3), so $E'$ is klt by [KM98, Proposition 5.20(4)]. The local index-one covers $Y' \to \mathcal{Y}$ form an étale atlas of $\mathcal{Y}$. By base change, the morphisms $E' = Y' \times_{\mathcal{Y}} \mathcal{E} \to \mathcal{E}$ form an étale atlas of $\mathcal{E}$, and the fiber product of two such charts is étale over each of them. Thus $\mathcal{E}$ has an étale atlas by varieties with klt singularities, which is the klt condition in Definition 2.12.

By (1) and (2), and since a positive power of $L$ descends to an ample line bundle on the coarse space $E$ and $\mathcal{E}$ has an étale atlas by varieties with klt singularities, $(\mathcal{E}, L, \theta_\omega)$ is a contact Fano orbifold (Definition 2.12). This proves (3).

**Step 6.** In this step we show that $\nu$ exists, that $\nu$ is non-degenerate, and that $A_{\omega-\nu}(E) \geq r + 1$.

Let $\mu_1, \dots, \mu_k$ generate $(\Omega_X^{[2]})_x$. As in Definition-Lemma 6.2(4), we can write $\omega = \sum_i h_i\mu_i$ near $x$ with holomorphic functions $h_i$. Then $\nu := \sum_i h_i(x)\mu_i$ is algebraic and

$$\omega - \nu = \sum_i (h_i - h_i(x))\mu_i \in \mathfrak{m} \cdot (\Omega_X^{[2]})_x^{\mathrm{an}}.$$

Thus a form $\nu$ as in the theorem exists.

Let $\nu$ be any such form. Write $\omega - \nu = \sum_i h_i'\mu_i$, where the $h_i'$ are holomorphic functions vanishing at $x$. Since $v = c\,\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$, Theorem 2.13(1) implies that $A_{\mu_i}(E) \geq A_X(E)/n = r$ for any $i$ with $\mu_i \neq 0$. Since $\mathrm{ord}_E(h_i') \geq 1$, we obtain

$$A_{\omega-\nu}(E) \geq r + 1. \tag{6.4}$$

In particular, $A_\nu(E) = r$ by (1).

We have $\omega^n - \nu^n \in \mathfrak{m} \cdot (\omega_X)_x^{\mathrm{an}}$, since any term of $\omega^n - \nu^n = \big(\nu + (\omega - \nu)\big)^n - \nu^n$ contains the factor $\omega - \nu$. Since $\omega^n$ generates $(\omega_X)_x^{\mathrm{an}}$ (**Step 1**), $\nu^n$ generates $(\omega_X)_x^{\mathrm{an}}$ by Nakayama's lemma, hence $\nu^n$ generates $\omega_{X,x}$. Possibly shrinking $X$ around $x$, $\nu^n$ generates $\omega_X$, so $\nu$ is non-degenerate on $X_{\mathrm{reg}}$, although $\nu$ need not be closed. Since $X$ is klt and

$$A_X(E) = nr = nA_\nu(E),$$

[XZ26b, Proposition 3.12] implies that $E$ is $\nu$-admissible [XZ26b, Definition 3.11], that is, $\omega_0$ is non-degenerate.

**Step 7.** In this step we compute $\omega_0$ on a dense open subset of $W$ and prove (4).

By Theorem 5.4, the grading of $W$ is induced by a faithful $\mathbb{G}_m$-action, so Proposition 5.1 can be applied to $v$, and the Kollár component in Proposition 5.1 is $E$ since $v = c\,\mathrm{ord}_E$. Let $L_1 := \mathcal{O}_\mathcal{Y}(-\mathcal{E})|_\mathcal{E}$ be the line bundle of Proposition 5.1, so that $L = L_1^{\otimes r}$.

Let $y_0 \in \mathcal{E}$ be a general point. Then $q$ is an isomorphism near $y_0$, and we take $y' = y_0$ and the coordinates $z, y_1, \dots, y_{2n-1}$ of **Step 3** on a neighborhood of $y_0$ in $Y$, so that (6.2) and (6.3) hold. By (6.4), possibly shrinking the neighborhood of $y_0$, we can write

$$\pi^*\nu = z^r\Big(\frac{dz}{z} \wedge \alpha_\nu + \beta_\nu\Big),$$

where $\alpha_\nu$ and $\beta_\nu$ are a 1-form and a 2-form in $dy_1, \dots, dy_{2n-1}$ with regular coefficients, $\alpha_\nu|_E = \alpha_0$, and $\beta_\nu|_E = \beta_0$.

Near $y_0$, the deformation to the normal cone of $E$ in $Y$ has coordinates $s, y_1, \dots, y_{2n-1}, t$ with $z = ts$, where $t$ is the parameter of the deformation, and the central fiber $(t = 0)$ is the normal line bundle of $E$, with fiber coordinate $s$. By [XZ26b, Lemma 2.9], the deformation to the normal cone is isomorphic to the degeneration of $X$ to $W$ defined by $E$ at the generic point of $W$. Let $h \in \mathcal{O}_{X,x}$ with $\mathrm{ord}_E(h) \geq m$, and write $\pi^*h = z^m u$ near $y_0$ with $u$ regular. Then $t^{-m}h = s^m u(ts, y)$ on the deformation to the normal cone, whose restriction to $(t = 0)$ is $s^m u(0, y)$. By Proposition 5.1(2), the class of $h$ in degree $m$ corresponds to the section $u|_E \cdot z^m$ of $L_1^{\otimes m}$. Under the isomorphisms of Proposition 5.1(2)(3), the functions $y_i|_E$ correspond to functions on $W\backslash\{o\}$ pulled back from $\mathcal{E}$, and $s$ is the homogeneous function of degree one corresponding to the local generator of $L_1$ given by the image of $z$. In relative differentials over $t$, we have $\frac{dz}{z} = \frac{ds}{s}$, so modulo $dt$,

$$\nu = t^r s^r\Big(\frac{ds}{s} \wedge \alpha_\nu(ts, y) + \beta_\nu(ts, y)\Big).$$

The form $t^{-r}\nu$ is regular near the generic point of $(t = 0)$ and restricts on $(t = 0)$ to the non-zero form below. Thus the exponent $m$ in [XZ26b, Definition 3.3] equals $-r$, and the specialization $\omega_0$ of $\nu$ is, on a dense open subset $W^\circ$ of $W$,

$$\omega_0 = s^r\Big(\frac{ds}{s} \wedge \alpha_0 + \beta_0\Big). \tag{6.5}$$

The right side of (6.5) depends only on $\omega$, and a reflexive form on $W$ is determined by the restriction to $W^\circ$. Thus $\omega_0$ does not depend on $\nu$. By (6.3),

$$d\omega_0 = s^{r-1}ds \wedge (r\beta_0 - d\alpha_0) + s^r d\beta_0 = 0$$

on $W^\circ$. Since $d\omega_0$ is a regular 3-form on $W_{\mathrm{reg}}$ which vanishes on the dense open subset $W^\circ \cap W_{\mathrm{reg}}$, $\omega_0$ is closed on $W_{\mathrm{reg}}$. By **Step 6**, $\omega_0$ is non-degenerate. The $\mathbb{G}_m$-action multiplies $s$ by $\lambda$ and fixes $y_1, \dots, y_{2n-1}$, so $\omega_0$ is homogeneous of degree $r$ on $W^\circ$, hence on $W$, and $\mathrm{eu} = s\partial_s$ on $W^\circ$. By Theorem 2.20, $W$ is klt, so $\omega_0$ extends regularly to any log resolution of $W$ by [GKKP11, Theorem 1.4].

Finally, by (6.5),

$$i_{\mathrm{eu}}\omega_0 = s^r\alpha_0$$

on $W^\circ$. The 1-form $i_{\mathrm{eu}}\omega_0$ on $W\backslash\{o\}$ is homogeneous of degree $r$ and is annihilated by $i_{\mathrm{eu}}$, so it descends to a section $\vartheta$ of $\Omega_\mathcal{E}^{[1]} \otimes L_1^{\otimes r} = \Omega_\mathcal{E}^{[1]} \otimes L$. Since $s$ corresponds to the image of $z$ in $L_1$, we have $\vartheta = \alpha_0 \otimes z^r = \theta_\omega$ near $y_0$. Since $\Omega_\mathcal{E}^{[1]} \otimes L$ is torsion free, $\theta_\omega = \vartheta$ on $\mathcal{E}$. This proves (4). □

**Remark 6.4.** When $\omega$ is algebraic, one may take $\nu = \omega$ in Theorem 6.3(4), and the closedness of $\omega_0$ also follows from [XZ26b, Proposition 3.5(1)].

With the local index-one covers $E' \to E'/\Gamma$ of **Step 5** of the proof of Theorem 6.3 as charts, the contact Fano orbifold $(\mathcal{E}, L, \theta_\omega)$ of Theorem 6.3(3) is a contact orbifold in the sense of [Nam11, (4.4)]. In that definition, the charts are allowed to be singular. Since klt singularities are rational [KM98, Theorem 5.22], the local index-one charts of $\mathcal{E}$ in Theorem 6.3(3) are contact varieties in the sense of [Smi22, Definition 2.1].

6.3. **Kähler differentials.** In this subsection, we show that Theorem 6.3 has to be formulated with reflexive differentials.

In the following example, $\theta_\omega$ is a reflexive form but not locally the image of a Kähler differential: on the open subset of $Y$ where $E$ is Cartier, the index-one cover is an isomorphism, so it does not resolve the $A_1$ singularities of $\mathcal{E}$ there.

**Example 6.5.** Let $\sigma_0 := dx_1 \wedge dx_2 + dx_3 \wedge dx_4$ on $\mathbb{A}^4$, let $G = \{1, g\}$ act by $g(x_1, x_2, x_3, x_4) = (-x_1, -x_2, x_3, x_4)$, let $X := \mathbb{A}^4/G$ with vertex $o$, and let $\omega$ be the form on $X_{\mathrm{reg}}$ induced by $\sigma_0$. Let $E$ be the image in $\mathrm{Bl}_0 \mathbb{A}^4/G$ of the exceptional divisor of $\mathrm{Bl}_0 \mathbb{A}^4$. Then $E$ is a Kollár component over $(X \ni o)$, $\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$, and

$$A_X(E) = 4, \qquad A_\omega(E) = 2, \qquad \mathrm{vol}(\mathrm{ord}_E) = \frac{1}{2}, \qquad \widehat{\mathrm{vol}}(X \ni o) = 128.$$

With the notation of Set-up 6.1 and Definition-Lemma 6.2, the residue $\theta_\omega$ of $\omega$ is an orbifold contact form with $L = \mathcal{O}_{\mathcal{Y}}(-2\mathcal{E})|_{\mathcal{E}}$, but $\theta_\omega$ is not locally contained in the image of $\Omega^1_{\mathcal{E}} \otimes L \to \Omega^{[1]}_{\mathcal{E}} \otimes L$, where $\Omega^1_{\mathcal{E}}$ is the sheaf of Kähler differentials.

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we show that $X$ is a symplectic singularity and that $E$ is a Kollár component with $A_X(E) = 4$.

Since $g^*\sigma_0 = \sigma_0$, the form $\omega$ is well defined, and $X \cong (\mathbb{A}^2/\{\pm 1\}) \times \mathbb{A}^2$, where $\mathbb{A}^2/\{\pm 1\}$ is the $A_1$ surface singularity. The minimal resolution of $\mathbb{A}^2/\{\pm 1\}$ is covered by two charts with coordinates $(u_1, u_2)$ and $(u_1', u_2')$, where $u_1 = x_1^2$, $u_2 = x_2/x_1$, $u_1' = x_2^2$, and $u_2' = x_1/x_2$, and

$$dx_1 \wedge dx_2 = \frac{1}{2} du_1 \wedge du_2 = -\frac{1}{2} du_1' \wedge du_2'.$$

Thus the pullback of $\omega$ extends to a regular form on the product of the minimal resolution with $\mathbb{A}^2$, which is a resolution of $X$. Since the extension condition in Definition 2.9 only needs to be checked on one resolution [Bea99, (1.2)], $X$ is a symplectic singularity of dimension 4 with symplectic form $\omega$. In particular, $n = 2$. Since $\omega^2$ vanishes nowhere on $X_{\mathrm{reg}}$ and $X$ is normal, $\omega^2$ generates $\omega_X$.

Let $\rho : \mathrm{Bl}_0 \mathbb{A}^4 \to \mathbb{A}^4$ be the blow-up of the origin, let $\widetilde{E} \cong \mathbb{P}^3$ be the exceptional divisor, let $Y := \mathrm{Bl}_0 \mathbb{A}^4/G$, let $p : \mathrm{Bl}_0 \mathbb{A}^4 \to Y$ be the quotient morphism, and let $f : Y \to X$ be the induced morphism. Then $f$ is projective and birational, $f$ is an isomorphism over $X \backslash \{o\}$, and $E = p(\widetilde{E})$ is the only $f$-exceptional prime divisor. The fixed locus of $g$ on $\mathrm{Bl}_0 \mathbb{A}^4$ is the union of the strict transform of the plane $(x_1 = x_2 = 0)$ and the lines $(x_1 = x_2 = 0)$ and $(x_3 = x_4 = 0)$ in $\widetilde{E} \cong \mathbb{P}^3$, which is of codimension 2. Thus $p$ is quasi-étale and $p^*E = \widetilde{E}$. The variety $\mathrm{Bl}_0 \mathbb{A}^4$ is covered by the $G$-stable charts $U_j$ $(1 \le j \le 4)$ on which $x_j$ is a local equation of $\widetilde{E}$. Since $g^*x_j = \pm x_j$, the function $x_j^2$ is $G$-invariant, and since $p^*E = \widetilde{E}$, it is a local equation of $2E$ on $U_j/G$. Thus $2E$ is Cartier. Since $p$ is finite and surjective and $-\widetilde{E}$ is ample over $\mathbb{A}^4$, $-E$ is $f$-ample.

By [KM98, Lemma 2.29], $\rho^*(\sigma_0^2)$ vanishes to order 3 along $\widetilde{E}$. Since $\sigma_0^2$ is the pullback of $\omega^2$, which generates $\omega_X$, and $p$ is étale at the generic point of $\widetilde{E}$, $f^*(\omega^2)$ vanishes to order 3 along $E$. Thus $A_X(E) = 4$, and $K_Y + E = f^*K_X + 4E$ is $\mathbb{Q}$-Cartier.

Since $p$ is quasi-étale and $p^*E = \widetilde{E}$, we have $K_{\mathrm{Bl}_0 \mathbb{A}^4} + \widetilde{E} = p^*(K_Y + E)$. Since $\widetilde{E}$ is a smooth divisor on the smooth variety $\mathrm{Bl}_0 \mathbb{A}^4$, $\mathrm{discrep}(\mathrm{Bl}_0 \mathbb{A}^4, \widetilde{E}) = 0$ [KM98, Corollary 2.31(3)]. By [KM98, Proposition 5.20(2)],

$$2\big(\mathrm{discrep}(Y, E) + 1\big) \ge \mathrm{discrep}(\mathrm{Bl}_0 \mathbb{A}^4, \widetilde{E}) + 1 > 0,$$

so $(Y, E)$ is plt [KM98, Definition 2.34]. Thus $E$ is a Kollár component over $(X \ni o)$.

**Step 2.** In this step we compute $\mathrm{vol}(\mathrm{ord}_E)$ and $\widehat{\mathrm{vol}}(X \ni o)$, and show that $\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$.

Since $p$ is étale at the generic point of $\widetilde{E}$, $\mathrm{ord}_E$ is the restriction of $\mathrm{ord}_{\widetilde{E}}$ to $\mathbb{C}(X)$. Since $\mathrm{ord}_{\widetilde{E}}$ is the order of vanishing at the origin, $\mathrm{ord}_E$ of a $G$-invariant polynomial is the least total degree of the monomials in it. The ring of $G$-invariant polynomials is spanned by the monomials $x_1^{i_1}x_2^{i_2}x_3^{i_3}x_4^{i_4}$ with $i_1+i_2$ even. For a positive integer $m$, let $N_m$ be the colength of the valuation ideal $\{h \in \mathcal{O}_{X,o} \mid \mathrm{ord}_E(h) \geq m\}$. The quotient is supported at $o$, so localization at $o$ does not change it, and $N_m$ is the number of monomials $x_1^{i_1}x_2^{i_2}x_3^{i_3}x_4^{i_4}$ with $i_1+i_2$ even and $i_1+i_2+i_3+i_4 < m$. The generating series of the number of monomials of degree $k$ with $i_1+i_2$ even minus the number of monomials of degree $k$ with $i_1+i_2$ odd is

$$\frac{1}{(1+t)^2(1-t)^2} = \frac{1}{(1-t^2)^2} = \sum_{h\geq 0}(h+1)t^{2h},$$

and the number of all monomials of degree less than $m$ is $\binom{m+3}{4}$. The number of monomials of degree $k$ with $i_1+i_2$ even is half the sum of the number of all monomials of degree $k$ and of the coefficient of $t^k$ in $(1-t^2)^{-2}$, so

$$N_m = \frac{1}{2}\left(\binom{m+3}{4} + \sum_{h\geq 0, 2h<m}(h+1)\right) = \frac{1}{2}\binom{m+3}{4} + O(m^2).$$

Thus

$$\mathrm{vol}(\mathrm{ord}_E) = \lim_{m\to+\infty}\frac{4!N_m}{m^4} = \frac{1}{2}, \qquad \widehat{\mathrm{vol}}(\mathrm{ord}_E) = 4^4\cdot\frac{1}{2} = 128. \tag{6.6}$$

Let $\widehat{\mathrm{vol}}^G(\mathbb{A}^4 \ni 0)$ be the infimum of $\widehat{\mathrm{vol}}$ over the $G$-invariant valuations centered at 0 [LX17, Section 2.2]. By [LX17, Theorem 1.6], $\widehat{\mathrm{vol}}(\mathbb{A}^4 \ni 0) = 4^4 = 256$. The valuation $\mathrm{ord}_0$ is $G$-invariant, $A_{\mathbb{A}^4}(\mathrm{ord}_0) = 4$ by [KM98, Lemma 2.29], and $\mathrm{vol}(\mathrm{ord}_0) = 1$, so $\widehat{\mathrm{vol}}(\mathrm{ord}_0) = 256$ and $\widehat{\mathrm{vol}}^G(\mathbb{A}^4 \ni 0) = 256$. Since $G$ acts faithfully on $\mathbb{A}^4$ and freely in codimension one, [LX17, Theorem 2.7(1)] implies

$$\widehat{\mathrm{vol}}(X \ni o) = \frac{1}{|G|}\widehat{\mathrm{vol}}^G(\mathbb{A}^4 \ni 0) = 128.$$

By (6.6), $\mathrm{ord}_E$ is a minimizer of $\widehat{\mathrm{vol}}$. By Theorem 2.20, any minimizer of $\widehat{\mathrm{vol}}$ is a positive multiple of $\mathrm{ord}_E$, so $E$ is the Kollár component of Set-up 6.1 and $r = A_X(E)/n = 2$. By Theorem 6.3(1), $A_\omega(E) = 2$, and $\theta_\omega$ is an orbifold contact form with $L = \mathcal{O}_{\mathcal{Y}}(-2\mathcal{E})|_{\mathcal{E}}$.

**Step 3.** In this step we compute $\theta_\omega$ on a chart and show that $\theta_\omega$ is not locally contained in the image of $\Omega^1_{\mathcal{E}} \otimes L$.

On the chart $(x_3 \neq 0)$ of $\mathrm{Bl}_0\,\mathbb{A}^4$, we have coordinates $a, b, t, z$ with

$$x_3 = z, \quad x_1 = za, \quad x_2 = zb, \quad x_4 = zt,$$

and $g(a,b,t,z) = (-a,-b,t,z)$. Thus the corresponding chart of $Y$ is

$$\mathrm{Spec}\,\mathbb{C}[w_1,w_2,w_3,t,z]/(w_1w_3 - w_2^2), \qquad w_1 = a^2, \quad w_2 = ab, \quad w_3 = b^2,$$

and $E$ is the Cartier divisor $(z=0)$ on it. Since the index-one cover of $E$ is an isomorphism over the locus where $E$ is Cartier, $q$ is an isomorphism over the chart, and $\mathcal{E}$ is $V \times \mathbb{A}^1_t$ over the chart, where $V := \mathrm{Spec}\,\mathbb{C}[w_1,w_2,w_3]/(w_1w_3 - w_2^2)$. Over the chart, $\pi = f$ and $p^*f^*\omega = \rho^*\sigma_0$, and a direct calculation shows that

$$\rho^*\sigma_0 = z^2\left(\frac{dz}{z}\wedge(dt + adb - bda) + da\wedge db\right).$$

Trivializing $L$ by the image of $z^2$, we obtain

$$\theta_\omega = (dt + adb - bda)\otimes z^2,$$

where $adb - bda$ is invariant under $g$ and defines a regular form on $V_{\mathrm{reg}}$, hence a reflexive 1-form on $V$.

Suppose that $\theta_\omega$ is contained in the image of $\Omega^1_{\mathcal{E}} \otimes L$ near the point $w_1 = w_2 = w_3 = t = 0$ of $\mathcal{E}$. Then there exist regular functions $\varphi_0, \varphi_1, \varphi_2, \varphi_3$ near $w_1 = w_2 = w_3 = t = 0$ such that $dt + adb - bda = \varphi_0 dt + \varphi_1 dw_1 + \varphi_2 dw_2 + \varphi_3 dw_3$ as reflexive forms. Since equality of reflexive forms is checked on the smooth locus, subtracting $dt$ and restricting to the slice $(t = 0)$, we obtain

$$adb - bda = \psi_1 dw_1 + \psi_2 dw_2 + \psi_3 dw_3$$

on $V_{\mathrm{reg}}$ near the vertex of $V$, where $\psi_i := \varphi_i|_{t=0}$ are regular functions near the vertex. Pulling back to the $(a, b)$-plane, we have $dw_1 = 2ada$, $dw_2 = adb + bda$, and $dw_3 = 2bdb$, and the pullbacks of the $\psi_i$ are power series in $a, b$ whose terms of positive degree have degree at least 2. Comparing the terms of degree one in the coefficients of $da$ and $db$, we obtain

$$-b = 2\psi_1(0)a + \psi_2(0)b \quad \text{and} \quad a = \psi_2(0)a + 2\psi_3(0)b.$$

Thus $\psi_2(0) = -1$ and $\psi_2(0) = 1$, a contradiction. Hence $\theta_\omega$ is not contained in the image of $\Omega^1_{\mathcal{E}} \otimes L \to \Omega^{[1]}_{\mathcal{E}} \otimes L$ near the point $w_1 = w_2 = w_3 = t = 0$. □

## 7. Minimizing symplectic cones and their covers

In this section, we prove that a cone whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$ has no non-zero derivation of negative degree preserving the maximal ideal of the vertex (Lemma 7.1), that for a conical symplectic variety whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$, the part of degree $r/2$ splits off as a polynomial symplectic factor (Proposition 7.3), and that when the symplectic form has degree $2s$, the derivations of degree $-s$ are exactly the Hamiltonian derivations of the elements of $R_s$ (Lemma 7.4). We also prove that a quasi-étale cover of a product is dominated by a product of quasi-étale covers of the factors, which is étale outside a closed subset of codimension at least 4 (Lemma 7.5), and that the lifted degree valuation of a graded quasi-étale cover of such a cone is again a minimizer of $\widehat{\mathrm{vol}}$ (Proposition 7.6). The central fiber of the K-polystable degeneration in Section 10, the quasi-étale covers of the central fiber, and the factors of the central fiber are such cones, and the results of this section are used in Sections 8, 9, and 10.

### 7.1. Negative derivations on minimizing cones.

In this subsection, we study homogeneous derivations of negative degree on cones whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$.

A minimizer of $\widehat{\mathrm{vol}}$ is invariant under the automorphisms fixing the vertex, while the exponential of a non-zero derivation of negative degree preserving the maximal ideal of the vertex does not preserve the degree valuation. Thus such derivations do not exist. Without the minimizing assumption, the conclusion fails, see Example 7.2 below.

**Lemma 7.1.** *Let $X = \operatorname{Spec} R$ be a normal affine variety with klt singularities, let $R = \bigoplus_{j\geq 0} R_j$ be a grading with $R_0 = \mathbb{C}$, let $\mathfrak{m} := \bigoplus_{j>0} R_j$, and let $o \in X$ be the point defined by $\mathfrak{m}$. Assume that the degree valuation of the grading is a minimizer of $\widehat{\mathrm{vol}}$ at $o$. Then any homogeneous derivation $D$ of $R$ of negative degree with $D(\mathfrak{m}) \subset \mathfrak{m}$ is zero.*

*Proof.* Let $v$ be the degree valuation of the grading, so that $v(\sum_j f_j) = \min\{j \mid f_j \neq 0\}$ for $f_j \in R_j$. Let $g$ be an automorphism of the $\mathbb{C}$-algebra $R$ with $g(\mathfrak{m}) = \mathfrak{m}$. Then $g$ induces an automorphism of $X$ fixing $o$, and since $v$ is a minimizer of $\widehat{\mathrm{vol}}$ at $o$, we have

$$v(g(f)) = v(f) \quad \text{for any } f \in R$$

by [XZ20, Theorem 1.1 and Corollary 1.2].

Let $D$ be a homogeneous derivation of $R$ of degree $-d$ with $d > 0$ and $D(\mathfrak{m}) \subset \mathfrak{m}$. Since $D(R_0) \subset R_{-d} = 0$, $D$ is $\mathbb{C}$-linear. For $f \in R_j$ we have $D^k(f) \in R_{j-kd} = 0$ when $kd > j$. Thus, for $c \in \mathbb{C}$, the sum

$$\exp(cD)(f) := \sum_{k\geq 0} \frac{c^k}{k!} D^k(f)$$

is finite for any $f \in R$, and $\exp(cD)$ is a $\mathbb{C}$-linear map $R \to R$. By the Leibniz formula

$$D^k(fh) = \sum_{i=0}^{k} \binom{k}{i} D^i(f) D^{k-i}(h),$$

$\exp(cD)(fh) = \exp(cD)(f)\exp(cD)(h)$ for any $f, h \in R$. Since $\sum_{k+l=m} c^k(-c)^l/(k!l!) = 0$ for any $m \geq 1$, we have $\exp(cD) \circ \exp(-cD) = \exp(-cD) \circ \exp(cD) = \mathrm{id}_R$. Since $D(\mathfrak{m}) \subset \mathfrak{m}$, both $\exp(cD)$ and $\exp(-cD)$ map $\mathfrak{m}$ into $\mathfrak{m}$, hence $\exp(cD)(\mathfrak{m}) = \mathfrak{m}$. Therefore $\exp(cD)$ is an automorphism of the $\mathbb{C}$-algebra $R$ fixing $\mathfrak{m}$, and $v(\exp(cD)(f)) = v(f)$ for any $f \in R$.

Suppose that $D \neq 0$. Since $R$ is spanned by homogeneous elements, there exist an integer $j \geq 0$ and $f \in R_j$ with $D(f) \neq 0$. Let $N \geq 1$ be the largest integer with $D^N(f) \neq 0$. Then

$$\exp(D)(f) = \sum_{k=0}^{N} \frac{1}{k!} D^k(f), \qquad D^k(f) \in R_{j-kd},$$

and the component of $\exp(D)(f)$ of degree $j - Nd$ is $D^N(f)/N! \neq 0$. Hence

$$v(\exp(D)(f)) = j - Nd < j = v(f),$$

a contradiction. Thus $D = 0$. □

**Example 7.2.** Let $R = \mathbb{C}[x, y]$ with the grading given by $\deg x = 1$ and $\deg y = 3$, let $\mathfrak{m} := (x, y)$, let $\zeta$ be the $\mathbb{G}_m$-action $\lambda \cdot (x, y) = (\lambda x, \lambda^3 y)$, which induces the grading of $R$, let $\mathrm{wt}_\zeta$ be the degree valuation of the grading, and let $D := x^2 \partial/\partial y$. Then $D$ is a homogeneous derivation of degree $-1$. Since $R_1 = \mathbb{C}x$ and $D(x) = 0$, we have $D(R_1) = 0$ and $D(R_j) \subset R_{j-1} \subset \mathfrak{m}$ for $j \geq 2$, so $D(\mathfrak{m}) \subset \mathfrak{m}$. The automorphism $\exp(D)$ maps $y$ to $y + x^2$, and

$$\mathrm{wt}_\zeta(y + x^2) = 2 < 3 = \mathrm{wt}_\zeta(y).$$

Since $\mathrm{wt}_\zeta$ is the monomial valuation with weights 1 and 3 on $x$ and $y$, a direct calculation shows that $A_{\mathbb{A}^2}(\mathrm{wt}_\zeta) = 1 + 3 = 4$ and $\mathrm{vol}(\mathrm{wt}_\zeta) = 1/(1 \cdot 3) = 1/3$. Hence

$$\widehat{\mathrm{vol}}(\mathrm{wt}_\zeta) = \frac{16}{3} > 4 = \widehat{\mathrm{vol}}(\mathrm{ord}_0),$$

and $\mathrm{wt}_\zeta$ is not a minimizer of $\widehat{\mathrm{vol}}$ at $0 \in \mathbb{A}^2$. In particular, Lemma 7.1 does not hold without the assumption that the degree valuation is a minimizer of $\widehat{\mathrm{vol}}$.

**Proposition 7.3.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety of dimension $2n$, $n \geq 1$, whose symplectic form $\omega$ has degree $r$ and whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex, and write $R = \bigoplus_{j \geq 0} R_j$. Let $\{,\}$ be the Poisson bracket of $\omega$, and set $R_{r/2} := 0$ if $r$ is odd. Then $R_j = 0$ for any integer $0 < j < r$ with $j \neq r/2$, and $\{,\}$ restricts to a non-degenerate alternating form on $R_{r/2}$. Write $\dim R_{r/2} = 2b$. Then there exist a symplectic basis $p_1, q_1, \ldots, p_b, q_b$ of $R_{r/2}$ and a finitely generated graded normal Poisson subalgebra $S \subset R$ with $S_0 = \mathbb{C}$ and $S_j = 0$ for $0 < j < r$ such that multiplication induces a graded Poisson isomorphism*

$$\mathbb{C}[p_1, q_1, \ldots, p_b, q_b] \otimes_{\mathbb{C}} S \cong R,$$

*where $\{p_i, q_j\} = \delta_{ij}$, $\{p_i, p_j\} = \{q_i, q_j\} = 0$, and $\{p_i, S\} = \{q_i, S\} = 0$. The variety $\operatorname{Spec} S$ is a point or a conical symplectic variety of dimension $2n - 2b$ whose symplectic form has degree $r$.*

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step, we recall basic properties of the Poisson bracket and prove that $R_j = 0$ for any integer $0 < j < r$ with $j \neq r/2$.

Let $\mathfrak{m} := \bigoplus_{j>0} R_j$, and for $f \in R$ let $H_f := \{f, \cdot\}$ be the Hamiltonian derivation of $f$. Since $\omega$ is homogeneous of degree $r$, $H_f$ is homogeneous of degree $j - r$ for $f \in R_j$, and hence

$$\{R_j, R_l\} \subset R_{j+l-r} \quad \text{for any } j, l \geq 0,$$

where $R_t := 0$ for $t < 0$. Since $\omega$ is closed, $\{,\}$ satisfies the Jacobi identity, so $[H_f, H_h] = H_{\{f,h\}}$ for any $f, h \in R$. Moreover, $H_{fh} = fH_h + hH_f$ for any $f, h \in R$, and $H_c = 0$ for $c \in \mathbb{C}$. If $H_f = 0$ for some $f \in R$, then $df = 0$ on $X_{\mathrm{reg}}$ as $i_{H_f}\omega = -df$, so $f \in \mathbb{C}$ since $X_{\mathrm{reg}}$ is connected. By Lemma 2.10, $X$ has klt singularities, so Lemma 7.1 holds for $R$.

Suppose that $0 < j < r/2$ and $0 \neq f \in R_j$. Let $D := H_{f^2} = 2fH_f$. Then $D$ is homogeneous of degree $2j - r < 0$, $D \neq 0$ since $f^2 \notin \mathbb{C}$, and $D(R) \subset fR \subset \mathfrak{m}$. In particular, $D(\mathfrak{m}) \subset \mathfrak{m}$, which contradicts Lemma 7.1. Thus $R_j = 0$ for any $0 < j < r/2$.

Suppose that $r/2 < j < r$ and $0 \neq f \in R_j$. Since $R_l = 0$ for $0 < l < r/2$, any $l > 0$ with $R_l \neq 0$ satisfies $l \geq r/2$, so $j + l - r > 0$ and $H_f(R_l) \subset R_{j+l-r} \subset \mathfrak{m}$. Thus $H_f(\mathfrak{m}) \subset \mathfrak{m}$. Since $H_f \neq 0$ is homogeneous of degree $j - r < 0$, this contradicts Lemma 7.1. Thus $R_j = 0$ for any $0 < j < r$ with $j \neq r/2$.

**Step 2.** In this step, we prove that $\{,\}$ restricts to a non-degenerate alternating form on $R_{r/2}$.

If $r$ is odd, then $R_{r/2} = 0$ and there is nothing to prove. Suppose that $r = 2s$. Then $\{R_s, R_s\} \subset R_0 = \mathbb{C}$ by **Step 1**, so $\{,\}$ restricts to an alternating bilinear form on $R_s$, and $\dim R_s < +\infty$ since $R$ is finitely generated. Suppose that $0 \neq f \in R_s$ and $\{f, R_s\} = 0$. Then $H_f(R_s) = 0$. For any $l > 0$ with $l \neq s$ and $R_l \neq 0$, **Step 1** implies that $l \geq 2s$, so $H_f(R_l) \subset R_{l-s} \subset \mathfrak{m}$. Thus $H_f(\mathfrak{m}) \subset \mathfrak{m}$. Since $H_f \neq 0$ is homogeneous of degree $-s < 0$, this contradicts Lemma 7.1. Therefore the alternating form $\{,\}$ on $R_s$ is non-degenerate. In particular, $\dim R_s$ is even.

**Step 3.** In this step, we construct $p_1, q_1, \ldots, p_b, q_b$ and $S$, and prove that multiplication induces a graded Poisson isomorphism $\mathbb{C}[p_1, q_1, \ldots, p_b, q_b] \otimes_{\mathbb{C}} S \cong R$.

If $b = 0$, we let $S := R$. Suppose that $b \geq 1$. Then $r = 2s$ for some integer $s \geq 1$. By **Step 2**, there exists a basis $p_1, q_1, \ldots, p_b, q_b$ of $R_s$ with $\{p_i, q_j\} = \delta_{ij}$ and $\{p_i, p_j\} = \{q_i, q_j\} = 0$ for any $i, j$. Let

$$(u_1, \ldots, u_{2b}) := (p_1, q_1, \ldots, p_b, q_b),$$

and let $D_{2i-1} := -H_{q_i}$ and $D_{2i} := H_{p_i}$ for $1 \leq i \leq b$. Then $D_k(u_l) = \delta_{kl}$ for any $k, l$. By **Step 1**, any $D_k$ is homogeneous of degree $-s$, so $D_k^m(R_j) \subset R_{j-ms} = 0$ when $ms > j$, and $D_k$ is locally nilpotent. Each $D_k$ is $H_f$ or $-H_f$ for some $f \in \{p_1, q_1, \ldots, p_b, q_b\} \subset R_s$. Since $\{R_s, R_s\} \subset \mathbb{C}$ by **Step 2** and $H_c = 0$ for $c \in \mathbb{C}$, the identity $[H_f, H_h] = H_{\{f,h\}}$ of **Step 1** implies that $[D_k, D_l] = 0$ for any $k, l$.

For $0 \leq k \leq 2b$, let $R^{(k)} := \bigcap_{l=1}^{k} \operatorname{Ker} D_l$, so that $R^{(0)} = R$, and let $S := R^{(2b)}$. Since $D_1, \ldots, D_{2b}$ are homogeneous, any $R^{(k)}$ is a graded subalgebra of $R$, and it is an integral domain containing $\mathbb{C}$. Let $1 \leq k \leq 2b$. Since $D_k$ commutes with $D_1, \ldots, D_{k-1}$, we have $D_k(R^{(k-1)}) \subset R^{(k-1)}$, and $D_k|_{R^{(k-1)}}$ is a locally nilpotent derivation of $R^{(k-1)}$ whose kernel is $R^{(k)}$. Moreover, $u_k \in R^{(k-1)}$ and $D_k(u_k) = 1$. By the slice theorem [Fre17, Corollary 1.26], applied to the integral domain $R^{(k-1)} \supset \mathbb{C}$, the locally nilpotent derivation $D_k|_{R^{(k-1)}}$ and the element $u_k$ with $D_k(u_k) = 1$, we have $R^{(k-1)} = R^{(k)}[u_k]$, a polynomial ring in $u_k$ over $R^{(k)}$. By induction on $k$,

$$R = S[u_1, \ldots, u_{2b}]$$

is a polynomial ring in $u_1, \ldots, u_{2b}$ over $S$.

For $f \in S$ we have

$$\{p_i, f\} = D_{2i}(f) = 0, \qquad \{q_i, f\} = -D_{2i-1}(f) = 0$$

for any $i$, so $\{u_k, f\} = 0$ for any $k$. Hence, for $f, h \in S$, the Jacobi identity implies that $\{u_k, \{f, h\}\} = \{\{u_k, f\}, h\} + \{f, \{u_k, h\}\} = 0$ for any $k$, so $\{f, h\} \in S$. Thus $S$ is a Poisson subalgebra of $R$ with $\{p_i, S\} = \{q_i, S\} = 0$. Since $S$ is graded and $\deg p_i = \deg q_i = s = r/2$, multiplication induces a graded isomorphism of $\mathbb{C}$-algebras

$$\mu : \mathbb{C}[p_1, q_1, \ldots, p_b, q_b] \otimes_{\mathbb{C}} S \to R.$$

Let $\{,\}'$ be the Poisson bracket on $\mathbb{C}[p_1, q_1, \ldots, p_b, q_b] \otimes_{\mathbb{C}} S$ given by $\{p_i, q_j\}' = \delta_{ij}$, $\{p_i, p_j\}' = \{q_i, q_j\}' = 0$, $\{p_i, S\}' = \{q_i, S\}' = 0$, and the restriction of $\{,\}$ to $S$. Then $\{,\}'$ and the pullback

of $\{,\}$ by $\mu$ are biderivations which coincide on pairs of generators, hence they coincide. Thus $\mu$ is a graded Poisson isomorphism.

**Step 4.** In this step, we prove that $S$ is finitely generated and normal, $S_0 = \mathbb{C}$, and $S_j = 0$ for $0 < j < r$.

If $b = 0$, then $S = R$, and the claims follow from **Step 1**. Suppose that $b \geq 1$. Since $S_0 \subset R_0 = \mathbb{C}$, $S_0 = \mathbb{C}$. By **Step 1**, $S_j \subset R_j = 0$ for any $0 < j < r$ with $j \neq s$. If $f \in S_s$, then $\{p_i, f\} = \{q_i, f\} = 0$ for any $i$, so $\{f, R_s\} = 0$, and $f = 0$ by **Step 2**. Thus $S_j = 0$ for any $0 < j < r$. Since $R = S[u_1, \dots, u_{2b}]$, we have $S \cong R/(u_1, \dots, u_{2b})$, so $S$ is finitely generated.

Let $f \in \operatorname{Frac} S$ be integral over $S$. Then $f$ is integral over $R$, so $f \in R$ since $R$ is normal. Write $f = \sum_\alpha f_\alpha u^\alpha$ with $f_\alpha \in S$, where $u^\alpha := u_1^{\alpha_1} \cdots u_{2b}^{\alpha_{2b}}$, and write $f = h/c$ with $h, c \in S$ and $c \neq 0$. Then $\sum_\alpha c f_\alpha u^\alpha = h$ in $S[u_1, \dots, u_{2b}]$, so $c f_\alpha = 0$ for any $\alpha \neq 0$. Since $S$ is an integral domain, $f_\alpha = 0$ for any $\alpha \neq 0$, and $f = f_0 \in S$. Thus $S$ is normal.

**Step 5.** In this step, we prove that $\operatorname{Spec} S$ is a point or a conical symplectic variety of dimension $2n - 2b$ whose symplectic form has degree $r$.

Let $Y := \operatorname{Spec} S$. If $b = 0$, then $Y = X$ and there is nothing to prove. Suppose that $b \geq 1$. Since $R$ is a polynomial ring in $2b$ variables over $S$, $\dim Y = 2n - 2b$, and $X \cong \mathbb{A}^{2b} \times Y$. If $\dim Y = 0$, then $S$ is a finitely generated $\mathbb{C}$-algebra which is a field, so $S = \mathbb{C}$ and $Y$ is a point. Suppose that $\dim Y > 0$. By **Step 4**, $Y$ is a normal affine variety and $S = \bigoplus_{j \geq 0} S_j$ with $S_0 = \mathbb{C}$. Since the projection $\mathbb{A}^{2b} \times Y \to Y$ is smooth, $X_{\mathrm{reg}} = \mathbb{A}^{2b} \times Y_{\mathrm{reg}}$.

Let $\Pi$ be the bivector field on $X_{\mathrm{reg}}$ with $\{f, h\} = \Pi(df, dh)$ for $f, h \in R$, and let $\Pi_Y$ be the bivector field on $Y_{\mathrm{reg}}$ with $\{f, h\} = \Pi_Y(df, dh)$ for $f, h \in S$. Since $\omega$ is non-degenerate on $X_{\mathrm{reg}}$, $\Pi$ is non-degenerate on $X_{\mathrm{reg}}$. By **Step 3**,

$$\Pi = \sum_{i=1}^{b} \frac{\partial}{\partial p_i} \wedge \frac{\partial}{\partial q_i} + \Pi_Y \quad \text{on } \mathbb{A}^{2b} \times Y_{\mathrm{reg}},$$

since both sides take equal values on any pair of differentials among $dp_i$, $dq_i$, and $df$ for $f \in S$, which generate $\Omega^1_{X_{\mathrm{reg}}}$. Since $\Pi$ is non-degenerate at any point of $\mathbb{A}^{2b} \times Y_{\mathrm{reg}}$, $\Pi_Y$ is non-degenerate on $Y_{\mathrm{reg}}$. For $f \in S$, let $H_f := \Pi_Y(df, \cdot)$, and let $\omega_Y$ be the 2-form on $Y_{\mathrm{reg}}$ with $i_{H_f}\omega_Y = -df$ for any $f \in S$, which exists and is unique since $\Pi_Y$ is non-degenerate and the $df$, $f \in S$, generate $\Omega^1_{Y_{\mathrm{reg}}}$. This is the convention of Definition 2.9, under which $\omega$ is inverse to $\Pi$. Let $\omega_{\mathbb{A}^{2b}}$ be the 2-form on $\mathbb{A}^{2b}$ with constant coefficients inverse to $\sum_i \partial/\partial p_i \wedge \partial/\partial q_i$. Since the inverse of a direct sum of non-degenerate bivectors is the direct sum of the inverses,

$$\omega = \omega_{\mathbb{A}^{2b}} + \mathrm{pr}_2^* \omega_Y \quad \text{on } \mathbb{A}^{2b} \times Y_{\mathrm{reg}},$$

where $\mathrm{pr}_2 : \mathbb{A}^{2b} \times Y_{\mathrm{reg}} \to Y_{\mathrm{reg}}$ is the projection. Let $\iota : Y \to X$ be the closed embedding with image $\{0\} \times Y$, which is defined by the ideal $(p_1, q_1, \dots, p_b, q_b)$ and is equivariant for the gradings. Since $\iota^* dp_i = \iota^* dq_i = 0$, we have $\iota^*\omega = \omega_Y$ on $Y_{\mathrm{reg}}$. Thus $\omega_Y$ is closed and homogeneous of degree $r$.

Let $h : Y' \to Y$ be a resolution. Then $\mathrm{id}_{\mathbb{A}^{2b}} \times h : \mathbb{A}^{2b} \times Y' \to X$ is a resolution, so $(\mathrm{id}_{\mathbb{A}^{2b}} \times h)^*\omega$ extends to a holomorphic 2-form $\omega'$ on $\mathbb{A}^{2b} \times Y'$. Let $\iota' : Y' \to \mathbb{A}^{2b} \times Y'$ be the embedding with image $\{0\} \times Y'$. Then $\iota'^*\omega'$ is a holomorphic 2-form on $Y'$ which coincides with $h^*\omega_Y$ on $h^{-1}(Y_{\mathrm{reg}})$. Thus $Y$ is a conical symplectic variety of dimension $2n - 2b$ whose symplectic form $\omega_Y$ has degree $r$. □

Lemma 7.4 is used in the proof of Corollary 9.19 in Section 9 to verify condition (ii) of Theorem 9.18, which splits off the factor $\mathbb{A}^{2b}$ of Proposition 7.3 in a family.

**Lemma 7.4.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety whose symplectic form has degree $2s$ with $s \geq 1$ and whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex. Then restriction to $R_s$ induces an isomorphism*

$$\mathrm{Der}_{\mathbb{C}}(R)_{-s} \to \mathrm{Hom}_{\mathbb{C}}(R_s, \mathbb{C}),$$

*any element of* $\mathrm{Der}_{\mathbb{C}}(R)_{-s}$ *is the Hamiltonian derivation* $\{f,\cdot\}$ *of a unique* $f \in R_s$*, and any two elements of* $\mathrm{Der}_{\mathbb{C}}(R)_{-s}$ *commute. In particular, if* $R$ *is the polynomial ring* $S[u_1,\dots,u_{2b}]$ *over a graded subalgebra* $S \subset R$ *with* $S_j = 0$ *for* $0 < j \le s$*, where* $\deg u_j = s$*, then any derivation of* $R$ *of degree* $-s$ *is a* $\mathbb{C}$*-linear combination of* $\partial/\partial u_1,\dots,\partial/\partial u_{2b}$*.*

*Proof.* Let $\mathfrak{m} := \bigoplus_{j>0} R_j$, and for $f \in R$ let $H_f := \{f,\cdot\}$. By Lemma 2.10, $X$ has klt singularities. By Proposition 7.3 with $r = 2s$, $R_j = 0$ for any $0 < j < 2s$ with $j \neq s$, and $\{,\}$ restricts to a non-degenerate alternating form on $R_s$. Since the symplectic form is homogeneous of degree $2s$, $\{R_j, R_l\} \subset R_{j+l-2s}$ for any $j,l \ge 0$, where $R_t := 0$ for $t < 0$. In particular, $H_f \in \mathrm{Der}_{\mathbb{C}}(R)_{-s}$ for $f \in R_s$, and $\{R_s, R_s\} \subset \mathbb{C}$.

Let $D \in \mathrm{Der}_{\mathbb{C}}(R)_{-s}$ with $D(R_s) = 0$. For any $j > 0$ with $j \neq s$ and $R_j \neq 0$, we have $j \ge 2s$, so $D(R_j) \subset R_{j-s} \subset \mathfrak{m}$. Thus $D(\mathfrak{m}) \subset \mathfrak{m}$, and $D = 0$ by Lemma 7.1. Therefore the restriction map

$$\mathrm{Der}_{\mathbb{C}}(R)_{-s} \to \mathrm{Hom}_{\mathbb{C}}(R_s,\mathbb{C}), \qquad D \to D|_{R_s},$$

is injective. For $f \in R_s$, the image of $H_f$ is the linear form $g \to \{f,g\}$ on $R_s$. Since $\{,\}$ is non-degenerate on $R_s$, any element of $\mathrm{Hom}_{\mathbb{C}}(R_s,\mathbb{C})$ is equal to $g \to \{f,g\}$ for a unique $f \in R_s$. Hence the restriction map is an isomorphism. For any $D \in \mathrm{Der}_{\mathbb{C}}(R)_{-s}$, there exists a unique $f \in R_s$ with $D|_{R_s} = H_f|_{R_s}$, and the injectivity of the restriction map implies that $D = H_f$.

Let $f,h \in R_s$. Since $\{f,h\} \in \mathbb{C}$, the Jacobi identity implies that

$$[H_f, H_h] = H_{\{f,h\}} = 0.$$

Thus any two elements of $\mathrm{Der}_{\mathbb{C}}(R)_{-s}$ commute.

Now assume that $R = S[u_1,\dots,u_{2b}]$ is a polynomial ring over a graded subalgebra $S \subset R$ with $S_j = 0$ for $0 < j \le s$ and $\deg u_j = s$. Since $\mathbb{C} \subset S_0 \subset R_0 = \mathbb{C}$, we have $S_0 = \mathbb{C}$, so $R_s$ is spanned by $u_1,\dots,u_{2b}$, and $u_1,\dots,u_{2b}$ form a basis of $R_s$. The $S$-linear derivations $\partial/\partial u_1,\dots,\partial/\partial u_{2b}$ belong to $\mathrm{Der}_{\mathbb{C}}(R)_{-s}$, and their restrictions to $R_s$ form the dual basis of $u_1,\dots,u_{2b}$. Let $D$ be a derivation of $R$ of degree $-s$. Then $D(u_k) \in R_0 = \mathbb{C}$ for any $k$, and $D|_{R_s} = \big(\sum_k D(u_k)\partial/\partial u_k\big)|_{R_s}$. Since the restriction map is injective,

$$D = \sum_{k=1}^{2b} D(u_k)\frac{\partial}{\partial u_k}.$$

□

7.2. **Product covers.** In this subsection, we prove Lemma 7.5, which is used in the proof of Theorem 8.7, and through it in the proof of Theorem 8.8, to provide the hypothesis of Theorem 10.4 on the non-étale locus in Section 10.

**Lemma 7.5.** *Let* $k \ge 2$*, let* $Z_1,\dots,Z_k$ *be irreducible normal affine varieties with* $\dim Z_i \ge 1$*, let* $X$ *be an irreducible normal affine variety, and let* $q : X \to Z_1 \times \cdots \times Z_k$ *be a finite surjective quasi-étale morphism. Then there exist irreducible normal affine varieties* $X_1',\dots,X_k'$*, finite surjective quasi-étale morphisms* $p_i : X_i' \to Z_i$*, a finite group* $Q$ *acting faithfully on* $X' := X_1' \times \cdots \times X_k'$ *by automorphisms over* $Z_1 \times \cdots \times Z_k$ *which preserve every factor, and a finite surjective* $Q$*-invariant morphism* $h : X' \to X$ *with* $q \circ h = p_1 \times \cdots \times p_k$ *which induces an isomorphism* $X'/Q \cong X$*, such that the following hold.*

*(1) Any element of* $Q\backslash\{1\}$ *acts non-trivially on at least two factors, and there exists a closed subset* $\Sigma \subset X$ *of codimension at least* 4 *such that* $h^{-1}(X\backslash\Sigma) \to X\backslash\Sigma$ *is a finite étale* $Q$*-torsor.*

*(2) If* $\mathcal{O}(X)$ *and every* $\mathcal{O}(Z_i)$ *carry good positive gradings such that* $q^*$ *is graded for the diagonal grading of* $\mathcal{O}(Z_1) \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} \mathcal{O}(Z_k)$*, then there exist an integer* $l \ge 1$ *and good positive gradings of the* $\mathcal{O}(X_i')$ *such that, for the diagonal grading of* $\mathcal{O}(X')$*, we have* $p_i^*\mathcal{O}(Z_i)_d \subset \mathcal{O}(X_i')_{ld}$ *and* $h^*\mathcal{O}(X)_d \subset \mathcal{O}(X')_{ld}$ *for any* $i$ *and* $d \ge 0$*, and* $Q$ *preserves the grading of* $\mathcal{O}(X')$*.*

(3) *If in addition every $Z_i$ is a conical symplectic variety whose symplectic form $\omega_i$ has degree $w_i$ for the given grading, then every $X'_i$, with the grading in (2), is a conical symplectic variety whose symplectic form is the reflexive extension of $p_i^*\omega_i$, of degree $lw_i$.*

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we define a finite-index subgroup $H$ of the fundamental group of the regular locus of $Z_1 \times \cdots \times Z_k$, the subgroups $H_i = H \cap G_i$ of the factors, and the group $Q$.

Let $Z := Z_1 \times \cdots \times Z_k$. For $1 \leq i \leq k$, let $U_i := (Z_i)_{\mathrm{reg}}$, and let $U := U_1 \times \cdots \times U_k$. Then $U = Z_{\mathrm{reg}}$. Since $Z_i$ and $Z$ are normal [Gro65, Corollaire 6.14.3], we have

$$\mathrm{codim}_{Z_i}(Z_i \backslash U_i) \geq 2 \text{ and } \mathrm{codim}_Z(Z \backslash U) \geq 2.$$

Since $q$ is quasi-étale, there exists a closed subset $\Sigma_Z \subset Z$ of codimension at least 2 such that $q$ is étale over $Z \backslash \Sigma_Z$. By the Zariski–Nagata purity theorem [Gro03, Exposé X, Théorème 3.1], applied to the finite dominant morphism $q^{-1}(U) \to U$ from the normal variety $q^{-1}(U)$ to the smooth variety $U$, the non-étale locus of $q^{-1}(U) \to U$ is either empty or of pure codimension one in $q^{-1}(U)$. The non-étale locus is contained in $q^{-1}(\Sigma_Z)$, and $\mathrm{codim}_X q^{-1}(\Sigma_Z) \geq 2$ since $q$ is finite. Thus $q^{-1}(U) \to U$ is finite étale. Moreover, $q^{-1}(U)$ is connected since $X$ is irreducible.

The complex manifolds $U_i^{\mathrm{an}}$ are connected. Fix $u_i \in U_i$, let $u := (u_1, \ldots, u_k)$, fix $x_0 \in q^{-1}(u)$, and let $G_i := \pi_1(U_i^{\mathrm{an}}, u_i)$. By [Hat02, Proposition 1.12], we have

$$G := \pi_1(U^{\mathrm{an}}, u) = G_1 \times \cdots \times G_k,$$

and we regard $G_i$ as the subgroup of $G$ consisting of the elements whose $j$-th coordinate is 1 for any $j \neq i$. Let $H \subset G$ be the image of $\pi_1(q^{-1}(U)^{\mathrm{an}}, x_0)$. By [Hat02, Proposition 1.32], $[G : H] = \deg q < +\infty$. Let

$$H_i := H \cap G_i \text{ and } P := H_1 \times \cdots \times H_k.$$

Since $H_1, \ldots, H_k$ are contained in $H$ and commute with each other, $P \subset H$. Since the map $G_i/H_i \to G/H$ is injective, $[G_i : H_i] \leq [G : H] < +\infty$.

Let $g = (g_1, \ldots, g_k) \in H$, let $1 \leq i \leq k$, and let $\gamma \in H_i$. Then

$$g\gamma g^{-1} = (1, \ldots, 1, g_i \gamma g_i^{-1}, 1, \ldots, 1) \in H \cap G_i = H_i.$$

Thus $g_i H_i g_i^{-1} \subset H_i$. Since $g^{-1} = (g_1^{-1}, \ldots, g_k^{-1}) \in H$, replacing $g$ with $g^{-1}$, we also have $g_i^{-1} H_i g_i \subset H_i$, hence

$$g_i \in N_{G_i}(H_i) \text{ for any } g = (g_1, \ldots, g_k) \in H \text{ and any } 1 \leq i \leq k.$$

In particular, $P$ is a normal subgroup of $H$. Let $Q := H/P$, which is a finite group.

**Step 2.** In this step we construct $X'_i$, $p_i$, the action of $Q$ on $X'$, and $h$.

By [Hat02, Theorem 1.38, Proposition 1.32], the subgroup $H_i \subset G_i$ corresponds to a connected covering of $U_i^{\mathrm{an}}$ with $[G_i : H_i]$ sheets. The covering space has a unique complex structure for which the covering map is locally biholomorphic, and by [Gro03, Exposé XII, Théorème 5.1] the covering map is the analytification of a connected finite étale cover $V_i \to U_i$. Fix $v_i \in V_i$ over $u_i$ such that the image of $\pi_1(V_i^{\mathrm{an}}, v_i)$ in $G_i$ is $H_i$. Since the functor of [Gro03, Exposé XII, Théorème 5.1] is an equivalence, any automorphism of $V_i^{\mathrm{an}}$ over $U_i^{\mathrm{an}}$ is algebraic. Hence the automorphisms of $V_i$ over $U_i$ form the group $N_{G_i}(H_i)/H_i$ [Hat02, Proposition 1.39].

Let $V := V_1 \times \cdots \times V_k$ and $v := (v_1, \ldots, v_k)$. Then $V^{\mathrm{an}} \to U^{\mathrm{an}}$ is a connected finite covering, and by [Hat02, Proposition 1.12] the image of $\pi_1(V^{\mathrm{an}}, v)$ in $G$ is $P$. Since $P \subset H$, by the lifting criterion [Hat02, Proposition 1.33] there exists a map from $V^{\mathrm{an}}$ to $q^{-1}(U)^{\mathrm{an}}$ over $U^{\mathrm{an}}$ which sends $v$ to $x_0$. It is a morphism of finite étale covers of $U^{\mathrm{an}}$, hence it is the analytification of a finite étale morphism

$$h_V : V \to q^{-1}(U)$$

[Gro03, Exposé XII, Théorème 5.1]. The map $h_V^{\mathrm{an}}$ is a connected finite covering, and the image of $\pi_1(V^{\mathrm{an}}, v)$ in $\pi_1(q^{-1}(U)^{\mathrm{an}}, x_0) \cong H$ is $P$. Since $P$ is normal in $H$, the deck transformations of $h_V^{\mathrm{an}}$ form the group $H/P = Q$, which acts simply transitively on any fiber of $h_V^{\mathrm{an}}$ [Hat02,

Proposition 1.39]. By [Gro03, Exposé XII, Théorème 5.1] again, the deck transformations of $h_V^{\mathrm{an}}$ are algebraic, so $Q$ acts on $V$ over $q^{-1}(U)$.

For $\varphi = gP \in Q$ with $g = (g_1, \dots, g_k) \in H$, we have $g_i \in N_{G_i}(H_i)$ by **Step 1**, and we let $\varphi_i$ be the automorphism of $V_i$ over $U_i$ corresponding to $g_i H_i \in N_{G_i}(H_i)/H_i$. Then $\varphi = \varphi_1 \times \cdots \times \varphi_k$ on $V$, since both sides send $v$ to $\tilde{\gamma}(1)$, where $\tilde{\gamma}$ is the lift starting at $v$ of a loop $\gamma$ representing $g$, and two deck transformations of the connected covering $V^{\mathrm{an}} \to U^{\mathrm{an}}$ which coincide at $v$ are equal [Hat02, Proposition 1.34]. The homomorphism

$$Q \to \prod_{i=1}^{k} N_{G_i}(H_i)/H_i, \quad gP \to (g_1 H_1, \dots, g_k H_k)$$

is injective, since $g_i \in H_i$ for any $i$ means $g \in P$. Thus $Q$ acts faithfully on $V$ over $U$ and preserves the factors $V_i$.

Let $X_i'$ be the normalization of $Z_i$ in the function field $\mathbb{C}(V_i)$, and let $p_i : X_i' \to Z_i$ be the induced morphism. By [Har77, Theorem I.3.9A], $p_i$ is finite, and $X_i'$ is an irreducible normal affine variety. Since $V_i$ is normal and finite over $U_i$, we have $p_i^{-1}(U_i) = V_i$. Thus $p_i$ is finite and surjective, and $p_i$ is quasi-étale since $p_i$ is étale over $U_i$ and $\operatorname{codim}_{Z_i}(Z_i \backslash U_i) \geq 2$. The product $X' = X_1' \times \cdots \times X_k'$ is an irreducible [Har77, Exercise I.3.15(a)] normal [Gro65, Corollaire 6.14.3] affine variety which is finite over $Z$, and

$$(p_1 \times \cdots \times p_k)^{-1}(U) = V.$$

Any automorphism $\varphi_i$ of $V_i$ over $U_i$ induces an automorphism of $\mathbb{C}(X_i') = \mathbb{C}(V_i)$ over $\mathbb{C}(Z_i)$, which preserves the integral closure $\mathcal{O}(X_i')$ of $\mathcal{O}(Z_i)$ in $\mathbb{C}(X_i')$. Thus $\varphi_i$ extends to an automorphism of $X_i'$ over $Z_i$, which we also denote by $\varphi_i$. Therefore, $Q$ acts faithfully on $X'$ over $Z$ by $\varphi \to \varphi_1 \times \cdots \times \varphi_k$, and the action preserves any factor of $X'$.

The morphism $h_V$ induces an inclusion $\mathbb{C}(X) \subset \mathbb{C}(V) = \mathbb{C}(X')$ over $\mathbb{C}(Z)$ with $\mathbb{C}(X) \subset \mathbb{C}(X')^Q$. Since $h_V$ is finite étale of degree $|Q|$ and $Q$ acts faithfully on $\mathbb{C}(X')$, we have

$$[\mathbb{C}(X') : \mathbb{C}(X)] = |Q| = [\mathbb{C}(X') : \mathbb{C}(X')^Q],$$

hence $\mathbb{C}(X) = \mathbb{C}(X')^Q$. Since $X'$ is normal and finite over $Z$, $\mathcal{O}(X')$ is the integral closure of $\mathcal{O}(Z)$ in $\mathbb{C}(X')$. Since $X$ is normal and finite over $Z$, $\mathcal{O}(X)$ is the integral closure of $\mathcal{O}(Z)$ in $\mathbb{C}(X)$. Thus

$$\mathcal{O}(X) = \mathcal{O}(X') \cap \mathbb{C}(X) = \mathcal{O}(X')^Q.$$

Let $h : X' \to X$ be the morphism corresponding to the inclusion $\mathcal{O}(X) \subset \mathcal{O}(X')$. Then $h$ is $Q$-invariant and induces an isomorphism $X'/Q \cong X$. Since $\mathcal{O}(X')$ is finite over $\mathcal{O}(Z) \subset \mathcal{O}(X)$, $h$ is finite, hence surjective. Since $q^*$ and $(p_1 \times \cdots \times p_k)^*$ both correspond to the inclusion $\mathcal{O}(Z) \subset \mathcal{O}(X')$, we have $q \circ h = p_1 \times \cdots \times p_k$. In particular, $h^{-1}(q^{-1}(U)) = V$, and $h|_V = h_V$ since $h|_V$ and $h_V$ both induce the inclusion $\mathbb{C}(X) \subset \mathbb{C}(X')$ of function fields.

**Step 3.** In this step we prove (1).

For $\varphi = gP \in Q$, recall from **Step 2** that $\varphi$ acts on the $i$-th factor $X_i'$ by $\varphi_i$, and $\varphi_i = \mathrm{id}_{X_i'}$ if and only if $g_i \in H_i$. Let $\varphi = gP \in Q \backslash \{1\}$, and suppose that there exists $1 \leq j \leq k$ such that $g_i \in H_i$ for any $i \neq j$. Let $g' \in P$ be the element whose $j$-th coordinate is 1 and whose $i$-th coordinate is $g_i^{-1}$ for $i \neq j$. Since $g \in H$ and $g' \in P \subset H$,

$$gg' = (1, \dots, 1, g_j, 1, \dots, 1) \in H \cap G_j = H_j.$$

Thus $g_j \in H_j$, hence $g \in P$ and $\varphi = 1$, a contradiction. Therefore, $\varphi$ acts non-trivially on at least two factors.

Let $\gamma$ be an automorphism of $V_i$ over $U_i$ with $\gamma \neq \mathrm{id}_{V_i}$. Then $\gamma$ has no fixed point on $V_i$, since $\gamma$ and $\mathrm{id}_{V_i}$ are lifts of $V_i \to U_i$ through the connected covering $V_i^{\mathrm{an}} \to U_i^{\mathrm{an}}$, and two such lifts which coincide at one point are equal [Hat02, Proposition 1.34]. Hence the fixed locus of the extension of $\gamma$ to $X_i'$ is contained in $X_i' \backslash V_i = p_i^{-1}(Z_i \backslash U_i)$. Since $p_i$ is finite and surjective,

$$\dim p_i^{-1}(Z_i \backslash U_i) = \dim(Z_i \backslash U_i) \leq \dim X_i' - 2.$$

Thus $\operatorname{codim}_{X'_i} \operatorname{Fix}_{X'_i}(\varphi_i) \geq 2$ for any $\varphi \in Q$ with $\varphi_i \neq \mathrm{id}_{X'_i}$.

For $\varphi \in Q$, we have

$$\operatorname{Fix}_{X'}(\varphi) = \operatorname{Fix}_{X'_1}(\varphi_1) \times \cdots \times \operatorname{Fix}_{X'_k}(\varphi_k),$$

where $\operatorname{Fix}_{X'_i}(\varphi_i) = X'_i$ if $\varphi_i = \mathrm{id}_{X'_i}$. If $\varphi \neq 1$, then there exist $j_1 \neq j_2$ such that $\varphi_{j_1} \neq \mathrm{id}_{X'_{j_1}}$ and $\varphi_{j_2} \neq \mathrm{id}_{X'_{j_2}}$. Thus, if $\operatorname{Fix}_{X'}(\varphi) \neq \emptyset$, then

$$\dim \operatorname{Fix}_{X'}(\varphi) \leq \sum_{i=1}^{k} \dim X'_i - 4 = \dim X' - 4.$$

Let

$$F := \bigcup_{\varphi \in Q \setminus \{1\}} \operatorname{Fix}_{X'}(\varphi),$$

where $F = \emptyset$ if $Q = \{1\}$. Then $F$ is a closed subset of $X'$ of codimension at least 4, and $F$ is $Q$-invariant since $\psi(\operatorname{Fix}_{X'}(\varphi)) = \operatorname{Fix}_{X'}(\psi\varphi\psi^{-1})$ for any $\varphi, \psi \in Q$. Let $\Sigma := h(F)$. Since $h$ is finite and surjective and $\dim X' = \dim X$, $\Sigma$ is a closed subset of $X$ with $\operatorname{codim}_X \Sigma = \operatorname{codim}_{X'} F \geq 4$. Since the fibers of the quotient map $h$ by the finite group $Q$ are the $Q$-orbits and $F$ is $Q$-invariant, $h^{-1}(\Sigma) = F$.

By the definition of $F$, $Q$ acts freely on the closed points of $X' \setminus F$. We show that the inertia group of any point of $X' \setminus F$ is trivial. Let $x \in X' \setminus F$ be a point, and let $\psi \in Q$ be an element with $\psi(x) = x$ which acts trivially on the residue field $\kappa(x)$. Then $\psi$ restricts to an automorphism of the closure $\overline{\{x\}}$ with the reduced structure which is the identity on the function field $\kappa(x)$, hence $\psi$ is the identity on $\overline{\{x\}}$. Since $\overline{\{x\}} \cap (X' \setminus F)$ is a non-empty open subset of $\overline{\{x\}}$, it contains a closed point of $X' \setminus F$, which is fixed by $\psi$. Thus $\psi = 1$.

Let $f \in \mathcal{O}(X)$ with $\{f \neq 0\} \subset X \setminus \Sigma$. Then $h^{-1}(\{f \neq 0\}) = \operatorname{Spec} \mathcal{O}(X')_{h^*f}$ is contained in $X' \setminus F$. Since $h^*f$ is $Q$-invariant and $\mathcal{O}(X')$ is a domain, $(\mathcal{O}(X')_{h^*f})^Q = (\mathcal{O}(X')^Q)_{h^*f} = \mathcal{O}(X)_f$. Thus, since the inertia groups are trivial, $h^{-1}(\{f \neq 0\}) \to \{f \neq 0\}$ is a finite étale $Q$-torsor by [Gro03, Exposé V, Corollaire 2.3 and Proposition 2.6]. Since $X \setminus \Sigma$ is covered by such open subsets $\{f \neq 0\}$, $h^{-1}(X \setminus \Sigma) \to X \setminus \Sigma$ is a finite étale $Q$-torsor. This proves (1).

**Step 4.** In this step we lift the gradings to the $X'_i$ and prove (2).

Assume that $\mathcal{O}(X)$ and $\mathcal{O}(Z_1), \ldots, \mathcal{O}(Z_k)$ carry good positive gradings as in (2). Let $\zeta_i$ be the $\mathbb{G}_m$-action on $Z_i$ inducing the grading of $\mathcal{O}(Z_i)$, that is,

$$\mathcal{O}(Z_i)_d = \{f \in \mathcal{O}(Z_i) \mid f(\zeta_i(a, z)) = a^d f(z) \text{ for } a \in \mathbb{C}^* \text{ and } z \in Z_i\},$$

let $\zeta_X$ be the $\mathbb{G}_m$-action on $X$ inducing the grading of $\mathcal{O}(X)$, and let $\zeta$ be the diagonal action on $Z$ given by $\zeta(a, (z_1, \ldots, z_k)) = (\zeta_1(a, z_1), \ldots, \zeta_k(a, z_k))$. Since $q^*$ is graded,

$$q(\zeta_X(a, x)) = \zeta(a, q(x)) \text{ for any } a \in \mathbb{C}^* \text{ and } x \in X.$$

Since automorphisms preserve regular loci, $U_i$ is $\zeta_i$-invariant and $U$ is $\zeta$-invariant. Hence $q^{-1}(U)$ is $\zeta_X$-invariant.

Recall that $u_i \in U_i$ is the base point of $G_i = \pi_1(U_i^{\mathrm{an}}, u_i)$ and that $V_i = p_i^{-1}(U_i)$ is the cover of $U_i$ corresponding to $H_i$, with base point $v_i$. Let $c_i \in G_i$ be the class of the loop $t \to \zeta_i(e^{2\pi\sqrt{-1}t}, u_i)$, $t \in [0, 1]$. Since $[G_i : H_i] < +\infty$, the cosets $c_i^n H_i$, $n \geq 0$, are not pairwise distinct, so there exists an integer $l_i \geq 1$ such that $c_i^{l_i} \in H_i$. By [Hat02, Proposition 1.12],

$$\pi_1(\mathbb{C}^* \times V_i^{\mathrm{an}}, (1, v_i)) = \mathbb{Z} \times \pi_1(V_i^{\mathrm{an}}, v_i),$$

and the map

$$\mathbb{C}^* \times V_i \to U_i, \quad (a, x) \to \zeta_i(a^{l_i}, p_i(x))$$

sends the generator of $\mathbb{Z}$ to $c_i^{l_i} \in H_i$ and $\pi_1(V_i^{\mathrm{an}}, v_i)$ onto $H_i$. By the lifting criterion [Hat02, Proposition 1.33], the map lifts to a unique continuous map $\zeta'_i : \mathbb{C}^* \times V_i \to V_i$ with $\zeta'_i(1, v_i) = v_i$,

and $\zeta_i'(1,\cdot) = \mathrm{id}_{V_i}$ by [Hat02, Proposition 1.34]. The map $\zeta_i'$ is an algebraic action of $\mathbb{G}_m$ on $V_i$ with

$$p_i(\zeta_i'(a,x)) = \zeta_i(a^{l_i}, p_i(x)).$$

The map $\zeta_i'$ is holomorphic since $V_i^{\mathrm{an}} \to U_i^{\mathrm{an}}$ is locally biholomorphic, $\zeta_i'$ is algebraic since the graph of $\zeta_i'$ is a section of the pullback of the finite étale cover $V_i \to U_i$ along an algebraic map and the functor of [Gro03, Exposé XII, Théorème 5.1] is an equivalence, and the action law holds since both sides of $\zeta_i'(a,\zeta_i'(b,x)) = \zeta_i'(ab,x)$ are lifts of the map $(a,b,x) \to \zeta_i((ab)^{l_i}, p_i(x))$ on the connected space $\mathbb{C}^* \times \mathbb{C}^* \times V_i^{\mathrm{an}}$ which coincide at $(1,1,v_i)$ [Hat02, Propositions 1.33 and 1.34].

We show that the action $\zeta_i'$ extends to $X_i'$. The variety $\mathbb{C}^* \times X_i'$ is normal [Gro65, Corollaire 6.14.3] and finite over $\mathbb{C}^* \times Z_i$, so $\mathcal{O}(\mathbb{C}^* \times X_i')$ is the integral closure of $\mathcal{O}(\mathbb{C}^* \times Z_i)$ in $\mathbb{C}(\mathbb{C}^* \times X_i')$. The automorphism $(a,x) \to (a,\zeta_i'(a,x))$ of $\mathbb{C}^* \times V_i$ is over the automorphism $(a,z) \to (a,\zeta_i(a^{l_i},z))$ of $\mathbb{C}^* \times Z_i$, so the induced automorphism of $\mathbb{C}(\mathbb{C}^* \times X_i')$ preserves the integral closure of $\mathcal{O}(\mathbb{C}^* \times Z_i)$. Thus it extends to an automorphism of $\mathbb{C}^* \times X_i'$, whose second coordinate is a morphism $\zeta_i' : \mathbb{C}^* \times X_i' \to X_i'$ extending $\zeta_i'$. The action law extends from the dense open subset $\mathbb{C}^* \times \mathbb{C}^* \times V_i$. We obtain a grading $\mathcal{O}(X_i') = \bigoplus_{j\in\mathbb{Z}} \mathcal{O}(X_i')_j$ induced by $\zeta_i'$, and

$$p_i^*\mathcal{O}(Z_i)_d \subset \mathcal{O}(X_i')_{l_i d} \text{ for any } d \geq 0.$$

We show that the grading of $\mathcal{O}(X_i')$ is good and positive. Let $b \in \mathcal{O}(X_i')_s$ with $b \neq 0$. Since $\mathcal{O}(X_i')$ is integral over $p_i^*\mathcal{O}(Z_i)$, there exist $f_1, \ldots, f_m \in p_i^*\mathcal{O}(Z_i)$ such that

$$b^m + f_1 b^{m-1} + \cdots + f_m = 0.$$

For any $j$, let $f_{j,js}$ be the component of $f_j$ in $\mathcal{O}(X_i')_{js}$. Then $f_{j,js} \in p_i^*\mathcal{O}(Z_i)_{js/l_i}$ if $l_i \mid js$, and $f_{j,js} = 0$ otherwise. Taking the components in $\mathcal{O}(X_i')_{ms}$, we have $b^m + \sum_{j=1}^m f_{j,js} b^{m-j} = 0$. If $s < 0$, then $f_{j,js} = 0$ for any $j$, hence $b^m = 0$, which is not possible since $\mathcal{O}(X_i')$ is a domain. If $s = 0$, then $f_{j,0} \in p_i^*\mathcal{O}(Z_i)_0 = \mathbb{C}$, so $b$ is a root of a monic polynomial with coefficients in $\mathbb{C}$. Since the polynomial is a product of linear factors and $\mathcal{O}(X_i')$ is a domain, $b \in \mathbb{C}$. Thus $\mathcal{O}(X_i')_j = 0$ for $j < 0$ and $\mathcal{O}(X_i')_0 = \mathbb{C}$. Moreover, since $\mathcal{O}(X_i')$ is a finite $\mathcal{O}(Z_i)$-module, it is generated by finitely many homogeneous elements $b_1, \ldots, b_n$, of degrees $e_1, \ldots, e_n$. Then

$$\mathcal{O}(X_i')_j = \sum_{t : l_i \mid j-e_t,\ e_t \leq j} p_i^*\mathcal{O}(Z_i)_{(j-e_t)/l_i} \cdot b_t,$$

hence $\dim \mathcal{O}(X_i')_j < +\infty$ for any $j$.

Let $l$ be the least common multiple of $l_1, \ldots, l_k$. Possibly replacing $\zeta_i'$ with the action $(a,x) \to \zeta_i'(a^{l/l_i}, x)$, which multiplies the degrees of the grading of $\mathcal{O}(X_i')$ by $l/l_i$, we may assume that

$$p_i(\zeta_i'(a,x)) = \zeta_i(a^l, p_i(x)) \text{ and } p_i^*\mathcal{O}(Z_i)_d \subset \mathcal{O}(X_i')_{ld}$$

for any $i$ and $d \geq 0$, and the grading of $\mathcal{O}(X_i')$ is still good and positive. Let $\zeta'$ be the diagonal action on $X'$ given by $\zeta'(a,(x_1,\ldots,x_k)) = (\zeta_1'(a,x_1),\ldots,\zeta_k'(a,x_k))$. Then $\zeta'$ induces the diagonal grading of $\mathcal{O}(X') = \mathcal{O}(X_1') \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} \mathcal{O}(X_k')$, which is good and positive.

We show that $h^*\mathcal{O}(X)_d \subset \mathcal{O}(X')_{ld}$. Since $V = \prod_i V_i$ is $\zeta'$-invariant and $q^{-1}(U)$ is $\zeta_X$-invariant, the maps

$$\mathbb{C}^* \times V \to q^{-1}(U), \quad (a,x) \to h(\zeta'(a,x)) \text{ and } (a,x) \to \zeta_X(a^l, h(x))$$

are well defined. Since $q \circ h = p_1 \times \cdots \times p_k$, we have

$$q(h(\zeta'(a,x))) = \zeta(a^l, q(h(x))) = q(\zeta_X(a^l, h(x))).$$

Thus both maps are lifts through the covering $q^{-1}(U)^{\mathrm{an}} \to U^{\mathrm{an}}$ of the map $(a,x) \to \zeta(a^l, q(h(x)))$ from the connected space $\mathbb{C}^* \times V^{\mathrm{an}}$, and they coincide at $(1,v)$. By [Hat02, Proposition 1.34], they are equal, and since $\mathbb{C}^* \times V$ is dense in $\mathbb{C}^* \times X'$,

$$h(\zeta'(a,y)) = \zeta_X(a^l, h(y)) \text{ for any } a \in \mathbb{C}^* \text{ and } y \in X'.$$

Thus $h^*\mathcal{O}(X)_d \subset \mathcal{O}(X')_{ld}$ for any $d \geq 0$.

We show that $Q$ preserves the grading of $\mathcal{O}(X')$. Let $\varphi \in Q$. For $a \in \mathbb{C}^*$, let $\psi_a := \zeta'(a^{-1}, \cdot) \circ \varphi \circ \zeta'(a, \cdot)$, which is an automorphism of $V$. By the equality $h(\zeta'(a, y)) = \zeta_X(a^l, h(y))$ and the $Q$-invariance of $h$,

$$h(\psi_a(x)) = \zeta_X(a^{-l}, h(\varphi(\zeta'(a, x)))) = \zeta_X(a^{-l}, h(\zeta'(a, x))) = h(x)$$

for any $x \in V$. Thus $\psi_a$ is a deck transformation of $h_V = h|_V$, hence $\psi_a \in Q$ by **Step 2**. The sets $\{a \in \mathbb{C}^* \mid \psi_a = \psi\}$ for $\psi \in Q$ are finitely many pairwise disjoint closed subsets of $\mathbb{C}^*$ which cover $\mathbb{C}^*$. A deck transformation of the connected cover $h_V$ is determined by the image of $v$ [Hat02, Proposition 1.34], so $\{a \mid \psi_a = \psi\} = \{a \mid \psi_a(v) = \psi(v)\}$, which is closed since $a \to \psi_a(v) = \zeta'(a^{-1}, \varphi(\zeta'(a, v)))$ is continuous. Since $\mathbb{C}^*$ is connected and $\psi_1 = \varphi$, we have $\psi_a = \varphi$ for any $a \in \mathbb{C}^*$, that is, $\varphi$ commutes with $\zeta'(a, \cdot)$ on $V$, hence on $X'$ by density. Thus $\varphi^* \mathcal{O}(X')_j = \mathcal{O}(X')_j$ for any $j$. This proves (2).

**Step 5.** In this step we prove (3).

Assume in addition that $Z_i$ is a conical symplectic variety whose symplectic form $\omega_i$ has degree $w_i$. By **Step 2**, $p_i$ is finite, surjective, and quasi-étale, and $X_i'$ is normal. By Lemma 2.11, $p_i^* \omega_i$ extends to a 2-form $\omega_i'$ on $(X_i')_{\mathrm{reg}}$, and $(X_i', \omega_i')$ is a symplectic variety. In particular, $\omega_i'$ is the reflexive extension of $p_i^* \omega_i$.

Finally, by (2), $\mathcal{O}(X_i')$ carries the good positive grading induced by $\zeta_i'$, with $\mathcal{O}(X_i')_0 = \mathbb{C}$, and $p_i(\zeta_i'(a, x)) = \zeta_i(a^l, p_i(x))$. Since $\zeta_i(a, \cdot)^* \omega_i = a^{w_i} \omega_i$, we have

$$\zeta_i'(a, \cdot)^* p_i^* \omega_i = p_i^* \zeta_i(a^l, \cdot)^* \omega_i = a^{l w_i} p_i^* \omega_i$$

on $V_i$ for any $a \in \mathbb{C}^*$. Since $V_i$ is dense in $(X_i')_{\mathrm{reg}}$, $\zeta_i'(a, \cdot)^* \omega_i' = a^{l w_i} \omega_i'$. Thus $\omega_i'$ is homogeneous of degree $l w_i > 0$ (Definition 2.8), and $X_i'$ is a conical symplectic variety. This proves (3). □

7.3. **Covers of minimizing cones.** In this subsection, we show that on a graded quasi-étale cover of a conical symplectic variety whose degree valuation minimizes $\widehat{\mathrm{vol}}$, the lifted degree valuation is again a minimizer of $\widehat{\mathrm{vol}}$. Proposition 7.6 is used in the proof of Theorem 10.5 in Section 10. The key input is the finite degree formula [Liu26b, Theorem 1.1 and Corollary 1.2], which implies that the normalized volume of the vertex of the cover is the degree of the cover times the normalized volume of the vertex of the base.

**Proposition 7.6.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety of dimension $2n$, $n \geq 1$, with vertex $o$ and symplectic form $\omega$ of degree $r$, whose grading is induced by a faithful $\mathbb{G}_m$-action, let $h : X' = \operatorname{Spec} R' \to X$ be a finite surjective quasi-étale morphism from a normal variety, and let $l \geq 1$ be an integer. Assume that the degree valuation $v$ of $X$ is a minimizer of $\widehat{\mathrm{vol}}$ at $o$, and that $R'$ carries a $\mathbb{Z}$-grading with $h^* R_d \subset R'_{ld}$ for any $d \geq 0$. Then the following hold.*

1. *The grading of $R'$ is a good positive grading, $h^{-1}(o)$ is a single point $o'$, $(X', h^*\omega)$ is a conical symplectic variety whose symplectic form $h^*\omega$ has degree $lr$, the canonical divisors $K_X$ and $K_{X'}$ are Cartier, and $K_{X'} = h^* K_X$.*
2. *The degree valuation $v'$ of $X'$ satisfies $v'|_{\mathbb{C}(X)} = lv$, $A_{X'}(v') = l A_X(v) = lnr$, $\mathrm{vol}(v') = \deg(h) l^{-2n} \mathrm{vol}(v)$, and $\widehat{\mathrm{vol}}(v') = \deg(h) \widehat{\mathrm{vol}}(v) = \widehat{\mathrm{vol}}(X' \ni o')$. In particular, $v'$ is a minimizer of $\widehat{\mathrm{vol}}$ at $o'$.*

*Proof.* The proof is long, so we divide it into several steps. Since $h$ is finite and surjective, $h^* : R \to R'$ is injective, and we regard $R$ as a graded subring of $R'$ and $\mathbb{C}(X)$ as a subfield of $\mathbb{C}(X')$. Thus $R_d \subset R'_{ld}$ for any $d \geq 0$. Let $\zeta$ be the faithful $\mathbb{G}_m$-action on $X$ inducing the grading of $R$, and let $\zeta'$ be the $\mathbb{G}_m$-action on $X'$ defined by the grading of $R'$, that is, $\zeta'(\lambda)$ acts on $R'_j$ by multiplication by $\lambda^j$. For $f \in R_d$, $\lambda \in \mathbb{C}^*$, and $y \in X'$, we have

$$f(h(\zeta'(\lambda) \cdot y)) = \lambda^{ld} f(h(y)) = f(\zeta(\lambda^l) \cdot h(y)).$$

Since $R$ is generated by homogeneous elements, we obtain

$$h(\zeta'(\lambda) \cdot y) = \zeta(\lambda^l) \cdot h(y) \quad \text{for any } \lambda \in \mathbb{C}^* \text{ and } y \in X'. \tag{$*$}$$

**Step 1.** In this step we show that the grading of $R'$ is a good positive grading and that $h^{-1}(o) = \{o'\}$, where $o'$ is the vertex of the grading of $R'$.

Let $j \in \mathbb{Z}$ and let $s \in R'_j$ be a non-zero element. Since $R'$ is finite over $R$, there exist an integer $e \geq 1$ and elements $a_1, \dots, a_e \in R$ such that

$$s^e + a_1 s^{e-1} + \cdots + a_e = 0.$$

Let $a'_i$ be the homogeneous component of $a_i$ of degree $ij$ with respect to the grading of $R'$. Since $R_d \subset R'_{ld}$, we have $a'_i \in R_{ij/l}$ if $l$ divides $ij$, and $a'_i = 0$ otherwise. Taking the homogeneous component of degree $ej$ of the equality $s^e + a_1 s^{e-1} + \cdots + a_e = 0$, we obtain

$$s^e + a'_1 s^{e-1} + \cdots + a'_e = 0.$$

If $j < 0$, then $a'_i \in R_{ij/l} = 0$ for any $i$, so $s^e = 0$, which is not possible as $R'$ is a domain. If $j = 0$, then $a'_i \in R_0 = \mathbb{C}$ for any $i$. Write $t^e + a'_1 t^{e-1} + \cdots + a'_e = \prod_{k=1}^e (t - c_k)$ with $c_k \in \mathbb{C}$. Then $\prod_{k=1}^e (s - c_k) = 0$, and since $R'$ is a domain, $s = c_k$ for some $k$. Thus $R' = \bigoplus_{j \geq 0} R'_j$ and $R'_0 = \mathbb{C}$, that is, the grading of $R'$ is a good positive grading. Let $o'$ be the vertex of the grading of $R'$.

Let $f_1, \dots, f_t$ be homogeneous generators of the $\mathbb{C}$-algebra $R'$ of positive degrees $j_1, \dots, j_t$. For any $y \in X'$, we have $f_i(\zeta'(\lambda) \cdot y) = \lambda^{j_i} f_i(y) \to 0$ as $\lambda \to 0$, so $o'$ belongs to the closure of the orbit $\zeta'(\mathbb{G}_m) \cdot y$. If $y \in X'$ is fixed by $\zeta'(\mathbb{G}_m)$, then $f(y) = \lambda^j f(y)$ for any $f \in R'_j$ with $j > 0$ and any $\lambda \in \mathbb{C}^*$, so $f(y) = 0$, and hence $y = o'$. Therefore, the action of $T' := \zeta'(\mathbb{G}_m)$ on $X'$ is good with vertex $o'$. Since $o$ is fixed by $\zeta(\mathbb{G}_m)$, $(*)$ implies that the finite set $h^{-1}(o)$ is $T'$-invariant. Since $T'$ is connected, any point of $h^{-1}(o)$ is fixed by $T'$, so $h^{-1}(o) \subset \{o'\}$. Since $h$ is surjective, $h^{-1}(o) = \{o'\}$.

**Step 2.** In this step we show that $(X', h^*\omega)$ is a conical symplectic variety with $h^*\omega$ of degree $lr$, that $K_X$ and $K_{X'}$ are Cartier with $K_{X'} = h^* K_X$, and that $X$ and $X'$ are canonical.

By Lemma 2.11, $h^*\omega$ extends to a 2-form on $X'_{\mathrm{reg}}$, again denoted $h^*\omega$, and $(X', h^*\omega)$ is a symplectic variety. By $(*)$,

$$\zeta'(\lambda)^* h^*\omega = h^* \zeta(\lambda^l)^* \omega = \lambda^{lr} h^*\omega \quad \text{for any } \lambda \in \mathbb{C}^*,$$

so $h^*\omega$ is algebraic and homogeneous of degree $lr > 0$. Since the grading of $R'$ is a good positive grading (**Step 1**), $(X', h^*\omega)$ is a conical symplectic variety whose symplectic form has degree $lr$.

Since $\omega^n$ and $(h^*\omega)^n$ have no zero on $X_{\mathrm{reg}}$ and $X'_{\mathrm{reg}}$ respectively, and since $\mathrm{codim}_X(X \backslash X_{\mathrm{reg}}) \geq 2$ and $\mathrm{codim}_{X'}(X' \backslash X'_{\mathrm{reg}}) \geq 2$, we have

$$\Omega_X^{[2n]} = \mathcal{O}_X \cdot \omega^n \quad \text{and} \quad \Omega_{X'}^{[2n]} = \mathcal{O}_{X'} \cdot (h^*\omega)^n.$$

Thus $K_X \sim 0$ and $K_{X'} \sim 0$ are Cartier, and $K_{X'} = h^* K_X$ since $(h^*\omega)^n = h^*(\omega^n)$.

By Lemma 2.10, $X$ and $X'$ are canonical, and in particular klt.

**Step 3.** In this step we show that $v'|_{\mathbb{C}(X)} = lv$ and $A_{X'}(v') = lA_X(v) = lnr$.

Let $f \in R$ be a non-zero element and write $f = \sum_d f_d$ with $f_d \in R_d \subset R'_{ld}$. Then

$$v'(f) = \min\{ld \mid f_d \neq 0\} = lv(f).$$

Since $\mathbb{C}(X)$ is the fraction field of $R$, we have $v'|_{\mathbb{C}(X)} = lv$.

Let $T := \zeta(\mathbb{G}_m)$. By **Step 1**, $T'$ acts faithfully on $X'$ with a good action and vertex $o'$, and $\zeta'$ is a cocharacter of $T'$. Let $\alpha$ be a non-zero character of $T'$ with $R'_\alpha \neq 0$. Since $\zeta' : \mathbb{G}_m \to T'$ is surjective, $\alpha \circ \zeta'$ is non-trivial, so $R'_\alpha = R'_j$ for $j = \langle \alpha, \zeta' \rangle \neq 0$, and $j > 0$ by **Step 1**. Thus $\zeta'$ is a Reeb vector of $T'$, and $v' = \mathrm{wt}_{\zeta'}$. By **Step 2**, $\Omega_{X'}^{[2n]}$ is generated by $(h^*\omega)^n$, which is a $T'$-eigenvector of character $\kappa'$ with

$$\langle \kappa', \zeta' \rangle = nlr.$$

Since $X'$ is klt (**Step 2**), Lemma 2.14 implies that $A_{X'}(v') = nlr$. Since $\zeta$ is faithful, $\zeta : \mathbb{G}_m \to T$ is an isomorphism. Let $\alpha$ be a non-zero character of $T$ with $R_\alpha \neq 0$. Then $R_\alpha = R_j$ for

$j = \langle \alpha, \zeta \rangle \neq 0$, and $j > 0$ as $R = \bigoplus_{j \geq 0} R_j$. Thus $\zeta$ is a Reeb vector of $T$, and $v = \mathrm{wt}_\zeta$. By **Step 2**, $\Omega_X^{[2n]}$ is generated by $\omega^n$, which is a $T$-eigenvector of character $\kappa$ with

$$\langle \kappa, \zeta \rangle = nr.$$

Since $X$ is klt (**Step 2**), Lemma 2.14 implies that $A_X(v) = nr$. Thus $A_{X'}(v') = lA_X(v) = lnr$.

**Step 4.** In this step we show that $\mathrm{vol}(v') = \deg(h) l^{-2n} \mathrm{vol}(v)$ and conclude the proof.

For a real number $x$, let

$$H_R(x) := \sum_{0 \leq j < x} \dim R_j \quad \text{and} \quad H_{R'}(x) := \sum_{0 \leq j < x} \dim R'_j,$$

where $j$ runs over integers, so that $H_R(x) = H_{R'}(x) = 0$ for $x \leq 0$. Since the gradings of $R$ and $R'$ are good positive gradings and $R$, $R'$ are finitely generated, $\dim R_j < +\infty$ and $\dim R'_j < +\infty$ for any $j$. Let $m$ be a positive integer and let $I_m := \bigoplus_{j \geq m} R_j$. Any element of $\mathcal{O}_{X,o}$ can be written as $f/u$ with $f, u \in R$ and $u(o) \neq 0$. Since $u(o)$ is the degree zero component of $u$, $v(u) = 0$, and hence $v(f/u) = v(f)$. Thus $\mathfrak{a}_m(v) = I_m \mathcal{O}_{X,o}$. Since $I_m$ contains the $m$-th power of $\bigoplus_{j>0} R_j$, $R/I_m$ is supported at $o$, so

$$\ell(\mathcal{O}_{X,o}/\mathfrak{a}_m(v)) = \dim R/I_m = H_R(m).$$

Possibly replacing $R$, $X$, $o$, and $v$ with $R'$, $X'$, $o'$, and $v'$, the argument proving $\ell(\mathcal{O}_{X,o}/\mathfrak{a}_m(v)) = H_R(m)$ shows that $\ell(\mathcal{O}_{X',o'}/\mathfrak{a}_m(v')) = H_{R'}(m)$. Therefore,

$$\mathrm{vol}(v) = \lim_{m \to +\infty} \frac{(2n)! H_R(m)}{m^{2n}} \quad \text{and} \quad \mathrm{vol}(v') = \lim_{m \to +\infty} \frac{(2n)! H_{R'}(m)}{m^{2n}}.$$

Since $H_R(x) = H_R(\lceil x \rceil)$, we have $(2n)! H_R(x)/x^{2n} \to \mathrm{vol}(v)$ as the real number $x$ tends to $+\infty$. In particular, for any real numbers $c$ and $c'$,

$$H_R\Big(\frac{m-c}{l}\Big) = \frac{\mathrm{vol}(v)}{(2n)!} \cdot \frac{m^{2n}}{l^{2n}} + o(m^{2n}) \quad \text{and} \quad H_R\Big(\frac{m-c}{l}\Big) - H_R\Big(\frac{m-c}{l} - c'\Big) = o(m^{2n}).$$

Since $R'$ is finite over $R$, any element of $\mathbb{C}(X')$ can be written as $f'/f$ with $f' \in R'$ and $0 \neq f \in R$. Since $R'$ is spanned over $\mathbb{C}$ by homogeneous elements, there exist homogeneous elements $e_1, \dots, e_{\deg(h)} \in R'$ which form a basis of $\mathbb{C}(X')$ over $\mathbb{C}(X)$. Let $k_i$ be the degree of $e_i$. The morphism of $R$-modules

$$\phi : R^{\oplus \deg(h)} \to R', \qquad (f_1, \dots, f_{\deg(h)}) \to \sum_{i=1}^{\deg(h)} f_i e_i$$

is injective, and it sends $R_d$ in the $i$-th summand into $R'_{ld+k_i}$. Let $Q := R'/\phi(R^{\oplus \deg(h)})$, with the grading induced by $R'$, and let $Q_j$ be the degree $j$ part of $Q$. Then $Q$ is a finitely generated graded $R$-module with $Q \otimes_R \mathbb{C}(X) = 0$, so the annihilator of $Q$ in $R$ is a non-zero homogeneous ideal. If $Q \neq 0$, let $a \in R_{d_0}$ be a non-zero element with $aQ = 0$, where $d_0 > 0$ since $R_0 = \mathbb{C}$. Let $q_1, \dots, q_t$ be homogeneous generators of $Q$ of degrees $c_1, \dots, c_t$. Since multiplication by $a$ is injective from $R_{d-d_0}$ to $R_d$, $\dim(R/aR)_d = \dim R_d - \dim R_{d-d_0}$ for any $d$, where $R_d = 0$ for $d < 0$. Since $Q$ is a quotient of $\bigoplus_{i=1}^t (R/aR) q_i$, we obtain

$$\begin{aligned} \sum_{j<m} \dim Q_j &\leq \sum_{i=1}^{t} \sum_{0 \leq d < (m-c_i)/l} \big(\dim R_d - \dim R_{d-d_0}\big) \\ &= \sum_{i=1}^{t} \Big( H_R\Big(\frac{m-c_i}{l}\Big) - H_R\Big(\frac{m-c_i}{l} - d_0\Big)\Big) = o(m^{2n}). \end{aligned}$$

Since $\dim \phi(R^{\oplus \deg(h)}) \cap R'_j = \sum_i \dim R_{(j-k_i)/l}$, where the sum runs over the indices $i$ such that $l$ divides $j-k_i$, we have

$$H_{R'}(m) = \sum_{i=1}^{\deg(h)} H_R\Big(\frac{m-k_i}{l}\Big) + \sum_{j<m} \dim Q_j = \deg(h)\frac{\mathrm{vol}(v)}{(2n)!}\cdot\frac{m^{2n}}{l^{2n}} + o(m^{2n}).$$

Thus $\mathrm{vol}(v') = \deg(h) l^{-2n}\,\mathrm{vol}(v)$. By **Step 3**,

$$\widehat{\mathrm{vol}}(v') = A_{X'}(v')^{2n}\,\mathrm{vol}(v') = (lnr)^{2n}\deg(h) l^{-2n}\,\mathrm{vol}(v) = \deg(h) A_X(v)^{2n}\,\mathrm{vol}(v) = \deg(h)\widehat{\mathrm{vol}}(v).$$

Since $v'(f) \geq 0$ for any $f \in R'$ and $\{f \in R' \mid v'(f) > 0\} = \bigoplus_{j>0} R'_j$, the center of $v'$ on $X'$ is $o'$, so $v' \in \mathrm{Val}_{X',o'}$. By [Liu26b, Corollary 1.2], if $g : X_1 \to X_2$ is a finite surjective morphism between klt varieties with $K_{X_1} = g^* K_{X_2}$, then $\sum_{x\in g^{-1}(y)} \widehat{\mathrm{vol}}(X_1 \ni x) = \deg(g)\widehat{\mathrm{vol}}(X_2 \ni y)$ for any closed point $y \in X_2$. By **Step 2**, $X$ and $X'$ are klt and $K_{X'} = h^* K_X$, and by **Step 1**, $h^{-1}(o) = \{o'\}$. Hence, for $g = h$ and $y = o$, the sum in [Liu26b, Corollary 1.2] has the single term $o'$, and

$$\widehat{\mathrm{vol}}(X' \ni o') = \deg(h)\widehat{\mathrm{vol}}(X \ni o).$$

Since $v$ is a minimizer of $\widehat{\mathrm{vol}}$ at $o$, $\widehat{\mathrm{vol}}(v) = \widehat{\mathrm{vol}}(X \ni o)$. Thus

$$\widehat{\mathrm{vol}}(v') = \deg(h)\widehat{\mathrm{vol}}(v) = \deg(h)\widehat{\mathrm{vol}}(X \ni o) = \widehat{\mathrm{vol}}(X' \ni o'),$$

and $v'$ is a minimizer of $\widehat{\mathrm{vol}}$ at $o'$. □

## 8. Splitting K-polystable symplectic cones

The goal of this section is to prove Theorem 8.8. As a first step, we show that parallel projectors on a cone satisfying the metric hypotheses induce a finite quasi-étale morphism onto a product of conical symplectic varieties.

### 8.1. **Parallel projectors.** In this subsection, we prove Theorem 8.2.

**Set-up 8.1.** Let $X$, $\zeta$, $a$, $\varpi_1, \dots, \varpi_k$, and $\tau_1, \dots, \tau_k$ be as follows.

(1) $X = \operatorname{Spec} R$, $T$, $\xi$, $g = d\rho^2 + \rho^2 h$, $M$, and $N$ are as in Set-up 2.31, and conditions (1)–(4) of Set-up 2.31 hold.
(2) $\dim X = 2n$ with $n \geq 2$, and $\xi = a\zeta$ for a real number $a > 0$ and a cocharacter $\zeta$ of $T$ inducing a good positive grading $R = \bigoplus_{j\geq 0} R_j$ with Euler derivation eu.
(3) $k \geq 2$, and $\varpi_1, \dots, \varpi_k$ are endomorphisms of the reflexive tangent sheaf $T_X$ whose restrictions to $M$ are non-zero parallel complex-linear projectors of ranks $2n_1, \dots, 2n_k$, self-adjoint with respect to $g$, with $\varpi_i \varpi_j = 0$ for $i \neq j$ and $\sum_i \varpi_i = \mathrm{id}_{T_X}$.
(4) $\tau_1, \dots, \tau_k$ are reflexive 2-forms on $X$ whose restrictions to $M$ are parallel, such that $\tau_i$ is non-degenerate on $\operatorname{im}\varpi_i$ and has kernel $\bigoplus_{j\neq i} \operatorname{im}\varpi_j$.

Let $\delta_i := \varpi_i(a\,\mathrm{eu})$, let $S_i := \{f \in R \mid \delta_j(f) = 0 \text{ for any } j \neq i\}$, let $X_i := \operatorname{Spec} S_i$, and let $q = (q_1, \dots, q_k) : X \to X_1 \times \cdots \times X_k$ be the morphism induced by the inclusions $S_i \subset R$.

We require that $\tau_i$ be non-degenerate on $\operatorname{im}\varpi_i$ with kernel exactly the sum of the other images, because the symplectic form $\omega_i$ of the factor $X_i$ in Theorem 8.2 descends from $\tau_i$. Projectors $\varpi_i$ and forms $\tau_i$ satisfying Set-up 8.1 are provided by Theorem 8.7.

We do not assume that the link $N$ is complete, and the factors are defined by the subrings $S_i$.

**Theorem 8.2.** *Notation and conditions as in Set-up 8.1. Then the following hold.*

*(1) Each $S_i$ is a finitely generated normal graded algebra with $(S_i)_0 = \mathbb{C}$, and $X_i$ is a conical symplectic variety of dimension $2n_i$ with klt singularities, with a closed non-degenerate reflexive 2-form $\omega_i$ which extends regularly to any resolution of $X_i$.*
*(2) The morphism $q$ is finite, surjective, and quasi-étale, and $q_i^*\omega_i = \tau_i$ as rational forms.*
*(3) The number $2/a$ is a positive integer, and it is the degree of $\omega_i$ for any $i$.*

*(4) The derivations $\delta_1, \dots, \delta_k$ pairwise commute and commute with* eu, *and*

$$R = \bigoplus_{\lambda \in \mathbb{R}^k_{\geq 0}} R^\lambda, \qquad R^\lambda := \{f \in R \mid \delta_j(f) = \lambda_j f \text{ for } 1 \leq j \leq k\}.$$

*Each $S_i$ is a graded subalgebra of $R$, the projection of $R$ onto*

$$S_i = \bigoplus_{\lambda:\ \lambda_j = 0 \text{ for } j \neq i} R^\lambda$$

*along the other summands is a homomorphism of graded $\mathbb{C}$-algebras, and the induced closed embedding $s_i : X_i \to X$ satisfies $q_i \circ s_i = \mathrm{id}_{X_i}$ and $\rho^2 = \sum_i \rho^2 \circ s_i \circ q_i$ on $X$. In particular, $q$ is equivariant for the grading of $X$ and the diagonal grading of $X_1 \times \cdots \times X_k$.*

*Proof.* The proof is long, so we divide it into several steps. Since $j$ denotes degrees, in this proof we write $i'$ for a second index in $\{1, \dots, k\}$.

**Step 1.** In this step we show that the derivations $\delta_i$ commute with each other and with eu, and that $R = \bigoplus_{\lambda \in \mathbb{R}^k_{\geq 0}} R^\lambda$.

Since $\xi = a\zeta$ and $R_j = \bigoplus_{\langle \alpha, \zeta \rangle = j} R_\alpha$, condition (1) of Set-up 2.31 implies

$$\rho \partial_\rho f = ajf \quad \text{for } f \in R_j. \tag{8.1}$$

Since $X$ is normal and affine, the global sections of $T_X$ are the $\mathbb{C}$-derivations of $R$. Thus each $\delta_i$ is a derivation of $R$, and $\sum_i \delta_i = a\,\mathrm{eu}$.

Let $\nabla$ be the Levi-Civita connection of $g$ and $I$ the complex structure of $M$. For a real vector field $V$ on $M$ whose flow preserves $I$, the vector fields $V$ and $V^{1,0} := (V - \sqrt{-1}IV)/2$ satisfy $Vf = V^{1,0}f$ for any holomorphic function $f$, and $(IV)^{1,0} = \sqrt{-1}V^{1,0}$. Since $g$ is a cone metric, $\nabla(\rho\partial_\rho) = \mathrm{id}_{T_M}$ and $\mathrm{grad}_g(\rho^2/2) = \rho\partial_\rho$. Let

$$V_i := \varpi_i(\rho\partial_\rho).$$

Since $\varpi_i$ is parallel and $\varpi_i \varpi_{i'} = 0$ for $i \neq i'$, we obtain

$$\nabla V_i = \varpi_i, \qquad [V_i, V_{i'}] = \nabla_{V_i} V_{i'} - \nabla_{V_{i'}} V_i = \varpi_{i'}\varpi_i \rho\partial_\rho - \varpi_i \varpi_{i'} \rho\partial_\rho = 0, \qquad \sum_i V_i = \rho\partial_\rho.$$

For a vector field $Y$ on $M$, since $\nabla I = 0$,

$$(L_{V_i} I)(Y) = -\nabla_{IY} V_i + I \nabla_Y V_i = -\varpi_i IY + I\varpi_i Y = 0,$$

so the flow of $V_i$ preserves $I$. Since $\nabla(\rho\partial_\rho) = \mathrm{id}_{T_M}$, we have $(L_{\rho\partial_\rho} I)(Y) = -IY + IY = 0$, so the flow of $\rho\partial_\rho$ preserves $I$. By (8.1), the holomorphic vector fields $(\rho\partial_\rho)^{1,0}$ and $a\,\mathrm{eu}$ satisfy $(\rho\partial_\rho)^{1,0} f = a\,\mathrm{eu}(f)$ for $f \in R$. Since the differentials of the elements of $R$ span the cotangent space at any point of $X_{\mathrm{reg}}$, $(\rho\partial_\rho)^{1,0} = a\,\mathrm{eu}$ on $M$. Since $\varpi_i$ is complex-linear, $V_i^{1,0} = \varpi_i(a\,\mathrm{eu}) = \delta_i$ on $M$. Thus

$$V_i f = \delta_i f \quad \text{and} \quad (IV_i) f = \sqrt{-1}\delta_i f \qquad \text{for } f \in R.$$

Since $[V_i, V_{i'}] = 0$ and $[\rho\partial_\rho, V_i] = \sum_{i'} [V_{i'}, V_i] = 0$, the derivations $\delta_1, \dots, \delta_k$ pairwise commute and commute with eu. In particular, $\delta_i(R_j) \subset R_j$ for any $i$ and $j$.

Since $R$ is finitely generated and $R_0 = \mathbb{C}$, we have $\dim R_j < +\infty$. For $z \in \mathbb{C}$, the matrix exponential $\exp(z\delta_i)$ on each $R_j$ defines a graded $\mathbb{C}$-algebra automorphism of $R$ with inverse $\exp(-z\delta_i)$: the Leibniz rule for $\delta_i$ implies $\exp(z\delta_i)(fg) = \exp(z\delta_i)(f)\exp(z\delta_i)(g)$ by comparing the power series in $z$. Let $\psi_{i,z} : X \to X$ be the automorphism with $f \circ \psi_{i,z} = \exp(z\delta_i) f$ for $f \in R$. Since $\exp(z\delta_i)$ preserves $\bigoplus_{j>0} R_j$, we have $\psi_{i,z}(o) = o$, and hence $\psi_{i,z}(M) = M$. For $y \in X$ and $f \in R$,

$$\frac{d}{ds}\Big|_{s=0} f(\psi_{i,s}(y)) = (\delta_i f)(y), \qquad \frac{d}{dt}\Big|_{t=0} f(\psi_{i,\sqrt{-1}t}(y)) = \sqrt{-1}(\delta_i f)(y).$$

Since $R$ defines a closed embedding of $X$ into an affine space, $s \to \psi_{i,s}$ for $s \in \mathbb{R}$ is the flow of $V_i$ on $M$, and $t \to \psi_{i,\sqrt{-1}t}$ for $t \in \mathbb{R}$ is the flow of $IV_i$ on $M$. In particular, both flows are complete on $M$.

Since $I$ is skew-adjoint with respect to $g$ and $\varpi_i$ is self-adjoint and commutes with $I$, the endomorphism $I\varpi_i$ is skew-adjoint. Thus

$$(IV_i)(\rho^2) = 2g(\rho\partial_\rho, I\varpi_i\rho\partial_\rho) = 0.$$

Hence $\rho \circ \psi_{i,\sqrt{-1}t} = \rho$ on $M$ for $t \in \mathbb{R}$, and then on $X$ since $\rho$ is continuous and $M$ is dense in $X$.

Fix $j$. For $f \in R_j$ let $\|f\| := \sup_{\rho\le 1} |f|$. By condition (2) of Set-up 2.31, the set $\{\rho \le 1\}$ is compact, so $\|f\| < +\infty$. If $\|f\| = 0$, then $f$ vanishes on the non-empty open subset $\{0 < \rho < 1\}$ of $M$, and hence $f = 0$ since $X$ is irreducible. Thus $\|\cdot\|$ is a norm on $R_j$. Since $\psi_{i,\sqrt{-1}t}$ preserves $\{\rho \le 1\}$, we have $\|\exp(\sqrt{-1}t\delta_i)f\| = \|f\|$ for $t \in \mathbb{R}$. If $\delta_i|_{R_j}$ had an eigenvector $f$ with eigenvalue $\mu \notin \mathbb{R}$, then $\|\exp(\sqrt{-1}t\delta_i)f\| = e^{-t\operatorname{Im}\mu}\|f\|$ would be unbounded on $\mathbb{R}$. If $\delta_i|_{R_j}$ had a Jordan block of size at least 2 with eigenvalue $\mu \in \mathbb{R}$, there would exist $f, f' \in R_j$ with $f \ne 0$, $\delta_i f = \mu f$, and $\delta_i f' = \mu f' + f$, and then $\|\exp(\sqrt{-1}t\delta_i)f'\| = \|f' + \sqrt{-1}tf\|$ would be unbounded on $\mathbb{R}$. Thus $\delta_i|_{R_j}$ is diagonalizable with real eigenvalues. Since the $\delta_i$ pairwise commute, they are simultaneously diagonalizable on $R_j$.

Let $b_i := |V_i|_g^2$ on $M$. Since $\varpi_i$ is self-adjoint and $\varpi_i\varpi_{i'} = 0$ for $i \ne i'$, the images of the $\varpi_i$ are pairwise orthogonal, so $\rho^2 = |\rho\partial_\rho|_g^2 = \sum_i b_i$. Moreover,

$$V_{i'}b_i = 2g(\nabla_{V_{i'}}V_i, V_i) = 2g(\varpi_i\varpi_{i'}\rho\partial_\rho, V_i) = \begin{cases} 2b_i & \text{if } i' = i, \\ 0 & \text{if } i' \ne i. \end{cases}$$

Thus for $y \in M$ and $s \in \mathbb{R}$,

$$\rho^2(\psi_{i,s}(y)) = e^{2s}b_i(y) + \sum_{i'\ne i} b_{i'}(y). \tag{8.2}$$

In particular, $\psi_{i,s}(y) \in \{\rho \le \rho(y)\}$ for $s \le 0$. Let $f \in R_j$ be non-zero with $\delta_{i'}f = \lambda_{i'}f$ for $1 \le i' \le k$, and suppose that $\lambda_i < 0$. Choose $y \in M$ with $f(y) \ne 0$. Then

$$f(\psi_{i,s}(y)) = e^{s\lambda_i}f(y)$$

is unbounded as $s \to -\infty$, which contradicts the compactness of $\{\rho \le \rho(y)\}$. Thus $\lambda_i \ge 0$, and $\sum_i \lambda_i = aj$ since $\sum_i \delta_i = a\,\mathrm{eu}$.

Therefore $R = \bigoplus_{\lambda\in\mathbb{R}^k_{\ge 0}} R^\lambda$, and $R^\lambda = \bigoplus_j (R^\lambda \cap R_j)$ for any $\lambda$ since eu commutes with the $\delta_i$. Take a finite set of homogeneous generators of $R$ and replace any element $f$ of the set by the components of $f$ in the spaces $R^\lambda \cap R_j$. Thus we obtain a finite set $\mathcal{G}$ of generators of $R$ such that any $f \in \mathcal{G}$ belongs to $R^\lambda \cap R_j$ for some $\lambda$ and $j$.

**Step 2.** In this step we prove the remaining claims of (4), the claims of (1) on $S_i$, and the finiteness of $q$.

Write $f \in R$ as $f = \sum_\lambda f_\lambda$ with $f_\lambda \in R^\lambda$. Then $\delta_{i'}f = \sum_\lambda \lambda_{i'}f_\lambda$, so

$$S_i = \bigoplus_{\lambda:\lambda_{i'}=0 \text{ for } i'\ne i} R^\lambda.$$

Since each $\delta_{i'}$ is a derivation, $R^\lambda R^\mu \subset R^{\lambda+\mu}$. Since $\lambda, \mu \in \mathbb{R}^k_{\ge 0}$, the vector $\lambda + \mu$ has vanishing $i'$-th components for all $i' \ne i$ if and only if both $\lambda$ and $\mu$ do. Thus a monomial in the elements of $\mathcal{G}$ belongs to $S_i$ if and only if all factors of the monomial belong to $S_i$. Hence $S_i$ is generated by $\mathcal{G} \cap S_i$, and in particular $S_i$ is finitely generated. Since each $R^\lambda$ is graded, $S_i$ is a graded subalgebra of $R$ with $(S_i)_0 = S_i \cap R_0 = \mathbb{C}$.

We show that the algebra $S_i$ is normal. Let $u \in \operatorname{Frac}(S_i)$ be integral over $S_i$. Then $u$ is integral over $R$, so $u \in R$ since $R$ is normal. Write $u = u_1/u_2$ with $u_1, u_2 \in S_i$. For $i' \ne i$, the derivation $\delta_{i'}$ extends to $\operatorname{Frac}(R)$ by the quotient rule, and $\delta_{i'}(u) = 0$ since $\delta_{i'}(u_1) = \delta_{i'}(u_2) = 0$. Thus $u \in S_i$.

Let $p_i : R \to S_i$ be the projection along $\bigoplus R^\lambda$, where $\lambda$ runs over the vectors with $\lambda_{i'} \neq 0$ for some $i' \neq i$. For $f \in R^\lambda$ and $f' \in R^\mu$, the product $ff'$ belongs to $R^{\lambda+\mu}$, and $p_i(ff') = p_i(f)p_i(f')$, since $\lambda + \mu$ has vanishing $i'$-th components for all $i' \neq i$ if and only if both $\lambda$ and $\mu$ do. Thus $p_i$ is a homomorphism of graded $\mathbb{C}$-algebras with $p_i|_{S_i} = \mathrm{id}_{S_i}$. Since $p_i$ is surjective, it induces a closed embedding $s_i : X_i \to X$, and $q_i \circ s_i = \mathrm{id}_{X_i}$. Since the inclusions $S_i \subset R$ are graded, $q$ is equivariant for the grading of $X$ and the diagonal grading of $X_1 \times \cdots \times X_k$.

Let $y \in X$ and $s \in \mathbb{R}$. For $f \in \mathcal{G} \cap R^\lambda$,

$$f\Big(\prod_{i' \neq i} \psi_{i',s}(y)\Big) = \exp\Big(s \sum_{i' \neq i} \lambda_{i'}\Big) f(y),$$

where the product denotes the composition of the commuting automorphisms $\psi_{i',s}$. As $s \to -\infty$, the right-hand side tends to $f(y)$ if $f \in S_i$ and to 0 otherwise. In both cases the limit is $(p_i f)(y) = f(s_i(q_i(y)))$. Since $\mathcal{G}$ defines a closed embedding of $X$ into an affine space,

$$\lim_{s \to -\infty} \prod_{i' \neq i} \psi_{i',s}(y) = s_i(q_i(y))$$

in the analytic topology. Now let $y \in M$. Applying (8.2) to each $i' \neq i$, we obtain

$$\rho^2\Big(\prod_{i' \neq i} \psi_{i',s}(y)\Big) = b_i(y) + e^{2s} \sum_{i' \neq i} b_{i'}(y).$$

Since $\rho$ is continuous, $\rho^2(s_i(q_i(y))) = b_i(y)$. Summing over $i$ and using $\rho^2 = \sum_i b_i$, we obtain

$$\rho^2 = \sum_i \rho^2 \circ s_i \circ q_i \tag{8.3}$$

on $M$, and hence on $X$ since both sides are continuous and $M$ is dense in $X$. This proves (4).

We show that $q$ is finite. Let $o_i \in X_i$ be the vertex of the grading of $S_i$. Since $p_i(\bigoplus_{j>0} R_j) = \bigoplus_{j>0}(S_i)_j$, we have $s_i(o_i) = o$. If $q(y) = (o_1, \dots, o_k)$, then (8.3) implies $\rho^2(y) = k\rho^2(o) = 0$, so $y = o$ by condition (2) of Set-up 2.31. Let $\mathfrak{a} \subset R$ be the ideal generated by $\bigoplus_{j>0}(S_i)_j$ for $1 \leq i \leq k$. The zero locus of $\mathfrak{a}$ is $q^{-1}(o_1, \dots, o_k) = \{o\}$, so $\sqrt{\mathfrak{a}} = \bigoplus_{j>0} R_j$ by the Nullstellensatz [Har77, Theorem I.1.3A]. Since $R$ is finitely generated, $(\bigoplus_{j>0} R_j)^m \subset \mathfrak{a}$ for some $m$, and hence $\dim R/\mathfrak{a} < +\infty$. By the graded Nakayama lemma [BH93, Section 1.5], homogeneous lifts of a basis of $R/\mathfrak{a}$ generate $R$ as a module over the subalgebra of $R$ generated by $S_1, \dots, S_k$. Thus $R$ is finite over $S_1 \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} S_k$, and $q$ is finite.

**Step 3.** In this step we show that for $f \in S_i$ and $i' \neq i$, the differential $df$ vanishes on $\operatorname{im} \varpi_{i'}$ at any point of $M$.

Since $S_i$ is graded, we may assume that $f \in S_i \cap R_j$ for some $j$. Recall that $M = (0, +\infty) \times N$, that $N = \{\rho = 1\}$, and that $\dim_{\mathbb{R}} M = 4n$, so $d\operatorname{Vol}_g = \rho^{4n-1} d\rho d\operatorname{Vol}_h$. By (8.1), $f(\rho, x) = \rho^{aj} f(1, x)$ for $(\rho, x) \in (0, +\infty) \times N$. Since $\rho^{-1}(1) \subset X$ is compact by condition (2) of Set-up 2.31 and $f$ is continuous on $X$, $C_f := \sup_N |f| \leq \max_{\rho^{-1}(1)} |f| < +\infty$, and

$$|f(\rho, x)| \leq C_f \rho^{aj} \quad \text{for } (\rho, x) \in (0, +\infty) \times N. \tag{8.4}$$

For a real smooth function $u$ on $M$, let $\Delta_{i'} u := \operatorname{div}_g(\varpi_{i'} \operatorname{grad}_g u)$. Since $\varpi_{i'}$ is parallel, $\Delta_{i'} u$ is the trace of $\operatorname{Hess}_g u$ on $\operatorname{im} \varpi_{i'}$. Since $\operatorname{im} \varpi_{i'}$ is $I$-invariant, the trace is a sum of terms $\operatorname{Hess}_g u(e, e) + \operatorname{Hess}_g u(Ie, Ie)$, which are multiples of the Levi form of $u$ at $e^{1,0}$ since $g$ is Kähler. Since $f$ is holomorphic, $\operatorname{Re} f$ and $\operatorname{Im} f$ are pluriharmonic, so $\Delta_{i'} \operatorname{Re} f = \Delta_{i'} \operatorname{Im} f = 0$. Moreover, $V_{i'} f = \delta_{i'} f = 0$ by **Step 1**, so $V_{i'} \operatorname{Re} f = V_{i'} \operatorname{Im} f = 0$. Since $\operatorname{grad}_g(-\rho^2/2) = -\rho \partial_\rho$ and $\varpi_{i'}$ is self-adjoint, for $u \in \{\operatorname{Re} f, \operatorname{Im} f\}$ we obtain

$$\operatorname{div}_g\left(e^{-\rho^2/2} \varpi_{i'} \operatorname{grad}_g u\right) = e^{-\rho^2/2}\big(\Delta_{i'} u - g(V_{i'}, \operatorname{grad}_g u)\big) = e^{-\rho^2/2}(\Delta_{i'} u - V_{i'} u) = 0.$$

Let $\eta$ be a real smooth function with compact support on $M$. Multiplying the last equation by $\eta^2 u$ and integrating by parts, and using $g(\operatorname{grad}_g u, \varpi_{i'} \operatorname{grad}_g u) = |\varpi_{i'} \operatorname{grad}_g u|_g^2$, we obtain

$$\int_M \eta^2 |\varpi_{i'} \operatorname{grad}_g u|_g^2 e^{-\rho^2/2} d\operatorname{Vol}_g = -2 \int_M \eta u g(\operatorname{grad}_g \eta, \varpi_{i'} \operatorname{grad}_g u) e^{-\rho^2/2} d\operatorname{Vol}_g.$$

By the Cauchy–Schwarz inequality, the right-hand side is at most

$$2\Big( \int_M \eta^2 |\varpi_{i'} \operatorname{grad}_g u|_g^2 e^{-\rho^2/2} d\operatorname{Vol}_g \Big)^{1/2} \Big( \int_M u^2 |d\eta|_g^2 e^{-\rho^2/2} d\operatorname{Vol}_g \Big)^{1/2}.$$

Let

$$|\varpi_{i'} \operatorname{grad}_g f|_g^2 := |\varpi_{i'} \operatorname{grad}_g \operatorname{Re} f|_g^2 + |\varpi_{i'} \operatorname{grad}_g \operatorname{Im} f|_g^2.$$

Adding the inequalities for $u = \operatorname{Re} f$ and $u = \operatorname{Im} f$, we obtain

$$\int_M \eta^2 |\varpi_{i'} \operatorname{grad}_g f|_g^2 e^{-\rho^2/2} d\operatorname{Vol}_g \le 4 \int_M |f|^2 |d\eta|_g^2 e^{-\rho^2/2} d\operatorname{Vol}_g. \tag{8.5}$$

Both sides of (8.5) are finite since $\eta$ has compact support.

By condition (4) of Set-up 2.31, there exists a sequence of $T_c$-invariant $\Phi$-cutoffs $\beta_l$ on $N$. Thus $\beta_l$ is smooth with compact support, $0 \le \beta_l \le 1$, $\beta_l = 1$ on any given compact subset of $N$ for $l \gg 0$, and $\int_N \Phi(|d\beta_l|_h) d\operatorname{Vol}_h \to 0$. Since $\Phi(t) \ge t^2$, we have $\int_N |d\beta_l|_h^2 d\operatorname{Vol}_h \to 0$ (Remark 2.30).

Fix a real number $0 < \epsilon < 1/4$. Let $\chi : (0, +\infty) \to [0, 1]$ be a smooth function with $\chi = 1$ on $[2\epsilon, \epsilon^{-1}]$, $\operatorname{Supp} \chi \subset [\epsilon, 2\epsilon^{-1}]$, $|\chi'| \le C/\epsilon$ on $[\epsilon, 2\epsilon]$, and $|\chi'| \le C\epsilon$ on $[\epsilon^{-1}, 2\epsilon^{-1}]$, where $C$ is a constant independent of $\epsilon$. Let $\eta := \chi(\rho)\beta_l(x)$, which has compact support on $M$. Since $g = d\rho^2 + \rho^2 h$,

$$|d\eta|_g^2 = \beta_l^2 \chi'(\rho)^2 + \chi(\rho)^2 \rho^{-2} |d\beta_l|_h^2.$$

By (8.4), the contribution of the second term to the right-hand side of (8.5) is at most

$$4C_f^2 \int_\epsilon^{2\epsilon^{-1}} \rho^{2aj+4n-3} d\rho \cdot \int_N |d\beta_l|_h^2 d\operatorname{Vol}_h,$$

which tends to 0 as $l \to +\infty$. The contribution of the first term is at most

$$4 \int_M |f|^2 \chi'(\rho)^2 e^{-\rho^2/2} d\operatorname{Vol}_g,$$

which is finite by (8.4) and condition (3) of Set-up 2.31. Since $\beta_l \to 1$ pointwise on $N$, Fatou's lemma implies

$$\int_M \chi(\rho)^2 |\varpi_{i'} \operatorname{grad}_g f|_g^2 e^{-\rho^2/2} d\operatorname{Vol}_g \le 4 \int_M |f|^2 \chi'(\rho)^2 e^{-\rho^2/2} d\operatorname{Vol}_g. \tag{8.6}$$

By (8.4), the right-hand side of (8.6) is at most

$$4C_f^2 C^2 \operatorname{Vol}_h(N) \Big( \epsilon^{-2} \int_\epsilon^{2\epsilon} \rho^{2aj+4n-1} d\rho + \epsilon^2 \int_{\epsilon^{-1}}^{2\epsilon^{-1}} \rho^{2aj+4n-1} e^{-\rho^2/2} d\rho \Big).$$

The first summand is at most a constant times $\epsilon^{2aj+4n-2}$, which tends to 0 as $\epsilon \to 0$ since $aj \ge 0$ and $4n - 2 > 0$. The second summand is at most $\epsilon^2 \int_0^{+\infty} \rho^{2aj+4n-1} e^{-\rho^2/2} d\rho$, which is finite and independent of $\epsilon$, so it also tends to 0 as $\epsilon \to 0$. Let $\epsilon \to 0$ along a sequence. Then the corresponding functions $\chi$ tend to 1 pointwise on $(0, +\infty)$, and Fatou's lemma implies

$$\int_M |\varpi_{i'} \operatorname{grad}_g f|_g^2 e^{-\rho^2/2} d\operatorname{Vol}_g = 0.$$

Thus $\varpi_{i'} \operatorname{grad}_g \operatorname{Re} f = \varpi_{i'} \operatorname{grad}_g \operatorname{Im} f = 0$ on $M$. Since

$$df(v) = g(\operatorname{grad}_g \operatorname{Re} f, v) + \sqrt{-1} g(\operatorname{grad}_g \operatorname{Im} f, v)$$

and $g(w, v) = g(\varpi_{i'} w, v)$ for $v \in \operatorname{im} \varpi_{i'}$, the differential $df$ vanishes on $\operatorname{im} \varpi_{i'}$.

**Step 4.** In this step we show that $\dim X_i = 2n_i$ and that $q$ is surjective.

Since $S_i \subset R$, the morphism $q_i$ is dominant. By generic smoothness [Har77, Corollary III.10.7] applied to $q_i|_{X_{\mathrm{reg}}}$, there exists a dense open subset of $M$ on which $q_i$ is smooth over $(X_i)_{\mathrm{reg}}$, so the differential of $q_i$ has rank $\dim X_i$ there. Since $X_i$ is embedded in an affine space by generators of $S_i$, the differential of $q_i$ at $y \in M$ vanishes on a tangent vector $v$ if and only if $df(v) = 0$ for $f \in S_i$. By **Step 3**, the differential of $q_i$ vanishes on $\bigoplus_{i' \neq i} \operatorname{im} \varpi_{i'}$, so $\dim X_i \leq 2n_i$. Since $q$ is finite by **Step 2**,

$$2n = \dim X = \dim q(X) \leq \sum_i \dim X_i \leq \sum_i 2n_i = 2n.$$

Thus $\dim X_i = 2n_i$ for any $i$, and the closed subset $q(X)$ of the irreducible variety $X_1 \times \cdots \times X_k$ satisfies $\dim q(X) = 2n = \dim(X_1 \times \cdots \times X_k)$, so $q(X) = X_1 \times \cdots \times X_k$. Moreover, there exists a dense open subset $M^\circ \subset M$ such that for $y \in M^\circ$ and any $i$, the point $q_i(y)$ belongs to $(X_i)_{\mathrm{reg}}$ and the differential of $q_i$ at $y$ restricts to an isomorphism from $\operatorname{im} \varpi_i$ onto the tangent space of $X_i$ at $q_i(y)$.

For each $i$, let $M_i^\circ$ be the dense open subset of $M \cap q_i^{-1}((X_i)_{\mathrm{reg}})$ on which $q_i$ is smooth, given by generic smoothness [Har77, Corollary III.10.7] applied to $q_i|_{X_{\mathrm{reg}}}$, and let $M^\circ := \bigcap_i M_i^\circ$. For $y \in M^\circ$, the differential of $q_i$ at $y$ is surjective onto the tangent space of $X_i$ at $q_i(y)$. Since $\dim X = 2n$ and $\dim X_i = 2n_i$, the kernel of the differential is a subspace of dimension $2n - 2n_i$. By **Step 3**, the kernel contains $\bigoplus_{i' \neq i} \operatorname{im} \varpi_{i'}$, whose dimension is $\sum_{i' \neq i} 2n_{i'} = 2n - 2n_i$. Hence the kernel equals $\bigoplus_{i' \neq i} \operatorname{im} \varpi_{i'}$. Since the tangent space of $X$ at $y$ is $\bigoplus_{i'} \operatorname{im} \varpi_{i'}$ by Set-up 8.1(3), the differential restricts to an isomorphism from $\operatorname{im} \varpi_i$ onto the tangent space of $X_i$ at $q_i(y)$.

**Step 5.** In this step we construct $\omega_i$, show that $q_i^* \omega_i = \tau_i$, and show that $q$ is quasi-étale.

Let $\tau := \sum_i \tau_i$. Since $\tau_i$ is non-degenerate on $\operatorname{im} \varpi_i$ with kernel $\bigoplus_{i' \neq i} \operatorname{im} \varpi_{i'}$, the form $\tau$ is non-degenerate on $M$. Since the $\tau_i$ are parallel on $M$, $d\tau = 0$ on $X_{\mathrm{reg}}$. If $o \in X_{\mathrm{reg}}$, then $\tau^n$ is a section of the canonical bundle near $o$ whose zero locus is contained in $\{o\}$, so it has no zero by Krull's principal ideal theorem [Har77, Theorem I.1.11A], since $\dim X \geq 2$. Thus $\tau$ is non-degenerate on $X_{\mathrm{reg}}$. Let $\pi$ be the inverse bivector of $\tau$ on $X_{\mathrm{reg}}$. For $f, f' \in R$, let $\{f, f'\} := \pi(df, df')$. It is regular on $X_{\mathrm{reg}}$, and hence belongs to $R$ by [Har77, Proposition II.6.3A] since $X$ is normal.

On $M$, write $\pi = \sum_i \pi_i$, where $\pi_i$ is a section of $\bigwedge^2 \operatorname{im} \varpi_i$ whose restriction to $\operatorname{im} \varpi_i$ is the inverse of $\tau_i|_{\operatorname{im} \varpi_i}$. For vector fields $Y, Y'$ on $M$, since $\nabla \tau_i = 0$ and $\nabla V_{i'} = \varpi_{i'}$,

$$(L_{V_{i'}} \tau_i)(Y, Y') = \tau_i(\varpi_{i'} Y, Y') + \tau_i(Y, \varpi_{i'} Y').$$

If $i' \neq i$, then $\varpi_{i'} Y$ and $\varpi_{i'} Y'$ belong to the kernel of $\tau_i$. If $i' = i$, then $\tau_i(\varpi_i Y, Y') = \tau_i(Y, Y')$ since $Y - \varpi_i Y$ belongs to the kernel of $\tau_i$. Thus

$$L_{V_{i'}} \tau_i = \begin{cases} 2\tau_i & \text{if } i' = i, \\ 0 & \text{if } i' \neq i, \end{cases} \qquad L_{V_{i'}} \pi_i = \begin{cases} -2\pi_i & \text{if } i' = i, \\ 0 & \text{if } i' \neq i. \end{cases} \tag{8.7}$$

Let $f, f' \in S_i$. By **Step 3**, $\{f, f'\} = \pi_i(df, df')$. For $i' \neq i$, (8.7) and $V_{i'} f = V_{i'} f' = 0$ imply

$$\delta_{i'}\{f, f'\} = V_{i'}\big(\pi_i(df, df')\big) = (L_{V_{i'}} \pi_i)(df, df') + \pi_i(d(V_{i'} f), df') + \pi_i(df, d(V_{i'} f')) = 0,$$

so $\{f, f'\} \in S_i$. If $f \in S_i$ and $f' \in S_{i'}$ with $i \neq i'$, then by **Step 3**, $\pi_{i''}(df, df') = 0$ for any $i''$, so $\{f, f'\} = 0$. Thus each $S_i$ is closed under $\{,\}$, and $q$ is compatible with the brackets, where $X_1 \times \cdots \times X_k$ carries the product bracket.

Fix $i$. The bracket of $S_i$ defines a section $\pi_{X_i}$ of $\bigwedge^2 T_{X_i}$ on $(X_i)_{\mathrm{reg}}$. Let $y \in M^\circ$, where $M^\circ$ is the dense open subset of $M$ from the end of **Step 4**. Since the differential of $q_i$ at $y$ maps $\operatorname{im} \varpi_i$ isomorphically onto the tangent space of $X_i$ at $q_i(y)$, and $\pi_i$ is non-degenerate on $\operatorname{im} \varpi_i$, the bivector $\pi_{X_i}$ is non-degenerate at $q_i(y)$. Let $\omega_i$ be the rational 2-form on $X_i$ inverse to $\pi_{X_i}$. At $y$, both $q_i^* \omega_i$ and $\tau_i$ vanish on $\bigoplus_{i' \neq i} \operatorname{im} \varpi_{i'}$, and both restrict to the inverse of $\pi_i$ on $\operatorname{im} \varpi_i$. Let $\sigma := \sum_i \mathrm{pr}_i^* \, \omega_i$ on $X_1 \times \cdots \times X_k$, where $\mathrm{pr}_i$ is the $i$-th projection. Thus

$$q_i^* \omega_i = \tau_i \quad \text{and} \quad q^* \sigma = \tau \tag{8.8}$$

on $M^\circ$, and hence as rational forms on $X$. Since $q$ is finite and surjective, the pullback by $q$ of rational forms is injective. Thus $d\sigma = 0$ since $q^* d\sigma = d\tau = 0$, and hence $d\omega_i = 0$ for any $i$.

Let $\Pi := \sum_i \mathrm{pr}_i^* \pi_{X_i}$, which is a regular section of $\bigwedge^2 T_{X_1 \times \cdots \times X_k}$ on the smooth locus of $X_1 \times \cdots \times X_k$. Since $\sigma$ is inverse to $\Pi$, the regular section $\Pi^n$ of the anti-canonical bundle on the smooth locus is a non-zero constant multiple of the inverse of $\sigma^n$. Let $P$ be a prime divisor on $X_1 \times \cdots \times X_k$. Since $X_1 \times \cdots \times X_k$ is normal [Gro65, Corollaire 6.14.3], the generic point of $P$ belongs to the smooth locus. Let $b := \mathrm{ord}_P(\Pi^n) \geq 0$, so $\mathrm{ord}_P(\sigma^n) = -b$. Since $q$ is finite and surjective, there exists a prime divisor $P'$ on $X$ with $q(P') = P$. Let $e \geq 1$ be the ramification index of $q$ along $P'$. By the Hurwitz formula,

$$\mathrm{ord}_{P'}(q^*\sigma^n) = e\,\mathrm{ord}_P(\sigma^n) + (e-1).$$

By (8.8), $q^*\sigma^n = \tau^n$, and $\mathrm{ord}_{P'}(\tau^n) = 0$ since $\tau$ is non-degenerate on $X_{\mathrm{reg}}$. Thus

$$0 = -eb + e - 1.$$

Since $b$ is a non-negative integer and $e \geq 1$, we obtain $e = 1$ and $b = 0$.

Since $b = 0$ for any prime divisor $P$, the zero locus of $\Pi^n$ on the smooth locus of $X_1 \times \cdots \times X_k$ contains no divisor, so it is empty by Krull's principal ideal theorem [Har77, Theorem I.1.11A]. Thus $\sigma$ is a regular closed non-degenerate 2-form on the smooth locus of $X_1 \times \cdots \times X_k$. Since the smooth locus of $X_1 \times \cdots \times X_k$ is $(X_1)_{\mathrm{reg}} \times \cdots \times (X_k)_{\mathrm{reg}}$ and $\sigma$ is the sum of the forms $\mathrm{pr}_i^* \omega_i$, each $\omega_i$ is a regular closed non-degenerate 2-form on $(X_i)_{\mathrm{reg}}$.

Since $e = 1$ for any prime divisor $P'$ on $X$ over any prime divisor $P$, and the residue field extensions are separable, $q$ is étale over the generic point of any prime divisor on $X_1 \times \cdots \times X_k$. Let $\Sigma' \subset X$ be the set of points at which $q$ is not étale, which is closed. Since $q$ is finite, $\Sigma := q(\Sigma')$ is closed, and $\Sigma$ contains no prime divisor. Thus $\mathrm{codim}\,\Sigma \geq 2$, and $q$ is étale over $(X_1 \times \cdots \times X_k) \backslash \Sigma$. Since $X$ and $X_1 \times \cdots \times X_k$ are normal, $q$ is quasi-étale (Definition 2.4). Together with **Steps 2 and 4** and (8.8), this proves (2).

**Step 6.** In this step we prove (3) and the remaining claims of (1).

Since $\tau^n$ has no zero on $X_{\mathrm{reg}}$ and $\sigma^n$ has no zero on the smooth locus of $X_1 \times \cdots \times X_k$, the canonical divisors $K_X$ and $K_{X_1 \times \cdots \times X_k}$ are Cartier and trivial. Since $q^*\sigma^n = \tau^n$ by (8.8),

$$K_X = q^* K_{X_1 \times \cdots \times X_k}.$$

Since $X$ is klt, $X_1 \times \cdots \times X_k$ is klt by [KM98, Proposition 5.20(4)].

We show that each $X_i$ is klt. Fix $i$. Since $\omega_i^{n_i}$ has no zero on $(X_i)_{\mathrm{reg}}$, $K_{X_i}$ is Cartier and trivial. Let $\gamma : \widetilde{X}_i \to X_i$ be a higher model and $E$ a prime divisor on $\widetilde{X}_i$. Then $A_{X_i}(E) = 1 + \mathrm{ord}_E(\gamma^* \omega_i^{n_i})$. Let $U := \prod_{i' \neq i} (X_{i'})_{\mathrm{reg}}$. Since $\sigma^n$ is a non-zero constant multiple of the wedge product of the forms $\mathrm{pr}_{i'}^* \omega_{i'}^{n_{i'}}$, and $\omega_{i'}^{n_{i'}}$ has no zero on $(X_{i'})_{\mathrm{reg}}$, the prime divisor $E \times U$ on $\widetilde{X}_i \times U$ over the open subset $X_i \times U$ of $X_1 \times \cdots \times X_k$ satisfies

$$A_{X_1 \times \cdots \times X_k}(E \times U) = 1 + \mathrm{ord}_{E \times U}\left((\gamma \times \mathrm{id}_U)^* \sigma^n\right) = 1 + \mathrm{ord}_E(\gamma^* \omega_i^{n_i}) = A_{X_i}(E).$$

Since $X_1 \times \cdots \times X_k$ is klt, $A_{X_i}(E) > 0$. Thus $X_i$ is klt.

By [GKKP11, Theorem 1.4], $\omega_i$ extends to a regular 2-form on any log resolution of $X_i$. Let $X' \to X_i$ be a resolution. There exists a log resolution $X'' \to X_i$ which factors through $X'$. Since $X'$ is smooth, the image in $X'$ of the exceptional set of $X'' \to X'$ has codimension at least 2, and $X'' \to X'$ is an isomorphism outside the image. Thus the pullback of $\omega_i$ to $X'$ is regular outside a closed subset of codimension at least 2, and hence regular on $X'$ by [Har77, Proposition II.6.3A]. Hence $X_i$ is a symplectic variety with the form $\omega_i$ (Definition 2.9).

It remains to compute the degree of $\omega_i$. By (8.7) and $\sum_{i'} V_{i'} = \rho \partial_\rho$, we have $L_{\rho\partial_\rho} \tau_i = 2\tau_i$ on $M$. Since $\tau_i$ is holomorphic and closed, $L_W \tau_i = 0$ for any vector field $W$ of type $(0,1)$. Thus $L_{(\rho\partial_\rho)^{1,0}} \tau_i = 2\tau_i$, and by **Step 1**,

$$L_{\mathrm{eu}} \tau_i = \frac{2}{a} \tau_i.$$

Since $S_i \subset R$ is a graded subalgebra, the Euler derivation of $S_i$ is the restriction of eu, and the vector fields defined by them are $q_i$-related. By (8.8) and the injectivity of $q_i^*$ on rational forms, $L_{\mathrm{eu}}\omega_i = (2/a)\omega_i$ on $X_i$. The regular 2-forms on $(X_i)_{\mathrm{reg}}$ form a graded $S_i$-module, and $L_{\mathrm{eu}}$ acts on a homogeneous component of degree $m \in \mathbb{Z}$ by multiplication by $m$. Writing $\omega_i$ as a sum of homogeneous components, we see that $\omega_i$ is homogeneous of degree $2/a$. In particular, $2/a$ is an integer since $\omega_i \neq 0$, and $2/a > 0$ since $a > 0$. Thus $X_i$ is a conical symplectic variety. Together with **Steps 2 and 4**, this proves (1) and (3). □

**Remark 8.3.** The morphism $q$ in Theorem 8.2 need not be an isomorphism. Let $X = \mathbb{A}^4/\{\pm 1\}$ with the flat cone metric, where $\mathbb{A}^4$ has coordinates $x_1, y_1, x_2, y_2$, $T$ is the image of the scalar $\mathbb{G}_m$-action, $a = 2$, and $\zeta$ induces the grading of $R = \mathbb{C}[x_1, y_1, x_2, y_2]^{\{\pm 1\}}$ by half the polynomial degree. The link $N = S^7/\{\pm 1\}$ is compact, so $\beta_l = 1$ are $\Phi$-cutoffs and Set-up 8.1 holds. Let $\varpi_1$ and $\varpi_2$ be the projections onto the factors $\mathbb{A}^2$ with coordinates $x_1, y_1$ and $x_2, y_2$ respectively, and let $\tau_i = dx_i \wedge dy_i$. Then

$$q : \mathbb{A}^4/\{\pm 1\} \to (\mathbb{A}^2/\{\pm 1\})^2$$

is the quotient by $\{\pm 1\}^2/\{\pm 1\}$, and $\deg q = 2$. The morphism $q$ is étale outside the image of $\{x_1 = y_1 = 0\} \cup \{x_2 = y_2 = 0\}$, which has codimension 2. Thus $q$ is quasi-étale but not an isomorphism.

8.2. **Factor metrics and two-forms on products.** In this subsection, we construct Ricci-flat Kähler cone metrics on the factors of Theorem 8.2, and we compute the 2-forms of degree equal to the degree of the symplectic form on products of conical symplectic varieties whose degree valuations minimize $\widehat{\mathrm{vol}}$.

To repeat the constructions of this section on the factors $X_i$ and on finite covers of $X_i$, we need the quotient log Fano pairs of $X_i$ and of the covers to be K-polystable: then, by Lemma 2.24, $X_i$ and the covers are K-polystable, hence K-semistable (Definition 2.16), for the torus of the grading, so they carry weak Ricci-flat Kähler cone potentials by Theorem 2.27, and their degree valuations minimize $\widehat{\mathrm{vol}}$ by Theorem 2.21. In the next proposition, the cone potential $\rho^2$ splits along the finite morphism $q$ into potentials on the factors, and Proposition 3.12 turns the factor potentials into K-polystability of the factors and of their covers.

**Proposition 8.4.** *Notation and conditions as in Set-up 8.1, let $o_i$ be the vertex of $X_i$, and let $\omega_i$ be the reflexive 2-form on $X_i$ with $q_i^*\omega_i = \tau_i$, which exists by Theorem 8.2 and is unique since $q_i$ is dominant. Then the following hold.*

*(1) There exist continuous proper functions $p_i : X_i \to [0, +\infty)$ with $p_i^{-1}(0) = \{o_i\}$ such that $\rho^2 = \sum_i p_i \circ q_i$. The form $\frac{\sqrt{-1}}{2}\partial\bar{\partial}p_i$ is a smooth Ricci-flat Kähler cone metric on $(X_i)_{\mathrm{reg}}\backslash\{o_i\}$ whose radial vector field acts on the degree $j$ part of $S_i$ as multiplication by $aj$, the pullback by $q$ of the product metric of the forms $\frac{\sqrt{-1}}{2}\partial\bar{\partial}p_i$ coincides with $g$ on $q^{-1}((X_1 \times \cdots \times X_k)_{\mathrm{reg}}) \cap M$, and $\omega_i$ is parallel.*

*(2) For the faithful $\mathbb{G}_m$-action whose image is that of the grading of $X_i$, the quotient log Fano pair of $X_i$ is K-polystable.*

*(3) Let $h_i : X_i' \to X_i$ be a finite surjective quasi-étale morphism from a normal affine variety $X_i'$ with a good positive grading such that $h_i$ is equivariant for the grading of $X_i'$ and a positive power of the grading of $X_i$. Then, for the faithful $\mathbb{G}_m$-action whose image is that of the grading of $X_i'$, the quotient log Fano pair of $X_i'$ is K-polystable.*

*Proof.* The proof is long, so we divide it into several steps. Let $Y := X_1 \times \cdots \times X_k$, let $\zeta_i$ be the $\mathbb{G}_m$-action on $X_i$ induced by the grading of $S_i$, and let $M_i := (X_i)_{\mathrm{reg}}\backslash\{o_i\}$.

**Step 1.** In this step we construct $p_i$ and show that $p_i$ is homogeneous of degree $2/a$.

Let $s_i : X_i \to X$ be the closed embedding of Theorem 8.2(4), and let $p_i := \rho^2 \circ s_i$. By Set-up 2.31(2), $\rho$ is continuous and proper with $\rho^{-1}(0) = \{o\}$. Since $s_i$ is a closed embedding, $p_i$ is continuous and proper. The homomorphism $R \to S_i$ defining $s_i$ is graded and $R_0 = (S_i)_0 = \mathbb{C}$,

so the preimage of the ideal $\bigoplus_{j>0}(S_i)_j$ of $o_i$ is the ideal $\bigoplus_{j>0} R_j$ of $o$. Thus $s_i(o_i) = o$, and since $s_i$ is injective, $p_i^{-1}(0) = s_i^{-1}(o) = \{o_i\}$. By Theorem 8.2(4),

$$\rho^2 = \sum_{i=1}^{k} \rho^2 \circ s_i \circ q_i = \sum_{i=1}^{k} p_i \circ q_i.$$

Since $\xi = a\zeta$, Set-up 2.31(1) implies that $\rho\partial_\rho f = ajf$ for $f \in R_j$. The real vector field on $M$ whose flow at time $t \in \mathbb{R}$ is $\zeta(e^{at})$ also sends $f \in R_j$ to $ajf$. Since $X$ is affine, around any point of $M$ some elements of $R$ form local holomorphic coordinates, so a real vector field $\upsilon$ on $M$ is determined by the functions $\upsilon f$ for $f \in R$, so the flow of $\rho\partial_\rho$ at time $t$ is $\zeta(e^{at})$. Since $\rho\partial_\rho(\rho^2) = 2\rho^2$, we have $\rho^2(\zeta(e^{at})y) = e^{2t}\rho^2(y)$ for $y \in M$. Moreover, $\zeta(\{|\lambda| = 1\}) \subset T_c$ and $\rho$ is $T_c$-invariant. Since $M$ is dense in $X$ and $\rho$ is continuous, we obtain

$$\rho^2(\zeta(\lambda)y) = |\lambda|^{2/a}\rho^2(y) \quad \text{for any } \lambda \in \mathbb{C}^* \text{ and } y \in X.$$

Since $s_i$ is graded, $s_i \circ \zeta_i(\lambda) = \zeta(\lambda) \circ s_i$. Thus

$$p_i(\zeta_i(\lambda)x) = |\lambda|^{2/a}p_i(x) \quad \text{for any } \lambda \in \mathbb{C}^* \text{ and } x \in X_i. \tag{8.9}$$

**Step 2.** In this step we show that $p_i$ is smooth on $(X_i)_{\mathrm{reg}}$, that $\theta_i := \frac{\sqrt{-1}}{2}\partial\bar\partial p_i$ is a Ricci-flat Kähler form on $(X_i)_{\mathrm{reg}}$ for which $\omega_i$ is parallel, and that $q$ pulls back the product metric to $g$.

By Theorem 8.2(2), $q$ is finite and quasi-étale. By the purity of the branch locus [Gro03, Exposé X, Théorème 3.1], the branch locus of $q$ over the smooth variety $Y_{\mathrm{reg}} = (X_1)_{\mathrm{reg}} \times \cdots \times (X_k)_{\mathrm{reg}}$ is empty or of pure codimension one, and it has codimension at least two since $q$ is quasi-étale. Thus $q$ is étale over $Y_{\mathrm{reg}}$. Let $X^\circ := q^{-1}(Y_{\mathrm{reg}})$. Then $X^\circ \subset X_{\mathrm{reg}}$. Let $\theta_g$ be the Kähler form of $g$. Since $g$ is a Kähler cone metric with radius $\rho$, $\theta_g = \frac{\sqrt{-1}}{2}\partial\bar\partial\rho^2$ on $M$ by Definition 2.26.

Fix $i$ and $x_i \in (X_i)_{\mathrm{reg}}$. Since $k \geq 2$, we may pick $j \neq i$ and points $z_l \in (X_l)_{\mathrm{reg}}$ for $l \neq i$ with $z_j \neq o_j$. Let $\iota : X_i \to Y$ be the morphism with $\mathrm{pr}_i \circ \iota = \mathrm{id}_{X_i}$ and $\mathrm{pr}_l \circ \iota$ the constant map to $z_l$ for $l \neq i$, and pick $y \in q^{-1}(\iota(x_i))$. Since $q$ is graded, $q(o) = (o_1, \ldots, o_k)$, and since $z_j \neq o_j$, we have $y \neq o$. Thus $y \in X^\circ\backslash\{o\} \subset M$. Since $q$ is étale at $y$, there exist open neighborhoods $W \subset M$ of $y$ and $W' \subset Y_{\mathrm{reg}}$ of $\iota(x_i)$ in the Euclidean topology such that $q|_W : W \to W'$ is biholomorphic. By **Step 1**,

$$\sum_{l=1}^{k} p_l \circ \mathrm{pr}_l = \rho^2 \circ (q|_W)^{-1} \quad \text{on } W',$$

which is smooth. Pulling back by $\iota$, the function $p_i + \sum_{l\neq i} p_l(z_l)$ is smooth near $x_i$, so $p_i$ is smooth near $x_i$. Since $i$ and $x_i$ are arbitrary, $p_l$ is smooth on $(X_l)_{\mathrm{reg}}$ for any $l$. Now

$$\sum_{l=1}^{k} \mathrm{pr}_l^*\,\theta_l = \frac{\sqrt{-1}}{2}\partial\bar\partial\Big(\rho^2 \circ (q|_W)^{-1}\Big) = \big((q|_W)^{-1}\big)^*\theta_g \quad \text{on } W'$$

is a Kähler form. Since $\iota^*\,\mathrm{pr}_l^*\,\theta_l = 0$ for $l \neq i$ and $\iota$ is an immersion, $\theta_i = \iota^*\big(\sum_l \mathrm{pr}_l^*\,\theta_l\big)$ is a Kähler form near $x_i$. Hence $\theta_i$ is a Kähler form on $(X_i)_{\mathrm{reg}}$, and $\sum_l \mathrm{pr}_l^*\,\theta_l$ is the Kähler form of the product metric on $Y_{\mathrm{reg}}$. Since $\rho^2 = q^*\big(\sum_l \mathrm{pr}_l^*\,p_l\big)$, we have

$$q^*\Big(\sum_{l=1}^{k} \mathrm{pr}_l^*\,\theta_l\Big) = \theta_g \quad \text{on } X^\circ\backslash\{o\} = q^{-1}(Y_{\mathrm{reg}}) \cap M.$$

A Kähler metric is determined by the Kähler form and the complex structure, and $q$ is holomorphic. Thus the pullback by $q$ of the product metric coincides with $g$ on $q^{-1}(Y_{\mathrm{reg}}) \cap M$. In particular, $q|_W$ is an isometry.

By Theorem 8.2(2), $\tau_i = (q|_W)^*\,\mathrm{pr}_i^*\,\omega_i$ on $W$. Since $\tau_i$ is parallel for $g$ by Set-up 8.1(4) and $q|_W$ is an isometry, $\mathrm{pr}_i^*\,\omega_i$ is parallel for the product metric on $W'$. The Levi-Civita connection

of the product metric is the sum of the Levi-Civita connections of $\theta_1, \dots, \theta_k$, so $\omega_i$ is parallel for $\theta_i$ near $x_i$. Hence $\omega_i$ is parallel on $(X_i)_{\mathrm{reg}}$.

Let $\Omega_i := \omega_i^{n_i}$. By Theorem 8.2(1), $\omega_i$ is non-degenerate on $(X_i)_{\mathrm{reg}}$, so $\Omega_i$ is a nowhere vanishing holomorphic $2n_i$-form on $(X_i)_{\mathrm{reg}}$, and $\Omega_i$ is parallel for $\theta_i$. Thus the positive function $\theta_i^{2n_i}/((\sqrt{-1})^{(2n_i)^2}\Omega_i \wedge \bar{\Omega}_i)$ is parallel. Since $X_i$ is irreducible, $(X_i)_{\mathrm{reg}}$ is connected, so there exists a constant $C_i > 0$ with

$$\theta_i^{2n_i} = C_i(\sqrt{-1})^{(2n_i)^2}\Omega_i \wedge \bar{\Omega}_i \quad \text{on } (X_i)_{\mathrm{reg}}. \tag{8.10}$$

In local holomorphic coordinates $z_1, \dots, z_{2n_i}$ with $\Omega_i = u dz_1 \wedge \cdots \wedge dz_{2n_i}$, (8.10) shows that the determinant of $\theta_i$ is a constant multiple of $|u|^2$. Since $u$ is holomorphic, the Ricci form $-dd^c \log \det(\theta_i)$ vanishes, and $\theta_i$ is Ricci-flat.

**Step 3.** In this step we show that $\theta_i$ is a Kähler cone metric on $M_i$ whose radial vector field acts on $(S_i)_j$ as multiplication by $aj$, and that $dd^c \log p_i \geq 0$ on $M_i$. With **Step 2**, this proves (1).

By **Step 1**, $p_i > 0$ on $M_i$. Let $I$ be the complex structure, and let $\upsilon_i$ be the real vector field on $(X_i)_{\mathrm{reg}}$ whose flow at time $t \in \mathbb{R}$ is $\zeta_i(e^{at})$. For $f \in (S_i)_j$ we have $f(\zeta_i(e^{at})x) = e^{ajt}f(x)$, so $\upsilon_i f = ajf$. Since $\zeta_i(\lambda)$ is holomorphic in $\lambda$, $I\upsilon_i$ generates the flow $\zeta_i(e^{\sqrt{-1}at})$, and $L_{\upsilon_i} I = 0$. By (8.9), $\upsilon_i(p_i) = 2p_i$ and $(I\upsilon_i)(p_i) = 0$. Since $\theta_i = -\frac{1}{4}d(dp_i \circ I)$, Cartan's formula implies

$$i_{\upsilon_i}\theta_i = -\frac{1}{4}\Big(d\big(\upsilon_i(p_i)\big) \circ I - d\big((I\upsilon_i)(p_i)\big)\Big) = -\frac{1}{2}dp_i \circ I \quad \text{and} \quad L_{\upsilon_i}\theta_i = di_{\upsilon_i}\theta_i = 2\theta_i.$$

Let $g_i := \theta_i(\cdot, I\cdot)$ be the Kähler metric of $\theta_i$, and let $r_i := \sqrt{p_i}$ on $M_i$. Then

$$g_i(\upsilon_i, \cdot) = \frac{1}{2}dp_i = r_i dr_i, \quad g_i(\upsilon_i, \upsilon_i) = \frac{1}{2}\upsilon_i(p_i) = r_i^2, \quad L_{\upsilon_i} g_i = 2g_i.$$

Let $N_i := \{r_i = 1\} \subset M_i$, which is a smooth real hypersurface since $dr_i(\upsilon_i) = r_i \neq 0$. By (8.9), $r_i(\zeta_i(e^{at})x) = e^t r_i(x)$, so any orbit of the flow of $\upsilon_i$ in $M_i$ intersects $N_i$ in exactly one point. Thus the map

$$\psi_i : (0, +\infty) \times N_i \to M_i, \quad \psi_i(s, x) := \zeta_i(s^a)x,$$

is a diffeomorphism with $r_i \circ \psi_i(s, x) = s$. The vector $\psi_{i*}\partial_s = s^{-1}\upsilon_i$ has length one, it is $g_i$-orthogonal to the level sets of $r_i$ by $g_i(\upsilon_i, \cdot) = r_i dr_i$, and $\zeta_i(s^a)^* g_i = s^2 g_i$ since $L_{\upsilon_i} g_i = 2g_i$. Hence

$$\psi_i^* g_i = ds^2 + s^2 g_i|_{N_i}.$$

Thus $\theta_i$ is a Kähler cone metric on $M_i$ with radius $r_i$ and radial vector field $r_i\partial_{r_i} = \upsilon_i$, which acts on $(S_i)_j$ as multiplication by $aj$. By **Step 2**, $\theta_i$ is Ricci-flat. For a Kähler cone metric with radius $r_i$, $dd^c \log r_i^2$ is the pullback of the transverse Kähler form [MSY06, Section 2], so $dd^c \log p_i \geq 0$ on $M_i$.

**Step 4.** In this step we prove (2).

Let $d_i$ be the greatest common divisor of the integers $j > 0$ with $(S_i)_j \neq 0$. Since $S_i$ is a domain, the kernel of $\zeta_i$ is the group $\mu_{d_i}$ of $d_i$-th roots of unity, and the faithful $\mathbb{G}_m$-action $\bar{\zeta}_i$ with $\bar{\zeta}_i(\lambda^{d_i}) = \zeta_i(\lambda)$ satisfies $\bar{\zeta}_i(\mathbb{G}_m) = \zeta_i(\mathbb{G}_m)$. By Theorem 8.2(3), $\omega_i$ has degree $2/a$ for $\zeta_i$, so $\Omega_i$ has degree $2n_i/a$ for $\zeta_i$. Since $\mu_{d_i}$ acts trivially on $X_i$, hence on $\Omega_i$, $d_i$ divides $2n_i/a$, and $\Omega_i$ is homogeneous of degree $2n_i/(ad_i)$ for $\bar{\zeta}_i$. By (8.9),

$$p_i(\bar{\zeta}_i(\lambda)x) = |\lambda|^{2/(ad_i)}p_i(x), \quad \text{and} \quad \frac{2}{ad_i} = \frac{2}{2n_i} \cdot \frac{2n_i}{ad_i}.$$

Thus $p_i$ has the homogeneity of Proposition 3.12 for the form $\Omega_i$ of degree $2n_i/(ad_i)$ on the $2n_i$-dimensional variety $X_i$.

We check the other hypotheses of Proposition 3.12 for $X_i$, $\bar{\zeta}_i$, $\Omega_i$, $p_i$, and $U := M_i$. By Theorem 8.2(1), $X_i$ is a normal affine klt variety with $\dim X_i = 2n_i \geq 2$, and $\bar{\zeta}_i$ induces a good positive grading with vertex $o_i$. Since $\Omega_{X_i}^{[2n_i]}$ is reflexive of rank one and $\Omega_i$ is nowhere vanishing on $(X_i)_{\mathrm{reg}}$, $\Omega_i$ is a nowhere vanishing reflexive $2n_i$-form. By **Step 1**, $p_i$ is continuous and positive

on $X_i\backslash\{o_i\}$. The set $M_i$ is a non-empty $\bar{\zeta}_i$-invariant Zariski open subset of $(X_i)_{\mathrm{reg}}\backslash\{o_i\}$. On $M_i$, $p_i$ is smooth, $dd^c p_i = 2\theta_i$ is a Kähler form, $dd^c \log p_i \geq 0$ by **Step 3**, and by (8.10),

$$(dd^c p_i)^{2n_i} = 2^{2n_i} C_i (\sqrt{-1})^{(2n_i)^2} \Omega_i \wedge \bar{\Omega}_i.$$

Since $dd^c = \sqrt{-1}\partial\bar{\partial}$ and $\frac{\sqrt{-1}}{2}\partial\bar{\partial}$ differ by the factor 2, only the constant changes from $C_i$ to $2^{2n_i}C_i$. By Proposition 3.12, the quotient log Fano pair of $X_i$ for $\bar{\zeta}_i$ is K-polystable. This proves (2).

**Step 5.** In this step we prove (3).

Let $\zeta_i'$ be the $\mathbb{G}_m$-action on $X_i'$ induced by the grading, let $o_i'$ be the vertex of $X_i'$, and let $c_i \geq 1$ be the integer with $h_i \circ \zeta_i'(\lambda) = \zeta_i(\lambda^{c_i}) \circ h_i$. The set $h_i^{-1}(o_i)$ is finite and $\zeta_i'$-invariant, hence pointwise fixed by $\zeta_i'$ since $\mathbb{G}_m$ is connected. The only fixed point of a good positive grading is the vertex, so $h_i^{-1}(o_i) = \{o_i'\}$.

Since $h_i$ is quasi-étale, $K_{X_i'} = h_i^* K_{X_i}$ by the Hurwitz formula, so $X_i'$ is klt by [KM98, Proposition 5.20]. By the purity of the branch locus [Gro03, Exposé X, Théorème 3.1], $h_i$ is étale over $(X_i)_{\mathrm{reg}}$. Thus $h_i^{-1}((X_i)_{\mathrm{reg}}) \subset (X_i')_{\mathrm{reg}}$, and the complement of $h_i^{-1}((X_i)_{\mathrm{reg}})$ in $X_i'$ has codimension at least two since $h_i$ is finite. The form $h_i^*\Omega_i$ is regular and nowhere vanishing on $h_i^{-1}((X_i)_{\mathrm{reg}})$, hence extends to a reflexive $2n_i$-form $\Omega_i'$ on $X_i'$. On $(X_i')_{\mathrm{reg}}$, $\Omega_i'$ is a section of the line bundle of $2n_i$-forms whose zero locus has codimension at least two, so the zero locus is empty. Hence $\Omega_i'$ is a nowhere vanishing reflexive $2n_i$-form. Since

$$\zeta_i'(\lambda)^*\Omega_i' = h_i^*\zeta_i(\lambda^{c_i})^*\Omega_i = \lambda^{2c_i n_i/a}\Omega_i',$$

$\Omega_i'$ has degree $2c_i n_i/a$ for $\zeta_i'$.

Let $d_i'$ be the greatest common divisor of the positive degrees of the grading of $X_i'$, and let $\bar{\zeta}_i'$ be the faithful $\mathbb{G}_m$-action with $\bar{\zeta}_i'(\lambda^{d_i'}) = \zeta_i'(\lambda)$. As in **Step 4**, $\Omega_i'$ is homogeneous of degree $2c_i n_i/(ad_i')$ for $\bar{\zeta}_i'$. Let $p_i' := p_i \circ h_i$. Then $p_i'$ is continuous and positive on $X_i'\backslash\{o_i'\}$, and by (8.9),

$$p_i'(\bar{\zeta}_i'(\lambda)x) = |\lambda|^{2c_i/(ad_i')} p_i'(x), \quad \text{and} \quad \frac{2c_i}{ad_i'} = \frac{2}{2n_i} \cdot \frac{2c_i n_i}{ad_i'}.$$

Let $U' := h_i^{-1}(M_i)$, which is a non-empty $\bar{\zeta}_i'$-invariant Zariski open subset of $(X_i')_{\mathrm{reg}}\backslash\{o_i'\}$, and $h_i|_{U'}$ is étale. On $U'$, $p_i'$ is smooth, $dd^c p_i' = h_i^*(2\theta_i)$ is a Kähler form since $h_i|_{U'}$ is a local biholomorphism, $dd^c \log p_i' = h_i^* dd^c \log p_i \geq 0$ by **Step 3**, and by (8.10),

$$(dd^c p_i')^{2n_i} = 2^{2n_i} C_i (\sqrt{-1})^{(2n_i)^2} \Omega_i' \wedge \bar{\Omega}_i'.$$

Moreover, $X_i'$ is a normal affine klt variety with $\dim X_i' = 2n_i \geq 2$, and $\bar{\zeta}_i'$ induces a good positive grading with vertex $o_i'$. By Proposition 3.12, the quotient log Fano pair of $X_i'$ for $\bar{\zeta}_i'$ is K-polystable. This proves (3). □

The next lemma computes the space of 2-forms of degree $w$ on the product covers in Theorem 8.8(3). Only the constant forms on the plane factor and the forms of the single factors occur, without mixed terms. Restricted to $(X_1)_{\mathrm{reg}} \times \cdots \times (X_k)_{\mathrm{reg}}$, Lemma 8.5 implies condition (5) of Set-up 9.8 in Section 9. The vanishing $\mathcal{O}(X_i)_j = 0$ for $0 < j < w$ and the lower bound on the degrees of forms (Theorem 2.13) exclude the mixed terms.

**Lemma 8.5.** *Let $w \geq 1$ and $b, k \geq 0$ be integers. For $1 \leq i \leq k$ let $X_i$ be a conical symplectic variety whose symplectic form $\omega_i$ has degree $w$ and whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex, such that $\mathcal{O}(X_i)_j = 0$ for $0 < j < w$ and $H^0((X_i)_{\mathrm{reg}}, \Omega^2)_w = \mathbb{C}\omega_i$. If $b > 0$, assume that $w$ is even, and grade $\mathbb{A}^{2b}$ so that every linear coordinate has degree $w/2$. Let $X' := \mathbb{A}^{2b} \times X_1 \times \cdots \times X_k$ with the diagonal grading. Then*

$$H^0(X'_{\mathrm{reg}}, \Omega^2)_w = \bigwedge^2 (\mathbb{C}^{2b})^* \oplus \bigoplus_{i=1}^k \mathbb{C}\,\mathrm{pr}_i^*\,\omega_i,$$

*where $\mathrm{pr}_i : X' \to X_i$ is the projection and $\bigwedge^2(\mathbb{C}^{2b})^*$ is the space of constant 2-forms on $\mathbb{A}^{2b}$.*

*Proof.* Let $X_0 := \mathbb{A}^{2b}$, so that $X' = X_0 \times X_1 \times \cdots \times X_k$ and $\mathrm{pr}_0 : X' \to X_0$ is the projection. For $0 \le i \le k$ and $0 \le p \le 2$, the spaces $H^0((X_i)_{\mathrm{reg}}, \Omega^p)$ are graded as in Definition 2.8, where $\Omega^0 = \mathcal{O}$. Since $X_i$ is normal, $H^0((X_i)_{\mathrm{reg}}, \mathcal{O}) = \mathcal{O}(X_i)$ [Har77, Proposition II.6.3A].

We first show that, for $1 \le i \le k$, $1 \le p \le 2$, and a non-zero $\sigma \in H^0((X_i)_{\mathrm{reg}}, \Omega^p)_j$, we have $j \ge pw/2$. Let $\dim X_i = 2m_i$ and let $v_i$ be the degree valuation of $X_i$. By Lemma 2.10, $X_i$ is klt. The form $\omega_i^{m_i}$ is nowhere vanishing on $(X_i)_{\mathrm{reg}}$, so $\omega_i^{m_i}$ generates the reflexive rank one sheaf $\Omega_{X_i}^{[2m_i]}$, and $\omega_i^{m_i}$ has degree $m_i w$. By Lemma 2.14 applied to $\mathbb{G}_m$ acting through the grading, $A_{X_i}(v_i) = m_i w$. By Theorem 2.13(2), $A_\sigma(v_i) = j$. By Theorem 2.13(1),

$$\frac{w}{2} = \frac{m_i w}{2m_i} \le \frac{j}{p},$$

that is,

$$j \ge \frac{pw}{2} \quad \text{for any non-zero } \sigma \in H^0((X_i)_{\mathrm{reg}}, \Omega^p)_j,\ 1 \le i \le k,\ 1 \le p \le 2. \tag{8.11}$$

Next we show that, for $1 \le i \le k$ and a non-zero $\eta \in H^0((X_i)_{\mathrm{reg}}, \Omega^1)_j$, we have $j \ge w$. By the previous paragraph, $j \ge w/2 > 0$. Suppose that $j < w$. Then $d\eta \in H^0((X_i)_{\mathrm{reg}}, \Omega^2)_j$ with $j < w$, so $d\eta = 0$ by the previous paragraph. Let eu be the Euler vector field of the grading of $X_i$. Since $\eta$ is homogeneous of degree $j$, Cartan's formula implies

$$j\eta = L_{\mathrm{eu}}\eta = d(i_{\mathrm{eu}}\eta) + i_{\mathrm{eu}} d\eta = d(i_{\mathrm{eu}}\eta).$$

The function $i_{\mathrm{eu}}\eta$ belongs to $H^0((X_i)_{\mathrm{reg}}, \mathcal{O})_j = \mathcal{O}(X_i)_j = 0$. Thus $j\eta = 0$, and since $j > 0$, $\eta = 0$, a contradiction. Hence

$$j \ge w \quad \text{for any non-zero } \eta \in H^0((X_i)_{\mathrm{reg}}, \Omega^1)_j,\ 1 \le i \le k. \tag{8.12}$$

On $X_0 = \mathbb{A}^{2b}$ with linear coordinates $x_1, \dots, x_{2b}$ of degree $w/2$, a homogeneous form is a sum of terms $u dx_{l_1} \wedge \cdots \wedge dx_{l_p}$ with $u$ a homogeneous polynomial, and such a term has degree $\deg u + pw/2$. Hence

$$\begin{aligned} &j \ge \frac{w}{2} \quad \text{for any non-zero } u \in \mathcal{O}(X_0)_j \text{ with } j > 0 \text{ and any non-zero } \eta \in H^0(X_0, \Omega^1)_j, \\ &H^0(X_0, \Omega^2)_j = 0 \quad \text{for } j < w, \quad \text{and} \quad H^0(X_0, \Omega^2)_w = \textstyle\bigwedge^2 (\mathbb{C}^{2b})^*. \end{aligned} \tag{8.13}$$

We have $X'_{\mathrm{reg}} = (X_0)_{\mathrm{reg}} \times (X_1)_{\mathrm{reg}} \times \cdots \times (X_k)_{\mathrm{reg}}$, and

$$\Omega^2_{X'_{\mathrm{reg}}} = \bigoplus_{i=0}^{k} \mathrm{pr}_i^*\, \Omega^2_{(X_i)_{\mathrm{reg}}} \oplus \bigoplus_{0 \le i < l \le k} \mathrm{pr}_i^*\, \Omega^1_{(X_i)_{\mathrm{reg}}} \otimes \mathrm{pr}_l^*\, \Omega^1_{(X_l)_{\mathrm{reg}}}.$$

By the Künneth formula [Sta, Tag 0BED], the global sections of the summands are

$$H^0\big((X_i)_{\mathrm{reg}}, \Omega^2\big) \otimes \bigotimes_{l' \ne i} \mathcal{O}(X_{l'}) \quad \text{and} \quad H^0\big((X_i)_{\mathrm{reg}}, \Omega^1\big) \otimes H^0\big((X_l)_{\mathrm{reg}}, \Omega^1\big) \otimes \bigotimes_{l' \ne i,l} \mathcal{O}(X_{l'}),$$

and since the grading of $X'$ is diagonal, the degree of a tensor product of homogeneous elements is the sum of the degrees. Since the gradings of $X_0, \dots, X_k$ are good positive gradings, $\mathcal{O}(X_{l'})_0 = \mathbb{C}$ and $\mathcal{O}(X_{l'})_j = 0$ for $j < 0$. By (8.11), (8.12), and (8.13), non-zero homogeneous 2-forms on any $(X_i)_{\mathrm{reg}}$ have degree at least $w$, and non-zero homogeneous 1-forms on $(X_i)_{\mathrm{reg}}$ have degree at least $w$ for $i \ge 1$ and at least $w/2$ for $i = 0$. Thus the degree $w$ part of the first summand is $H^0((X_i)_{\mathrm{reg}}, \Omega^2)_w \otimes \mathbb{C}$, which is $\bigwedge^2 (\mathbb{C}^{2b})^*$ for $i = 0$ and $\mathbb{C}\, \mathrm{pr}_i^*\, \omega_i$ for $i \ge 1$. For $0 \le i < l \le k$, non-zero homogeneous elements of the second summand have degree at least

$$\min\left\{\frac{w}{2} + w,\ 2w\right\} = \frac{3w}{2} > w,$$

so the degree $w$ part of the second summand is zero. This proves the lemma. □

We conclude this subsection with the conical symplectic varieties which are smooth or carry a non-zero function of positive degree below the degree of the symplectic form.

**Lemma 8.6.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety of dimension $2m$ with $m \geq 1$ whose symplectic form $\omega$ has degree $w$ and whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex, and assume that $H^0(X_{\mathrm{reg}}, \Omega^2)_w = \mathbb{C}\omega$. If $R_j \neq 0$ for some $0 < j < w$, or if $X$ is smooth, then $m = 1$ and $R$ is a polynomial ring in two variables of degree $w/2$.*

*Proof.* First assume that $R_j \neq 0$ for some $0 < j < w$. By Proposition 7.3 with $r = w$, $w$ is even, $j = w/2$, and the Poisson bracket $\{,\}$ of $\omega$ is a non-degenerate alternating form on $R_{w/2} \neq 0$. Write $\dim R_{w/2} = 2b$ with $b \geq 1$. By Proposition 7.3, there exist a symplectic basis $x_1, y_1, \ldots, x_b, y_b$ of $R_{w/2}$ and a finitely generated graded normal Poisson subalgebra $S \subset R$ such that multiplication induces an isomorphism $\mathbb{C}[x_1, y_1, \ldots, x_b, y_b] \otimes_{\mathbb{C}} S \cong R$, and $\operatorname{Spec} S$ is a point or a conical symplectic variety of dimension $2m - 2b$. Let $x := x_1$ and $y := y_1$, so that $\{x, y\} = 1$.

We show that $dx \wedge dy$ is non-zero at any point of $X_{\mathrm{reg}}$. Otherwise there exist a point $z \in X_{\mathrm{reg}}$ and $(\alpha, \beta) \in \mathbb{C}^2 \backslash \{0\}$ with $\alpha dx + \beta dy = 0$ at $z$. Since $\omega$ is non-degenerate at $z$, $\alpha H_x + \beta H_y = 0$ at $z$. Since $H_x(x) = \{x, x\} = 0$, $H_y(x) = \{y, x\} = -1$, $H_x(y) = \{x, y\} = 1$, and $H_y(y) = 0$, applying $\alpha H_x + \beta H_y$ to $x$ and to $y$ implies $\beta = 0$ and $\alpha = 0$, a contradiction. Thus $dx \wedge dy$ is a 2-form of rank 2 at any point of $X_{\mathrm{reg}}$, and $dx \wedge dy \in H^0(X_{\mathrm{reg}}, \Omega^2)_w$ since $x$ and $y$ have degree $w/2$. By assumption, $dx \wedge dy = c\omega$ for some $c \in \mathbb{C}$, and $c \neq 0$. Since $\omega$ has rank $2m$ at any point of $X_{\mathrm{reg}}$, we obtain $m = 1$. Then $2m - 2b = 2 - 2b \leq 0$, so $b = 1$ and $\operatorname{Spec} S$ is a point. Since $S$ is graded with $S_0 = \mathbb{C}$, $S = \mathbb{C}$, and $R = \mathbb{C}[x, y]$ with $x$ and $y$ of degree $w/2$.

Now assume that $X$ is smooth. Let $o$ be the vertex and $\mathfrak{m} := \bigoplus_{j>0} R_j$ the ideal of $o$. Pick homogeneous $f_1, \ldots, f_N \in \mathfrak{m}$ whose images form a basis of $\mathfrak{m}/\mathfrak{m}^2$, and let $d_l > 0$ be the degree of $f_l$. By the graded Nakayama lemma [BH93, Section 1.5], $f_1, \ldots, f_N$ generate $R$ as a $\mathbb{C}$-algebra. Since $X$ is smooth at $o$, $N = \dim X = 2m$, so $R$ is a quotient of the polynomial ring $\mathbb{C}[t_1, \ldots, t_{2m}]$ by a prime ideal $P$ with $\dim \mathbb{C}[t_1, \ldots, t_{2m}]/P = 2m$. Thus $P$ has height zero, $P = 0$, and $R = \mathbb{C}[f_1, \ldots, f_{2m}]$. Write

$$\omega = \sum_{1 \leq l < l' \leq 2m} u_{ll'} df_l \wedge df_{l'}$$

with $u_{ll'} \in R$. Since $\omega$ is homogeneous of degree $w$, $u_{ll'}$ is homogeneous of degree $w - d_l - d_{l'}$, so $u_{ll'}(o) = 0$ unless $d_l + d_{l'} = w$. Since $\omega$ is non-degenerate at $o$ and $m \geq 1$, there exists $l' > 1$ with $u_{1l'}(o) \neq 0$, hence $d_1 + d_{l'} = w$. Since $d_{l'} > 0$, we have $0 < d_1 < w$, and $f_1 \in R_{d_1}$ is non-zero. By the first case, $m = 1$ and $R$ is a polynomial ring in two variables of degree $w/2$. □

8.3. **The binary split.** In this subsection, we prove Theorem 8.7. If a reflexive 2-form whose degree equals the degree of the symplectic form is not a multiple of the symplectic form, then the pencil spanned by the symplectic form and the given form contains a degenerate member, which is parallel and has constant rank on the punctured regular locus. The kernel of the degenerate member may be Lagrangian for the symplectic form, so in the proof we construct, by a Schur complement, two parallel forms whose kernels are the kernel of the degenerate member and its orthogonal complement with respect to the Ricci-flat Kähler cone metric, instead of restricting the symplectic form to the kernel of the degenerate member and to its orthogonal complement.

**Theorem 8.7.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety of dimension $2n$, $n \geq 1$, with vertex $o$ and symplectic form $\omega$ of degree $r$, whose grading is induced by a $\mathbb{G}_m$-action $\zeta$, and assume that $(X, \zeta)$ is K-polystable for a torus which acts faithfully on $X$ and contains $\zeta(\mathbb{G}_m)$. Assume that there exists a reflexive 2-form $\beta$ of degree $r$ on $X$ which is not a multiple of $\omega$. Then $n \geq 2$, and the following hold.*

*(1) There exist conical symplectic varieties $X_1, X_2$ of dimensions $2n_1, 2n_2$ with $n_1, n_2 \geq 1$ and $n_1 + n_2 = n$, whose symplectic forms have degree $r$, and a finite surjective quasi-étale morphism $q : X \to X_1 \times X_2$ which is equivariant for the gradings.*

*(2) There exist an integer $l \geq 1$, normal affine varieties $X_1', X_2'$ with good positive gradings, finite surjective quasi-étale morphisms $p_i : X_i' \to X_i$ and $h : X_1' \times X_2' \to X$ with $q \circ h = p_1 \times p_2$ which are equivariant for the gradings of $X_i'$ and the $l$-th powers of the*

*gradings of $X_i$ and $X$, and a closed subset $\Sigma \subset X$ of codimension at least 4 such that $h$ is étale over $X\backslash\Sigma$.*

*(3) Each $X_i'$ is a conical symplectic variety whose symplectic form has degree $lr$, whose degree valuation is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex, and which is K-polystable for the torus defined by the image of the $\mathbb{G}_m$-action of the grading.*

*Proof.* The proof is long, so we divide it into several steps. For $\lambda \in \mathbb{G}_m$, we also write $\zeta(\lambda)$ for the automorphism of $X$ defined by $\lambda$, so that $\zeta(\lambda)^* f = \lambda^j f$ for $f \in R_j$.

**Step 1.** In this step we show that $n \geq 2$, and that any reflexive 2-form of degree $r$ on $X$ is parallel with respect to a Ricci-flat Kähler cone metric $g$ on $M := X_{\mathrm{reg}}\backslash\{o\}$.

Suppose that $n = 1$. Since $\omega$ is non-degenerate on the surface $X_{\mathrm{reg}}$, there exists a regular function $f$ on $X_{\mathrm{reg}}$ such that $\beta = f\omega$. Since $X$ is normal, $f \in R$. Since $\beta$ and $\omega$ are homogeneous of degree $r$, $f$ is homogeneous of degree 0, so $f \in R_0 = \mathbb{C}$. This contradicts the assumption that $\beta$ is not a multiple of $\omega$. Thus $n \geq 2$.

Let $T$ be a torus which acts faithfully on $X$ and contains $\zeta(\mathbb{G}_m)$ such that $(X, \zeta)$ is K-polystable for $T$. We regard $\zeta$ as a cocharacter of $T$. Since $R_0 = \mathbb{C}$, $\zeta$ is a Reeb vector of $T$, and the action of $T$ on $X$ is good. Let $m$ be the order of the kernel of $\zeta$. Since $T$ acts faithfully, $\zeta = m\chi$ for a primitive cocharacter $\chi$ of $T$. The degree $j$ part of $R$ for the grading induced by $\chi$ is $R_{mj}$. Thus $\omega$ has degree $r/m$ for $\chi$, the reflexive 2-forms of degree $r$ for $\zeta$ are exactly the reflexive 2-forms of degree $r/m$ for $\chi$, and

$$\xi := \frac{2}{r}\zeta = \frac{2m}{r}\chi.$$

In particular, passing from $\zeta$ to the faithful action $\chi$ changes neither $\xi$ nor the space of reflexive 2-forms of degree $r$. Since $\chi$ is faithful, $(X, \chi)$ is a Fano cone (Lemma 2.10), and hence so are $(X, \zeta)$ and $(X, \xi)$. By Lemma 2.19 applied with $T_0 := T$ and the real number $2/r$, $(X, \xi)$ is K-polystable for $T$.

Since $\omega^n$ is nowhere zero on $X_{\mathrm{reg}}$, $\omega^n$ generates $\Omega_X^{[2n]}$. The form $\omega^n$ is homogeneous of degree $nr$, so $\omega^n$ is an eigenvector of the torus $\zeta(\mathbb{G}_m)$ whose character $\kappa$ satisfies $\langle \kappa, \zeta \rangle = nr$. Since $\xi$ is a Reeb vector of $\zeta(\mathbb{G}_m)$, Lemma 2.14 applied to the torus $\zeta(\mathbb{G}_m)$ implies that

$$A_X(\mathrm{wt}_\xi) = \langle \kappa, \xi \rangle = \frac{2}{r} \cdot nr = 2n.$$

Thus the hypotheses of Theorem 2.27 hold for $(X, \xi)$, $T$, and $\chi$. Let $\rho$ and $\rho_\chi$ be the functions of Theorem 2.27 (see Remark 2.28), let $g$ be the Ricci-flat Kähler cone metric on $M$ of Theorem 2.27(1), and let $N := \{\rho = 1\}$. Then the conditions of Set-up 3.1 hold, and by Corollary 3.11, $X$, $\xi$, and $g$ satisfy conditions (1)–(4) of Set-up 2.31. In particular, $M = (0, +\infty) \times N$ with $N$ connected, so $M$ is connected. Since $R_j$ is the sum of the spaces $R_\alpha$ with $\langle \alpha, \zeta \rangle = j$, and $\langle \alpha, \xi \rangle = 2j/r$ for such $\alpha$, Set-up 2.31(1) implies that

$$\rho\partial_\rho s = \frac{2j}{r} s \quad \text{for any } s \in R_j. \tag{8.14}$$

Let $\sigma'$ be a reflexive 2-form of degree $r$ on $X$, and fix a point $p \in M$. Let $x_1, \ldots, x_{2n} \in R$ be homogeneous elements whose differentials at $p$ form a basis of the cotangent space at $p$. Then $dx_1, \ldots, dx_{2n}$ form a basis of the rational 1-forms on $X$, so $\sigma' = \sum_{k<l} f_{kl} dx_k \wedge dx_l$ with rational functions $f_{kl}$ regular at $p$. Comparing $\zeta(\lambda)^*$ of both sides and using the uniqueness of the coefficients, $f_{kl}$ is homogeneous of degree $r - \deg x_k - \deg x_l$. By (8.14) and the Leibniz rule, $\rho\partial_\rho$ acts on a homogeneous rational function of degree $j$ as multiplication by $2j/r$. Since the Lie derivative commutes with $d$, we obtain $L_{\rho\partial_\rho}\sigma' = 2\sigma'$ on $M$. By Proposition 4.3, $\sigma'$ is parallel with respect to $g$ on $M$. In particular, $\omega$ and $\beta$ are parallel on $M$.

**Step 2.** In this step we find a degenerate member $\gamma$ of the pencil spanned by $\beta$ and $\omega$, and we construct the orthogonal projectors onto $\ker\gamma$ and onto the orthogonal complement of $\ker\gamma$.

Since $\omega^n$ is nowhere zero on $X_{\mathrm{reg}}$, there exist regular functions $f_0, \dots, f_{n-1}$ on $X_{\mathrm{reg}}$ such that

$$(\beta + s\omega)^n = \Big(s^n + \sum_{k=0}^{n-1} f_k s^k\Big)\omega^n \quad \text{for any } s \in \mathbb{C}.$$

Each $f_k$ is homogeneous of degree 0. Since $X$ is normal, $f_k \in R_0 = \mathbb{C}$. Let $s_0 \in \mathbb{C}$ be a root of the polynomial $s^n + \sum_{k=0}^{n-1} f_k s^k$, and let $\gamma := \beta + s_0\omega$. Then $\gamma^n = 0$ on $X_{\mathrm{reg}}$, and $\gamma \neq 0$ since $\beta$ is not a multiple of $\omega$. The form $\gamma$ is a reflexive 2-form of degree $r$, so $\gamma$ is parallel on $M$ by **Step 1**. Parallel transport along paths in the connected manifold $M$ preserves $\gamma$, so the rank of $\gamma$ at $p \in M$ does not depend on $p$. Since $\gamma \neq 0$ and $\gamma^n = 0$, the rank of $\gamma$ is $2n_1$ for an integer $n_1$ with $1 \leq n_1 \leq n-1$. Let $n_2 := n - n_1$, so that $n_2 \geq 1$.

We regard $T_M$ as a complex vector bundle via the complex structure $J$. Let $\mathcal{K} := \ker \gamma \subset T_M$, a complex subbundle of rank $2n_2$, and let $\mathcal{K}^\perp$ be the orthogonal complement of $\mathcal{K}$ with respect to $g$, a complex subbundle of rank $2n_1$ since $g$ is Hermitian. Since $\gamma$, $g$, and $J$ are parallel, $\mathcal{K}$ and $\mathcal{K}^\perp$ are parallel. Let $\varpi_1$ and $\varpi_2$ be the orthogonal projections of $T_M$ onto $\mathcal{K}^\perp$ and $\mathcal{K}$ respectively. Then $\varpi_1$ and $\varpi_2$ are parallel complex-linear projectors, self-adjoint with respect to $g$, of ranks $2n_1$ and $2n_2$, with

$$\varpi_1\varpi_2 = \varpi_2\varpi_1 = 0, \qquad \varpi_1 + \varpi_2 = \mathrm{id}_{T_M}.$$

Since $g$ is Kähler, the Levi-Civita connection on $T_M$ is the Chern connection, and the $(0,1)$-part of the Chern connection is $\bar\partial$. Thus $\varpi_1$ and $\varpi_2$ are holomorphic sections of $\mathcal{H}om(T_M, T_M)$. Moreover, $\gamma|_{\mathcal{K}^\perp}$ is non-degenerate: if $u \in \mathcal{K}^\perp$ satisfies $\gamma(u, u') = 0$ for any $u' \in \mathcal{K}^\perp$, then $\gamma(u, \cdot) = 0$ since $T_M = \mathcal{K} \oplus \mathcal{K}^\perp$ and $\mathcal{K} = \ker\gamma$, so $u \in \mathcal{K} \cap \mathcal{K}^\perp = 0$.

We show that $\varpi_1$ and $\varpi_2$ are invariant under the $\mathbb{G}_m$-action $\zeta$. Since $\zeta(\lambda)^*\gamma = \lambda^r\gamma$ for $\lambda \in \mathbb{G}_m$, $d\zeta(\lambda)$ maps $\mathcal{K}$ onto $\mathcal{K}$. If $|\lambda| = 1$, then $\zeta(\lambda) \in T_c$, so $\zeta(\lambda)$ is an isometry of $g$ by Set-up 2.31(1). By (8.14), the vector field $\rho\partial_\rho$ and the vector field generated by the flow $t \to \zeta(e^{2t/r})$ both act on $R_j$ as multiplication by $2j/r$. Since the differentials of elements of $R$ span the cotangent spaces of $M$, $\rho\partial_\rho$ coincides with the vector field generated by $t \to \zeta(e^{2t/r})$. On $M = (0, +\infty) \times N$, the flow of $\rho\partial_\rho$ at time $t$ is $(\rho, y) \to (e^t\rho, y)$, which multiplies $g$ by $e^{2t}$. Thus $\zeta(\lambda)$ is a homothety of $g$ for $\lambda \in \mathbb{R}_{>0}$. Since $\mathbb{G}_m$ is generated by the unit circle and $\mathbb{R}_{>0}$, $d\zeta(\lambda)$ is conformal for $g$ and maps $\mathcal{K}$ onto $\mathcal{K}$ for any $\lambda \in \mathbb{G}_m$. Hence $d\zeta(\lambda)$ maps $\mathcal{K}^\perp$ onto $\mathcal{K}^\perp$, and $\varpi_1$ and $\varpi_2$ commute with $d\zeta(\lambda)$.

**Step 3.** In this step we show that $\varpi_1$ and $\varpi_2$ are algebraic.

We have $\dim X = 2n \geq 4$, and the grading of $X$ is a good positive grading induced by $\zeta$. By **Step 2** and Lemma 4.2, $\varpi_1$ and $\varpi_2$ are the restrictions to $M$ of unique sections of $\mathcal{H}om(T_X, T_X)$, which we also denote by $\varpi_1$ and $\varpi_2$. Since $\mathcal{H}om(T_X, T_X)$ is torsion-free and $M$ is dense in $X$, the identities $\varpi_1\varpi_2 = \varpi_2\varpi_1 = 0$ and $\varpi_1 + \varpi_2 = \mathrm{id}_{T_X}$ hold on $X$. By **Step 2**, $\varpi_1$ and $\varpi_2$ satisfy condition (3) of Set-up 8.1 with $k = 2$. In particular, $\mathcal{K} = \operatorname{im}\varpi_2$ and $\mathcal{K}^\perp = \operatorname{im}\varpi_1$ are algebraic subbundles of $T_M$.

**Step 4.** In this step we construct parallel reflexive 2-forms $\tau_1$ and $\tau_2$ on $X$ such that $\tau_1$ is non-degenerate on $\mathcal{K}^\perp$ with kernel $\mathcal{K}$, and $\tau_2$ is non-degenerate on $\mathcal{K}$ with kernel $\mathcal{K}^\perp$.

For $z \in \mathbb{C}$, let $\sigma_z := \gamma + z\omega$. Fix a point $p \in M$. In a basis of $T_pM$, the determinant of $\sigma_z(p)$ is a polynomial in $z$ of degree $2n$ whose leading coefficient is the determinant of $\omega(p)$, which is non-zero. In a basis of $\mathcal{K}_p^\perp$, the determinant of $\sigma_z(p)|_{\mathcal{K}_p^\perp}$ is a polynomial in $z$ whose value at $z = 0$ is the determinant of $\gamma(p)|_{\mathcal{K}_p^\perp}$, which is non-zero by **Step 2**. Thus there exists $z \in \mathbb{C}$ such that $\sigma_z(p)$ is non-degenerate on $T_pM$ and on $\mathcal{K}_p^\perp$. Fix such a $z$, and let $\sigma := \sigma_z$. Since $\sigma$ and $\mathcal{K}^\perp$ are parallel and $M$ is connected, $\sigma$ is non-degenerate on $T_M$ and on $\mathcal{K}^\perp$ at any point of $M$.

Since $\sigma|_{\mathcal{K}^\perp}$ is non-degenerate, there exists a unique bundle map $\varphi : \mathcal{K} \to \mathcal{K}^\perp$ such that

$$\sigma(\varphi(v), u) = \sigma(v, u) \quad \text{for any } v \in \mathcal{K} \text{ and } u \in \mathcal{K}^\perp.$$

Since $\sigma$, $\varpi_1$, and $\varpi_2$ are algebraic on $M$ and $\sigma|_{\mathcal{K}^\perp}$ is non-degenerate at any point of $M$, $\varphi$ is algebraic by Cramer's rule. Since $\sigma$, $\mathcal{K}$, and $\mathcal{K}^\perp$ are parallel, the uniqueness of $\varphi$ implies that $\varphi$

is parallel. Let $\psi : \mathcal{K} \to T_M$ be the bundle map with $\psi(v) := v - \varphi(v)$. Then $\psi$ is injective, and

$$\sigma(\psi(v), u) = 0 \quad \text{for any } v \in \mathcal{K} \text{ and } u \in \mathcal{K}^\perp.$$

Since $\sigma$ is non-degenerate, the orthogonal complement of $\mathcal{K}^\perp$ with respect to $\sigma$ has rank $2n - 2n_1 = 2n_2$, so it is equal to $\psi(\mathcal{K})$. Since $\sigma|_{\mathcal{K}^\perp}$ is non-degenerate, $T_M = \mathcal{K}^\perp \oplus \psi(\mathcal{K})$, and $\sigma$ is the sum of $\sigma|_{\mathcal{K}^\perp}$ and $\sigma|_{\psi(\mathcal{K})}$ with respect to the decomposition $T_M = \mathcal{K}^\perp \oplus \psi(\mathcal{K})$. Since $\sigma$ is non-degenerate, $\sigma|_{\psi(\mathcal{K})}$ is non-degenerate.

Define 2-forms $\tau_1$ and $\tau_2$ on $M$ by

$$\tau_1(u, v) := \sigma(\varpi_1 u, \varpi_1 v), \qquad \tau_2(u, v) := \sigma(\psi\varpi_2 u, \psi\varpi_2 v).$$

For $v, v' \in \mathcal{K}$, $\tau_2(v, v') = \sigma(\psi v, \psi v')$, which is the Schur complement of the block $\sigma|_{\mathcal{K}^\perp}$ in the matrix of $\sigma$ with respect to $T_M = \mathcal{K}^\perp \oplus \mathcal{K}$. Since $\sigma|_{\mathcal{K}^\perp}$ and $\sigma|_{\psi(\mathcal{K})}$ are non-degenerate and $\psi$ is injective, $\tau_1$ is non-degenerate on $\mathcal{K}^\perp = \operatorname{im} \varpi_1$ with kernel $\mathcal{K} = \ker \varpi_1$, and $\tau_2$ is non-degenerate on $\mathcal{K} = \operatorname{im} \varpi_2$ with kernel $\mathcal{K}^\perp = \ker \varpi_2$. The forms $\tau_1$ and $\tau_2$ are algebraic and parallel on $M$. Since $\Omega_X^{[2]}$ is reflexive and $X \backslash M$ has codimension at least 2 in $X$, $\tau_1$ and $\tau_2$ extend to reflexive 2-forms on $X$, which we also denote by $\tau_1$ and $\tau_2$. Thus $\tau_1$ and $\tau_2$ satisfy condition (4) of Set-up 8.1.

**Step 5.** In this step we prove (1) and (2), and that $X_1'$ and $X_2'$ are conical symplectic varieties whose symplectic forms have degree $lr$.

Let $a := 2/r$, so that $\xi = a\zeta$. By **Steps 1, 3, and 4**, the conditions of Set-up 8.1 hold for $X$, $T$, $\xi$, $g$, $M$, $N$, $\zeta$, $a$, $k = 2$, $\varpi_1, \varpi_2$, and $\tau_1, \tau_2$. Let $X_1, X_2$ and $q : X \to X_1 \times X_2$ be as in Set-up 8.1. By Theorem 8.2(1) and (2), $X_i$ is a conical symplectic variety of dimension $2n_i$ with klt singularities and symplectic form $\omega_i$, and $q$ is finite, surjective, and quasi-étale. By Theorem 8.2(3), $\omega_1$ and $\omega_2$ have degree $2/a = r$. By Theorem 8.2(4), $q$ is equivariant for the grading of $X$ and the diagonal grading of $X_1 \times X_2$. This proves (1).

The varieties $X$, $X_1$, and $X_2$ are irreducible, normal, and affine, $\dim X_i = 2n_i \geq 2$, and $q$ is finite, surjective, and quasi-étale. By Theorem 8.2(4), $q^*$ is graded for the diagonal grading of $\mathcal{O}(X_1) \otimes_{\mathbb{C}} \mathcal{O}(X_2)$. By Lemma 7.5(1) and (2) applied with $k = 2$, $Z_i := X_i$, and $q$, there exist irreducible normal affine varieties $X_1', X_2'$, finite surjective quasi-étale morphisms $p_i : X_i' \to X_i$, a finite surjective morphism $h : X_1' \times X_2' \to X$ with $q \circ h = p_1 \times p_2$, a closed subset $\Sigma \subset X$ of codimension at least 4, an integer $l \geq 1$, and good positive gradings of $\mathcal{O}(X_1')$ and $\mathcal{O}(X_2')$ such that $h$ is étale over $X \backslash \Sigma$ and

$$p_i^* \mathcal{O}(X_i)_d \subset \mathcal{O}(X_i')_{ld}, \qquad h^* \mathcal{O}(X)_d \subset \mathcal{O}(X_1' \times X_2')_{ld} \quad \text{for } i = 1, 2 \text{ and any } d \geq 0.$$

In particular, $h$ is finite, surjective, and quasi-étale, and $p_i$ and $h$ are equivariant for the gradings of $X_i'$ and the $l$-th powers of the gradings of $X_i$ and $X$. This proves (2). By Lemma 7.5(3), $X_i'$ with the grading of $\mathcal{O}(X_i')$ is a conical symplectic variety whose symplectic form is the reflexive extension of $p_i^* \omega_i$, of degree $lr$.

**Step 6.** In this step we complete the proof of (3).

Fix $i \in \{1, 2\}$. Let $\zeta_i'$ be the $\mathbb{G}_m$-action inducing the grading of $X_i'$, let $o_i'$ be the vertex of $X_i'$, and let $m_i$ be the order of the kernel of $\zeta_i'$. Then $\zeta_i' = m_i \chi_i'$ for a faithful $\mathbb{G}_m$-action $\chi_i'$ with $\chi_i'(\mathbb{G}_m) = \zeta_i'(\mathbb{G}_m)$, and $\chi_i'$ is a primitive cocharacter of the torus $\zeta_i'(\mathbb{G}_m)$. By **Step 5** and Lemma 2.10, $(X_i', \chi_i')$ is a Fano cone, and hence so is $(X_i', \zeta_i')$.

By **Step 5**, $p_i : X_i' \to X_i$ is a finite surjective quasi-étale morphism from a normal affine variety with a good positive grading, equivariant for the grading of $X_i'$ and the $l$-th power of the grading of $X_i$. By Proposition 8.4(3) applied to $h_i := p_i$, the quotient log Fano pair of $(X_i', \chi_i')$ is K-polystable. By Lemma 2.24 applied with $\xi := \zeta_i'$, $c := m_i$, and the primitive cocharacter $\chi_i'$, $(X_i', \zeta_i')$ is K-polystable for the torus $\zeta_i'(\mathbb{G}_m)$.

By Definition 2.16, $(X_i', \zeta_i')$ is K-semistable for $\zeta_i'(\mathbb{G}_m)$. By Theorem 2.21 applied with $T_0 := \zeta_i'(\mathbb{G}_m)$, the degree valuation $\mathrm{wt}_{\zeta_i'}$ of $X_i'$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_i', o_i'}$. Together with **Step 5**, this proves (3). □

8.4. **Product covers étale outside codimension four.** In this subsection, we prove Theorem 8.8. The cover is constructed in at most $n-1$ steps, and in each step one factor of a product of conical symplectic varieties is split by Theorem 8.7. Singular factors of dimension 2 are kept as factors (Remark 8.9).

**Theorem 8.8.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety of dimension $2n$, $n \geq 1$, with vertex $o$ and symplectic form $\omega$ of degree $r$, whose grading is induced by a $\mathbb{G}_m$-action $\zeta$, and assume that $(X, \zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$. Then there exist integers $l \geq 1$ and $b, k \geq 0$, singular conical symplectic varieties $X_1, \dots, X_k$ of dimension at least 2, a finite surjective morphism*

$$h : X' := \mathbb{A}^{2b} \times X_1 \times \cdots \times X_k \to X,$$

*equivariant for a good positive diagonal grading on $X'$ and the $l$-th power of the grading of $X$, and a closed $\zeta(\mathbb{G}_m)$-invariant subset $\Sigma \subset X$ of codimension at least 4 such that $h$ is étale over $X \backslash \Sigma$. Moreover, with $w := lr$, the following hold.*

1. *Each $X_i$ is K-polystable for the torus defined by the image of the $\mathbb{G}_m$-action of the grading, the degree valuation of $X_i$ is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex, and $X_i$ carries a symplectic form $\omega_i$ of degree $w$ with $H^0((X_i)_{\mathrm{reg}}, \Omega^2)_w = \mathbb{C}\omega_i$ and $\mathcal{O}(X_i)_j = 0$ for $0 < j < w$.*
2. *If $b > 0$, then $w$ is even and any linear coordinate of $\mathbb{A}^{2b}$ has degree $w/2$.*
3. *The space of reflexive 2-forms of degree $w$ on $X'$ is*

$$H^0(X'_{\mathrm{reg}}, \Omega^2)_w = \textstyle\bigwedge^2 (\mathbb{C}^{2b})^* \oplus \bigoplus_{i=1}^k \mathbb{C} \operatorname{pr}_i^* \omega_i,$$

   *where $\operatorname{pr}_i : X' \to X_i$ is the projection and $\bigwedge^2 (\mathbb{C}^{2b})^*$ is the space of constant 2-forms on $\mathbb{A}^{2b}$.*
4. *After rescaling the $\omega_i$ and changing the linear coordinates $x_1, y_1, \dots, x_b, y_b$ of $\mathbb{A}^{2b}$, we have*

$$h^* \omega = \sum_{j=1}^b dx_j \wedge dy_j + \sum_{i=1}^k \operatorname{pr}_i^* \omega_i.$$

5. *If $n = 1$, then one can take $h = \mathrm{id}_X$ and $l = 1$.*

*Proof.* The proof is long, so we divide it into several steps.

For a conical symplectic variety $F$, let $\zeta_F$ be the $\mathbb{G}_m$-action inducing the grading of $F$, and let $T_F := \zeta_F(\mathbb{G}_m)$, a torus acting faithfully on $F$. For an integer $c \geq 1$, to multiply the grading of $F$ by $c$ means to assign the degree $cj$ to the elements of $\mathcal{O}(F)_j$ for any $j \geq 0$. The new grading is good and positive, it is induced by the $\mathbb{G}_m$-action $u \to \zeta_F(u^c)$, and the degree of the symplectic form of $F$ is multiplied by $c$.

**Step 1.** In this step we list the properties of the products which we construct by induction.

For an integer $\nu \geq 0$, we construct integers $l_1, \dots, l_\nu \geq 1$, conical symplectic varieties $X_1^{(\nu)}, \dots, X_{\nu+1}^{(\nu)}$, a morphism

$$h_\nu : X^{(\nu)} := X_1^{(\nu)} \times \cdots \times X_{\nu+1}^{(\nu)} \to X,$$

and a closed subset $\Sigma_\nu \subset X$, such that, with $w_\nu := l_1 \cdots l_\nu r$, the following hold.

(i) For any $1 \leq i \leq \nu + 1$, $\dim X_i^{(\nu)} > 0$, the symplectic form $\omega_i^{(\nu)}$ of $X_i^{(\nu)}$ has degree $w_\nu$, and $(X_i^{(\nu)}, \zeta_{X_i^{(\nu)}})$ is K-polystable for the torus $T_{X_i^{(\nu)}}$.
(ii) $h_\nu$ is finite and surjective, and $h_\nu^* R_d \subset \mathcal{O}(X^{(\nu)})_{l_1 \cdots l_\nu d}$ for any $d \geq 0$, where $\mathcal{O}(X^{(\nu)})$ carries the diagonal grading.
(iii) $\operatorname{codim}_X \Sigma_\nu \geq 4$, and $h_\nu$ is étale over $X \backslash \Sigma_\nu$.

For $\nu = 0$, let $X_1^{(0)} := X$, $h_0 := \mathrm{id}_X$, and $\Sigma_0 := \emptyset$. Then (i) is the assumption of the theorem, and (ii) and (iii) hold since $h_0 = \mathrm{id}_X$.

**Step 2.** Let $\nu \geq 1$, and assume that $X^{(\nu-1)}$, $h_{\nu-1}$, $l_1, \dots, l_{\nu-1}$, and $\Sigma_{\nu-1}$ are constructed and satisfy (i), (ii), and (iii), and that there exists a factor $F := X_{i_0}^{(\nu-1)}$ of $X^{(\nu-1)}$ with

$$H^0(F_{\mathrm{reg}}, \Omega^2)_{w_{\nu-1}} \neq \mathbb{C}\omega_{i_0}^{(\nu-1)}.$$

In this step we construct $X^{(\nu)}$, $h_\nu$, $l_\nu$, and $\Sigma_\nu$.

Since $\omega_{i_0}^{(\nu-1)} \in H^0(F_{\mathrm{reg}}, \Omega^2)_{w_{\nu-1}}$, there exists a reflexive 2-form $\beta$ of degree $w_{\nu-1}$ on $F$ which is not a multiple of $\omega_{i_0}^{(\nu-1)}$. By (i), $(F, \zeta_F)$ is K-polystable for $T_F$. By Theorem 8.7(1)(2) for $F$, the form $\omega_{i_0}^{(\nu-1)}$ of degree $w_{\nu-1}$, the $\mathbb{G}_m$-action $\zeta_F$, the torus $T_F$, and $\beta$, there exist conical symplectic varieties $F_1, F_2$ of positive dimension, a finite surjective morphism $q_F : F \to F_1 \times F_2$, an integer $l_\nu \geq 1$, normal affine varieties $F_1', F_2'$ with good positive gradings, finite surjective morphisms $p_i : F_i' \to F_i$ and $g : F_1' \times F_2' \to F$ with $q_F \circ g = p_1 \times p_2$, such that

$$g^*\mathcal{O}(F)_d \subset \mathcal{O}(F_1' \times F_2')_{l_\nu d} \quad \text{for any } d \geq 0,$$

and a closed subset $\Sigma_F \subset F$ with $\operatorname{codim}_F \Sigma_F \geq 4$ such that $g$ is étale over $F \backslash \Sigma_F$. By Theorem 8.7(3), each $F_i'$ is a conical symplectic variety whose symplectic form has degree $l_\nu w_{\nu-1} = w_\nu$, and $(F_i', \zeta_{F_i'})$ is K-polystable for $T_{F_i'}$. Since $p_i$ is finite and surjective, $\dim F_i' = \dim F_i > 0$.

Let $W$ be the product of the factors $X_i^{(\nu-1)}$ with $i \neq i_0$ (a point if $\nu = 1$), so that $X^{(\nu-1)} = F \times W$ after reordering the factors. We let

$$X_1^{(\nu)} := F_1', \quad X_2^{(\nu)} := F_2',$$

and we let $X_3^{(\nu)}, \dots, X_{\nu+1}^{(\nu)}$ be the factors $X_i^{(\nu-1)}$ with $i \neq i_0$, ordered as in $X^{(\nu-1)}$, each with grading multiplied by $l_\nu$. Then $X^{(\nu)} = F_1' \times F_2' \times W$. Let

$$h_\nu := h_{\nu-1} \circ (g \times \mathrm{id}_W) \quad \text{and} \quad \Sigma_\nu := \Sigma_{\nu-1} \cup h_{\nu-1}(\Sigma_F \times W).$$

We check (i). For $X_1^{(\nu)}$ and $X_2^{(\nu)}$, (i) was shown above. Let $F'$ be a factor $X_i^{(\nu-1)}$ with $i \neq i_0$, and let $F''$ be $F'$ with grading multiplied by $l_\nu$. Then $\dim F'' > 0$, and the symplectic form of $F''$ has degree $l_\nu w_{\nu-1} = w_\nu$. Since $\zeta_{F''}(u) = \zeta_{F'}(u^{l_\nu})$ and the map $\mathbb{G}_m \to \mathbb{G}_m$, $u \to u^{l_\nu}$ is surjective, we have $T_{F''} = T_{F'}$, and the Reeb vector of $\zeta_{F''}$ is $l_\nu$ times the Reeb vector of $\zeta_{F'}$. Since $(F', \zeta_{F'})$ is K-polystable for $T_{F'}$, $(F'', \zeta_{F''})$ is K-polystable for $T_{F''}$ by Lemma 2.19. This implies (i).

We check (ii). The morphism $g \times \mathrm{id}_W$ is finite and surjective, so $h_\nu$ is finite and surjective. For the diagonal gradings, $\mathcal{O}(X^{(\nu-1)})_d$ is spanned by products of homogeneous elements of $\mathcal{O}(F)$ and of the $\mathcal{O}(X_i^{(\nu-1)})$, $i \neq i_0$, whose degrees sum to $d$, and $g \times \mathrm{id}_W$ multiplies the degree of any such element by $l_\nu$. Thus

$$(g \times \mathrm{id}_W)^*\mathcal{O}(X^{(\nu-1)})_d \subset \mathcal{O}(X^{(\nu)})_{l_\nu d} \quad \text{for any } d \geq 0.$$

Together with (ii) for $h_{\nu-1}$, we obtain $h_\nu^* R_d \subset \mathcal{O}(X^{(\nu)})_{l_1 \cdots l_\nu d}$ for any $d \geq 0$.

We check (iii). The morphism $g \times \mathrm{id}_W$ is the base change of $g$ by the projection $X^{(\nu-1)} \to F$, so $g \times \mathrm{id}_W$ is étale over $X^{(\nu-1)} \backslash (\Sigma_F \times W)$, and

$$\operatorname{codim}_{X^{(\nu-1)}}(\Sigma_F \times W) = \operatorname{codim}_F \Sigma_F \geq 4.$$

Since $h_{\nu-1}$ is finite, $h_{\nu-1}(\Sigma_F \times W)$ is closed in $X$ and $\dim h_{\nu-1}(\Sigma_F \times W) \leq \dim(\Sigma_F \times W)$. Since $\dim X^{(\nu-1)} = \dim X$, $\Sigma_\nu$ is closed and $\operatorname{codim}_X \Sigma_\nu \geq 4$. Let $x \in X \backslash \Sigma_\nu$ and let $y \in h_{\nu-1}^{-1}(x)$. Since $x \notin \Sigma_{\nu-1}$, $h_{\nu-1}$ is étale at $y$. Since $x \notin h_{\nu-1}(\Sigma_F \times W)$, $y \notin \Sigma_F \times W$, so $g \times \mathrm{id}_W$ is étale at any point of $(g \times \mathrm{id}_W)^{-1}(y)$. Thus $h_\nu$ is étale at any point of $h_\nu^{-1}(x)$, which implies (iii).

**Step 3.** In this step we show that **Step 2** can be performed at most $n-1$ times.

Let $\nu \geq 0$ be an integer such that $X^{(\nu)}$ is constructed. Since $h_\nu$ is finite and surjective, $\dim X^{(\nu)} = \dim X = 2n$. Any factor $X_i^{(\nu)}$ carries the symplectic form $\omega_i^{(\nu)}$, so $\dim X_i^{(\nu)}$ is even, and $\dim X_i^{(\nu)} \geq 2$ by (i). Since $X^{(\nu)}$ has $\nu+1$ factors, $2(\nu+1) \leq 2n$, that is, $\nu \leq n-1$. Therefore

there exists an integer $0 \leq N \leq n-1$ such that $X^{(N)}$, $h_N$, $l_1, \dots, l_N$, and $\Sigma_N$ are constructed and

$$H^0((X_i^{(N)})_{\mathrm{reg}}, \Omega^2)_{w_N} = \mathbb{C}\omega_i^{(N)} \quad \text{for any } 1 \leq i \leq N+1.$$

We let $l := l_1 \cdots l_N$, so that $w = lr = w_N$. In particular, $N = 0$ if $n = 1$.

**Step 4.** In this step we construct $X'$ and $h$, and we prove (1), (2), (3), and (5).

Let $F$ be a factor $X_i^{(N)}$ with vertex $o_F$. Let $m_F$ be the order of the kernel of $\zeta_F$. Then $\zeta_F = m_F\chi_F$ for a faithful $\mathbb{G}_m$-action $\chi_F$ with $\chi_F(\mathbb{G}_m) = T_F$, and for the grading induced by $\chi_F$, $F$ is conical with symplectic form of degree $w/m_F$. By Lemma 2.10, $(F, \chi_F)$ is a Fano cone with torus $T_F$, hence so is $(F, \zeta_F)$, and $\mathrm{wt}_{\zeta_F}$ is the degree valuation of $F$ (Definition 2.6). By (i), $(F, \zeta_F)$ is K-polystable for $T_F$, hence K-semistable for $T_F$ (Definition 2.16). By Theorem 2.21, the degree valuation of $F$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{F,o_F}$. By **Step 3**, $H^0(F_{\mathrm{reg}}, \Omega^2)_w = \mathbb{C}\omega_i^{(N)}$. Thus, by Lemma 8.6, if $F$ is smooth or $\mathcal{O}(F)_j \neq 0$ for some $0 < j < w$, then $\dim F = 2$, $w$ is even, and $\mathcal{O}(F)$ is a polynomial ring in two variables of degree $w/2$.

Let $b$ be the number of smooth factors of $X^{(N)}$, and let $X_1, \dots, X_k$ be the singular factors of $X^{(N)}$, with symplectic forms $\omega_1, \dots, \omega_k$. By the previous paragraph, any smooth factor of $X^{(N)}$ is $\mathbb{A}^2$ with linear coordinates of degree $w/2$, and $\mathcal{O}(X_i)_j = 0$ for any $1 \leq i \leq k$ and any $0 < j < w$. Reordering the factors, we obtain an isomorphism

$$f : X' := \mathbb{A}^{2b} \times X_1 \times \cdots \times X_k \to X^{(N)}$$

which preserves the diagonal gradings, where any linear coordinate of $\mathbb{A}^{2b}$ has degree $w/2$. Let $h := h_N \circ f : X' \to X$. By (ii), $h$ is finite and surjective, and $h$ is equivariant for the diagonal grading on $X'$, which is good and positive, and the $l$-th power of the grading of $X$.

Now (1) follows from (i), the previous paragraph, and **Step 3**, and $\dim X_i \geq 2$ for any $i$ by (i). If $b > 0$, then $w$ is even by the previous paragraph, which implies (2). By (1), (2), and Lemma 8.5 for $w$, $b$, $k$, and $X_1, \dots, X_k$, (3) holds. If $n = 1$, then $N = 0$ by **Step 3**, so $X^{(0)} = X$, $f = \mathrm{id}_X$, $h = h_0 = \mathrm{id}_X$, and $l = 1$, which implies (5).

**Step 5.** In this step we construct $\Sigma$ and we prove (4).

Let $B' \subset X'$ be the set of points at which $h$ is not étale, and let $\Sigma := h(B')$. Since the étale locus of $h$ is open and $h$ is finite, $\Sigma$ is closed in $X$, and $h$ is étale over $X \backslash \Sigma$. Recall that $\Sigma_N \subset X$ is a closed subset with $\mathrm{codim}_X \Sigma_N \geq 4$ such that $h_N$ is étale over $X \backslash \Sigma_N$ (**Step 1**(iii)). Thus $B' \subset h^{-1}(\Sigma_N)$, so $\Sigma \subset \Sigma_N$ and $\mathrm{codim}_X \Sigma \geq 4$. Let $\zeta'$ be the $\mathbb{G}_m$-action inducing the diagonal grading of $X'$. Since $h^*R_d \subset \mathcal{O}(X')_{ld}$ for any $d \geq 0$ and $R$ is generated by homogeneous elements,

$$h \circ \zeta'(u) = \zeta(u^l) \circ h \quad \text{for any } u \in \mathbb{G}_m.$$

Since $\zeta'(u)$ and $\zeta(u^l)$ are automorphisms, $B'$ is $\zeta'(\mathbb{G}_m)$-invariant, so $\zeta(u^l)(\Sigma) = \Sigma$ for any $u \in \mathbb{G}_m$. Since the map $\mathbb{G}_m \to \mathbb{G}_m$, $u \to u^l$ is surjective, $\Sigma$ is $\zeta(\mathbb{G}_m)$-invariant.

Since $h$ is finite, $\dim h^{-1}(\Sigma) \leq \dim \Sigma$, so $\mathrm{codim}_{X'} h^{-1}(\Sigma) \geq 4$. Let $U' := X'_{\mathrm{reg}} \backslash h^{-1}(\Sigma)$ and let $y \in U'$. Since $h$ is étale at $y$, the completions of the local rings of $X'$ at $y$ and of $X$ at $h(y)$ are isomorphic, so $h(y) \in X_{\mathrm{reg}}$. Thus $h^*\omega$ is a regular 2-form on $U'$, and $h^*\omega$ is non-degenerate on $U'$ since $h$ is étale on $U'$. Since $\mathrm{codim}_{X'_{\mathrm{reg}}}(X'_{\mathrm{reg}} \backslash U') \geq 4 \geq 2$ and $X'_{\mathrm{reg}}$ is smooth, $h^*\omega$ extends to a 2-form on $X'_{\mathrm{reg}}$, which we still denote by $h^*\omega$. By the equality $h \circ \zeta'(u) = \zeta(u^l) \circ h$, we have $\zeta'(u)^*h^*\omega = u^{lr}h^*\omega$ for any $u \in \mathbb{G}_m$, so

$$h^*\omega \in H^0(X'_{\mathrm{reg}}, \Omega^2)_w.$$

Let $V \subset X'_{\mathrm{reg}}$ be the zero set of $(h^*\omega)^{\wedge n}$. Since $(h^*\omega)^{\wedge n}$ is a section of the canonical bundle of $X'_{\mathrm{reg}}$, $V$ is either empty or a divisor in $X'_{\mathrm{reg}}$. Since $V \subset X'_{\mathrm{reg}} \backslash U'$, $V = \emptyset$, that is, $h^*\omega$ is non-degenerate on $X'_{\mathrm{reg}}$.

By (3), there exist a constant 2-form $\sigma_0$ on $\mathbb{A}^{2b}$ and $c_1, \dots, c_k \in \mathbb{C}$ such that

$$h^*\omega = \mathrm{pr}_0^*\,\sigma_0 + \sum_{i=1}^{k} c_i\,\mathrm{pr}_i^*\,\omega_i,$$

where $\mathrm{pr}_0 : X' \to \mathbb{A}^{2b}$ is the projection. Let $a \in \mathbb{A}^{2b}$ and $z_i \in (X_i)_{\mathrm{reg}}$ for $1 \le i \le k$. At the point $(a, z_1, \dots, z_k) \in X'_{\mathrm{reg}}$, the tangent space of $X'$ is $\mathbb{C}^{2b} \oplus \bigoplus_{i=1}^k T_{z_i} X_i$, and $h^*\omega$ is the direct sum of $\sigma_0$ and the forms $c_i\omega_i(z_i)$. Since $h^*\omega$ is non-degenerate at $(a, z_1, \dots, z_k)$, $\sigma_0$ is non-degenerate if $b > 0$, and $c_i \neq 0$ for any $i$. We replace $\omega_i$ by $c_i\omega_i$ for any $i$, which does not affect (1) and (3). Since $\sigma_0$ is a non-degenerate alternating form on $\mathbb{C}^{2b}$, there exist linear coordinates $x_1, y_1, \dots, x_b, y_b$ of $\mathbb{A}^{2b}$ with $\sigma_0 = \sum_{j=1}^b dx_j \wedge dy_j$, and they have degree $w/2$, so (2) still holds. This implies (4). □

**Remark 8.9.** In Theorem 8.8, a singular factor $X_i$ of dimension 2 is not replaced by a smooth finite cover. Indeed, the cover $\mathbb{A}^2 \to \{xy = z^2\}$ is branched over the vertex, so in the product the branch locus of the cover has codimension 2, while Theorem 10.4 in Section 10 needs a cover which is étale outside a subset of codimension at least 3.

## 9. Rigidity of test configurations with product central fibers

The goal of this section is to prove Corollary 9.19: a marked test configuration as in Set-up 2.15 whose central fiber is one of the products in Corollary 9.19, and whose general fiber $X$ carries on $X_{\mathrm{reg}}$ a closed non-degenerate 2-form of the degree of the symplectic form of the central fiber, is a product. The difficulty is that the limit on the central fiber of a symplectic form on the general fiber may be degenerate. When any 2-form on the smooth locus of the central fiber whose degree equals the degree of the symplectic form of the central fiber is a multiple of the symplectic form, the limit is non-degenerate (Lemma 9.5), and otherwise we use the product structure of the central fiber and lift the factors one at a time.

### 9.1. Poisson rigidity and descent.

In this subsection, we show that a marked test configuration with conical symplectic central fiber is a product as soon as it carries formal Poisson brackets of the degree of the central bracket, and we descend formal products from finite covers after a base change of the parameter. We start with a formal version of the slice theorem [Fre17, Corollary 1.26], which is used in Lemma 9.4.

**Lemma 9.1.** *Let $\widehat{R}$ be a $\mathbb{C}[[s]]$-algebra which is $s$-adically separated and complete, that is, $\widehat{R} = \varprojlim_N \widehat{R}/s^N\widehat{R}$, and on which $s$ is a nonzerodivisor, and let $\delta$ be a $\mathbb{C}$-derivation of $\widehat{R}$ with $\delta(s) = 1$. Then the following hold.*

*(1) For any $y \in \widehat{R}$ the series $\iota(y) := \sum_{j\ge 0}(-s)^j\delta^j(y)/j!$ converges, $\iota$ is a ring homomorphism with $\iota(s) = 0$ and $\iota(y) - y \in s\widehat{R}$ for any $y$, and $\iota$ induces an isomorphism of $\mathbb{C}[[s]]$-algebras $(\widehat{R}/s\widehat{R})[[s]] \to \widehat{R}$.*

*(2) If $g$ is an automorphism of the $\mathbb{C}$-algebra $\widehat{R}$ with $g(s) = cs$ and $g \circ \delta \circ g^{-1} = c^{-1}\delta$ for some $c \in \mathbb{C}^*$, then $g \circ \iota = \iota \circ g$.*

*Proof.* Since $\delta(s^N y) = Ns^{N-1}y + s^N\delta(y)$ for any $y \in \widehat{R}$ and any $N \ge 1$, we have $\delta(s^N\widehat{R}) \subset s^{N-1}\widehat{R}$, so $\delta$ is continuous for the $s$-adic topology.

(1) For any $y \in \widehat{R}$ and $j \ge 0$, the $j$-th term $(-s)^j\delta^j(y)/j!$ belongs to $s^j\widehat{R}$. Thus the partial sums of the series form a Cauchy sequence for the $s$-adic topology, and the series converges to a unique limit because $\widehat{R}$ is separated and complete. Since $(-1)^j s^{j-1}\delta^j(y)/j! \in s^{j-1}\widehat{R}$ for $j \ge 1$, the series $\sum_{j\ge 1}(-1)^j s^{j-1}\delta^j(y)/j!$ converges to some $w \in \widehat{R}$ by the argument above, and $\iota(y) - y = sw \in s\widehat{R}$.

The map $\iota$ is $\mathbb{C}$-linear and $\iota(1)=1$. For $y,z\in\widehat{R}$, the Leibniz rule implies

$$\delta^j(yz)=\sum_{a+b=j}\binom{j}{a}\delta^a(y)\delta^b(z),$$

hence

$$\iota(y)\iota(z)=\sum_{a,b\geq 0}\frac{(-s)^a\delta^a(y)}{a!}\cdot\frac{(-s)^b\delta^b(z)}{b!}=\sum_{j\geq 0}\frac{(-s)^j}{j!}\sum_{a+b=j}\binom{j}{a}\delta^a(y)\delta^b(z)=\iota(yz).$$

Here the double series may be summed by total index, since the terms with $a+b\geq N$ belong to $s^N\widehat{R}$. Since $\delta(s)=1$, we have $\delta^j(s)=0$ for $j\geq 2$, so $\iota(s)=s-s=0$.

Since $\iota$ is a ring homomorphism with $\iota(s)=0$, it vanishes on $s\widehat{R}$ and induces a ring homomorphism $\kappa:\widehat{R}/s\widehat{R}\to\widehat{R}$. By the previous paragraphs, $\kappa(\bar{y})-y\in s\widehat{R}$ for any $y\in\widehat{R}$ with image $\bar{y}$ in $\widehat{R}/s\widehat{R}$. Let

$$\Phi:(\widehat{R}/s\widehat{R})[[s]]\to\widehat{R},\quad\sum_{k\geq 0}c_ks^k\to\sum_{k\geq 0}\kappa(c_k)s^k,$$

where the right-hand side converges because $\widehat{R}$ is complete. Since $\kappa$ is a ring homomorphism and multiplication on $\widehat{R}$ is continuous, $\Phi$ is a homomorphism of $\mathbb{C}[[s]]$-algebras.

We show that $\Phi$ is surjective. Let $y\in\widehat{R}$ and let $y_0:=y$. Inductively, let $c_k$ be the image of $y_k$ in $\widehat{R}/s\widehat{R}$, and choose $y_{k+1}\in\widehat{R}$ with $y_k-\kappa(c_k)=sy_{k+1}$. Then

$$y-\sum_{k<N}\kappa(c_k)s^k=s^Ny_N\in s^N\widehat{R}$$

for any $N$, hence $y=\Phi(\sum_k c_ks^k)$ because $\widehat{R}$ is separated.

We show that $\Phi$ is injective. Suppose that $\Phi(\sum_k c_ks^k)=0$ and that some $c_k$ is non-zero, and let $k_0$ be the smallest such $k$. Then

$$s^{k_0}\Big(\kappa(c_{k_0})+s\sum_{k>k_0}\kappa(c_k)s^{k-k_0-1}\Big)=0.$$

Since $s$ is a nonzerodivisor on $\widehat{R}$, $\kappa(c_{k_0})\in s\widehat{R}$. Since the image of $\kappa(c_{k_0})$ in $\widehat{R}/s\widehat{R}$ is $c_{k_0}$, we obtain $c_{k_0}=0$, a contradiction. Thus $\Phi$ is an isomorphism, which proves (1).

(2) Since $g(s^N\widehat{R})=c^Ns^N\widehat{R}=s^N\widehat{R}$, $g$ is continuous for the $s$-adic topology. The relation $g\circ\delta\circ g^{-1}=c^{-1}\delta$ implies $\delta\circ g=cg\circ\delta$, hence $\delta^j\circ g=c^jg\circ\delta^j$ for any $j\geq 0$. Therefore, for any $y\in\widehat{R}$,

$$\iota(g(y))=\sum_{j\geq 0}\frac{(-s)^jc^jg(\delta^j(y))}{j!}=\sum_{j\geq 0}\frac{g\big((-s)^j\delta^j(y)\big)}{j!}=g(\iota(y)),$$

where the second equality follows from $g(s^j)=c^js^j$ and the last one from the continuity of $g$. □

The result [XZ26b, Theorem 4.1], first proved in [NO25, Theorems 3.3 and 3.5], treats the degeneration induced by a Kollár component. It assumes a $\mathbb{G}_m$-equivariant relative 2-form which is symplectic on each fiber, and concludes that the singularity and the central fiber of the degeneration are formally isomorphic as symplectic singularities. The proof of Theorem 10.5, through Corollary 9.19 and Lemma 9.4, needs a different statement for a marked test configuration as in Set-up 2.15. There we are only given compatible Poisson brackets of $\zeta$-degree $-r$ modulo $t^{N+1}$, or a closed relative 2-form on the formal completion along $(X_0)_{\mathrm{reg}}$, and neither the brackets nor the form need be homogeneous for $\Lambda$. The conclusion we need is a $T\times\Lambda$-equivariant isomorphism of $\mathbb{C}[t]$-algebras $\mathcal{R}\cong(\mathcal{R}/t\mathcal{R})[t]$ over the original parameter $t$, which implies a product in the sense of Definition 2.16 (Remark 9.3). Theorem 9.2 proves such an isomorphism.

**Theorem 9.2.** *Notation and conditions as in Set-up 2.15, and assume that $X_0$ is a conical symplectic variety for the $\zeta$-grading whose symplectic form $\omega_0$ has degree $r > 0$. Assume that one of the following holds.*

*(i) For any $N \geq 0$ there exists a $\mathbb{C}[t]/(t^{N+1})$-linear Poisson bracket on $\mathcal{R}/t^{N+1}\mathcal{R}$ of $\zeta$-degree $-r$ such that the brackets are compatible with the quotient maps and the bracket for $N = 0$ is the Poisson bracket of $\omega_0$.*

*(ii) The formal completion of the smooth locus of $\mathcal{X} \to \mathbb{A}^1$ along $(X_0)_{\mathrm{reg}}$ carries a closed relative 2-form of $\zeta$-degree $r$ whose restriction to $(X_0)_{\mathrm{reg}}$ is $\omega_0$.*

*Then there exists a $T \times \Lambda$-equivariant isomorphism of $\mathbb{C}[t]$-algebras $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ inducing the identity modulo $t$. In particular, $X \cong X_0$ $T$-equivariantly.*

*Proof of Theorem 9.2.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we show that the $\Lambda$-weights in each $\zeta$-degree of $\mathcal{R}$ are bounded above, and that reduction modulo $t^N$ is bijective on weight spaces of large $\Lambda$-weight.

For a character $\chi$ of $T$ and $q \in \mathbb{Z}$, let $\mathcal{R}_{\chi,q}$ be the space of $a \in \mathcal{R}$ with $(\tau, \lambda) \cdot a = \chi(\tau)\lambda^q a$ for any $(\tau, \lambda) \in T \times \Lambda$, and let $(\mathcal{R}/t^N\mathcal{R})_{\chi,q}$ be the space of $a \in \mathcal{R}/t^N\mathcal{R}$ with $(\tau, \lambda) \cdot a = \chi(\tau)\lambda^q a$ for any $(\tau, \lambda) \in T \times \Lambda$. Since the actions of $T$ and $\Lambda$ are algebraic and commute, $\mathcal{R} = \bigoplus_{\chi,q} \mathcal{R}_{\chi,q}$, and $\mathcal{R}_{\chi,q} \subset \mathcal{R}_j$ for $j = \langle \chi, \zeta \rangle$. By Set-up 2.15(3), $t \in \mathcal{R}_{0,-1}$ and $\mathcal{R}_0 = \mathbb{C}[t]$. Since $\mathcal{R}$ is finitely generated, it is generated by finitely many elements of the spaces $\mathcal{R}_{\chi,q}$, and those of $\zeta$-degree 0 belong to $\mathbb{C}[t]$. Thus there exist $z_1, \dots, z_p$ with $z_i \in \mathcal{R}_{\chi_i,q_i}$ and $\langle \chi_i, \zeta \rangle > 0$ such that $\mathcal{R} = \mathbb{C}[t, z_1, \dots, z_p]$. For any $j \geq 0$, only finitely many monomials in $z_1, \dots, z_p$ have $\zeta$-degree $j$, and they generate $\mathcal{R}_j$ as a $\mathbb{C}[t]$-module. Let $M_j$ be the largest $\Lambda$-weight of the monomials of $\zeta$-degree $j$, with $M_j := -\infty$ if no such monomial exists. Since $t$ has $\Lambda$-weight $-1$, we have

$$\mathcal{R}_{\chi,q} = 0 \quad \text{if } q > M_{\langle \chi,\zeta \rangle}.$$

Let $N \geq 1$. We claim that the reduction map $\mathcal{R}_{\chi,q} \to (\mathcal{R}/t^N\mathcal{R})_{\chi,q}$ is surjective for any $\chi$ and $q$, and injective if $q + N > M_{\langle \chi,\zeta \rangle}$. For surjectivity, any element of $(\mathcal{R}/t^N\mathcal{R})_{\chi,q}$ is the image of some $a \in \mathcal{R}$, and it is also the image of the $(\chi, q)$-component of $a$, because the quotient map is equivariant. For injectivity, let $a \in \mathcal{R}_{\chi,q} \cap t^N\mathcal{R}$ and write $a = t^N b$. Decomposing $b$ into components in the spaces $\mathcal{R}_{\chi',q'}$ and using that $t$ is a nonzerodivisor on the domain $\mathcal{R}$, we obtain $b \in \mathcal{R}_{\chi,q+N}$, and $\mathcal{R}_{\chi,q+N} = 0$ since $q + N > M_{\langle \chi,\zeta \rangle}$. Thus $a = 0$.

**Step 2.** In this step we assume (i), and show that for any $N \geq 0$ there exists an isomorphism of $\mathbb{C}[t]/(t^{N+1})$-algebras

$$\phi_N : \mathcal{R}/t^{N+1}\mathcal{R} \to (\mathcal{R}/t\mathcal{R})[t]/(t^{N+1})$$

inducing the identity modulo $t$. **Step 5** shows that (ii) implies (i).

Let $\{,\}_N$ be the brackets of (i). Since the $\zeta$-grading of $\mathcal{R}/t\mathcal{R}$ is non-negative with degree-zero part $\mathbb{C}$, the action of $\zeta$ contracts $X_0^{\mathrm{an}}$ to the vertex, so $H^1(X_0^{\mathrm{an}}, \mathbb{C}) = 0$. By [Nam06, Corollary (2.5)(2)], the Poisson deformation functor of $(X_0, \{,\}_0)$ is prorepresented by a complete local noetherian $\mathbb{C}$-algebra $\mathcal{A}$ with maximal ideal $\mathfrak{m}_{\mathcal{A}}$. We use the natural $\mathbb{C}^*$-action on the universal formal Poisson deformation constructed in [Nam26a, Theorem (2.1)], induced by the action of $\zeta$ on $X_0$. Since $r > 0$, $\mathbb{C}^*$ acts on $\mathfrak{m}_{\mathcal{A}}/\mathfrak{m}_{\mathcal{A}}^k$ with positive weights for any $k \geq 1$ [Nam06, (5.4)].

Let $\zeta$ act on $\mathcal{R}/t^{N+1}\mathcal{R}$ through $T$, and trivially on $\mathbb{C}[t]/(t^{N+1})$, which is possible since $T$ fixes $t$. Since $\mathcal{R}$ is flat over $\mathbb{C}[t]$ and $\{,\}_N$ has $\zeta$-degree $-r$, $(\mathcal{R}/t^{N+1}\mathcal{R}, \{,\}_N)$ is a $\mathbb{C}^*$-equivariant Poisson deformation of $(X_0, \{,\}_0)$ over $\mathbb{C}[t]/(t^{N+1})$. By [Nam26a, Corollary (2.4)], it is the pullback of the universal formal Poisson deformation along a $\mathbb{C}^*$-equivariant local homomorphism $h_N : \mathcal{A} \to \mathbb{C}[t]/(t^{N+1})$. Since $h_N(\mathfrak{m}_{\mathcal{A}}^{N+1}) \subset (t^{N+1}) = 0$, $h_N$ factors through $\mathcal{A}/\mathfrak{m}_{\mathcal{A}}^{N+1}$. Since $\mathbb{C}^*$ acts on $\mathfrak{m}_{\mathcal{A}}/\mathfrak{m}_{\mathcal{A}}^{N+1}$ with positive weights and trivially on $\mathbb{C}[t]/(t^{N+1})$, we have $h_N(\mathfrak{m}_{\mathcal{A}}) = 0$. Thus $h_N$ factors through the residue field $\mathbb{C}$ of $\mathcal{A}$, and $\mathcal{R}/t^{N+1}\mathcal{R}$ is isomorphic, as a deformation of $X_0$, to the trivial deformation $(\mathcal{R}/t\mathcal{R})[t]/(t^{N+1})$. Forgetting the brackets, we obtain $\phi_N$. We do not need $\phi_N$ to be compatible for different $N$, or to be equivariant.

**Step 3.** In this step we construct a $\mathbb{C}$-derivation $\delta$ of $\mathcal{R}$ such that $\delta(t)=1$ and $\delta(\mathcal{R}_{\chi,q})\subset\mathcal{R}_{\chi,q+1}$ for any $\chi$ and $q$.

Let $N\geq 1$, and let $\bar{\phi}_N:\mathcal{R}/t^N\mathcal{R}\to(\mathcal{R}/t\mathcal{R})[t]/(t^N)$ be the isomorphism induced by $\phi_N$. The derivative $\partial_t$ with respect to $t$ induces a map $(\mathcal{R}/t\mathcal{R})[t]/(t^{N+1})\to(\mathcal{R}/t\mathcal{R})[t]/(t^N)$. Let

$$\delta_N:\mathcal{R}\to\mathcal{R}/t^N\mathcal{R},\quad a\to\bar{\phi}_N^{-1}\big(\partial_t(\phi_N(a \bmod t^{N+1}))\big).$$

Then $\delta_N$ is $\mathbb{C}$-linear, $\delta_N(t)=1$, and $\delta_N(ab)=\bar{a}\delta_N(b)+\bar{b}\delta_N(a)$ for any $a,b\in\mathcal{R}$, where $\bar{a},\bar{b}$ are the images of $a,b$ in $\mathcal{R}/t^N\mathcal{R}$. Let

$$\delta'_N(a):=\sum_{\chi,q}\mathrm{pr}_{\chi,q+1}\left(\delta_N(a_{\chi,q})\right),$$

where $a_{\chi,q}$ is the $(\chi,q)$-component of $a$ and $\mathrm{pr}_{\chi,q+1}$ is the projection of $\mathcal{R}/t^N\mathcal{R}$ onto $(\mathcal{R}/t^N\mathcal{R})_{\chi,q+1}$. Comparing components on both sides of the Leibniz rule for homogeneous $a$ and $b$, we see that $\delta'_N(ab)=\bar{a}\delta'_N(b)+\bar{b}\delta'_N(a)$ for any $a,b\in\mathcal{R}$. Since $t\in\mathcal{R}_{0,-1}$, we have $\delta'_N(t)=\mathrm{pr}_{0,0}(1)=1$. By construction, $\delta'_N(\mathcal{R}_{\chi,q})\subset(\mathcal{R}/t^N\mathcal{R})_{\chi,q+1}$.

Let $z_1,\dots,z_p$ and $M_j$ be as in **Step 1**. The $T\times\Lambda$-equivariant surjection $\pi:\mathbb{C}[t,z_1,\dots,z_p]\to\mathcal{R}$ from the polynomial ring, where $t$ and the variables $z_i$ are given the weights $(0,-1)$ and $(\chi_i,q_i)$, has a kernel $I$ generated by finitely many elements $F_1,\dots,F_k$ with $F_l$ of weight $(\psi_l,q'_l)$, because $I$ is stable under $T\times\Lambda$. Fix $N\geq 1$ with

$$q_i+1+N>M_{\langle\chi_i,\zeta\rangle}\quad\text{and}\quad q'_l+1+N>M_{\langle\psi_l,\zeta\rangle}$$

for any $i$ and $l$. By **Step 1**, there exists a unique $y_i\in\mathcal{R}_{\chi_i,q_i+1}$ whose image in $\mathcal{R}/t^N\mathcal{R}$ is $\delta'_N(z_i)$. Let $\tilde{\delta}:\mathbb{C}[t,z_1,\dots,z_p]\to\mathcal{R}$ be the derivation along $\pi$ given by

$$\tilde{\delta}(P):=\pi\Big(\frac{\partial P}{\partial t}\Big)+\sum_{i=1}^{p}\pi\Big(\frac{\partial P}{\partial z_i}\Big)y_i.$$

Then $\tilde{\delta}$ maps polynomials of weight $(\chi,q)$ into $\mathcal{R}_{\chi,q+1}$, so $\tilde{\delta}(F_l)\in\mathcal{R}_{\psi_l,q'_l+1}$. The reduction of $\tilde{\delta}$ modulo $t^N$ and $\delta'_N\circ\pi$ are derivations along $\mathbb{C}[t,z_1,\dots,z_p]\to\mathcal{R}/t^N\mathcal{R}$ which coincide on $t,z_1,\dots,z_p$, hence they coincide. Thus the image of $\tilde{\delta}(F_l)$ in $\mathcal{R}/t^N\mathcal{R}$ is $\delta'_N(\pi(F_l))=0$, and **Step 1** implies $\tilde{\delta}(F_l)=0$. Since $\tilde{\delta}(PF_l)=\pi(P)\tilde{\delta}(F_l)+\pi(F_l)\tilde{\delta}(P)=0$ for any polynomial $P$, we have $\tilde{\delta}(I)=0$, and $\tilde{\delta}$ descends to a $\mathbb{C}$-derivation $\delta$ of $\mathcal{R}$ with $\delta(t)=1$ and $\delta(\mathcal{R}_{\chi,q})\subset\mathcal{R}_{\chi,q+1}$.

**Step 4.** In this step we conclude the proof under (i).

Let $\delta$ be the derivation of **Step 3**. For $a\in\mathcal{R}_{\chi,q}$ and $k\geq 0$, we have $\delta^k(a)\in\mathcal{R}_{\chi,q+k}$, which is zero if $q+k>M_{\langle\chi,\zeta\rangle}$ by **Step 1**. Since any element of $\mathcal{R}$ is a finite sum of such $a$, $\delta$ is locally nilpotent. Since $\mathcal{R}$ is a domain and $\delta(t)=1$, the slice theorem [Fre17, Corollary 1.26] implies

$$\mathcal{R}=\mathcal{R}^\delta[t],$$

where $\mathcal{R}^\delta:=\operatorname{Ker}\delta$ and $t$ is algebraically independent over $\mathcal{R}^\delta$. In particular, $\mathcal{R}=\mathcal{R}^\delta\oplus t\mathcal{R}$, so the quotient map restricts to an isomorphism $\mathcal{R}^\delta\to\mathcal{R}/t\mathcal{R}$.

For $(\tau,\lambda)\in T\times\Lambda$ and $a\in\mathcal{R}_{\chi,q}$, we have

$$(\tau,\lambda)\cdot\delta\big((\tau,\lambda)^{-1}\cdot a\big)=\chi(\tau)^{-1}\lambda^{-q}\chi(\tau)\lambda^{q+1}\delta(a)=\lambda\delta(a),$$

so $(\tau,\lambda)\circ\delta\circ(\tau,\lambda)^{-1}=\lambda\delta$, and $\mathcal{R}^\delta$ is stable under $T\times\Lambda$. Thus the map

$$\mathcal{R}=\mathcal{R}^\delta[t]\to(\mathcal{R}/t\mathcal{R})[t],\quad\sum_k c_kt^k\to\sum_k\bar{c}_kt^k,$$

where $\bar{c}_k$ is the image of $c_k\in\mathcal{R}^\delta$ in $\mathcal{R}/t\mathcal{R}$, is a $T\times\Lambda$-equivariant isomorphism of $\mathbb{C}[t]$-algebras. It induces the identity modulo $t$, since $a=\sum_k c_kt^k$ has image $\bar{c}_0$ in $\mathcal{R}/t\mathcal{R}$.

Since $T$ fixes $t$, the isomorphism induces a $T$-equivariant isomorphism $\mathcal{R}/(t-1)\mathcal{R}\cong\mathcal{R}/t\mathcal{R}$. Since $t$ is a unit modulo $t-1$, Set-up 2.15(4) implies $\mathcal{R}/(t-1)\mathcal{R}\cong R[t,t^{-1}]/(t-1)=R$ $T$-equivariantly. Thus $X\cong X_0$ $T$-equivariantly.

**Step 5.** In this step we show that (ii) implies (i).

Let $\mathcal{U} \subset \mathcal{X}$ be the smooth locus of $\mathcal{X} \to \mathbb{A}^1$. Since $\mathcal{X}$ is flat over $\mathbb{A}^1$, $\mathcal{U} \cap X_0 = (X_0)_{\mathrm{reg}}$. For $N \geq 0$, let $\mathcal{X}_N := \mathrm{Spec}(\mathcal{R}/t^{N+1}\mathcal{R})$, $S_N := \mathrm{Spec}\,\mathbb{C}[t]/(t^{N+1})$, and $\mathcal{U}_N := \mathcal{U} \cap \mathcal{X}_N$, an open subscheme of $\mathcal{X}_N$ with underlying space $(X_0)_{\mathrm{reg}}$. By (ii), there exist closed relative 2-forms $\sigma_N \in H^0(\mathcal{U}_N, \Omega^2_{\mathcal{U}_N/S_N})$ of $\zeta$-degree $r$ such that $\sigma_N|_{\mathcal{U}_{N-1}} = \sigma_{N-1}$ and $\sigma_0 = \omega_0$.

Since $\mathcal{U}_N \to S_N$ is smooth, $\sigma_N$ defines a homomorphism $T_{\mathcal{U}_N/S_N} \to \Omega^1_{\mathcal{U}_N/S_N}$ of locally free sheaves of rank $\dim X_0$, which is an isomorphism modulo $t$ because $\omega_0$ is non-degenerate on $(X_0)_{\mathrm{reg}}$. Since $t$ is nilpotent on $\mathcal{U}_N$, it is an isomorphism by Nakayama's lemma. Thus for any regular function $f$ on an open subset of $\mathcal{U}_N$ there exists a unique relative vector field $H_f$ with $i_{H_f}\sigma_N = -d_{\mathrm{rel}}f$, and we let $\{f,g\}_N := H_f(g)$. The bracket $\{,\}_N$ is $\mathbb{C}[t]/(t^{N+1})$-bilinear since $d_{\mathrm{rel}}t = 0$, it is antisymmetric since $\{f,g\}_N = \sigma_N(H_f, H_g)$, and it satisfies the Leibniz rule since $H_f$ is a derivation. Expanding $(d_{\mathrm{rel}}\sigma_N)(H_f, H_g, H_h)$ by the formula for the exterior derivative of a 2-form and using $\sigma_N(V, H_u) = V(u)$ for any relative vector field $V$ and $u \in \{f,g,h\}$, we obtain

$$\{f,\{g,h\}_N\}_N + \{g,\{h,f\}_N\}_N + \{h,\{f,g\}_N\}_N = -(d_{\mathrm{rel}}\sigma_N)(H_f, H_g, H_h),$$

so the Jacobi identity holds since $d_{\mathrm{rel}}\sigma_N = 0$. Since $\sigma_N$ has $\zeta$-degree $r$, $\{,\}_N$ has $\zeta$-degree $-r$. Since $\sigma_N|_{\mathcal{U}_{N-1}} = \sigma_{N-1}$, the brackets are compatible with restriction, and $\{,\}_0$ is the Poisson bracket of $\omega_0$ by Definition 2.9.

We show that the restriction $\mathcal{R}/t^{N+1}\mathcal{R} \to H^0(\mathcal{U}_N, \mathcal{O}_{\mathcal{U}_N})$ is an isomorphism. Let $\iota : (X_0)_{\mathrm{reg}} \to X_0$ be the inclusion, and let $Z$ be the closed subset $X_0 \backslash (X_0)_{\mathrm{reg}}$ of $X_0$. Since $X_0$ is normal, $\mathrm{codim}_{X_0} Z \geq 2$ and $\mathcal{O}_{X_0}$ satisfies $S_2$ [Har77, Theorem II.8.22A], so $\mathrm{depth}_Z \mathcal{O}_{X_0} \geq 2$. By [Har67, Theorem 3.8], $\mathcal{H}^i_Z(\mathcal{O}_{X_0}) = 0$ for $i = 0, 1$. Since $t$ is a nonzerodivisor on $\mathcal{R}$, multiplication by $t^N$ induces an isomorphism $\mathcal{O}_{X_0} \to t^N\mathcal{O}_{\mathcal{X}_N}$, and we have an exact sequence

$$0 \to \mathcal{O}_{X_0} \to \mathcal{O}_{\mathcal{X}_N} \to \mathcal{O}_{\mathcal{X}_{N-1}} \to 0$$

of sheaves on the topological space $X_0$. By the long exact sequence of local cohomology and induction on $N$, $\mathcal{H}^i_Z(\mathcal{O}_{\mathcal{X}_N}) = 0$ for $i = 0, 1$. By [Har67, Corollary 1.9], $\mathcal{O}_{\mathcal{X}_N} \to \iota_*\mathcal{O}_{\mathcal{U}_N}$ is an isomorphism, and taking global sections proves the claim.

For $f, g \in \mathcal{R}/t^{N+1}\mathcal{R}$, let $\{f,g\}_N$ be the element of $\mathcal{R}/t^{N+1}\mathcal{R}$ whose restriction to $\mathcal{U}_N$ is $\{f|_{\mathcal{U}_N}, g|_{\mathcal{U}_N}\}_N$. Since restriction to $\mathcal{U}_N$ is injective, the bilinearity, the antisymmetry, the Leibniz rule, the Jacobi identity, the compatibility for different $N$, and the $\zeta$-degree $-r$ hold for the brackets $\{,\}_N$ on $\mathcal{R}/t^{N+1}\mathcal{R}$, and $\{,\}_0$ is the Poisson bracket of $\omega_0$. Thus (i) holds, and **Steps 2–4** conclude the proof. □

**Remark 9.3.** If $(X, \xi)$ is a Fano cone, $T$ acts faithfully on $X$ and $\xi$ is a Reeb vector of $T$, and $(\mathcal{X}, \eta)$, where $\eta$ is the vector field generating $\Lambda$, is a $T$-equivariant test configuration of $(X, \xi)$ in the sense of [LWX18, Definition 2.14], then the conclusion of Theorem 9.2 and Lemma 2.18 imply that $(\mathcal{X}, \eta)$ is a product in the sense of Definition 2.16.

The finite covers of Section 10 need a base change $t = s^m$. The normalized trace takes a derivation $\delta'$ of the completion of $\mathcal{R}'$ with $\delta'(s) = 1$ down to the completion of $\widetilde{\mathcal{R}}$, and the mean over $\mu_m$ of the conjugates of the derivation on the completion of $\widetilde{\mathcal{R}}$ brings the problem back to $\mathcal{R}$.

**Lemma 9.4.** *Notation and conditions as in Set-up 2.15, and assume that $X_0$ is a conical symplectic variety for the $\zeta$-grading whose symplectic form has positive degree. Let $m, e \geq 1$ be integers, let $\widetilde{\mathcal{R}} := \mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$ with $t = s^m$, graded so that $s$ has degree 0 and an element of $\zeta$-degree $j$ has degree $ej$, and let $\widetilde{\mathcal{R}} \subset \mathcal{R}'$ be a finite extension of domains such that $\mathcal{R}'$ carries a non-negative grading extending the grading of $\widetilde{\mathcal{R}}$. Assume that $\widetilde{\mathcal{R}}$ is a normal domain, and that there exists an isomorphism of $\mathbb{C}[[s]]$-algebras $\phi : (\mathcal{R}'/s\mathcal{R}')[[s]] \to \widehat{\mathcal{R}'}$ onto the $s$-adic completion of $\mathcal{R}'$ which induces the identity modulo $s$ and whose reduction modulo $s^N$ is a graded isomorphism $(\mathcal{R}'/s\mathcal{R}')[s]/(s^N) \to \mathcal{R}'/s^N\mathcal{R}'$ for any $N \geq 1$. Then there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ inducing the identity modulo $t$.*

*Proof.* The proof is long, so we divide it into several steps. Let $\widehat{\mathcal{R}}$ be the $t$-adic completion of $\mathcal{R}$, and let $\widehat{\widetilde{\mathcal{R}}}$ be the $s$-adic completion of $\widetilde{\mathcal{R}}$. The rings $\mathcal{R}$, $\widetilde{\mathcal{R}}$, and $\mathcal{R}'$ are Noetherian, since $\mathcal{R}$ is finitely generated over $\mathbb{C}$ (Set-up 2.15(2)) and $\mathcal{R} \subset \widetilde{\mathcal{R}} \subset \mathcal{R}'$ are finite. For each of the pairs $(\mathcal{R}, t)$, $(\widetilde{\mathcal{R}}, s)$, and $(\mathcal{R}', s)$, the kernel of the map from the completion onto the quotient modulo the $N$-th power of the parameter is generated by the $N$-th power of the parameter, and completion is exact on finitely generated modules [Har77, Theorem II.9.3A]. Thus the completion is separated and complete for the adic topology of the parameter, and the parameter, which is a nonzerodivisor on the ring, is a nonzerodivisor on the completion.

For $\lambda \in \mathbb{C}^*$, let $\rho_\lambda$ be the automorphism of $\mathcal{R}'$ which multiplies elements of degree $k$ by $\lambda^k$. It preserves $\widetilde{\mathcal{R}}$ and $\mathcal{R} \subset \widetilde{\mathcal{R}}$, it acts on $\mathcal{R}_j$ by $\lambda^{ej}$, and it fixes $s$. Thus $\rho_\lambda$ extends to continuous automorphisms of the three completions, again denoted by $\rho_\lambda$, and we let $\rho_\lambda$ act on $(\mathcal{R}'/s\mathcal{R}')[[s]]$ coefficientwise. Since the reduction of $\phi$ modulo $s^N$ is graded for any $N$, it commutes with $\rho_\lambda$ for any $N$, hence $\phi \circ \rho_\lambda = \rho_\lambda \circ \phi$ because $\widehat{\mathcal{R}'}$ is separated.

**Step 1.** In this step we construct the normalized trace $\mathrm{tr} : \mathcal{R}' \to \widetilde{\mathcal{R}}$ and extend it to the completions.

Let $d := [\mathrm{Frac}(\mathcal{R}') : \mathrm{Frac}(\widetilde{\mathcal{R}})]$, which is finite since $\mathcal{R}'$ is finite over $\widetilde{\mathcal{R}}$, and let

$$\mathrm{tr} := d^{-1} \operatorname{Tr}_{\mathrm{Frac}(\mathcal{R}')/\mathrm{Frac}(\widetilde{\mathcal{R}})}.$$

Since any element of $\mathcal{R}'$ is integral over the normal domain $\widetilde{\mathcal{R}}$, $\mathrm{tr}(\mathcal{R}') \subset \widetilde{\mathcal{R}}$ [AM69, Proposition 5.15]. Thus $\mathrm{tr} : \mathcal{R}' \to \widetilde{\mathcal{R}}$ is $\widetilde{\mathcal{R}}$-linear and restricts to the identity on $\widetilde{\mathcal{R}}$. In particular, $\mathrm{tr}(s^N\mathcal{R}') \subset s^N\widetilde{\mathcal{R}}$, so tr induces maps $\mathrm{tr}_N : \mathcal{R}'/s^N\mathcal{R}' \to \widetilde{\mathcal{R}}/s^N\widetilde{\mathcal{R}}$ and a continuous $\widehat{\widetilde{\mathcal{R}}}$-linear map

$$\widehat{\mathrm{tr}} : \widehat{\mathcal{R}'} \to \widehat{\widetilde{\mathcal{R}}}$$

which restricts to the identity on the image of $\widehat{\widetilde{\mathcal{R}}}$. Thus the map $\widehat{\widetilde{\mathcal{R}}} \to \widehat{\mathcal{R}'}$ is injective, and we regard $\widehat{\widetilde{\mathcal{R}}}$ as a subring of $\widehat{\mathcal{R}'}$. Since $\rho_\lambda$ induces an automorphism of $\mathrm{Frac}(\mathcal{R}')$ mapping $\mathrm{Frac}(\widetilde{\mathcal{R}})$ onto itself, tr commutes with $\rho_\lambda$, so each $\mathrm{tr}_N$ preserves degrees and $\widehat{\mathrm{tr}}$ commutes with $\rho_\lambda$.

**Step 2.** In this step we construct a derivation $\delta$ of $\widehat{\widetilde{\mathcal{R}}}$ with $\delta(s) = 1$ which commutes with $\rho_\lambda$ for any $\lambda \in \mathbb{C}^*$.

Let $\partial_s$ be the derivative with respect to $s$ on $(\mathcal{R}'/s\mathcal{R}')[[s]]$, and let $\delta' := \phi \circ \partial_s \circ \phi^{-1}$. Then $\delta'$ is a derivation of $\widehat{\mathcal{R}'}$ with $\delta'(s) = 1$, since $\phi(s) = s$. Since $\partial_s$ commutes with $\rho_\lambda$, so does $\delta'$. In other words, $\delta'$ has degree 0: it maps the closure in $\widehat{\mathcal{R}'}$ of any graded piece of $\mathcal{R}'$ into itself. For $x \in \widehat{\widetilde{\mathcal{R}}}$, let $\delta(x) := \widehat{\mathrm{tr}}(\delta'(x))$. For $x, y \in \widehat{\widetilde{\mathcal{R}}}$, the $\widehat{\widetilde{\mathcal{R}}}$-linearity of $\widehat{\mathrm{tr}}$ implies

$$\delta(xy) = \widehat{\mathrm{tr}}\big(x\delta'(y) + y\delta'(x)\big) = x\delta(y) + y\delta(x).$$

Thus $\delta$ is a $\mathbb{C}$-derivation of $\widehat{\widetilde{\mathcal{R}}}$, $\delta(s) = \widehat{\mathrm{tr}}(1) = 1$, and $\delta$ commutes with $\rho_\lambda$ by **Step 1**.

**Step 3.** In this step we modify $\delta$ so that it is compatible with the action of $\mu_m$, and we construct an isomorphism of $\mathbb{C}[[t]]$-algebras $\widehat{\Psi} : (\mathcal{R}/t\mathcal{R})[[t]] \to \widehat{\mathcal{R}}$ inducing the identity modulo $t$ and commuting with $\rho_\lambda$ for any $\lambda \in \mathbb{C}^*$.

Let $\epsilon := \exp(2\pi\sqrt{-1}/m)$, and let $g$ be the automorphism of $\widetilde{\mathcal{R}}$ which fixes $\mathcal{R}$ and maps $s$ to $\epsilon s$. It is well defined since $(\epsilon s)^m = t$, it generates an action of $\mu_m$ on $\widetilde{\mathcal{R}}$, and it commutes with $\rho_\lambda$. Since $g(s^N\widetilde{\mathcal{R}}) = s^N\widetilde{\mathcal{R}}$, $g$ extends to a continuous automorphism of $\widehat{\widetilde{\mathcal{R}}}$. Let

$$\delta_* := \frac{1}{m}\sum_{j=0}^{m-1} \epsilon^j g^j \circ \delta \circ g^{-j}.$$

Each $g^j \circ \delta \circ g^{-j}$ is a derivation commuting with $\rho_\lambda$, and $(g^j \circ \delta \circ g^{-j})(s) = g^j(\delta(\epsilon^{-j}s)) = \epsilon^{-j}$. Thus $\delta_*$ is a derivation commuting with $\rho_\lambda$ and $\delta_*(s) = 1$. Since $g^m = \mathrm{id}_{\widehat{\widetilde{\mathcal{R}}}}$ and $\epsilon^m = 1$, reindexing the sum implies $g \circ \delta_* \circ g^{-1} = \epsilon^{-1}\delta_*$.

Since $\widetilde{\mathcal{R}} \cong \bigoplus_{i=0}^{m-1} s^i\mathcal{R}$ as $\mathcal{R}$-modules and $s^{mN} = t^N$, the $s$-adic and $t$-adic topologies on $\widetilde{\mathcal{R}}$ coincide, and

$$\widehat{\widetilde{\mathcal{R}}} = \bigoplus_{i=0}^{m-1} s^i\widehat{\mathcal{R}},$$

where $g$ acts on $s^i\widehat{\mathcal{R}}$ by $\epsilon^i$. Thus the $\mu_m$-invariant part of $\widehat{\widetilde{\mathcal{R}}}$ is $\widehat{\mathcal{R}}$, we have $\widehat{\mathcal{R}} \cap s\widehat{\widetilde{\mathcal{R}}} = t\widehat{\mathcal{R}}$, and $\widehat{\widetilde{\mathcal{R}}}/s\widehat{\widetilde{\mathcal{R}}} = \mathcal{R}/t\mathcal{R}$, on which $g$ acts trivially.

By Lemma 9.1(1) applied to $\widehat{\widetilde{\mathcal{R}}}$ and $\delta_*$, we obtain a ring homomorphism $\iota : \widehat{\widetilde{\mathcal{R}}} \to \widehat{\widetilde{\mathcal{R}}}$ vanishing on $s\widehat{\widetilde{\mathcal{R}}}$, the induced homomorphism $\kappa : \mathcal{R}/t\mathcal{R} \to \widehat{\widetilde{\mathcal{R}}}$, and an isomorphism

$$\widetilde{\Phi} : (\mathcal{R}/t\mathcal{R})[[s]] \to \widehat{\widetilde{\mathcal{R}}}, \quad \sum_k c_k s^k \to \sum_k \kappa(c_k)s^k,$$

where $\kappa(u) - y \in s\widehat{\widetilde{\mathcal{R}}}$ for any $u \in \mathcal{R}/t\mathcal{R}$ and any lift $y \in \widehat{\widetilde{\mathcal{R}}}$ of $u$. By Lemma 9.1(2) with $c = \epsilon$, $g \circ \iota = \iota \circ g$. For $u \in \mathcal{R}/t\mathcal{R}$ with lift $y \in \widehat{\widetilde{\mathcal{R}}}$, we have $g(y) - y \in s\widehat{\widetilde{\mathcal{R}}}$ since $g$ acts trivially modulo $s$, hence

$$g(\kappa(u)) = g(\iota(y)) = \iota(g(y)) = \iota(y) = \kappa(u).$$

Thus $\kappa(\mathcal{R}/t\mathcal{R}) \subset \widehat{\mathcal{R}}$, and $g(\widetilde{\Phi}(\sum_k c_k s^k)) = \widetilde{\Phi}(\sum_k c_k\epsilon^k s^k)$ by the continuity of $g$. Since $\widetilde{\Phi}$ is an isomorphism, the $\mu_m$-invariant part $\widehat{\mathcal{R}}$ of $\widehat{\widetilde{\mathcal{R}}}$ is the image of $(\mathcal{R}/t\mathcal{R})[[s^m]] = (\mathcal{R}/t\mathcal{R})[[t]]$. Let $\widehat{\Psi}$ be the restriction of $\widetilde{\Phi}$ to $(\mathcal{R}/t\mathcal{R})[[t]]$. Then $\widehat{\Psi}$ is an isomorphism of $\mathbb{C}[[t]]$-algebras onto $\widehat{\mathcal{R}}$. Since $\kappa(u) \in \widehat{\mathcal{R}}$ is congruent modulo $s\widehat{\widetilde{\mathcal{R}}}$ to any lift of $u$ in $\widehat{\mathcal{R}}$, and $\widehat{\mathcal{R}} \cap s\widehat{\widetilde{\mathcal{R}}} = t\widehat{\mathcal{R}}$, $\widehat{\Psi}$ induces the identity modulo $t$. By Lemma 9.1(2) applied to the automorphism $\rho_\lambda$ with $c = 1$, $\iota$ commutes with $\rho_\lambda$, so $\kappa$ and $\widehat{\Psi}$ commute with $\rho_\lambda$.

**Step 4.** In this step we construct, for any $N \geq 0$, a Poisson bracket on $\mathcal{R}/t^{N+1}\mathcal{R}$ which satisfies condition (i) of Theorem 9.2, and conclude the proof.

Let $\omega_0$ be the symplectic form of $X_0$, let $r > 0$ be the degree of $\omega_0$, and let $\{,\}$ be the Poisson bracket of $\omega_0$ on $\mathcal{R}/t\mathcal{R}$. For $N \geq 0$, the reduction of $\widehat{\Psi}$ modulo $t^{N+1}$ is an isomorphism

$$\Psi_N : (\mathcal{R}/t\mathcal{R})[t]/(t^{N+1}) \to \mathcal{R}/t^{N+1}\mathcal{R}$$

of $\mathbb{C}[t]/(t^{N+1})$-algebras inducing the identity modulo $t$, and the maps $\Psi_N$ are compatible with the quotient maps. Since $\Psi_N$ commutes with $\rho_\lambda$ for any $\lambda \in \mathbb{C}^*$, and $\rho_\lambda$ acts on the parts of $\zeta$-degree $j$ of both sides by $\lambda^{ej}$ with $e \geq 1$, $\Psi_N$ preserves the $\zeta$-degree. Let $\{,\}_t$ be the $\mathbb{C}[t]$-bilinear extension of $\{,\}$ to $(\mathcal{R}/t\mathcal{R})[t]/(t^{N+1})$, and let $\{,\}_N$ be the $\mathbb{C}[t]/(t^{N+1})$-linear Poisson bracket on $\mathcal{R}/t^{N+1}\mathcal{R}$ given by $\{f_1, f_2\}_N := \Psi_N(\{\Psi_N^{-1}(f_1), \Psi_N^{-1}(f_2)\}_t)$ for $f_1, f_2 \in \mathcal{R}/t^{N+1}\mathcal{R}$. Then $\{,\}_N$ has $\zeta$-degree $-r$, the brackets are compatible with the quotient maps, and $\{,\}_0$ is the Poisson bracket of $\omega_0$. Thus Theorem 9.2(i) holds, and Theorem 9.2 implies the lemma. □

When $H^0((X_0)_{\mathrm{reg}}, \Omega^2)_r = \mathbb{C}\omega_0$, the limit on the central fiber of a closed non-degenerate 2-form of degree $r$ on $X_{\mathrm{reg}}$, normalized as in Definition-Lemma 4.1(1), is a non-zero multiple of $\omega_0$. Thus it is non-degenerate, and Theorem 9.2 can be used directly.

**Lemma 9.5.** *Notation and conditions as in Set-up 2.15, and assume that $X_0$ is a conical symplectic variety for the $\zeta$-grading whose symplectic form $\omega_0$ has degree $r > 0$, that $H^0((X_0)_{\mathrm{reg}}, \Omega^2)_r = \mathbb{C}\omega_0$, and that $X_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form of $\zeta$-degree $r$. Then there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ inducing the identity modulo $t$.*

*Proof.* Let $\sigma$ be a closed non-degenerate algebraic 2-form of $\zeta$-degree $r$ on $X_{\mathrm{reg}}$, and let $\mathcal{U} \subset \mathcal{X}$ be the smooth locus of $\mathcal{X} \to \mathbb{A}^1$. We check the assumptions of Definition-Lemma 4.1. Since $X_0$ is a symplectic variety, it is integral and normal and $\dim X_0$ is even. Since $\mathcal{X}$ is integral and flat over $\mathbb{A}^1$, and the fiber of $\mathcal{X}$ over $t = 1$ is $X$ by Set-up 2.15(4), we have $\dim X = \dim \mathcal{X} - 1 = \dim X_0$.

Since $\sigma$ is a non-zero closed reflexive 2-form on $X$ of $\zeta$-degree $r$, Definition-Lemma 4.1(1) implies that there exists an integer $b$ such that $\gamma := t^{-b}\sigma$ extends to a regular relative 2-form on $\mathcal{U}$ whose restriction $\gamma_0$ to $(X_0)_{\mathrm{reg}}$ is non-zero. Since $t$ has $\zeta$-degree 0, $\gamma$ has $\zeta$-degree $r$, so $\gamma_0 \in H^0((X_0)_{\mathrm{reg}}, \Omega^2)_r = \mathbb{C}\omega_0$, and $\gamma_0 = c\omega_0$ for some $c \in \mathbb{C}^*$. Since $\sigma$ is closed, $d_{\mathrm{rel}}\gamma = t^{-b} d_{\mathrm{rel}}\sigma = 0$ on the dense open subset $\mathcal{U}\backslash X_0$ of the integral scheme $\mathcal{U}$, hence $d_{\mathrm{rel}}\gamma = 0$ on $\mathcal{U}$.

Thus $c^{-1}\gamma$ is a closed relative 2-form on $\mathcal{U}$ of $\zeta$-degree $r$ whose restriction to $(X_0)_{\mathrm{reg}}$ is $\omega_0$. The restriction of $c^{-1}\gamma$ to the formal completion of $\mathcal{U}$ along $(X_0)_{\mathrm{reg}}$ satisfies Theorem 9.2(ii), and Theorem 9.2 implies the lemma. □

9.2. **Lifting a factor projection.** In this subsection, we prove Lemma 9.6 and Lemma 9.7.

To split a factor off a marked test configuration whose central fiber is a product, we first lift to the family the projection from the central fiber $\mathrm{Spec}(S_1 \otimes_{\mathbb{C}} S_2)$ to its second factor. In the notation of Lemma 9.7, the obstruction to extending such a lift from $\mathcal{R}/t^N\mathcal{R}$ to $\mathcal{R}/t^{N+1}\mathcal{R}$ is a $T$-invariant element of $S_1 \otimes_{\mathbb{C}} T^1(S_2)$ on which $\Lambda$ acts with weight $N$. Hence the obstructions vanish when $\Lambda$ has no positive weights on $(S_1 \otimes_{\mathbb{C}} T^1(S_2))^T$.

The next lemma is used in the proofs of Lemma 9.7, Lemma 9.15, and Theorem 9.18.

**Lemma 9.6.** *Let $T$ be a torus, $\Lambda := \mathbb{G}_m$, and let $M$ be a finitely generated flat $\mathbb{C}[t]$-module with a rational action of $T \times \Lambda$ such that, for any eigenvector $y \in M$ of weight $(\chi, w)$, the element $ty$ is an eigenvector of weight $(\chi, w-1)$. Then the following hold.*

*(1) The weights of $\Lambda$ on $M$ are bounded from above.*

*(2) Let $M'$ be a finitely generated flat $\mathbb{C}[t]$-module with a rational action of $T \times \Lambda$ such that, for any eigenvector $y \in M'$ of weight $(\chi, w)$, the element $ty$ is an eigenvector of weight $(\chi, w-1)$. Then any $T \times \Lambda$-equivariant $\mathbb{C}[t]$-linear map $\Phi : M' \to M$ whose reduction $M'/tM' \to M/tM$ is an isomorphism is an isomorphism.*

*(3) If $e_1, \dots, e_h \in M$ are eigenvectors whose images form a basis of $M/tM$, then $e_1, \dots, e_h$ form a basis of the $\mathbb{C}[t]$-module $M$. In particular, $\bigcap_{N \geq 1} t^N M = 0$.*

*(4) Let $y_N \in M/t^N M$ for $N \geq 1$ be eigenvectors of a common weight $(\chi, w)$ such that $y_{N+1}$ reduces to $y_N$ modulo $t^N$ for any $N$. Then there exists $y \in M$ such that $y$ reduces to $y_N$ modulo $t^N$ for any $N$.*

*Proof.* Here an eigenvector is a simultaneous eigenvector of $T \times \Lambda$, and the weight of an eigenvector is the pair $(\chi, w)$, where $\chi$ is the character of $T$ and $w$ is the weight of $\Lambda$ on it. Since $M$ is finitely generated and torsion free over the principal ideal domain $\mathbb{C}[t]$, $M$ is free [Lan02, Chapter III, Theorem 7.3], of rank $\dim_{\mathbb{C}} M/tM$.

For (1), $M$ is generated by finitely many eigenvectors, say of weights $(\chi'_s, w'_s)$. Then $M$ is spanned over $\mathbb{C}$ by the elements $t^c y$, where $c \geq 0$ and $y$ is a generator, and $\Lambda$ acts on $t^c y$ with weight at most $\max_s w'_s$.

For (2), let $Q := \operatorname{coker} \Phi$. Then $Q/tQ = 0$, and $Q$ is a rational representation whose $\Lambda$-weights are bounded from above by (1). Let $y \in Q$ be an eigenvector of weight $(\chi, w)$. Since $Q = tQ$, we may write $y = ty_1$, and replacing $y_1$ by the component of $y_1$ of weight $(\chi, w+1)$, we may assume that $y_1$ is an eigenvector of weight $(\chi, w+1)$. Repeating this, we obtain $y = t^c y_c$ for any $c \geq 0$, where $y_c$ is an eigenvector of weight $(\chi, w+c)$. For $c \gg 0$, $w + c$ exceeds the bound of (1), so $y_c = 0$ and $y = 0$. Hence $Q = 0$. Thus $\Phi$ is a surjective map between free $\mathbb{C}[t]$-modules $M'$ and $M$ of equal rank $\dim_{\mathbb{C}} M'/tM' = \dim_{\mathbb{C}} M/tM$, so $\Phi$ has a right inverse, the product of the determinants of $\Phi$ and of the right inverse is 1, and $\Phi$ is an isomorphism.

For (3), let $M' := \bigoplus_{i=1}^h \mathbb{C}[t]e'_i$, with the action of $T \times \Lambda$ for which $t^c e'_i$ is an eigenvector whose weight is the weight of $e_i$ with the $\Lambda$-weight lowered by $c$. By (2), the map $\Phi : M' \to M$ with $\Phi(e'_i) = e_i$ is an isomorphism. The last claim holds for any free $\mathbb{C}[t]$-module.

For (4), the submodule $tM$ is $T \times \Lambda$-stable, so the surjection $M \to M/tM$ maps the weight space of weight $(\chi, w)$ in $M$ onto the weight space of weight $(\chi, w)$ in $M/tM$ for any $(\chi, w)$ [Bor91, Chapter III, §8.4, Proposition]. Hence there exist eigenvectors $e_1, \dots, e_h$ as in (3). Let $(\chi_i, w_i)$ be the weight of $e_i$. Since $t^p e_i$ has weight $(\chi_i, w_i - p)$, the eigenvector $y_N$ is of the form

$$y_N = \sum_{i:\ \chi_i = \chi,\ 0 \le w_i - w < N} a_{i,N} t^{w_i - w} e_i \bmod t^N, \quad a_{i,N} \in \mathbb{C}.$$

Since $y_{N+1}$ reduces to $y_N$, by (3) we have $a_{i,N+1} = a_{i,N}$ whenever $w_i - w < N$. Thus $y := \sum_{i:\ \chi_i = \chi,\ w_i \ge w} a_{i,w_i - w + 1} t^{w_i - w} e_i$ reduces to $y_N$ modulo $t^N$ for any $N$.

□

**Lemma 9.7.** *Let $T$ be a torus, $\Lambda := \mathbb{G}_m$, and $\zeta$ a cocharacter of $T$. Let $S_1, S_2$ be finitely generated $\mathbb{C}$-algebras with commuting algebraic actions of $T$ and $\Lambda$ whose $\zeta$-gradings are non-negative with degree-zero parts $\mathbb{C}$. Let $\mathcal{R}$ be a finitely generated flat $\mathbb{C}[t]$-algebra with commuting algebraic actions of $T$ and $\Lambda$ such that $T$ fixes $t$, $\Lambda$ acts on $t$ with weight $-1$, and the $\zeta$-grading of $\mathcal{R}$ is non-negative with degree-zero part $\mathbb{C}[t]$. Fix a $T \times \Lambda$-equivariant graded isomorphism $\mathcal{R}/t\mathcal{R} \cong S_1 \otimes_{\mathbb{C}} S_2$. Write $S_2 = P/I$ with $P$ a polynomial ring in finitely many variables on which $T$ and $\Lambda$ act linearly and compatibly with their actions on $S_2$, and let*

$$T^1(S_2) := \operatorname{coker}\left(\operatorname{Der}_{\mathbb{C}}(P, S_2) \to \operatorname{Hom}_{S_2}(I/I^2, S_2)\right)$$

*with the induced $T \times \Lambda$-action, which does not depend on the presentation. Assume that $\Lambda$ acts on $(S_1 \otimes_{\mathbb{C}} T^1(S_2))^T$ with non-positive weights. Then there exists a $T \times \Lambda$-equivariant graded injective homomorphism of $\mathbb{C}[t]$-algebras $j : S_2[t] \to \mathcal{R}$, where $T$ and $\Lambda$ act on $S_2[t]$ through their actions on $S_2$, $T$ fixes $t$, and $\Lambda$ acts on $t$ with weight $-1$, such that $j$ reduces modulo $t$ to the inclusion of the second factor $S_2 \to S_1 \otimes_{\mathbb{C}} S_2$, and $\mathcal{R}$ is a free $S_2[t]$-module with a basis of homogeneous $T \times \Lambda$-eigenvectors whose reductions modulo $t$ form a basis of $S_1 \otimes 1$.*

*Proof.* The proof is long, so we divide it into several steps. In this proof, an *eigenvector* is a simultaneous eigenvector of $T \times \Lambda$, and the *weight* of an eigenvector $y$ is the pair $(\chi, w)$, where $\chi$ is the character of $T$ on $y$ and $w$ is the weight of $\Lambda$ on $y$. Since $\zeta$ is a cocharacter of $T$, any eigenvector is homogeneous. The algebras $S_1$, $S_2$, and $\mathcal{R}$, and their quotients by $T \times \Lambda$-stable ideals, are rational representations of $T \times \Lambda$. Thus any element of them is a finite sum of eigenvectors, and an equivariant surjection between them maps the weight space of weight $(\chi, w)$ onto the weight space of weight $(\chi, w)$ [Bor91, Chapter III, §8.4, Proposition]. The torus $T \times \Lambda$ acts on spaces of homomorphisms and derivations by $(\gamma \cdot \varphi)(y) := \gamma(\varphi(\gamma^{-1} y))$ for $\gamma \in T \times \Lambda$.

Since $T \times \Lambda$ acts linearly on the span of the variables of $P$, we may write $P = \mathbb{C}[x_1, \dots, x_q]$, where $x_l$ is an eigenvector of weight $(\chi_l, w_l)$ for $1 \le l \le q$ [Bor91, Chapter III, §8.4, Proposition]. Let $\bar{x}_l$ be the image of $x_l$ in $S_2$. For an $S_2$-module $M$ with a compatible action of $T \times \Lambda$, let

$$T^1(S_2; M) := \operatorname{coker}\left(\operatorname{Der}_{\mathbb{C}}(P, M) \to \operatorname{Hom}_{S_2}(I/I^2, M)\right),$$

where the map sends $\theta$ to $\theta|_I$. Thus $T^1(S_2) = T^1(S_2; S_2)$. That $T^1(S_2)$ does not depend on the presentation follows from [Har10, Lemmas 3.2 and 3.3, Proposition 3.10]. Since the variables of the presentations can be taken to be $T \times \Lambda$-eigenvectors, the comparison isomorphisms between two presentations can be taken to be $T \times \Lambda$-equivariant, so the identification respects the action of $T \times \Lambda$. We identify $\mathcal{R}/t\mathcal{R}$ with $S_1 \otimes_{\mathbb{C}} S_2$ by the fixed isomorphism, and we regard $S_1 \otimes_{\mathbb{C}} S_2$ as an $S_2$-module through the second factor.

**Step 1.** In this step, we construct $T \times \Lambda$-equivariant homomorphisms of $\mathbb{C}$-algebras $j_N : S_2 \to \mathcal{R}/t^N\mathcal{R}$ for $N \ge 1$ such that $j_1(f) = 1 \otimes f$ for any $f \in S_2$ and $j_{N+1}$ reduces to $j_N$ modulo $t^N$.

First, we show that there exists a $T \times \Lambda$-equivariant isomorphism

$$T^1(S_2; S_1 \otimes_{\mathbb{C}} S_2) \cong S_1 \otimes_{\mathbb{C}} T^1(S_2).$$

Since $S_2$ is Noetherian, $I/I^2$ is the cokernel of a homomorphism $S_2^{\oplus p_1} \to S_2^{\oplus p_0}$. Hence $\operatorname{Hom}_{S_2}(I/I^2, M)$ is the kernel of the induced map $M^{\oplus p_0} \to M^{\oplus p_1}$. Since $S_1$ is flat over $\mathbb{C}$,

tensoring with $S_1$ preserves kernels, so $\mathrm{Hom}_{S_2}(I/I^2, S_1\otimes_{\mathbb{C}} S_2) = S_1\otimes_{\mathbb{C}}\mathrm{Hom}_{S_2}(I/I^2, S_2)$. Similarly, $\theta \to (\theta(x_l))_l$ is an isomorphism $\mathrm{Der}_{\mathbb{C}}(P, M) \cong M^{\oplus q}$, so $\mathrm{Der}_{\mathbb{C}}(P, S_1 \otimes_{\mathbb{C}} S_2) = S_1 \otimes_{\mathbb{C}} \mathrm{Der}_{\mathbb{C}}(P, S_2)$. Since tensoring with $S_1$ preserves cokernels, the isomorphism follows.

Let $j_1(f) := 1 \otimes f$. Suppose that $j_N$ has been constructed for some $N \geq 1$. The element $j_N(\bar{x}_l)$ is an eigenvector of weight $(\chi_l, w_l)$, so there exists an eigenvector $y_l \in \mathcal{R}/t^{N+1}\mathcal{R}$ of weight $(\chi_l, w_l)$ lifting $j_N(\bar{x}_l)$. Let $\varphi : P \to \mathcal{R}/t^{N+1}\mathcal{R}$ be the homomorphism of $\mathbb{C}$-algebras with $\varphi(x_l) = y_l$ for any $l$. It is $T \times \Lambda$-equivariant. For $F \in I$, the image of $\varphi(F)$ in $\mathcal{R}/t^N\mathcal{R}$ is $j_N(F + I) = 0$. Since $\mathcal{R}$ is flat over $\mathbb{C}[t]$, $t$ is a nonzerodivisor on $\mathcal{R}$, and multiplication by $t^N$ induces an isomorphism

$$\mathcal{R}/t\mathcal{R} \to t^N\mathcal{R}/t^{N+1}\mathcal{R}$$

which is $T$-equivariant and lowers $\Lambda$-weights by $N$. Thus there exists a unique $\epsilon(F) \in S_1 \otimes_{\mathbb{C}} S_2$ with $\varphi(F) = t^N\epsilon(F)$. For $G \in P$ and $F, F' \in I$, we have $\varphi(G) \equiv 1 \otimes (G + I)$ modulo $t$, so $\varphi(GF) = t^N(1 \otimes (G + I))\epsilon(F)$, and $\varphi(FF') \in t^{2N}\mathcal{R}/t^{N+1}\mathcal{R} = 0$. Hence $\epsilon$ is an element of $\mathrm{Hom}_{S_2}(I/I^2, S_1 \otimes_{\mathbb{C}} S_2)$. If $F \in I$ is an eigenvector of weight $(\chi, w)$, then so is $\varphi(F)$, and $\epsilon(F)$ is an eigenvector of weight $(\chi, w+N)$. Thus $\epsilon$ is $T$-invariant and $\Lambda$ acts on $\epsilon$ with weight $N$. By the isomorphism $T^1(S_2; S_1 \otimes_{\mathbb{C}} S_2) \cong S_1 \otimes_{\mathbb{C}} T^1(S_2)$, the class of $\epsilon$ in $T^1(S_2; S_1 \otimes_{\mathbb{C}} S_2)$ is an element of $(S_1 \otimes_{\mathbb{C}} T^1(S_2))^T$ of $\Lambda$-weight $N > 0$, hence it is zero by assumption. Therefore $\epsilon = \theta|_I$ for some $\theta \in \mathrm{Der}_{\mathbb{C}}(P, S_1 \otimes_{\mathbb{C}} S_2)$. The map $\theta \to \theta|_I$ is equivariant, so replacing $\theta(x_l)$ by the component of $\theta(x_l)$ of weight $(\chi_l, w_l + N)$ for any $l$, we may assume that $\theta(x_l)$ is an eigenvector of weight $(\chi_l, w_l + N)$.

Let $\varphi' : P \to \mathcal{R}/t^{N+1}\mathcal{R}$ be the homomorphism of $\mathbb{C}$-algebras with $\varphi'(x_l) = y_l - t^N\theta(x_l)$, where $t^N\theta(x_l)$ is the image of $\theta(x_l) \in S_1 \otimes_{\mathbb{C}} S_2 = \mathcal{R}/t\mathcal{R}$ under multiplication by $t^N$. Then $\varphi'(x_l)$ is an eigenvector of weight $(\chi_l, w_l)$, so $\varphi'$ is equivariant. Since $(t^N)^2 = 0$ in $\mathcal{R}/t^{N+1}\mathcal{R}$, the Taylor expansion of $F \in P$ implies

$$\varphi'(F) = \varphi(F) - \sum_{l=1}^{q} \varphi\Big(\frac{\partial F}{\partial x_l}\Big)t^N\theta(x_l) = \varphi(F) - t^N\theta(F).$$

For $F \in I$, we obtain $\varphi'(F) = t^N\epsilon(F) - t^N\theta(F) = 0$. Thus $\varphi'$ induces an equivariant homomorphism $j_{N+1} : S_2 \to \mathcal{R}/t^{N+1}\mathcal{R}$, which reduces to $j_N$ modulo $t^N$.

**Step 2.** In this step, we construct $j$. For $d \geq 0$, let $\mathcal{R}_d$ be the part of $\zeta$-degree $d$ of $\mathcal{R}$. Since $T$ fixes $t$ and $\Lambda$ commutes with $T$, $\mathcal{R} = \bigoplus_d \mathcal{R}_d$ is a decomposition into $T \times \Lambda$-stable $\mathbb{C}[t]$-submodules. Since $\mathcal{R}$ is generated over $\mathcal{R}_0 = \mathbb{C}[t]$ by finitely many eigenvectors of positive $\zeta$-degree, $\mathcal{R}_d$ is finite over $\mathbb{C}[t]$, and $\mathcal{R}_d$ is flat as a direct summand of $\mathcal{R}$. Thus $\mathcal{R}_d$ satisfies the conditions of Lemma 9.6.

Let $d_l$ be the $\zeta$-degree of $x_l$. By **Step 1**, the elements $j_N(\bar{x}_l) \in \mathcal{R}_{d_l}/t^N\mathcal{R}_{d_l}$ are eigenvectors of weight $(\chi_l, w_l)$ compatible under reduction. By Lemma 9.6(4), there exists $y_l \in \mathcal{R}_{d_l}$ which reduces to $j_N(\bar{x}_l)$ modulo $t^N$ for any $N$. The components of $y_l$ of weights other than $(\chi_l, w_l)$ belong to $\bigcap_N t^N\mathcal{R}_{d_l}$, which is zero by Lemma 9.6(3). Thus $y_l$ is an eigenvector of weight $(\chi_l, w_l)$. Let $\psi : P \to \mathcal{R}$ be the homomorphism of $\mathbb{C}$-algebras with $\psi(x_l) = y_l$. It is equivariant. For $F \in I$, the element $\psi(F)$ reduces to $j_N(F + I) = 0$ modulo $t^N$ for any $N$. Since the homogeneous components of $\psi(F)$ belong to $\bigcap_N t^N\mathcal{R}_d = 0$, we have $\psi(F) = 0$. Hence $\psi$ induces an equivariant graded homomorphism $S_2 \to \mathcal{R}$, and we let $j : S_2[t] \to \mathcal{R}$ be the induced homomorphism of $\mathbb{C}[t]$-algebras. Since $\Lambda$ acts on $t$ with weight $-1$ on both sides, $j$ is $T \times \Lambda$-equivariant, and $j$ reduces modulo $t$ to $j_1$, which is the inclusion of the second factor.

We show that $j$ is injective. Suppose that $0 \neq h \in \mathrm{Ker}\, j$. Write $h = t^c h'$ with $c \geq 0$ and $h' \in S_2[t]$ such that $h'(0) \neq 0$. Since $t$ is a nonzerodivisor on $\mathcal{R}$, we have $j(h') = 0$. Reducing modulo $t$, we obtain $1 \otimes h'(0) = 0$ in $S_1 \otimes_{\mathbb{C}} S_2$, so $h'(0) = 0$, a contradiction.

**Step 3.** In this step, we prove that $\mathcal{R}$ is free over $S_2[t]$. Let $\{u_\alpha\}$ be a basis of $S_1$ consisting of eigenvectors. For any $\alpha$, let $e_\alpha \in \mathcal{R}$ be an eigenvector lifting $u_\alpha \otimes 1$ whose weight is the weight of $u_\alpha \otimes 1$. Let $\mathcal{M} := \bigoplus_\alpha S_2[t]e'_\alpha$ be the free $S_2[t]$-module with the grading and the $T \times \Lambda$-action

for which $e'_\alpha$ has the degree and the weight of $u_\alpha$, and let $\Phi : \mathcal{M} \to \mathcal{R}$ be the $S_2[t]$-linear map with $\Phi(e'_\alpha) = e_\alpha$, where $S_2[t]$ acts on $\mathcal{R}$ through $j$. Then $\Phi$ is $T \times \Lambda$-equivariant and graded.

Fix $d \geq 0$. Since the gradings of $S_1$ and $S_2$ are non-negative with degree-zero parts $\mathbb{C}$ and $S_1, S_2$ are finitely generated, we have $\dim_{\mathbb{C}}(S_1)_e < +\infty$ and $\dim_{\mathbb{C}}(S_2)_e < +\infty$ for any $e \geq 0$. Thus $\mathcal{M}_d$ is a free $\mathbb{C}[t]$-module of finite rank, and $\mathcal{M}_d$ satisfies the conditions of Lemma 9.6. The reduction $\mathcal{M}_d/t\mathcal{M}_d \to (\mathcal{R}/t\mathcal{R})_d$ of $\Phi_d$ sends $fe'_\alpha$ to $u_\alpha \otimes f$ for $f \in S_2$, so it is an isomorphism, as $\{u_\alpha\}$ is a basis of $S_1$. By Lemma 9.6(2), $\Phi_d : \mathcal{M}_d \to \mathcal{R}_d$ is an isomorphism. Taking the direct sum over $d$, $\Phi$ is an isomorphism. Hence $\mathcal{R}$ is a free $S_2[t]$-module with basis $\{e_\alpha\}$, and the reductions $u_\alpha \otimes 1$ of the $e_\alpha$ form a basis of $S_1 \otimes 1$. □

9.3. **Non-degenerate two-forms on degenerations to products.** In this subsection, we prove Lemma 9.10 on the derivations of degree zero which preserve a homogeneous closed non-degenerate 2-form, Lemma 9.11 on the derivations of degree zero of a product of conical symplectic varieties, and Lemma 9.12 on the degeneration of a closed non-degenerate 2-form along a marked test configuration whose central fiber is such a product.

**Set-up 9.8.** Let $k$, $r$, $Y_i = \operatorname{Spec} S_i$, $n_i$, $\omega_i$, $S$, $S_{(j)}$, $Y$, $U$, and $\tau_i$ be as follows.

(1) $k \geq 0$ and $r \geq 1$ are integers.
(2) For $1 \leq i \leq k$, $Y_i = \operatorname{Spec} S_i$ is a conical symplectic variety of dimension $2n_i$ with $n_i \geq 1$ such that $S_i$ is Cohen–Macaulay and the symplectic form $\omega_i$ of $Y_i$ has degree $r$.
(3) $S := S_1 \otimes \cdots \otimes S_k$ with the total grading, where $S := \mathbb{C}$ if $k = 0$, $S_{(j)}$ is the part of $S$ of degree $j$ for $j \geq 0$, $Y := Y_1 \times \cdots \times Y_k$, and $U := (Y_1)_{\mathrm{reg}} \times \cdots \times (Y_k)_{\mathrm{reg}}$.
(4) $\tau_i := \mathrm{pr}_i^* \omega_i$ for $1 \leq i \leq k$.
(5) $H^0(U, \Omega^2_U)_r = \bigoplus_{i=1}^k \mathbb{C}\tau_i$.

Condition (5) of Set-up 9.8 excludes constant 2-forms of degree $r$ on affine-space factors other than the multiples of $\omega_i$, and 2-forms of degree $r$ mixing two factors. It implies that any derivation of degree zero preserves each factor (Lemma 9.11(1)), and that a 2-form on $U$ obtained by dividing a degenerating form by a power of $t$ is a combination of the forms $\tau_i$.

The condition in (2) that $S_i$ is Cohen–Macaulay is not an additional restriction. Indeed, $Y_i$ has rational singularities by Lemma 2.10, so $S_i$ is Cohen–Macaulay [KM98, Theorem 5.10]. We include it in Set-up 9.8 because Claim 9.17 in the proof of Theorem 9.16 uses that $S = S_1 \otimes \cdots \otimes S_k$ is Cohen–Macaulay.

**Set-up 9.9.** Let $k$, $r$, $Y_i = \operatorname{Spec} S_i$, $n_i$, $\omega_i$, $S$, $Y$, $U$, $\tau_i$, $(\mathcal{X}, T, \Lambda, \zeta)$, and $X = \operatorname{Spec} R$ be as follows.

(1) $k$, $r$, $Y_i = \operatorname{Spec} S_i$, $n_i$, $\omega_i$, $S$, $Y$, $U$, and $\tau_i$ are as in Set-up 9.8, and $k \geq 2$.
(2) $(\mathcal{X}, T, \Lambda, \zeta)$ is a marked test configuration of $X = \operatorname{Spec} R$ as in Set-up 2.15 together with a graded isomorphism $X_0 \cong Y$.
(3) $X_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form of $\zeta$-degree $r$.

For $1 \leq i \leq k$, let $\mathrm{eu}_i$ be the Euler derivation of $S_i$, extended by zero to $S$. Let

$$\mathfrak{h}_0 := \{\delta \in \mathrm{Der}_{\mathbb{C}}(S)_0 \mid L_\delta(\textstyle\sum_i \tau_i) = 0\}.$$

**Lemma 9.10.** *Let $A$ be a finitely generated normal $\mathbb{C}$-domain with a grading $A = \bigoplus_{j \geq 0} A_j$ with $A_0 = \mathbb{C}$ and Euler derivation* eu *(Definition 2.6), and let $\sigma$ be a closed non-degenerate algebraic 2-form on $(\operatorname{Spec} A)_{\mathrm{reg}}$ which is homogeneous of degree $r > 0$. For $f \in A_r$ let $H_f$ be the vector field on $(\operatorname{Spec} A)_{\mathrm{reg}}$ with $i_{H_f}\sigma = -df$. Then $H_f$ is a derivation of $A$ of degree 0 with $L_{H_f}\sigma = 0$, and $f \to H_f$ is a linear bijection from $A_r$ onto $\{\delta \in \mathrm{Der}_{\mathbb{C}}(A)_0 \mid L_\delta\sigma = 0\}$, whose inverse is $\delta \to -\frac{1}{r} i_{\mathrm{eu}} i_\delta \sigma$.*

*Proof.* Let $V := (\operatorname{Spec} A)_{\mathrm{reg}}$. Since $A$ is normal, $\operatorname{Spec} A \backslash V$ has codimension at least 2 in $\operatorname{Spec} A$, so $H^0(V, \mathcal{O}_V) = A$ [Har77, Proposition II.6.3A]. Hence any vector field on $V$ is a derivation of $A$,

and any derivation of $A$ defines a vector field on the smooth variety $V$. Since $\sigma$ is non-degenerate on $V$, $H_f$ exists and is unique for $f \in A_r$. As $L_{\mathrm{eu}}\sigma = r\sigma$ and $L_{\mathrm{eu}}df = d(\mathrm{eu}(f)) = rdf$, we have

$$i_{[\mathrm{eu},H_f]}\sigma = L_{\mathrm{eu}}(i_{H_f}\sigma) - i_{H_f}L_{\mathrm{eu}}\sigma = -rdf + rdf = 0,$$

so $[\mathrm{eu}, H_f] = 0$ and $H_f$ has degree 0. Since $\sigma$ is closed, $L_{H_f}\sigma = d(i_{H_f}\sigma) = -ddf = 0$. Moreover,

$$-\frac{1}{r}i_{\mathrm{eu}}i_{H_f}\sigma = \frac{1}{r}i_{\mathrm{eu}}df = \frac{1}{r}\,\mathrm{eu}(f) = f,$$

so $f \to H_f$ is injective.

Conversely, let $\delta \in \mathrm{Der}_{\mathbb{C}}(A)_0$ with $L_\delta\sigma = 0$. Since $\sigma$ is closed, $d(i_\delta\sigma) = L_\delta\sigma = 0$. Since $\delta$ has degree 0, the 1-form $i_\delta\sigma$ is homogeneous of degree $r$, and the Cartan formula implies

$$ri_\delta\sigma = L_{\mathrm{eu}}(i_\delta\sigma) = d(i_{\mathrm{eu}}i_\delta\sigma) + i_{\mathrm{eu}}d(i_\delta\sigma) = d(i_{\mathrm{eu}}i_\delta\sigma).$$

Let $f := -\frac{1}{r}i_{\mathrm{eu}}i_\delta\sigma$. Then $f \in H^0(V, \mathcal{O}_V) = A$ is homogeneous of degree $r$, so $f \in A_r$, and $i_\delta\sigma = -df$. Since $\sigma$ is non-degenerate, $\delta = H_f$. Thus $f \to H_f$ is a linear bijection onto $\{\delta \in \mathrm{Der}_{\mathbb{C}}(A)_0 \mid L_\delta\sigma = 0\}$ with inverse $\delta \to -\frac{1}{r}i_{\mathrm{eu}}i_\delta\sigma$. □

**Lemma 9.11.** *Notation and conditions as in Set-up 9.9. Then the following hold.*

1. *For any derivation $\delta$ of $S$ of degree 0 and any $i$, we have $L_\delta\tau_i \in \mathbb{C}\tau_i$, and $\delta$ preserves each subalgebra $S_i \subset S$.*
2. *The actions of $T$ and $\Lambda$ on $S$ preserve each $S_i$ and each line $\mathbb{C}\tau_i$, and the infinitesimal generator of $\Lambda$ on $S$ equals $\sum_i (l_i/r)\,\mathrm{eu}_i + \delta_H$, where $l_i$ is the weight of $\Lambda$ on $\tau_i$ and $\delta_H \in \mathfrak{h}_0$.*
3. *We have $\mathfrak{h}_0 = \{\delta \in \mathrm{Der}_{\mathbb{C}}(S)_0 \mid L_\delta\tau_i = 0 \text{ for every } i\}$ and $\dim \mathfrak{h}_0 = \dim S_{(r)}$.*

*Proof.* Since $Y_i$ is normal, $S_i$ is a normal domain. By induction on $k$, $S$ is a domain [Har77, Exercise I.3.15(a)(b)], and $S$ is normal [Gro65, Corollaire 6.14.3]. Since $\mathrm{codim}_{Y_i}\mathrm{Sing}(Y_i) \geq 2$ for any $i$, the complement $Y\backslash U = \bigcup_i \mathrm{pr}_i^{-1}(\mathrm{Sing}(Y_i))$ has codimension at least 2 in $Y$, so $H^0(U, \mathcal{O}_U) = S$ by [Har77, Proposition II.6.3A]. Since $U$ is smooth, it follows that $\mathrm{Der}_{\mathbb{C}}(S) = H^0(U, T_U)$. For $1 \leq i \leq k$, let $\mathcal{T}_i := \mathrm{pr}_i^*\, T_{(Y_i)_{\mathrm{reg}}}$. Then $T_U = \bigoplus_i \mathcal{T}_i$, and for any $p \geq 0$ we have the decomposition

$$\Omega_U^p = \bigoplus_{p_1+\cdots+p_k=p} \mathrm{pr}_1^*\,\Omega^{p_1}_{(Y_1)_{\mathrm{reg}}} \otimes \cdots \otimes \mathrm{pr}_k^*\,\Omega^{p_k}_{(Y_k)_{\mathrm{reg}}}$$

into summands of multidegree $(p_1, \ldots, p_k)$.

(1) Let $\delta \in \mathrm{Der}_{\mathbb{C}}(S)_0$ and fix $i$. Since $\delta$ has degree 0, we have $L_\delta\tau_i \in H^0(U, \Omega_U^2)_r$, so by Set-up 9.8(5) there exist $c_{ij} \in \mathbb{C}$ with $L_\delta\tau_i = \sum_j c_{ij}\tau_j$. Since $\dim(Y_i)_{\mathrm{reg}} = 2n_i$, we have $\tau_i^{n_i+1} = 0$, hence

$$0 = L_\delta(\tau_i^{n_i+1}) = (n_i+1)\tau_i^{n_i} \wedge L_\delta\tau_i = (n_i+1)\sum_{j\neq i} c_{ij}\tau_i^{n_i} \wedge \tau_j.$$

For $j \neq i$, the form $\tau_i^{n_i} \wedge \tau_j$ is non-zero, because $\omega_i^{n_i}$ vanishes nowhere on $(Y_i)_{\mathrm{reg}}$ and $\omega_j \neq 0$, and $\tau_i^{n_i} \wedge \tau_j$ belongs to the summand of multidegree $2n_i$ in the $i$-th place, 2 in the $j$-th place, and 0 elsewhere. The multidegree of $\tau_i^{n_i} \wedge \tau_j$ determines $j$, so $c_{ij} = 0$ for $j \neq i$. We let $c_i(\delta) := c_{ii}$, so that

$$L_\delta\tau_i = c_i(\delta)\tau_i.$$

Next, the kernel of $\partial \to i_\partial\tau_i$ on $T_U$ is $\bigoplus_{j\neq i}\mathcal{T}_j$. For a local section $\partial$ of $\bigoplus_{j\neq i}\mathcal{T}_j$, we have

$$i_{[\delta,\partial]}\tau_i = L_\delta(i_\partial\tau_i) - i_\partial L_\delta\tau_i = -c_i(\delta)i_\partial\tau_i = 0.$$

Thus $[\delta, \bigoplus_{j\neq i}\mathcal{T}_j] \subset \bigoplus_{j\neq i}\mathcal{T}_j$ for any $i$. Since $\bigcap_{i\neq j}\bigoplus_{l\neq i}\mathcal{T}_l = \mathcal{T}_j$, we obtain $[\delta, \mathcal{T}_j] \subset \mathcal{T}_j$ for any $j$. Let $f \in S_i$. For $j \neq i$ and a local section $\partial$ of $\mathcal{T}_j$, we have $\partial f = 0$ and $[\delta, \partial]f = 0$, hence $\partial(\delta f) = \delta(\partial f) - [\delta, \partial]f = 0$.

It remains to show that an element $g \in S$ with $\partial g = 0$ for any local section $\partial$ of $\bigoplus_{j\neq i}\mathcal{T}_j$ belongs to $S_i$. Let $S_i' := \bigotimes_{j\neq i} S_j$ and $U_i' := \prod_{j\neq i}(Y_j)_{\mathrm{reg}}$. Fix a $\mathbb{C}$-basis $\{s_p\}$ of $S_i$ and write $g = \sum_p s_p \otimes g_p$ with $g_p \in S_i'$. For a vector field $\partial'$ on an affine open subset $V' \subset U_i'$, we have

$0 = \partial' g = \sum_p s_p \otimes \partial' g_p$ on $(Y_i)_{\mathrm{reg}} \times V'$. Since $S_i \otimes_{\mathbb{C}} \mathcal{O}(V')$ is free over $\mathcal{O}(V')$ with basis $\{s_p \otimes 1\}$ and embeds into the regular functions on the dense open subset $(Y_i)_{\mathrm{reg}} \times V'$ of the integral scheme $Y_i \times V'$, we have $\partial' g_p = 0$ for any $p$. Hence $dg_p = 0$ on the smooth connected variety $U_i'$, and $g_p \in \mathbb{C}$. Thus $g \in S_i$, and $\delta(S_i) \subset S_i$.

(2) Let $V$ be a rational representation of $T \times \Lambda$ with weight decomposition $V = \bigoplus_\chi V^\chi$ over the characters $\chi$ of $T \times \Lambda$. An element $\xi \in \mathrm{Lie}(T \times \Lambda)$ acts on $V^\chi$ by the scalar $d\chi(\xi)$, and distinct characters have distinct differentials. Hence a subspace $W \subset V$ stable under the infinitesimal action of $\mathrm{Lie}(T \times \Lambda)$ satisfies $W = \bigoplus_\chi (W \cap V^\chi)$, so $W$ is $T \times \Lambda$-stable.

For $\xi \in \mathrm{Lie}(T \times \Lambda)$, let $\delta_\xi$ be the derivation of $S$ given by the infinitesimal action of $\xi$. Since $T \times \Lambda$ commutes with $\zeta(\mathbb{G}_m) \subset T$, $\delta_\xi$ has degree 0, so $\delta_\xi(S_i) \subset S_i$ by (1). Taking $V = S$ and $W = S_i$ above, $T$ and $\Lambda$ preserve $S_i$. Thus $T \times \Lambda$ acts on $Y_i$, preserves $U$, and acts on $H^0(U, \Omega_U^2)_r = \bigoplus_i \mathbb{C}\tau_i$ with infinitesimal action $\xi \to L_{\delta_\xi}$. By (1), $L_{\delta_\xi}$ preserves $\mathbb{C}\tau_i$. Taking $V = H^0(U, \Omega_U^2)_r$ and $W = \mathbb{C}\tau_i$ above, $T \times \Lambda$ preserves $\mathbb{C}\tau_i$.

Let $\delta_\Lambda$ be the infinitesimal generator of $\Lambda$ on $S$, so that $L_{\delta_\Lambda}\tau_i = l_i\tau_i$. Since $\omega_i$ has degree $r$, we have $L_{\mathrm{eu}_i}\tau_i = r\tau_i$. For $j \neq i$, $\mathrm{eu}_j$ is a section of $\mathcal{T}_j$, so $\mathrm{eu}_j$ annihilates the pullbacks of regular functions on open subsets of $Y_i$, and $L_{\mathrm{eu}_j}\tau_i = 0$ because $\tau_i$ is locally a sum of pullbacks of forms $h dh' \wedge dh''$ with $h, h', h''$ regular functions on open subsets of $(Y_i)_{\mathrm{reg}}$. Let

$$\delta_H := \delta_\Lambda - \sum_i \frac{l_i}{r} \mathrm{eu}_i \,.$$

Then $\delta_H$ has degree 0 and $L_{\delta_H}\tau_j = l_j\tau_j - l_j\tau_j = 0$ for any $j$. Thus $\delta_H \in \mathfrak{h}_0$.

(3) For $\delta \in \mathrm{Der}_{\mathbb{C}}(S)_0$, (1) implies $L_\delta(\sum_i \tau_i) = \sum_i c_i(\delta)\tau_i$, and the $\tau_i$ are linearly independent by Set-up 9.8(5). Hence $\delta \in \mathfrak{h}_0$ if and only if $c_i(\delta) = 0$ for any $i$, which proves the first claim.

Since a point $(y_1, \dots, y_k) \in Y$ is a smooth point of $Y$ if and only if $Y_i$ is smooth at $y_i$ for any $i$, we have $U = Y_{\mathrm{reg}}$. By Lemma 9.10, applied to $A = S$ and $\sigma = \sum_i \tau_i$ on $U = Y_{\mathrm{reg}}$, we have $\dim \mathfrak{h}_0 = \dim S_{(r)}$. □

The limit on the central fiber of a closed non-degenerate 2-form on $X_{\mathrm{reg}}$ of $\zeta$-degree $r$, normalized as in Definition-Lemma 4.1(1), may be degenerate, since the factors $Y_i$ can occur at different orders in $t$. Lemma 9.12 determines the order in $t$ at which each factor $Y_i$ occurs, and in Section 9.4 we lift the factors one at a time.

**Lemma 9.12.** *Notation and conditions as in Set-up 9.9. Let $\sigma$ be a closed non-degenerate algebraic 2-form on $X_{\mathrm{reg}}$ of $\zeta$-degree $r$, let $b := b(\sigma)$ be the integer of Definition-Lemma 4.1, and let $\beta := t^{-b}\sigma$, a relative 2-form on the smooth locus $\mathcal{U}$ of $\mathcal{X} \to \mathbb{A}^1$. Let $I_\beta$ denote the morphism from relative tangent vectors to relative 1-forms given by contraction with $\beta$. Then there exist integers $a_1, \dots, a_k \geq 0$ and $v_1, \dots, v_k \in \mathbb{C}^*$ such that, with $m := \max_i a_i$, the following hold.*

*(1) The weight $l_i$ of $\Lambda$ on $\tau_i$ equals $b + a_i$ for any $i$.*

*(2) The morphism $I_\beta$ is an isomorphism over $\mathcal{U} \backslash X_0$. On the formal completion of $\mathcal{U}$ along $U$, the morphism $I_\beta$ becomes an isomorphism after $t$ is made invertible, $t^m I_\beta^{-1}$ extends to a regular morphism, and the restriction to $U$ of the extended morphism equals $\sum_{i:a_i=m} v_i^{-1} I_{\tau_i}^{-1}$, where the $i$-th summand is defined on the summand $\mathrm{pr}_i^* \Omega^1_{(Y_i)_{\mathrm{reg}}}$ of $\Omega_U^1$ and is zero on the other summands.*

*(3) The reduction modulo $t$ of any $\mathbb{C}[t]_{(t)}$-linear derivation $\delta$ of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$ of $\zeta$-degree 0 with $L_\delta \beta = 0$ on $\mathcal{U} \times_{\mathbb{A}^1} \mathrm{Spec}\, \mathbb{C}[t]_{(t)}$ belongs to $\mathfrak{h}_0$.*

*Proof.* The proof is long, so we divide it into several steps. Let $\mathcal{T}_i := \mathrm{pr}_i^* T_{(Y_i)_{\mathrm{reg}}}$ for $1 \leq i \leq k$, so that $T_U = \bigoplus_i \mathcal{T}_i$.

**Step 1.** In this step, we check the conditions of Definition-Lemma 4.1 and introduce the pairings $\gamma_a$. By induction on $k$, the central fiber $X_0 \cong Y$ is integral [Har77, Exercise I.3.15(a)(b)], and it is normal [Gro65, Corollaire 6.14.3], and $\dim X = \dim X_0 = \sum_i 2n_i$ since $\mathcal{X} \to \mathbb{A}^1$ is flat. Thus the conditions of Definition-Lemma 4.1 hold, $b = b(\sigma)$ is defined, and $\beta$ extends to a

regular relative 2-form on $\mathcal{U}$ whose restriction to $(X_0)_{\mathrm{reg}}$ is non-zero. Since $\mathcal{X} \to \mathbb{A}^1$ is flat, $\mathcal{U} \cap X_0 = (X_0)_{\mathrm{reg}}$ [Gro67, Théorème 17.5.1], and $(X_0)_{\mathrm{reg}} = U$ under $X_0 \cong Y$, since a point $(y_1, \dots, y_k) \in Y$ is a smooth point of $Y$ if and only if $Y_i$ is smooth at $y_i$ for any $i$. Through the marking, $\mathcal{U} \backslash X_0 = X_{\mathrm{reg}} \times (\mathbb{A}^1 \backslash \{0\})$, and $\Lambda$ acts trivially on $R$. Hence $\sigma$ is $\Lambda$-invariant, and $\Lambda$ acts on $\beta$ with weight $b$. By Lemma 9.11(2), $T$ and $\Lambda$ preserve $U$ and each $\mathcal{T}_i$.

Let $\widehat{\mathcal{U}}$ be the formal completion of $\mathcal{U}$ along $U = \mathcal{U} \cap X_0$, with structure sheaf $\widehat{\mathcal{O}}$, and let $\mathcal{E}$ be the completion of the relative tangent sheaf $T_{\mathcal{U}/\mathbb{A}^1}$. Since $\mathcal{U}$ is smooth and flat over $\mathbb{A}^1$, $\mathcal{E}$ is locally free, $\mathcal{E}/t\mathcal{E} = T_U$, and $\widehat{\mathcal{O}}$ has no $t$-torsion. The form $\beta$ is an alternating pairing $\mathcal{E} \times \mathcal{E} \to \widehat{\mathcal{O}}$. For an integer $a \geq 0$, let

$$\mathcal{E}^a := \{\partial \in \mathcal{E} \mid \beta(\partial, \mathcal{E}) \subset t^a \widehat{\mathcal{O}}\},$$

and let $W_a \subset T_U$ be the image of $\mathcal{E}^a$. For local sections $\bar{\partial}, \bar{\partial}'$ of $W_a$ with lifts $\partial, \partial'$ in $\mathcal{E}^a$, let

$$\gamma_a(\bar{\partial}, \bar{\partial}') := t^{-a}\beta(\partial, \partial') \bmod t.$$

Two lifts of $\bar{\partial}$ in $\mathcal{E}^a$ differ by $t\partial''$ with $\partial'' \in \mathcal{E}$, and $\beta(t\partial'', \partial') = -t\beta(\partial', \partial'') \in t^{a+1}\widehat{\mathcal{O}}$. Thus $\gamma_a$ is a well-defined alternating $\mathcal{O}_U$-bilinear pairing on $W_a$. Since $t^a\widehat{\mathcal{O}}$ is $T \times \Lambda$-stable and $\beta$ is an eigenvector of $\zeta(\mathbb{G}_m)$ (of $\zeta$-degree $r$) and of $\Lambda$ (of weight $b$), $\mathcal{E}^a$ and $W_a$ are $\zeta(\mathbb{G}_m) \times \Lambda$-stable, $\gamma_a$ has $\zeta$-degree $r$, and $\Lambda$ acts on $\gamma_a$ with weight $b + a$.

**Step 2.** In this step, we show by induction on $a$ that $W_a$ is a sum of some of the summands $\mathcal{T}_i$, and we construct the integers $a_i$ and the scalars $v_i$. Let $\mathcal{V}$ be the set of affine open subsets $V \subset \mathcal{U}$ such that $V \cap X_0$ is a non-empty subset of $U$, $T_{V/\mathbb{A}^1}$ is free, and $\mathcal{T}_1, \dots, \mathcal{T}_k$ are free on $V \cap X_0$. The sets $V \cap X_0$ with $V \in \mathcal{V}$ cover $U$. For $V \in \mathcal{V}$, let $\widehat{A}_V := \widehat{\mathcal{O}}(V \cap X_0)$, the $t$-adic completion of $\mathcal{O}(V)$, and let $\mathcal{E}(V) := \mathcal{E}(V \cap X_0)$, a free $\widehat{A}_V$-module. We prove the following claim for any $a \geq 0$ by induction on $a$.

**Claim 9.13.** *There exists a subset $J_a \subset \{1, \dots, k\}$ with $W_a = \bigoplus_{i \in J_a} \mathcal{T}_i$. For $i \notin J_a$, the integer $a_i$ with $0 \leq a_i < a$ and the scalar $v_i \in \mathbb{C}^*$ have been defined in the induction below. For any $V \in \mathcal{V}$, there exists a $\beta$-orthogonal decomposition into free $\widehat{A}_V$-modules*

$$\mathcal{E}(V) = \Big( \bigoplus_{0 \leq d < a} \mathcal{E}_d \Big) \oplus \mathcal{E}'_a$$

*such that $\beta|_{\mathcal{E}_d} = t^d\beta_d$ for a non-degenerate alternating pairing $\beta_d$ on $\mathcal{E}_d$, $\beta|_{\mathcal{E}'_a} = t^a\beta'_a$ for an alternating pairing $\beta'_a$ on $\mathcal{E}'_a$, $\mathcal{E}_d/t\mathcal{E}_d = \bigoplus_{i:a_i=d} \mathcal{T}_i(V \cap X_0)$ with $\beta_d \bmod t = \sum_{i:a_i=d} v_i\tau_i$, and $\mathcal{E}'_a/t\mathcal{E}'_a = \bigoplus_{i \in J_a} \mathcal{T}_i(V \cap X_0)$.*

For $a = 0$, Claim 9.13 holds with $J_0 = \{1, \dots, k\}$, $\mathcal{E}'_0 = \mathcal{E}(V)$, and $\beta'_0 = \beta$.

Suppose that Claim 9.13 holds for $a$, and fix $V \in \mathcal{V}$. Let $\partial = \sum_d \partial_d + \partial'$ with $\partial_d \in \mathcal{E}_d$ and $\partial' \in \mathcal{E}'_a$. Since $\beta_d$ is non-degenerate and $\widehat{A}_V$ has no $t$-torsion, $\beta(\partial, \mathcal{E}_d) = t^d\beta_d(\partial_d, \mathcal{E}_d) \subset t^a\widehat{A}_V$ if and only if $\partial_d \in t^{a-d}\mathcal{E}_d$, and $\beta(\partial, \mathcal{E}'_a) \subset t^a\widehat{A}_V$ always holds. Thus

$$\mathcal{E}^a(V \cap X_0) = \Big( \bigoplus_{0 \leq d < a} t^{a-d}\mathcal{E}_d \Big) \oplus \mathcal{E}'_a.$$

For $\partial = \sum_d t^{a-d}\partial_d + \partial'$ and $\partial^\bullet = \sum_d t^{a-d}\partial_d^\bullet + \partial'^\bullet$ in $\mathcal{E}^a(V \cap X_0)$, we have

$$t^{-a}\beta(\partial, \partial^\bullet) = \sum_{0 \leq d < a} t^{a-d}\beta_d(\partial_d, \partial_d^\bullet) + \beta'_a(\partial', \partial'^\bullet),$$

so $\gamma_a$ is the reduction of $\beta'_a$ modulo $t$ on $W_a = \bigoplus_{i \in J_a} \mathcal{T}_i$.

Let $\tilde{\gamma}_a$ be the 2-form on $U$ which equals $\gamma_a$ on $W_a$ and vanishes on $\bigoplus_{i \notin J_a} \mathcal{T}_i$, that is, $\tilde{\gamma}_a(\partial, \partial^\bullet) := \gamma_a(p(\partial), p(\partial^\bullet))$ for the projection $p : T_U \to W_a$. By **Step 1**, $\tilde{\gamma}_a \in H^0(U, \Omega^2_U)_r$, so $\tilde{\gamma}_a \in \bigoplus_i \mathbb{C}\tau_i$ by Set-up 9.8(5). Since $\tau_j|_{\mathcal{T}_i} = 0$ for $j \neq i$ and $\tau_i|_{\mathcal{T}_i} \neq 0$, and $\tilde{\gamma}_a|_{\mathcal{T}_i} = 0$ for $i \notin J_a$, we may write

$$\tilde{\gamma}_a = \sum_{i \in J_a} c_i\tau_i, \quad c_i \in \mathbb{C}.$$

For $i \in J_a$ with $c_i \neq 0$, let $a_i := a$ and $v_i := c_i$, and let $J_{a+1} := \{i \in J_a \mid c_i = 0\}$. Since $\tau_i$ is non-degenerate on $\mathcal{T}_i$, the pairing $\gamma_a$ is non-degenerate on $\mathcal{N} := \bigoplus_{i:a_i=a} \mathcal{T}_i$, and the radical of $\gamma_a$ is $\bigoplus_{i\in J_{a+1}} \mathcal{T}_i$.

Choose free submodules $\mathcal{F}, \mathcal{F}' \subset \mathcal{E}'_a$ with $\mathcal{E}'_a = \mathcal{F} \oplus \mathcal{F}'$ such that $\mathcal{F}/t\mathcal{F} = \mathcal{N}(V \cap X_0)$ and $\mathcal{F}'/t\mathcal{F}' = \bigoplus_{i\in J_{a+1}} \mathcal{T}_i(V \cap X_0)$, by lifting bases. The Gram matrix of $\beta'_a|_{\mathcal{F}}$ reduces modulo $t$ to the Gram matrix of $\gamma_a|_{\mathcal{N}}$, which is invertible, so $\beta'_a|_{\mathcal{F}}$ is non-degenerate because $t$ belongs to the Jacobson radical of $\widehat{A}_V$. Let $\phi : \mathcal{F}' \to \mathcal{F}$ be the $\widehat{A}_V$-linear map with $\beta'_a(\phi(\partial'), \partial) = \beta'_a(\partial', \partial)$ for any $\partial \in \mathcal{F}$ and $\partial' \in \mathcal{F}'$. Since $\beta'_a(\mathcal{F}', \mathcal{F}) \subset t\widehat{A}_V$, we have $\phi(\mathcal{F}') \subset t\mathcal{F}$. Let $\mathcal{F}'' := \{\partial' - \phi(\partial') \mid \partial' \in \mathcal{F}'\}$. Then $\mathcal{E}'_a = \mathcal{F} \oplus \mathcal{F}''$, $\mathcal{F}''$ is $\beta'_a$-orthogonal to $\mathcal{F}$, and $\mathcal{F}''/t\mathcal{F}'' = \bigoplus_{i\in J_{a+1}} \mathcal{T}_i(V \cap X_0)$. Since the reduction of $\beta'_a$ to $\mathcal{F}''/t\mathcal{F}''$ is zero and $\widehat{A}_V$ has no $t$-torsion, $\beta'_a|_{\mathcal{F}''} = t\beta'_{a+1}$ for an alternating pairing $\beta'_{a+1}$ on $\mathcal{F}''$. Let $\mathcal{E}_a := \mathcal{F}$, $\beta_a := \beta'_a|_{\mathcal{F}}$, and $\mathcal{E}'_{a+1} := \mathcal{F}''$. Then $\beta_a \bmod t = \sum_{i:a_i=a} v_i\tau_i$. By the description of $\mathcal{E}^{a+1}(V \cap X_0)$ above with $a+1$ in place of $a$, the image of $\mathcal{E}^{a+1}(V \cap X_0)$ in $T_U(V \cap X_0)$ is $\mathcal{E}'_{a+1}/t\mathcal{E}'_{a+1}$. Since the sets $V \cap X_0$ with $V \in \mathcal{V}$ cover $U$, we obtain $W_{a+1} = \bigoplus_{i\in J_{a+1}} \mathcal{T}_i$. This proves Claim 9.13 for $a + 1$.

**Step 3.** In this step, we show that $J_a = \emptyset$ for $a \gg 0$, and we prove (1). Fix $V \in \mathcal{V}$ and a basis $\partial_1, \dots, \partial_{\dim X}$ of $T_{V/\mathbb{A}^1}$, and let $f := \det(\beta(\partial_s, \partial_{s'}))_{s,s'} \in \mathcal{O}(V)$. Since $\beta$ is non-degenerate on $V \backslash X_0 \neq \emptyset$, we have $f \neq 0$. By Claim 9.13, in $\widehat{A}_V$ we have

$$f = u \prod_{0\le d<a} t^{d \operatorname{rank} \mathcal{E}_d} \det(\beta_d) \cdot t^{a \operatorname{rank} \mathcal{E}'_a} \det(\beta'_a)$$

for a unit $u$, so $f \in t^{a \operatorname{rank} \mathcal{E}'_a} \widehat{A}_V$. Since $\mathcal{O}(V)/t^s\mathcal{O}(V) = \widehat{A}_V/t^s\widehat{A}_V$ for any $s \ge 1$, we obtain $f \in t^{a \operatorname{rank} \mathcal{E}'_a} \mathcal{O}(V)$. Let $\eta$ be the generic point of $U$. Since $\mathcal{U}$ is regular and $X_0$ is integral, $\mathcal{O}_{\mathcal{U},\eta}$ is a discrete valuation ring with uniformizer $t$. Let $e := \operatorname{ord}_\eta(f) < +\infty$. Then $a \operatorname{rank} \mathcal{E}'_a \le e$ for any $a$. Thus $\mathcal{E}'_a = 0$ and $J_a = \emptyset$ for $a > e$. Hence $a_i$ and $v_i$ are defined for any $i$, and $J_{m+1} = \emptyset$ for $m := \max_i a_i$.

For any $a$, $\tilde{\gamma}_a = \sum_{i:a_i=a} v_i\tau_i$. By **Step 1** and since the projection $p$ is $\Lambda$-equivariant, $\Lambda$ acts on $\tilde{\gamma}_a$ with weight $b + a$. By Lemma 9.11(2), $\Lambda$ acts on $\tau_i$ with weight $l_i$, and the $\tau_i$ are linearly independent. Since $v_i \neq 0$, we obtain $l_i = b + a_i$ for any $i$. This proves (1).

**Step 4.** In this step, we prove (2). Through the marking, $\beta = t^{-b}\sigma$ on $\mathcal{U}\backslash X_0 = X_{\mathrm{reg}} \times (\mathbb{A}^1\backslash\{0\})$, and $\sigma$ is non-degenerate on $X_{\mathrm{reg}}$. Thus $I_\beta$ is an isomorphism over $\mathcal{U}\backslash X_0$.

Fix $V \in \mathcal{V}$. By Claim 9.13 for $a = m + 1$, we have $\mathcal{E}(V) = \bigoplus_{0\le d\le m} \mathcal{E}_d$ with $\beta = \sum_d t^d\beta_d$ with respect to the decomposition. Let $\mathcal{E}_d^\vee \subset \mathcal{E}(V)^\vee$ be the annihilator of $\bigoplus_{d'\neq d} \mathcal{E}_{d'}$, so that $\mathcal{E}(V)^\vee = \bigoplus_d \mathcal{E}_d^\vee$ is the completion of the relative 1-forms on $V$, and

$$I_\beta = \bigoplus_{0\le d\le m} t^d I_{\beta_d}, \quad I_{\beta_d} : \mathcal{E}_d \to \mathcal{E}_d^\vee.$$

Each $I_{\beta_d}$ is an isomorphism. Hence $I_\beta$ becomes an isomorphism after $t$ is made invertible, and

$$t^m I_\beta^{-1} = \bigoplus_{0\le d\le m} t^{m-d} I_{\beta_d}^{-1}$$

is regular. Modulo $t$, the summands with $d < m$ vanish. Since $\mathcal{E}_d/t\mathcal{E}_d = \bigoplus_{i:a_i=d} \mathcal{T}_i(V \cap X_0)$ for any $d$, we have $\mathcal{E}_m^\vee/t\mathcal{E}_m^\vee = \bigoplus_{i:a_i=m} \mathrm{pr}_i^* \Omega^1_{(Y_i)_{\mathrm{reg}}}(V \cap X_0)$, and $\beta_m \bmod t = \sum_{i:a_i=m} v_i\tau_i$. Thus the reduction of $t^m I_\beta^{-1}$ modulo $t$ equals $\sum_{i:a_i=m} v_i^{-1} I_{\tau_i}^{-1}$ on $V \cap X_0$, where the $i$-th summand is defined on $\mathrm{pr}_i^* \Omega^1_{(Y_i)_{\mathrm{reg}}}$ and is zero on the other summands of $\Omega^1_U$. Since the sets $V \cap X_0$ with $V \in \mathcal{V}$ cover $U$, (2) follows.

**Step 5.** In this step, we prove (3). Let $\delta$ be a $\mathbb{C}[t]_{(t)}$-linear derivation of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$ of $\zeta$-degree 0 with $L_\delta\beta = 0$ on $\mathcal{U} \times_{\mathbb{A}^1} \operatorname{Spec} \mathbb{C}[t]_{(t)}$, and let $\delta_0 \in \mathrm{Der}_{\mathbb{C}}(S)_0$ be the reduction of $\delta$ modulo $t$.

Since $U$ is contained in the fiber $X_0$ over $t=0$ and $\delta(t)=0$, $\delta$ defines a continuous derivation of $\widehat{\mathcal{O}}$ and acts on $\mathcal{E}$ by $\partial \to [\delta,\partial]$. By continuity, $L_\delta \beta = 0$ holds on $\widehat{\mathcal{U}}$, that is,

$$\beta([\delta,\partial],\partial') = \delta(\beta(\partial,\partial')) - \beta(\partial,[\delta,\partial']) \tag{9.1}$$

for local sections $\partial,\partial'$ of $\mathcal{E}$. Let $\partial \in \mathcal{E}^a$. Since $\delta(t)=0$, we have $\delta(t^a\widehat{\mathcal{O}}) \subset t^a\widehat{\mathcal{O}}$, so $\beta([\delta,\partial],\mathcal{E}) \subset t^a\widehat{\mathcal{O}}$ and $[\delta,\partial] \in \mathcal{E}^a$. Since $[\delta,\partial]$ reduces to $[\delta_0,\bar{\partial}]$ modulo $t$, $\delta_0$ preserves $W_a$, and dividing (9.1) by $t^a$ and reducing modulo $t$, we obtain

$$\gamma_a([\delta_0,\bar{\partial}],\bar{\partial}') + \gamma_a(\bar{\partial},[\delta_0,\bar{\partial}']) = \delta_0(\gamma_a(\bar{\partial},\bar{\partial}'))$$

for local sections $\bar{\partial},\bar{\partial}'$ of $W_a$. By Lemma 9.11(1), $\delta_0$ preserves each $S_j$. For $\partial \in \mathcal{T}_j$ and $g \in S_l$ with $l \neq j$, we have $[\delta_0,\partial]g = \delta_0(\partial g) - \partial(\delta_0 g) = 0$ because $\delta_0 g \in S_l$. Since $\mathcal{T}_j$ is the subsheaf of $T_U$ annihilating $S_l$ for any $l \neq j$, it follows that $[\delta_0,\mathcal{T}_j] \subset \mathcal{T}_j$ for any $j$. Thus $\delta_0$ commutes with the projection $p : T_U \to W_a$, and $L_{\delta_0}\tilde{\gamma}_a = 0$. By Lemma 9.11(1), there exist $\kappa_i \in \mathbb{C}$ with $L_{\delta_0}\tau_i = \kappa_i\tau_i$ for any $i$. By **Step 3**, we obtain

$$0 = L_{\delta_0}\tilde{\gamma}_a = \sum_{i:a_i=a} v_i\kappa_i\tau_i.$$

Since the $\tau_i$ are linearly independent and $v_i \neq 0$, we have $\kappa_i = 0$ for any $i$ with $a_i = a$. As $a$ is arbitrary and any $i$ has an integer $a_i$, $L_{\delta_0}\tau_i = 0$ for any $i$. Thus $\delta_0 \in \mathfrak{h}_0$ by Lemma 9.11(3). □

9.4. **Products with several factors.** In this subsection, we prove Theorem 9.16.

**Lemma 9.14.** *Notation and conditions as in Set-up 9.9. Then any $\delta_0 \in \mathfrak{h}_0$ is the reduction modulo $t$ of a $\mathbb{C}[t]_{(t)}$-linear derivation of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$ of $\zeta$-degree 0.*

*Proof.* Let $\mathcal{R}_\circ := \mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$. For any $j \geq 0$, the $\mathbb{C}[t]$-module $\mathcal{R}_j$ is finite, because $\mathcal{R}$ is generated over $\mathcal{R}_0 = \mathbb{C}[t]$ by finitely many homogeneous elements of positive degree, and torsion-free, because $\mathcal{R}$ is flat over $\mathbb{C}[t]$. Thus $\mathcal{R}_j$ is a free $\mathbb{C}[t]$-module of finite rank [Lan02, Chapter III, Theorem 7.3]. Since the marking is $T$-equivariant, $\mathcal{R}_j[t^{-1}] \cong R_j \otimes_{\mathbb{C}} \mathbb{C}[t,t^{-1}]$, and $\mathcal{R}_j/t\mathcal{R}_j \cong S_{(j)}$. Thus

$$\operatorname{rank}_{\mathbb{C}[t]} \mathcal{R}_j = \dim R_j = \dim S_{(j)}.$$

Choose homogeneous generators $x_1,\dots,x_N$ of the $\mathbb{C}[t]$-algebra $\mathcal{R}$ of positive degrees $d_1,\dots,d_N$. Let $\mathbb{C}[t][y_1,\dots,y_N]$ be graded by $\deg t = 0$ and $\deg y_q = d_q$, and let $f_1,\dots,f_M$ be homogeneous generators of the kernel of the surjection $\mathbb{C}[t][y_1,\dots,y_N] \to \mathcal{R}$, $y_q \to x_q$, of degrees $e_1,\dots,e_M$. Let

$$P := \bigoplus_{q=1}^{N} (\mathcal{R}_\circ)_{d_q}, \qquad Q := \bigoplus_{\mu=1}^{M} (\mathcal{R}_\circ)_{e_\mu},$$

which are free $\mathbb{C}[t]_{(t)}$-modules of finite rank, and let $\Phi : P \to Q$ be the $\mathbb{C}[t]_{(t)}$-linear map

$$\Phi\big((p_q)_q\big) := \Big(\sum_{q=1}^{N} \frac{\partial f_\mu}{\partial y_q}(x_1,\dots,x_N)p_q\Big)_\mu.$$

A $\mathbb{C}[t]_{(t)}$-linear derivation $\delta$ of $\mathcal{R}_\circ$ of degree 0 is determined by $(\delta(x_q))_q \in P$, and an element $p \in P$ equals $(\delta(x_q))_q$ for such a derivation $\delta$ if and only if $\Phi(p) = 0$. Likewise, $\mathbb{C}(t)$-linear derivations of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}(t)$ of degree 0 correspond to the elements of the kernel of $\Phi \otimes \mathbb{C}(t)$ on $P \otimes \mathbb{C}(t)$, and derivations of $S = \mathcal{R}/t\mathcal{R}$ of degree 0 correspond to the elements of the kernel of $\Phi \otimes \mathbb{C}$ on $P/tP = \bigoplus_q S_{(d_q)}$, because the reductions of the $f_\mu$ modulo $t$ generate the ideal of relations among the images of the $x_q$ in $S$. We regard derivations of degree 0 as elements of $P$, $P \otimes \mathbb{C}(t)$, and $P/tP$ in this way.

Let $\sigma$ be a closed non-degenerate algebraic 2-form on $X_{\mathrm{reg}}$ of $\zeta$-degree $r$, and let $b := b(\sigma)$ and the relative 2-form $\beta := t^{-b}\sigma$ on the smooth locus $\mathcal{U}$ of $\mathcal{X} \to \mathbb{A}^1$ be as in Lemma 9.12. Let

$\mathcal{H} \subset P \otimes \mathbb{C}(t)$ be the $\mathbb{C}(t)$-vector space of $\mathbb{C}(t)$-linear derivations $\delta$ of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}(t)$ of degree 0 with $L_\delta \beta = 0$ on the generic fiber of $\mathcal{U} \to \mathbb{A}^1$. We show that

$$\dim_{\mathbb{C}(t)} \mathcal{H} = \dim S_{(r)}. \tag{9.2}$$

Through the marking $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}(t) \cong R \otimes_{\mathbb{C}} \mathbb{C}(t)$, and since $L_\delta \beta = t^{-b} L_\delta \sigma$ for a $\mathbb{C}(t)$-linear derivation $\delta$, flat base change to $\mathbb{C}(t)$ implies

$$\mathcal{H} = \mathbb{C}(t) \otimes_{\mathbb{C}} \{\delta \in \mathrm{Der}_{\mathbb{C}}(R)_0 \mid L_\delta \sigma = 0\}.$$

By Lemma 9.10, applied to $A = R$ and $\sigma$, we have $\dim\{\delta \in \mathrm{Der}_{\mathbb{C}}(R)_0 \mid L_\delta \sigma = 0\} = \dim R_r$, and (9.2) follows from $\dim R_r = \dim S_{(r)}$.

Let $\mathcal{L} := P \cap \mathcal{H}$ inside $P \otimes \mathbb{C}(t)$. For $p \in \mathcal{L}$, we have $\Phi(p) = 0$ in $Q \otimes \mathbb{C}(t)$, so $\Phi(p) = 0$ because $Q$ is torsion-free. Thus $\mathcal{L}$ consists of $\mathbb{C}[t]_{(t)}$-linear derivations of $\mathcal{R}_\circ$ of degree 0. If $p \in P$ and $tp \in \mathcal{L}$, then $p \in \mathcal{H}$, hence $p \in \mathcal{L}$. Thus $P/\mathcal{L}$ is a finitely generated torsion-free $\mathbb{C}[t]_{(t)}$-module, hence free [Lan02, Chapter III, Theorem 7.3], and $\mathcal{L}$ is a direct summand of $P$. Moreover, any element of $\mathcal{H}$ becomes an element of $\mathcal{L}$ after multiplication by a power of $t$, so $\mathcal{L} \otimes \mathbb{C}(t) = \mathcal{H}$. By (9.2), $\mathcal{L}$ is free of rank $\dim S_{(r)}$, and the image $\mathcal{L}_0$ of $\mathcal{L}/t\mathcal{L} \to P/tP$ satisfies $\dim \mathcal{L}_0 = \dim S_{(r)}$. The elements of $\mathcal{L}_0$ are the reductions modulo $t$ of the derivations in $\mathcal{L}$.

Let $\delta \in \mathcal{L}$. The relative 2-form $L_\delta \beta$ on $\mathcal{U} \times_{\mathbb{A}^1} \operatorname{Spec} \mathbb{C}[t]_{(t)}$ vanishes on the generic fiber, and $\Omega^2_{\mathcal{U}/\mathbb{A}^1}$ is locally free, so $L_\delta \beta = 0$. Thus the reduction of $\delta$ modulo $t$ belongs to $\mathfrak{h}_0$ by Lemma 9.12(3). Thus $\mathcal{L}_0 \subset \mathfrak{h}_0$. Since $\dim \mathfrak{h}_0 = \dim S_{(r)}$ (Lemma 9.11(3)), we have $\mathcal{L}_0 = \mathfrak{h}_0$. □

**Lemma 9.15.** *Notation and conditions as in Set-up 9.9. Let $\delta_0$ be a semisimple derivation of $S$ of degree 0 with integral eigenvalues which commutes with $T$ and $\Lambda$ and which is the reduction modulo $t$ of a $\mathbb{C}[t]_{(t)}$-linear derivation of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$. Then there exists an algebraic $\mathbb{G}_m$-action on $\mathcal{R}$ which fixes $t$, commutes with $T$ and $\Lambda$, and whose infinitesimal generator reduces modulo $t$ to $\delta_0$.*

*Proof.* Since $\zeta$ is a cocharacter of $T$ and $T$ is commutative, $T$ preserves the grading of $\mathcal{R}$.

First we find a $\mathbb{C}[t]$-linear derivation $\delta_3$ of $\mathcal{R}$ of degree 0 which commutes with $T$ and $\Lambda$ and reduces modulo $t$ to $\delta_0$. Let $\delta_1$ be a $\mathbb{C}[t]_{(t)}$-linear derivation of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$ reducing modulo $t$ to $\delta_0$. Since $\mathcal{R}$ is torsion-free over $\mathbb{C}[t]$, we regard $\mathcal{R}$ as a subring of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$. Since the action of $T \times \Lambda$ on $\mathcal{R}$ is algebraic, there exist generators $x_1, \dots, x_N$ of the $\mathbb{C}[t]$-algebra $\mathcal{R}$ which are eigenvectors of $T \times \Lambda$, of characters $\chi_1, \dots, \chi_N$. There exists $g \in \mathbb{C}[t]$ with $g(0) \neq 0$ such that $g\delta_1(x_q) \in \mathcal{R}$ for any $q$. By the Leibniz rule, $\delta_2 := g(0)^{-1} g \delta_1$ restricts to a $\mathbb{C}[t]$-linear derivation of $\mathcal{R}$, which reduces modulo $t$ to $\delta_0$. Let $\mathcal{D}$ be the $\mathbb{C}$-vector space of $\mathbb{C}[t]$-linear derivations of $\mathcal{R}$, with $\gamma \in T \times \Lambda$ acting by $\delta \to \gamma \circ \delta \circ \gamma^{-1}$. The injective linear map $\mathcal{D} \to \mathcal{R}^{\oplus N}$, $\delta \to (\delta(x_q))_q$, is equivariant for the action $\gamma \cdot (y_q)_q := (\chi_q(\gamma)^{-1} \gamma(y_q))_q$ on $\mathcal{R}^{\oplus N}$, which is algebraic. Thus the action on $\mathcal{D}$ is algebraic, and $\delta_2$ is a finite sum of eigenvectors of $T \times \Lambda$ with distinct characters. Let $\delta_3$ be the summand with trivial character. Then $\delta_3$ commutes with $T$ and $\Lambda$, and it has degree 0 because $\zeta$ is a cocharacter of $T$. The reduction map $\mathcal{D} \to \mathrm{Der}_{\mathbb{C}}(S)$ is equivariant and $\delta_0$ is invariant, so $\delta_3$ reduces modulo $t$ to $\delta_0$.

Next we choose bases of the graded pieces. Fix $j \geq 0$. The $\mathbb{C}[t]$-module $\mathcal{R}_j$ is finite, being generated by monomials of degree $j$ in finitely many homogeneous generators of $\mathcal{R}$ of positive degree, it is flat because it is a direct summand of the flat $\mathbb{C}[t]$-module $\mathcal{R}$, and $\mathcal{R}_j/t\mathcal{R}_j \cong S_{(j)}$. Since $\delta_0$ commutes with $\Lambda$, both are diagonalizable on $S_{(j)}$, and $\dim S_{(j)} < +\infty$, there exists a basis $\bar{e}_1, \dots, \bar{e}_h$ of $S_{(j)}$ consisting of common eigenvectors of $\Lambda$ and $\delta_0$. Let $w_\nu$ be the $\Lambda$-weight and $\lambda_\nu \in \mathbb{Z}$ the $\delta_0$-eigenvalue of $\bar{e}_\nu$. Since the reduction map $\mathcal{R}_j \to S_{(j)}$ is surjective and $\Lambda$-equivariant, we may lift $\bar{e}_\nu$ to a $\Lambda$-eigenvector $e_\nu \in \mathcal{R}_j$ of weight $w_\nu$. By Lemma 9.6(3), applied to $M = \mathcal{R}_j$, on which $\Lambda$ acts algebraically and $t$ has $\Lambda$-weight $-1$, with the trivial torus in place of $T$, the elements $e_1, \dots, e_h$ form a basis of the $\mathbb{C}[t]$-module $\mathcal{R}_j$.

Next we show that $\delta_3$ is locally finite with integral eigenvalues. For an integer $w$, let $\mathcal{R}_{j,w} \subset \mathcal{R}_j$ be the subspace of $\Lambda$-weight $w$. It has the basis $\{t^{w_\nu - w} e_\nu \mid w_\nu \geq w\}$, so $\dim \mathcal{R}_{j,w} < +\infty$, and $\mathcal{R} = \bigoplus_{j,w} \mathcal{R}_{j,w}$. Since $\delta_3$ has degree 0 and commutes with $\Lambda$, it preserves every $\mathcal{R}_{j,w}$. Write

$\delta_3(e_\nu) = \sum_\mu g_{\mu\nu}(t)e_\mu$ with $g_{\mu\nu} \in \mathbb{C}[t]$. Since $\delta_3(e_\nu)$ has weight $w_\nu$ and $t^a e_\mu$ has weight $w_\mu - a$, we have $g_{\mu\nu} = \kappa_{\mu\nu} t^{w_\mu - w_\nu}$ with $\kappa_{\mu\nu} \in \mathbb{C}$, and $\kappa_{\mu\nu} = 0$ if $w_\mu < w_\nu$. Reducing modulo $t$, for $w_\mu = w_\nu$ we obtain $\kappa_{\mu\nu} = \lambda_\nu$ if $\mu = \nu$ and $\kappa_{\mu\nu} = 0$ otherwise. Thus

$$\delta_3(t^{w_\nu - w} e_\nu) = \lambda_\nu t^{w_\nu - w} e_\nu + \sum_{\mu : w_\mu > w_\nu} \kappa_{\mu\nu} t^{w_\mu - w} e_\mu,$$

so the matrix of $\delta_3$ on $\mathcal{R}_{j,w}$ is triangular, for the basis ordered by decreasing $w_\nu$, with diagonal entries $\lambda_\nu \in \mathbb{Z}$.

Finally we take the semisimple part. For $\lambda \in \mathbb{Z}$, let $\mathcal{R}^{(\lambda)}$ be the sum over all $j, w$ of the generalized $\lambda$-eigenspaces of $\delta_3$ on $\mathcal{R}_{j,w}$. Then $\mathcal{R} = \bigoplus_{\lambda \in \mathbb{Z}} \mathcal{R}^{(\lambda)}$. For $f \in \mathcal{R}^{(\lambda)} \cap \mathcal{R}_{j,w}$, $f' \in \mathcal{R}^{(\lambda')} \cap \mathcal{R}_{j',w'}$ and $c \geq \dim \mathcal{R}_{j,w} + \dim \mathcal{R}_{j',w'}$, the Leibniz rule gives

$$(\delta_3 - \lambda - \lambda')^c (ff') = \sum_{a=0}^{c} \binom{c}{a} (\delta_3 - \lambda)^a (f) (\delta_3 - \lambda')^{c-a} (f') = 0,$$

hence $\mathcal{R}^{(\lambda)} \mathcal{R}^{(\lambda')} \subset \mathcal{R}^{(\lambda+\lambda')}$. Thus $\mathcal{R} = \bigoplus_\lambda \mathcal{R}^{(\lambda)}$ is a $\mathbb{Z}$-grading of the algebra $\mathcal{R}$, and $z \cdot f := z^\lambda f$ for $f \in \mathcal{R}^{(\lambda)}$ defines an algebraic $\mathbb{G}_m$-action on $\mathcal{R}$. The infinitesimal generator $\delta_s$ of the $\mathbb{G}_m$-action is the semisimple part of $\delta_3$ on each $\mathcal{R}_{j,w}$. Since $\delta_3(t) = 0$, we have $t \in \mathcal{R}^{(0)}$, so the action fixes $t$. The actions of $T$ and $\Lambda$ preserve every $\mathcal{R}_{j,w}$ and commute with $\delta_3$, so they preserve every $\mathcal{R}^{(\lambda)}$ and commute with the $\mathbb{G}_m$-action. On $\mathcal{R}_{j,w}$ we have $\delta_s = F(\delta_3)$ for a polynomial $F$. The surjection from $\mathcal{R}_{j,w}$ onto the $\Lambda$-weight $w$ part of $S_{(j)}$ intertwines $\delta_3$ and $\delta_0$, so the operator induced by $\delta_s$ is $F(\delta_0)$, which is semisimple, commutes with $\delta_0$, and $\delta_0 - F(\delta_0)$ is nilpotent. By the uniqueness of the Jordan decomposition and since $\delta_0$ is semisimple, $F(\delta_0) = \delta_0$. Thus $\delta_s$ reduces modulo $t$ to $\delta_0$. □

We prove Theorem 9.16 by an iteration. After the base change $t = s^r$ of the parameter, we replace $\Lambda$ with a $\mathbb{G}_m$-action $\Lambda'$ which acts on $s$ with weight $-1$ and acts on each factor $S_i$ through a multiple $c_i$ of the grading of $S_i$. At each stage we take a factor $S_p$ with $c_p$ maximal, lift the $\mathbb{G}_m$-action generated by $\mathrm{eu}_p$, and lower $c_p$. After finitely many stages we obtain a closed relative 2-form whose restriction to $U$ is non-degenerate, and we conclude by Theorem 9.2.

**Theorem 9.16.** *Notation and conditions as in Set-up 9.9. Then there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong S[t]$ inducing the identity modulo $t$, where $T$ and $\Lambda$ act on $S$ through the isomorphism $X_0 \cong Y$ and $\Lambda$ acts on $t$ with weight $-1$.*

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we construct a $\mathbb{G}_m$-action $\Gamma$ on $\mathcal{R}$ which fixes $t$, commutes with $T$ and $\Lambda$, and whose infinitesimal generator reduces modulo $t$ to $r\delta_\Lambda - \sum_i l_i \,\mathrm{eu}_i$, where $\delta_\Lambda$ is the infinitesimal generator of $\Lambda$ on $S$ and $l_i$ is the weight of $\Lambda$ on $\tau_i$.

By Lemma 9.11(2), $T$ and $\Lambda$ preserve every $S_i$, and $\delta_\Lambda = \sum_i (l_i/r) \,\mathrm{eu}_i + \delta_H$ with $\delta_H \in \mathfrak{h}_0$. Thus

$$\delta_0 := r\delta_\Lambda - \sum_{i=1}^{k} l_i \,\mathrm{eu}_i = r\delta_H \in \mathfrak{h}_0.$$

Since $T$ and $\Lambda$ preserve every $S_i$ and the grading of $S$, every subspace $(S_1)_{j_1} \otimes \cdots \otimes (S_k)_{j_k}$ of $S$ is $T \times \Lambda$-stable, and $\mathrm{eu}_i$ acts on it by $j_i$. Choosing in each such subspace a basis of $T \times \Lambda$-eigenvectors, we see that $\delta_0$ is semisimple with integral eigenvalues, has degree 0, and commutes with $T$ and $\Lambda$. By Lemma 9.14, $\delta_0$ is the reduction modulo $t$ of a $\mathbb{C}[t]_{(t)}$-linear derivation of $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$. Thus Lemma 9.15 implies the existence of $\Gamma$.

**Step 2.** In this step we pass to the base change $t = s^r$ and construct a $\mathbb{G}_m$-action $\Lambda'$ on $\mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$ which acts on $s$ with weight $-1$ and acts on each $S_i$ through a multiple of the grading of $S_i$.

Let $\mathcal{R}' := \mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$ with $t = s^r$, graded by $\mathcal{R}'_j := \mathcal{R}_j \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$, let $\mathcal{X}' := \operatorname{Spec} \mathcal{R}'$, and let $\mathcal{U}'$ be the smooth locus of $\mathcal{X}' \to \mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[s]$. Then $\mathcal{R}'/s\mathcal{R}' = \mathcal{R}/t\mathcal{R} \cong S$. Let $T$ and $\Gamma$ act on $\mathcal{R}'$ through $\mathcal{R}$ and trivially on $s$, and let $T' := T \times \mathbb{G}_m$ act on $\mathcal{R}'$ through $T$ and $\Gamma$. Let $\Lambda'$ be the $\mathbb{G}_m$-action on $\mathcal{R}'$ such that $\Lambda'(z)$ acts on $\mathcal{R}$ by $\Lambda(z^r)\Gamma(z)^{-1}$ and on $s$ with weight $-1$. It is well defined because $\Lambda(z^r)\Gamma(z)^{-1}$ acts on $t = s^r$ with weight $-r$, and it commutes with $T'$. By **Step 1**, the infinitesimal generator of $\Lambda'$ on $S$ is

$$r\delta_\Lambda - \Big(r\delta_\Lambda - \sum_{i=1}^{k} l_i \operatorname{eu}_i\Big) = \sum_{i=1}^{k} l_i \operatorname{eu}_i . \tag{9.3}$$

**Claim 9.17.** *Let $T'$ be a torus containing $T$ which acts algebraically on $\mathcal{R}'$, fixes $s$, and restricts on $T$ to the action of $T$ on $\mathcal{R}'$ defined above. Let $\Lambda'$ be an algebraic $\mathbb{G}_m$-action on $\mathcal{R}'$ which commutes with $T'$ and acts on $s$ with weight $-1$. Then $S$ is Cohen–Macaulay, $\mathcal{R}'$ is a normal domain, and $(\mathcal{X}', T', \Lambda', \zeta)$, with the $T' \times \Lambda'$-equivariant isomorphism $\mathcal{R}'[s^{-1}] \cong (\mathcal{R}'/(s-1)\mathcal{R}')[s, s^{-1}]$ defined by $\Lambda'$, the isomorphism $\mathcal{R}'/(s-1)\mathcal{R}' = \mathcal{R}/(t-1)\mathcal{R} \cong R$, and the graded isomorphism $\mathcal{R}'/s\mathcal{R}' \cong S$, satisfies the conditions of Set-up 9.9.*

*Proof.* Since $\mathcal{R}'$ is flat over $\mathbb{C}[s]$, $s$ is a nonzerodivisor on $\mathcal{R}'$, and

$$\mathcal{R}' \subset \mathcal{R}'[s^{-1}] = \mathcal{R}[t^{-1}] \otimes_{\mathbb{C}[t,t^{-1}]} \mathbb{C}[s, s^{-1}] \cong R[s, s^{-1}],$$

which is a normal domain. Thus $\mathcal{R}'$ is a domain. We show that $\mathcal{R}'$ is normal by Serre's criterion [Mat89, Theorem 23.8]. Let $\mathfrak{p}$ be a prime ideal of $\mathcal{R}'$. If $s \notin \mathfrak{p}$, then $\mathcal{R}'_{\mathfrak{p}}$ is a localization of $R[s, s^{-1}]$, hence normal. Assume that $s \in \mathfrak{p}$. The ring $S$ is Cohen–Macaulay: by induction on $k$, it suffices to treat $S_1 \otimes_{\mathbb{C}} S''$ for finitely generated Cohen–Macaulay $\mathbb{C}$-algebras $S_1$ and $S''$, and for a maximal ideal $\mathfrak{n}$ of $S_1 \otimes_{\mathbb{C}} S''$ with contraction $\mathfrak{m}$ to $S''$, the homomorphism $S''_{\mathfrak{m}} \to (S_1 \otimes_{\mathbb{C}} S'')_{\mathfrak{n}}$ is flat and local with closed fiber a localization of $S_1$, so $(S_1 \otimes_{\mathbb{C}} S'')_{\mathfrak{n}}$ is Cohen–Macaulay [Mat89, Corollary of Theorem 23.3]. Since $s$ is a nonzerodivisor on $\mathcal{R}'_{\mathfrak{p}}$ and $\mathcal{R}'_{\mathfrak{p}}/s\mathcal{R}'_{\mathfrak{p}}$ is a localization of $S$, the local ring $\mathcal{R}'_{\mathfrak{p}}$ is Cohen–Macaulay [BH93, Theorem 2.1.3], so it satisfies $(S_2)$. If moreover $\mathfrak{p}$ has height one, then $\mathfrak{p}$ is minimal over $s\mathcal{R}'$, so $\mathcal{R}'_{\mathfrak{p}}/s\mathcal{R}'_{\mathfrak{p}}$ is the fraction field of the domain $S$. Thus the maximal ideal of $\mathcal{R}'_{\mathfrak{p}}$ is generated by $s$, and $\mathcal{R}'_{\mathfrak{p}}$ is a discrete valuation ring. Thus $\mathcal{R}'$ satisfies $(R_1)$ and $(S_2)$, and it is normal.

The ring $\mathcal{R}'$ is finitely generated and flat over $\mathbb{C}[s]$, and $\mathcal{R}'_0 = \mathbb{C}[s]$. Since $\zeta$ is a cocharacter of $T \subset T'$ and $\Lambda'$ commutes with $T'$, both $T'$ and $\Lambda'$ preserve the grading. The fiber of $\mathcal{X}'$ over $s = 1$ is $\operatorname{Spec}(\mathcal{R}/(t-1)\mathcal{R}) \cong X$, with the action of $T'$ induced from $\mathcal{R}'$ and the grading defined by $\zeta$. Since $\Lambda'$ acts on $s$ with weight $-1$, the induced action on $\mathbb{A}^1 \backslash \{0\}$ is simply transitive, so the morphism $X \times \mathbb{G}_m \to \mathcal{X}' \backslash X'_0$, $(x, z) \to \Lambda'(z)x$, is an isomorphism. It is $T'$-equivariant because $T'$ commutes with $\Lambda'$ and fixes $s$, and under it $\Lambda'$ acts trivially on $X$ and on $s$ with weight $-1$. Thus $(\mathcal{X}', T', \Lambda', \zeta)$ is a marked test configuration of $X$ as in Set-up 2.15, with the graded isomorphism $X'_0 \cong Y$. Finally, $X_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form of $\zeta$-degree $r$ by Set-up 9.9(3). □

The torus $T' = T \times \mathbb{G}_m$ and the action $\Lambda'$ above satisfy the conditions of Claim 9.17, and by (9.3) the infinitesimal generator of $\Lambda'$ on $S$ equals $\sum_i c_i \operatorname{eu}_i$ with $c_i := l_i \in \mathbb{Z}$.

**Step 3.** In this step and in **Step 4**, let $T'$ be a torus containing $T$ which acts algebraically on $\mathcal{R}'$, fixes $s$, and restricts on $T$ to the action of $T$ on $\mathcal{R}'$ defined in **Step 2**, let $\Lambda'$ be an algebraic $\mathbb{G}_m$-action on $\mathcal{R}'$ which commutes with $T'$ and acts on $s$ with weight $-1$, and assume that the infinitesimal generator of $\Lambda'$ on $S$ equals $\sum_i c_i \operatorname{eu}_i$ with integers $c_1, \dots, c_k$. Let $p$ be an index with $c_p = \max_i c_i$. In this step we construct a closed relative 2-form $\alpha_p$ on $\mathcal{U}'$ of $\zeta$-degree $r$ with $\alpha_p|_U = \tau_p$.

By Claim 9.17 and Lemma 9.11(2), $T'$ and $\Lambda'$ preserve every $S_i$. In particular, $\Lambda'$ acts on $(S_i)_d$ with weight $c_i d$. Let $P$ be a polynomial ring whose finitely many variables map onto homogeneous $T'$-eigenvectors generating $S_p$, and let $I$ be the kernel of $P \to S_p$. Let $T'$ act on each variable through the character of the image of the variable, and let $\Lambda'$ act on a variable of

degree $d$ with weight $c_p d$. The actions of $T'$ and $\Lambda'$ on $P$ are compatible with the actions on $S_p$, and $\Lambda'$ acts with weight $c_p d$ on the degree-$d$ part $T^1(S_p)_d$ of the graded module

$$T^1(S_p) = \operatorname{coker}\big(\operatorname{Der}_{\mathbb{C}}(P, S_p) \to \operatorname{Hom}_{S_p}(I/I^2, S_p)\big).$$

The space

$$\Big(\bigotimes_{i\neq p} S_i\Big) \otimes_{\mathbb{C}} T^1(S_p)$$

is the direct sum of the subspaces $(\bigotimes_{i\neq p}(S_i)_{q_i}) \otimes T^1(S_p)_d$ with $q_i \geq 0$ and $d \in \mathbb{Z}$, and $T'$ preserves each of them, so the $T'$-invariant part of $(\bigotimes_{i\neq p} S_i) \otimes_{\mathbb{C}} T^1(S_p)$ is the direct sum of the $T'$-invariant parts of the subspaces $(\bigotimes_{i\neq p}(S_i)_{q_i}) \otimes T^1(S_p)_d$. The subspace indexed by $(q_i)_{i\neq p}$ and $d$ has $\zeta$-degree $\sum_{i\neq p} q_i + d$ and $\Lambda'$-weight $\sum_{i\neq p} c_i q_i + c_p d$. Since $\zeta$ is a cocharacter of $T'$, the $T'$-invariant part is zero unless $d = -\sum_{i\neq p} q_i$, and then the $\Lambda'$-weight is

$$\sum_{i\neq p} c_i q_i + c_p d = \sum_{i\neq p} q_i(c_i - c_p) \leq 0.$$

Thus $\Lambda'$ acts on the $T'$-invariant part of $(\bigotimes_{i\neq p} S_i) \otimes_{\mathbb{C}} T^1(S_p)$ with non-positive weights. Lemma 9.7, applied to $(\mathcal{R}', T', \Lambda')$ and the decomposition $S \cong (\bigotimes_{i\neq p} S_i) \otimes_{\mathbb{C}} S_p$, with $\bigotimes_{i\neq p} S_i$ in place of $S_1$ and $S_p$ in place of $S_2$, implies that there exists a $T' \times \Lambda'$-equivariant graded injective homomorphism of $\mathbb{C}[s]$-algebras $j_p : S_p[s] \to \mathcal{R}'$ which reduces modulo $s$ to the inclusion $S_p \to S$, and that $\mathcal{R}'$ is a free $S_p[s]$-module. In particular, the morphism $f_p : \mathcal{X}' \to Y_p \times \mathbb{A}^1$ defined by $j_p$ is flat.

Let $Z_p := f_p^{-1}(\operatorname{Sing}(Y_p) \times \mathbb{A}^1)$. Since $Y_p$ is normal, $\operatorname{Sing}(Y_p) \times \mathbb{A}^1$ has codimension at least 2 in $Y_p \times \mathbb{A}^1$. Since $f_p$ is flat, going down holds for $f_p$ [Mat89, Theorem 9.5], so $Z_p$ has codimension at least 2 in $\mathcal{X}'$. On $\mathcal{U}'\backslash Z_p$, let $\alpha_p$ be the image in $\Omega^2_{\mathcal{U}'/\mathbb{A}^1}$ of the pullback under $f_p$ of the pullback of $\omega_p$ to $(Y_p)_{\mathrm{reg}} \times \mathbb{A}^1$. The scheme $\mathcal{U}'$ is regular, being smooth over $\mathbb{A}^1$, and $\Omega^2_{\mathcal{U}'/\mathbb{A}^1}$ is locally free, so $\alpha_p$ extends to a regular relative 2-form on $\mathcal{U}'$ [Har77, Proposition II.6.3A], which we also denote by $\alpha_p$. The relative form $d\alpha_p$ vanishes on the dense open subset $\mathcal{U}'\backslash Z_p$ because $\omega_p$ is closed, so $\alpha_p$ is closed. It has $\zeta$-degree $r$ because $j_p$ is graded and $\omega_p$ has degree $r$. Since $X'_0 \cong Y$ is smooth along $U$ and $\mathcal{X}' \to \mathbb{A}^1$ is flat, $U \subset \mathcal{U}'$ [Gro67, Théorème 17.5.1]. Since $f_p$ restricts over $s = 0$ to the projection $\mathrm{pr}_p : Y \to Y_p$, we have $\alpha_p|_U = \mathrm{pr}_p^* \omega_p = \tau_p$.

**Step 4.** In this step we construct a $\mathbb{G}_m$-action $\Gamma_p$ on $\mathcal{R}'$ which fixes $s$, commutes with $T'$ and $\Lambda'$, and whose infinitesimal generator reduces modulo $s$ to $\mathrm{eu}_p$.

Let $\sigma$ be a closed non-degenerate algebraic 2-form on $X_{\mathrm{reg}}$ of $\zeta$-degree $r$, regarded as a relative form on $\mathcal{U}'\backslash X'_0$ through the marking defined by $\Lambda'$. By Claim 9.17, we may use Lemma 9.12 for $(\mathcal{X}', T', \Lambda', \zeta)$. Let $b := b(\sigma)$ be the integer of Definition-Lemma 4.1, let $\beta := s^{-b}\sigma$, let $I_\beta$ be the morphism from relative tangent vectors to relative 1-forms given by contraction with $\beta$, and let $a_1, \dots, a_k \geq 0$, $v_1, \dots, v_k \in \mathbb{C}^*$, and $m := \max_i a_i$ be as in Lemma 9.12. By Lemma 9.12(2), on the formal completion of $\mathcal{U}'$ along $U$, $s^m I_\beta^{-1}$ extends to a regular morphism whose restriction to $U$ is $\sum_{i:a_i=m} v_i^{-1} I_{\tau_i}^{-1}$, where $I_{\tau_i}$ is contraction with $\tau_i$ on $\mathrm{pr}_i^*$ of the tangent sheaf of $(Y_i)_{\mathrm{reg}}$, and the $i$-th summand is zero on the summands $\mathrm{pr}_j^* \Omega^1_{(Y_j)_{\mathrm{reg}}}$ of $\Omega^1_U$ with $j \neq i$. Since $\sum_j c_j \mathrm{eu}_j$ acts on $\tau_i$ by $L_{\sum_j c_j \mathrm{eu}_j}\tau_i = r c_i \tau_i$, the weight of $\Lambda'$ on $\tau_i$ is $rc_i$, and Lemma 9.12(1) implies

$$a_i = r c_i - b \quad \text{for any } i.$$

Thus $a_p = m$. Let eu be the Euler vector field of $\zeta$ on $\mathcal{U}'$, which is a relative vector field because $\zeta$ fixes $s$, and whose restriction to $U$ is $\sum_i \mathrm{eu}_i$. By Lemma 9.12(2), $I_\beta$ is an isomorphism over $\mathcal{U}'\backslash X'_0$, and we let

$$\delta := v_p s^m I_\beta^{-1}(i_{\mathrm{eu}} \alpha_p), \tag{9.4}$$

a relative vector field on $\mathcal{U}'\backslash X'_0$. Through the marking, $\mathcal{U}'\backslash X'_0 \cong X_{\mathrm{reg}} \times \mathbb{G}_m$, whose complement in the normal variety $\mathcal{X}'\backslash X'_0 \cong X \times \mathbb{G}_m$ has codimension at least 2, so $\delta$ is a $\mathbb{C}[s]$-linear derivation of $\mathcal{R}'[s^{-1}]$ [Har77, Proposition II.6.3A].

We show that $\delta(\mathcal{R}') \subset \mathcal{R}'$. Let $x \in \mathcal{R}'$. Since $\mathcal{R}'$ is a normal domain (Claim 9.17), $\mathcal{R}'$ is the intersection of the localizations of $\mathcal{R}'$ at height-one primes [Har77, Proposition II.6.3A], and $\delta(x)$ belongs to the localization at any height-one prime not containing $s$. Since $\mathcal{R}'/s\mathcal{R}' \cong S$ is a domain, the only height-one prime containing $s$ is $s\mathcal{R}'$, and $\mathcal{R}'_{s\mathcal{R}'}$ is a discrete valuation ring with uniformizer $s$, because $\mathcal{R}'_{s\mathcal{R}'}/s\mathcal{R}'_{s\mathcal{R}'}$ is the fraction field of $S$. The local ring of $\mathcal{X}'$ at the generic point of $X'_0$ is $\mathcal{R}'_{s\mathcal{R}'}$, and since $U$ contains the generic point of $X'_0$, restriction to the generic point of $X'_0$ maps regular functions on the formal completion of $\mathcal{U}'$ along $U$ to the $s$-adic completion $\widehat{\mathcal{R}'_{s\mathcal{R}'}}$, compatibly with $\mathcal{R}'[s^{-1}] \subset \mathrm{Frac}(\widehat{\mathcal{R}'_{s\mathcal{R}'}})$. Since $s^m I_\beta^{-1}$ extends to a regular morphism on that formal completion, the image of $\delta(x) = v_p s^m I_\beta^{-1}(i_{\mathrm{eu}}\alpha_p)(dx)$ in $\mathrm{Frac}(\widehat{\mathcal{R}'_{s\mathcal{R}'}})$ belongs to $\widehat{\mathcal{R}'_{s\mathcal{R}'}}$. Since the $s$-adic valuation of $\widehat{\mathcal{R}'_{s\mathcal{R}'}}$ restricts to the valuation of $\mathcal{R}'_{s\mathcal{R}'}$, we have $\mathrm{Frac}(\mathcal{R}') \cap \widehat{\mathcal{R}'_{s\mathcal{R}'}} = \mathcal{R}'_{s\mathcal{R}'}$, so $\delta(x) \in \mathcal{R}'_{s\mathcal{R}'}$. Thus $\delta(x) \in \mathcal{R}'$.

We compute the reduction of $\delta$ modulo $s$. On $U$, we have $i_{\mathrm{eu}}\alpha_p|_U = i_{\sum_j \mathrm{eu}_j}\tau_p = i_{\mathrm{eu}_p}\tau_p$, which is a section of the summand $\mathrm{pr}_p^*\,\Omega^1_{(Y_p)_{\mathrm{reg}}}$ of $\Omega^1_U$. Since $a_p = m$ and the extension of $s^m I_\beta^{-1}$ restricts on $U$ to $\sum_{i:a_i=m} v_i^{-1} I_{\tau_i}^{-1}$, we obtain

$$\delta|_U = v_p v_p^{-1} I_{\tau_p}^{-1}(i_{\mathrm{eu}_p}\tau_p) = \mathrm{eu}_p\,.$$

Since $S \subset H^0(U, \mathcal{O}_U)$, the reduction of $\delta$ modulo $s$ equals $\mathrm{eu}_p$ as a derivation of $S$. The derivation $\mathrm{eu}_p$ of $S$ has degree 0, is semisimple with integral eigenvalues, since it acts on $(S_1)_{j_1} \otimes \cdots \otimes (S_k)_{j_k}$ by $j_p$, and commutes with $T'$ and $\Lambda'$, since they preserve every $S_i$ and the grading. By Claim 9.17, Lemma 9.15, applied to $(\mathcal{X}', T', \Lambda', \zeta)$ and $\delta_0 = \mathrm{eu}_p$, implies the existence of $\Gamma_p$.

**Step 5.** In this step we construct, for any $1 \le i \le k$, a closed relative 2-form $\alpha_i$ on $\mathcal{U}'$ of $\zeta$-degree $r$ with $\alpha_i|_U = \tau_i$.

Assume that for a subset $I \subsetneq \{1, \dots, k\}$ we have $T'$ and $\Lambda'$ as at the beginning of **Step 3**, with integers $c_1, \dots, c_k$, and, for any $i \in I$, a closed relative 2-form $\alpha_i$ on $\mathcal{U}'$ of $\zeta$-degree $r$ with $\alpha_i|_U = \tau_i$ and a $\mathbb{G}_m$-action $\Gamma_i$ on $\mathcal{R}'$ which is the restriction of the action of $T'$ to a cocharacter of $T'$ and whose infinitesimal generator reduces modulo $s$ to $\mathrm{eu}_i$. For $I = \emptyset$, the data of **Step 2** satisfy this. Choose integers $e_i \ge 0$ for $i \in I$ such that

$$c_i - e_i < c_j \quad \text{for any } i \in I \text{ and } j \notin I,$$

and let $\Lambda''$ be the $\mathbb{G}_m$-action $z \to \Lambda'(z)\prod_{i\in I}\Gamma_i(z)^{-e_i}$ on $\mathcal{R}'$. Since the $\Gamma_i$ factor through $T'$, which commutes with $\Lambda'$, $\Lambda''$ is an algebraic $\mathbb{G}_m$-action which commutes with $T'$ and acts on $s$ with weight $-1$, and the infinitesimal generator of $\Lambda''$ on $S$ equals

$$\sum_{i\in I}(c_i - e_i)\,\mathrm{eu}_i + \sum_{j\notin I} c_j\,\mathrm{eu}_j\,.$$

Thus a maximal coefficient occurs at an index $p \notin I$. **Steps 3 and 4** applied to $T'$, $\Lambda''$, and $p$ construct $\alpha_p$ and a $\mathbb{G}_m$-action $\Gamma_p$ on $\mathcal{R}'$ which fixes $s$, commutes with $T'$ and $\Lambda''$, and whose infinitesimal generator reduces modulo $s$ to $\mathrm{eu}_p$. We replace $T'$ with the torus $T' \times \mathbb{G}_m$ acting through $T'$ and $\Gamma_p$, $\Lambda'$ with $\Lambda''$, the $c_i$ with the coefficients of $\Lambda''$ above, and $I$ with $I \cup \{p\}$. The new torus contains $T$, fixes $s$, and commutes with $\Lambda''$, and any $\Gamma_i$ with $i \in I \cup \{p\}$ is the restriction of the action of $T' \times \mathbb{G}_m$ to a cocharacter. The forms $\alpha_i$ with $i \in I$ are relative forms on $\mathcal{U}'$, which depends only on $\mathcal{R}'$ as a $\mathbb{C}[s]$-algebra, so they are not affected by the replacement. After at most $k$ replacements, $I = \{1, \dots, k\}$.

**Step 6.** In this step we conclude the proof.

Let $T'$ and $\Lambda'$ be the torus and the action obtained at the end of **Step 5**, and let

$$\Omega' := \sum_{i=1}^{k} \alpha_i,$$

a closed relative 2-form of $\zeta$-degree $r$ on $\mathcal{U}'$, the smooth locus of $\mathcal{X}' \to \mathbb{A}^1$, with $\Omega'|_U = \sum_i \tau_i$. By Claim 9.17, $(\mathcal{X}', T', \Lambda', \zeta)$ is a marked test configuration of $X$ as in Set-up 2.15 with central fiber $X'_0 \cong Y$. The open subset $U$ is smooth, and a point $y = (y_1, \dots, y_k)$ of $Y$ is regular only if every $y_i$ is a regular point of $Y_i$, because $\mathcal{O}_{Y,y}$ is flat over $\mathcal{O}_{Y_i,y_i}$ [Mat89, Theorem 23.7]. Thus $Y_{\mathrm{reg}} = U$. The variety $Y$ is normal by Serre's criterion [Mat89, Theorem 23.8], since $S$ is Cohen–Macaulay (Claim 9.17) and $Y \backslash U$ has codimension at least 2. The product of resolutions of the $Y_i$ is a resolution of $Y$, the pullback of $\sum_i \tau_i$ to it is regular, and it suffices to check the extension of the pullback on one resolution [Bea99, (1.2)]. Thus $Y$ with $\sum_i \tau_i$ is a conical symplectic variety for the $\zeta$-grading whose symplectic form has degree $r$. The restriction of $\Omega'$ to the formal completion of $\mathcal{U}'$ along $Y_{\mathrm{reg}} = U$ satisfies Theorem 9.2(ii). Thus Theorem 9.2 implies that there exists a $T' \times \Lambda'$-equivariant isomorphism of $\mathbb{C}[s]$-algebras $\psi : \mathcal{R}' \to S[s]$ inducing the identity modulo $s$. Since $\zeta$ is a cocharacter of $T'$, $\psi$ is graded, so $\psi^{-1}$ induces an isomorphism of $\mathbb{C}[[s]]$-algebras $\phi : S[[s]] \to \widehat{\mathcal{R}'}$ onto the $s$-adic completion of $\mathcal{R}'$ which induces the identity modulo $s$ and whose reduction modulo $s^N$ is a graded isomorphism $S[s]/(s^N) \to \mathcal{R}'/s^N\mathcal{R}'$ for any $N \geq 1$.

Consider $(\mathcal{X}, T, \Lambda, \zeta)$ in Lemma 9.4 with $r$ in place of $m$ and 1 in place of $e$, and take both the base change $\widetilde{\mathcal{R}} = \mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$ and the finite extension of $\widetilde{\mathcal{R}}$ in that lemma to be $\mathcal{R}'$. The central fiber $X_0 \cong Y$ is a conical symplectic variety for the $\zeta$-grading whose symplectic form $\sum_i \tau_i$ has degree $r > 0$, the ring $\mathcal{R}'$ is a normal domain by Claim 9.17, and $\phi$ satisfies the condition of Lemma 9.4. By Lemma 9.4, there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ inducing the identity modulo $t$, where, as in Set-up 2.15, $T$ and $\Lambda$ act on $(\mathcal{R}/t\mathcal{R})[t]$ through their actions on $X_0$, $T$ fixes $t$, and $\Lambda$ acts on $t$ with weight $-1$. Composing with $\mathcal{R}/t\mathcal{R} \cong S$, which is $T \times \Lambda$-equivariant through $X_0 \cong Y$, we obtain the isomorphism $\mathcal{R} \cong S[t]$ of Theorem 9.16. □

9.5. **Smooth factors.** In this subsection, we prove Theorem 9.18. Theorem 9.16 does not apply directly when the central fiber has a factor $\operatorname{Spec} \mathbb{C}[u_1, \dots, u_{2b}]$ with $b \geq 2$, since then the constant 2-forms $du_i \wedge du_j$ are not linear combinations of the pullbacks of the symplectic forms of the factors. Theorem 9.18 shows that the factor $\operatorname{Spec} \mathbb{C}[u_1, \dots, u_{2b}]$ splits off in the family for any $b \geq 1$.

**Theorem 9.18.** *Notation and conditions as in Set-up 9.8 with $r = 2s$ for an integer $s \geq 1$. Let $b \geq 1$ be an integer, let $R_c := S[u_1, \dots, u_{2b}]$ be the polynomial ring over $S$ graded by $\deg u_j = s$, let $(\mathcal{X}, T, \Lambda, \zeta)$ be a marked test configuration of $X = \operatorname{Spec} R$ as in Set-up 2.15 together with a graded isomorphism $\mathcal{R}/t\mathcal{R} \cong R_c$, and let $\mathcal{K} \subset \mathcal{R}$ be the common kernel of the derivations in $\operatorname{Der}_{\mathbb{C}[t]}(\mathcal{R})_{-s}$. Assume that the following hold.*

- *(i) The degree-$d$ part of $S$ is zero for $0 < d \leq s$.*
- *(ii) Any derivation of $R_c$ of degree $-s$ is a $\mathbb{C}$-linear combination of $\partial/\partial u_1, \dots, \partial/\partial u_{2b}$.*
- *(iii) $X_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form of $\zeta$-degree $2s$.*

*Then the following hold.*

- *(1) Restriction to $\mathcal{R}_s$, the $\zeta$-degree $s$ part of $\mathcal{R}$, induces an isomorphism*

$$\operatorname{Der}_{\mathbb{C}[t]}(\mathcal{R})_{-s} \to \operatorname{Hom}_{\mathbb{C}[t]}(\mathcal{R}_s, \mathbb{C}[t]).$$

- *(2) There exists a basis $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$ of $\mathcal{R}_s$ over $\mathbb{C}[t]$ consisting of $T \times \Lambda$-eigenvectors such that $\mathcal{R} = \mathcal{K}[\widetilde{u}_1, \dots, \widetilde{u}_{2b}]$, and the natural map $\mathcal{K}/t\mathcal{K} \to \mathcal{R}/t\mathcal{R} \cong R_c$ is injective with image $S$.*
- *(3) There exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong R_c[t]$ inducing the identity modulo $t$, where $T$ and $\Lambda$ act on $R_c$ through $\mathcal{R}/t\mathcal{R} \cong R_c$ and $\Lambda$ acts on $t$ with weight $-1$.*

*Proof.* The proof is long, so we divide it into several steps. Let $\sigma$ be a closed non-degenerate algebraic 2-form on $X_{\mathrm{reg}}$ of $\zeta$-degree $2s$, which exists by (iii). Let $\{\ ,\ \}$ and $H_f = \{f, \cdot\}$ $(f \in R)$

be the Poisson bracket and the Hamiltonian derivations of $\sigma$, defined as in Definition 2.9 with $\sigma$ in place of $\omega$. Since $X$ is normal, $H_f$ is a derivation of $R$. Since $\sigma$ has $\zeta$-degree $2s$, we have $\{R_d, R_e\} \subset R_{d+e-2s}$, where $R_j := 0$ for $j < 0$. Since $\sigma$ is closed, $\{\ ,\ \}$ satisfies the Jacobi identity, because for $f, g, h \in R$ the sum

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\}$$

equals $-d\sigma(H_f, H_g, H_h)$ on $X_{\mathrm{reg}}$. Equivalently,

$$[H_f, H_g] = H_{\{f,g\}} \quad \text{for any } f, g \in R.$$

For a $\mathbb{C}[t]$-module $M$, let $M_{(t)} := M \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t)}$, and similarly for a homomorphism of $\mathbb{C}[t]$-modules.

**Step 1.** In this step, we show that the graded pieces of $\mathcal{R}$ are finite free $\mathbb{C}[t]$-modules, we construct the basis $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$ of $\mathcal{R}_s$, and we show that $S \subset R_c$ is $T \times \Lambda$-stable.

Since $\mathcal{R}$ is a finitely generated $\mathbb{C}$-algebra with a non-negative $\zeta$-grading and $\mathcal{R}_0 = \mathbb{C}[t]$, $\mathcal{R}$ is generated over $\mathbb{C}[t]$ by finitely many homogeneous elements of positive degree. There exist only finitely many monomials of a fixed degree $d$ in the generators, so $\mathcal{R}_d$ is a finite $\mathbb{C}[t]$-module. Since $\mathcal{R}$ is flat over $\mathbb{C}[t]$, $\mathcal{R}_d$ is torsion-free, hence free over the principal ideal domain $\mathbb{C}[t]$. Since $t$ has degree 0, $\mathcal{R}_d/t\mathcal{R}_d$ is the degree-$d$ part of $\mathcal{R}/t\mathcal{R} \cong R_c$, and since the marking is $T$-equivariant, $\mathcal{R}_d[t^{-1}] \cong R_d \otimes \mathbb{C}[t, t^{-1}]$. Hence

$$\operatorname{rank}_{\mathbb{C}[t]} \mathcal{R}_d = \dim_{\mathbb{C}} R_d = \dim_{\mathbb{C}}(\text{degree-}d \text{ part of } R_c).$$

The degree-$s$ part of $S$ is zero by (i), and the degree-0 part of $S$ is $\mathbb{C}$. Thus the degree-$s$ part of $R_c$ is $\mathbb{C}u_1 \oplus \cdots \oplus \mathbb{C}u_{2b}$, and

$$\operatorname{rank}_{\mathbb{C}[t]} \mathcal{R}_s = \dim_{\mathbb{C}} R_s = 2b.$$

Since the actions of $T$ and $\Lambda$ commute and $\zeta$ is a cocharacter of $T$, both actions preserve the $\zeta$-gradings of $\mathcal{R}$ and of $R_c$. For $g \in T \times \Lambda$ and a derivation $D$ of $R_c$ of degree $-s$, $g \circ D \circ g^{-1}$ is again a derivation of $R_c$ of degree $-s$. By (ii), and since $\partial/\partial u_1, \dots, \partial/\partial u_{2b}$ are derivations of degree $-s$, the common kernel of the derivations of $R_c$ of degree $-s$ is the common kernel of $\partial/\partial u_1, \dots, \partial/\partial u_{2b}$, which is $S$. Hence $S$ is $T \times \Lambda$-stable. Let $u'_1, \dots, u'_{2b}$ be a basis of $\mathbb{C}u_1 \oplus \cdots \oplus \mathbb{C}u_{2b}$ consisting of $T \times \Lambda$-eigenvectors. Then $R_c = S[u'_1, \dots, u'_{2b}]$.

The action of $T \times \Lambda$ on $\mathcal{R}_s$ is rational, so $\mathcal{R}_s$ is the direct sum of the $T \times \Lambda$-eigenspaces [Bor91, Chapter III, §8.4, Proposition]. For any character $\chi$ of $T \times \Lambda$, the $T \times \Lambda$-equivariant surjection

$$\mathcal{R}_s \to \mathcal{R}_s/t\mathcal{R}_s \cong \mathbb{C}u_1 \oplus \cdots \oplus \mathbb{C}u_{2b}$$

maps the $\chi$-eigenspace of $\mathcal{R}_s$ onto the $\chi$-eigenspace of $\mathbb{C}u_1 \oplus \cdots \oplus \mathbb{C}u_{2b}$. Thus there exist $T \times \Lambda$-eigenvectors $\widetilde{u}_1, \dots, \widetilde{u}_{2b} \in \mathcal{R}_s$ such that $\widetilde{u}_j$ reduces to $u'_j$ modulo $t$ and the characters of $\widetilde{u}_j$ and $u'_j$ coincide for any $j$.

The $\mathbb{C}[t]$-module $\mathcal{R}_s$ is finite free, multiplication by $t$ preserves the $T$-weight and lowers the $\Lambda$-weight by 1, and the images $u'_1, \dots, u'_{2b}$ of $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$ form a basis of $\mathcal{R}_s/t\mathcal{R}_s$. Thus $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$ is a basis of $\mathcal{R}_s$ over $\mathbb{C}[t]$ by Lemma 9.6(3), applied to $M = \mathcal{R}_s$.

**Step 2.** In this step, we show that $\operatorname{Der}_{\mathbb{C}[t]_{(t)}}(\mathcal{R}_{(t)})_{-s}$ is a free $\mathbb{C}[t]_{(t)}$-module of rank $2b$, that reduction modulo $t$ maps it isomorphically onto $\operatorname{Der}_{\mathbb{C}}(R_c)_{-s}$, and that restriction maps it isomorphically onto $\operatorname{Hom}_{\mathbb{C}[t]_{(t)}}((\mathcal{R}_s)_{(t)}, \mathbb{C}[t]_{(t)})$.

Choose a presentation $\mathcal{R} = \mathbb{C}[t][Z_1, \dots, Z_q]/(F_1, \dots, F_p)$, where $Z_a$ is homogeneous of degree $d_a > 0$ and $F_l$ is homogeneous of degree $e_l$, and set $\mathcal{R}_j := 0$ for $j < 0$. Let

$$P := \bigoplus_{a=1}^{q} \mathcal{R}_{d_a - s}, \qquad Q := \bigoplus_{l=1}^{p} \mathcal{R}_{e_l - s}, \qquad M : P \to Q, \quad (v_a)_a \to \Big( \sum_{a=1}^{q} \frac{\partial F_l}{\partial Z_a} v_a \Big)_l.$$

By **Step 1**, $P$ and $Q$ are finite free $\mathbb{C}[t]$-modules. A homogeneous derivation $D$ of degree $-s$ is determined by $(D(Z_a))_a \in P$, and a tuple $(v_a)_a \in P$ is of the form $(D(Z_a))_a$ if and only

if $M((v_a)_a) = 0$. For any $\mathbb{C}[t]$-algebra $A'$, we have $\mathcal{R} \otimes A' = A'[Z_1, \dots, Z_q]/(F_1, \dots, F_p)$ and $(\mathcal{R} \otimes A')_j = \mathcal{R}_j \otimes A'$. Hence

$$\operatorname{Der}_{A'}(\mathcal{R} \otimes A')_{-s} \cong \ker(M \otimes A').$$

For $A' = \mathbb{C} = \mathbb{C}[t]/(t)$, $\ker(M \otimes \mathbb{C}) \cong \operatorname{Der}_{\mathbb{C}}(R_c)_{-s}$, which has basis $\partial/\partial u_1, \dots, \partial/\partial u_{2b}$ by (ii). Thus $\dim_{\mathbb{C}} \ker(M \otimes \mathbb{C}) = 2b$.

For $A' = \mathbb{C}(t)$, the marking induces a graded isomorphism $\mathcal{R} \otimes \mathbb{C}(t) \cong R \otimes \mathbb{C}(t)$. The map $R_s \to \operatorname{Der}_{\mathbb{C}}(R)_{-s}$, $f \to H_f$, is injective: if $H_f = 0$, then $df = 0$ on $X_{\mathrm{reg}}$ since $\sigma$ is non-degenerate, and $sf = \mathrm{eu}(f) = df(\mathrm{eu}) = 0$. Moreover, $\mathbb{C}$-linearly independent derivations $D'_1, \dots, D'_n$ of $R$ remain $\mathbb{C}(t)$-linearly independent as derivations of $R \otimes \mathbb{C}(t)$: a relation $\sum_i c_i D'_i = 0$ with $c_i \in \mathbb{C}[t]$ not all zero implies, by comparing coefficients of a power of $t$, a non-trivial $\mathbb{C}$-linear relation among $D'_1, \dots, D'_n$. Since $\dim_{\mathbb{C}} R_s = 2b$ (**Step 1**), we obtain

$$\dim_{\mathbb{C}(t)} \ker(M \otimes \mathbb{C}(t)) \geq 2b.$$

On the other hand, write $M$ as a matrix with entries in $\mathbb{C}[t]$ with respect to bases of $P$ and $Q$. Any minor of $M \otimes \mathbb{C}$ is the value at $t = 0$ of the corresponding minor of $M$, so $\operatorname{rank}(M \otimes \mathbb{C}) \leq \operatorname{rank}(M \otimes \mathbb{C}(t))$. Hence

$$\dim_{\mathbb{C}(t)} \ker(M \otimes \mathbb{C}(t)) = \operatorname{rank} P - \operatorname{rank}(M \otimes \mathbb{C}(t)) \leq \operatorname{rank} P - \operatorname{rank}(M \otimes \mathbb{C}) = 2b.$$

Thus $\operatorname{rank}(M \otimes \mathbb{C}(t)) = \operatorname{rank}(M \otimes \mathbb{C}) = \operatorname{rank} P - 2b$.

By the elementary divisor theorem [Lan02, Chapter III, Theorem 7.8], there exist bases of $P_{(t)}$ and $Q_{(t)}$ in which $M_{(t)}$ is diagonal with non-zero entries $f_1, \dots, f_m$ and all other entries zero. Then $m = \operatorname{rank}(M \otimes \mathbb{C}(t))$, and the number of units among $f_1, \dots, f_m$ is $\operatorname{rank}(M \otimes \mathbb{C})$. Thus $f_1, \dots, f_m$ are units. Hence

$$L_{(t)} := \operatorname{Der}_{\mathbb{C}[t]_{(t)}}(\mathcal{R}_{(t)})_{-s} \cong \ker(M_{(t)})$$

is a free $\mathbb{C}[t]_{(t)}$-module of rank $2b$, and reduction modulo $t$ induces an isomorphism

$$L_{(t)} \otimes \mathbb{C} \to \operatorname{Der}_{\mathbb{C}}(R_c)_{-s},$$

because in the bases above both kernels are spanned by the last $\operatorname{rank} P - m$ basis vectors.

Any $D \in L_{(t)}$ maps $(\mathcal{R}_s)_{(t)}$ into $(\mathcal{R}_0)_{(t)} = \mathbb{C}[t]_{(t)}$. Let

$$\mathrm{res}_{(t)} : L_{(t)} \to \operatorname{Hom}_{\mathbb{C}[t]_{(t)}}((\mathcal{R}_s)_{(t)}, \mathbb{C}[t]_{(t)})$$

be the restriction map. Since $\mathcal{R}_s$ is free, reduction modulo $t$ of $\mathrm{res}_{(t)}$ is the restriction map

$$\operatorname{Der}_{\mathbb{C}}(R_c)_{-s} \to \operatorname{Hom}_{\mathbb{C}}(\mathbb{C}u_1 \oplus \cdots \oplus \mathbb{C}u_{2b}, \mathbb{C}),$$

which is an isomorphism by (ii) since $\partial/\partial u_1, \dots, \partial/\partial u_{2b}$ restrict to the dual basis of $u_1, \dots, u_{2b}$. By Nakayama's lemma, $\mathrm{res}_{(t)}$ is surjective. Since $L_{(t)}$ and $\operatorname{Hom}_{\mathbb{C}[t]_{(t)}}((\mathcal{R}_s)_{(t)}, \mathbb{C}[t]_{(t)})$ are free of rank $2b$, $\mathrm{res}_{(t)}$ is an isomorphism.

**Step 3.** In this step, we prove (1), we construct pairwise commuting locally nilpotent derivations $D_1, \dots, D_{2b}$ dual to $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$, and we show that $\mathcal{R} = \mathcal{K}[\widetilde{u}_1, \dots, \widetilde{u}_{2b}]$.

Let $L := \operatorname{Der}_{\mathbb{C}[t]}(\mathcal{R})_{-s} \cong \ker M$ and let

$$\mathrm{res} : L \to \operatorname{Hom}_{\mathbb{C}[t]}(\mathcal{R}_s, \mathbb{C}[t])$$

be the restriction map. Since localization is exact, $L_{(t)} = \operatorname{Der}_{\mathbb{C}[t]_{(t)}}(\mathcal{R}_{(t)})_{-s}$, and $\mathrm{res}_{(t)}$ is the restriction map of **Step 2**, which is an isomorphism. In particular, $\mathrm{res} \otimes \mathbb{C}(t)$ is injective. Since $L \subset P$ is torsion-free, res is injective. For $g \in T \times \Lambda$, the automorphism $g$ of $\mathcal{R}$ preserves the grading and maps $\mathbb{C}[t]$ to itself, so $T \times \Lambda$ acts on $L$ by $D \to g \circ D \circ g^{-1}$ and on $\operatorname{Hom}_{\mathbb{C}[t]}(\mathcal{R}_s, \mathbb{C}[t])$ by $\phi \to g \circ \phi \circ g^{-1}$.

We use the following fact. Let $N$ be a finitely generated $\mathbb{C}[t]$-module with an action of $\Lambda$ by additive bijections such that $\lambda(fn) = \lambda(f)\lambda(n)$ for any $\lambda \in \Lambda$, $f \in \mathbb{C}[t]$, and $n \in N$, and assume that $N_{(t)} = 0$. The annihilator of $N$ is $\Lambda$-stable, so $\operatorname{Supp} N$ is a finite $\Lambda$-stable subset of $\mathbb{A}^1 \backslash \{0\}$. Since $\Lambda$ acts transitively on $\mathbb{A}^1 \backslash \{0\}$, $\operatorname{Supp} N = \emptyset$, that is, $N = 0$.

The map res is equivariant, and $\lambda(fD) = \lambda(f)\lambda(D)$ for $\lambda \in \Lambda$ and $f \in \mathbb{C}[t]$. Thus $\Lambda$ acts on the finitely generated $\mathbb{C}[t]$-module $\operatorname{coker}(\mathrm{res})$ as in the fact above, and $\operatorname{coker}(\mathrm{res})_{(t)} = 0$. Hence $\operatorname{coker}(\mathrm{res}) = 0$. This proves (1).

By (1), there exists a basis $D_1, \dots, D_{2b}$ of $L$ over $\mathbb{C}[t]$ such that $D_j(\widetilde{u}_a) = \delta_{ja}$ for any $j$ and $a$. Hence

$$\mathcal{K} = \ker D_1 \cap \cdots \cap \ker D_{2b}.$$

Since $D_j^q(\mathcal{R}_d) \subset \mathcal{R}_{d-qs} = 0$ for $qs > d$, any $D_j$ is locally nilpotent.

We show that $[D_i, D_j] = 0$ for any $i, j$. Through the marking, extend $\{\ ,\ \}$ and $f \to H_f$ to $R \otimes \mathbb{C}(t) \cong \mathcal{R} \otimes \mathbb{C}(t)$ by $\mathbb{C}(t)$-linearity. Let $v_1, \dots, v_{2b}$ be a basis of $R_s$ over $\mathbb{C}$. By **Step 2**, $H_{v_1}, \dots, H_{v_{2b}}$ are $\mathbb{C}(t)$-linearly independent elements of the $2b$-dimensional $\mathbb{C}(t)$-vector space $\mathrm{Der}_{\mathbb{C}(t)}(\mathcal{R} \otimes \mathbb{C}(t))_{-s}$, hence a basis of it. Thus for any $j$ there exists $f_j \in R_s \otimes \mathbb{C}(t)$ such that $D_j = H_{f_j}$ on $\mathcal{R} \otimes \mathbb{C}(t)$. Then

$$[D_i, D_j] = [H_{f_i}, H_{f_j}] = H_{\{f_i, f_j\}}.$$

Since $\{f_i, f_j\}$ has degree $s + s - 2s = 0$, we have $\{f_i, f_j\} \in \mathbb{C}(t)$, and $H_c = 0$ for $c \in \mathbb{C}(t)$. Thus $[D_i, D_j] = 0$ on $\mathcal{R} \otimes \mathbb{C}(t)$, hence on $\mathcal{R} \subset \mathcal{R} \otimes \mathbb{C}(t)$.

Since $\mathcal{R}$ is a domain, $D_1$ is locally nilpotent, and $D_1(\widetilde{u}_1) = 1$, the slice theorem [Fre17, Corollary 1.26] implies that $\mathcal{R} = (\ker D_1)[\widetilde{u}_1]$, and $\widetilde{u}_1$ is transcendental over $\ker D_1$, since applying $D_1$ to a non-trivial polynomial relation of minimal degree $n \geq 1$ gives a non-trivial relation of degree $n - 1$ (the leading coefficient is multiplied by $n \neq 0$). For $j \geq 2$, $D_j$ commutes with $D_1$, so $D_j(\ker D_1) \subset \ker D_1$, and $\widetilde{u}_j \in \ker D_1$ since $D_1(\widetilde{u}_j) = 0$. Thus $D_2|_{\ker D_1}, \dots, D_{2b}|_{\ker D_1}$ are pairwise commuting locally nilpotent derivations with $D_j(\widetilde{u}_a) = \delta_{ja}$ for $2 \leq j, a \leq 2b$, and the common kernel of $D_2|_{\ker D_1}, \dots, D_{2b}|_{\ker D_1}$ is $\mathcal{K}$. By induction on the number of derivations, $\ker D_1 = \mathcal{K}[\widetilde{u}_2, \dots, \widetilde{u}_{2b}]$. Therefore

$$\mathcal{R} = \mathcal{K}[\widetilde{u}_1, \dots, \widetilde{u}_{2b}],$$

where $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$ are algebraically independent over $\mathcal{K}$.

**Step 4.** In this step, we prove the rest of (2), and we show that $\mathcal{K}$ defines a marked test configuration of a normal affine variety $X'$ whose central fiber is isomorphic to $Y$.

By **Step 3**, $L$ is stable under $D \to g \circ D \circ g^{-1}$ for $g \in T \times \Lambda$, so $\mathcal{K}$ is $T \times \Lambda$-stable. Since $D_1, \dots, D_{2b}$ are homogeneous, $\mathcal{K}$ is a graded subring of $\mathcal{R}$, and $\mathcal{K}_0 = \mathbb{C}[t]$ since $D_j(\mathcal{R}_0) \subset \mathcal{R}_{-s} = 0$.

By **Step 3**, $\mathcal{R} = \bigoplus_\alpha \mathcal{K}\widetilde{u}^\alpha$, where $\alpha$ runs over $\mathbb{N}^{2b}$ and $\widetilde{u}^\alpha$ are the monomials in $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$. Hence

$$\mathcal{R}/t\mathcal{R} = \bigoplus_\alpha (\mathcal{K}/t\mathcal{K})u'^\alpha,$$

since $\widetilde{u}_j$ reduces to $u'_j$. Thus the natural map $\mathcal{K}/t\mathcal{K} \to \mathcal{R}/t\mathcal{R} \cong R_c$ is injective. Let $S' \subset R_c$ be the image. Then $R_c = S'[u'_1, \dots, u'_{2b}]$. The reductions $\bar{D}_1, \dots, \bar{D}_{2b}$ of $D_1, \dots, D_{2b}$ modulo $t$ are derivations of $R_c$ of degree $-s$ with $\bar{D}_j(u'_a) = \delta_{ja}$. They are linearly independent, so by (ii) they span the $\mathbb{C}$-span of $\partial/\partial u_1, \dots, \partial/\partial u_{2b}$. Since $\bar{D}_j(S') = 0$ for any $j$, $S'$ is contained in the common kernel of $\partial/\partial u_1, \dots, \partial/\partial u_{2b}$, which is $S$. Conversely, let $x \in S$ and write $x = \sum_\alpha c_\alpha u'^\alpha$ with $c_\alpha \in S' \subset S$. Since $u'_1, \dots, u'_{2b}$ are algebraically independent over $S$ and $x \in S$, $x = c_0 \in S'$. Thus $S' = S$. This proves (2). In particular, we obtain a graded isomorphism $\operatorname{Spec}(\mathcal{K}/t\mathcal{K}) \cong \operatorname{Spec} S = Y$.

The $\mathcal{K}$-algebra homomorphism $\mathcal{R} \to \mathcal{K}$ with $\widetilde{u}_j \to 0$ is surjective, so $\mathcal{K}$ is a finitely generated $\mathbb{C}$-algebra. The ring $\mathcal{K}$ is a domain since $\mathcal{K} \subset \mathcal{R}$, and $\mathcal{K}$ is flat over $\mathbb{C}[t]$ since $\mathcal{K}$ is torsion-free. We show that $\mathcal{K}$ is normal. Let $x \in \operatorname{Frac}(\mathcal{K})$ be integral over $\mathcal{K}$. Then $x \in \operatorname{Frac}(\mathcal{R})$ is integral over $\mathcal{R}$, so $x \in \mathcal{R}$ since $\mathcal{R}$ is normal. Any $D_j$ extends to a derivation of $\operatorname{Frac}(\mathcal{R})$ which vanishes on $\operatorname{Frac}(\mathcal{K})$ by the quotient rule. Thus $D_j(x) = 0$ for any $j$, so $x \in \mathcal{K}$.

Through the marking $\mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$, we have $L[t^{-1}] \cong \mathrm{Der}_{\mathbb{C}[t,t^{-1}]}(R[t, t^{-1}])_{-s}$. For $D \in \mathrm{Der}_{\mathbb{C}[t,t^{-1}]}(R[t, t^{-1}])_{-s}$ and $g \in R$, write $D(g) = \sum_p D_{(p)}(g)t^p$ with $D_{(p)}(g) \in R$. Comparing

coefficients of $t^p$ in the Leibniz rule shows that $D_{(p)} \in \mathrm{Der}_{\mathbb{C}}(R)_{-s}$, and $D_{(p)} \neq 0$ for only finitely many $p$ since $R$ is finitely generated. Hence

$$L[t^{-1}] \cong \mathrm{Der}_{\mathbb{C}}(R)_{-s} \otimes \mathbb{C}[t, t^{-1}].$$

Let $R' \subset R$ be the common kernel of the derivations in $\mathrm{Der}_{\mathbb{C}}(R)_{-s}$. Since localization is exact and $D_1, \dots, D_{2b}$ is a basis of $L$, $\mathcal{K}[t^{-1}]$ is the common kernel of the elements of $L[t^{-1}]$ in $\mathcal{R}[t^{-1}]$. Comparing coefficients of powers of $t$ again, the marking restricts to a $T \times \Lambda$-equivariant isomorphism

$$\mathcal{K}[t^{-1}] \cong R'[t, t^{-1}],$$

where $R'$ is $T$-stable since $T$ preserves $\mathrm{Der}_{\mathbb{C}}(R)_{-s}$ under conjugation, and $\Lambda$ acts trivially on $R'$. The ring $R' \cong \mathcal{K}[t^{-1}]/(t-1)$ is a finitely generated domain, graded with $R'_0 = \mathbb{C}$ since $R_0 = \mathbb{C}$. The ring $R'$ is normal: an element $x \in \mathrm{Frac}(R')$ integral over $R'$ belongs to $R$ since $R$ is normal, and $D(x) = 0$ for any $D \in \mathrm{Der}_{\mathbb{C}}(R)_{-s}$ by the quotient rule, so $x \in R'$. Let $X' := \operatorname{Spec} R'$. Thus $(\operatorname{Spec} \mathcal{K}, T, \Lambda, \zeta)$ is a marked test configuration of $X'$ (Set-up 2.15) with central fiber $X'_0 := \operatorname{Spec}(\mathcal{K}/t\mathcal{K})$, and $X'_0 \cong Y$ as graded varieties.

**Step 5.** In this step, we show that $X'_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form $\sigma'$ of $\zeta$-degree $2s$.

Let $w_j \in R_s$ be the image of $\widetilde{u}_j$ under $\mathcal{R} \to \mathcal{R}[t^{-1}] \cong R[t, t^{-1}] \to R$, where the last map sends $t$ to 1. By **Steps 3 and 4**, $R[t, t^{-1}]$ is a polynomial ring over $R'[t, t^{-1}]$ in the images of $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$, so $R = R'[w_1, \dots, w_{2b}]$ with $w_1, \dots, w_{2b}$ algebraically independent over $R'$. Since $\widetilde{u}_1, \dots, \widetilde{u}_{2b}$ is a basis of $\mathcal{R}_s$ (**Step 1**), $w_1, \dots, w_{2b}$ is a basis of $R_s$.

Base change of (1) along $\mathbb{C}[t] \to \mathbb{C}[t, t^{-1}] \to \mathbb{C}$, $t \to 1$, together with

$$L[t^{-1}] \cong \mathrm{Der}_{\mathbb{C}}(R)_{-s} \otimes \mathbb{C}[t, t^{-1}] \text{ (**Step 4**)}, \qquad \mathcal{R}_s[t^{-1}] \cong R_s \otimes \mathbb{C}[t, t^{-1}] \text{ (**Step 1**)},$$

implies that restriction induces an isomorphism

$$\mathrm{Der}_{\mathbb{C}}(R)_{-s} \to \mathrm{Hom}_{\mathbb{C}}(R_s, \mathbb{C}).$$

In particular, $\dim_{\mathbb{C}} \mathrm{Der}_{\mathbb{C}}(R)_{-s} = 2b$, so the injective map $R_s \to \mathrm{Der}_{\mathbb{C}}(R)_{-s}$, $f \to H_f$ (**Step 2**), is an isomorphism. The composite $R_s \to \mathrm{Hom}_{\mathbb{C}}(R_s, \mathbb{C})$ sends $f$ to $g \to \{f, g\}$. Thus the matrix $(c_{ij})_{1 \le i,j \le 2b}$ with

$$c_{ij} := \{w_i, w_j\} \in R_0 = \mathbb{C}$$

is invertible.

Since $\mathrm{Der}_{\mathbb{C}}(R)_{-s} = \{H_g \mid g \in R_s\}$, $R'$ is the common kernel of $H_g$ for $g \in R_s$, that is, $\{g, f\} = 0$ for any $g \in R_s$ and $f \in R'$. For $f, h \in R'$ and $g \in R_s$, the Jacobi identity implies

$$\{g, \{f, h\}\} = \{\{g, f\}, h\} + \{f, \{g, h\}\} = 0,$$

so $\{f, h\} \in R'$. Thus $\{\ ,\ \}$ restricts to a biderivation of $R'$, which defines a bivector field $\Pi'$ on $X'_{\mathrm{reg}}$. Since $R = R'[w_1, \dots, w_{2b}]$, we have $X \cong X' \times \mathbb{A}^{2b}$ with coordinates $w_1, \dots, w_{2b}$ on $\mathbb{A}^{2b}$, and $X_{\mathrm{reg}} = X'_{\mathrm{reg}} \times \mathbb{A}^{2b}$. Let $\Pi$ be the bivector field of $\sigma$ on $X_{\mathrm{reg}}$, so that $\Pi(df, dh) = \{f, h\}$. The sheaf $\Omega^1_{X_{\mathrm{reg}}}$ is generated by $df$ for $f \in R'$ and $dw_1, \dots, dw_{2b}$. Since $\{f, w_j\} = 0$ for $f \in R'$, we obtain

$$\Pi = \mathrm{pr}_1^* \Pi' + \sum_{i<j} c_{ij} \frac{\partial}{\partial w_i} \wedge \frac{\partial}{\partial w_j}.$$

At any point of $X_{\mathrm{reg}}$, $\Pi$ is non-degenerate and block diagonal with respect to $\Omega^1_{X_{\mathrm{reg}}} = \mathrm{pr}_1^* \Omega^1_{X'_{\mathrm{reg}}} \oplus \mathrm{pr}_2^* \Omega^1_{\mathbb{A}^{2b}}$. Hence $\Pi'$ is non-degenerate at any point of $X'_{\mathrm{reg}}$. Let $\sigma'$ be the 2-form on $X'_{\mathrm{reg}}$ inverse to $\Pi'$. Then

$$\sigma = \mathrm{pr}_1^* \sigma' + \sum_{i<j} c'_{ij} dw_i \wedge dw_j$$

for some $c'_{ij} \in \mathbb{C}$. Let $\iota : X'_{\mathrm{reg}} \to X_{\mathrm{reg}}$ be the zero section $x \to (x, 0)$. Then $\iota^* \sigma = \sigma'$ since $\iota^* dw_j = 0$. Thus $\sigma'$ is a closed non-degenerate algebraic 2-form on $X'_{\mathrm{reg}}$, and since $\iota$ is $\zeta$-equivariant, $\sigma'$ has $\zeta$-degree $2s$.

**Step 6.** In this step, we construct a $T \times \Lambda$-equivariant isomorphism $\mathcal{K} \cong S[t]$ inducing the identity modulo $t$, where $T$ and $\Lambda$ act on $S$ through $\mathcal{K}/t\mathcal{K} \cong S \subset R_c$ and on $t$ as in Set-up 2.15, and we deduce (3).

Recall that $\mathcal{K}$ is the common kernel of $D_1, \dots, D_{2b}$, that $(\operatorname{Spec}\mathcal{K}, T, \Lambda, \zeta)$ is a marked test configuration of $X' = \operatorname{Spec} R'$ with central fiber $X'_0 \cong Y$ (**Step 4**), and that $\sigma'$ is the form on $X'_{\mathrm{reg}}$ from **Step 5**. There are three cases.

**Case 1.** $k \geq 2$. We check that $(\operatorname{Spec}\mathcal{K}, T, \Lambda, \zeta)$ satisfies the conditions of Set-up 9.9 with $r = 2s$: condition (1) holds by assumption since $k \geq 2$, condition (2) holds by **Step 4**, which shows that $(\operatorname{Spec}\mathcal{K}, T, \Lambda, \zeta)$ is a marked test configuration of $X'$ with a graded isomorphism $X'_0 \cong Y$, and condition (3) holds by **Step 5**. By Theorem 9.16, there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{K} \cong S[t]$ inducing the identity modulo $t$.

**Case 2.** $k = 1$. Then $Y = Y_1$ is a conical symplectic variety whose symplectic form $\omega_1$ has degree $2s$, and $H^0((Y_1)_{\mathrm{reg}}, \Omega^2)_{2s} = \mathbb{C}\omega_1$ by Set-up 9.8(5), since $U = (Y_1)_{\mathrm{reg}}$ and $\tau_1 = \omega_1$. Through the graded isomorphism $X'_0 \cong Y_1$ of **Step 4**, $X'_0$ is a conical symplectic variety for the $\zeta$-grading whose symplectic form $\omega_0$ corresponds to $\omega_1$ and has degree $2s$, and $H^0((X'_0)_{\mathrm{reg}}, \Omega^2)_{2s} = \mathbb{C}\omega_0$. Together with **Step 4** and the form $\sigma'$ of **Step 5**, the conditions of Lemma 9.5 hold for $(\operatorname{Spec}\mathcal{K}, T, \Lambda, \zeta)$ with $r = 2s$. By Lemma 9.5, there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{K} \cong (\mathcal{K}/t\mathcal{K})[t] = S[t]$ inducing the identity modulo $t$.

**Case 3.** $k = 0$. Then $S = \mathbb{C}$. Possibly replacing $\mathcal{R}$ with $\mathcal{K}$, the argument of **Step 1** implies that $\mathcal{K}_d$ is a free $\mathbb{C}[t]$-module whose rank is the dimension of the degree-$d$ part of $\mathcal{K}/t\mathcal{K} \cong \mathbb{C}$. Hence $\mathcal{K}_d = 0$ for $d > 0$, and $\mathcal{K} = \mathcal{K}_0 = \mathbb{C}[t] = S[t]$.

In any case, let $\phi : \mathcal{K} \to S[t]$ be a $T \times \Lambda$-equivariant isomorphism of $\mathbb{C}$-algebras inducing the identity modulo $t$. Recall that $u'_1, \dots, u'_{2b}$ is the basis of $\mathbb{C}u_1 \oplus \cdots \oplus \mathbb{C}u_{2b}$ consisting of $T \times \Lambda$-eigenvectors from **Step 1**, so that $R_c[t] = S[t][u'_1, \dots, u'_{2b}]$ is a polynomial ring over $S[t]$. Since $\mathcal{R} = \mathcal{K}[\widetilde{u}_1, \dots, \widetilde{u}_{2b}]$ is a polynomial ring over $\mathcal{K}$ (**Step 3**), $\phi$ extends to an isomorphism of $\mathbb{C}$-algebras

$$\Phi : \mathcal{R} \to R_c[t], \qquad \Phi(\widetilde{u}_j) = u'_j.$$

Since $\phi$ is $T \times \Lambda$-equivariant and the characters of $\widetilde{u}_j$ and $u'_j$ coincide (**Step 1**), $\Phi$ is $T \times \Lambda$-equivariant. Since $\phi$ induces the identity modulo $t$ and $\widetilde{u}_j$ reduces to $u'_j$, $\Phi$ induces the identity modulo $t$. This proves (3). □

9.6. **Product central fibers.** In this subsection, we prove Corollary 9.19.

**Corollary 9.19.** *Notation and conditions as in Set-up 9.8. Let $b \geq 0$ be an integer with $b + k \geq 1$, and with $r$ even if $b > 0$. Let $R_c := S[u_1, \dots, u_{2b}]$ be the polynomial ring over $S$ with the grading of $S$ and $\deg u_j = r/2$, where $R_c = S$ if $b = 0$. Let $(\mathcal{X}, T, \Lambda, \zeta)$ be a marked test configuration of $X = \operatorname{Spec} R$ as in Set-up 2.15 together with a graded isomorphism $\mathcal{R}/t\mathcal{R} \cong R_c$. Assume that*

*(1) $(S_i)_j = 0$ for $0 < j < r$ and any $i$,*
*(2) if $b > 0$, then the vertex of $\operatorname{Spec} R_c$ is klt and the degree valuation of $R_c$ is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex, and*
*(3) $X_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form of $\zeta$-degree $r$.*

*Then there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong R_c[t]$ inducing the identity modulo $t$, where $T$ and $\Lambda$ act on $R_c$ through the isomorphism $\mathcal{R}/t\mathcal{R} \cong R_c$ and $\Lambda$ acts on $t$ with weight $-1$.*

*Proof.* Since $b + k \geq 1$, we have either $b > 0$, or $b = 0$ and $k \geq 2$, or $b = 0$ and $k = 1$.

**Case 1.** $b > 0$. Let $s := r/2$. Since $r \geq 1$ is even, $s \geq 1$. Let $\mathrm{pr}_i : \operatorname{Spec} R_c \to Y_i$ be the projections. By [Bea99, (2.2)],

$$\operatorname{Spec} R_c = \mathbb{A}^{2b} \times Y_1 \times \cdots \times Y_k$$

is a symplectic variety with the symplectic form

$$\sum_{j=1}^{b} du_{2j-1} \wedge du_{2j} + \sum_{i=1}^{k} \mathrm{pr}_i^* \, \omega_i.$$

Since $\deg u_j = s$ and $\deg \omega_i = r$, the symplectic form of $\operatorname{Spec} R_c$ above is homogeneous of degree $2s = r$. Moreover, since $\deg u_j > 0$,

$$(R_c)_0 = S_{(0)} = (S_1)_0 \otimes \cdots \otimes (S_k)_0 = \mathbb{C}.$$

Thus $\operatorname{Spec} R_c$ is a conical symplectic variety in the sense of Definition 2.9.

Next we show that $S_{(j)} = 0$ for $0 < j < r$. If $k = 0$, then $S = \mathbb{C}$ and there is nothing to prove. If $k \geq 1$, then

$$S_{(j)} = \bigoplus_{j_1 + \cdots + j_k = j} (S_1)_{j_1} \otimes \cdots \otimes (S_k)_{j_k}.$$

If $0 < j < r$, then any summand has an index $i$ with $0 < j_i \leq j < r$, and $(S_i)_{j_i} = 0$ by (1). Thus $S_{(j)} = 0$ for $0 < j < r$. Since $s < r$, in particular $S_{(j)} = 0$ for $0 < j \leq s$.

By (2), the degree valuation of $R_c$ is a minimizer of $\widehat{\mathrm{vol}}$ at the vertex. Since $S_{(j)} = 0$ for $0 < j \leq s$, Lemma 7.4 implies that any derivation of $R_c$ of degree $-s$ is a $\mathbb{C}$-linear combination of $\partial/\partial u_1, \ldots, \partial/\partial u_{2b}$. Therefore conditions (i) and (ii) of Theorem 9.18 hold, and condition (iii) of Theorem 9.18 is condition (3) of Corollary 9.19 since $r = 2s$. By conclusion (3) of Theorem 9.18, there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong R_c[t]$ inducing the identity modulo $t$, with the actions in the corollary.

**Case 2.** $b = 0$ and $k \geq 2$. Then $R_c = S$, so the isomorphism $\mathcal{R}/t\mathcal{R} \cong R_c$ is a graded isomorphism $X_0 \cong Y$. Together with (3), the conditions of Set-up 9.9 hold. By Theorem 9.16, there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong S[t] = R_c[t]$ inducing the identity modulo $t$, where $T$ and $\Lambda$ act on $S$ through $X_0 \cong Y$ and $\Lambda$ acts on $t$ with weight $-1$. The actions of $T$ and $\Lambda$ on $S[t]$ are the actions in the corollary.

**Case 3.** $b = 0$ and $k = 1$. Then $R_c = S_1$, so the isomorphism $\mathcal{R}/t\mathcal{R} \cong R_c$ is a graded isomorphism $X_0 \cong Y_1$. Let $\omega_0$ be the 2-form on $(X_0)_{\mathrm{reg}}$ corresponding to $\omega_1$. Then $X_0$ is a conical symplectic variety for the $\zeta$-grading with symplectic form $\omega_0$ of degree $r$. Since $k = 1$, we have $U = (Y_1)_{\mathrm{reg}}$ and $\tau_1 = \omega_1$, so Set-up 9.8(5) reads $H^0((Y_1)_{\mathrm{reg}}, \Omega^2)_r = \mathbb{C}\omega_1$. Hence

$$H^0((X_0)_{\mathrm{reg}}, \Omega^2)_r = \mathbb{C}\omega_0.$$

By (3) and Lemma 9.5, there exists a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ inducing the identity modulo $t$, where the actions on $(\mathcal{R}/t\mathcal{R})[t]$ are those of Set-up 2.15. Composing with the isomorphism $(\mathcal{R}/t\mathcal{R})[t] \cong R_c[t]$ induced by $\mathcal{R}/t\mathcal{R} \cong R_c$, we obtain a $T \times \Lambda$-equivariant isomorphism $\mathcal{R} \cong R_c[t]$ inducing the identity modulo $t$, with the actions in the corollary. □

## 10. K-polystable degenerations of symplectic cones

The goal of this section is to prove Theorem 10.5.

10.1. **Extending covers through test configurations.** In this subsection, we extend a quasi-étale cover of the central fiber of a marked test configuration to a finite cover of the total space, see Theorem 10.4. We first prove two lemmas.

**Set-up 10.1.** Let $\mathcal{A}$, $z$, and $m$ be as follows.

(1) $\mathcal{A} = \bigoplus_{j \geq 0} \mathcal{A}_j$ is a finitely generated $\mathbb{C}$-algebra which is flat over $\mathbb{C}[z]$, such that the grading is induced by a $\mathbb{G}_m$-action fixing $z$, $\mathcal{A}_0 = \mathbb{C}[z]$, and any $\mathcal{A}_j$ is a finite $\mathbb{C}[z]$-module.
(2) $m \geq 1$ is an integer, and a second algebraic $\mathbb{G}_m$-action on $\mathcal{A}$ commutes with the first one and has weight $-m$ on $z$.

The next lemma bounds the codimension of closed subsets of a family over $\mathbb{A}^1$ which are invariant under two commuting torus actions.

**Lemma 10.2.** *Notation and conditions as in Set-up 10.1. Assume that $\mathcal{A}/z\mathcal{A}$ is a domain. Then $\mathcal{A}$ is a domain and $\dim \mathcal{A} = \dim(\mathcal{A}/z\mathcal{A}) + 1$.*

*Moreover, let $c \geq 0$ be an integer, and let $W \subset \operatorname{Spec} \mathcal{A}$ be a closed subset which is invariant under both $\mathbb{G}_m$-actions. Assume that $W \cap \operatorname{Spec}(\mathcal{A}/z\mathcal{A})$ has codimension at least $c$ in $\operatorname{Spec}(\mathcal{A}/z\mathcal{A})$. Then:*

*(1) any irreducible component of $W$ which dominates $\mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[z]$ has codimension at least $c$ in $\operatorname{Spec} \mathcal{A}$;*

*(2) any other irreducible component of $W$ is contained in $\operatorname{Spec}(\mathcal{A}/z\mathcal{A})$ and has codimension at least $c+1$ in $\operatorname{Spec} \mathcal{A}$; and*

*(3) for any $a \in \mathbb{C}\backslash\{0\}$, $W \cap \operatorname{Spec}(\mathcal{A}/(z-a)\mathcal{A})$ has codimension at least $c$ in $\operatorname{Spec}(\mathcal{A}/(z-a)\mathcal{A})$.*

*In particular, $W$ has codimension at least $c$ in $\operatorname{Spec} \mathcal{A}$.*

*Proof.* Since $\mathcal{A}$ is flat over $\mathbb{C}[z]$, any $\mathcal{A}_j$ is a finite torsion-free, hence free, $\mathbb{C}[z]$-module. Thus $\bigcap_{v \geq 0} z^v \mathcal{A} = 0$, and any non-zero $f \in \mathcal{A}$ can be written as $f = z^v f'$ with $v \geq 0$ and $f' \notin z\mathcal{A}$. If $g = z^w g'$ is another non-zero element with $w \geq 0$ and $g' \notin z\mathcal{A}$, then the images of $f'$ and $g'$ in the domain $\mathcal{A}/z\mathcal{A}$ are non-zero, so $f'g' \notin z\mathcal{A}$ and $fg = z^{v+w} f'g' \neq 0$. Thus $\mathcal{A}$ is a domain.

For any $a \in \mathbb{C}$, the element $z - a$ has degree 0, so $\mathcal{A}/(z-a)\mathcal{A}$ is graded with degree-zero part $\mathbb{C}[z]/(z-a) = \mathbb{C}$. In particular, $z-a$ is neither zero nor a unit in $\mathcal{A}$. By [Har77, Theorems I.1.8A and I.1.11A], $\dim V = \dim \mathcal{A} - 1$ for any irreducible component $V$ of $\operatorname{Spec}(\mathcal{A}/(z-a)\mathcal{A})$. The case $a = 0$ implies $\dim \mathcal{A} = \dim(\mathcal{A}/z\mathcal{A}) + 1$.

Since $\mathbb{G}_m \times \mathbb{G}_m$ is connected, it preserves any irreducible component $W'$ of $W$. Let $\mathfrak{p} \subset \mathcal{A}$ be the prime ideal of $W'$. Then $\mathfrak{p}$ is a homogeneous ideal, and $\mathfrak{p} \cap \mathbb{C}[z]$ is invariant under the second action.

(1) Assume that $W'$ dominates $\mathbb{A}^1$, that is, $\mathfrak{p} \cap \mathbb{C}[z] = 0$. Since $\mathfrak{p}$ is homogeneous and $\mathfrak{p} \cap \mathcal{A}_0 = 0$, the degree-zero part of $\mathcal{A}/(\mathfrak{p} + z\mathcal{A})$ is $\mathbb{C}$. Thus the image of $z$ in the domain $\mathcal{A}/\mathfrak{p}$ is neither zero nor a unit, and $W' \cap \operatorname{Spec}(\mathcal{A}/z\mathcal{A}) \neq \emptyset$. By [Har77, Theorems I.1.8A and I.1.11A], $\dim V = \dim W' - 1$ for any irreducible component $V$ of $W' \cap \operatorname{Spec}(\mathcal{A}/z\mathcal{A})$. Since $W' \cap \operatorname{Spec}(\mathcal{A}/z\mathcal{A}) \subset W \cap \operatorname{Spec}(\mathcal{A}/z\mathcal{A})$, we obtain

$$\dim W' - 1 \leq \dim(\mathcal{A}/z\mathcal{A}) - c = \dim \mathcal{A} - 1 - c.$$

Thus $W'$ has codimension at least $c$ in $\operatorname{Spec} \mathcal{A}$.

(2) Assume that $\mathfrak{p} \cap \mathbb{C}[z] \neq 0$. Then $\mathfrak{p} \cap \mathbb{C}[z] = (z-a)$ for some $a \in \mathbb{C}$. Since the second action has weight $-m$ on $z$ and preserves $\mathfrak{p}$, we have $\lambda^{-m} z - a \in (z-a)$ for any $\lambda \in \mathbb{C}^*$. Evaluating at $z = a$, we obtain $\lambda^{-m} a = a$ for any $\lambda \in \mathbb{C}^*$, so $a = 0$. Thus $W' \subset W \cap \operatorname{Spec}(\mathcal{A}/z\mathcal{A})$, and

$$\dim W' \leq \dim(\mathcal{A}/z\mathcal{A}) - c = \dim \mathcal{A} - 1 - c.$$

(3) Let $a \in \mathbb{C}\backslash\{0\}$. By (2), $W \cap \operatorname{Spec}(\mathcal{A}/(z-a)\mathcal{A})$ is the union of the subsets $W' \cap \operatorname{Spec}(\mathcal{A}/(z-a)\mathcal{A})$, where $W'$ runs through the irreducible components of $W$ which dominate $\mathbb{A}^1$. For such $W'$ with prime ideal $\mathfrak{p}$, the image of $z-a$ in $\mathcal{A}/\mathfrak{p}$ is non-zero, so by [Har77, Theorems I.1.8A and I.1.11A] and (1), $\dim V \leq \dim W' - 1 \leq \dim \mathcal{A} - 1 - c$ for any irreducible component $V$ of $W' \cap \operatorname{Spec}(\mathcal{A}/(z-a)\mathcal{A})$. Since $\dim V = \dim \mathcal{A} - 1$ for any irreducible component $V$ of $\operatorname{Spec}(\mathcal{A}/(z-a)\mathcal{A})$, (3) follows.

The last claim follows from (1) and (2). □

The second lemma shows that the families of Set-up 10.1 inherit normality and Cohen–Macaulayness from the fiber over 0.

**Lemma 10.3.** *Notation and conditions as in Set-up 10.1. Assume that $\mathcal{A}/z\mathcal{A}$ is a normal Cohen–Macaulay domain. Then $\mathcal{A}$ is a normal Cohen–Macaulay domain.*

*Proof.* By Lemma 10.2, $\mathcal{A}$ is a domain.

Since both $\mathbb{G}_m$-actions are algebraic and commute, and $\mathcal{A}_0 = \mathbb{C}[z]$, the $\mathbb{C}[z]$-algebra $\mathcal{A}$ is generated by finitely many elements $f_1, \ldots, f_k$ such that $f_i \in \mathcal{A}_{d_i}$ for some integer $d_i \geq 1$ and $f_i$ has weight $b_i$ for the second action. Choose an integer $L \geq 1$ such that $Ld_i - b_i > 0$ for any $i$, and let $\gamma : \mathbb{G}_m \to \mathbb{G}_m \times \mathbb{G}_m$ be the homomorphism $\lambda \to (\lambda^L, \lambda^{-1})$. For the action of $\mathbb{G}_m$ on $\mathcal{A}$

through $\gamma$, the element $f_i$ has weight $Ld_i - b_i > 0$ and $z$ has weight $m > 0$. Let $o \in \operatorname{Spec} \mathcal{A}$ be the closed point defined by the maximal ideal $(z, f_1, \dots, f_k)$. Then $o$ belongs to the closure of the $\gamma(\mathbb{G}_m)$-orbit of any closed point of $\operatorname{Spec} \mathcal{A}$, so any non-empty closed $\gamma(\mathbb{G}_m)$-invariant subset of $\operatorname{Spec} \mathcal{A}$ contains $o$.

Since $o \in \operatorname{Spec}(\mathcal{A}/z\mathcal{A})$ and $\mathcal{A}/z\mathcal{A}$ is Cohen–Macaulay, $\mathcal{A}_o/z\mathcal{A}_o$ is Cohen–Macaulay. Since $\mathcal{A}$ is a domain, $z$ is a nonzerodivisor in $\mathcal{A}_o$, so $\mathcal{A}_o$ is Cohen–Macaulay by [BH93, Theorem 2.1.3]. By [Gro65, Corollaire 6.11.3], the set of points of $\operatorname{Spec} \mathcal{A}$ at which $\mathcal{A}$ is not Cohen–Macaulay is closed. It is $\gamma(\mathbb{G}_m)$-invariant and does not contain $o$, so it is empty. Thus $\mathcal{A}$ is Cohen–Macaulay.

Let $p : \operatorname{Spec} \mathcal{A} \to \mathbb{A}^1$ be the morphism induced by $\mathbb{C}[z] \subset \mathcal{A}$, and let $W \subset \operatorname{Spec} \mathcal{A}$ be the set of points at which $p$ is not smooth. Then $W$ is closed, and $W$ is invariant under both $\mathbb{G}_m$-actions since $p$ is equivariant for the trivial action and for the action of weight $-m$ on $\mathbb{A}^1$, respectively. Since $p$ is flat, a point $x \in \operatorname{Spec}(\mathcal{A}/z\mathcal{A})$ belongs to $W$ if and only if $\operatorname{Spec}(\mathcal{A}/z\mathcal{A})$ is not smooth at $x$ [Gro67, Théorème 17.5.1]. Thus $W \cap \operatorname{Spec}(\mathcal{A}/z\mathcal{A}) = \operatorname{Sing} \operatorname{Spec}(\mathcal{A}/z\mathcal{A})$, which has codimension at least 2 in $\operatorname{Spec}(\mathcal{A}/z\mathcal{A})$ since $\mathcal{A}/z\mathcal{A}$ is normal [Har77, Theorem II.8.22A]. By Lemma 10.2 with $c = 2$, $W$ has codimension at least 2 in $\operatorname{Spec} \mathcal{A}$. Since $\mathbb{A}^1$ is smooth, $\operatorname{Spec} \mathcal{A}$ is regular outside $W$, so $\mathcal{A}$ satisfies $R_1$. Since $\mathcal{A}$ is Cohen–Macaulay, $\mathcal{A}$ satisfies $S_2$. By Serre's criterion [Har77, Theorem II.8.22A], $\mathcal{A}$ is normal. □

The next theorem extends a quasi-étale cover of the central fiber, étale outside a closed subset of codimension at least 3, to a finite cover of the marked test configuration.

**Theorem 10.4.** *Notation and conditions as in Set-up 2.15. Assume that $X_0$ is a conical symplectic variety for the $\zeta$-grading with symplectic form $\omega_0$, and that $X$ is a symplectic variety with an algebraic symplectic form $\omega$ which is homogeneous of $\zeta$-degree $r > 0$. Let $h_0 : X_0' \to X_0$ be a finite surjective quasi-étale morphism from a normal affine variety $X_0'$, and let $\Sigma \subset X_0$ be a $T \times \Lambda$-invariant closed subset of codimension at least 3 such that $h_0$ is étale over $X_0 \backslash \Sigma$. Then there exist integers $e, m \geq 1$ and a finite injective homomorphism of $\mathcal{R}$-algebras $\mathcal{R} \to \mathcal{R}'$ with the following properties.*

*(1) The variety $X_0'$ is a symplectic variety. The ring $\mathcal{R}'$ is a finitely generated normal Cohen–Macaulay domain which is flat over $\mathbb{C}[t]$, and $\mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X_0')$ compatibly with $h_0$. There exist a torus $T'$ acting on $\mathcal{R}'$ and fixing $t$, an isogeny $T' \to T$, a cocharacter $\zeta'$ of $T'$ mapping to $e\zeta$, and an action of $\Lambda' := \mathbb{G}_m$ on $\mathcal{R}'$ which commutes with the action of $T'$ and has weight $-m$ on $t$, such that $\operatorname{Spec} \mathcal{R}' \to \mathcal{X}$ is equivariant with respect to $T' \to T$ and the $m$-th power map $\Lambda' \to \Lambda$. The $\zeta'$-grading of $\mathcal{R}'$ is non-negative with degree-zero part $\mathbb{C}[t]$, and any graded piece is a finite free $\mathbb{C}[t]$-module.*

*(2) The morphism $\operatorname{Spec} \mathcal{R}' \to \mathcal{X}$ is quasi-étale, it is étale over $X_0 \backslash \Sigma$, and the image of the locus where it is not étale has codimension at least $\operatorname{codim}_{X_0} \Sigma$ in $\mathcal{X}$.*

*(3) Let $t = s^m$ and*

$$\widetilde{\mathcal{R}} := \mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[s], \qquad \widetilde{\mathcal{R}}' := \mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[s].$$

*Let $T'$ act on $\widetilde{\mathcal{R}}$ and $\widetilde{\mathcal{R}}'$ through $T' \to T$ and the action on $\mathcal{R}$, and through the action on $\mathcal{R}'$, respectively, fixing $s$, and let $\Lambda'$ act on $\widetilde{\mathcal{R}}$ and $\widetilde{\mathcal{R}}'$ through $\Lambda' \to \Lambda$ and the action on $\mathcal{R}$, and through the action on $\mathcal{R}'$, respectively, with weight $-1$ on $s$. Then $\widetilde{\mathcal{R}}$ and $\widetilde{\mathcal{R}}'$ are normal domains which are flat over $\mathbb{C}[s]$, and there exist $T' \times \Lambda'$-equivariant isomorphisms $\widetilde{\mathcal{R}}[s^{-1}] \cong R[s, s^{-1}]$ and $\widetilde{\mathcal{R}}'[s^{-1}] \cong R'[s, s^{-1}]$ compatible with the inclusion $\widetilde{\mathcal{R}} \subset \widetilde{\mathcal{R}}'$, where $R' := \widetilde{\mathcal{R}}'/(s-1)\widetilde{\mathcal{R}}'$ and $\Lambda'$ acts trivially on $R$ and $R'$. The morphism $X' := \operatorname{Spec} R' \to X$ is finite, surjective, and quasi-étale, and $X'$ is a symplectic variety with the pullback of $\omega$, of $\zeta'$-degree $er$. Moreover, $(\operatorname{Spec} \widetilde{\mathcal{R}}', T', \Lambda', \zeta')$ is a marked test configuration of $X'$ as in Set-up 2.15.*

The condition $\operatorname{codim}_{X_0} \Sigma \geq 3$ is used in (10.1) in **Step 1**: the vanishing of $H^1$ there implies the surjectivity of $\mathcal{S}_N \to \mathcal{S}_{N-1}$ in **Step 2**, and the equalities of rings of functions there extend the covers across $\Sigma$. If $\Sigma$ has codimension 2, then $H^1(X_0 \backslash \Sigma, \mathcal{O})$ can be non-zero, for example

$H^1(\mathbb{A}^2\backslash\{0\},\mathcal{O}) \neq 0$, since $x^{-1}y^{-1}$ defines a non-zero Čech class for the cover by $\{x \neq 0\}$ and $\{y \neq 0\}$, and then the argument in **Steps 1 and 2** does not work.

*Proof of Theorem 10.4.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we show that $X_0'$ is a symplectic variety, compute the cohomology of the structure sheaves of $X_0\backslash\Sigma$ and $h_0^{-1}(X_0\backslash\Sigma)$, and lift the action of $T\times\Lambda$ to $X_0'$.

By Lemma 2.11 applied to $(X_0,\omega_0)$ and $h_0$, $X_0'$ is a symplectic variety. By Lemma 2.10, $X_0$ and $X_0'$ have rational Gorenstein singularities, in particular they are Cohen–Macaulay.

Let $U := X_0\backslash\Sigma$ and $U' := h_0^{-1}(U)$. Since $h_0$ is finite and surjective, $X_0'\backslash U' = h_0^{-1}(\Sigma)$ has codimension at least 3 in $X_0'$. Since $X_0$ is Cohen–Macaulay, $\mathcal{O}_{X_0}$ has depth at least 3 along $\Sigma$, so $H^i_\Sigma(X_0,\mathcal{O}_{X_0}) = 0$ for $i \leq 2$ by [Har67, Theorem 3.8]. Since $X_0$ is affine, the exact sequence of [Har67, Corollary 1.9] implies $\Gamma(U,\mathcal{O}_U) = \mathcal{O}(X_0)$ and $H^1(U,\mathcal{O}_U) \cong H^2_\Sigma(X_0,\mathcal{O}_{X_0}) = 0$. Since $X_0'$ is Cohen–Macaulay and affine, and $h_0^{-1}(\Sigma)$ has codimension at least 3 in $X_0'$, [Har67, Corollary 1.9 and Theorem 3.8] imply the corresponding statements for $U'$. Thus

$$\Gamma(U,\mathcal{O}_U) = \mathcal{O}(X_0),\quad H^1(U,\mathcal{O}_U) = 0,\quad \Gamma(U',\mathcal{O}_{U'}) = \mathcal{O}(X_0'),\quad H^1(U',\mathcal{O}_{U'}) = 0. \tag{10.1}$$

The torus $T\times\Lambda$ acts on $X_0 = \operatorname{Spec}(\mathcal{R}/t\mathcal{R})$, and $U$ is a non-empty $T\times\Lambda$-invariant open subset of $X_0$ over which $h_0$ is étale. By [Liu26b, Lemmas 3.2–3.3], there exist a torus $\widetilde{G}$ acting on $X_0'$ and an isogeny $q : \widetilde{G} \to T\times\Lambda$ such that $h_0$ is equivariant with respect to $q$. The homomorphism $q$ induces an injection of cocharacter lattices with finite cokernel. Let $e \geq 1$ be an integer which annihilates the cokernel. Then the $e$-th power map of $T\times\Lambda$ factors as $q\circ p$ for a homomorphism $p : T\times\Lambda \to \widetilde{G}$. Let $T'$ and $\Lambda' := \mathbb{G}_m$ be copies of $T$ and $\Lambda$, let $T'\times\Lambda'$ act on $X_0'$ through $p$, and let $m := e$. Let $T' \to T$ and $\Lambda' \to \Lambda$ be the $e$-th and the $m$-th power maps, and let $\zeta'$ be the cocharacter of $T'$ given by $\zeta$. Then $T' \to T$ is an isogeny, $\zeta'$ maps to $e\zeta$, the actions of $T'$ and $\Lambda'$ on $X_0'$ commute, and $h_0$ is equivariant with respect to $T'\times\Lambda' \to T\times\Lambda$. We keep the letters $e$ and $m$ although $m = e$, since they play different roles: $e$ relates the gradings, as $\zeta'$ maps to $e\zeta$, and $m$ relates the parameters, as $t = s^m$ in (3). They enter Lemma 9.4 in the proof of Theorem 10.5 in different places.

Let $\mathcal{O}(X_0') = \bigoplus_{j\in\mathbb{Z}} \mathcal{O}(X_0')_j$ be the $\zeta'$-grading. Since $h_0$ is equivariant and $\zeta'$ maps to $e\zeta$, we have $\mathcal{O}(X_0)_j \subset \mathcal{O}(X_0')_{ej}$ for any $j \geq 0$. Let $y \in \mathcal{O}(X_0')_j$ be non-zero with $j \leq 0$. Since $h_0$ is finite, there exist $a_1,\dots,a_k \in \mathcal{O}(X_0)$ with

$$y^k + a_1y^{k-1} + \cdots + a_k = 0.$$

Taking the component of $\zeta'$-degree $kj$, we may assume that $a_i$ has $\zeta'$-degree $ij$ for any $i$. If $j < 0$, then $a_i = 0$ for any $i$ since $\mathcal{O}(X_0)$ has non-negative $\zeta'$-degrees, so $y^k = 0$, which is impossible in the domain $\mathcal{O}(X_0')$. If $j = 0$, then $a_i \in \mathcal{O}(X_0)_0 = \mathbb{C}$ for any $i$, so $y$ is algebraic over $\mathbb{C}$, and $y \in \mathbb{C}$ since $\mathcal{O}(X_0')$ is a domain. Moreover, $\mathcal{O}(X_0')$ is generated as an $\mathcal{O}(X_0)$-module by finitely many homogeneous elements, and $\dim_{\mathbb{C}} \mathcal{O}(X_0)_j < +\infty$ for any $j$. Thus

$$\mathcal{O}(X_0') = \bigoplus_{j\geq 0} \mathcal{O}(X_0')_j,\quad \mathcal{O}(X_0')_0 = \mathbb{C},\quad r_j := \dim_{\mathbb{C}} \mathcal{O}(X_0')_j < +\infty \text{ for any } j \geq 0. \tag{10.2}$$

**Step 2.** In this step we construct finite étale covers of the infinitesimal neighborhoods of $U$ in $\mathcal{X}$, and compute their rings of functions $\mathcal{S}_N$.

For $N \geq 1$, let $U_N$ be the open subscheme of $\operatorname{Spec}(\mathcal{R}/t^N\mathcal{R})$ whose underlying space is $U$. By [Gro03, Exposé I, Théorème 8.3 and Corollaire 8.4], there exist finite étale morphisms $U_N' \to U_N$, unique up to unique isomorphism, such that $U_1' \to U_1$ is $h_0|_{U'} : U' \to U$ and $U_N' \times_{U_N} U_{N-1} \cong U_{N-1}'$ for $N \geq 2$. Let

$$\mathcal{S}_N := \Gamma(U_N',\mathcal{O}_{U_N'}).$$

Since $\mathcal{R}$ is flat over $\mathbb{C}[t]$ and $U_N' \to U_N$ is flat, $\mathcal{O}_{U_N'}$ is flat over $\mathbb{C}[t]/(t^N)$. Thus for $1 \leq l \leq N-1$, the kernel of multiplication by $t^l$ on $\mathcal{O}_{U_N'}$ is $t^{N-l}\mathcal{O}_{U_N'}$, and $\mathcal{O}_{U_N'}/t^{N-l}\mathcal{O}_{U_N'} = \mathcal{O}_{U_{N-l}'}$. Hence

multiplication by $t^l$ induces an exact sequence of sheaves on $U'$

$$0 \to \mathcal{O}_{U'_{N-l}} \to \mathcal{O}_{U'_N} \to \mathcal{O}_{U'_l} \to 0.$$

Applying it with $l = 1$ and using $H^1(U', \mathcal{O}_{U'}) = 0$ from (10.1), induction on $N$ implies $H^1(U', \mathcal{O}_{U'_N}) = 0$ for any $N \geq 1$. Thus we obtain exact sequences

$$0 \to \mathcal{S}_{N-l} \xrightarrow{t^l} \mathcal{S}_N \to \mathcal{S}_l \to 0 \qquad (1 \leq l \leq N-1). \tag{10.3}$$

In particular, $\mathcal{S}_N \to \mathcal{S}_l$ is surjective for any $l \leq N$. Since $\mathcal{S}_N \to \mathcal{S}_{N-l}$ is surjective, the image of the first map in (10.3) is $t^l\mathcal{S}_N$, so

$$\mathcal{S}_N/t^l\mathcal{S}_N \cong \mathcal{S}_l \quad (1 \leq l \leq N), \qquad \mathcal{S}_1 = \mathcal{O}(X'_0), \tag{10.4}$$

where the last equality is (10.1).

The pullback $\mathcal{R}/t^N\mathcal{R} \to \mathcal{S}_N$ is injective. Let $f \in \mathcal{R}/t^N\mathcal{R}$ map to 0 in $\mathcal{S}_N$. Since $U'_N \to U_N$ is finite, étale, and surjective, it is faithfully flat, so $f|_{U_N} = 0$. Then the image of $f$ in $\mathcal{O}(X_0)$ vanishes on the dense open subset $U$ of the variety $X_0$, so $f = tg$ for some $g \in \mathcal{R}/t^{N-1}\mathcal{R}$ since $\mathcal{R}$ is flat over $\mathbb{C}[t]$. Since multiplication by $t$ induces an injection $\mathcal{O}_{U_{N-1}} \to \mathcal{O}_{U_N}$, $g|_{U_{N-1}} = 0$, and induction on $N$ implies $f = 0$.

We lift the action of $T' \times \Lambda'$ to $U'_N$. Let $T' \times \Lambda'$ act on $U_N$ through $T' \times \Lambda' \to T \times \Lambda$, and let $a_N, p_N : (T' \times \Lambda') \times U_N \to U_N$ be the action morphism and the projection. The pullbacks $a_N^*U'_N$ and $p_N^*U'_N$ are finite étale over $(T' \times \Lambda') \times U_N$, and the action of $T' \times \Lambda'$ on $U'$ from **Step 1** is an isomorphism $\alpha_1 : p_1^*U' \to a_1^*U'$ over $(T' \times \Lambda') \times U$. By [Gro03, Exposé I, Théorème 8.3 and Corollaire 8.4], $\alpha_1$ lifts uniquely to an isomorphism $\alpha_N : p_N^*U'_N \to a_N^*U'_N$. The uniqueness of lifts over $(T' \times \Lambda') \times (T' \times \Lambda') \times U_N$ implies that $\alpha_N$ defines an action of $T' \times \Lambda'$ on $U'_N$ such that $U'_N \to U_N$ is equivariant, and the uniqueness of lifts over $(T' \times \Lambda') \times U_{N-1}$ implies that the actions on $U'_{N-1}$ and $U'_N$ are compatible. Since $U'_N$ is quasi-compact and separated, $\Gamma((T' \times \Lambda') \times U'_N, \mathcal{O}) = \mathcal{O}(T' \times \Lambda') \otimes_{\mathbb{C}} \mathcal{S}_N$ [Har77, Proposition III.9.3], so $T' \times \Lambda'$ acts algebraically on $\mathcal{S}_N$. The maps $\mathcal{S}_N \to \mathcal{S}_{N-1}$ are equivariant, and the maps $\mathcal{R}/t^N\mathcal{R} \to \mathcal{S}_N$ are equivariant with respect to $T' \times \Lambda' \to T \times \Lambda$. Moreover, $T'$ fixes $t$, and $\Lambda'$ acts on $t$ with weight $-m$, since $\Lambda$ acts on $t$ with weight $-1$.

**Step 3.** In this step we construct $\mathcal{R}'$ and the homomorphism $\mathcal{R} \to \mathcal{R}'$, and prove that $\mathcal{R}'$ is finite over $\mathcal{R}$.

Recall that $\mathcal{S}_N = \Gamma(U'_N, \mathcal{O}_{U'_N})$. Let $\mathcal{S}_N = \bigoplus_{j \in \mathbb{Z}} (\mathcal{S}_N)_j$ be the $\zeta'$-grading. Since $\zeta'$ fixes $t$, (10.3) with $l = 1$ and (10.4) imply that for any $j$ the sequence

$$0 \to (\mathcal{S}_{N-1})_j \xrightarrow{t} (\mathcal{S}_N)_j \to \mathcal{O}(X'_0)_j \to 0$$

is exact, and that $(\mathcal{S}_N)_j/t(\mathcal{S}_N)_j \cong \mathcal{O}(X'_0)_j$. By (10.2) and induction on $N$, $(\mathcal{S}_N)_j = 0$ for $j < 0$ and $\dim_{\mathbb{C}}(\mathcal{S}_N)_j = Nr_j$ for $j \geq 0$. By the structure theorem for finitely generated modules over $\mathbb{C}[t]$, $(\mathcal{S}_N)_j$ is a direct sum of $r_j$ modules of the form $\mathbb{C}[t]/(t^n)$ with $1 \leq n \leq N$, and $\dim_{\mathbb{C}}(\mathcal{S}_N)_j = Nr_j$ forces $n = N$ for all of them. Thus $(\mathcal{S}_N)_j$ is a free $\mathbb{C}[t]/(t^N)$-module of rank $r_j$. Since $r_0 = 1$ and $(\mathcal{S}_N)_0$ contains the image of $(\mathcal{R}/t^N\mathcal{R})_0 = \mathbb{C}[t]/(t^N)$, we obtain $(\mathcal{S}_N)_0 = \mathbb{C}[t]/(t^N)$.

Since $T' \times \Lambda'$ commutes with $\zeta'$, $T' \times \Lambda'$ acts on the space $\mathcal{O}(X'_0)_j$ with $\dim_{\mathbb{C}} \mathcal{O}(X'_0)_j = r_j$. For any $j \geq 0$, fix a basis of $\mathcal{O}(X'_0)_j$ consisting of $T' \times \Lambda'$-eigenvectors. The surjections $(\mathcal{S}_N)_j \to (\mathcal{S}_{N-1})_j$ are $T' \times \Lambda'$-equivariant maps of finite-dimensional algebraic representations, so any eigenvector in $(\mathcal{S}_{N-1})_j$ with character $\chi$ lifts to an eigenvector in $(\mathcal{S}_N)_j$ with character $\chi$. By induction on $N$, we obtain elements $x_{j,i}^{(N)} \in (\mathcal{S}_N)_j$, $1 \leq i \leq r_j$, such that $x_{j,i}^{(N)}$ maps to $x_{j,i}^{(N-1)}$, $x_{j,1}^{(1)}, \ldots, x_{j,r_j}^{(1)}$ is the fixed basis of $\mathcal{O}(X'_0)_j$, and $x_{j,i}^{(N)}$ is a $T' \times \Lambda'$-eigenvector whose character equals the character of $x_{j,i}^{(1)}$. By Nakayama's lemma, $x_{j,1}^{(N)}, \ldots, x_{j,r_j}^{(N)}$ is a $\mathbb{C}[t]/(t^N)$-basis of $(\mathcal{S}_N)_j$.

Let $b_{j,i}$ be the $\Lambda'$-weight of $x_{j,i}^{(N)}$, let

$$\widehat{\mathcal{S}} := \varprojlim_N \mathcal{S}_N,$$

and let $x_{j,i} := (x_{j,i}^{(N)})_N \in \widehat{\mathcal{S}}$. Then $\varprojlim_N (\mathcal{S}_N)_j$ is a free $\mathbb{C}[[t]]$-module with basis $x_{j,1}, \dots, x_{j,r_j}$.

Let $\mathcal{R}' \subset \widehat{\mathcal{S}}$ be the $\mathbb{C}$-linear span of the elements $x = (x_N)_N \in \widehat{\mathcal{S}}$ for which there exist integers $j$ and $b$ such that $x_N \in (\mathcal{S}_N)_j$ and $x_N$ has $\Lambda'$-weight $b$ for any $N$. The product of two such elements is again such an element, and so are 1 and $t$, so $\mathcal{R}'$ is a $\mathbb{C}[t]$-subalgebra of $\widehat{\mathcal{S}}$. Let $x \in \varprojlim_N (\mathcal{S}_N)_j$, and write $x = \sum_i a_i(t) x_{j,i}$ with $a_i(t) = \sum_{v \geq 0} c_{i,v} t^v \in \mathbb{C}[[t]]$. For $u \in \Lambda'$, we have

$$u \cdot x = \sum_i a_i(u^{-m}t) u^{b_{j,i}} x_{j,i} = \sum_i \sum_{v \geq 0} c_{i,v} u^{b_{j,i} - mv} t^v x_{j,i}.$$

Thus $u \cdot x = u^b x$ for any $u \in \Lambda'$ if and only if $c_{i,v} = 0$ whenever $b_{j,i} - mv \neq b$. In this case $a_i(t) = c_{i,v_i} t^{v_i}$ for the unique integer $v_i \geq 0$ with $b_{j,i} - mv_i = b$ if it exists, and $a_i(t) = 0$ otherwise. Conversely, $t^v x_{j,i}$ has $\zeta'$-degree $j$ and $\Lambda'$-weight $b_{j,i} - mv$. Hence

$$\mathcal{R}' = \bigoplus_{j \geq 0} \mathcal{R}'_j, \qquad \mathcal{R}'_j = \bigoplus_{i=1}^{r_j} \mathbb{C}[t] x_{j,i}. \tag{10.5}$$

By (10.5), $\mathcal{R}'_0 = \mathbb{C}[t]$, any $\mathcal{R}'_j$ is a free $\mathbb{C}[t]$-module of rank $r_j$, and $\mathcal{R}'$ is flat over $\mathbb{C}[t]$. Comparing the bases $x_{j,i}$ and $x_{j,i}^{(N)}$, the projection $\mathcal{R}' \to \mathcal{S}_N$ is surjective with kernel $t^N \mathcal{R}'$, so $\mathcal{R}'/t^N\mathcal{R}' \cong \mathcal{S}_N$. In particular, $\mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X'_0)$ by (10.4). The actions of $T'$ and $\Lambda'$ on $\widehat{\mathcal{S}}$ preserve $\mathcal{R}'$ since they commute with each other and with $\zeta'$. They are algebraic on $\mathcal{R}'$ since $\mathcal{R}'$ is spanned by the joint eigenvectors $t^v x_{j,i}$. Moreover, $T'$ fixes $t$, and $\Lambda'$ acts on $t$ with weight $-m$.

Since $\mathcal{R}$ is finitely generated and $\mathcal{R}_0 = \mathbb{C}[t]$, any $\mathcal{R}_j$ is a finite $\mathbb{C}[t]$-module, which is free since $\mathcal{R}$ is flat over $\mathbb{C}[t]$. Thus $\bigcap_N t^N \mathcal{R} = 0$. The compatible maps $\mathcal{R} \to \mathcal{R}/t^N\mathcal{R} \to \mathcal{S}_N$ induce a homomorphism $\mathcal{R} \to \widehat{\mathcal{S}}$, which is injective by **Step 2** and $\bigcap_N t^N \mathcal{R} = 0$. The ring $\mathcal{R}$ is spanned by elements $f \in \mathcal{R}_j$ of $\Lambda$-weight $b$ for some $j$ and $b$, and the image of such $f$ in $\mathcal{S}_N$ has $\zeta'$-degree $ej$ and $\Lambda'$-weight $mb$ for any $N$. Thus we obtain an injective homomorphism of $\mathbb{C}[t]$-algebras $\mathcal{R} \to \mathcal{R}'$, which is equivariant with respect to $T' \times \Lambda' \to T \times \Lambda$. The reduction of $\mathcal{R} \to \mathcal{R}'$ modulo $t$ is the pullback $\mathcal{O}(X_0) \to \Gamma(U', \mathcal{O}_{U'}) = \mathcal{O}(X'_0)$ by $h_0$, so the isomorphism $\mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X'_0)$ is compatible with $h_0$.

We show that $\mathcal{R}'$ is a finite $\mathcal{R}$-module. Choose $T' \times \Lambda'$-eigenvectors $y_1, \dots, y_k \in \mathcal{O}(X'_0)$ which generate $\mathcal{O}(X'_0)$ as an $\mathcal{O}(X_0)$-module, and lift them to $T' \times \Lambda'$-eigenvectors $\tilde{y}_1, \dots, \tilde{y}_k \in \mathcal{R}'$ such that $\tilde{y}_l$ and $y_l$ have equal characters for each $l$, which is possible since $\mathcal{R}' \to \mathcal{O}(X'_0)$ is surjective and equivariant and the actions are algebraic. Let $x \in \mathcal{R}'_j$ be a $T' \times \Lambda'$-eigenvector of $\Lambda'$-weight $b$. Write the image $\bar{x}$ of $x$ in $\mathcal{O}(X'_0)$ as $\bar{x} = \sum_l \bar{f}_l y_l$, where $\bar{f}_l \in \mathcal{O}(X_0)$ are $T \times \Lambda$-eigenvectors such that the character of $\bar{f}_l y_l$ equals the character of $\bar{x}$. This is possible by taking the components of the character of $\bar{x}$. Lift each $\bar{f}_l$ to a $T \times \Lambda$-eigenvector $f_l \in \mathcal{R}$ whose character equals the character of $\bar{f}_l$. Then $x - \sum_l f_l \tilde{y}_l$ is a $T' \times \Lambda'$-eigenvector in $\mathcal{R}'_j$ with zero image in $\mathcal{O}(X'_0)$, so $x - \sum_l f_l \tilde{y}_l = t x_1$ for some eigenvector $x_1 \in \mathcal{R}'_j$ of $\Lambda'$-weight $b + m$. By (10.5), the $\Lambda'$-weights of non-zero eigenvectors in $\mathcal{R}'_j$ are at most $\max_i b_{j,i}$. Repeating the argument for $x_1$, after finitely many steps we obtain $x \in \sum_l \mathcal{R} \tilde{y}_l$. Since $\mathcal{R}'$ is spanned by such $x$, $\mathcal{R}' = \sum_l \mathcal{R} \tilde{y}_l$ is a finite $\mathcal{R}$-module. In particular, $\mathcal{R}'$ is a finitely generated $\mathbb{C}$-algebra, and $\operatorname{Spec} \mathcal{R}' \to \mathcal{X}$ is equivariant with respect to $T' \to T$ and $\Lambda' \to \Lambda$.

**Step 4.** In this step we prove that $\mathcal{R}'$ is a normal Cohen–Macaulay domain, which completes the proof of (1).

By **Step 3**, $\mathcal{R}'$ is a finitely generated $\mathbb{C}$-algebra which is flat over $\mathbb{C}[t]$, the $\zeta'$-grading $\mathcal{R}' = \bigoplus_{j \geq 0} \mathcal{R}'_j$ is induced by a $\mathbb{G}_m$-action fixing $t$, $\mathcal{R}'_0 = \mathbb{C}[t]$, any $\mathcal{R}'_j$ is a finite free $\mathbb{C}[t]$-module, and the action of $\Lambda'$ commutes with $\zeta'$ and has weight $-m$ on $t$. Thus Set-up 10.1 holds

for $\mathcal{A} = \mathcal{R}'$, $z = t$, and $m$, with the $\zeta'$-grading and the action of $\Lambda'$. By **Step 1**, $\mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X_0')$ is a normal Cohen–Macaulay domain. By Lemma 10.3, $\mathcal{R}'$ is a normal Cohen–Macaulay domain. Together with **Steps 1 and 3**, this proves (1).

**Step 5.** In this step we prove (2).

Let $h : \operatorname{Spec}\mathcal{R}' \to \mathcal{X}$ be the morphism induced by $\mathcal{R} \to \mathcal{R}'$. Since $\mathcal{R} \to \mathcal{R}'$ is finite and injective, $h$ is finite and surjective. Recall that $U = X_0\backslash\Sigma$. Let $\mathfrak{p} \in U$, viewed as a prime ideal of $\mathcal{R}$ containing $t$, let $A := \mathcal{R}_{\mathfrak{p}}$, and let $M := \mathcal{R}' \otimes_{\mathcal{R}} A$, a finite $A$-module. Then $M/tM = \mathcal{O}(X_0') \otimes_{\mathcal{O}(X_0)} \mathcal{O}_{X_0,\mathfrak{p}}$ is flat over $A/tA = \mathcal{O}_{X_0,\mathfrak{p}}$ since $h_0$ is étale over $U$. Since $\mathcal{R}'$ is flat over $\mathbb{C}[t]$, $t$ is a nonzerodivisor on $M$, so $\operatorname{Tor}_1^A(M, A/tA) = 0$. By [Mat89, Theorem 22.3], $M$ is flat over $A$. By [Har77, Proposition II.8.10],

$$\Omega_{\mathcal{R}'/\mathcal{R}} \otimes_{\mathcal{R}} \mathcal{R}/t\mathcal{R} = \Omega_{\mathcal{O}(X_0')/\mathcal{O}(X_0)},$$

whose localization at $\mathfrak{p}$ is zero since $h_0$ is étale over $U$. Since $\Omega_{\mathcal{R}'/\mathcal{R}} \otimes_{\mathcal{R}} A$ is a finite $A$-module and $t \in \mathfrak{p}A$, Nakayama's lemma implies $\Omega_{\mathcal{R}'/\mathcal{R}} \otimes_{\mathcal{R}} A = 0$. Thus $h$ is flat and unramified, hence étale, at any point of $h^{-1}(\mathfrak{p})$. Hence $h$ is étale over $U = X_0\backslash\Sigma$.

Let $W \subset \mathcal{X}$ be the image of the closed subset of $\operatorname{Spec}\mathcal{R}'$ where $h$ is not étale. Since $h$ is finite, $W$ is closed. Since $h$ is equivariant with respect to the surjective homomorphisms $T' \to T$ and $\Lambda' \to \Lambda$, $W$ is $T \times \Lambda$-invariant, in particular invariant under the $\mathbb{G}_m$-action $\zeta$ and under $\Lambda$. Since $h$ is étale over $U$, $W \cap X_0 \subset \Sigma$. Let $c := \operatorname{codim}_{X_0} \Sigma \geq 3$. By **Step 3**, $\mathcal{R}$ is a finitely generated $\mathbb{C}$-algebra which is flat over $\mathbb{C}[t]$, $\mathcal{R}_0 = \mathbb{C}[t]$, any $\mathcal{R}_j$ is a finite $\mathbb{C}[t]$-module, and $\Lambda$ acts on $t$ with weight $-1$. Thus Set-up 10.1 holds for $\mathcal{A} = \mathcal{R}$, $z = t$, and $m = 1$, with the $\zeta$-grading and the action of $\Lambda$. Since $\mathcal{R}/t\mathcal{R} = \mathcal{O}(X_0)$ is a domain, Lemma 10.2 implies that $W$ has codimension at least $c$ in $\mathcal{X}$, and that for any $a \in \mathbb{C}\backslash\{0\}$, $W \cap \operatorname{Spec}(\mathcal{R}/(t-a)\mathcal{R})$ has codimension at least $c$ in $\operatorname{Spec}(\mathcal{R}/(t-a)\mathcal{R})$. Since $\mathcal{X}$ and $\operatorname{Spec}\mathcal{R}'$ are normal varieties by Set-up 2.15 and **Step 4**, and $c \geq 2$, $h$ is quasi-étale. This proves (2).

**Step 6.** In this step we prove (3).

Let $t = s^m$, $\widetilde{\mathcal{R}} := \mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$, and $\widetilde{\mathcal{R}}' := \mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$. Since $\mathbb{C}[s]$ is flat over $\mathbb{C}[t]$, $\widetilde{\mathcal{R}} \subset \widetilde{\mathcal{R}}'$, and $\widetilde{\mathcal{R}}$ and $\widetilde{\mathcal{R}}'$ are flat over $\mathbb{C}[s]$. Let $T'$ act on $\widetilde{\mathcal{R}}$ and $\widetilde{\mathcal{R}}'$ through $T' \to T$ and the action on $\mathcal{R}$, and through the action on $\mathcal{R}'$, respectively, fixing $s$. Let $u \in \Lambda'$ act on $\widetilde{\mathcal{R}}$ and $\widetilde{\mathcal{R}}'$ through $\Lambda' \to \Lambda$ and the action on $\mathcal{R}$, and through the action on $\mathcal{R}'$, respectively, and by $s \to u^{-1}s$. This is well defined since $(u^{-1}s)^m = u^{-m}s^m$ and $\Lambda'$ acts on $t$ by $t \to u^{-m}t$, so the action respects the relation $t = s^m$. The $\zeta$-grading and the $\zeta'$-grading

$$\widetilde{\mathcal{R}} = \bigoplus_{j\geq 0} \mathcal{R}_j \otimes_{\mathbb{C}[t]} \mathbb{C}[s], \qquad \widetilde{\mathcal{R}}' = \bigoplus_{j\geq 0} \mathcal{R}_j' \otimes_{\mathbb{C}[t]} \mathbb{C}[s]$$

have degree-zero parts $\mathbb{C}[s]$ and graded pieces which are finite free $\mathbb{C}[s]$-modules, and the actions of $\Lambda'$ commute with $\zeta$ and $\zeta'$. Thus Set-up 10.1 holds for $\mathcal{A} = \widetilde{\mathcal{R}}$ with the $\zeta$-grading and for $\mathcal{A} = \widetilde{\mathcal{R}}'$ with the $\zeta'$-grading, with $z = s$, $m = 1$, and the action of $\Lambda'$. Moreover,

$$\widetilde{\mathcal{R}}/s\widetilde{\mathcal{R}} = \mathcal{R}/t\mathcal{R} = \mathcal{O}(X_0), \qquad \widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}' = \mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X_0'),$$

which are normal Cohen–Macaulay domains by **Step 1**. By Lemma 10.3, the rings $\widetilde{\mathcal{R}}$ and $\widetilde{\mathcal{R}}'$ are normal Cohen–Macaulay domains.

We construct the isomorphisms over $s \neq 0$. Let $\mathcal{A}$ be $\widetilde{\mathcal{R}}$ or $\widetilde{\mathcal{R}}'$. Since $\Lambda'$ acts algebraically on $\mathcal{A}[s^{-1}]$, $\mathcal{A}[s^{-1}]$ is the direct sum of the $\Lambda'$-weight spaces $\mathcal{A}[s^{-1}]^{(b)}$, $b \in \mathbb{Z}$. Let $I_{\mathcal{A}} := \mathcal{A}[s^{-1}]^{(0)}$. Since $s$ is a unit of $\Lambda'$-weight $-1$, $\mathcal{A}[s^{-1}]^{(b)} = s^{-b}I_{\mathcal{A}}$, so the natural map $I_{\mathcal{A}}[s, s^{-1}] \to \mathcal{A}[s^{-1}]$ is an isomorphism. Since $s$ is a unit modulo $s-1$, reducing modulo $s-1$ implies that $I_{\mathcal{A}} \to \mathcal{A}/(s-1)\mathcal{A}$ is an isomorphism. Combining $I_{\mathcal{A}}[s, s^{-1}] \cong \mathcal{A}[s^{-1}]$ and $I_{\mathcal{A}} \cong \mathcal{A}/(s-1)\mathcal{A}$, we obtain an isomorphism

$$\mathcal{A}[s^{-1}] \cong \big(\mathcal{A}/(s-1)\mathcal{A}\big)[s, s^{-1}], \qquad f \to \bar{f}s^{-b} \ \text{ for } f \in \mathcal{A}[s^{-1}]^{(b)}, \tag{10.6}$$

where $\bar{f}$ is the image of $f$ in $\mathcal{A}/(s-1)\mathcal{A}$. It is $T' \times \Lambda'$-equivariant, where $\Lambda'$ acts trivially on $\mathcal{A}/(s-1)\mathcal{A}$, and $T'$ acts on $\mathcal{A}/(s-1)\mathcal{A}$ through the action on $\mathcal{A}$, since $T'$ fixes $s$ and preserves

$\Lambda'$-weights. Moreover, (10.6) commutes with $\widetilde{\mathcal{R}} \subset \widetilde{\mathcal{R}}'$. For $\mathcal{A} = \widetilde{\mathcal{R}}'$, (10.6) is an equivariant isomorphism $\widetilde{\mathcal{R}}'[s^{-1}] \cong R'[s, s^{-1}]$. For $\mathcal{A} = \widetilde{\mathcal{R}}$, we have $\widetilde{\mathcal{R}}/(s-1)\widetilde{\mathcal{R}} = \mathcal{R}/(t-1)\mathcal{R}$, which is isomorphic to $R$ by the isomorphism $\mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$ of Set-up 2.15. Under $\mathcal{R}/(t-1)\mathcal{R} \cong R$, (10.6) for $\widetilde{\mathcal{R}}$ is the base change of $\mathcal{R}[t^{-1}] \cong R[t, t^{-1}]$ along $t = s^m$, since both send an element $f \in \mathcal{R}$ of $\Lambda$-weight $b$ to $\bar{f} s^{-mb}$. This proves the existence of the compatible equivariant isomorphisms in (3).

Since $\widetilde{\mathcal{R}}'[s^{-1}]$ is a normal domain, so is $R'[s, s^{-1}]$, and hence so is $R'$: an element of the fraction field of $R'$ which is integral over $R'$ belongs to $R'[s, s^{-1}]$, hence to $R'$ by comparing degrees in $s$. Thus $X' = \operatorname{Spec} R'$ is a normal affine variety. The morphism $X' \to X$ is the reduction of $\operatorname{Spec} \widetilde{\mathcal{R}}' \to \operatorname{Spec} \widetilde{\mathcal{R}}$ modulo $s-1$, so it is finite. Since $R[s, s^{-1}] \to R'[s, s^{-1}]$ is injective, $R \to R'$ is injective, so $X' \to X$ is surjective. Moreover, $\widetilde{\mathcal{R}}'/(s-1)\widetilde{\mathcal{R}}' = \mathcal{R}'/(t-1)\mathcal{R}'$, so $X' \to X$ is the restriction of $h$ over the fiber $\operatorname{Spec}(\mathcal{R}/(t-1)\mathcal{R}) \cong X$ of $\mathcal{X}$ over $t = 1$. By **Step 5** and the invariance of étale morphisms under base change, $X' \to X$ is étale over the complement of $W \cap \operatorname{Spec}(\mathcal{R}/(t-1)\mathcal{R})$, which has codimension at least $c \geq 3$. Thus $X' \to X$ is quasi-étale.

By Lemma 2.11 applied to $(X, \omega)$ and $X' \to X$, $X'$ is a symplectic variety with the extension of the pullback of $\omega$. Since $X' \to X$ is equivariant with respect to $T' \to T$, $\zeta'$ maps to $e\zeta$, and $\omega$ has $\zeta$-degree $r$, the pullback of $\omega$ has $\zeta'$-degree $er$.

Finally, we check the conditions of Set-up 2.15 for $(\operatorname{Spec} \widetilde{\mathcal{R}}', T', \Lambda', \zeta')$. The ring $R'$ is graded by $\zeta'$ with $R'_j = \mathcal{R}'_j/(t-1)\mathcal{R}'_j$, so $R' = \bigoplus_{j \geq 0} R'_j$ and $R'_0 = \mathbb{C}$ by **Step 3**. The ring $\widetilde{\mathcal{R}}'$ is a finitely generated normal domain which is flat over $\mathbb{C}[s]$. The groups $T'$ and $\Lambda'$ act algebraically on $\widetilde{\mathcal{R}}'$ with commuting actions, $T'$ fixes $s$, $\Lambda'$ acts on $s$ with weight $-1$, and the $\zeta'$-grading of $\widetilde{\mathcal{R}}'$ is non-negative with degree-zero part $\mathbb{C}[s]$. The isomorphism $\widetilde{\mathcal{R}}'[s^{-1}] \cong R'[s, s^{-1}]$ is $T' \times \Lambda'$-equivariant, and $\Lambda'$ acts trivially on $R'$. Thus $(\operatorname{Spec} \widetilde{\mathcal{R}}', T', \Lambda', \zeta')$ is a marked test configuration of $X'$. This proves (3). □

10.2. **The K-polystable degeneration is a product.** In this subsection, we prove Theorem 10.5.

The proof of Theorem 10.5 has four parts. First, $X_0$ admits a finite cover which is a product (Theorem 8.8). The proof of Theorem 8.8 is in Section 8 and uses the metric. Second, the cover extends along the test configuration (Theorem 10.4). Third, the extended family is a product (Corollary 9.19, Section 9). Fourth, the product structure descends to $\mathcal{X}$ (Lemma 9.4). The new arguments are in the first and the third parts. The main point of the third part is that the symplectic forms of the factors can occur with different orders in $t$. In the second part, the lifting of the cover to the infinitesimal neighborhoods of the central fiber follows from [Gro03, Exposé I, Théorème 8.3 and Corollaire 8.4]. The new point is to extend the cover across the branch locus by using codimension at least 3. Passing to the $\Lambda$-invariant closure lowers the codimension by at most 1, and for this reason Theorem 8.8 asks the cover to be étale outside a closed subset of codimension at least 4. The fourth part is short. The normalized trace is used there only because the cover need not be Galois.

**Theorem 10.5.** *Let $X = \operatorname{Spec} R$ be a conical symplectic variety of dimension $2n$ with symplectic form $\omega$ of degree $r$, whose grading is induced by a faithful $\mathbb{G}_m$-action $\zeta$, let $T$ be a torus acting faithfully on $X$ which contains $\zeta(\mathbb{G}_m)$, and let $(\mathcal{X}, T, \Lambda, \zeta)$ be a marked test configuration of $X$ as in Set-up 2.15. Assume that*

*(i) $n \geq 2$,*
*(ii) $(X, \zeta)$ is a K-semistable Fano cone,*
*(iii) $\mathcal{X}$, with the vector field $\eta$ generating the action of $\Lambda$, is a special test configuration of $(X, \zeta)$ with Futaki invariant 0 [LWX18, Definitions 2.16 and 2.22], and*
*(iv) the central fiber $(X_0, \zeta)$ is K-polystable for a torus which acts faithfully on $X_0$ and contains $T$.*

*Then there exists a $T \times \Lambda$-equivariant isomorphism of $\mathbb{C}[t]$-algebras $\mathcal{R} \cong (\mathcal{R}/t\mathcal{R})[t]$ inducing the identity modulo $t$, where $T$ and $\Lambda$ act on $\mathcal{R}/t\mathcal{R}$ through their actions on $X_0$ and $\Lambda$ acts on $t$ with weight $-1$. In particular, $X \cong X_0$ $T$-equivariantly.*

*Proof.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we show that $A_{X_0}(\mathrm{wt}_\xi) = 2n$ for $\xi := \frac{2}{r}\zeta$.

Let $T_0$ be a torus which acts faithfully on $X_0$ and contains $T$ such that $(X_0, \zeta)$ is K-polystable for $T_0$. Then $(X_0, \zeta)$ is a Fano cone and $\zeta$ is a Reeb vector of $T_0$ (Definition 2.16). In particular, $X_0$ is an integral normal affine variety with klt singularities, the actions of $\zeta(\mathbb{G}_m)$, $T$, and $T_0$ on $X_0$ are good, and $\xi$ is a Reeb vector of $\zeta(\mathbb{G}_m)$, of $T$, and of $T_0$ (Definition 2.6). Let $o$ be the vertex of $X_0$. Since $\zeta$ acts faithfully on $X$, the cocharacter $\zeta : \mathbb{G}_m \to T$ is injective. Since $T_0$ acts faithfully on $X_0$, so do $T$ and $\zeta$.

Let $\mathcal{U} \subset \mathcal{X}$ be the smooth locus of $\mathcal{X} \to \mathbb{A}^1$. A flat morphism is smooth at any point of a fiber at which the fiber is smooth [Gro67, Théorème 17.5.1], so $(X_0)_{\mathrm{reg}} \subset \mathcal{U}$. The hypotheses of Definition-Lemma 4.1 hold for $\sigma := \omega$: $\dim X = 2n$, $X_0$ is integral and normal, and $\omega$ is a closed non-degenerate algebraic 2-form on $X_{\mathrm{reg}}$ of $\zeta$-degree $r > 0$. Let $\nu := \omega^n/n!$ and let $a$ be the integer of Definition-Lemma 4.1(3). Then the relative top form $t^{-a}\nu$ vanishes nowhere on $\mathcal{U}$. Let $\nu_0$ be the restriction of $t^{-a}\nu$ to $(X_0)_{\mathrm{reg}}$. Then $\nu_0$ is a nowhere vanishing algebraic $2n$-form on $(X_0)_{\mathrm{reg}}$. Since $t$ has $\zeta$-degree 0 and $\omega$ has $\zeta$-degree $r$, $\nu_0$ is homogeneous of $\zeta$-degree $nr$. Since $X_0$ is normal and $\Omega^{[2n]}_{X_0}$ is reflexive, $\nu_0$ generates $\Omega^{[2n]}_{X_0}$. Thus $\Omega^{[2n]}_{X_0}$ is generated by a $\zeta(\mathbb{G}_m)$-eigenvector whose character $\kappa$ satisfies $\langle \kappa, \zeta \rangle = nr$. By Lemma 2.14, applied to $X_0$ and the torus $\zeta(\mathbb{G}_m)$,

$$A_{X_0}(\mathrm{wt}_\xi) = \langle \kappa, \xi \rangle = \frac{2}{r} \cdot nr = 2n.$$

**Step 2.** In this step we show that $r \in \{1, 2\}$ and that $X_0$ is a conical symplectic variety with a symplectic form $\tau$ of $\zeta$-degree $r$.

We first check the hypotheses of Theorem 2.27 for $(X_0, \xi)$ and the torus $T_0$. The pair $(X_0, \xi)$ is a Fano cone of dimension $2n \geq 2$ with vertex $o$, $T_0$ acts on $X_0$ with a good faithful action, $\xi$ is a Reeb vector of $T_0$ with $A_{X_0}(\mathrm{wt}_\xi) = 2n$ by **Step 1**, and $(X_0, \xi)$ is K-polystable for $T_0$ by Lemma 2.19. Let $\chi := \zeta$. Since $\zeta : \mathbb{G}_m \to T_0$ is injective by **Step 1**, $\chi$ is a primitive cocharacter of $T_0$, and $\chi$ is a Reeb vector. Thus $X_0$, $T_0$, $\xi$, and $\chi$ satisfy the conditions of Set-up 3.1, where $g = d\rho^2 + \rho^2 h$ is the metric of Theorem 2.27 (see Remark 2.28). By Corollary 3.11, $X_0$, $\xi$, and $g$ satisfy the conditions of Set-up 2.31 for the torus $T_0$. The metric $g$ and the $\Phi$-cutoffs of Set-up 2.31(4) are invariant under the maximal compact subtorus of $T_0$, which contains the maximal compact subtorus $T_c$ of $T$. Since the action of $T$ on $X_0$ is good and faithful and $\xi$ is a Reeb vector of $T$, the variety $X_0$, the vector $\xi$, and the metric $g$ also satisfy the conditions of Set-up 2.31 for the torus $T$.

Next we check the hypotheses of Theorem 4.7 for $(\mathcal{X}, T, \Lambda, \zeta)$ with $r$ in place of $w$, $\xi$, and $g$: $\dim X = 2n$ with $n \geq 2$, $X$ is a symplectic variety whose symplectic form $\omega$ is algebraic of $\zeta$-degree $r > 0$, $T$ acts faithfully on $X$ and on $X_0$, $X_0$ is integral and normal with klt singularities, the $\zeta$-grading of $X_0$ is good, $(X_0, \xi)$ is a K-semistable Fano cone with $A_{X_0}(\mathrm{wt}_\xi) = 2n$, and $X_0$, $\xi$, and $g$ satisfy the conditions of Set-up 2.31. By Theorem 4.7(1), $X_0$ carries a closed reflexive 2-form $\tau$ which is non-degenerate on $(X_0)_{\mathrm{reg}}$ and extends regularly to a resolution of $X_0$. By Theorem 4.7(2), there exist a primitive cocharacter $\zeta_0$ of $T$ and $r' \in \{1, 2\}$ such that $\xi = \frac{2}{r'}\zeta_0$ and $\zeta_0(\lambda)^*\tau = \lambda^{r'}\tau$ for $\lambda \in \mathbb{C}^*$. Since $\zeta : \mathbb{G}_m \to T$ is injective, $\zeta$ is a primitive cocharacter of $T$. Since

$$\frac{2}{r}\zeta = \xi = \frac{2}{r'}\zeta_0,$$

the primitive cocharacter $\zeta_0 = \frac{r'}{r}\zeta$ is a positive rational multiple of the primitive cocharacter $\zeta$. Thus $\zeta_0 = \zeta$ and $r' = r$. In particular, $r \in \{1, 2\}$ and $\tau$ is homogeneous of $\zeta$-degree $r$.

Since $t$ has $\zeta$-degree 0, the $\zeta$-grading of $\mathcal{O}(X_0) = \mathcal{R}/t\mathcal{R}$ satisfies $(\mathcal{R}/t\mathcal{R})_j = \mathcal{R}_j/t\mathcal{R}_j$ for any $j \geq 0$. By Set-up 2.15(3), $(\mathcal{R}/t\mathcal{R})_0 = \mathbb{C}[t]/(t) = \mathbb{C}$. Thus $X_0$ is a conical symplectic variety with symplectic form $\tau$ of degree $r$ (Definition 2.9).

**Step 3.** In this step we construct a finite cover $h_0 : X_0' \to X_0$ by a product and show that the degree valuation of $X_0'$ is a minimizer of $\widehat{\mathrm{vol}}$.

By **Step 2**, $X_0$ is a conical symplectic variety of dimension $2n$ with symplectic form $\tau$ of degree $r$, whose grading is induced by the faithful $\mathbb{G}_m$-action $\zeta$, and $(X_0, \zeta)$ is K-polystable for the torus $T_0$, which acts faithfully on $X_0$ and contains $\zeta(\mathbb{G}_m)$. By Corollary 3.13, $(X_0, \zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$. By Theorem 8.8, applied to $X_0$, $\tau$, and $\zeta$, there exist integers $l \geq 1$ and $b, k \geq 0$, singular conical symplectic varieties $X_1, \dots, X_k$ of dimension at least 2 with symplectic forms $\omega_1, \dots, \omega_k$, a finite surjective morphism

$$h_0 : X_0' := \mathbb{A}^{2b} \times X_1 \times \cdots \times X_k \to X_0,$$

and a closed $\zeta(\mathbb{G}_m)$-invariant subset $\Sigma \subset X_0$ with $\operatorname{codim}_{X_0} \Sigma \geq 4$, such that $h_0$ is étale over $X_0 \backslash \Sigma$ and (1)–(4) of Theorem 8.8 hold with $lr$ in place of $w$. Let $\zeta_{X_0'}$ be the $\mathbb{G}_m$-action on $X_0'$ inducing the good positive diagonal grading of Theorem 8.8. Then

$$h_0 \circ \zeta_{X_0'}(\lambda) = \zeta(\lambda^l) \circ h_0 \quad \text{for any } \lambda \in \mathbb{G}_m.$$

Since $X_0'$ is a product of integral normal affine varieties, $X_0'$ is an integral normal affine variety. Since $\operatorname{codim}_{X_0} \Sigma \geq 4$, $h_0$ is quasi-étale (Definition 2.4).

Since $(X_0, \zeta)$ is K-semistable for $T_0$, Theorem 2.21 implies that the degree valuation $\mathrm{wt}_\zeta$ of $X_0$ is a minimizer of $\widehat{\mathrm{vol}}$ at $o$. Grade $\mathcal{O}(X_0')$ by $\zeta_{X_0'}$. Then $h_0^* \mathcal{O}(X_0)_d \subset \mathcal{O}(X_0')_{ld}$ for any $d \geq 0$. By Proposition 7.6, applied to $h_0$ and $l$, the $\zeta_{X_0'}$-grading of $\mathcal{O}(X_0')$ is good and positive, $h_0^{-1}(o)$ is a single point $o'$, and the degree valuation $v'$ of $X_0'$ for $\zeta_{X_0'}$ is a minimizer of $\widehat{\mathrm{vol}}$ at $o'$.

**Step 4.** In this step we extend $h_0$ to a finite cover of $\mathcal{X}$, and we show that the $l\zeta'$-grading of $\mathcal{O}(X_0')$ induced by the extension is $e$ times the $\zeta_{X_0'}$-grading.

Since $\Lambda$ acts on $t$ with weight $-1$, the ideal $t\mathcal{R}$ is $\Lambda$-invariant, so $\Lambda$ acts on $X_0$. The actions of $\Lambda$ and $\zeta(\mathbb{G}_m)$ on $X_0$ commute. Let $\Sigma'$ be the closure of $\Lambda \cdot \Sigma$ in $X_0$. Since $\Sigma$ is $\zeta(\mathbb{G}_m)$-invariant, $\Sigma'$ is $\zeta(\mathbb{G}_m) \times \Lambda$-invariant. Since $\Sigma'$ is the closure of the image of $\Lambda \times \Sigma$ under the action morphism, $\dim \Sigma' \leq \dim \Sigma + 1$. Thus $\operatorname{codim}_{X_0} \Sigma' \geq 3$, and $h_0$ is étale over $X_0 \backslash \Sigma'$.

Restricting the action of $T$ to $\zeta(\mathbb{G}_m)$, we obtain a marked test configuration $(\mathcal{X}, \zeta(\mathbb{G}_m), \Lambda, \zeta)$ of $X$ as in Set-up 2.15 with central fiber $X_0$. We check the hypotheses of Theorem 10.4 for $(\mathcal{X}, \zeta(\mathbb{G}_m), \Lambda, \zeta)$, $h_0$, and $\Sigma'$: $X_0$ is a conical symplectic variety for the $\zeta$-grading with symplectic form $\tau$ (**Step 2**), $X$ is a symplectic variety with the algebraic symplectic form $\omega$ of $\zeta$-degree $r > 0$, $h_0$ is a finite surjective quasi-étale morphism from the normal affine variety $X_0'$ (**Step 3**), and $\Sigma'$ is a $\zeta(\mathbb{G}_m) \times \Lambda$-invariant closed subset of codimension at least 3 such that $h_0$ is étale over $X_0 \backslash \Sigma'$. By Theorem 10.4, there exist integers $e, m \geq 1$ and a finite injective homomorphism of $\mathcal{R}$-algebras $\mathcal{R} \to \mathcal{R}'$ such that (1)–(3) of Theorem 10.4 hold, with $\zeta(\mathbb{G}_m)$ in place of $T$. In particular, there exist a torus $T'$ acting on $\mathcal{R}'$ and fixing $t$, an isogeny $T' \to \zeta(\mathbb{G}_m)$, a cocharacter $\zeta'$ of $T'$ mapping to $e\zeta$, an action of $\Lambda' := \mathbb{G}_m$ on $\mathcal{R}'$ of weight $-m$ on $t$, and an isomorphism $\mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X_0')$ compatible with $h_0$. Let

$$\widetilde{\mathcal{R}} := \mathcal{R} \otimes_{\mathbb{C}[t]} \mathbb{C}[s], \quad \widetilde{\mathcal{R}}' := \mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[s] \quad \text{with } t = s^m,$$

let $R' := \widetilde{\mathcal{R}}'/(s-1)\widetilde{\mathcal{R}}'$, and let $X' := \operatorname{Spec} R'$.

Through the isomorphism $\mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X_0')$, $\zeta'$ induces a $\mathbb{G}_m$-action on $X_0'$. Since $\operatorname{Spec} \mathcal{R}' \to \mathcal{X}$ is equivariant with respect to $T' \to \zeta(\mathbb{G}_m)$,

$$h_0 \circ \zeta'(\lambda) = \zeta(\lambda^e) \circ h_0 \quad \text{for any } \lambda \in \mathbb{G}_m. \tag{10.7}$$

We show that

$$\zeta'(\lambda^l) = \zeta_{X_0'}(\lambda^e) \quad \text{on } X_0' \text{ for any } \lambda \in \mathbb{G}_m. \tag{10.8}$$

For $\lambda \in \mathbb{G}_m$, let $g_\lambda := \zeta'(\lambda^l) \circ \zeta_{X_0'}(\lambda^{-e})$. By (10.7) and **Step 3**,

$$h_0 \circ g_\lambda = \zeta(\lambda^{el}) \circ h_0 \circ \zeta_{X_0'}(\lambda^{-e}) = \zeta(\lambda^{el}) \circ \zeta(\lambda^{-el}) \circ h_0 = h_0.$$

Thus $g_\lambda$ is an automorphism of $X_0'$ over $X_0$. Let $F$ be the group of automorphisms of $X_0'$ over $X_0$. Since $X_0'$ is integral and $h_0$ is finite and surjective, any element of $F$ is determined by the induced automorphism of the finite field extension $\mathbb{C}(X_0')/\mathbb{C}(X_0)$, so $F$ is finite. Since $(\lambda, x) \to g_\lambda(x)$ is a morphism $\mathbb{G}_m \times X_0' \to X_0'$, the subset $\{\lambda \in \mathbb{G}_m \mid g_\lambda = \gamma\}$ is closed in $\mathbb{G}_m$ for any $\gamma \in F$. Thus $\mathbb{G}_m$ is the disjoint union of finitely many closed subsets of the form $\{\lambda \in \mathbb{G}_m \mid g_\lambda = \gamma\}$. Since $\mathbb{G}_m$ is connected and $g_1 = \mathrm{id}_{X_0'}$, we have $g_\lambda = \mathrm{id}_{X_0'}$ for any $\lambda \in \mathbb{G}_m$, which is (10.8). In other words, an element of $\mathcal{O}(X_0')$ of $\zeta_{X_0'}$-degree $j$ has $l\zeta'$-degree $ej$.

**Step 5.** In this step we show that there exists a $T' \times \Lambda'$-equivariant isomorphism $\widetilde{\mathcal{R}}' \cong (\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}')[s]$ of $\mathbb{C}[s]$-algebras inducing the identity modulo $s$, where $\widetilde{\mathcal{R}}'$ is graded by $l\zeta'$ and $\Lambda'$ acts on $s$ with weight $-1$.

**Claim 10.6.** *Let $A$ be a $\mathbb{C}$-algebra, let $G$ be a torus, let $\zeta_G$ be a cocharacter of $G$, let $u$ be a variable, and let $\Lambda_u := \mathbb{G}_m$. Let $G$ and $\Lambda_u$ act on $A$, and on $A[u]$ through their actions on $A$, with $G$ fixing $u$ and $\Lambda_u$ acting on $u$ with weight $-1$. Assume that the $\zeta_G$-grading of $A$ is non-negative with $A_0 = \mathbb{C}$. Let $B$ be a $\mathbb{C}[u]$-algebra with an action of $G \times \Lambda_u$, where $G$ fixes $u$ and $\Lambda_u$ acts on $u$ with weight $-1$, and let $\psi : B \to A[u]$ be a $G \times \Lambda_u$-equivariant isomorphism of $\mathbb{C}$-algebras which induces an isomorphism $B/uB \to A[u]/uA[u] = A$. Then $\psi(u) = cu$ for some $c \in \mathbb{C}^*$. Let $\alpha$ be the automorphism of $A[u]$ which is the identity on $A$ and maps $u$ to $c^{-1}u$. Then $\alpha \circ \psi$ is a $G \times \Lambda_u$-equivariant isomorphism of $\mathbb{C}[u]$-algebras which induces, modulo $u$, the isomorphism $B/uB \to A$ induced by $\psi$.*

*Proof of Claim 10.6.* Since $G$ fixes $u$ and $\psi$ is $G$-equivariant, $\psi(u)$ is $G$-invariant, so $\psi(u)$ has $\zeta_G$-degree 0. Thus $\psi(u) \in A_0[u] = \mathbb{C}[u]$. Since $u$ has $\Lambda_u$-weight $-1$, so does $\psi(u)$, and hence $\psi(u) = cu$ for some $c \in \mathbb{C}$. Since $\psi$ is surjective and the induced map $B/uB \to A[u]/uA[u]$ is injective, $\psi(uB) = uA[u]$, that is, $cuA[u] = uA[u]$. Thus $c \neq 0$. Since $c^{-1}u$, like $u$, is fixed by $G$ and has $\Lambda_u$-weight $-1$, $\alpha$ is $G \times \Lambda_u$-equivariant, and $\alpha$ induces the identity modulo $u$. Since $\alpha \circ \psi(u) = u$, $\alpha \circ \psi$ is an isomorphism of $\mathbb{C}[u]$-algebras. □

Let $w := elr$, where $l$ is the integer of **Step 3** and $e$ is the integer of **Step 4**. By Theorem 10.4(3), $(\operatorname{Spec} \widetilde{\mathcal{R}}', T', \Lambda', \zeta')$ is a marked test configuration of $X'$ as in Set-up 2.15, with parameter $s$. Replacing $\zeta'$ by $l\zeta'$ multiplies all degrees by $l$, so $(\operatorname{Spec} \widetilde{\mathcal{R}}', T', \Lambda', l\zeta')$ is also a marked test configuration of $X'$ as in Set-up 2.15. Since $t = s^m$, we have

$$\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}' = \mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[s]/(s) = \mathcal{R}'/t\mathcal{R}' \cong \mathcal{O}(X_0').$$

Let $S_i := \mathcal{O}(X_i)$ for $1 \leq i \leq k$, let $S := S_1 \otimes \cdots \otimes S_k$, where $S := \mathbb{C}$ if $k = 0$, let $u_1, \dots, u_{2b}$ be the linear coordinates of $\mathbb{A}^{2b}$ of Theorem 8.8(2), and let $R_c := S[u_1, \dots, u_{2b}] = \mathcal{O}(X_0')$, graded by $l\zeta'$. Then $\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}' \cong R_c$ is a graded isomorphism. By **Step 4**, the $l\zeta'$-grading of $R_c$ is the $\zeta_{X_0'}$-grading multiplied by $e$. Thus Theorem 8.8(1)–(3) implies the following.

(a) For any $i$, $\omega_i$ has $l\zeta'$-degree $elr = w$, and $(S_i)_j = 0$ for $0 < j < w$.
(b) If $b > 0$, then $lr$ is even, so $w$ is even, and $u_1, \dots, u_{2b}$ have $l\zeta'$-degree $w/2$.
(c) Let $U := (X_1)_{\mathrm{reg}} \times \cdots \times (X_k)_{\mathrm{reg}}$, let $\tau_i := \mathrm{pr}_i^* \omega_i$ on $U$, and let $\mathrm{pr}_U : (X_0')_{\mathrm{reg}} = \mathbb{A}^{2b} \times U \to U$ be the projection. Let $\beta \in H^0(U, \Omega_U^2)_w$. By Theorem 8.8(3), $\mathrm{pr}_U^* \beta = \gamma_0 + \sum_{i=1}^k a_i \mathrm{pr}_i^* \omega_i$ for a constant 2-form $\gamma_0$ on $\mathbb{A}^{2b}$ and $a_1, \dots, a_k \in \mathbb{C}$. Evaluating both sides on pairs of tangent vectors to the fibers of $\mathrm{pr}_U$, we obtain $\gamma_0 = 0$. Since $\mathrm{pr}_U$ is smooth and surjective, $\beta = \sum_{i=1}^k a_i \tau_i$. Thus $H^0(U, \Omega_U^2)_w = \bigoplus_{i=1}^k \mathbb{C}\tau_i$.

Each $X_i$ is a symplectic variety, so $X_i$ has rational Gorenstein singularities by Lemma 2.10, and in particular $S_i$ is Cohen–Macaulay. By (a) and (c), $k$, $w$, $X_i = \operatorname{Spec} S_i$, $\omega_i$, $S$, $X_1 \times \cdots \times X_k$, $U$, and $\tau_i$ satisfy the conditions of Set-up 9.8, with $w$ in place of $r$.

We check the conditions of Corollary 9.19 for the marked test configuration $(\operatorname{Spec}\widetilde{\mathcal{R}}', T', \Lambda', l\zeta')$ of $X'$, the integer $b$, and the graded isomorphism $\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}' \cong R_c$. Since $\dim X'_0 = 2n > 0$, $b+k \geq 1$. By (b), $w$ is even if $b > 0$, and $\deg u_j = w/2$. Condition (1) of Corollary 9.19 is (a). For condition (2), assume that $b > 0$. By Theorem 10.4(1), $X'_0$ is a symplectic variety, so the vertex $o'$ of $\operatorname{Spec} R_c = X'_0$ is klt by Lemma 2.10. By **Step 4**, the degree valuation of $R_c$ for $l\zeta'$ is $ev'$. For any $\mu > 0$ and any valuation $v$ centered at $o'$, we have $\mathfrak{a}_\lambda(\mu v) = \mathfrak{a}_{\lambda/\mu}(v)$ for the valuation ideals, so

$$\operatorname{vol}(\mu v) = \mu^{-2n}\operatorname{vol}(v), \quad A_{X'_0}(\mu v) = \mu A_{X'_0}(v), \quad \widehat{\operatorname{vol}}(\mu v) = \widehat{\operatorname{vol}}(v)$$

(Definition 2.5 and [LWX18, Definition-Proposition 2.3 and the sentence after it]). Since $v'$ is a minimizer of $\widehat{\operatorname{vol}}$ at $o'$ by **Step 3**, $ev'$ is a minimizer of $\widehat{\operatorname{vol}}$ at $o'$. For condition (3), Theorem 10.4(3) implies that the pullback of $\omega$ is a symplectic form on $X'$ of $\zeta'$-degree $er$, so $X'_{\mathrm{reg}}$ carries a closed non-degenerate algebraic 2-form of $l\zeta'$-degree $elr = w$. By Corollary 9.19, there exists a $T' \times \Lambda'$-equivariant isomorphism $\psi : \widetilde{\mathcal{R}}' \to R_c[s]$ inducing the identity modulo $s$, where $\Lambda'$ acts on $s$ with weight $-1$. The $l\zeta'$-grading of $R_c$ is $e$ times the $\zeta_{X'_0}$-grading by **Step 4**, so it is non-negative with degree-zero part $\mathbb{C}$ by **Step 3**. By Claim 10.6, applied with $A = R_c$, $G = T'$, $\zeta_G = l\zeta'$, $u = s$, $\Lambda_u = \Lambda'$, and $B = \widetilde{\mathcal{R}}'$ with the graded isomorphism $\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}' \cong R_c$, we may assume that $\psi$ is an isomorphism of $\mathbb{C}[s]$-algebras. Composing $\psi$ with the inverse of the graded isomorphism $\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}' \cong R_c$, extended to $R_c[s]$ by $s \to s$, we obtain a $T' \times \Lambda'$-equivariant isomorphism $\psi' : \widetilde{\mathcal{R}}' \to (\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}')[s]$ of $\mathbb{C}[s]$-algebras inducing the identity modulo $s$.

**Step 6.** In this step we descend the product structure of **Step 5** to $\mathcal{R}$ over the parameter $t$.

We check the hypotheses of Lemma 9.4 for $(\mathcal{X}, T, \Lambda, \zeta)$, where the integer $m$ of Lemma 9.4 is the integer $m$ of **Step 4** with $t = s^m$, the integer $e$ of Lemma 9.4 is $el$, and the ring $\mathcal{R}'$ of Lemma 9.4 is $\widetilde{\mathcal{R}}'$ with the $l\zeta'$-grading.

(a) By **Step 2**, $X_0$ is a conical symplectic variety for the $\zeta$-grading whose symplectic form $\tau$ has degree $r > 0$.

(b) By Theorem 10.4(3), $\widetilde{\mathcal{R}} \subset \widetilde{\mathcal{R}}'$ are normal domains. Since $\mathcal{R} \to \mathcal{R}'$ is finite, $\widetilde{\mathcal{R}} \subset \widetilde{\mathcal{R}}'$ is a finite extension of domains.

(c) By Theorem 10.4(1), the $\zeta'$-grading of $\mathcal{R}'$ is non-negative. By Set-up 2.15(3) for $(\operatorname{Spec}\widetilde{\mathcal{R}}', T', \Lambda', \zeta')$, $T'$ fixes $s$. Thus the $l\zeta'$-grading of $\widetilde{\mathcal{R}}'$ is non-negative and $s$ has degree 0. Since $\operatorname{Spec}\mathcal{R}' \to \mathcal{X}$ is equivariant with respect to $T' \to \zeta(\mathbb{G}_m)$ and $\zeta'$ maps to $e\zeta$, an element of $\mathcal{R}$ of $\zeta$-degree $j$ has $l\zeta'$-degree $elj$. Thus the $l\zeta'$-grading of $\widetilde{\mathcal{R}}'$ extends the grading of $\widetilde{\mathcal{R}}$ of Lemma 9.4.

(d) Recall that $\psi' : \widetilde{\mathcal{R}}' \to (\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}')[s]$ is the isomorphism of **Step 5**. Since $\psi'$ is $T'$-equivariant, $\psi'$ is graded for $l\zeta'$. The $s$-adic completion of $(\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}')[s]$ is $(\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}')[[s]]$. Let $\phi : (\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}')[[s]] \to \widehat{\widetilde{\mathcal{R}}'}$ be the $s$-adic completion of the inverse of $\psi'$. Then $\phi$ is an isomorphism of $\mathbb{C}[[s]]$-algebras onto the $s$-adic completion of $\widetilde{\mathcal{R}}'$ which induces the identity modulo $s$, and the reduction of $\phi$ modulo $s^N$ is the graded isomorphism $(\widetilde{\mathcal{R}}'/s\widetilde{\mathcal{R}}')[s]/(s^N) \to \widetilde{\mathcal{R}}'/s^N\widetilde{\mathcal{R}}'$ induced by the inverse of $\psi'$ for any $N \geq 1$.

The hypotheses of Lemma 9.4 involve only the grading of $\widetilde{\mathcal{R}}'$, and not an action of $T$ on $\widetilde{\mathcal{R}}'$.

By Lemma 9.4, there exists a $T \times \Lambda$-equivariant isomorphism $f : \mathcal{R} \to (\mathcal{R}/t\mathcal{R})[t]$ inducing the identity modulo $t$. By **Step 2**, the $\zeta$-grading of $\mathcal{R}/t\mathcal{R}$ is non-negative with $(\mathcal{R}/t\mathcal{R})_0 = \mathbb{C}$. By Claim 10.6, applied with $A = \mathcal{R}/t\mathcal{R}$, $G = T$, $\zeta_G = \zeta$, $u = t$, $\Lambda_u = \Lambda$, and $B = \mathcal{R}$, we may assume that $f$ is an isomorphism of $\mathbb{C}[t]$-algebras. This is the isomorphism of $\mathbb{C}[t]$-algebras in Theorem 10.5.

Finally, by Set-up 2.15(4), there exist $T$-equivariant isomorphisms

$$\mathcal{R}/(t-1)\mathcal{R} \cong \mathcal{R}[t^{-1}]/(t-1) \cong R[t, t^{-1}]/(t-1) = R,$$

and $(\mathcal{R}/t\mathcal{R})[t]/(t-1) = \mathcal{R}/t\mathcal{R}$. Since $f$ is a $T$-equivariant isomorphism of $\mathbb{C}[t]$-algebras, $f$ induces, by setting $t = 1$, a $T$-equivariant isomorphism $R \cong \mathcal{R}/t\mathcal{R}$. Thus $X \cong X_0$ $T$-equivariantly. □

**Remark 10.7.** In the situation of Theorem 10.5, let $\eta$ be the vector field generating the action of $\Lambda$. By Theorem 10.5 and Lemma 2.18, $(\mathcal{X}, \eta)$ is a product in the sense of Definition 2.16 for the torus $T$.

## 11. Proofs of the main theorem and the corollary

In this section, we prove Theorem 1.2 and Corollary 1.3.

### 11.1. Proof of the main theorem.

In this subsection, we combine the results of Sections 3–10.

*Proof of Theorem 1.2.* The proof is long, so we divide it into several steps.

**Step 1.** In this step we prove (1) and (2).

By Theorem 5.4, there exist a real number $c > 0$ and a Kollár component $E$ over $(X \ni x)$ with $v = c\,\mathrm{ord}_E$ and $A_X(E) \in \{n, 2n\}$. Thus (1) holds, and (2) holds with $r := A_X(E)/n \in \{1, 2\}$. Since $\mathrm{ord}_E(\mathbb{C}(X)^*) = \mathbb{Z}$, we have

$$v(\mathbb{C}(X)^*) = c\mathbb{Z}. \tag{11.1}$$

In particular, $c$ is determined by $v$, and $E$ is determined by $\mathrm{ord}_E = v/c$.

**Step 2.** In this step we prove (3) and (5).

The germ $(X \ni x)$, the form $\omega$, the valuation $v$, the Kollár component $E$ of **Step 1**, and $r = A_X(E)/n$ satisfy the conditions of Set-up 6.1. The morphisms $f$, $q$, and $\pi$, the divisor $\mathcal{E}$, and the sheaf

$$\mathcal{F}_r = \Omega_{\mathcal{Y}}^{[2]}(\log \mathcal{E}) \otimes \mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})$$

of Set-up 6.1 are those of (3). By Definition-Lemma 6.2(2), $\pi^*\omega$ extends to a section $s_\omega$ of $\mathcal{F}_r^{\mathrm{an}}$ over a neighborhood of $x$. If $\omega$ is algebraic, then $s_\omega$ is the image of the pullback of $\omega$ under the morphism $\mathfrak{p}$ of Definition-Lemma 6.2(1), because $\mathcal{F}_r$ is torsion free and both sections coincide with $\pi^*\omega$ over the open subset of $X_{\mathrm{reg}}$ over which $\pi$ is an isomorphism. Hence $s_\omega$ is algebraic in this case. Thus there is a neighborhood $U$ of $x$ such that $\pi^*\omega \in H^0(\pi^{-1}(U), \mathcal{F}_r)$, understood analytically if $\omega$ is not algebraic.

Let $\theta := \theta_\omega = \mathrm{res}(s_\omega|_{\mathcal{E}})$ be the residue of $\omega$ (Definition-Lemma 6.2(4)), that is, the residue of $\pi^*\omega$. It is an algebraic section of $\Omega_{\mathcal{E}}^{[1]} \otimes \mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}$. By Theorem 6.3(1), $\theta \wedge (d\theta)^{n-1}$, computed in local trivializations of $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}$, vanishes nowhere on $\mathcal{E}_{\mathrm{reg}}$. Thus $\theta$ is an orbifold contact form with contact line bundle $\mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}$ (Definition 2.12), and (3) holds. By Theorem 6.3(3), $(\mathcal{E}, \mathcal{O}_{\mathcal{Y}}(-r\mathcal{E})|_{\mathcal{E}}, \theta)$ is a contact Fano orbifold, which is the first sentence of (5). Since $(X \ni x)$ is klt (Lemma 2.10), a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$ exists (Theorem 2.20), and applying (1)–(3) and the first sentence of (5) to it gives the second sentence of (5).

**Step 3.** In this step we reduce both assertions of (4) to the K-polystability of the quotient log Fano pair of $(W, \zeta)$.

Recall that $(X \ni x)$ is klt (**Step 2**), that $W = \mathrm{Spec}\,\mathrm{gr}_v \mathcal{O}_{X,x}$ is graded by the values of $v/c$, and that $\zeta$ is the $\mathbb{G}_m$-action defined by the grading. Let $o$ be the vertex of $W$. By Theorem 2.20, $(W, \xi_v)$ is a K-semistable Fano cone with a good action of the torus $T$ whose character lattice is $v(\mathbb{C}(X)^*) = c\mathbb{Z}$ by (11.1). Since $\xi_v$ is defined by the grading by the values of $v$ and $\zeta$ by the grading by the values of $v/c$, we have $T = \zeta(\mathbb{G}_m)$ and $\xi_v = c\zeta$. By Theorem 5.4, the action $\zeta$ is faithful. Thus $\zeta$ is a primitive cocharacter of $\zeta(\mathbb{G}_m)$ which is a Reeb vector, and

$$\mathrm{wt}_{\xi_v} = c\,\mathrm{wt}_\zeta\,.$$

Let $(Z, \Delta_Z)$ be the quotient log Fano pair of $(W, \zeta)$. By Lemma 2.7, applied with $W$ in place of $X$ and $\zeta(\mathbb{G}_m)$ in place of $T$, there exists a Kollár component $F$ over $(W \ni o)$ with $\mathrm{ord}_F = \mathrm{wt}_\zeta$ and $(F, \Delta_F) \cong (Z, \Delta_Z)$. In particular, $\mathrm{wt}_{\xi_v} = c\,\mathrm{ord}_F$ is divisorial, so the conditions of Proposition 5.1 hold for $(X \ni x)$ and $v$. By **Step 1**, the number $c$ and the Kollár component

$E$ of Proposition 5.1 are those of **Step 1**, and Proposition 5.1(4) implies $(E, \Delta_E) \cong (F, \Delta_F)$. Thus

$$(E, \Delta_E) \cong (F, \Delta_F) \cong (Z, \Delta_Z). \tag{11.2}$$

By Lemma 2.24, applied with $W$ in place of $X$, $\zeta$ in place of $\xi$, and 1 in place of $c$, the Fano cone $(W, \zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$ if and only if $(Z, \Delta_Z)$ is K-polystable. Hence, by (11.2), both assertions of (4) are equivalent to the K-polystability of $(Z, \Delta_Z)$, and it suffices to prove that $(Z, \Delta_Z)$ is K-polystable.

**Step 4.** In this step we assume that $n \geq 2$ and construct a marked test configuration of $W$ whose central fiber is K-polystable.

We first show that $(W, \zeta)$ is K-semistable for $\zeta(\mathbb{G}_m)$. By Theorem 2.20 and Theorem 2.21, applied with $W$ in place of $X$ and $\zeta(\mathbb{G}_m)$ in place of $T_0$, the valuation $\mathrm{wt}_{\xi_v}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{W,o}$. For $\lambda > 0$ and $w \in \mathrm{Val}_{W,o}$, we have $A_W(\lambda w) = \lambda A_W(w)$ and $\mathrm{vol}(\lambda w) = \lambda^{-2n} \mathrm{vol}(w)$, so

$$\widehat{\mathrm{vol}}(\lambda w) = \lambda^{2n} A_W(w)^{2n} \cdot \lambda^{-2n} \mathrm{vol}(w) = \widehat{\mathrm{vol}}(w).$$

Hence $\mathrm{wt}_\zeta = c^{-1} \mathrm{wt}_{\xi_v}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{W,o}$, and $(W, \zeta)$ is K-semistable for $\zeta(\mathbb{G}_m)$ by Theorem 2.21.

By Theorem 2.22 (see Remark 2.23), applied with $W$ in place of $X$, $\zeta$ in place of $\xi$, and $\zeta(\mathbb{G}_m)$ in place of $T$, there exists a $\zeta(\mathbb{G}_m)$-equivariant special test configuration $(\mathcal{X}, \eta)$ of $(W, \zeta)$ with Futaki invariant 0 such that the central fiber $X_0$ of the test configuration of $W$ satisfies the following: $(X_0, \zeta)$ is a Fano cone which is K-polystable for some torus $T'$ which acts faithfully on $X_0$ and contains $\zeta(\mathbb{G}_m)$.

Write $\mathcal{X} = \operatorname{Spec} \mathcal{R}$, let $t$ be the coordinate of $\mathbb{A}^1$, and let $\Lambda \cong \mathbb{G}_m$ be the one-dimensional torus generated by $\eta$. We check the conditions (1)–(4) of Set-up 2.15, applied with $W$ in place of $X$ and $\zeta(\mathbb{G}_m)$ in place of $T$.

(1) The variety $W$ is normal and affine (Theorem 2.20). The $\zeta$-grading of $\mathrm{gr}_v \mathcal{O}_{X,x}$ has non-negative degrees, and the degree-zero part is $\mathcal{O}_{X,x}/\mathfrak{m} = \mathbb{C}$, where $\mathfrak{m}$ is the maximal ideal, because $v > 0$ on $\mathfrak{m}$.

(2) The scheme $\mathcal{X}$ is an affine variety which is flat over $\mathbb{A}^1$ [LWX18, Definition 2.14(1)], so $\mathcal{R}$ is a finitely generated domain which is flat over $\mathbb{C}[t]$. Special test configurations are $\mathbb{Q}$-Gorenstein, and $\mathbb{Q}$-Gorenstein test configurations are normal by definition [LWX18, Definition 2.14, the last sentence, and Definition 2.16]. Thus $\mathcal{R}$ is normal.

Conditions (3) and (4) of Set-up 2.15. Since $\eta$ maps to $-t\partial_t$ under the projection $\mathcal{X} \to \mathbb{A}^1$, the torus $\Lambda$ acts on $t$ with weight $-1$, and the $\Lambda$-orbits induce an isomorphism $\mathcal{X} \times_{\mathbb{A}^1} \mathbb{C}^* \cong W \times \mathbb{C}^*$ [LWX18, Definition 2.14(2)]. The torus $\zeta(\mathbb{G}_m)$ acts fiberwise, commutes with $\Lambda$, and coincides with the action on the first factor of $W \times \mathbb{C}^*$ [LWX18, Definition 2.14(3)]. Hence $\zeta(\mathbb{G}_m)$ fixes $t$, and there exists a $\zeta(\mathbb{G}_m) \times \Lambda$-equivariant isomorphism

$$\mathcal{R}[t^{-1}] \cong \mathrm{gr}_v \mathcal{O}_{X,x}[t, t^{-1}],$$

where $\Lambda$ acts trivially on $\mathrm{gr}_v \mathcal{O}_{X,x}$. Since $\mathcal{R}$ is flat over $\mathbb{C}[t]$, we have $\mathcal{R} \subset \mathcal{R}[t^{-1}]$. Since $\mathrm{gr}_v \mathcal{O}_{X,x}$ has no negative degrees and $t$ has degree 0, the $\zeta$-grading $\mathcal{R} = \bigoplus_j \mathcal{R}_j$ has no negative degrees, and $\mathbb{C}[t] \subset \mathcal{R}_0 \subset \mathbb{C}[t, t^{-1}]$. Suppose that $\mathcal{R}_0$ contains a Laurent polynomial $g$ whose least exponent is $-k$ with $k > 0$. Then $t^{k-1} g \in \mathcal{R}_0$, and subtracting the polynomial part of $t^{k-1} g$ we obtain $at^{-1} \in \mathcal{R}$ for some $a \in \mathbb{C}^*$. Hence $t$ is a unit of $\mathcal{R}$ and $X_0 = \operatorname{Spec}(\mathcal{R}/t\mathcal{R})$ is empty, a contradiction. Thus $\mathcal{R}_0 = \mathbb{C}[t]$.

Therefore $(\mathcal{X}, \zeta(\mathbb{G}_m), \Lambda, \zeta)$ is a marked test configuration of $W$ with central fiber $X_0$, and it is a special test configuration of $(W, \zeta)$ with Futaki invariant 0.

**Step 5.** In this step we prove that $(Z, \Delta_Z)$ is K-polystable when $n \geq 2$.

Since Set-up 6.1 holds for $(X \ni x)$, $\omega$, $v$, and $E$ (**Step 2**) and $\mathrm{ord}_E = v/c$, the cone of Theorem 6.3 is $W$. By Theorem 6.3(4) and since $W$ is normal (Theorem 2.20), $W$ is a conical

symplectic variety of dimension $2n$ whose symplectic form $\omega_0$ has degree $r$ (Definition 2.9), and its grading is induced by the faithful $\mathbb{G}_m$-action $\zeta$ (**Step 3**). By **Step 4**, hypotheses (ii)–(iv) of Theorem 10.5 hold for $W$, the torus $\zeta(\mathbb{G}_m)$ in place of $T$, and the marked test configuration $(\mathcal{X}, \zeta(\mathbb{G}_m), \Lambda, \zeta)$, with the torus $T'$ of **Step 4** in (iv), and (i) is $n \geq 2$. Thus Theorem 10.5, applied with $W$ in place of $X$ and $\zeta(\mathbb{G}_m)$ in place of $T$, yields a $\zeta(\mathbb{G}_m)$-equivariant isomorphism $\iota : W \to X_0$.

Let $T'_W$ be the torus $T'$ acting on $W$ by $\lambda \cdot y := \iota^{-1}(\lambda \cdot \iota(y))$ for $\lambda \in T'$ and $y \in W$. The action of $T'_W$ is faithful, and $T'_W$ contains $\zeta(\mathbb{G}_m)$ because $\iota$ is $\zeta(\mathbb{G}_m)$-equivariant. The isomorphism $\iota$ of the Fano cones $(W, \zeta)$ and $(X_0, \zeta)$ intertwines the actions of $T'_W$ and $T'$, so pulling back by $\iota$ induces a bijection between the $T'$-equivariant special test configurations of $(X_0, \zeta)$ and the $T'_W$-equivariant special test configurations of $(W, \zeta)$ which preserves Futaki invariants and products. Hence $(W, \zeta)$ is K-polystable for $T'_W$ (Definition 2.16). By Corollary 3.13, applied with $W$ in place of $X$ and $T'_W$ in place of $T$, the pair $(Z, \Delta_Z)$ is K-polystable, and $(W, \zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$. By (11.2), $(E, \Delta_E)$ is K-polystable. Thus (4) holds when $n \geq 2$.

**Step 6.** In this step we prove that $(Z, \Delta_Z)$ is K-polystable when $n = 1$.

By Lemma 5.3, there exist a finite subgroup $G \subset \mathrm{SU}(2)$ and a Kollár component over $(X \ni x)$ with properties (1)–(3) of the lemma, where $X_1 := \mathbb{A}^2/G$, $x_1$ is the image of 0, $\ell := |G \cap \mathbb{G}_m \cdot \mathrm{id}_{\mathbb{A}^2}|$, and $\zeta_1$ is the $\mathbb{G}_m$-action on $X_1$ defined by the grading $\mathbb{C}[X_1] = \bigoplus_{j \geq 0} \mathbb{C}[X_1]_j$. By Theorem 2.20, the order of vanishing along the Kollár component of Lemma 5.3 is $\lambda v$ for some $\lambda > 0$. Since its value group is $\mathbb{Z}$, (11.1) implies $\lambda = c^{-1}$, so the Kollár component of Lemma 5.3 is the Kollár component $E$ of **Step 1**, which is determined by $\mathrm{ord}_E = v/c$. Hence, by Lemma 5.3(3), there exists an isomorphism of graded rings $\mathbb{C}[X_1] \cong \mathrm{gr}_v \mathcal{O}_{X,x}$, where the right side is graded by the values of $v/c$. Thus there exists an isomorphism $\psi : W \to X_1$ with $\psi \circ \zeta(\mu) = \zeta_1(\mu) \circ \psi$ for $\mu \in \mathbb{C}^*$. In particular, $X_1$ is normal and has klt singularities (Theorem 2.20), and $\psi$ induces an isomorphism between $(Z, \Delta_Z)$ and the quotient log Fano pair $(Z_1, \Delta_{Z_1})$ of $(X_1, \zeta_1)$. It suffices to prove that $(Z_1, \Delta_{Z_1})$ is K-polystable.

We verify the conditions of Proposition 3.12 with $X_1$ in place of $X$. Let $z_1, z_2$ be the coordinates of $\mathbb{A}^2$, let $g : \mathbb{A}^2 \to X_1$ be the quotient morphism, and let $U := X_1 \backslash \{x_1\}$. By Lemma 5.3(1), $g$ is étale over $U$, so $U$ is smooth. Let $\Omega_0 := dz_1 \wedge dz_2$ and $p_0 := |z_1|^2 + |z_2|^2$. Since $G \subset \mathrm{SU}(2)$, both $\Omega_0$ and $p_0$ are $G$-invariant. Since $g$ is étale over $U$, there exist a nowhere vanishing reflexive 2-form $\Omega$ on $X_1$ with $g^*\Omega = \Omega_0$ on $\mathbb{A}^2 \backslash \{0\}$, and a positive smooth function $p$ on $U$ with $p \circ g = p_0$. For $\mu \in \mathbb{C}^*$, the automorphism $\zeta_1(\mu^\ell)$ of $X_1$ is induced by the scalar multiplication by $\mu$ on $\mathbb{A}^2$ (Lemma 5.3(1)). Hence, for $\mu \in \mathbb{C}^*$ and $y \in U$, we have

$$\zeta_1(\mu^\ell)^*\Omega = \mu^2\Omega, \quad p(\zeta_1(\mu^\ell) \cdot y) = |\mu|^2 p(y) = |\mu^\ell|^{2b/2} p(y), \quad \text{where } b := 2/\ell.$$

Thus $\Omega$ is homogeneous of degree $b > 0$, and $p$ satisfies the homogeneity condition of Proposition 3.12 with $d = 2$. The open subset $U = X_{1,\mathrm{reg}} \backslash \{x_1\}$ is $\mathbb{G}_m$-invariant. On $U$, the form $dd^c p$ is Kähler, because

$$g^* dd^c p = dd^c p_0 = \sqrt{-1}(dz_1 \wedge d\bar{z}_1 + dz_2 \wedge d\bar{z}_2),$$

and $dd^c \log p \geq 0$, because $dd^c \log p_0$ is the pullback of the Fubini–Study form under $\mathbb{A}^2 \backslash \{0\} \to \mathbb{P}^1$. A direct calculation shows

$$(dd^c p_0)^2 = -2dz_1 \wedge d\bar{z}_1 \wedge dz_2 \wedge d\bar{z}_2 = 2(\sqrt{-1})^4 \Omega_0 \wedge \bar{\Omega}_0,$$

so $(dd^c p)^2 = 2(\sqrt{-1})^4 \Omega \wedge \bar{\Omega}$ on $U$. By Proposition 3.12, $(Z_1, \Delta_{Z_1})$ is K-polystable. Hence $(Z, \Delta_Z)$ is K-polystable, and (4) holds when $n = 1$ by **Step 3**. □

## 11.2. **Proof of the corollary.**

*Proof of Corollary 1.3.* Since $\omega$ is algebraic, $(X \ni x, \omega)$ is a symplectic singularity in the sense of [Bea99] used in [XZ26b], whose symplectic forms are algebraic reflexive 2-forms. Thus, by [XZ26b, Theorem 6.2], there exist a K-semistable Fano cone $(X_1, \xi_1)$ with vertex $x_1$ and a symplectic

form $\sigma_1$ on $X_1$ which is equivariant for the torus generated by $\xi_1$, that is, the smallest torus acting on $X_1$ whose Lie algebra contains $\xi_1$ [XZ26b, Definition 2.12], such that $(X \ni x, \omega)$ is formally isomorphic to $(X_1 \ni x_1, \sigma_1)$. In particular, there exists an isomorphism of $\mathbb{C}$-algebras $\varphi : \widehat{\mathcal{O}}_{X_1,x_1} \to \widehat{\mathcal{O}}_{X,x}$, and $\dim X_1 = 2n$. Since $X_1$ is a Fano cone, $X_1$ is klt. The form $\sigma_1$ is closed and non-degenerate on $X_{1,\mathrm{reg}}$, and it extends to a regular form on any log resolution of $X_1$ by [GKKP11, Theorem 1.4], hence on any resolution [Bea99, (1.2)]. Thus $X_1$ satisfies Definition 2.9, and $(X_1 \ni x_1)$ is a symplectic singularity of dimension $2n$.

Let $v_1 := \mathrm{wt}_{\xi_1}$. By Theorem 2.21, applied with $X_1$ in place of $X$, the valuation $v_1$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X_1,x_1}$. By Theorem 5.4, applied with $(X_1 \ni x_1)$ in place of $(X \ni x)$, there exist $c_1 > 0$ and a Kollár component $E_1$ over $(X_1 \ni x_1)$ with $v_1 = c_1 \,\mathrm{ord}_{E_1}$. By Lemma 5.2, applied with $(X \ni x)$ in place of $(X_2 \ni x_2)$ and $\varphi$ in place of $\phi$, there exists a Kollár component $E_2$ over $(X \ni x)$ such that $\mathrm{ord}_{E_2}$ is a minimizer of $\widehat{\mathrm{vol}}$ on $\mathrm{Val}_{X,x}$ and $\varphi$ induces an isomorphism of graded $\mathbb{C}$-algebras

$$\mathrm{gr}_{\mathrm{ord}_{E_1}} \mathcal{O}_{X_1,x_1} \cong \mathrm{gr}_{\mathrm{ord}_{E_2}} \mathcal{O}_{X,x}.$$

By Theorem 2.20, $\mathrm{ord}_{E_2} = \lambda v$ for some $\lambda > 0$. Since $\mathrm{ord}_{E_2}(\mathbb{C}(X)^*) = \mathbb{Z}$ and $v(\mathbb{C}(X)^*) = c\mathbb{Z}$, we have $\lambda = c^{-1}$. Hence $\mathrm{gr}_{\mathrm{ord}_{E_2}} \mathcal{O}_{X,x}$ is $\mathrm{gr}_v \mathcal{O}_{X,x}$ graded by the values of $v/c$.

Write $X_1 = \operatorname{Spec} R_1$, and let $R_1 = \bigoplus_\alpha R_{1,\alpha}$ be the decomposition into eigenspaces of the torus of the Fano cone $(X_1, \xi_1)$. For $h \in R_1$ with components $h_\alpha \in R_{1,\alpha}$, we have $v_1(h) = \min\{\langle \alpha, \xi_1 \rangle \mid h_\alpha \neq 0\}$. Hence, for any $s \in \mathbb{R}$, the valuation ideal $\{v_1 \geq s\}$ of $\mathcal{O}_{X_1,x_1}$ is generated by $\bigoplus_{\langle \alpha,\xi_1\rangle \geq s} R_{1,\alpha}$, and $\mathrm{gr}_{v_1} \mathcal{O}_{X_1,x_1} \cong R_1$ as graded rings, where $\mathrm{gr}_{v_1}$ is graded by the values of $v_1$ and $R_{1,\alpha}$ has degree $\langle \alpha, \xi_1 \rangle$. Since $\mathrm{ord}_{E_1} = c_1^{-1} v_1$, the grading of $\mathrm{gr}_{\mathrm{ord}_{E_1}} \mathcal{O}_{X_1,x_1}$ is the grading of $\mathrm{gr}_{v_1} \mathcal{O}_{X_1,x_1}$ divided by $c_1$, so $\mathrm{gr}_{\mathrm{ord}_{E_1}} \mathcal{O}_{X_1,x_1} \cong R_1$, where $R_{1,\alpha}$ has degree $\langle \alpha, \xi_1 \rangle / c_1$. Combining the isomorphism of graded $\mathbb{C}$-algebras of Lemma 5.2, the equality $\mathrm{gr}_{\mathrm{ord}_{E_2}} \mathcal{O}_{X,x} = \mathrm{gr}_v \mathcal{O}_{X,x}$ graded by the values of $v/c$, and the isomorphism $\mathrm{gr}_{\mathrm{ord}_{E_1}} \mathcal{O}_{X_1,x_1} \cong R_1$, we obtain an isomorphism $\mathrm{gr}_v \mathcal{O}_{X,x} \cong R_1$ which maps the part of $v/c$-degree $j$ onto $\bigoplus_{\langle \alpha,\xi_1\rangle = c_1 j} R_{1,\alpha}$ for any $j \geq 0$. Thus $W \cong X_1$, and the isomorphism sends the vertex $o$ of $W$ to $x_1$, because both points are the common zeros of the elements of positive degree. Therefore

$$\widehat{\mathcal{O}}_{W,o} \cong \widehat{\mathcal{O}}_{X_1,x_1} \cong \widehat{\mathcal{O}}_{X,x}.$$

By Theorem 1.2(1), there exist a real number $c' > 0$ and a Kollár component $E$ over $(X \ni x)$ with $v = c' \,\mathrm{ord}_E$. Since $\mathrm{ord}_E(\mathbb{C}(X)^*) = \mathbb{Z}$, we have $v(\mathbb{C}(X)^*) = c'\mathbb{Z}$, so $c' = c$ and $v = c\,\mathrm{ord}_E$. Since $A_X(v) = cA_X(E)$, Theorem 1.2(2) implies $r = A_X(v)/(cn) = A_X(E)/n \in \{1, 2\}$. With $E$ and $r$, Set-up 6.1 holds, and since $\mathrm{ord}_E = v/c$ the cone of Theorem 6.3 is $W$. By Theorem 6.3(4), Theorem 2.20 (normality of $W$), and Theorem 5.4 (faithfulness of $\zeta$), $W$ is a conical symplectic variety whose grading is induced by $\zeta$ and whose symplectic form $\omega_0$ has degree $r$ (Definition 2.9). By Theorem 1.2(4), the Fano cone $(W, \zeta)$ is K-polystable for the torus $\zeta(\mathbb{G}_m)$. □

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